Al-Azhar University-Gaza
Faculty of Applied Medical Sciences
Nursing Department

Radiology for Nursing

General Radiology

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Master of Public Health
Master of Medical Imaging

Introduction to Medical Imaging and Radiologic Sciences

- **Medical imaging** refers to techniques and processes used to create images of various parts of the human body for diagnostic and treatment purposes. The term, medical imaging, includes various radiological imaging techniques.
- Medical imaging seeks to reveal internal structures hidden by the skin and bones, as well as to diagnose and treat disease.
- Medical imaging also establishes a database of normal anatomy and physiology to make it possible to identify abnormalities.

History of Radiologic Technology

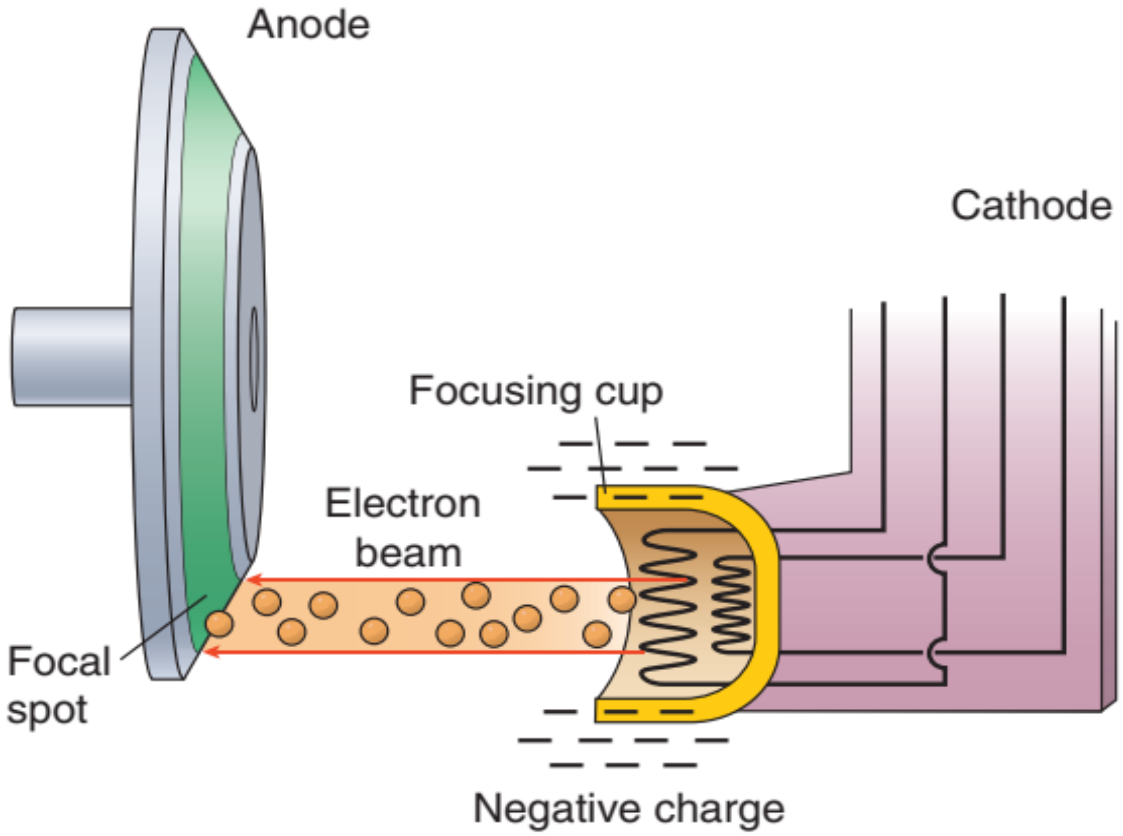
- The field of radiologic technology began on November 8, 1895, when Wilhelm Röntgen, a German physicist, was working with a vacuum tube and was exploring their properties outside glass tubes.
- He was working in a darkened room with black cardboard covering the tube. Coincidentally, there were many fluorescent plates in his laboratory at the time of his experiments with light.
- During this work, Röntgen observed that a fluorescent plate was emitting light. This effect had to be caused by invisible rays being emitted from the tube.
- Röntgen became the first radiographer when he produced a series of photographs of radiographic images, most notably the image of his wife's hand
- He termed these **invisible rays, x-rays** because **x** is the symbol for an unknown variable.

Hand mit Ringen 5. 2. 96.
Eigenthum von Prof. Zehnder
Freiburg i/B
PHYSIK. INSTITUT
DER UNIVERSITÄT
FREIBURG i. Br.

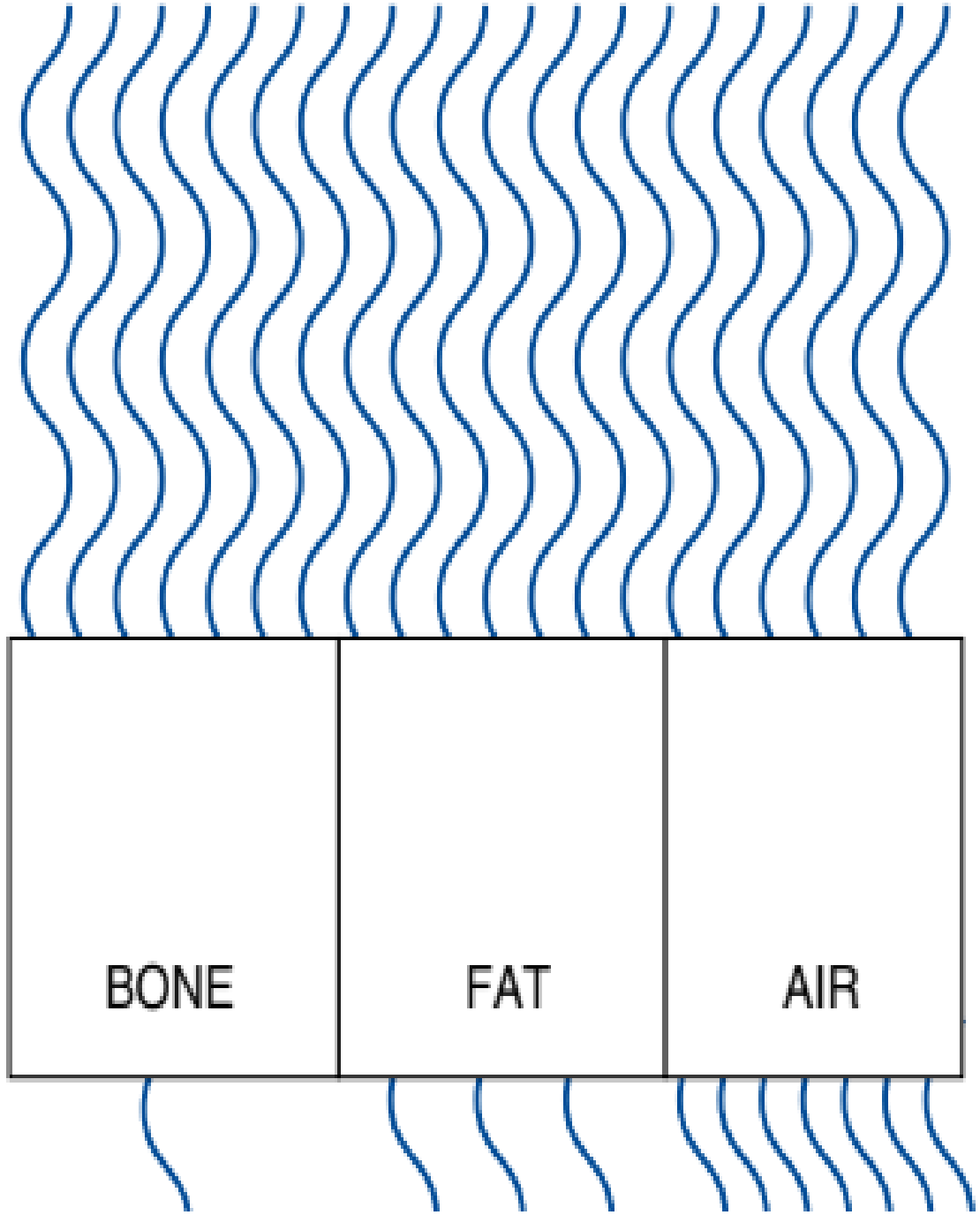
- **Radiation**
- Radiation is defined as the emission and propagation of energy through space or a substance in the form of waves or particles.
- **X-rays**
- Type of radiation that are invisible waves possessing certain properties that enable them to penetrate structures. X-rays are the energy source that are waves with no mass and are part of a grouping called electromagnetic radiation, which are actually waves of energy or the individual units of energy, called photons.
- **Ionization**
- When an orbiting electron is ejected from its shell in an electrically stable or neutral atom, the process is called ionization. The remainder of the atom has a positive charge, and the ejected electron has a negative charge. Electrons can be removed from atoms by heating or interaction with x-ray photons.

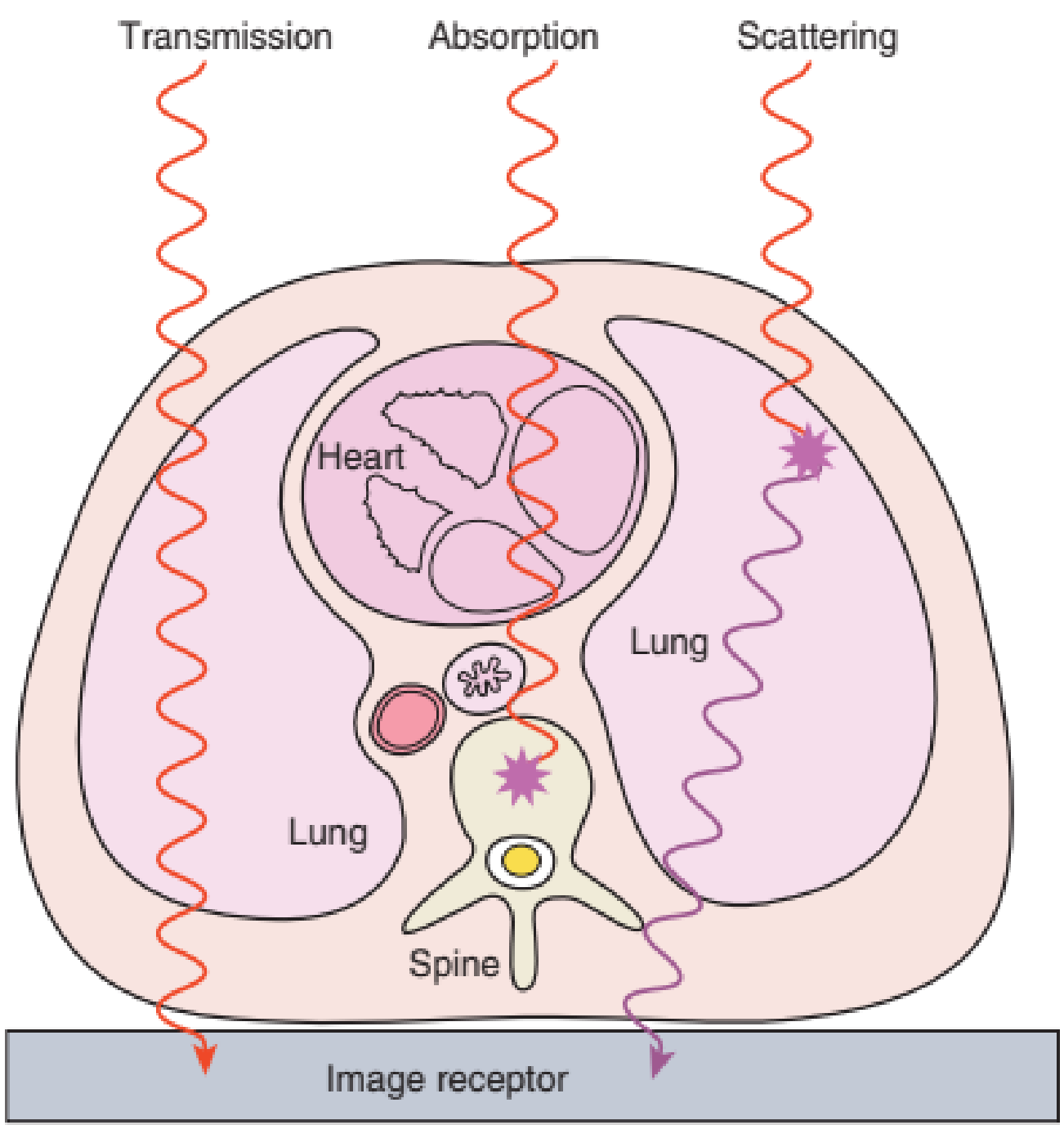
Radiography (X-rays, plain films)

- X-rays are produced in an X-ray tube by focusing a beam of high-energy electrons onto a tungsten target. X-rays are a form of electromagnetic radiation, able to pass through the human body and produce an image of internal structures.



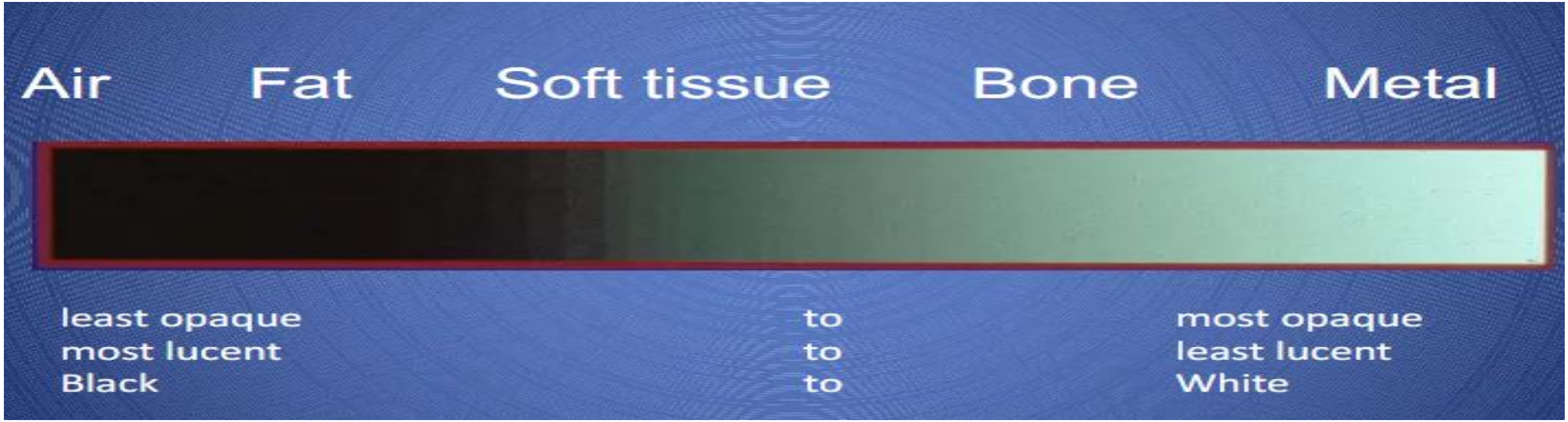
- The resulting image is called a radiograph, more commonly known as an ‘X-ray’ or ‘plain film’. As a beam of X-rays passes through the human body, some of the X-rays are absorbed or scattered producing reduction or attenuation of the beam.
- Tissues of high density and/or high atomic number cause more X-ray beam attenuation and are shown as lighter grey or white on a radiograph.
- Less dense tissues and structures cause less attenuation of the X-ray beam, and appear darker on radiographs than tissues of higher density.


BONE
FAT
AIR


Transmission
Absorption
Scattering
Heart
Lung
Lung
Spine
Image receptor

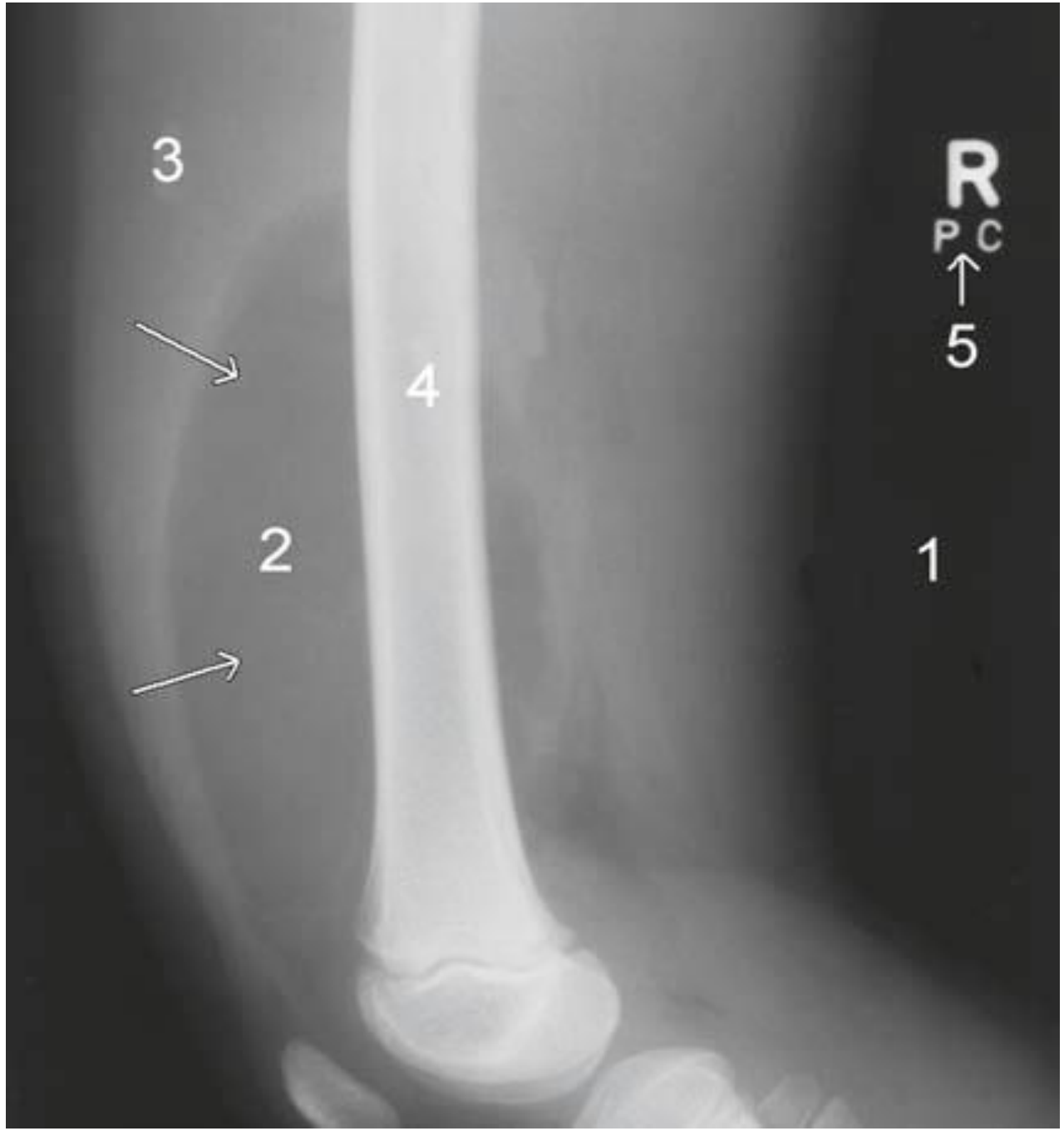
- **Five principal densities are recognized on plain radiographs**

1. **Air/gas: black**, e.g. lungs, bowel and stomach
2. **Fat: dark grey**, e.g. subcutaneous tissue layer, retroperitoneal fat
3. **Soft tissues/water: light grey**, e.g. solid organs, heart, blood vessels, muscle and fluid-filled organs such as bladder
4. **Bone: off-white**
5. **Contrast material/metal: bright white**



- **The five principal radiographic densities.**
- This radiograph of a benign lipoma (arrows) in a child's thigh demonstrates the five basic radiographic densities:

1. Air
2. Fat
3. Soft tissue
4. Bone
5. Metal

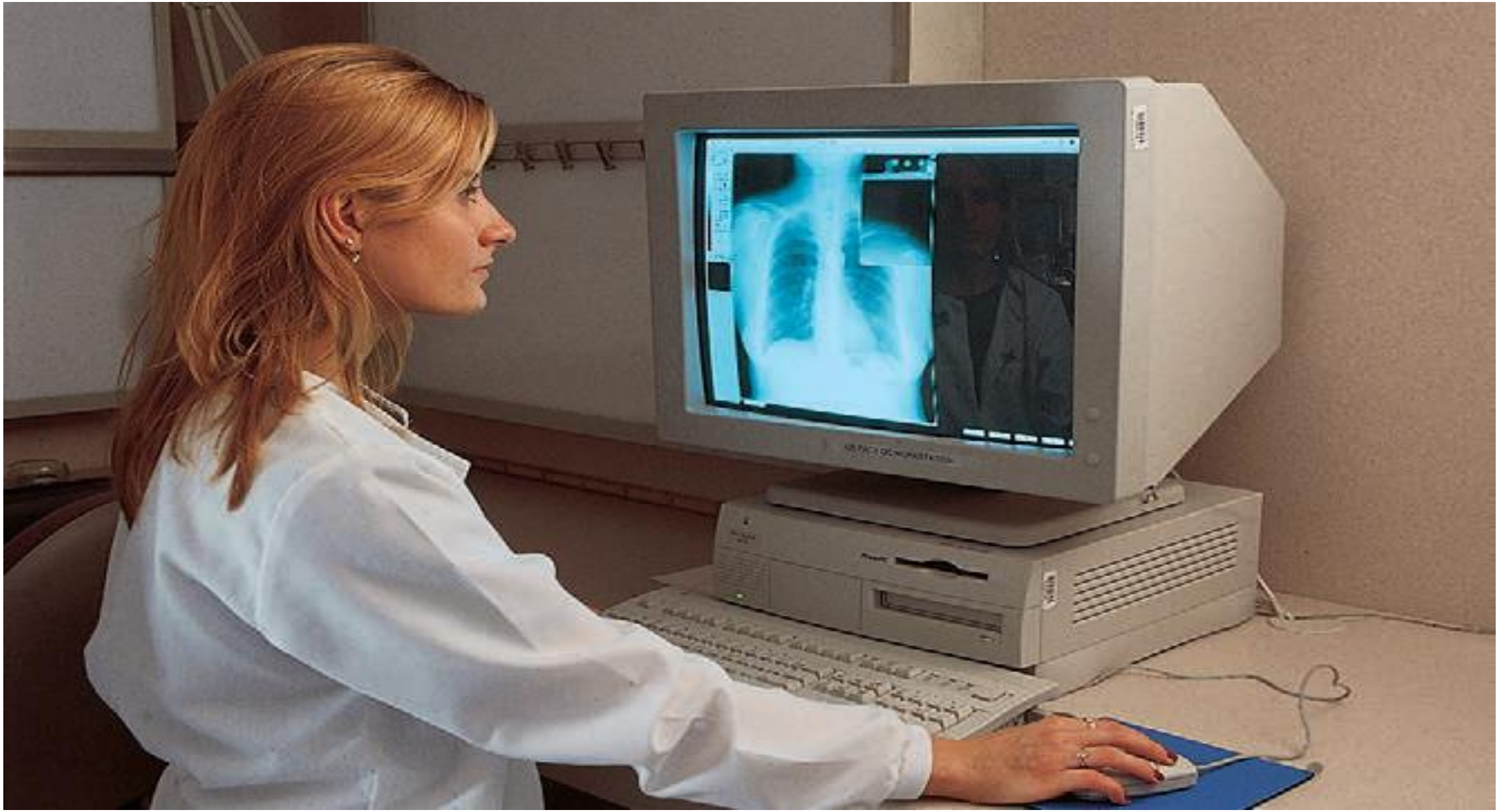


Computed radiography, Digital radiography and Picture Archiving and Communication Systems

- In the past, X-ray films were processed in a darkroom and processors. In modern practice, radiographic images are produced digitally using one of two processes, computed radiography (CR) and digital radiography (DR).
- Computed radiography (CR) involves conventional radiographic projection radiography in which the latent image (the unseen image) is produced in digital format using computer technology.
- The CR system uses a conventional radiography machine and conventional positioning and technical factors.

- The **Image Receptor (IR)** consists of a photostimulable phosphor material plate inside a closed cassette.
- These storage-phosphor IRs are often referred to as "plates" or "imaging plates" (IPs).
- After exposure, the CR cassette is inserted into an image reader device, where the IP is scanned by a laser beam, releasing the stored energy pattern, and the final image appears on a computer monitor.
- The radiographer can store the image in the computer system to be read directly from the monitor by the radiologist or print it on laser film.

Radiographer inserting IP into an image reader unit on CR system. The unit scans the plate with a laser beam and places the digitized image of the body part in a computer for reading on a monitor or, if necessary, for printing on a laser film.



Radiographer at the monitor uses the mouse to adjust the CR image of the body part to the proper size, density, and contrast before electronically sending the image for reading.

- The newest digital x-ray systems are called **digital radiography (DR)** systems.
- These are similar to CR systems, in that they convert x-ray energy patterns into digital signals, but the IR is not housed in any type of cassette. The IR does not need to be removed from its housing for the image information to be extracted or processed.
- **An analogue-digital converter (ADC)** produces a digital image. DR uses a detector screen containing silicon detectors that produce an electrical signal when exposed to X-rays. This signal is analyzed to produce a digital image.

- Digital images obtained by CR and DR are sent to viewing workstations for interpretation.

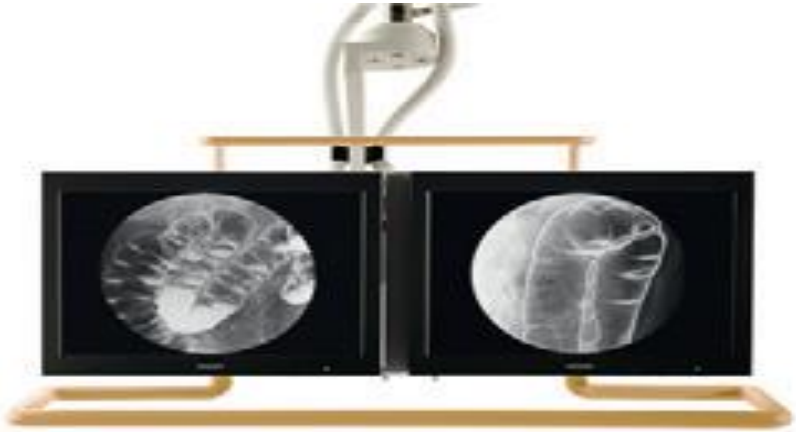
- Many medical imaging departments now employ large computer storage facilities and networks known as **picture archiving and communication systems (PACS).**

- Images obtained by CR and DR are stored digitally. Images can be displayed on monitors throughout the hospital in wards, meeting rooms and operating theatres as required .

NEC
R

Fluoroscopy

- Radiographic examination of the anatomy and motion of internal structures by a constant stream of X-rays is known as fluoroscopy.
- Fluoroscopy is a type of medical imaging that shows a continuous X-ray image on a monitor, much like an X-ray movie.
- During a fluoroscopy procedure, an X-ray beam is passed through the body. The image is transmitted to a monitor so the movement of a body part or of an instrument or contrast agent ("X-ray dye") through the body can be seen in detail.

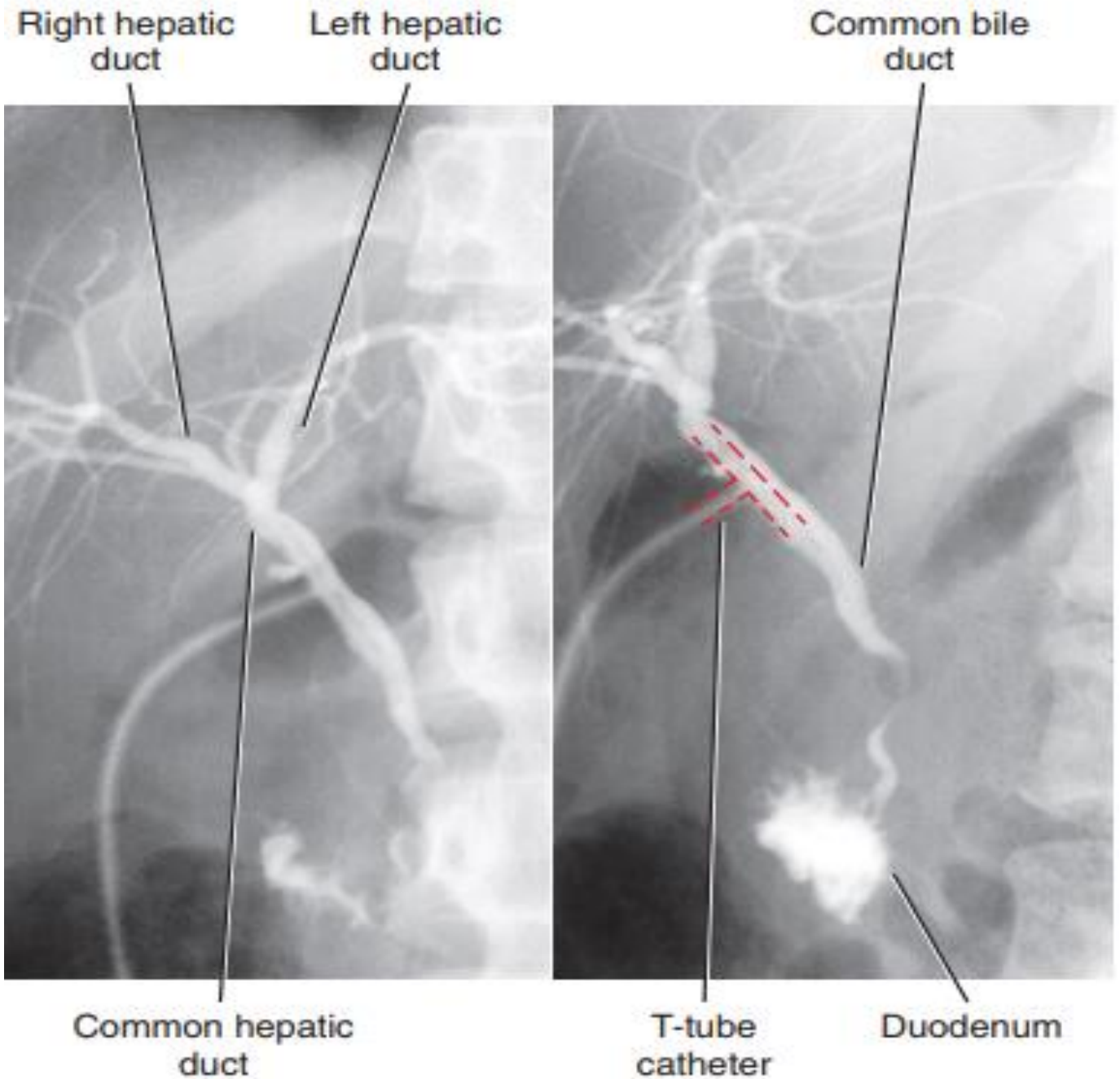


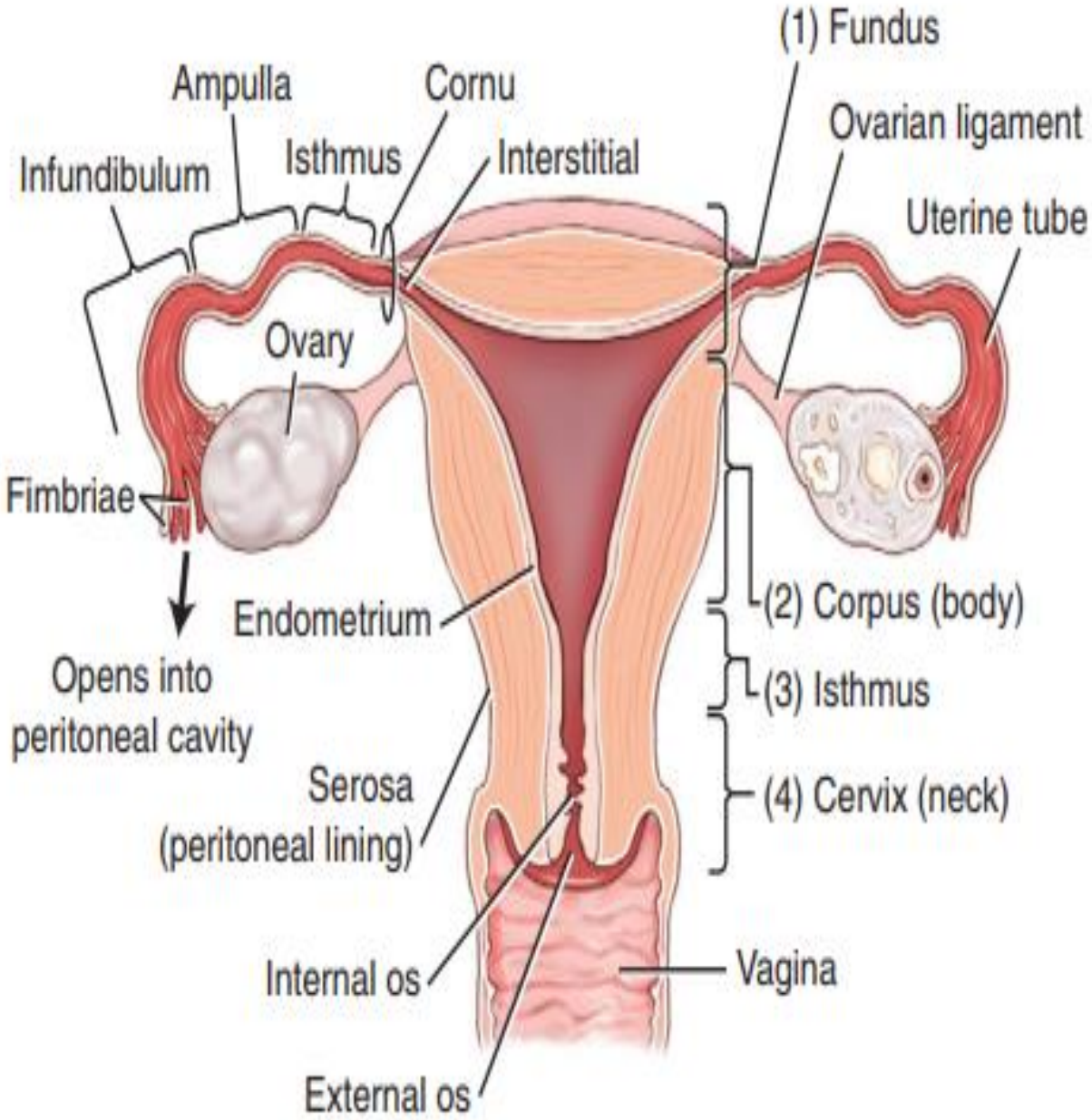
PHILIPS

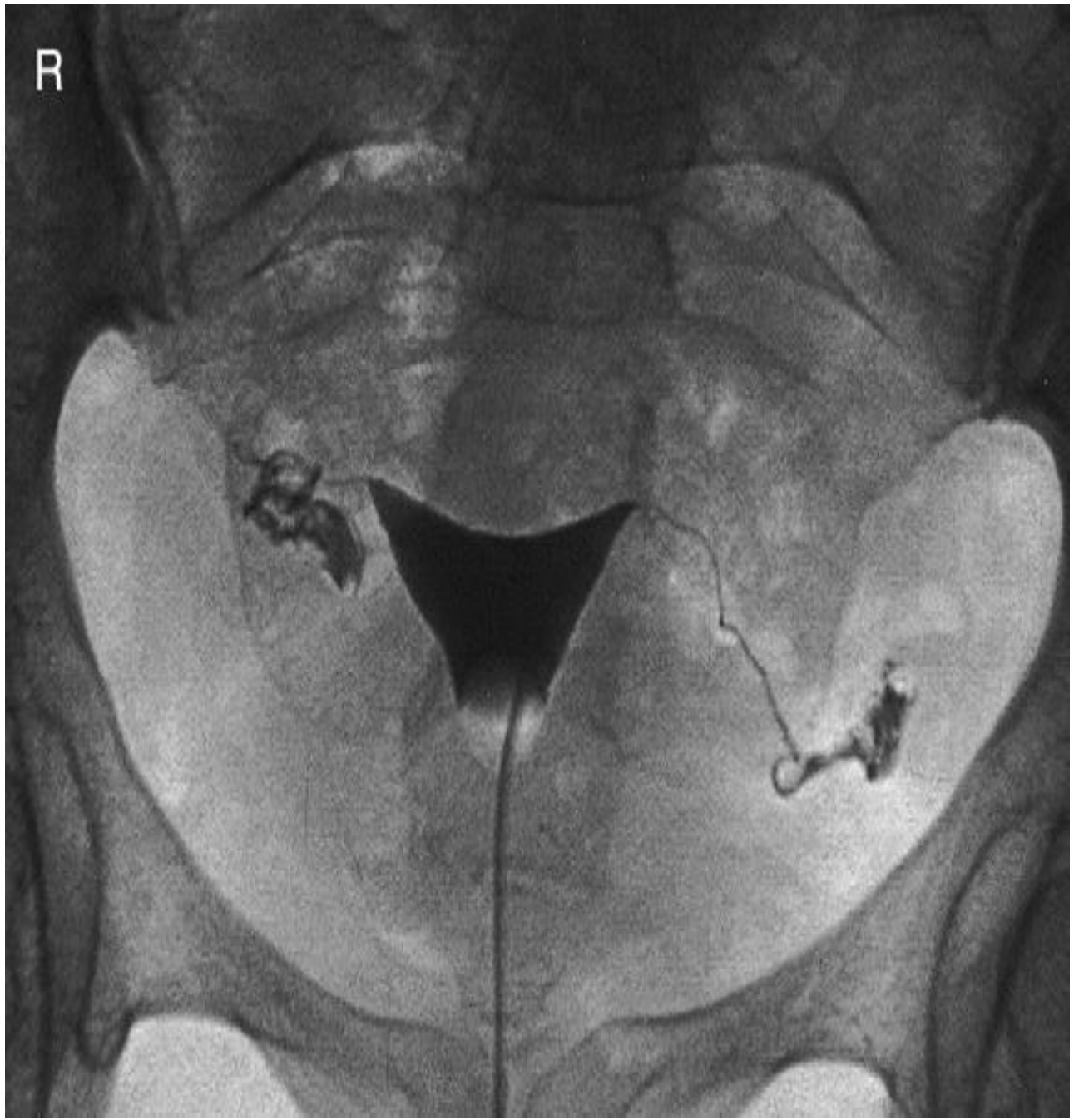
- **Uses of fluoroscopy include:**

1. Angiography and interventional radiology
2. Contrast studies of the gastrointestinal tract
3. General surgery, e.g. operative cholangiography
4. Urology, e.g. retrograde pyelography
5. Orthopaedic surgery, e.g. reduction and fixation of fractures, joint replacements

Upper GI Series


Right hepatic duct
Left hepatic duct
Common bile duct
Common hepatic duct
T-tube catheter
Duodenum


(1) Fundus
Ampulla
Cornu
Ovarian ligament
Infundibulum
Isthmus
Interstitial
Uterine tube
Ovary
Fimbriae
Endometrium
(2) Corpus (body)
Opens into
peritoneal cavity
(3) Isthmus
Serosa
(peritoneal lining)
(4) Cervix (neck)
Internal os
Vagina
External os


R

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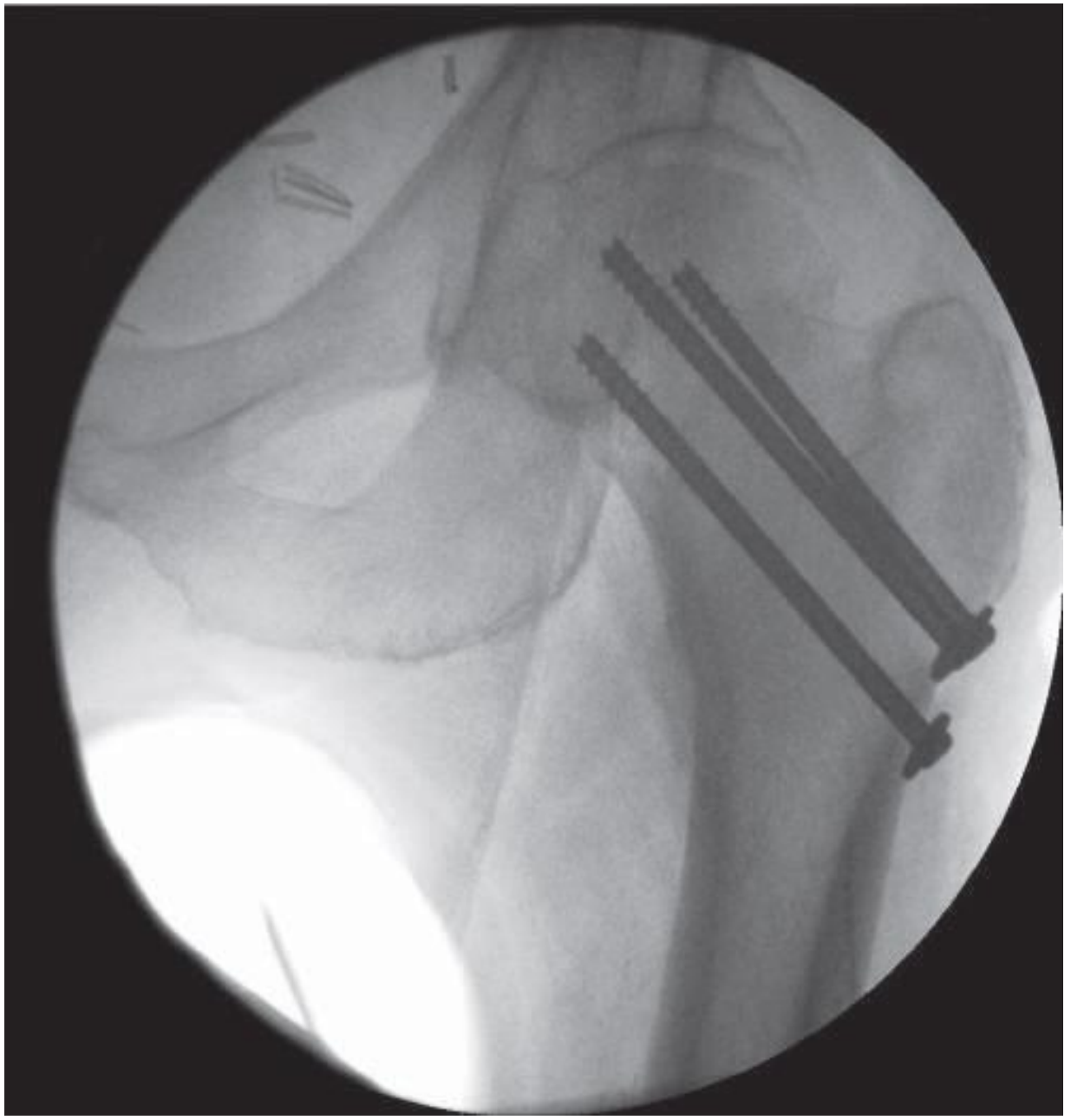

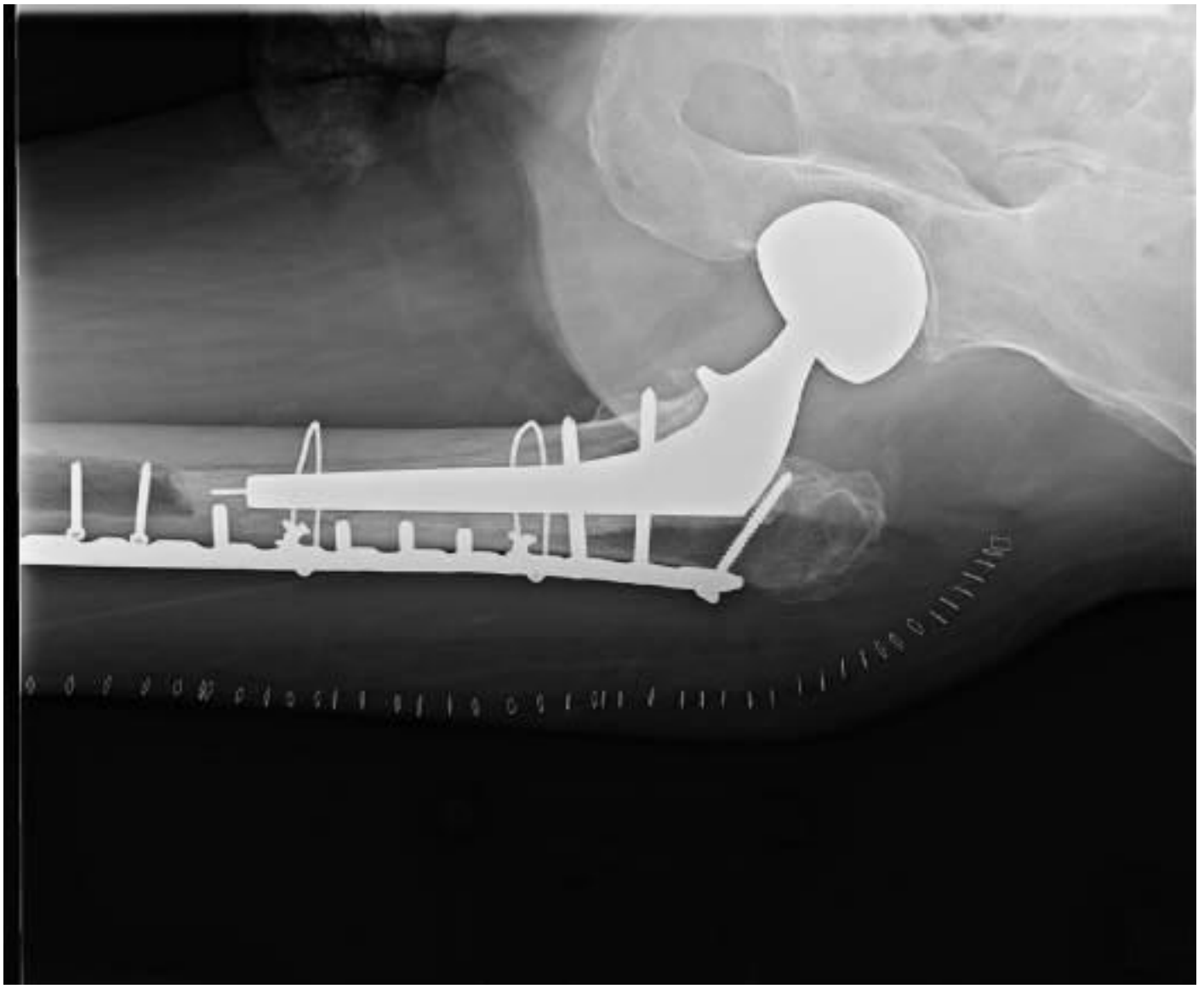

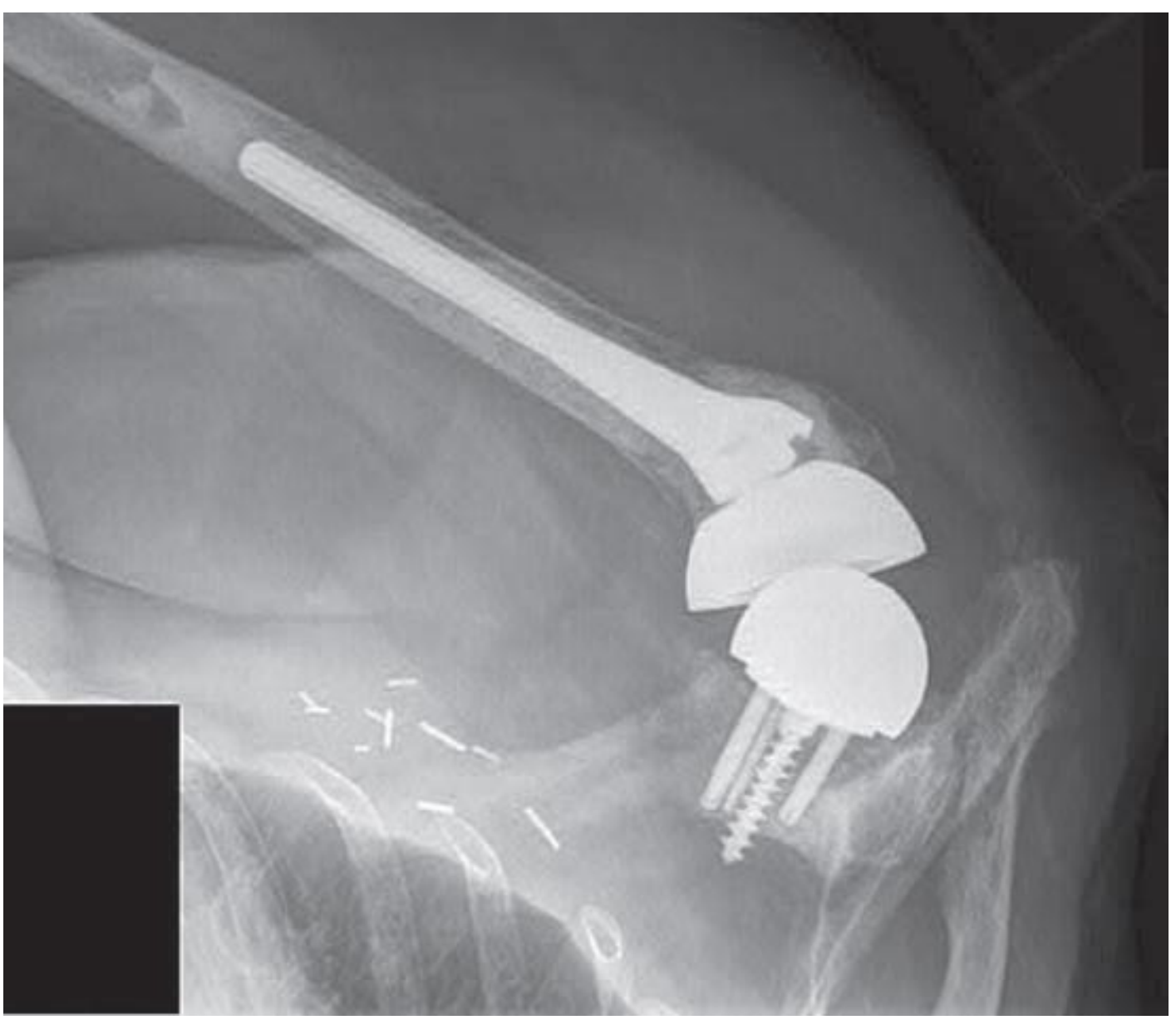

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Al-Azhar University-Gaza
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Radiology for Nursing

Advance Medical Imaging Modalities

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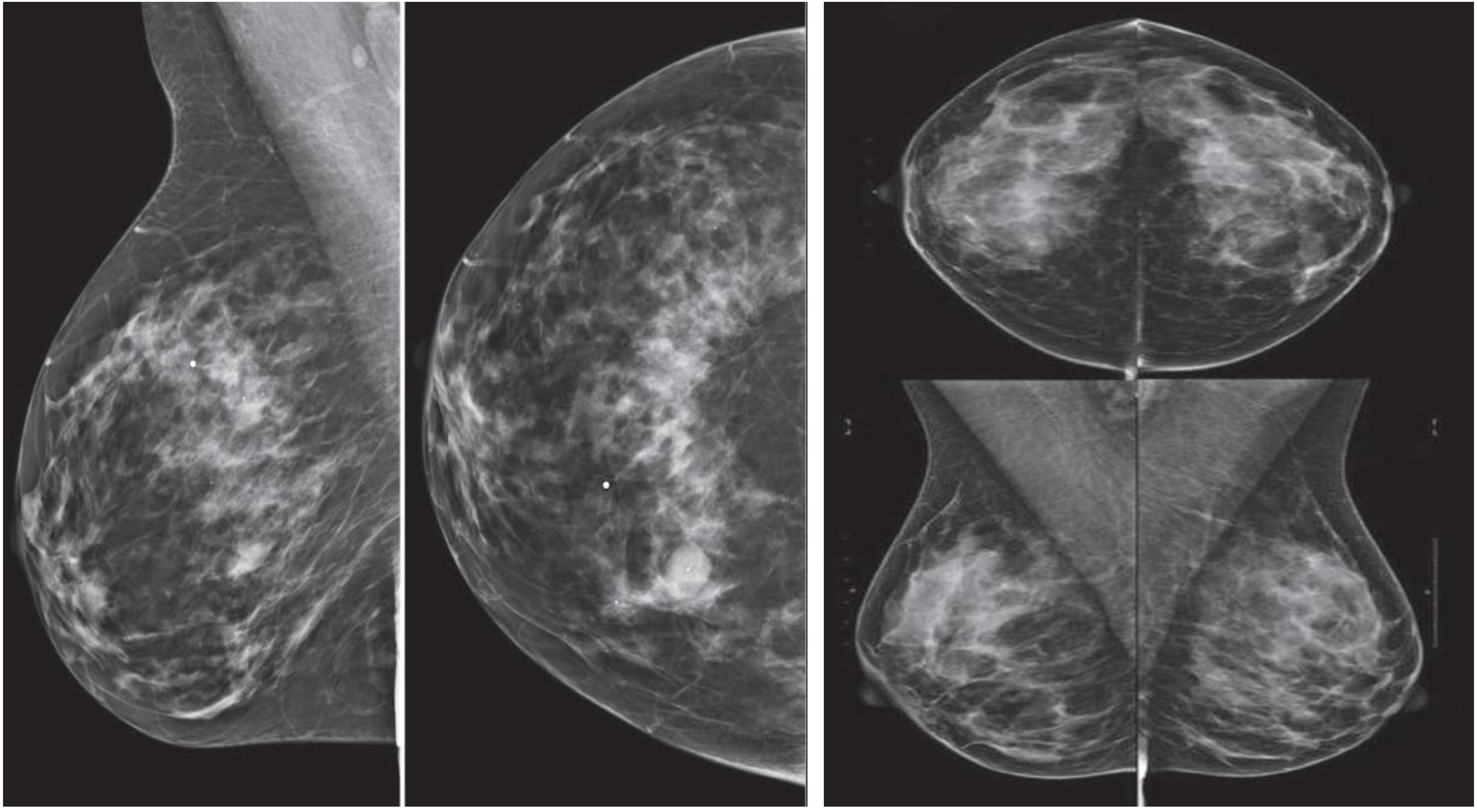
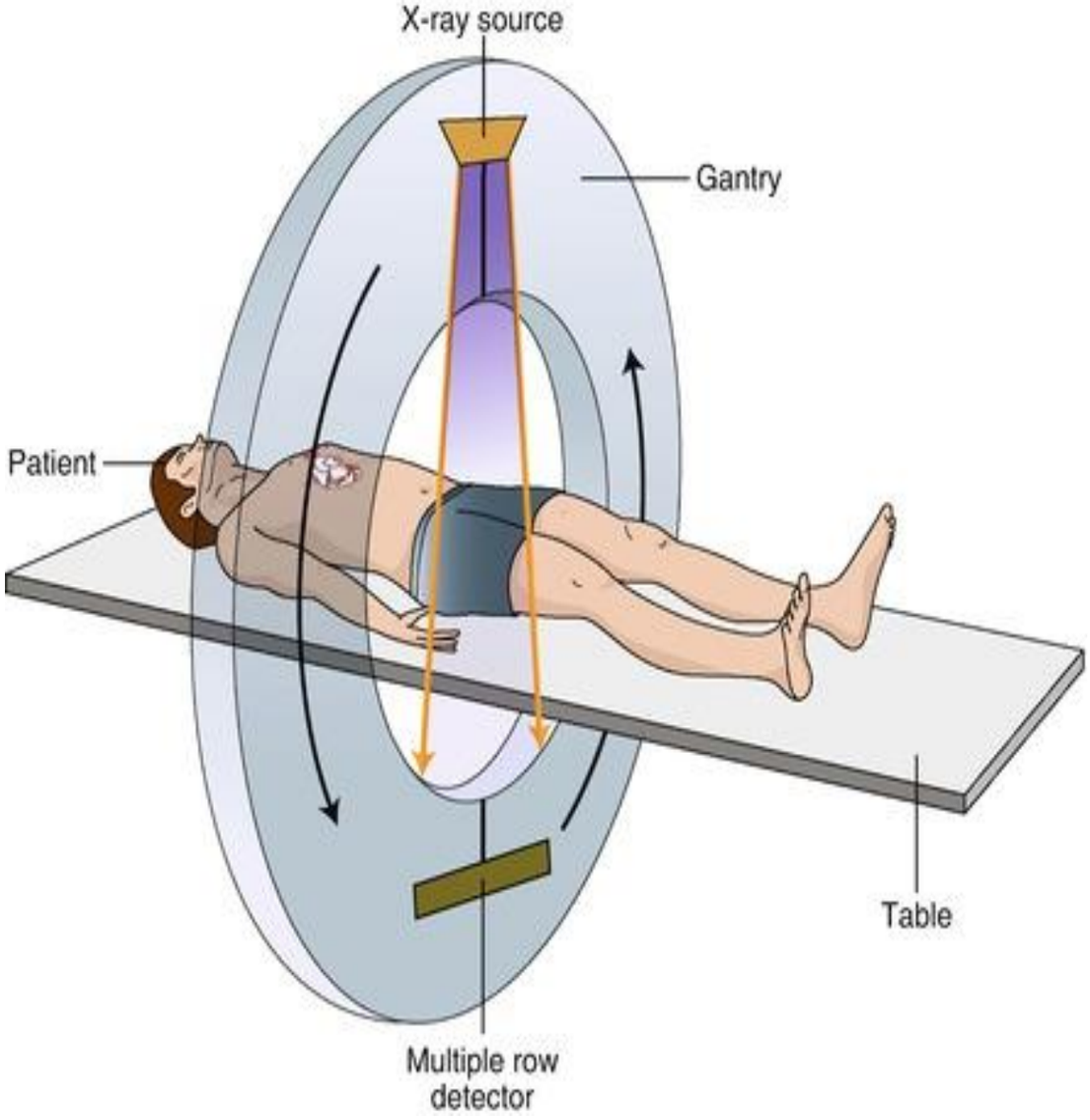
- **Mammography**
- Mammography (also called mastography) is the process of using low-energy X-rays to examine the human breast for diagnosis and screening.
- The goal of mammography is the early detection of breast cancer, typically through detection of characteristic masses or microcalcifications.
- **What is the difference between a screening mammogram and a diagnostic mammogram?**
- While screening mammograms are routinely administered to detect breast cancer in women who have no apparent symptoms, diagnostic mammograms are used after suspicious results on a screening mammogram or after some signs of breast cancer alert the physician to check the tissue.

- **American Cancer Society Guidelines for the Early Detection of Cancer**
- Women ages 40 to 44 should have the choice to start annual breast cancer screening with mammograms if they wish to do so.
- Women age 45 to 54 should get mammograms every year.
- Women 55 and older should switch to mammograms every 2 years, or can continue yearly screening.



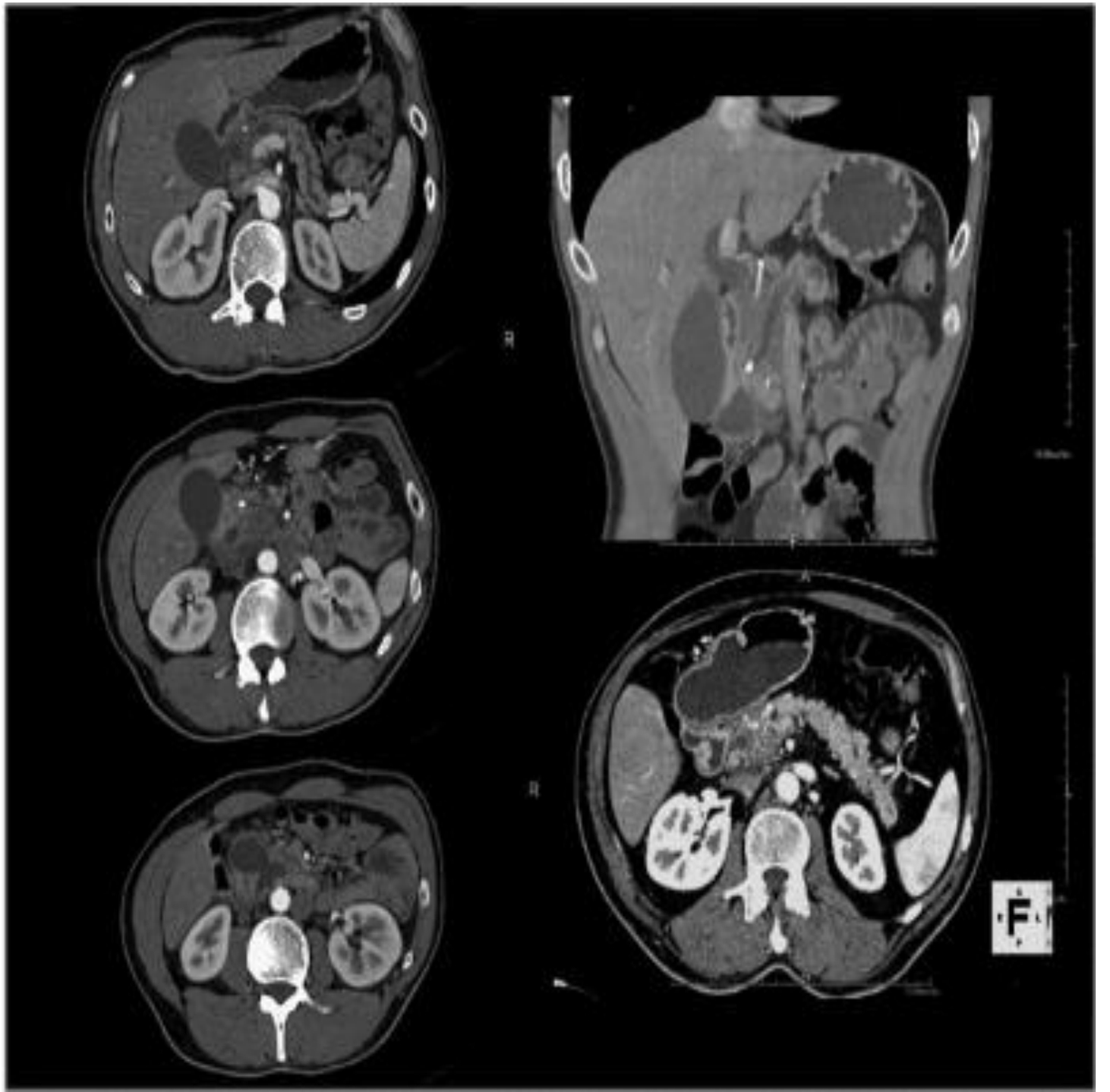
Computed Tomography (CT)



- CT is imaging technique that was first introduced for clinical use in the early 1970s. CT is an imaging technique whereby cross-sectional images are obtained with the use of X-rays.
- In CT scanning, the patient is passed through a rotating gantry that has an X-ray tube on one side and a set of detectors on the other.
- Information from the detectors is analyzed by computer and displayed as a grey-scale image.

- As with plain radiography, high- density objects cause more attenuation of the X-ray beam and are therefore displayed as lighter grey than objects of lower density.
- The relative density of an area of interest may be measured electronically. This density measurement is given as an attenuation value, expressed in Hounsfield units (HU) (named for Godfrey Hounsfield, the inventor of CT).

R

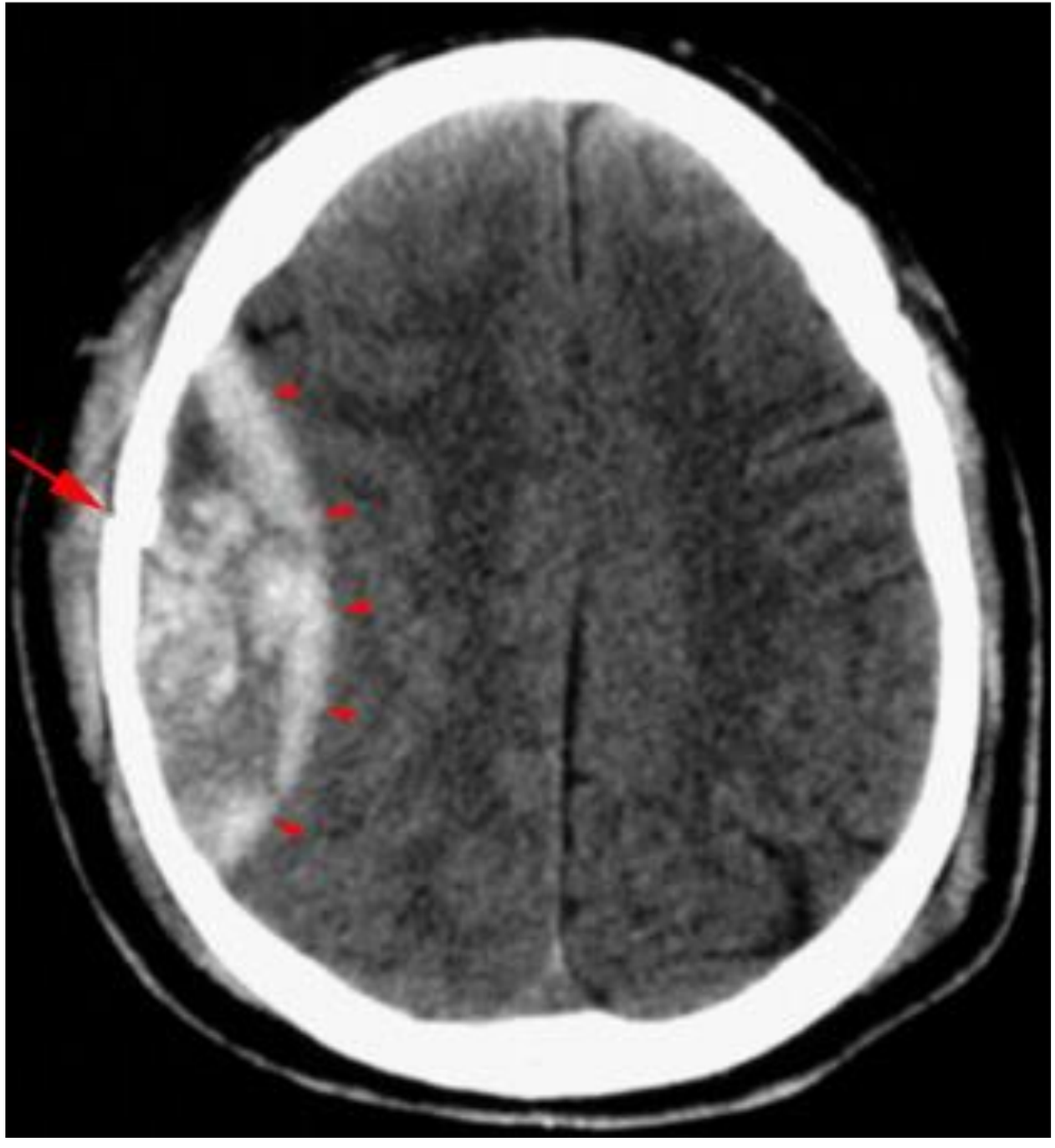

F

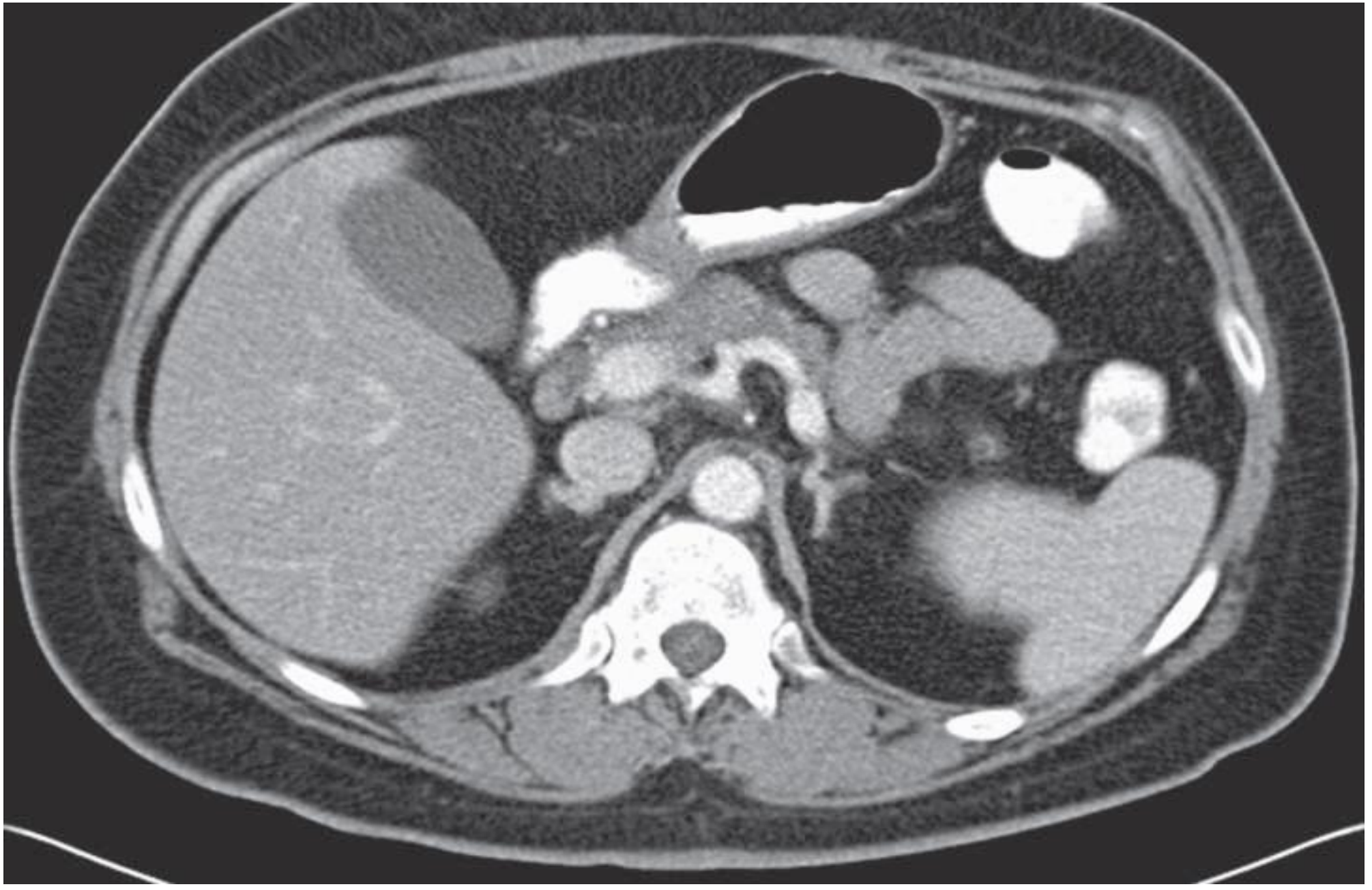
CT Procedures
Workstation
Post-Processing
Workstation

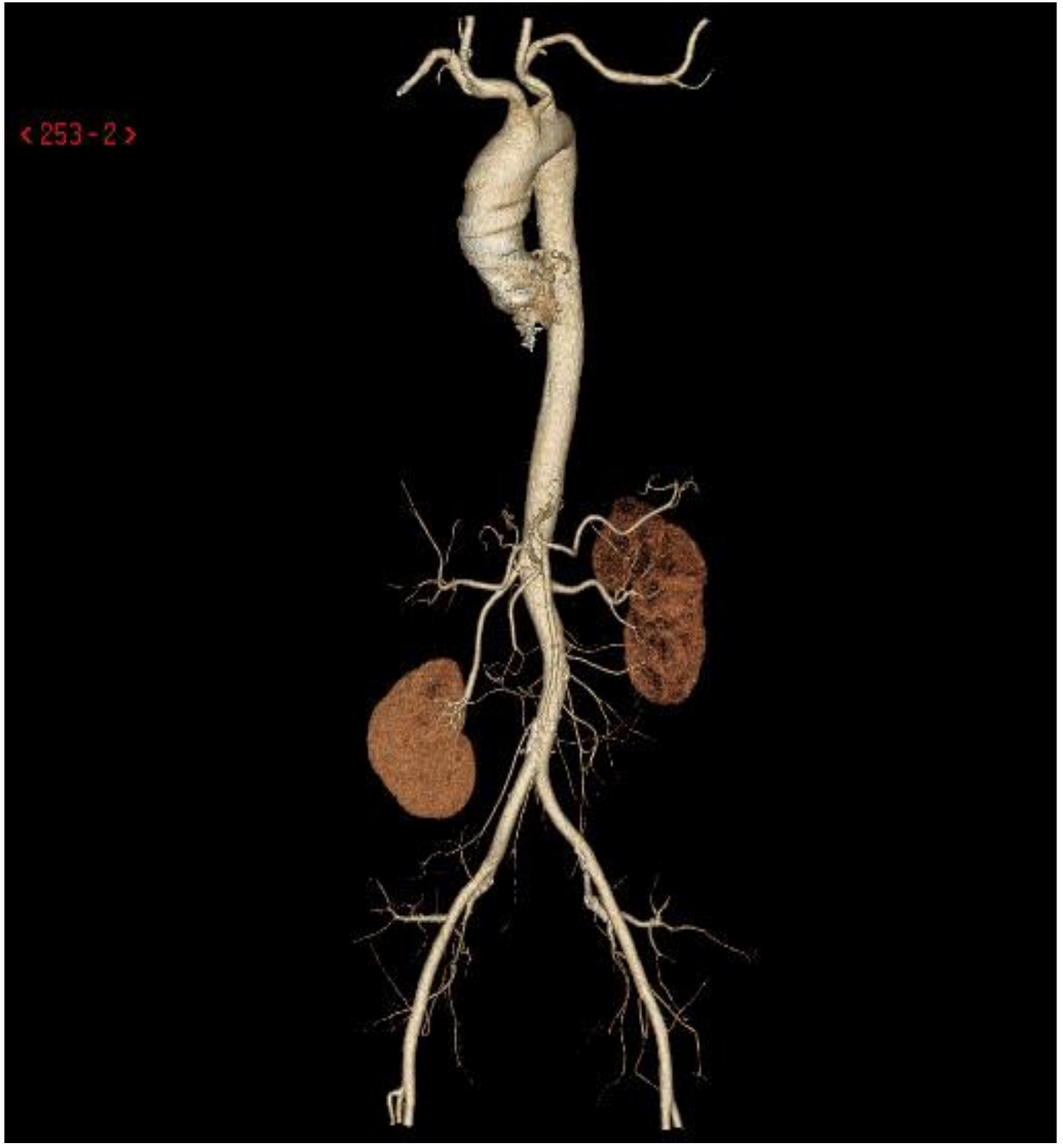
PHILIPS
Gantry
– X-ray tube
– Detectors
Patient table (couch)

Diagnostic Applications of CT

- The original CT studies were used primarily for diagnosing neurologic disorders. As scanner technology advanced, the range of applications was extended to other areas of the body. The most commonly requested procedures involve the head, chest, abdomen, and pelvis.
- CT is the examination of choice for head trauma; it clearly shows skull fractures and associated subdural hematomas. CT examinations of the head are one of the first exams performed on patients being evaluated for stroke or cerebrovascular accident where evidence of hemorrhage must be ruled out.



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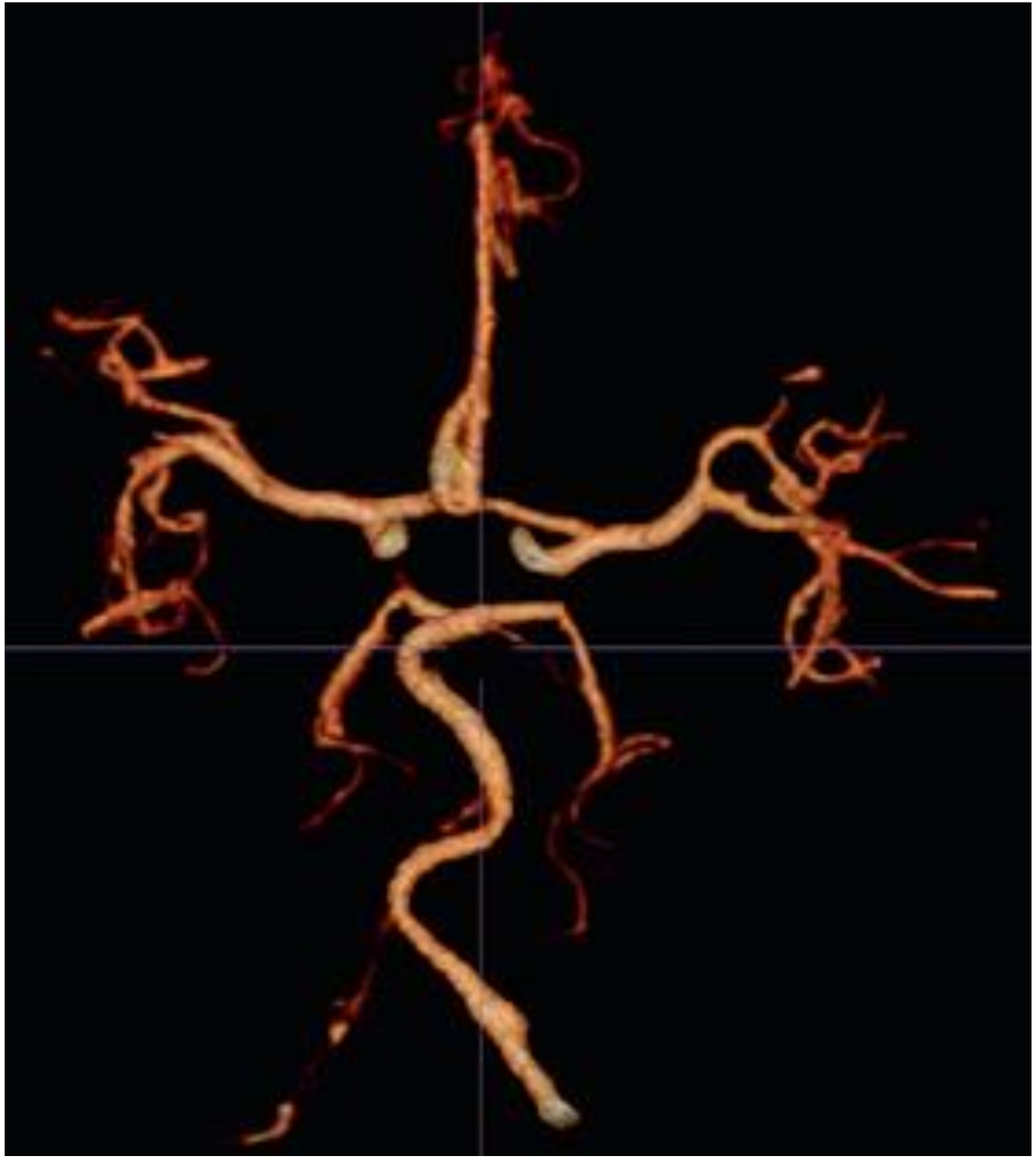

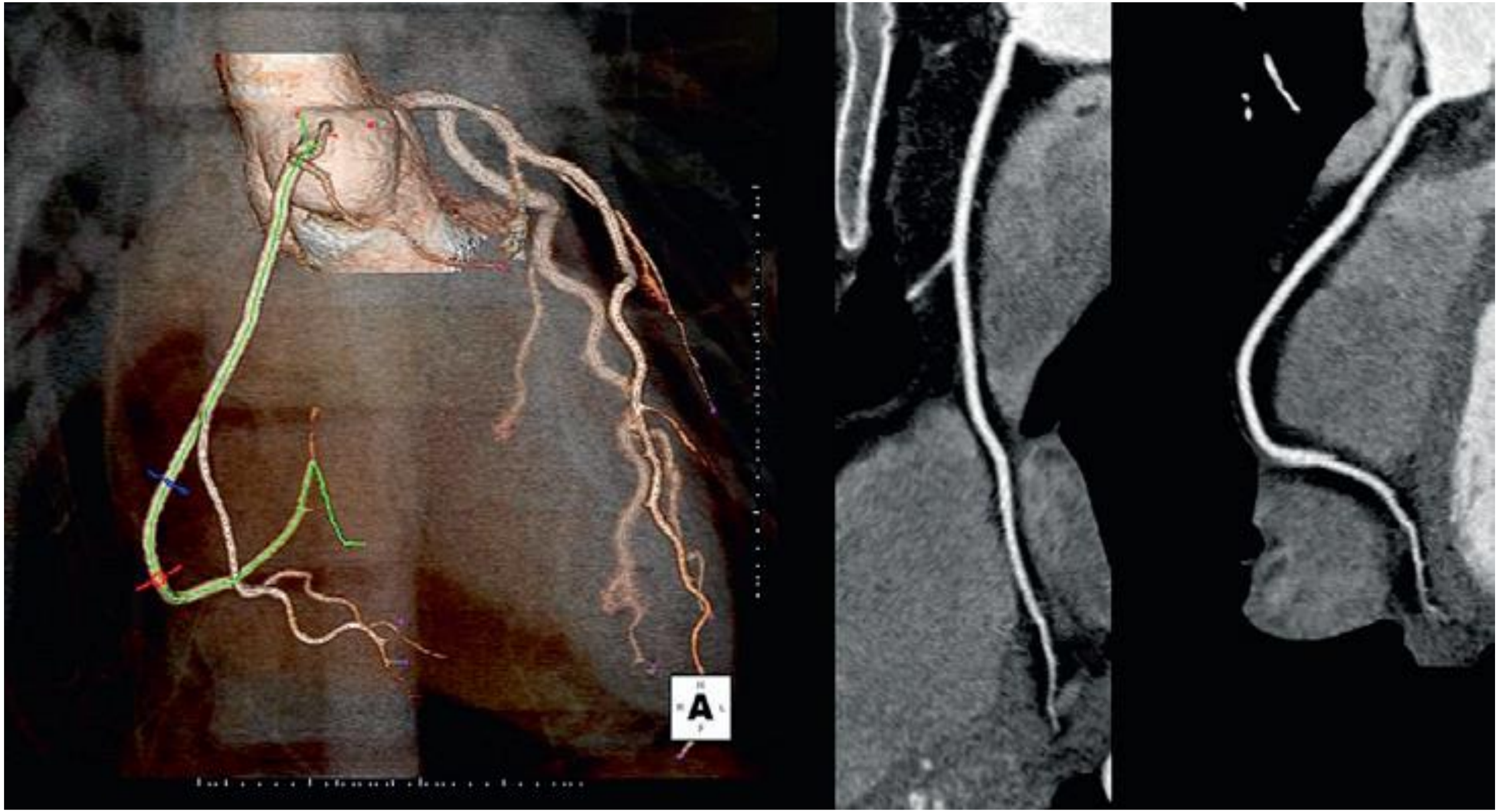
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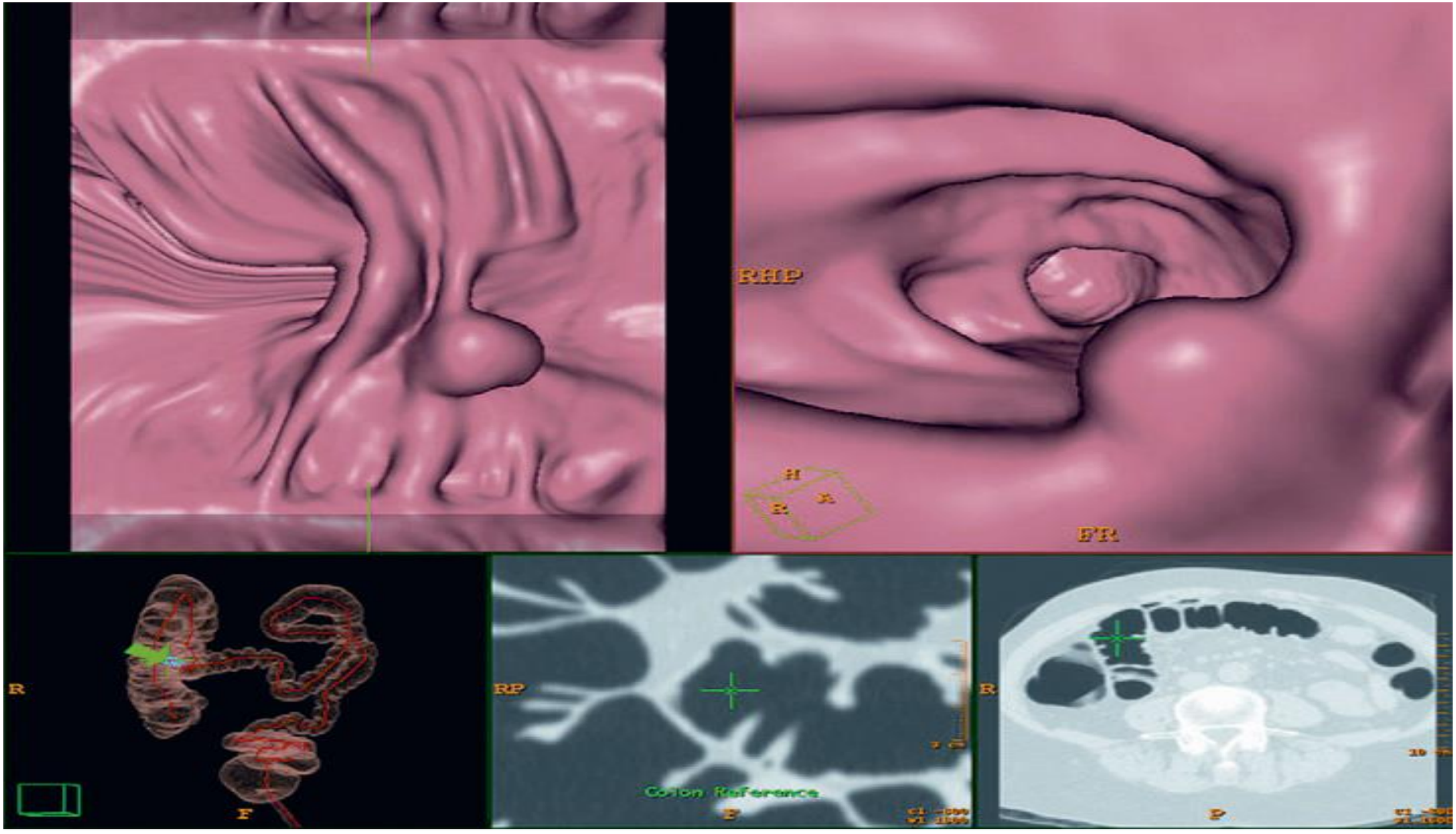
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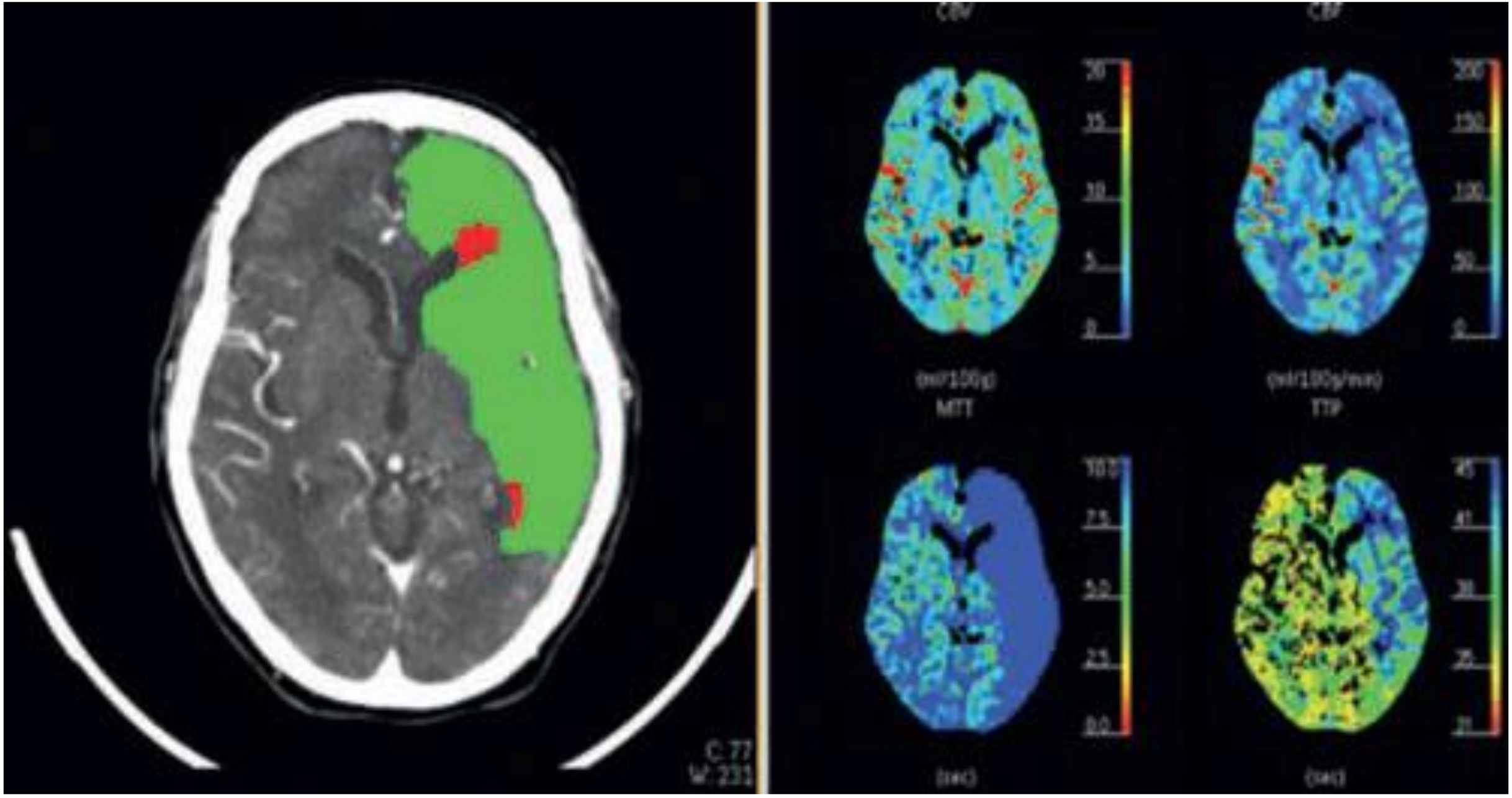
C: 77
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CBF
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MTT
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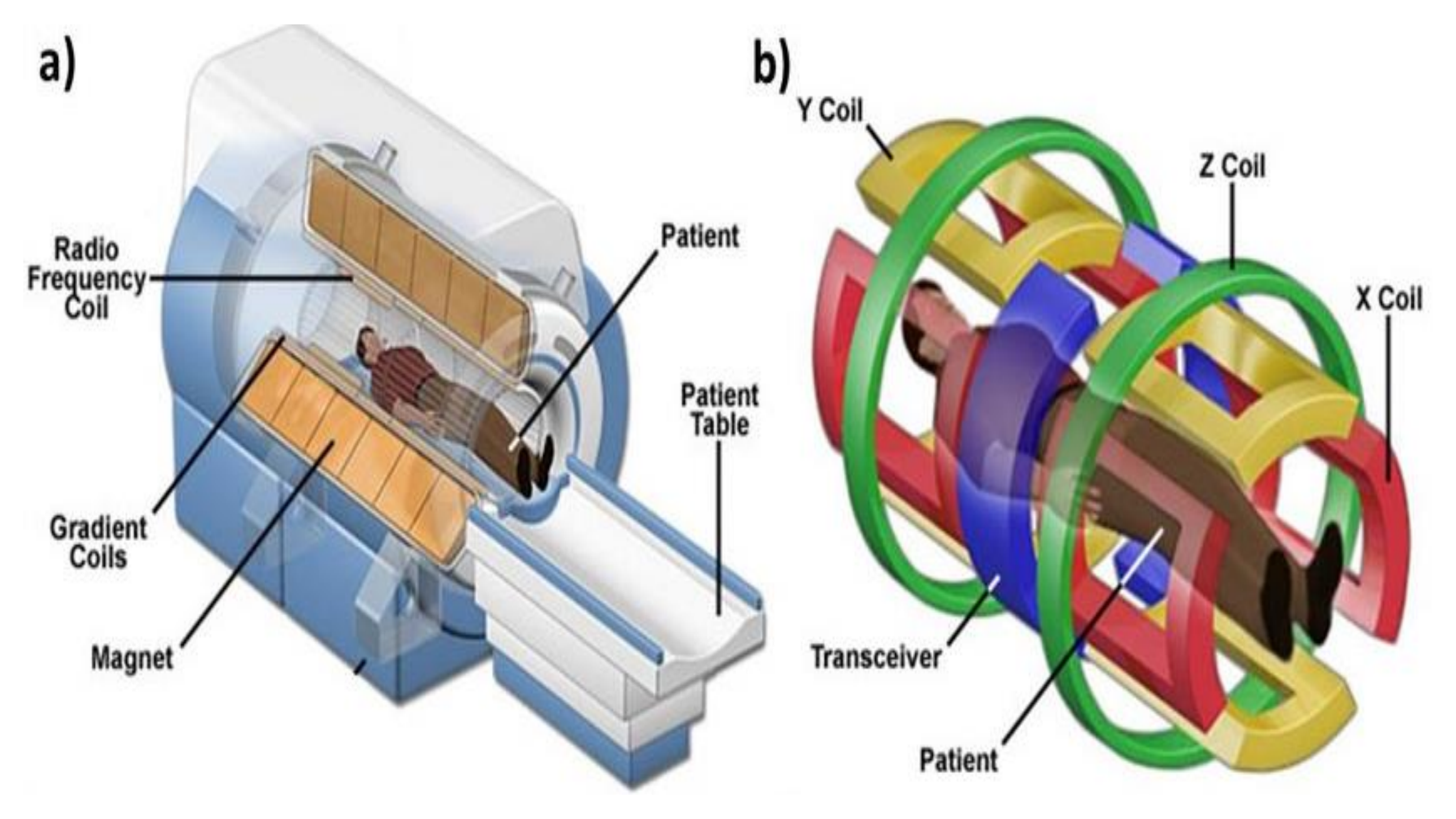
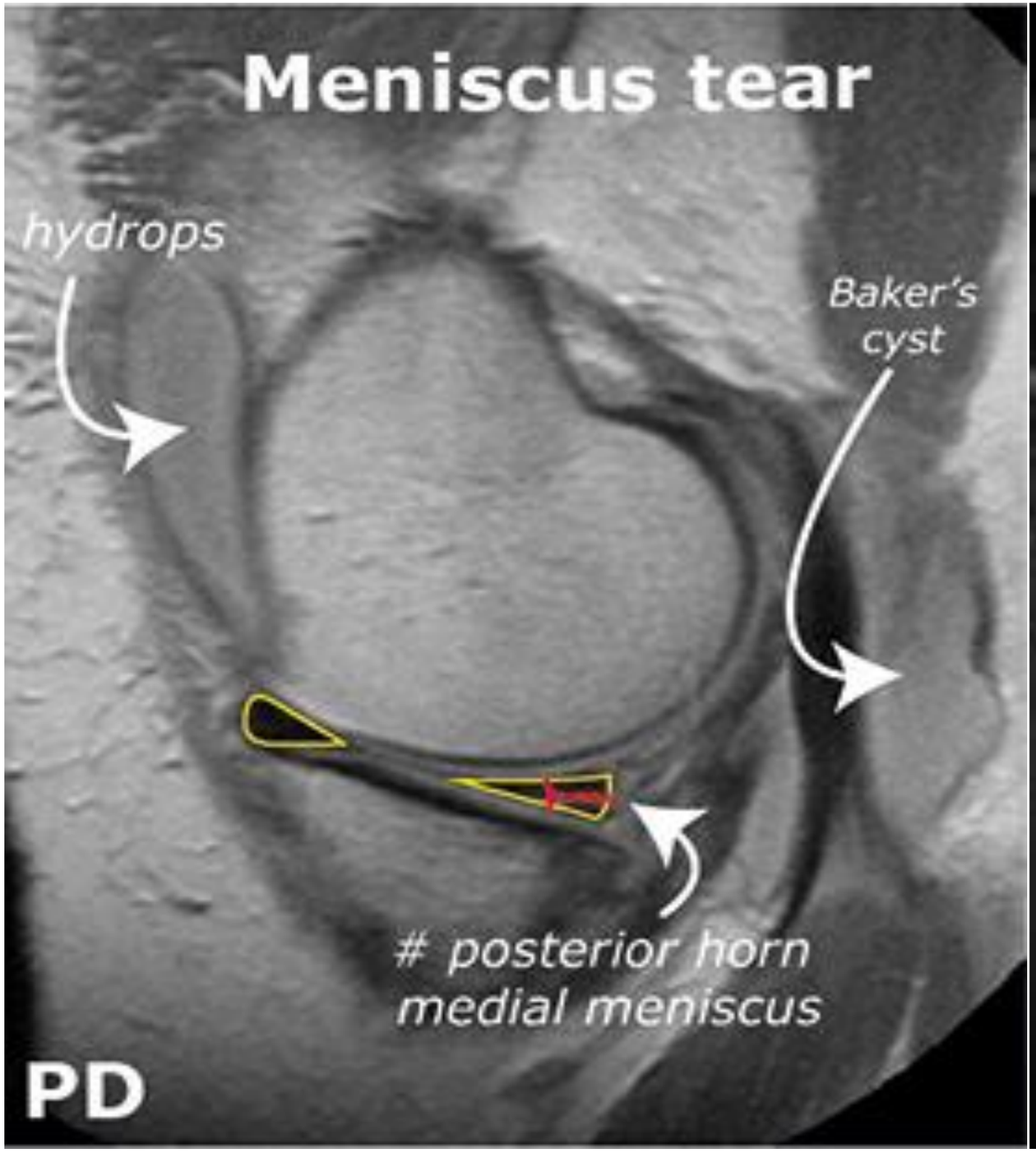
Magnetic Resonance Imaging (MRI)

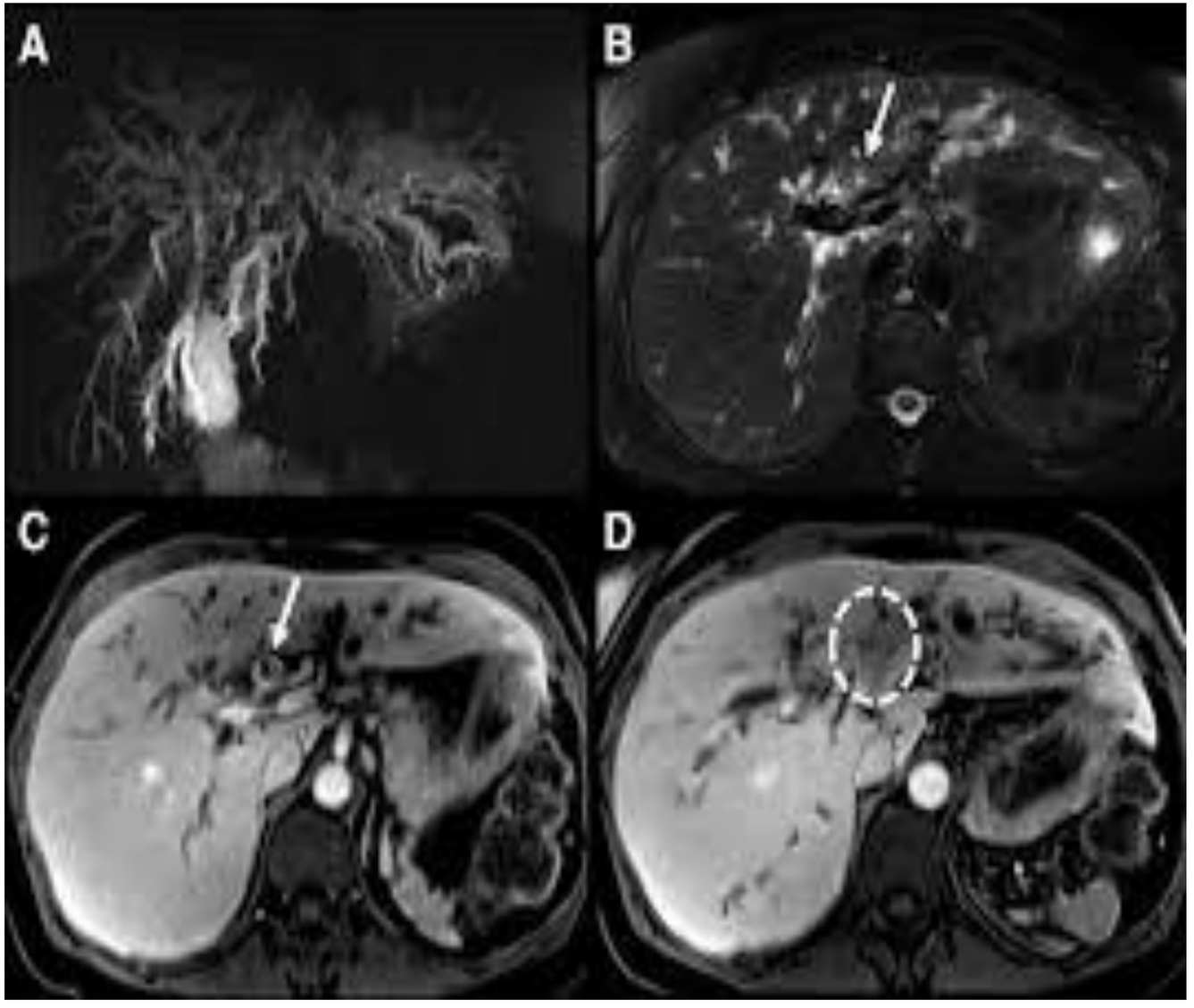
- Magnetic resonance imaging (MRI) is a medical imaging technique used in radiology to form pictures of the anatomy and the physiological processes of the body. The additional advantage of MRI is the lack of ionizing radiation and excellent soft-tissue contrast.
- MRI produces images of the body by utilizing the magnetic properties of certain nuclei, principally those of hydrogen in water molecules.
- The patient is placed in the scanner tunnel and subjected to a high intensity magnetic field. This forces the hydrogen atom nuclei to align with the magnetic field.


a)
Radio Frequency Coil
Patient
Patient Table
Gradient Coils
Magnet
b)
Y Coil
Z Coil
X Coil
Transceiver
Patient

- **Uses of Magnetic Resonance Imaging include:**

1. **Central nervous system (CNS):** technique of choice for brain and spinal imaging.
2. **Musculoskeletal:** accurate imaging of joints, tendons, ligaments and muscular abnormalities.
3. **Cardiac:** diagnosis of many cardiac conditions.
4. **Thorax:** assessment of vascular structures in the mediastinum.
5. **Abdomen:** abdominal organs are well visualized.
6. **Pelvis:** staging of prostate, bladder and pelvic neoplasms.


Meniscus tear
hydrops
Baker's cyst
posterior horn medial meniscus
PD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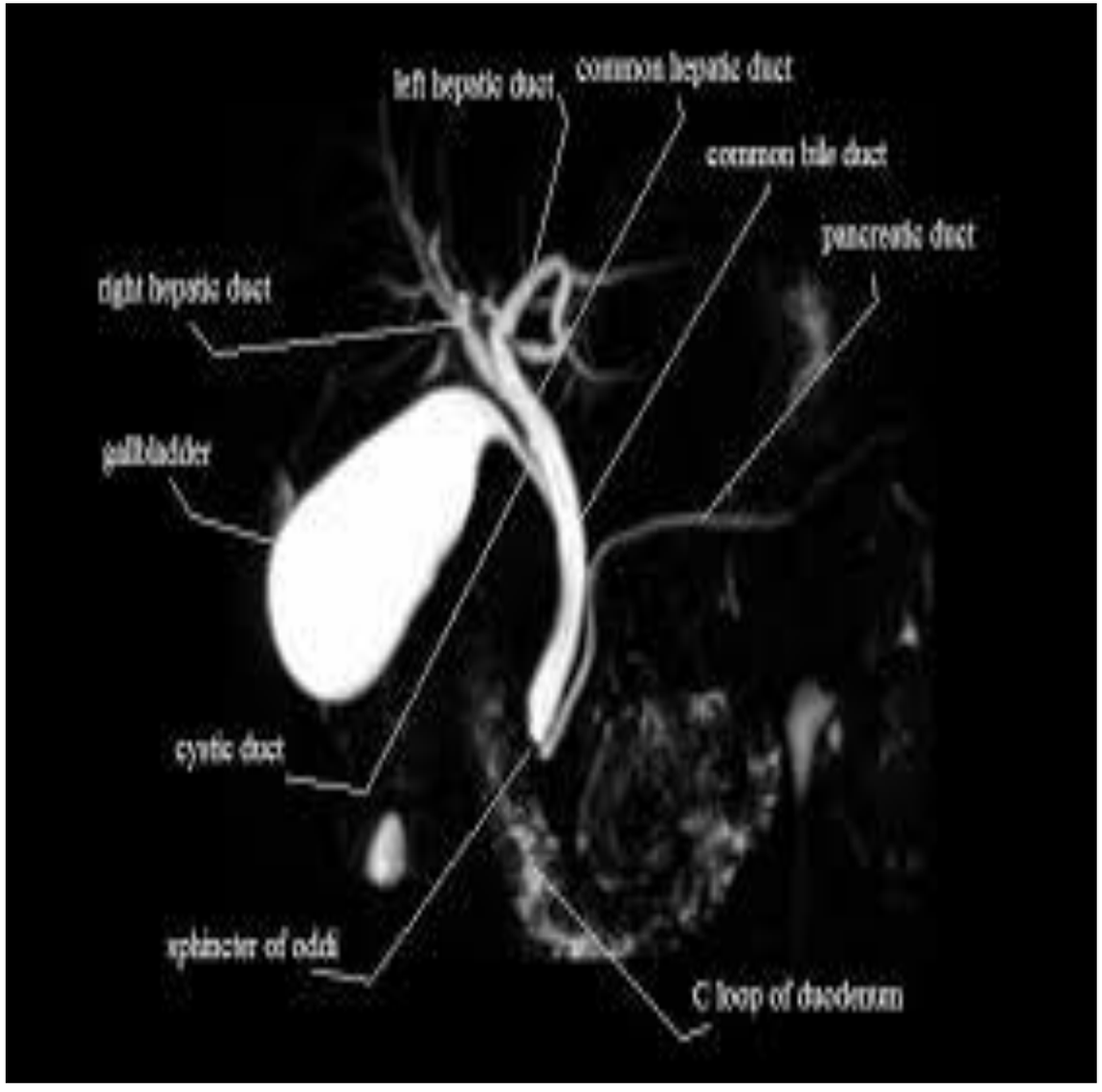
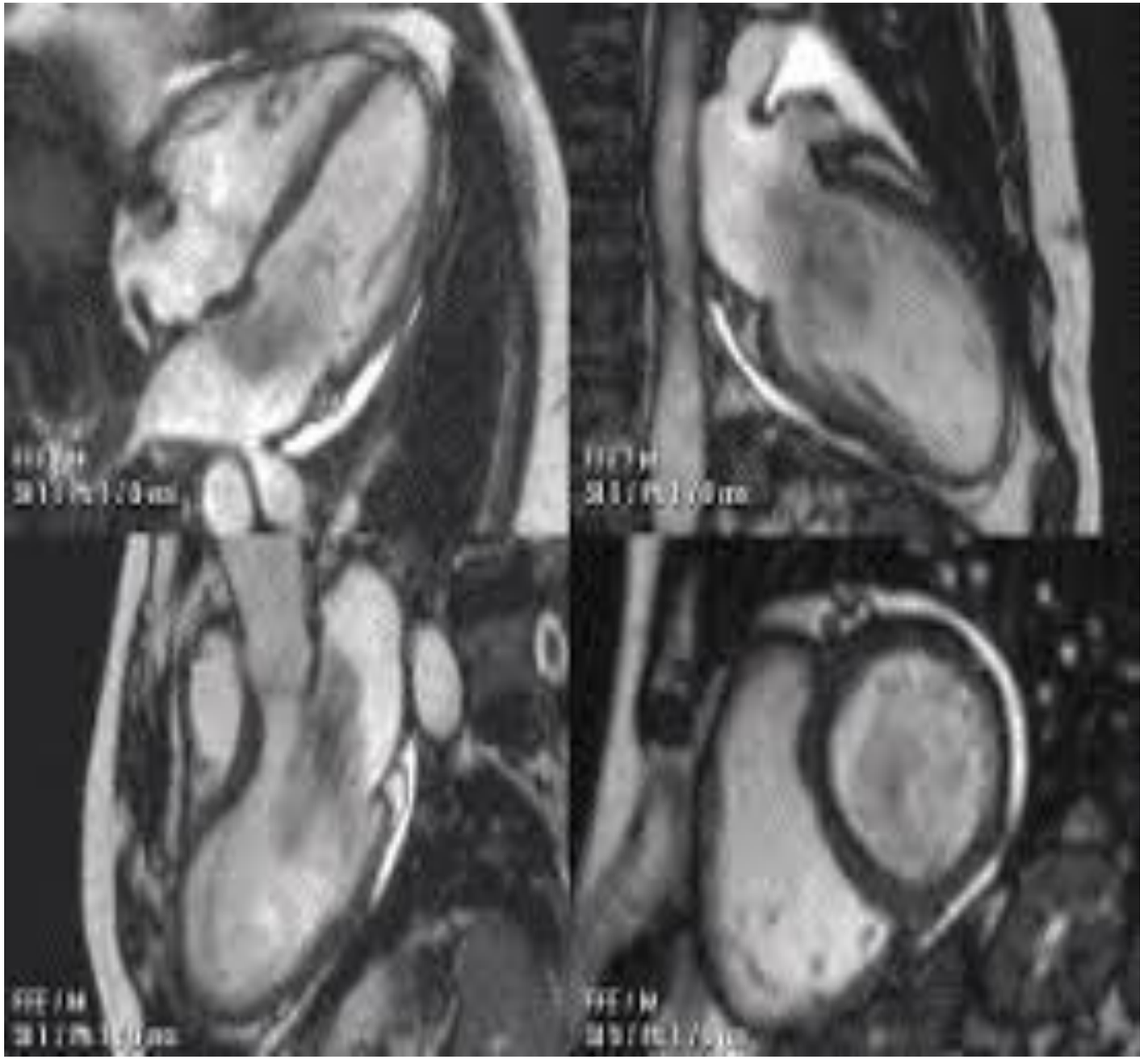
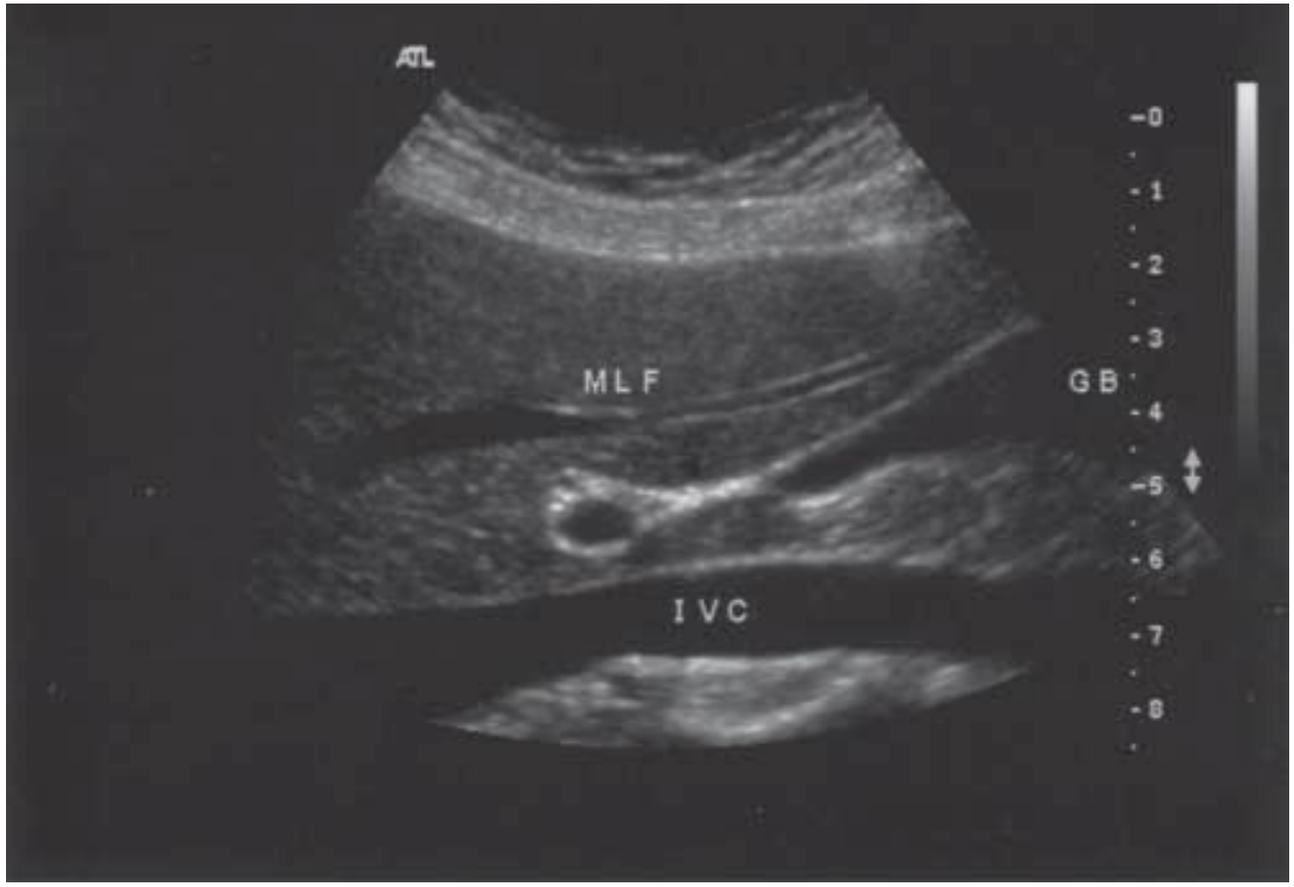
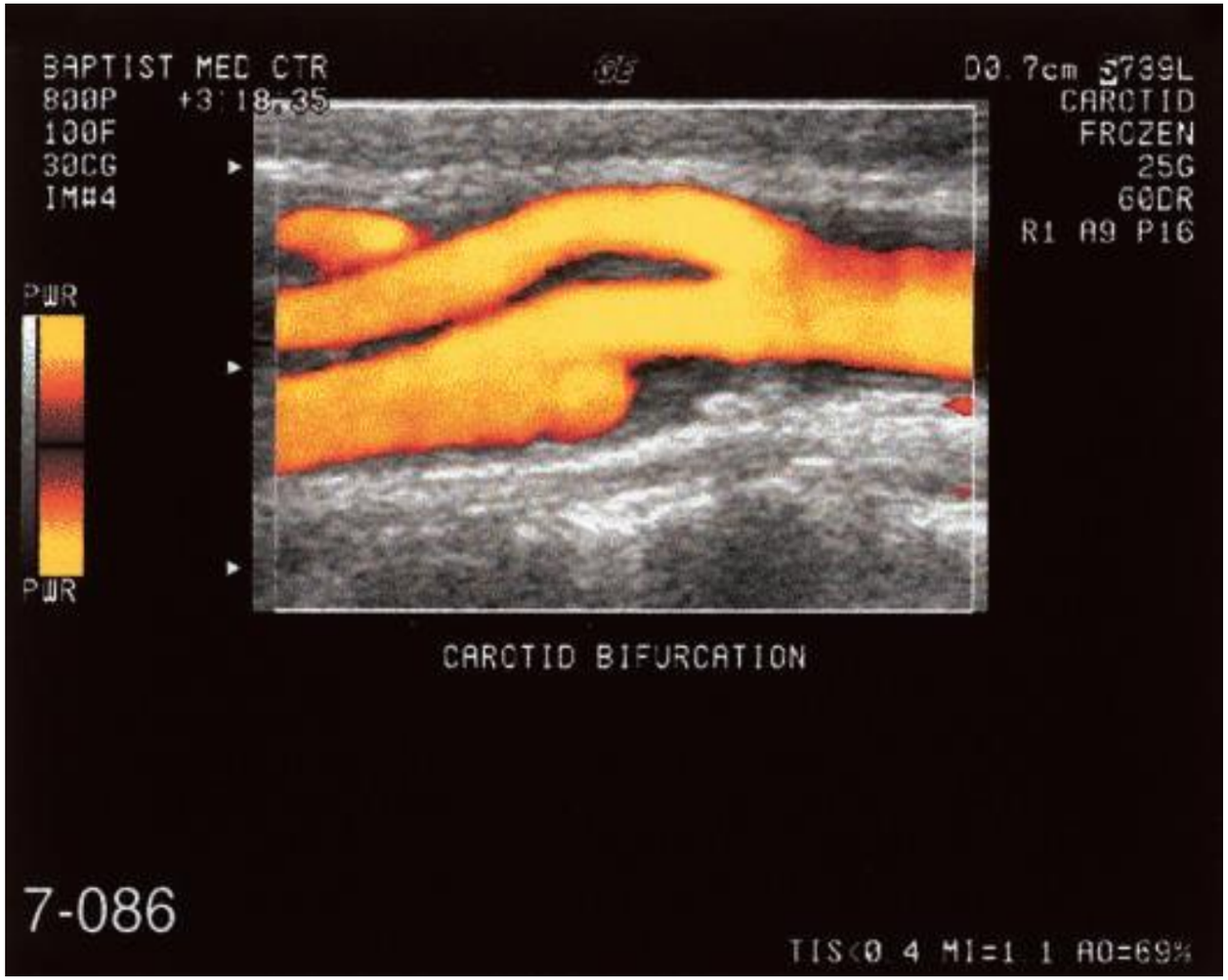

left hepatic duct
common hepatic duct
common bile duct
pancreatic duct
right hepatic duct
gallbladder
cystic duct
sphincter of oddi
C loop of duodenum

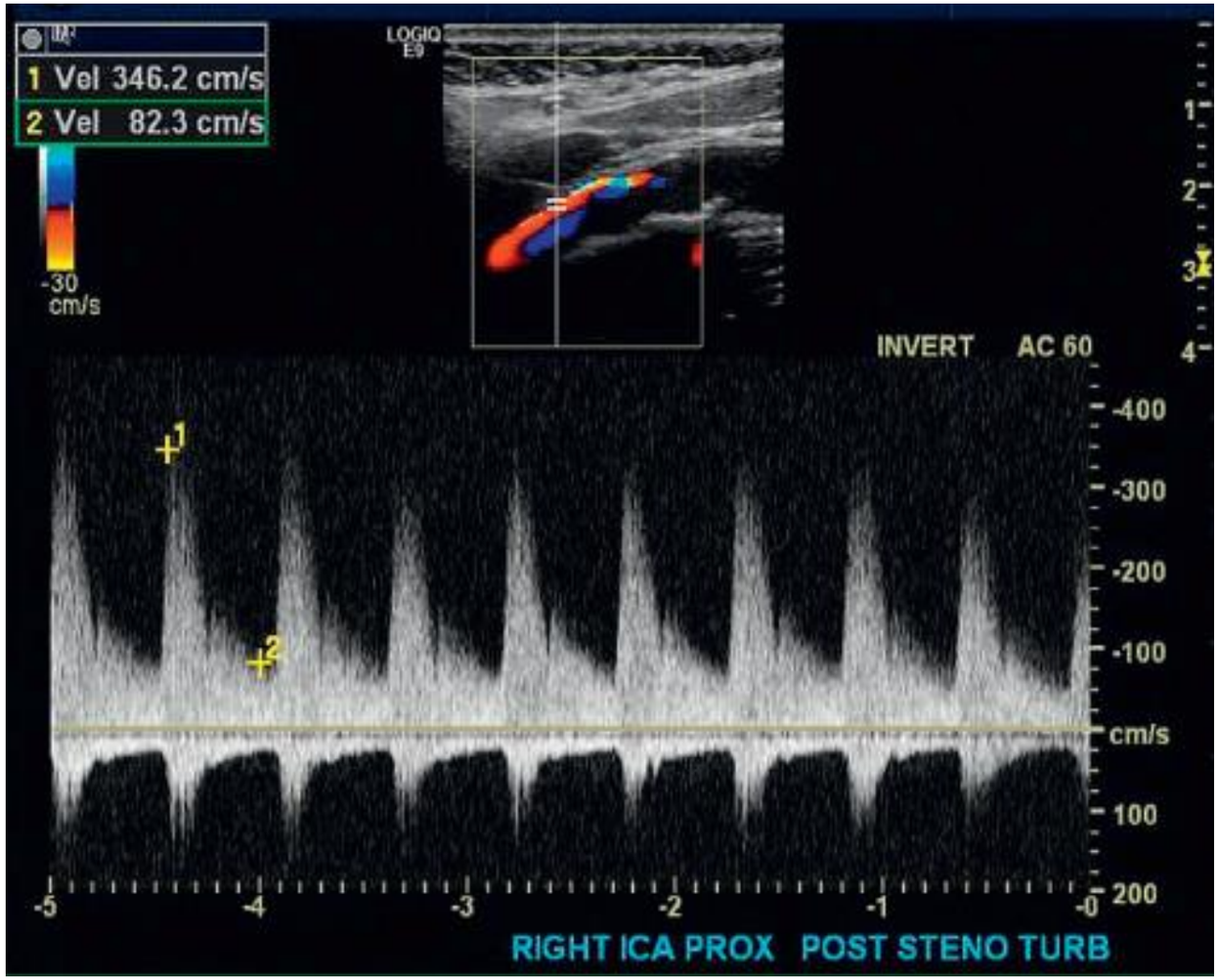


- **Diagnostic Ultrasound**
- Diagnostic medical sonography is a general term used to encompass abdominal, breast, cardiac, gynecologic, obstetric, and vascular sonography.
- Diagnostic medical sonography employs high-frequency transducers ranging from 2 to 30 MHz.
- The transducer emits short pulses of ultrasound (pulse waves) into the human body.
- The transducer receives real-time reflections or frequency shifts from structures or vessels along the sound waves path, and they are displayed as a grayscale, color Doppler, spectral, or duplex image.



A, Transverse sonogram of liver showing middle hepatic vein (MHV) dividing left and right hepatic lobes. LHV, left hepatic vein; RHV, right hepatic vein.
B, Sagittal image of main lobar fissure (MLF) and its relationship to gallbladder (GB).





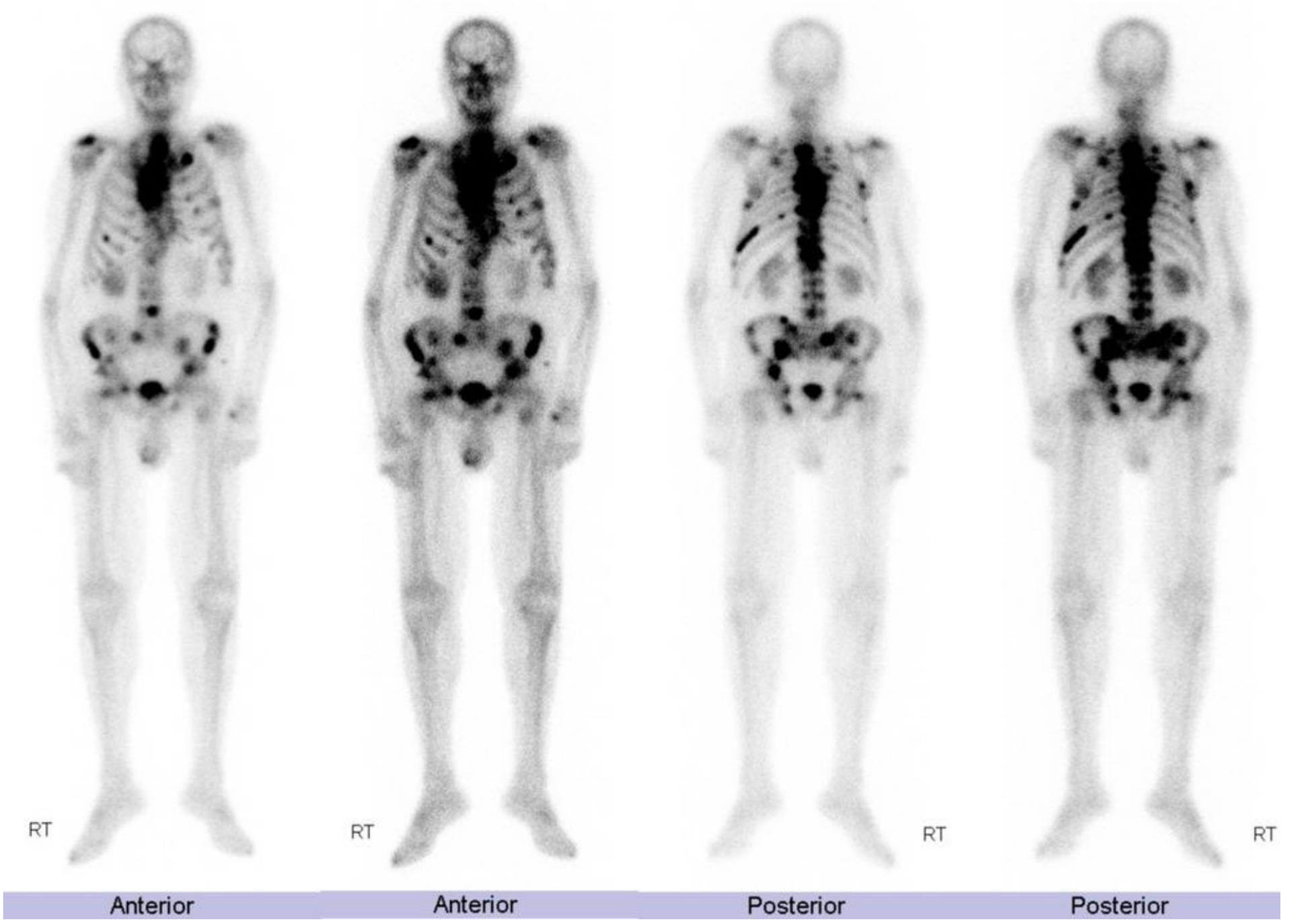
A, Sagittal sonogram of carotid artery and bifurcation (arrow) into internal and external carotid arteries.

B, Color Doppler and spectral analysis show increases in flow velocity in stenotic internal carotid artery.

- **Nuclear Medicine**
- Nuclear medicine is a medical specialty that focuses on the use of radioactive materials called radiopharmaceuticals for diagnosis, therapy, and medical research.
- In contrast to radiographic procedures that determine the presence of disease based on structural appearance, nuclear medicine determines the cause of a medical problem based on the physiologic function of organs or tissues.
- For a nuclear medicine procedure, the radioactive material, commonly referred to as a radiopharmaceutical or a radiotracer, is primarily introduced into the body by injection, ingestion, or inhalation.

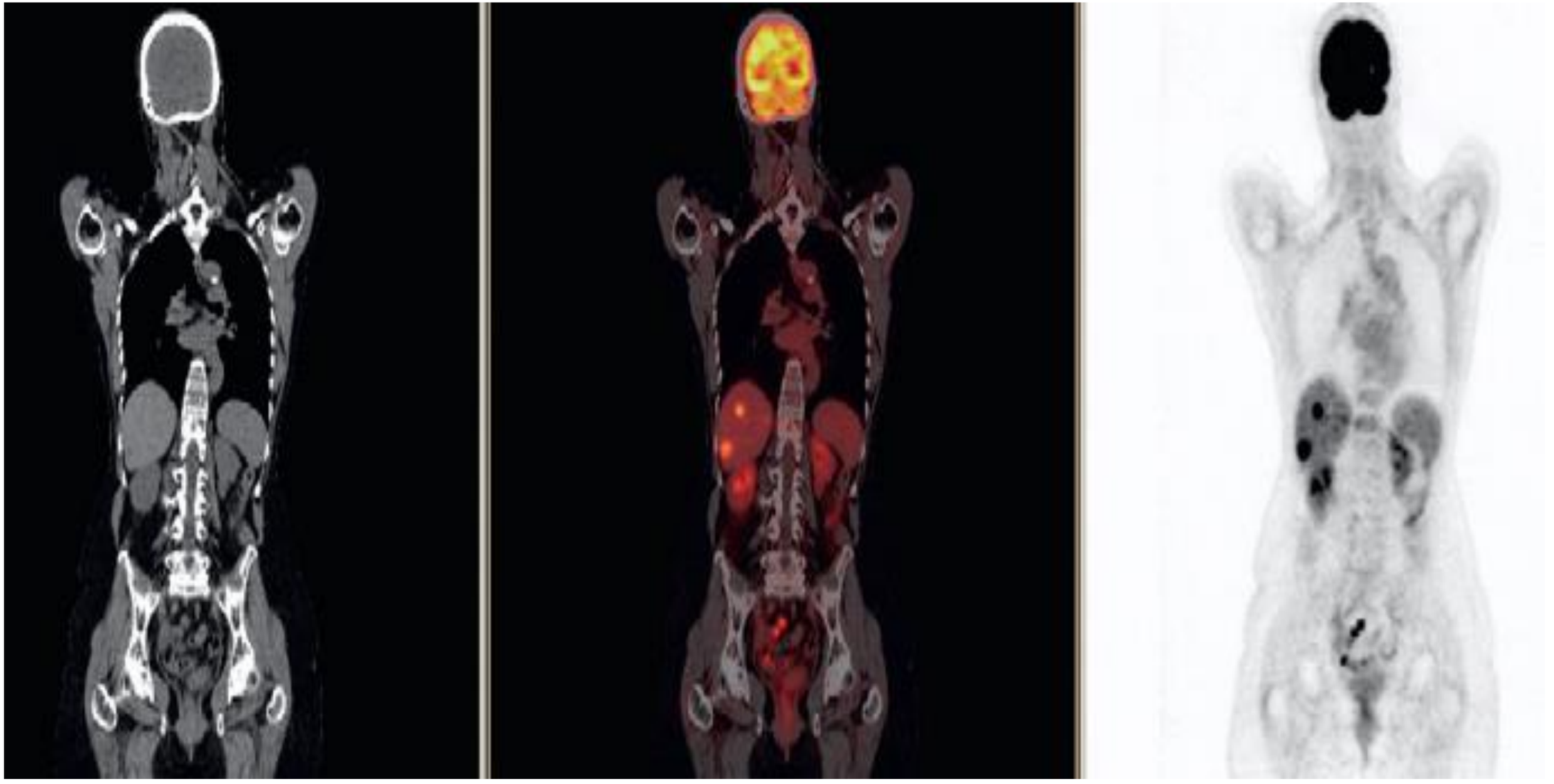
- **Radiopharmaceuticals**, defined as radioactive drugs used in the diagnosis and treatment of disease, tag a radionuclide with a pharmaceutical.
- The radionuclide emits a gamma ray as it decays, and the pharmaceutical is formulated to go to a specific organ.
- When a radiopharmaceutical is injected into a patient, the pharmaceutical carries the radionuclide to the organ being examined, and the gamma rays create an image that is digitally recorded by a gamma camera.
- These images provide an anatomic view of the organ structure and provide diagnostic insights regarding the function of the organ.

Whole body bone scan was performed. There are multiple foci of abnormal uptake throughout the skeleton, e.g. right acromion, bilateral scapulae, multiple ribs bilaterally (especially left 9th rib), thoracolumbar spine at multiple levels (especially mid- thoracic spine), bilateral bony pelvis and right femoral neck.



Hybrid(PET/CT) Positron Emission Tomography-Computed Tomography

- PET/CT is a nuclear medicine technique which combines, in a single gantry, a positron emission tomography (PET) scanner and an x-ray computed tomography (CT) scanner, to acquire sequential images from both devices in the same session, which are combined into a single superposed (co-registered) image.
- Thus, functional imaging obtained by PET, which depicts the spatial distribution of metabolic or biochemical activity in the body can be more precisely aligned or correlated with anatomic imaging obtained by CT scanning.
- Two and three-dimensional image reconstruction may be rendered as a function of a common software and control system.
- The combination of PET and CT into a single PET/CT has resulted in significant improvements in the diagnosis and staging of disease, particularly in the field of oncology.



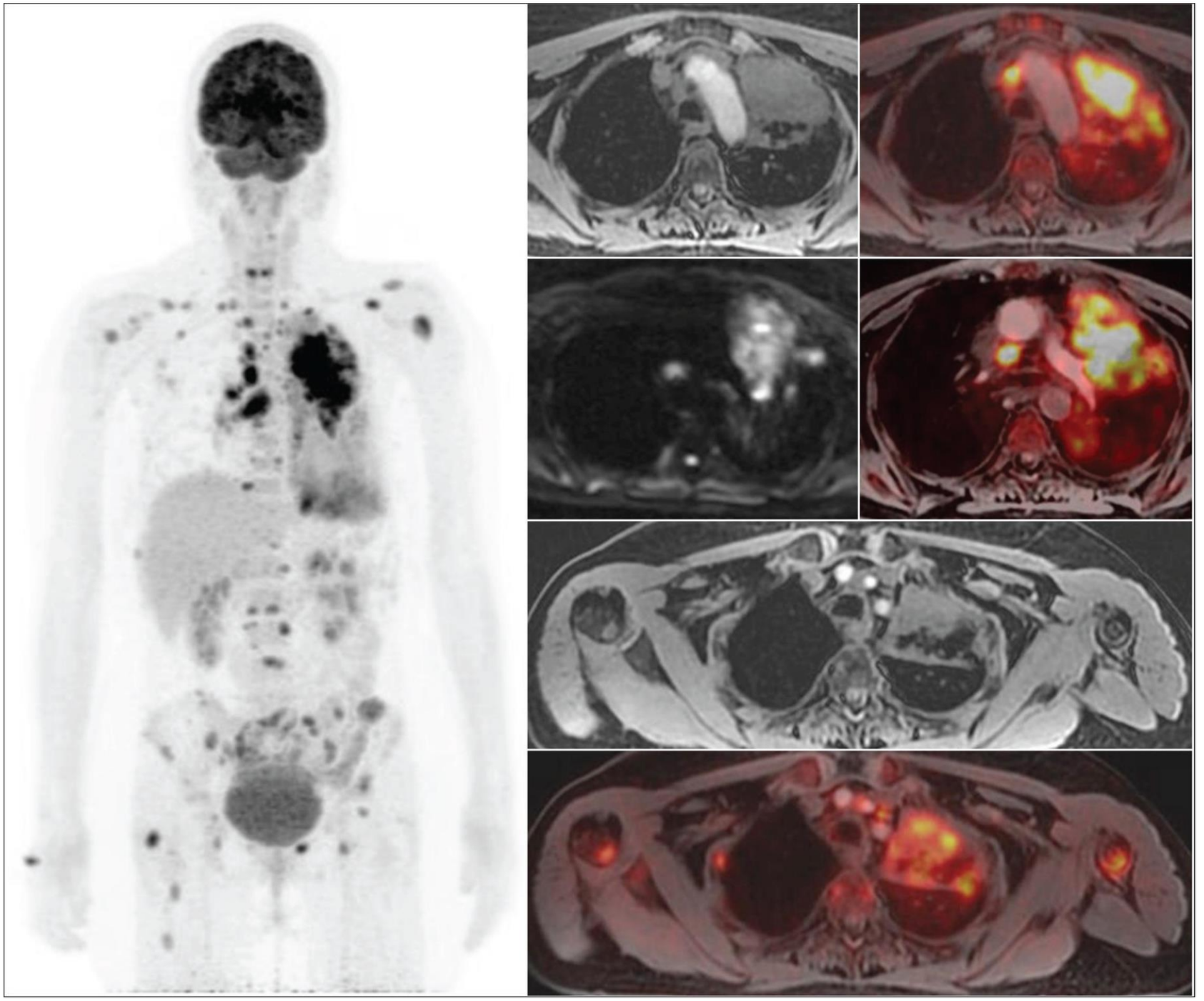
PET/CT fusion study—liver metastasis.

Hybrid(PET/MRI) Positron Emission Tomography-Magnetic Resonance Imaging

- MR/PET is the newest imaging modality falling within the category of hybrid imaging.
- PET-MR aims to combine the high-spatial resolution morphological and functional assessment afforded by MRI with the ability of PET for quantification of metabolism, perfusion, and inflammation.
- Benefits of PET/MRI scans
- **There are major benefits to PET/MRI scans:**

1. More accurate diagnosis and treatment options: PET/MRI scans of the brain can detect abnormal findings that PET/CT misses in more than 50% of patients scanned.
2. Improved safety from significantly reduced radiation exposure: Compared to PET/CT scans, PET/MRI exposes patients to about 50% less radiation.

- **Coronal MIP PET and axial slices of T1-weighted PET/MRI sequences and diffusion. Staging of non-small cell lung adenocarcinoma.**
- **A PET/MRI scan shows the primary pulmonary mass in the left upper lobe, with signs of thoracic wall infiltration (cortical in the second rib) with bilateral mediastinal lymph node metastases and multiple sparse bone metastases, as well as good morphological correlation in the MR component.**
- **This case shows the excellent performance of PET/MRI for staging of lung neoplasia.**



Al-Azhar University-Gaza
Faculty of Applied Medical Sciences
Nursing Department

Radiology for Nursing

Interventional Radiology

Lecturer: Husam Mansour
Master of Public Health
Master of Medical Imaging

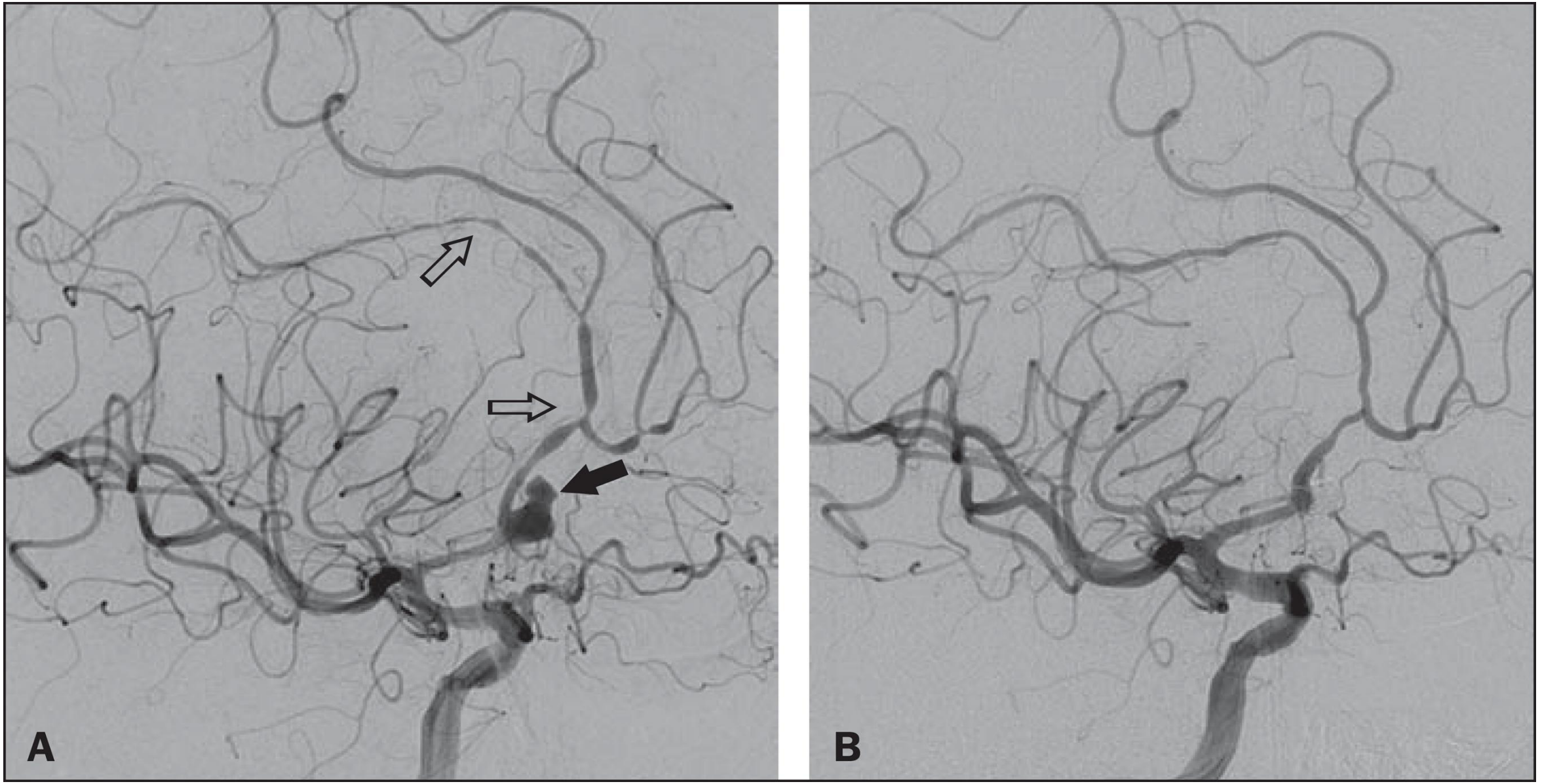
- **Interventional radiology (IR)** is a medical specialization that involves performing a range of imaging procedures to obtain images of the inside of the body.
- IR is a medical subspecialty that performs various invasive procedures using medical imaging guidance, such as x-ray fluoroscopy, computed tomography, magnetic resonance imaging, or ultrasound.
- IR performs both diagnostic and therapeutic procedures through very small incisions or body orifices.
- **Diagnostic IR** procedures are those intended to help make a diagnosis or guide further medical treatment, and include image-guided biopsy of a tumor or injection of an imaging contrast agent into a hollow structure, such as a blood vessel or a duct.
- **Therapeutic IR** procedures provide direct treatment include catheter-based medicine delivery, medical device placement (e.g., stents), and angioplasty of narrowed structures.

Diagnostic interventional radiology

- **Angiography:** Radiologic imaging of blood vessels after injection of contrast media.
- **Arteriography:** Imaging of the arteries.
- **Venography:** Imaging of the veins.
- **Angiocardiography:** Imaging of the heart and associated structures.
- **Cholangiography:** Imaging the bile ducts within the liver to look for areas of blockage.
- **Biopsy:** Taking of a tissue sample from the area of interest for pathological examination from a percutaneous or transvenous approach.

- **Angiography:**
- **Indications**
- Angiography is primarily used to identify the anatomy or pathologic process of blood vessels.
- Chronic cramping leg pain after physical exertion, a condition known as **claudication**, may prompt a physician to order an arteriogram of the lower limbs to determine whether atherosclerosis is diminishing the blood supply to the leg muscles.
- **Stenosis or occlusion** is commonly caused by **atherosclerosis** and is an indication for an arteriogram.
- Cerebral angiography is performed to detect and verify the existence and exact position of an intracranial vascular lesion such as an **aneurysm.**

- **Specific Angiographic Procedures**
- **Cerebral angiography**
- Cerebral angiography is a radiologic study of the blood vessels of the brain.
- Vessels commonly selected for cerebral angiography include the common carotid arteries, internal carotid arteries, external carotid arteries, and vertebral arteries.
- **Clinical Indications**
- Vascular stenosis and occlusions
- Aneurysms
- Trauma
- Arteriovenous malformations
- Neoplastic disease



Digital subtraction angiography of the cerebral vessels, demonstrating an aneurysm (black arrow) at right ACA. The aneurysm is irregular in appearance, with a rupture sac. B: Complete embolization of the aneurysm sac after coiling

• Thoracic angiography

Thoracic aortography is an angiographic study of the ascending aorta, the arch, the descending portion of the thoracic aorta, and the major branches.

• Pulmonary arteriography

Is an angiographic study of the pulmonary vessels that usually is performed to investigate for pulmonary embolus.

Clinical Indications

- Aneurysms
- Congenital abnormalities
- Vessel stenosis
- Embolus
- Trauma

- **Angiocardiography**

Refers specifically to radiologic imaging of the heart and associated structures.

- **Coronary arteriography**

Is performed at the same time to visualize the coronary arteries.

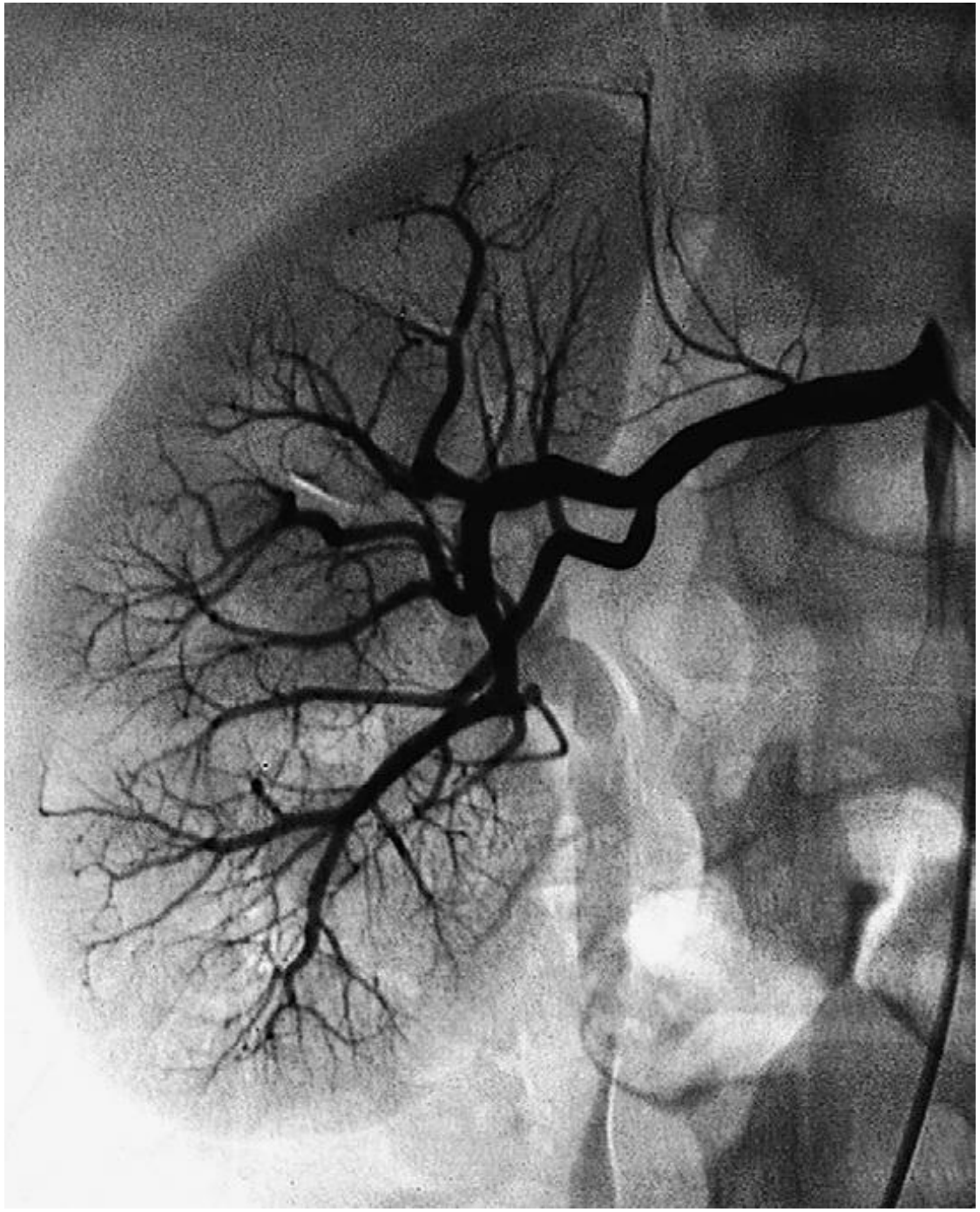
- **Cardiac catheterization**

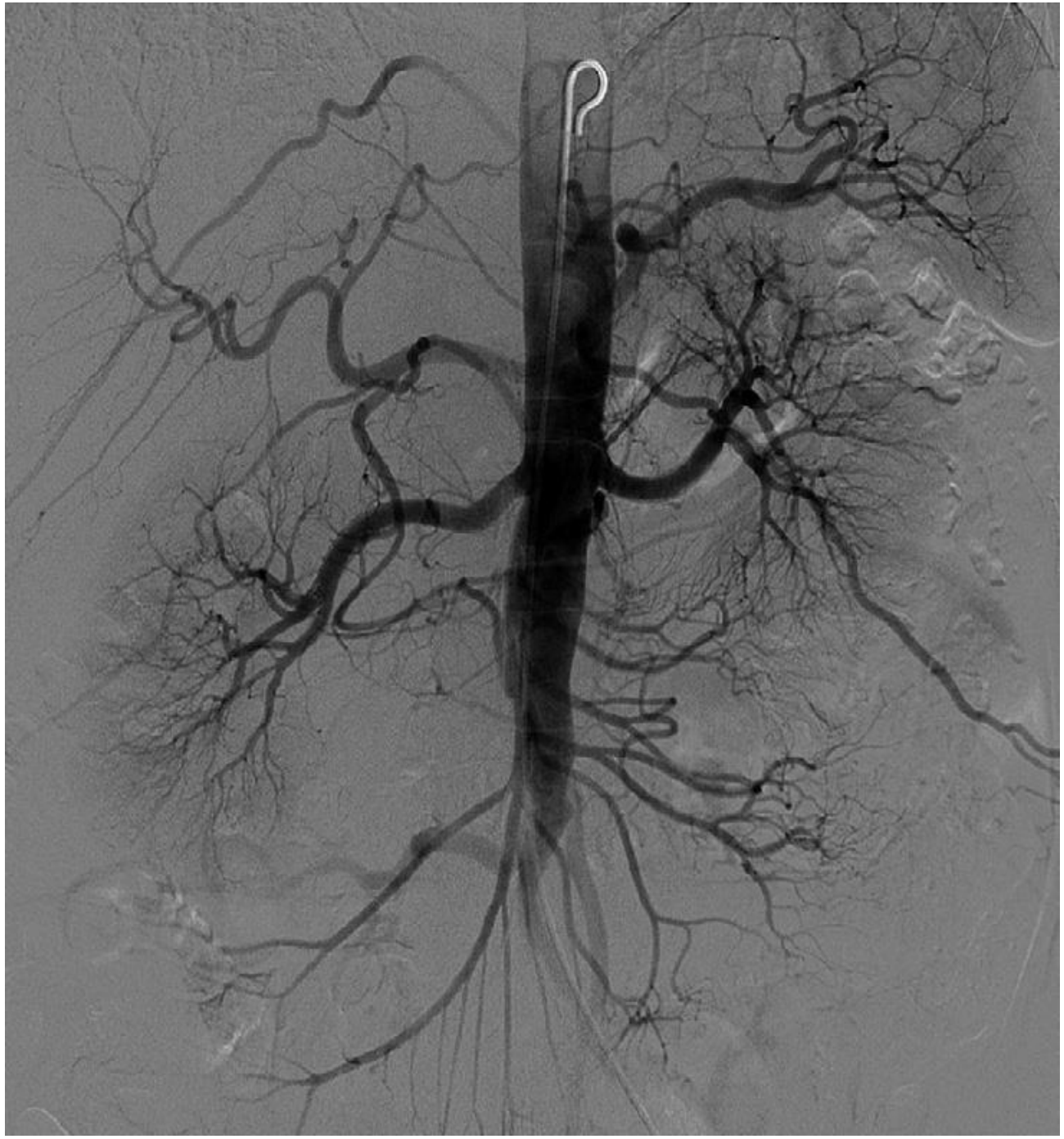
Is a more general term that is used to describe placing a catheter in the heart.

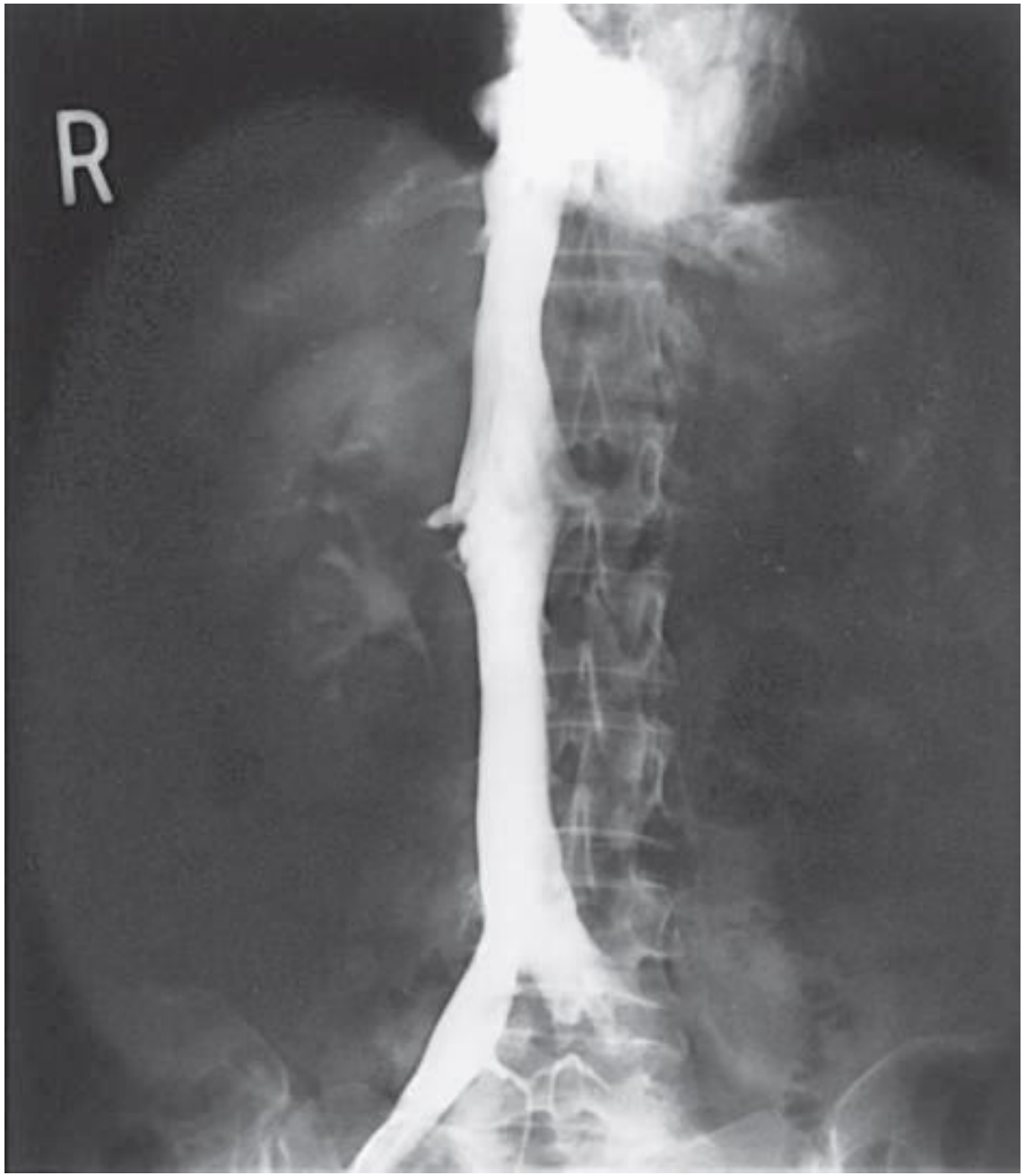
- **Clinical Indications**
- Coronary artery disease and angina
- Myocardial infarct
- Valvular disease
- Atypical chest pain
- Congenital heart anomaly

Abdominal angiography

- Demonstrates the contour and integrity of abdominal vasculature.
- **Aortography** refers to an angiographic study of the aorta, and selective studies refer to the catheterization of a specific vessel.
- **Venacavography** demonstrates the superior and inferior venae cavae.
- **Clinical Indications**
- Aneurysm
- Congenital abnormality
- Gastrointestinal bleed
- Stenosis or occlusion
- Trauma






R

- **Peripheral angiography**
- Is a radiologic examination of the peripheral vasculature
- **Arteriogram**
- In which case the injection is administered by a catheter in an artery.
- **Venogram**
- In which the injection is placed into a vein of the extremity being examined.
- **Clinical Indications**
- Atherosclerotic disease
- Vessel occlusion and stenosis
- Trauma
- Neoplasm
- Embolus and thrombus

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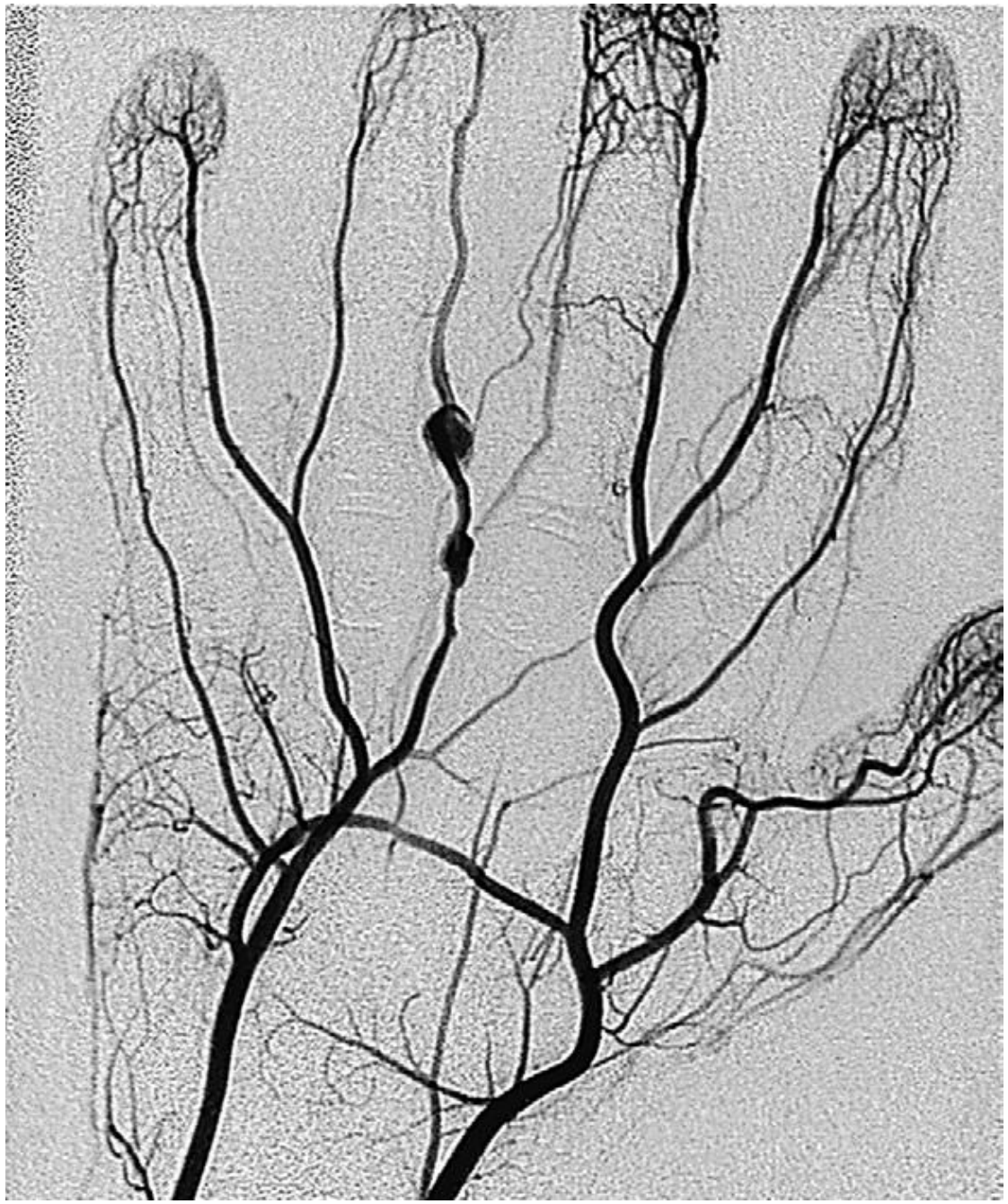
R

R

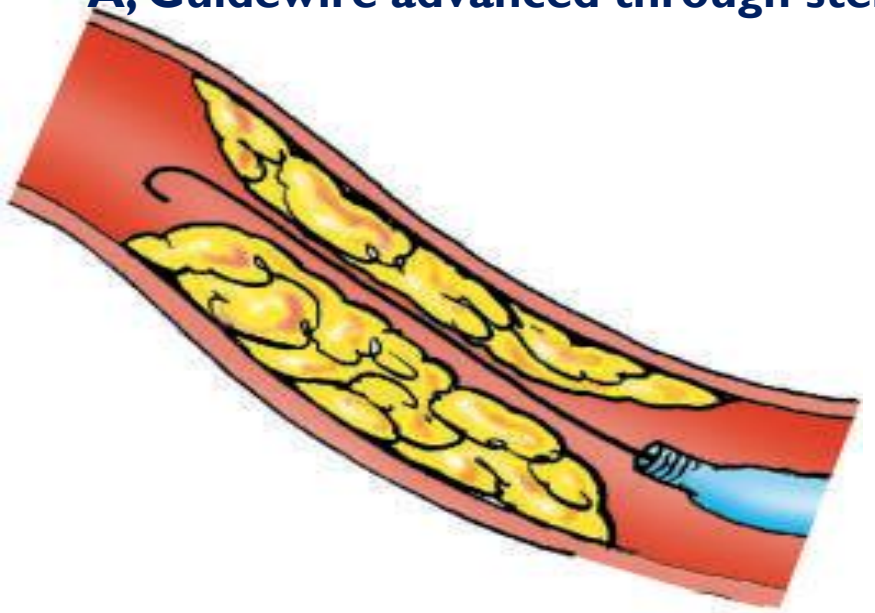
R



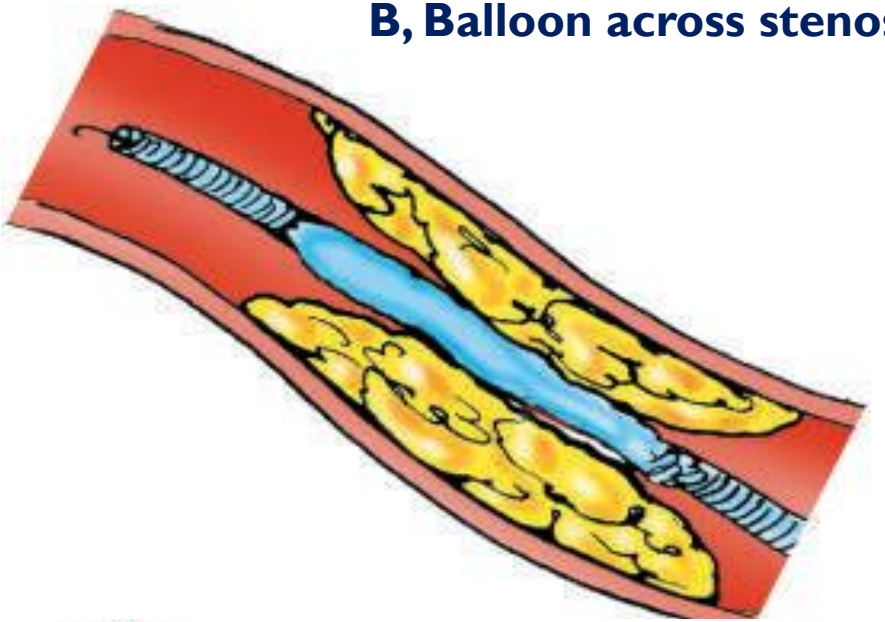
Therapeutic interventional radiology

- **Percutaneous Transluminal Angioplasty and Stent Placement**
- Percutaneous transluminal angioplasty (PTA) is a therapeutic radiologic procedure designed to dilate or reopen stenotic or occluded areas within a vessel using a catheter.
- First, a guidewire is passed through the narrowed area of a vessel.
- A smaller catheter then is passed over the guidewire through the stenosis to begin the dilation process.
- Finally, a larger catheter is passed over the smaller catheter to cause further dilation.
- **Stent placement**
- To assist in maintaining patency of the vessel, a stent is inserted across the treated area during the angioplasty.
- **A stent** is a cage like metal device that is placed in the lumen of a vessel to provide support.

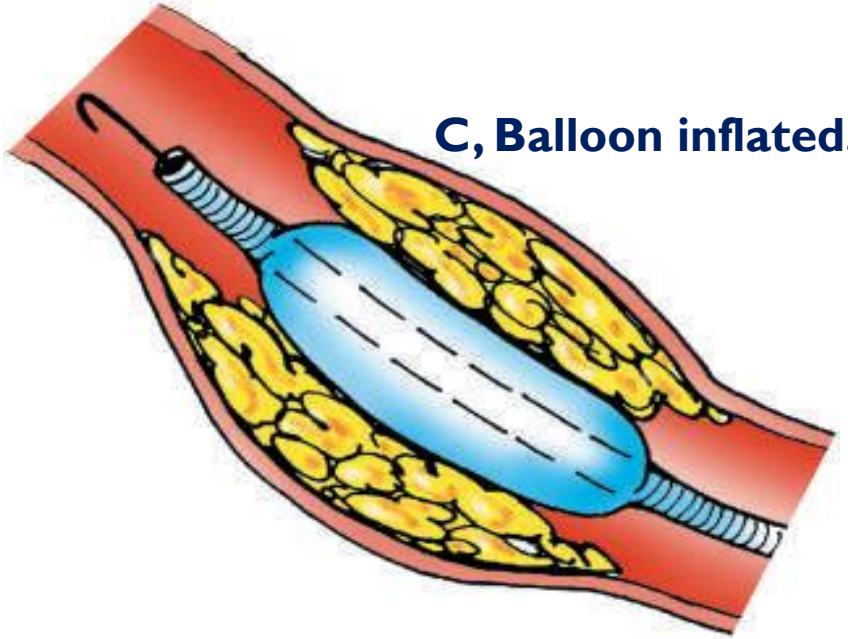
A, Guidewire advanced through stenosis.



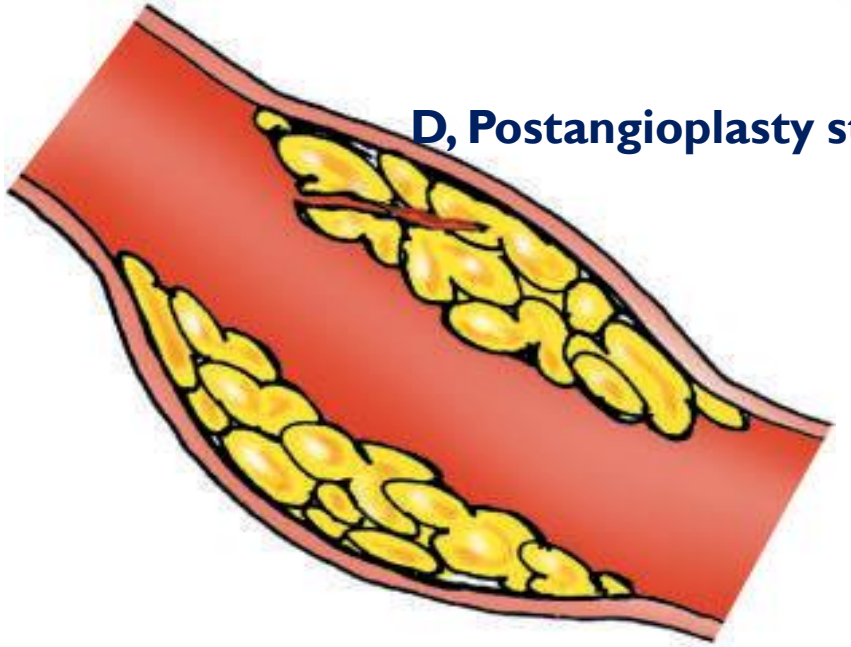
B, Balloon across stenosis.



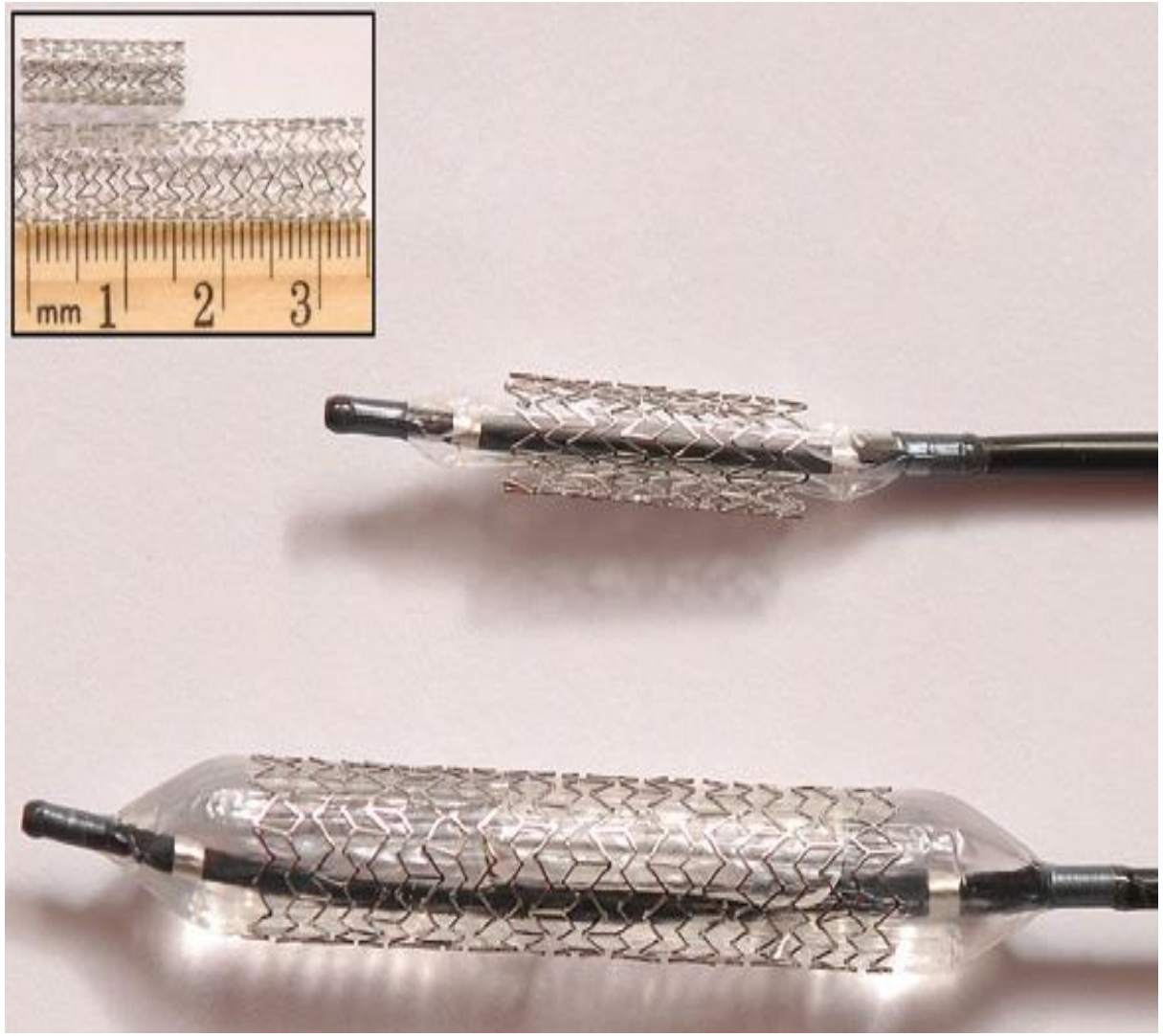
C, Balloon inflated.

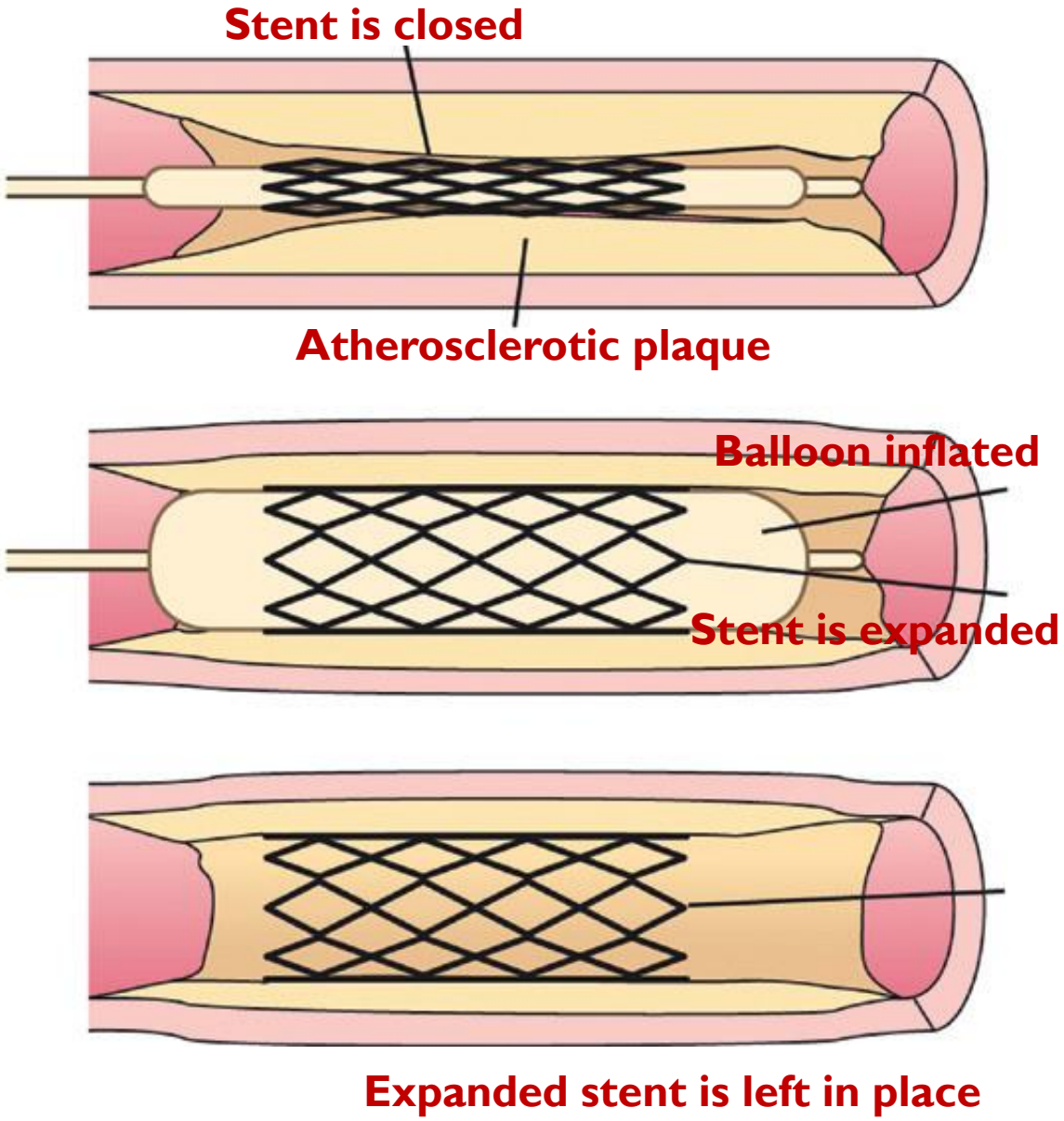


D, Postangioplasty stenotic area.



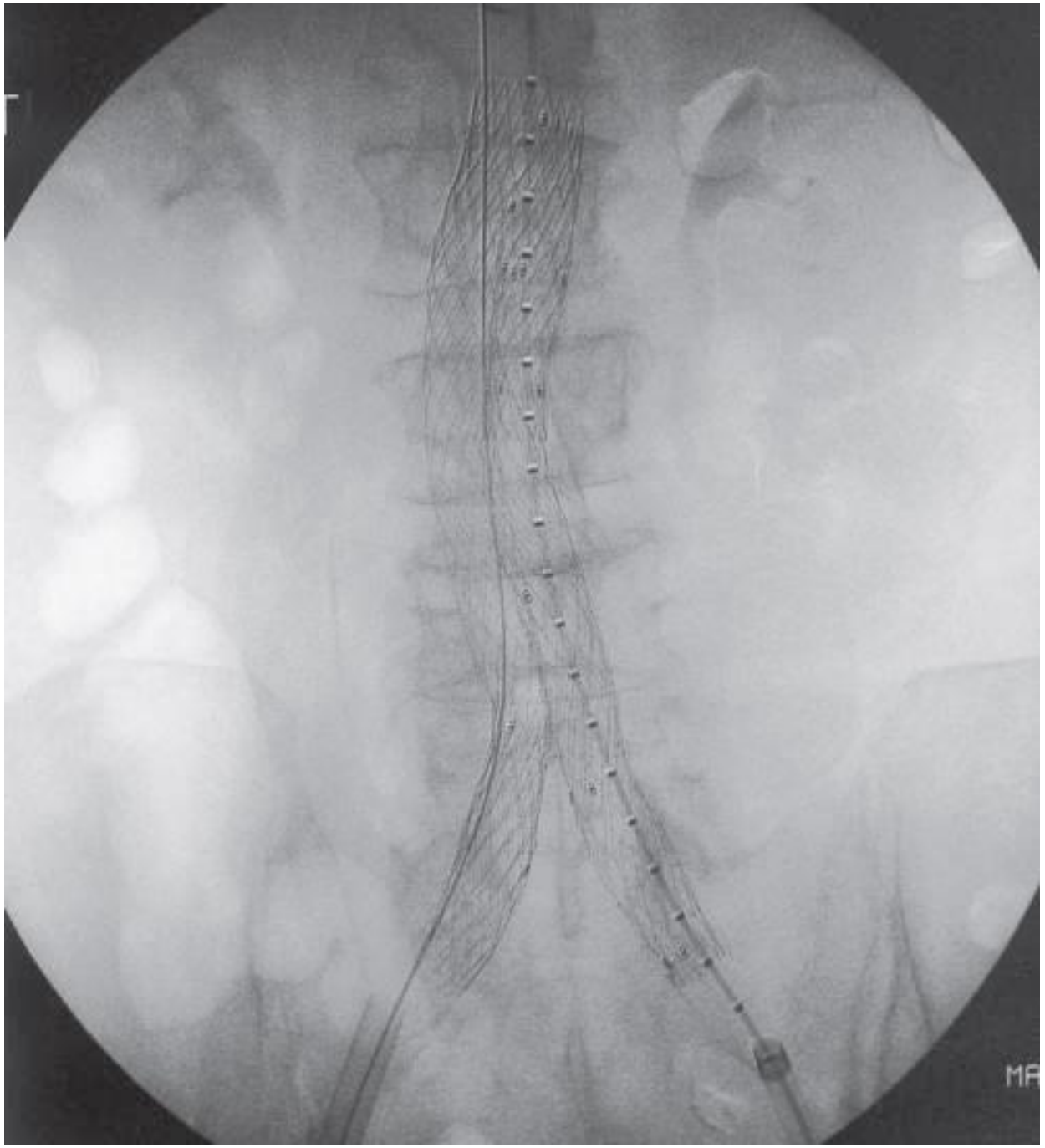
Balloon angioplasty of atherosclerotic stenosis.

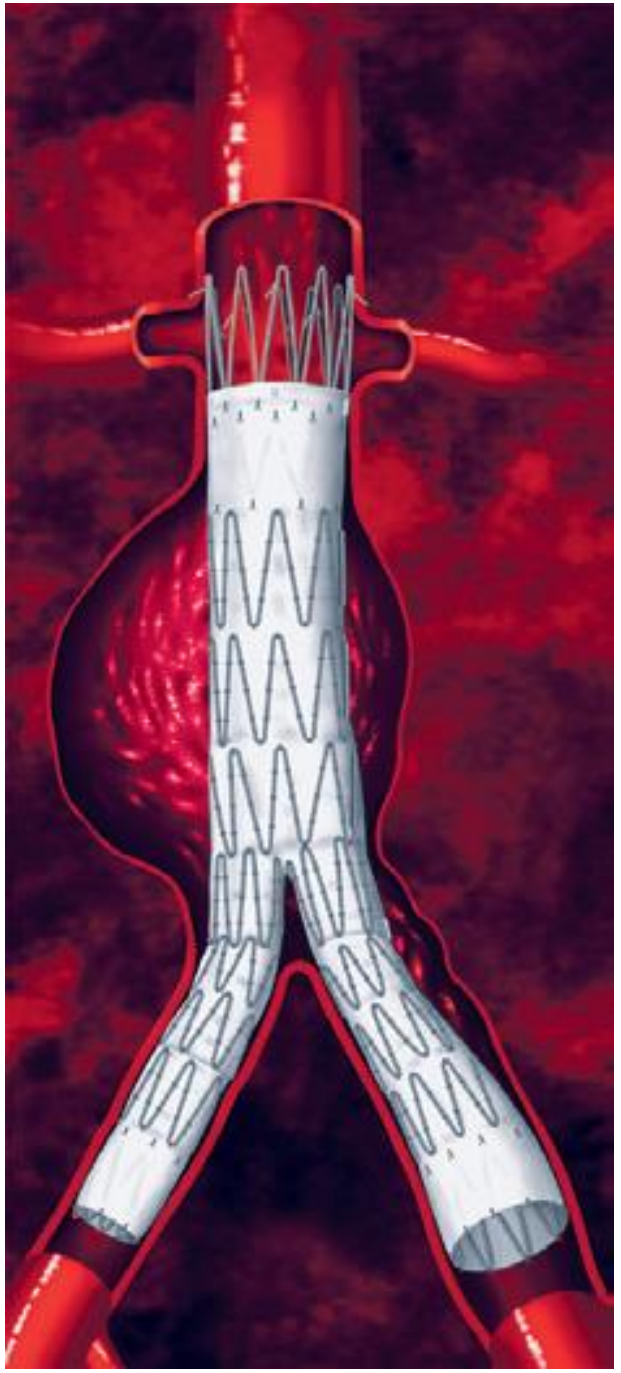


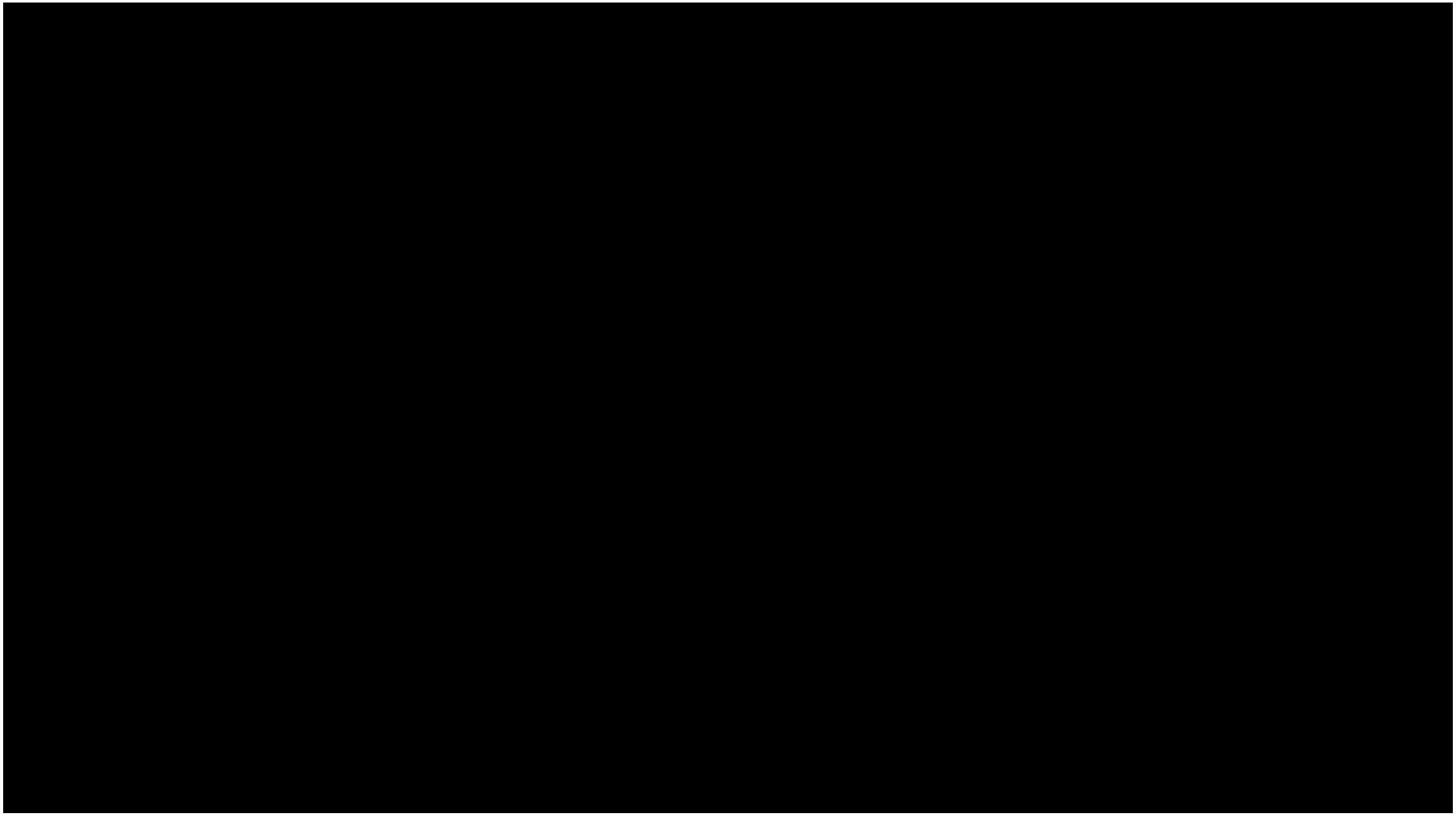


Balloon expanding stent

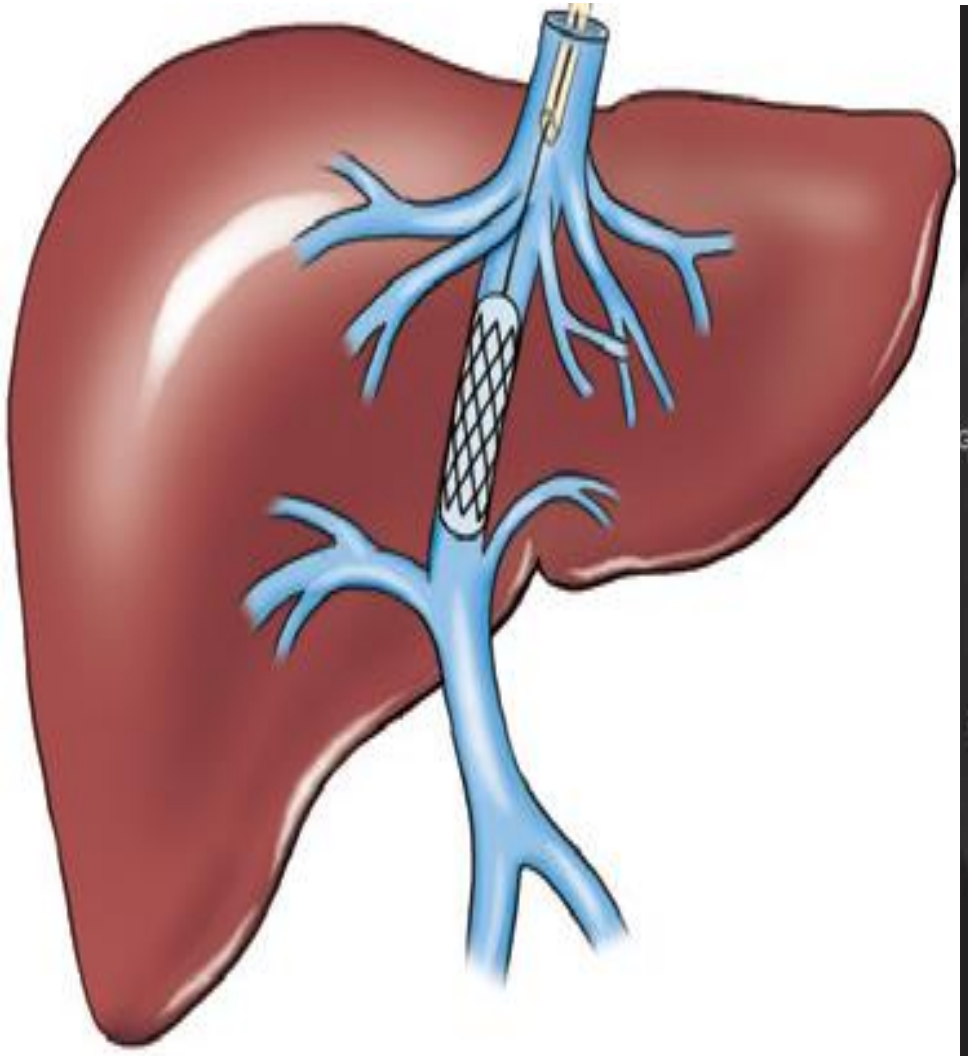
- **Abdominal Aortic Aneurysm Endografts**
- An interventional therapy treats Abdominal aortic aneurysm (AAA) with a transcatheter approach and stenting.
- Patients preferably should have an AAA that is infrarenal or occurring below the renal arteries.
- The stent is a covered device and would occlude the renal arteries. Some newer devices are designed to treat aneurysms that extend to involve the renal artery takeoffs.

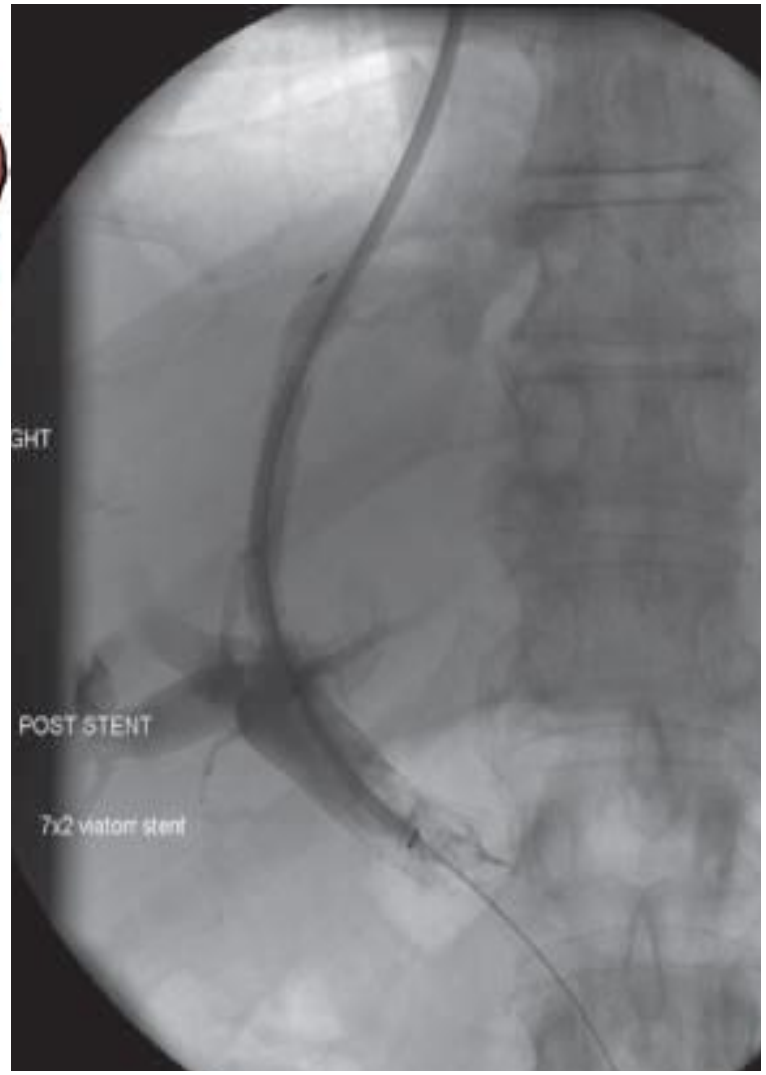


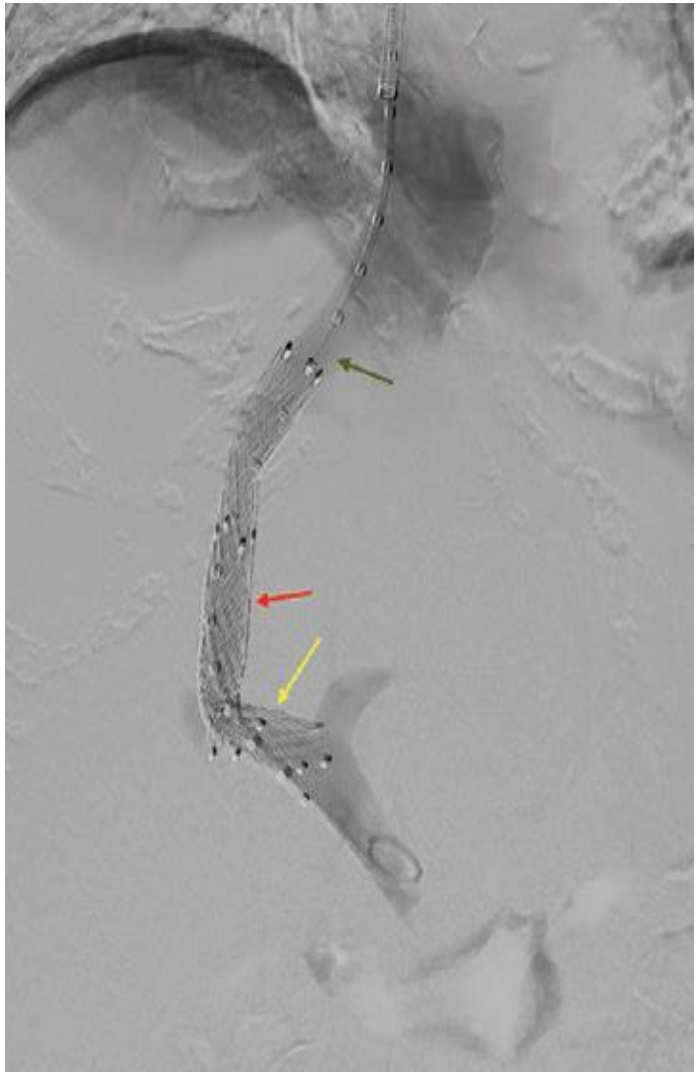




- **Transjugular Intrahepatic Portosystemic Shunt (TIPS).**
- TIPS a procedure that may be used to reduce portal hypertension and its complications. Percutaneous intervention for creating an artificial low-pressure pathway between the portal and hepatic veins.
- The most common approach for a TIPS procedure is from a right internal jugular venous puncture site to the middle or right hepatic vein.
- A special long needle is passed into the hepatic vein and advanced through the liver tissue into the portal vein. The needle is exchanged for an angioplasty balloon catheter, and the tract through the liver tissue is dilated.
- An intravascular stent is positioned across the tract to maintain its patency







Intravascular stent placement in a TIPS procedure

- **Insertion of Venous Access Devices**
- The placement of venous access devices has become a common procedure in vascular and interventional units because the insertion of a catheter may be followed under fluoroscopy.
- Ultrasound is used frequently to identify the location of vessels such as the internal jugular vein for line placement.
- These venous catheters are used for administering chemotherapy, large amounts of antibiotics, frequent blood tests, hemodialysis, or total parenteral nutrition.
- The catheters may remain in place for several months; this depends on the type of catheter used and the clinical indication.

- ***The three most common devices inserted include the following:***

1. Peripherally inserted central catheters (PICC lines)

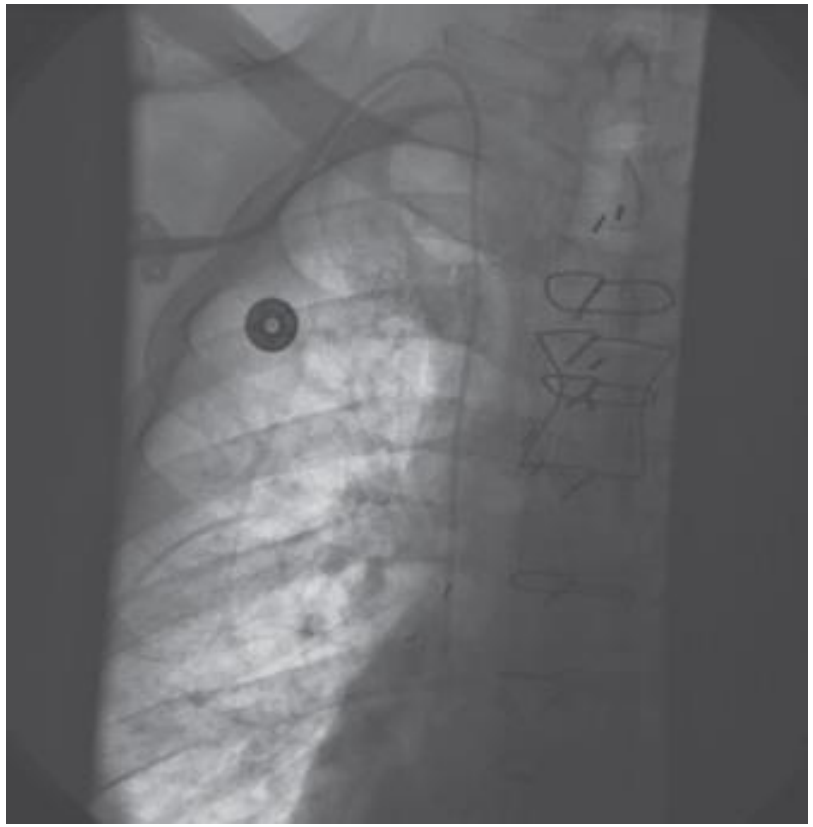
- May remain in place for 6 months if good care is taken of the line.
- The proximal catheter tip is positioned near the right atrium, and the distal end remains exposed (usually on the patient's upper extremity) and must be kept covered and dry to prevent infection.

2. Central lines

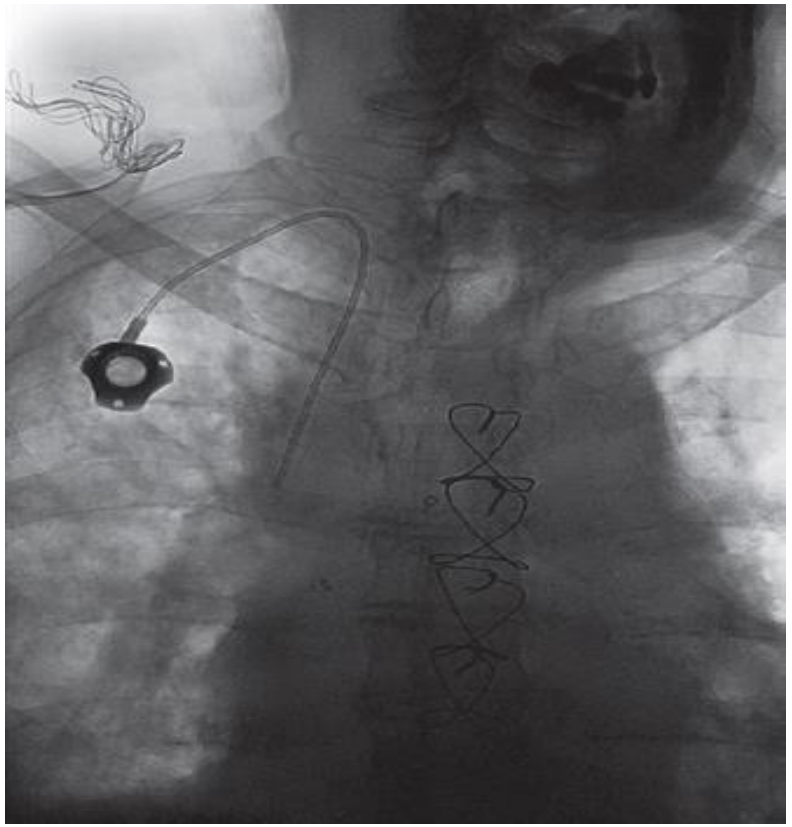
- Which may be tunneled under the chest wall, come in many brands and types. Although some of these central lines may be used for hemodialysis, other central lines are used for general venous access.

3. Subcutaneous ports

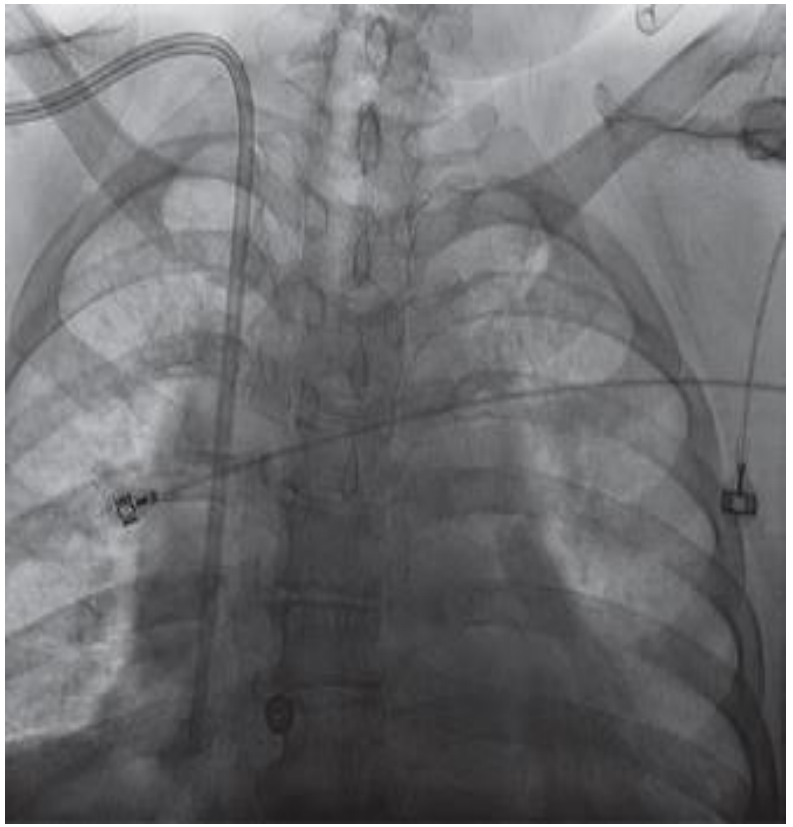
- The most permanent and generally the most expensive devices.
- The catheter tip is placed near the right atrium; the injection port for chemotherapy rests just beneath the chest wall.



Central venous catheter

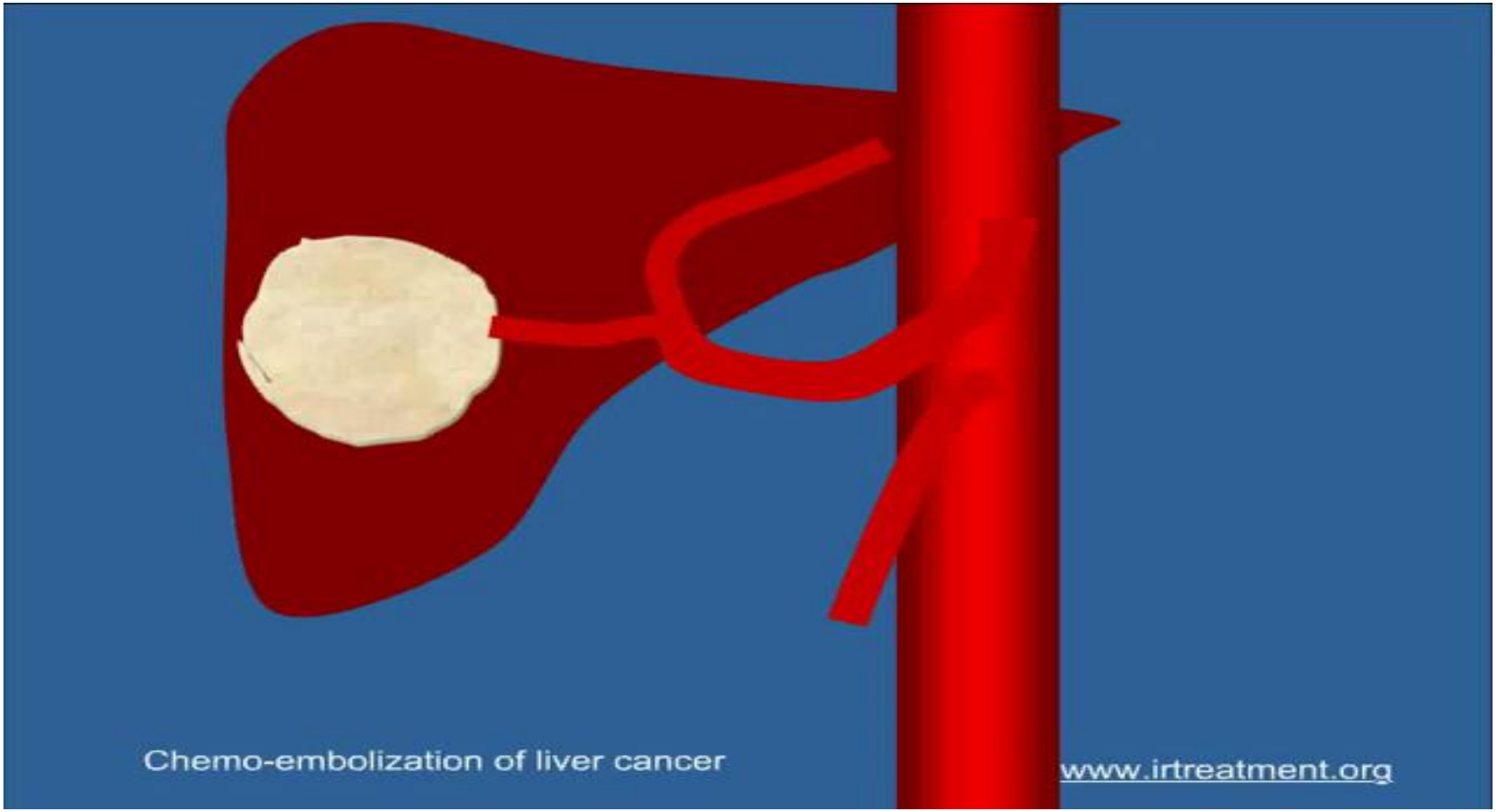


Power Port

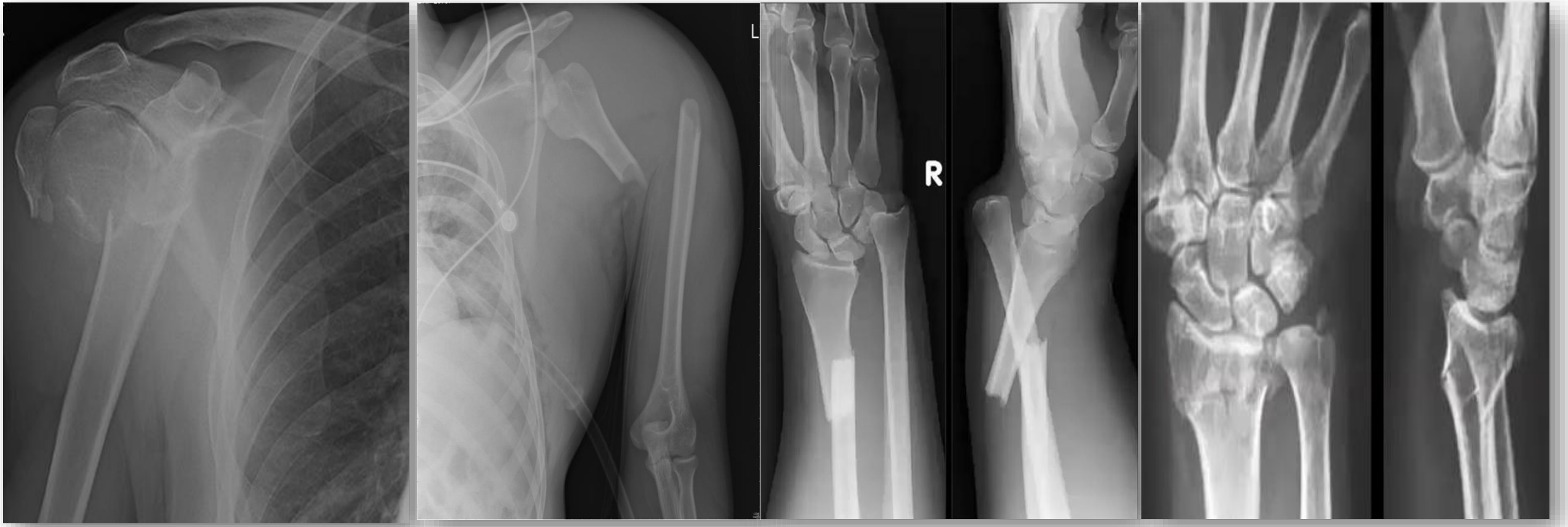


Hemodialysis catheter

- **Embolization Transcatheter**
- **Embolization**
- Obliteration of a vessel by introducing into the bloodstream an occlusive agent (foreign body, biological tissue, sclerosing fluid), which produces the deliberate interruption of vascular flow.
- **Applications**
- The treatment of active hemorrhages and, in many cases, embolization is the procedure that saves the patient's life.
- It is used as a treatment vascular malformations, palliative or therapeutic treatment of neoplasms, redistribution of preoperative blood flow, portal embolization, ablation of transplants and tumor reestablishment.


Chemo-embolization of liver cancer
www.irtreatment.org

Thank you



Upper Limb Fractures

Lecturer: Husam Mansour

Master of Public Health

Master of Medical Imaging

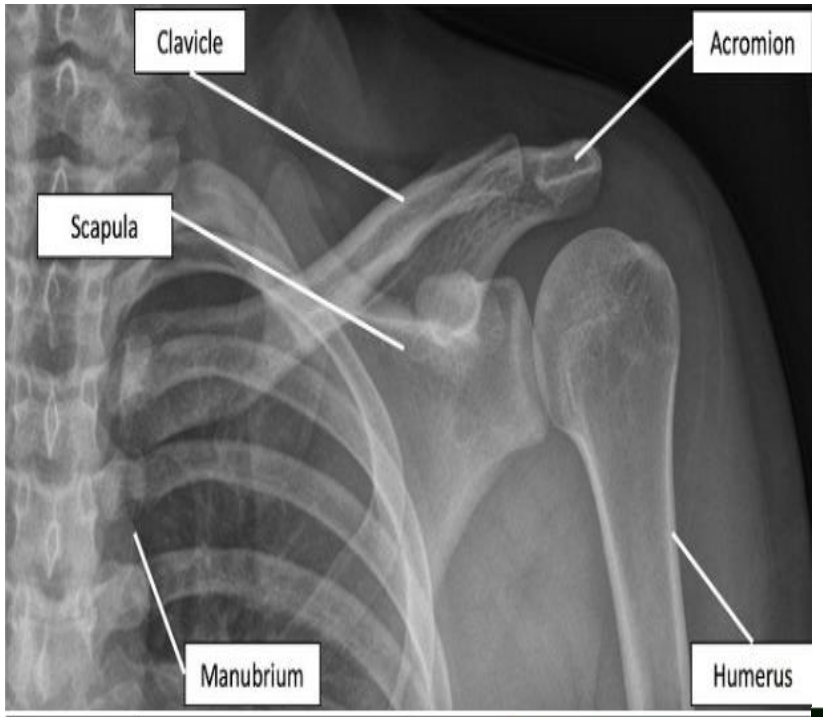
Part I

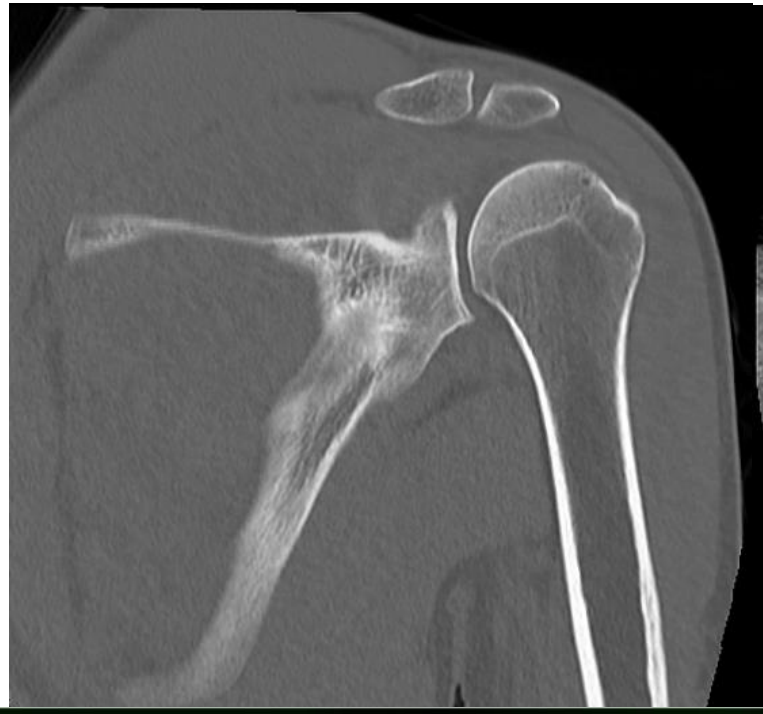
Shoulder truma

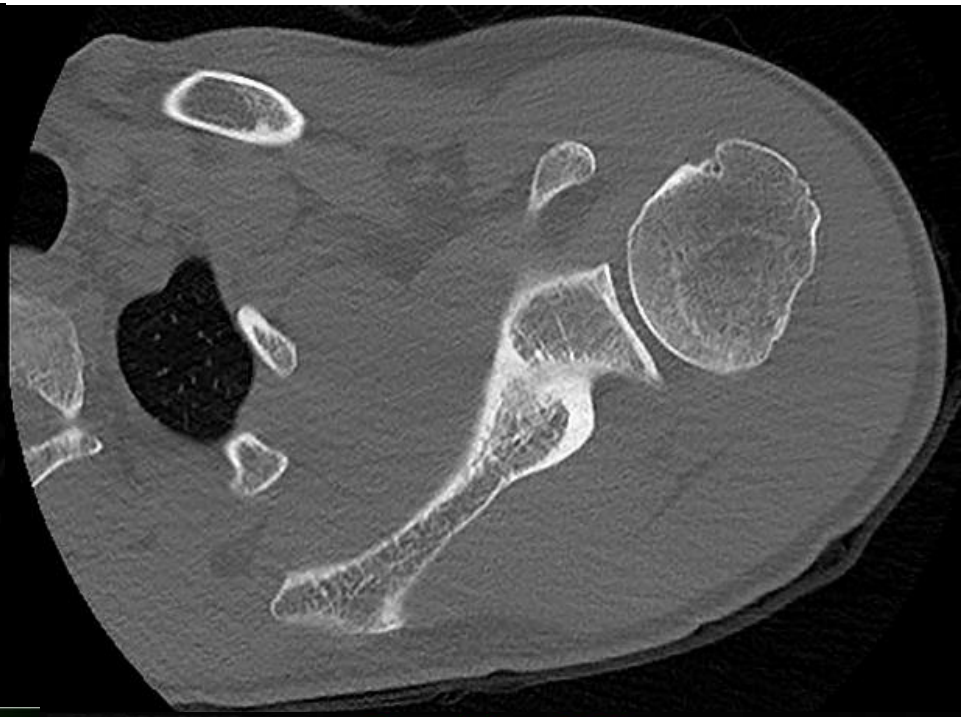
- **Fractures** commonly involve the clavicle, proximal humerus (top of the upper arm bone), and scapula (shoulder blade).
- **Dislocations** can involve any of three different joints.

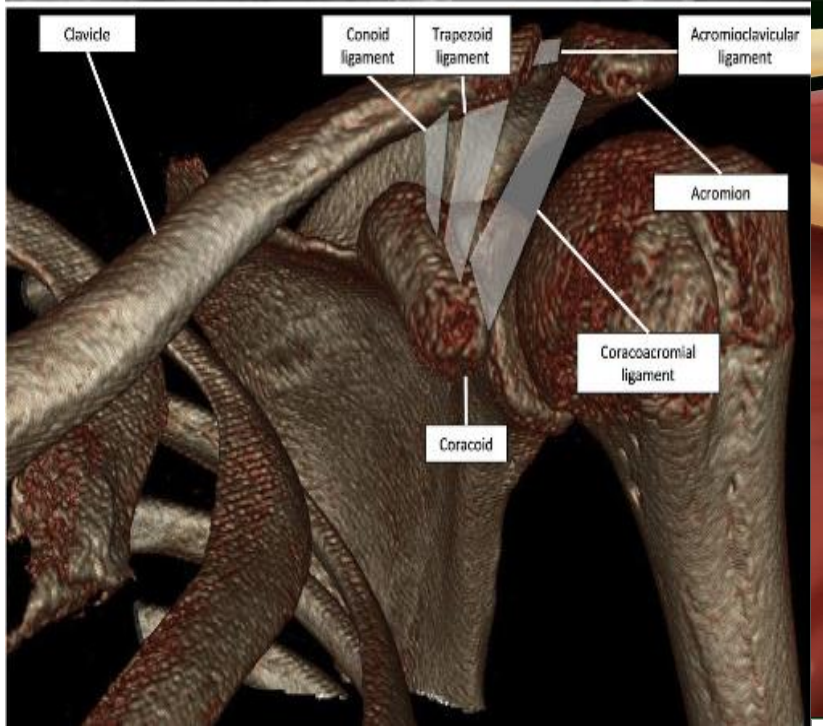
1. A dislocation of the acromioclavicular joint is called a "separated shoulder."
2. A dislocation of the sternoclavicular joint interrupts the connection between the clavicle and the sternum.
3. The glenohumeral joint (the ball and socket joint of the shoulder) can be dislocated toward the front (anteriorly) or toward the back (posteriorly).

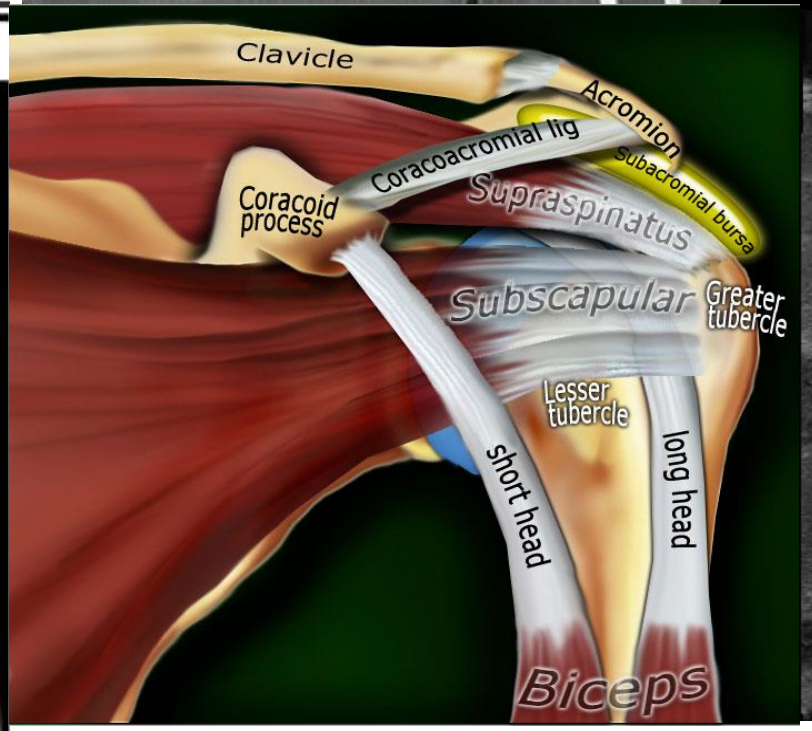
- **Soft-tissue injuries** are tears of the ligaments, tendons, muscles, and joint capsule of the shoulder, such as rotator cuff tears and labral tears.

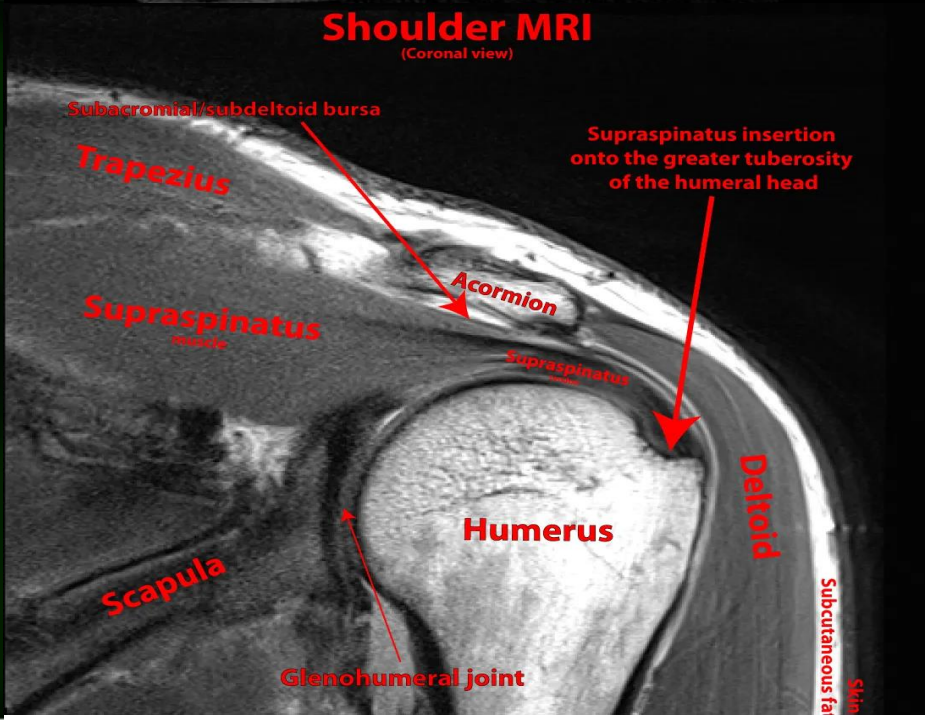

Clavicle
Acromion
Scapula
Manubrium
Humerus



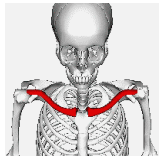



Clavicle
Conoid ligament
Trapezoid ligament
Acromioclavicular ligament
Acromion
Coracoacromial ligament
Coracoid

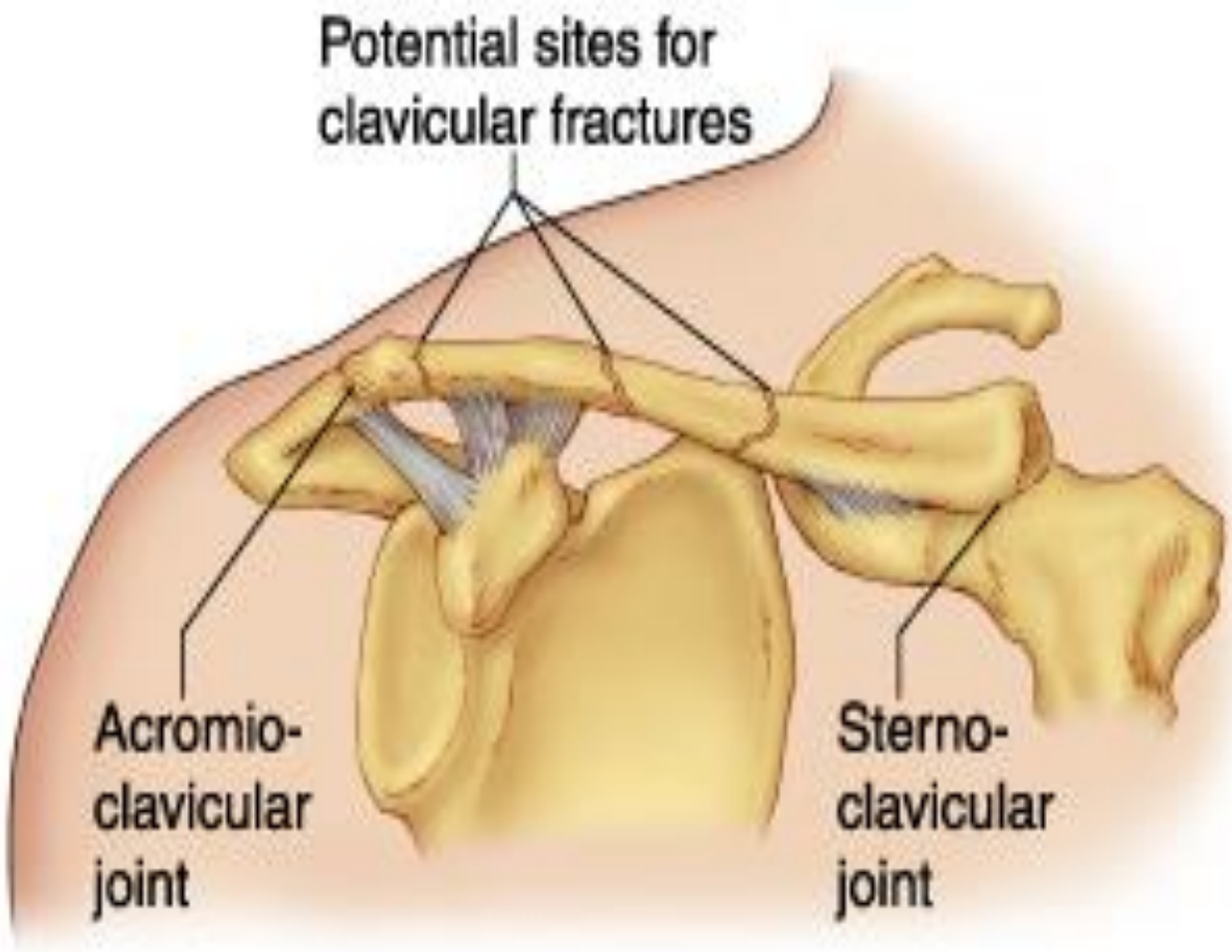

Clavicle
Acromion
Coracoacromial lig
Subacromial bursa
Coracoid process
Supraspinatus
Subscapular
Greater tubercle
Lesser tubercle
short head
long head
Biceps

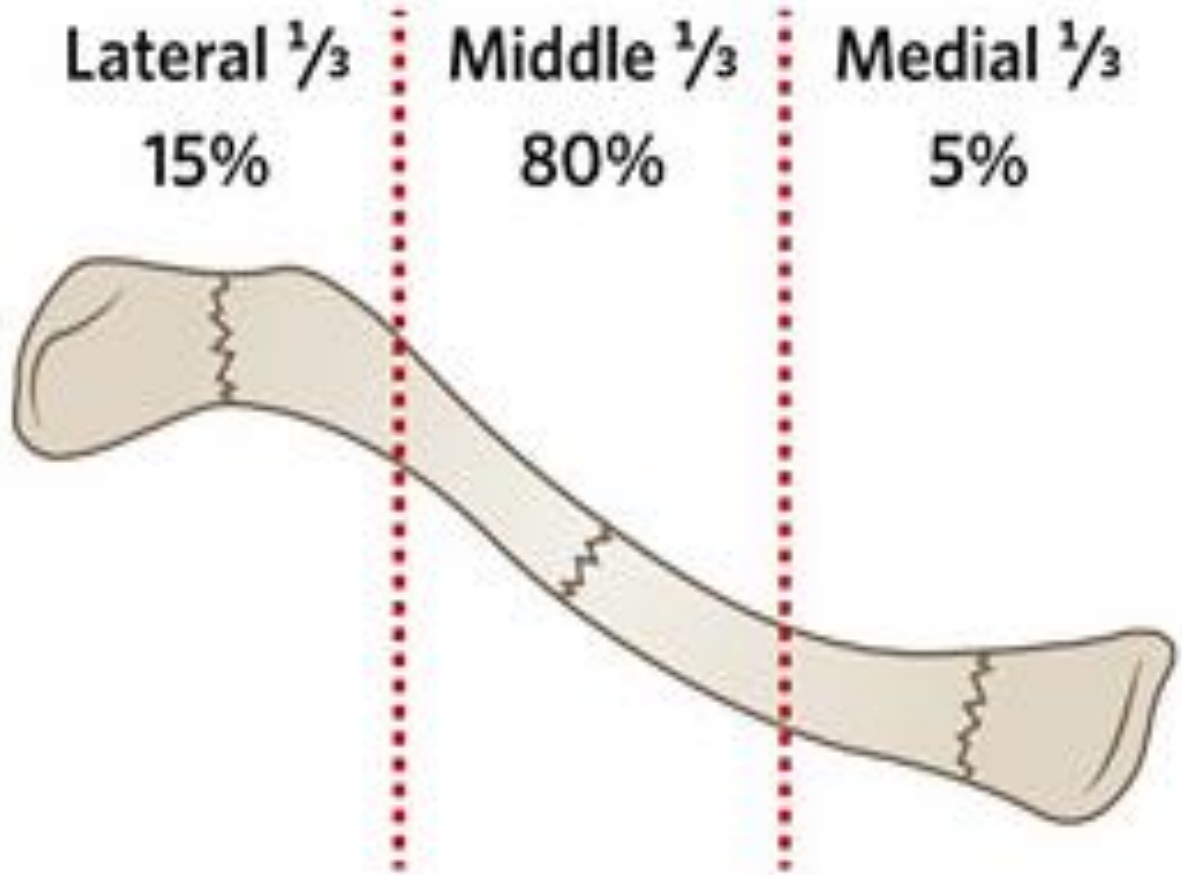

Shoulder MRI
(Coronal view)
Subacromial/subdeltoid bursa
Trapezius
Supraspinatus insertion onto the greater tuberosity of the humeral head
Acormion
Supraspinatus
Supraspinatus
Deltoid
Humerus
Scapula
Glenohumeral joint

Clavicular fracture

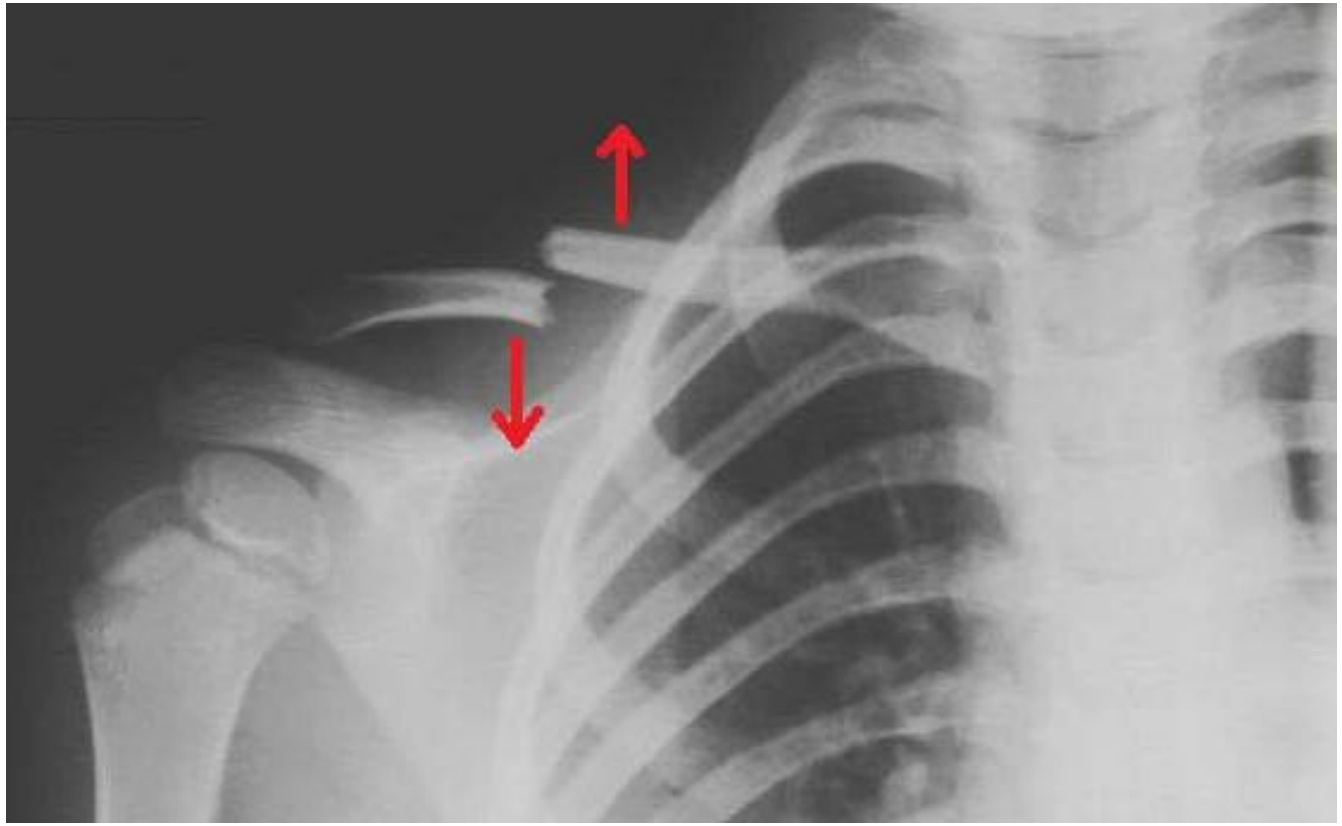


- Fractures can occur at any part of the clavicle. However, the vast majority (80%) occur in the midshaft, at or near the junction of the middle and outer third.
- This is due to two factors: firstly this is the thinnest part of the bone, and secondly, it is the only part of the bone not reinforced by attached musculature and ligaments.
- In most instances, the fracture is evident clinically and easily identified on radiographs.
- Plain radiograph with any trauma situation, two views are better than one. In most clavicle trauma situations, AP and axial views (cephalic tilt 15-45 degree) are obtained.


Potential sites for clavicular fractures
Acromio-clavicular joint
Sterno-clavicular joint


Lateral 1/3
15%
Middle 1/3
80%
Medial 1/3
5%

- After a fracture, the lateral end of the clavicle is displaced inferiorly by the weight of the arm and displaced medially by the pectoralis major.
- The medial end is pulled superiorly by the sternocleidomastoid muscle.

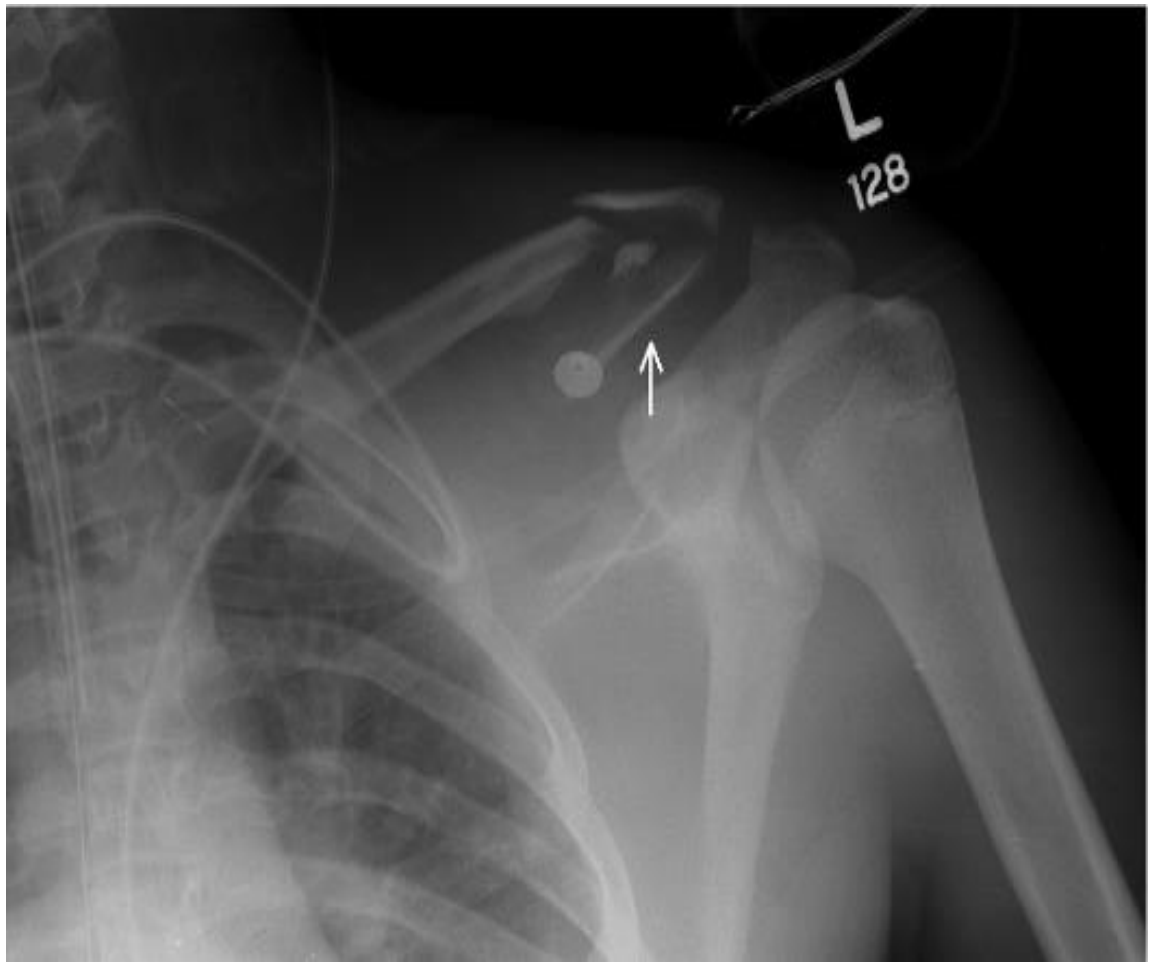


X-Ray of a clavicular fracture. Note how the medial end is raised and the lateral end lowered

Comminuted midshaft fractures of the right clavicle, with cephalic angulation.

3 D CT shows midshaft fractures of the clavicle,

Radiograph demonstrates
Group 3 fracture (arrow)



Comminuted avulsion of
coracoclavicular ligament

Coronal and 3D CT shows a fracture of the medial third of the right clavicle, which is displaced anteriorly. Sternoclavicular joint is also subluxed.

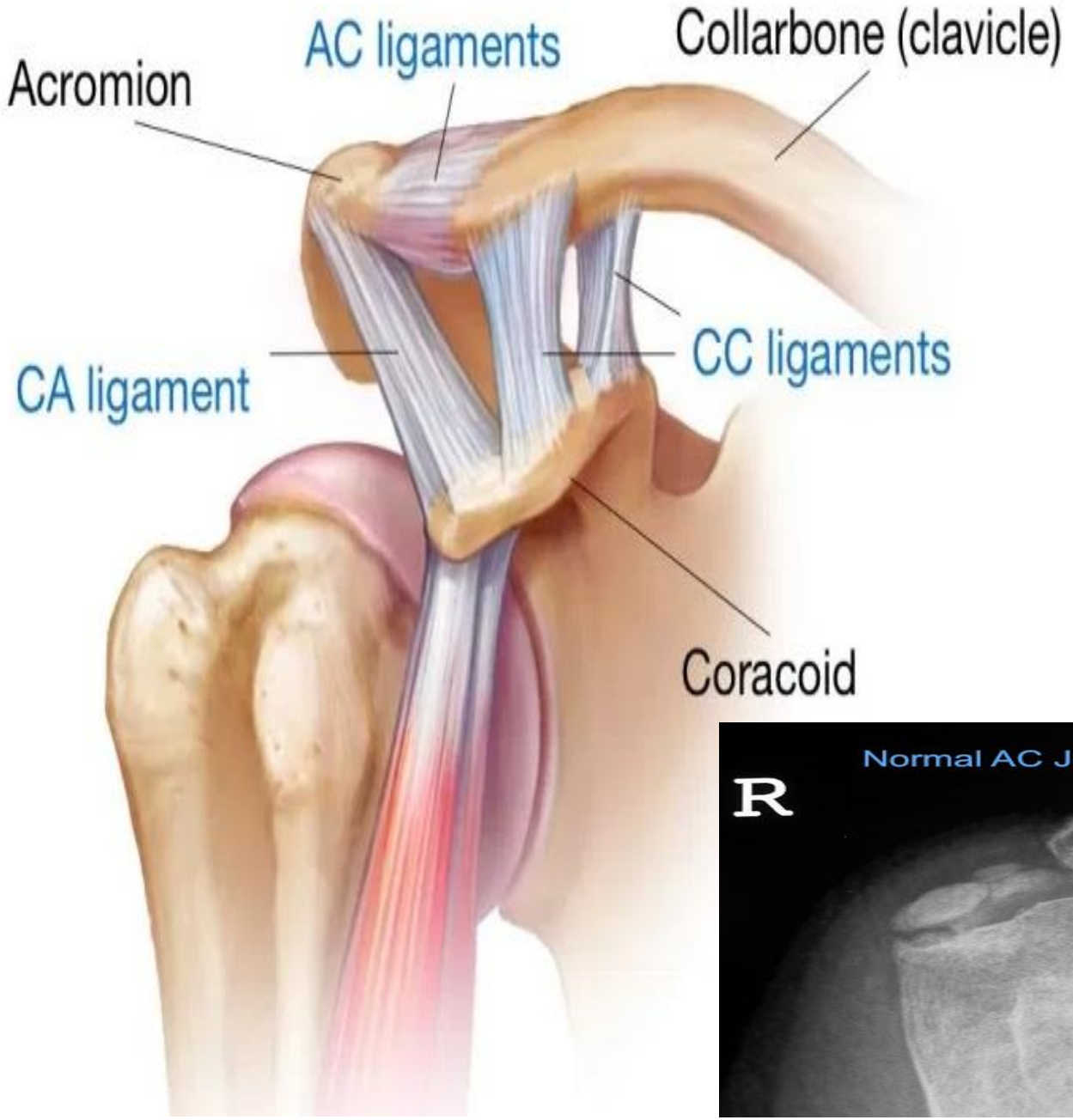
Un-displaced lateral third fracture of the clavicle in a 12 year old boy.

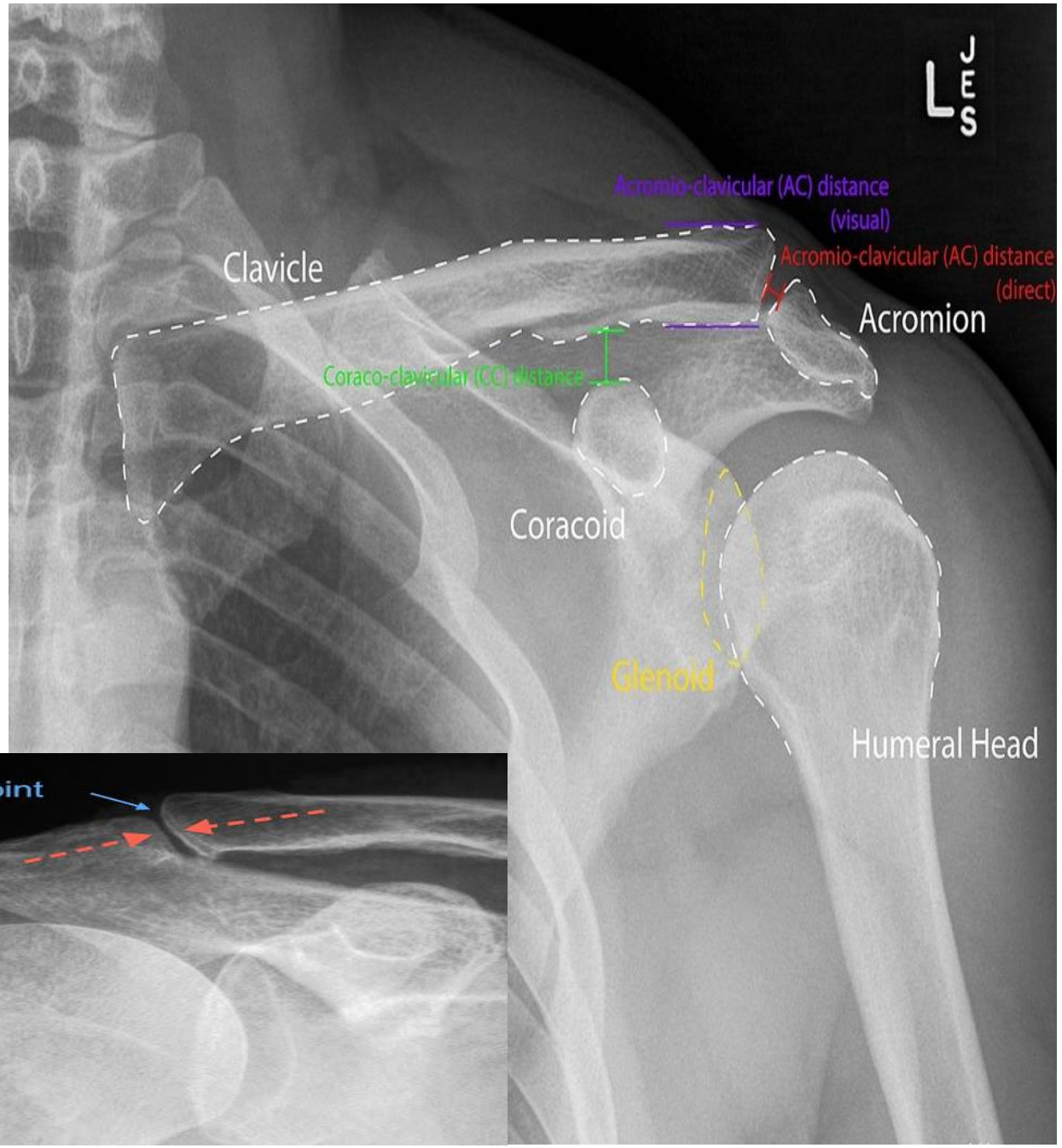
Seven year old boy with fracture of the middle third of the clavicle.

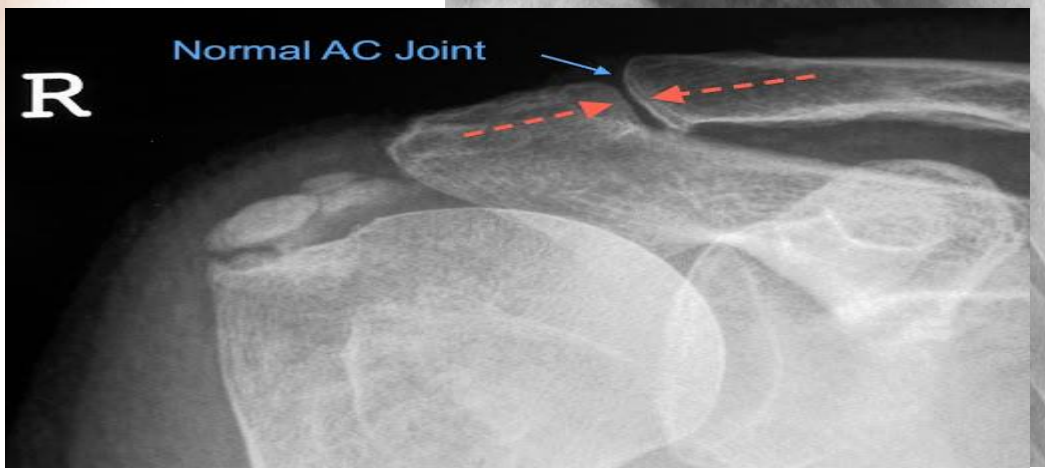
Acromioclavicular joint (ACJ) dislocations

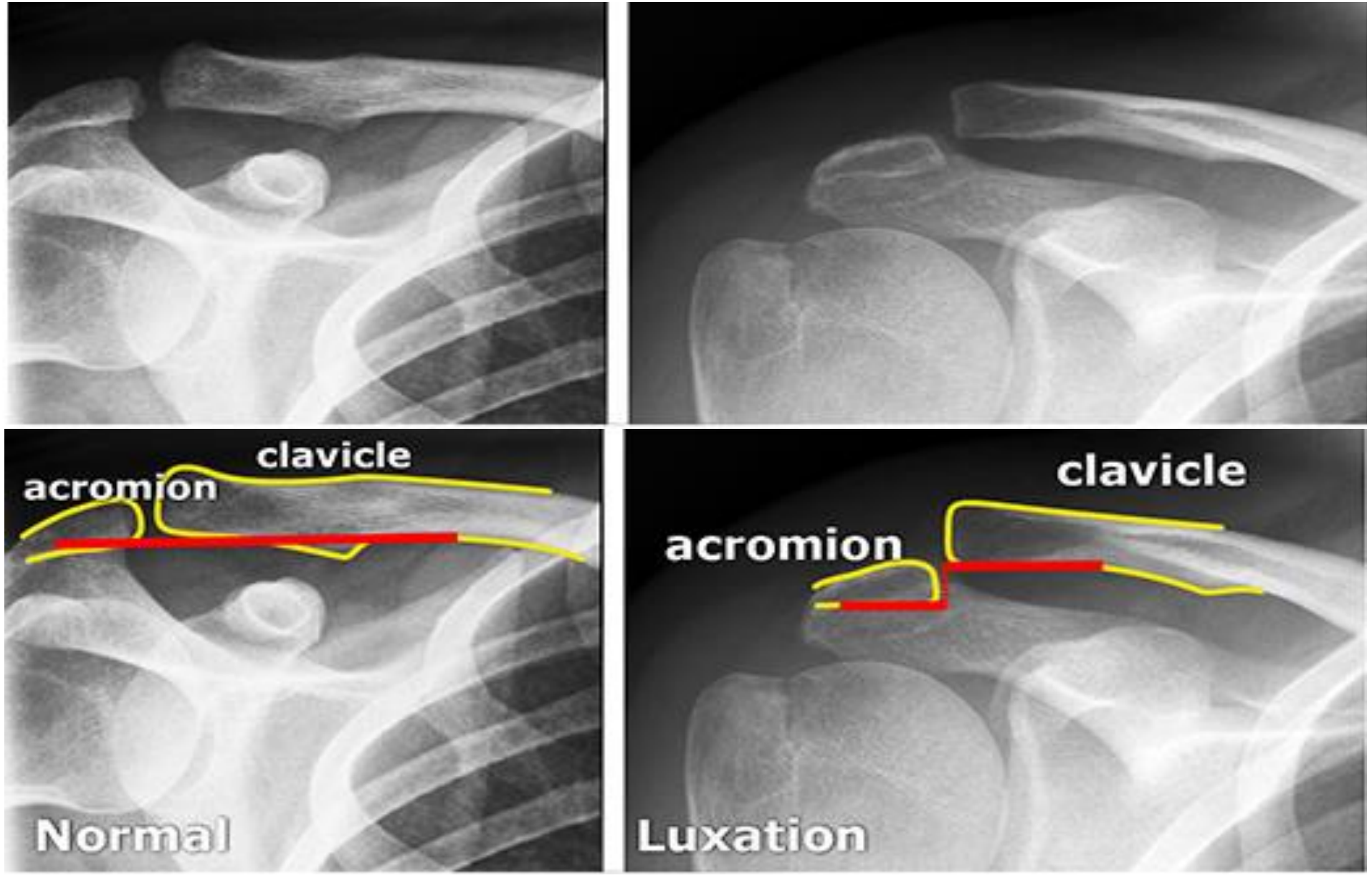
- ACJ, is a joint at the top of the shoulder. It is the junction between the acromion and the clavicle.
- AC joint separation, often called a shoulder separation, is a dislocation of the clavicle from the acromion.
- Three main ligaments stabilize the ACJ.

1. The acromioclavicular ligament (AC) which attaches the clavicle to the acromion of the scapula.
2. The coracoacromial ligament, which runs from the coracoid process to the acromion.
3. The coracoclavicular ligaments (CC) which consists of two ligaments, the conoid and trapezoid ligaments.


Acromion
AC ligaments
Collarbone (clavicle)
CA ligament
CC ligaments
Coracoid


L
J
E
S
Acromio-clavicular (AC) distance (visual)
Acromio-clavicular (AC) distance (direct)
Clavicle
Acromion
Coraco-clavicular (CC) distance
Coracoid
Glenoid
Humeral Head


R
Normal AC Joint


clavicle
acromion
Normal
clavicle
acromion
Luxation

Rockwood classification of ACJ dislocations.

Grade 1. Partial tear of the acromioclavicular (AC) ligaments.

Grade 2. Complete tear of the AC ligaments and superior displacement of clavicle.

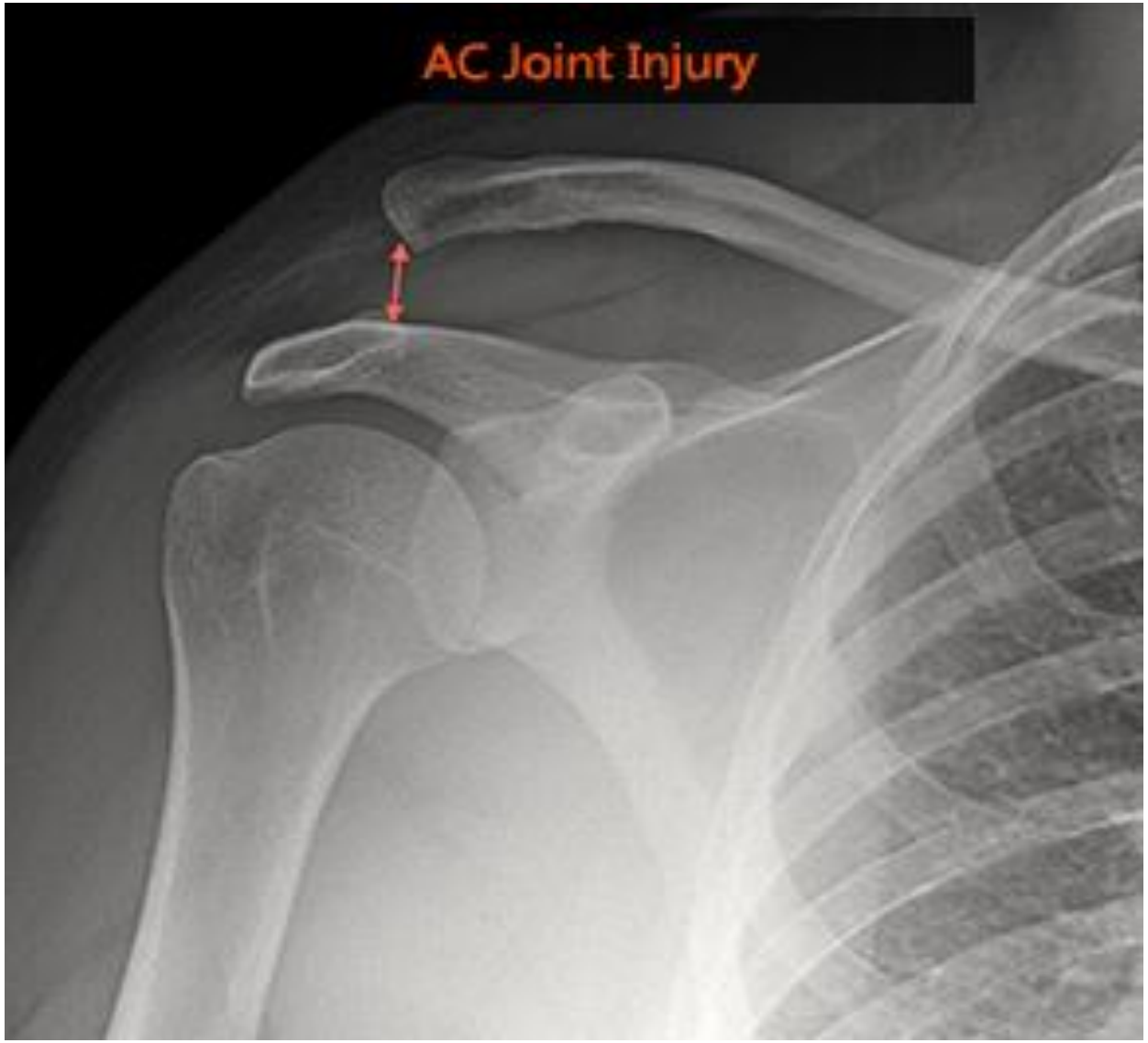
Grade 3. Complete tear of the AC and coracoclavicular (CC) ligaments plus superior displacement of clavicle and complete loss of contact with acromion.

Grade 4. Complete tear of the AC and CC ligaments plus posterior displacement of clavicle into or through the trapezius muscle.

Grade 5. Complete tear of the AC and CC ligaments plus superior displacement of clavicle and complete detachment of deltoid and trapezius from distal clavicle.

Grade 6. Complete tear of the AC and CC ligaments plus clavicle displaced inferior to acromion and coracoid process.

Type 3 AC Separation
AC Separation
CC
Separation

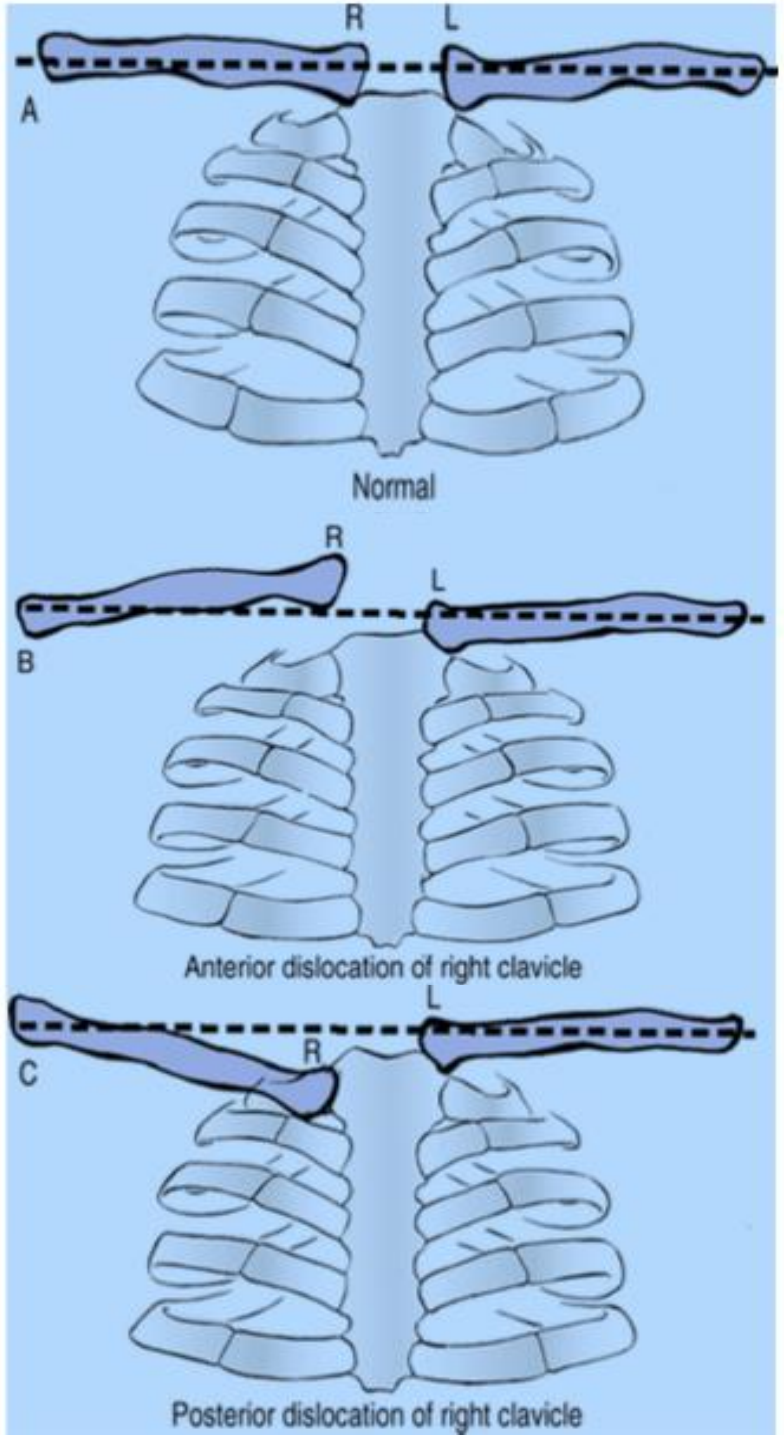

AC Joint Injury

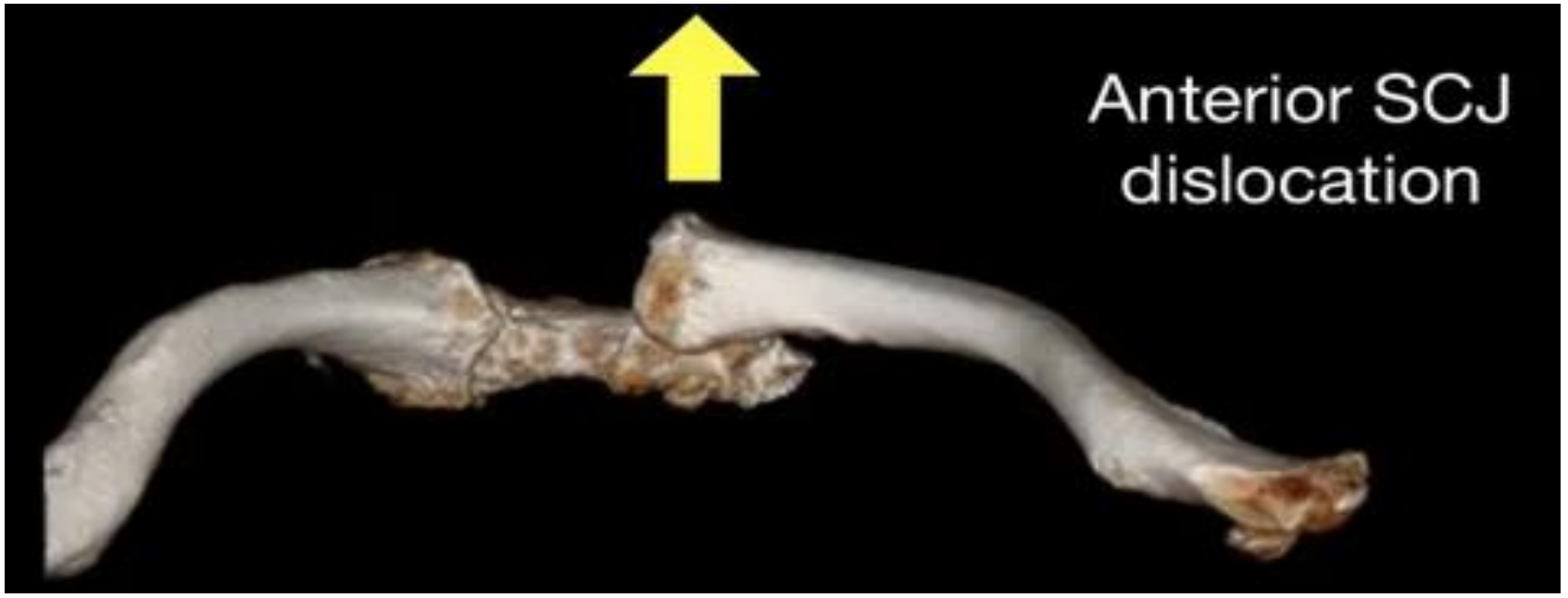
Sternoclavicular joint dislocation

- Sternoclavicular joint dislocation is a relatively uncommon injury that can be easily missed or misdiagnosed.
- Specialized sternoclavicular X-ray views should be supplemented by CT/MRI if clinical suspicion is high.
- Sternoclavicular dislocations account for 3% of all shoulder girdle injuries.
- 95% of SCJ dislocations are unilateral and anterior dislocations are far more common than posterior dislocations due to the weaker anterior sternoclavicular ligament.
- Dislocations of the SCJ generally occur following a fall on the outstretched hand or a direct blow to the shoulder.

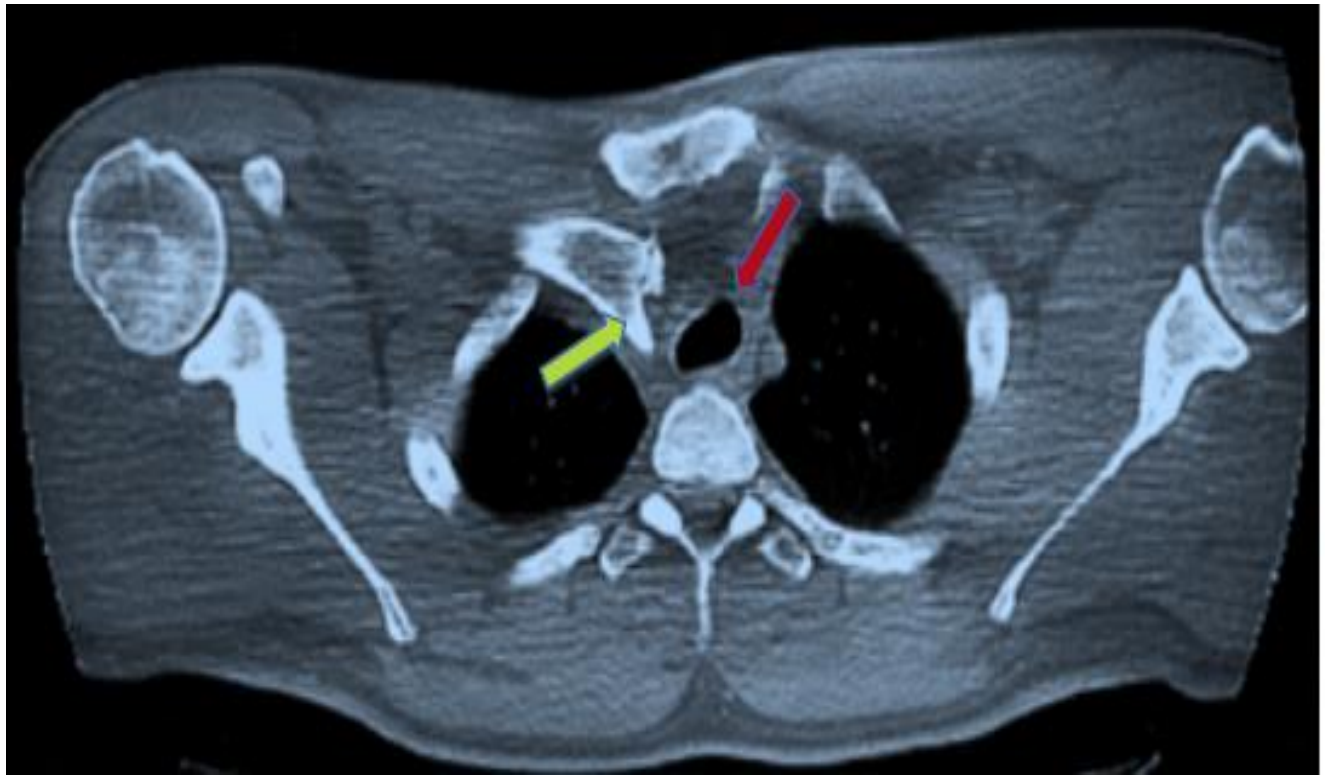
Sternoclavicular joint dislocation Radiology

- A chest x-ray may be normal and additional views of the clavicle may be needed.
- Normally, the medial ends of the clavicles will appear at the same level.
- Posteriorly dislocated clavicular head will appear lower, and an anteriorly dislocated clavicular head will appear higher than its counterpart.
- Plain radiograph indicate joint space widening at the sternoclavicular joint more easily visualized on an angled view.
- CT is the preferred imaging modality because of its ability to easily demonstrate the dislocation as well as soft issue abnormalities

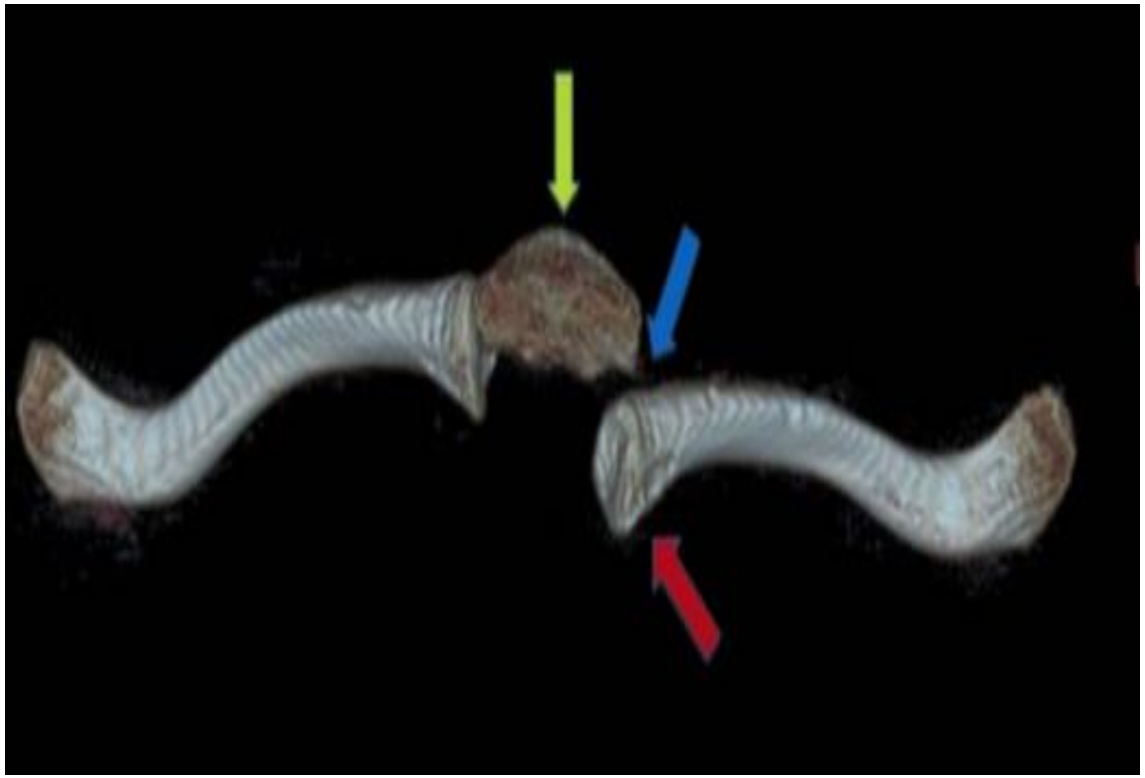

R
L
A
Normal
R
L
B
Anterior dislocation of right clavicle
L
R
C
Posterior dislocation of right clavicle


Anterior SCJ
dislocation

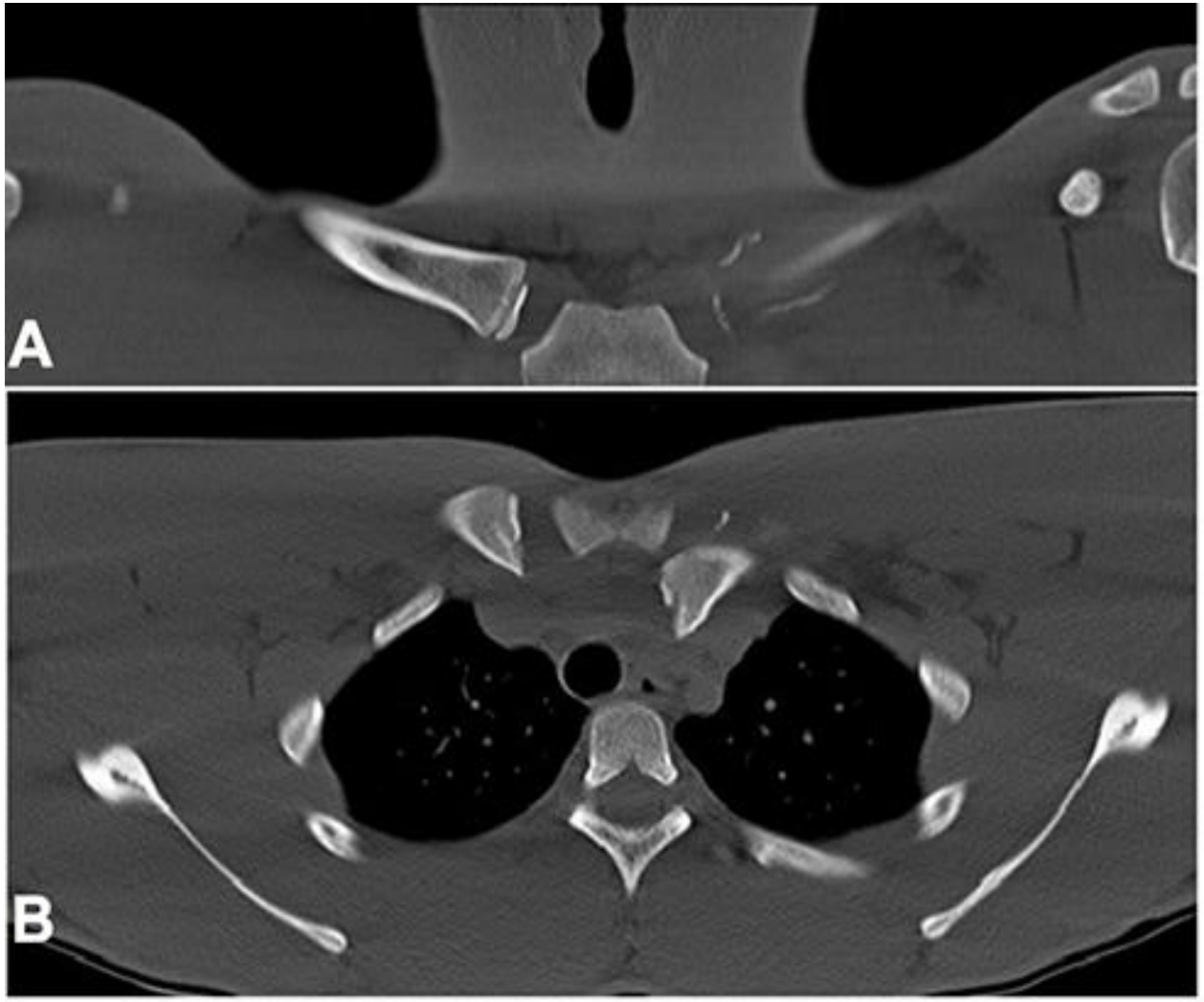
Posterior SCJ
dislocation



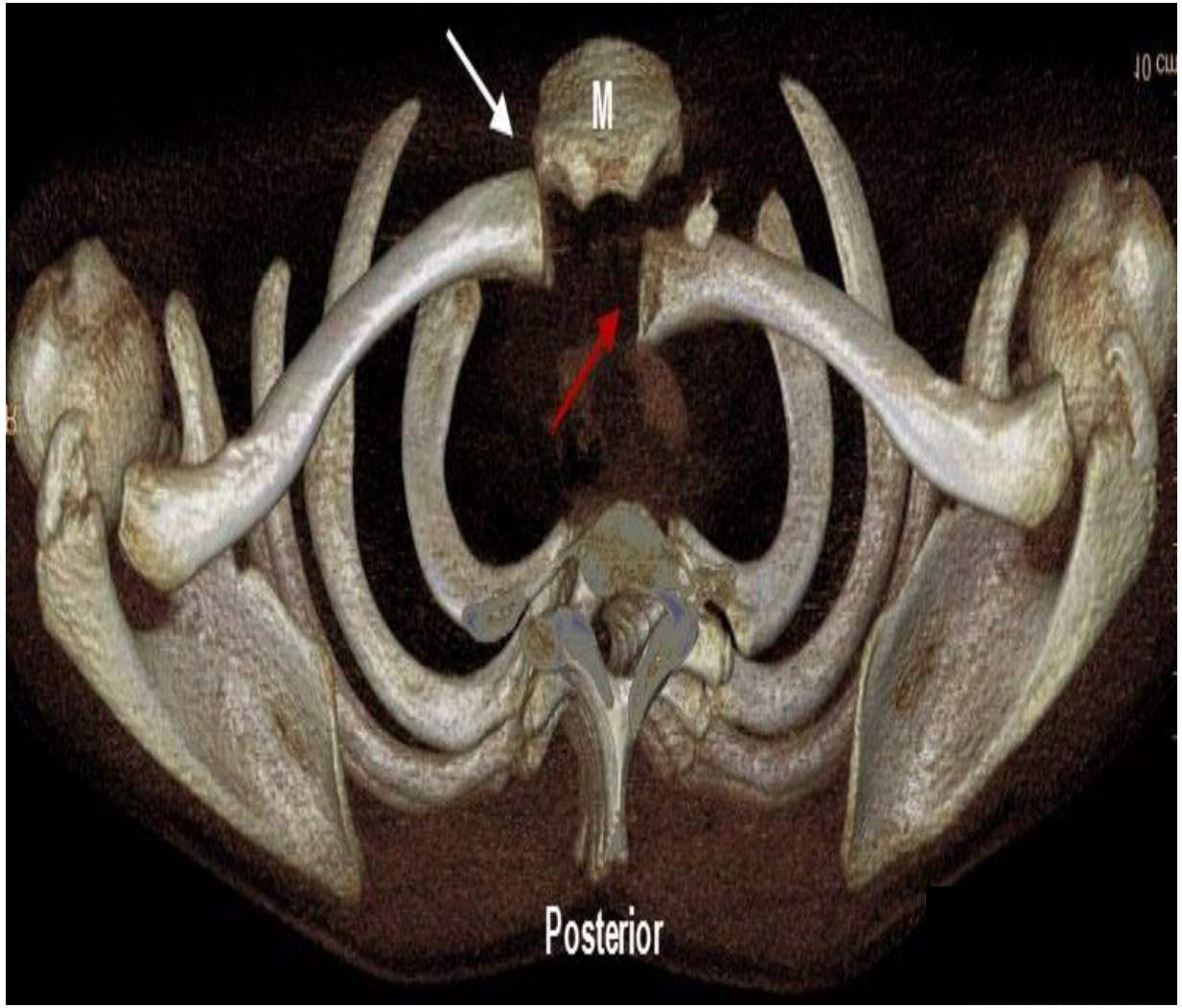
CT of a posterior sternoclavicular dislocation. Note the clavicle is pushed deeper into the chest (yellow arrow), placing the trachea (breathing tube) in danger (red arrow: trachea). The trachea is pushed deeper into the chest, away from its usual spot; this may cause breathing difficulty and can be life-threatening



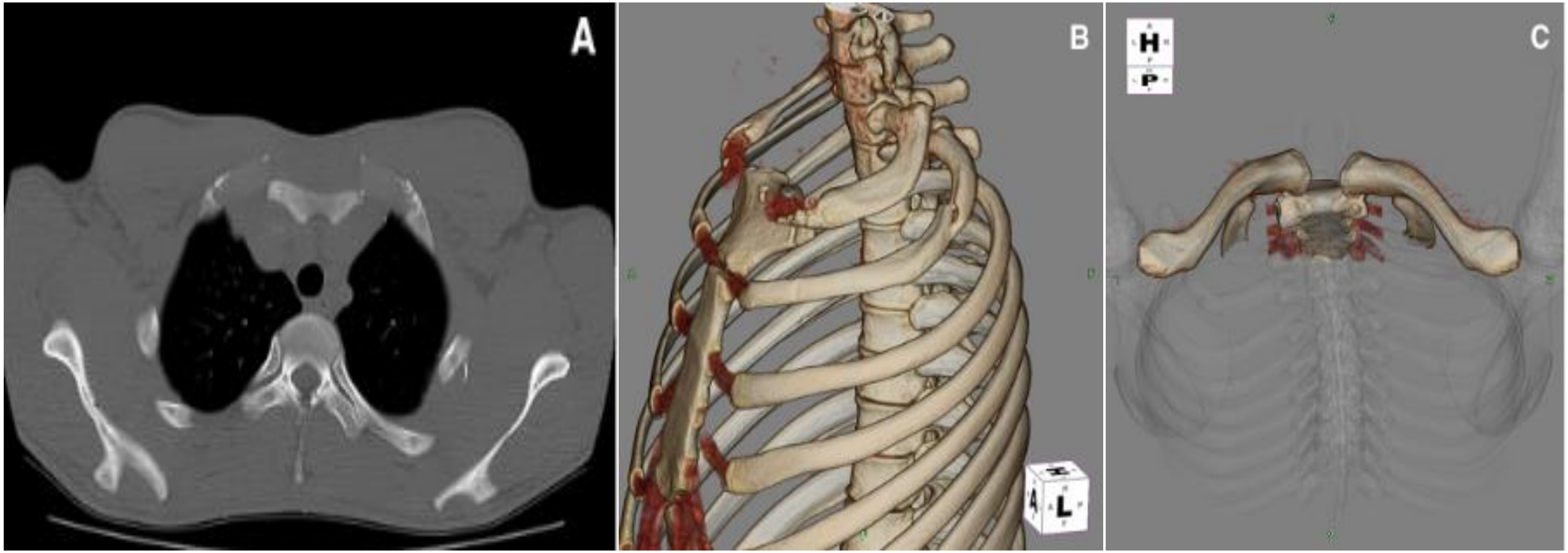
3D CT scan of a posterior sternoclavicular dislocation. The clavicle (red arrow) is pushed deeper into the chest behind the breastbone (yellow arrow) in a posterior dislocation. There is now a gap (blue arrow) between the breastbone and the clavicle.



CT scan depicting coronal (A) and axial (B) views of a left posterior sternoclavicular physeal fracture dislocation.



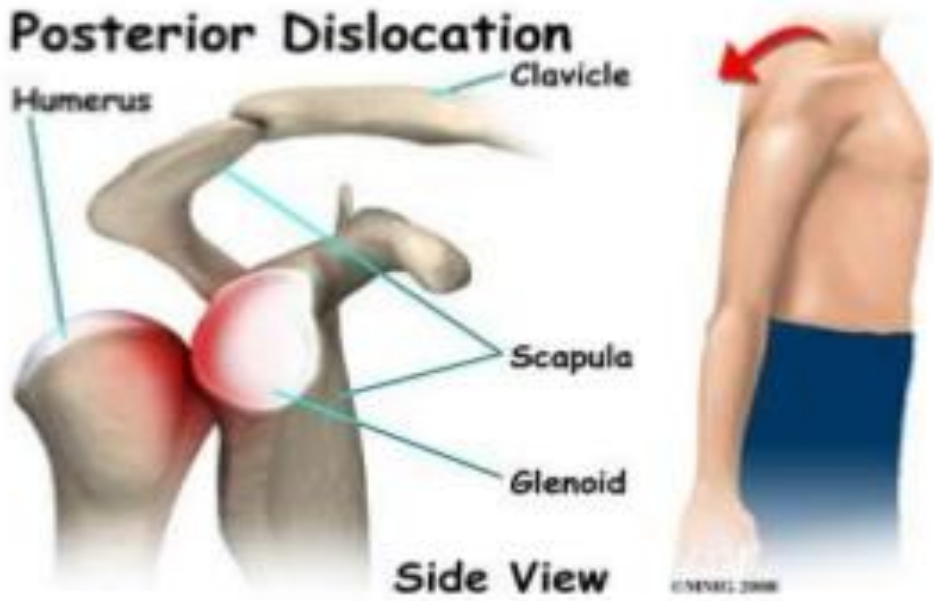
Posterior Dislocation of the Left Sternoclavicular Joint. A 3D reconstruction in the axial plane shows posterior dislocation of the medial head of the left clavicle Red arrow) relative to the manubrium of the sternum (M). The opposite SC joint is intact (white arrow).



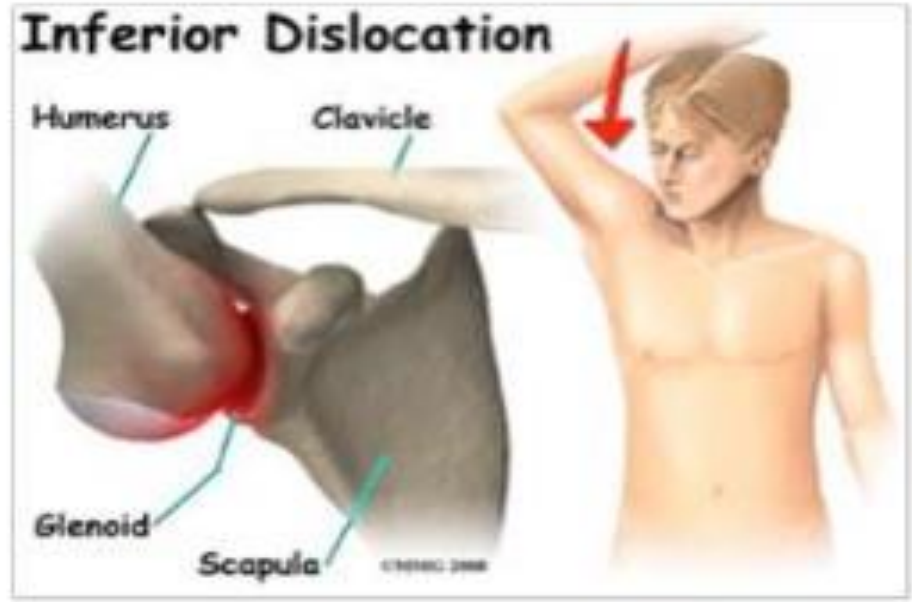
A Initial computed tomography scan showing the anteriorly dislocated clavicle. **B** 3-D reconstruction graphically demonstrates displaced medial end of the both clavicle and 1st rib to anterior. **C** superior view of 3-D computed tomography scan with the dislocated both sternoclavicular joints to anterior

Glenohumeral Dislocation (Luxation)

- A dislocated shoulder is when the head of the humerus is out of the shoulder joint.
- Symptoms include shoulder pain and instability. Complications may include a Bankart lesion, Hill-Sachs lesion, rotator cuff tear, or injury to the axillary nerve.
- A shoulder dislocation often occurs as a result of a fall onto an outstretched arm or onto the shoulder. Diagnosis is typically based on symptoms and confirmed by X-rays.
- They are classified as anterior, posterior, inferior, and superior with most being anterior.


Posterior Dislocation
Clavicle
Humerus
Scapula
Glenoid
Side View

Anterior Dislocation
Clavicle
Glenoid
Scapula
Humerus
Side View


Inferior Dislocation
Humerus
Clavicle
Glenoid
Scapula

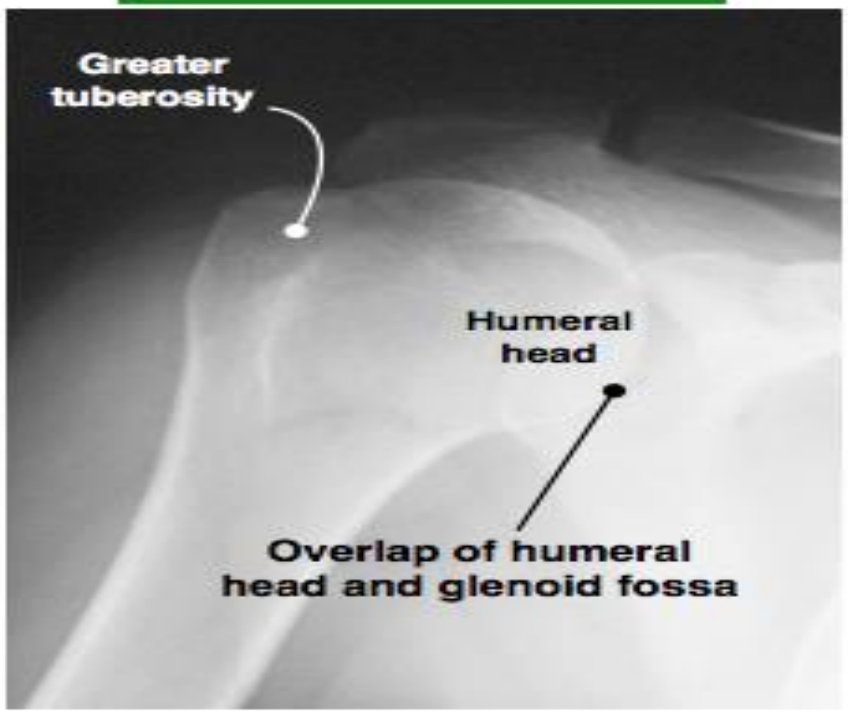
Anterior shoulder dislocation

- Anterior or subcoracoid shoulder dislocation (**96%**)
- External rotation and abduction
- 40% recurrent
- May be associated with:

1. **Greater Tuberosity Fracture** (15%)
2. **Bankart lesion**: Fracture of anterior glenoid rim.
3. **Hill-Sachs defect** (50%): Impaction fracture of posterolateral surface of humeral head due to impaction of humeral head against anterior rim of glenoid during dislocation
4. **Peripheral nerve injuries** is common because of the proximity of the brachial plexus

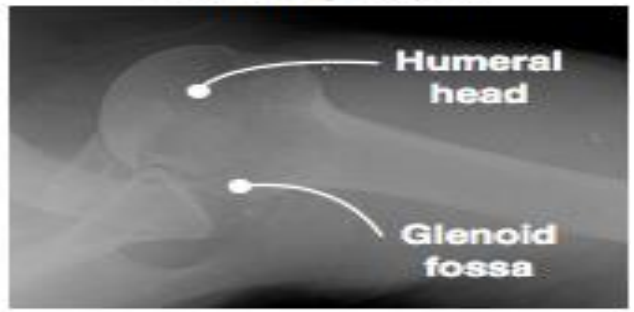
Anterior Shoulder Dislocation

Normal

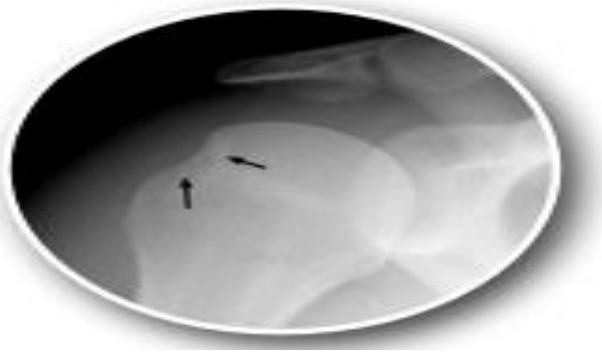


Anterior Dislocation

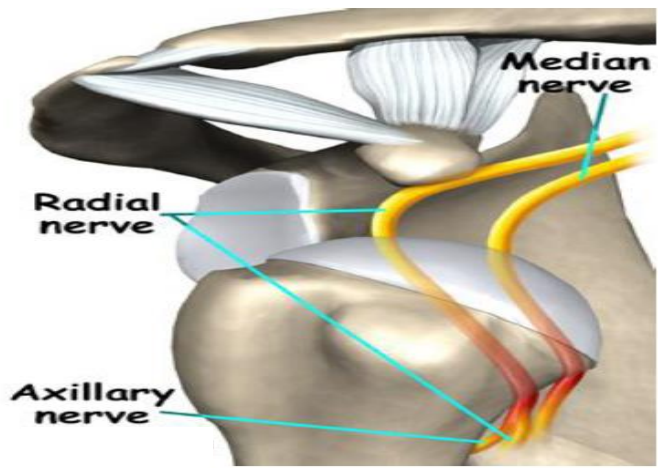
Axillary view



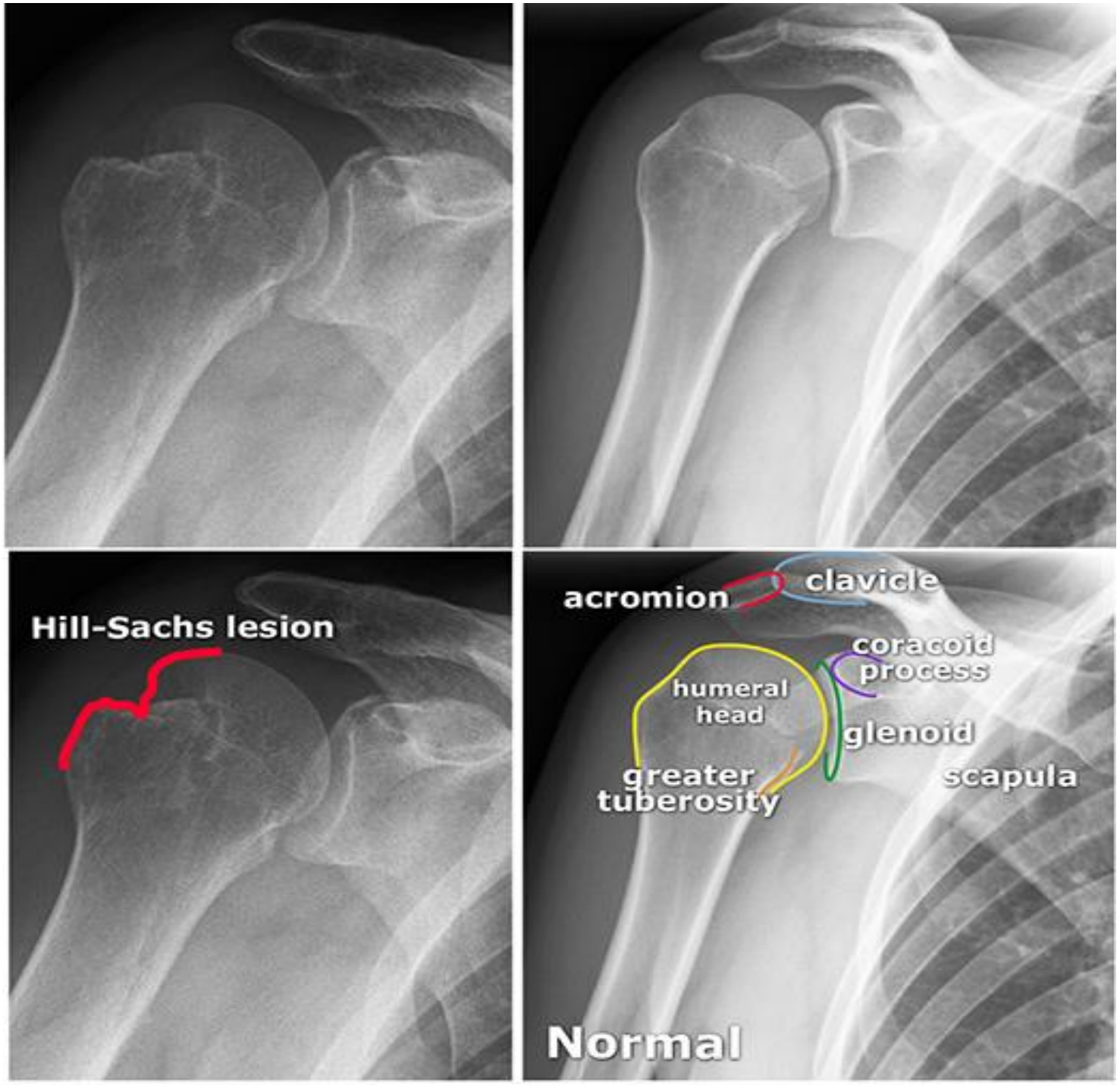
Hill-Sachs Lesion
Posterolateral humeral head compression fracture



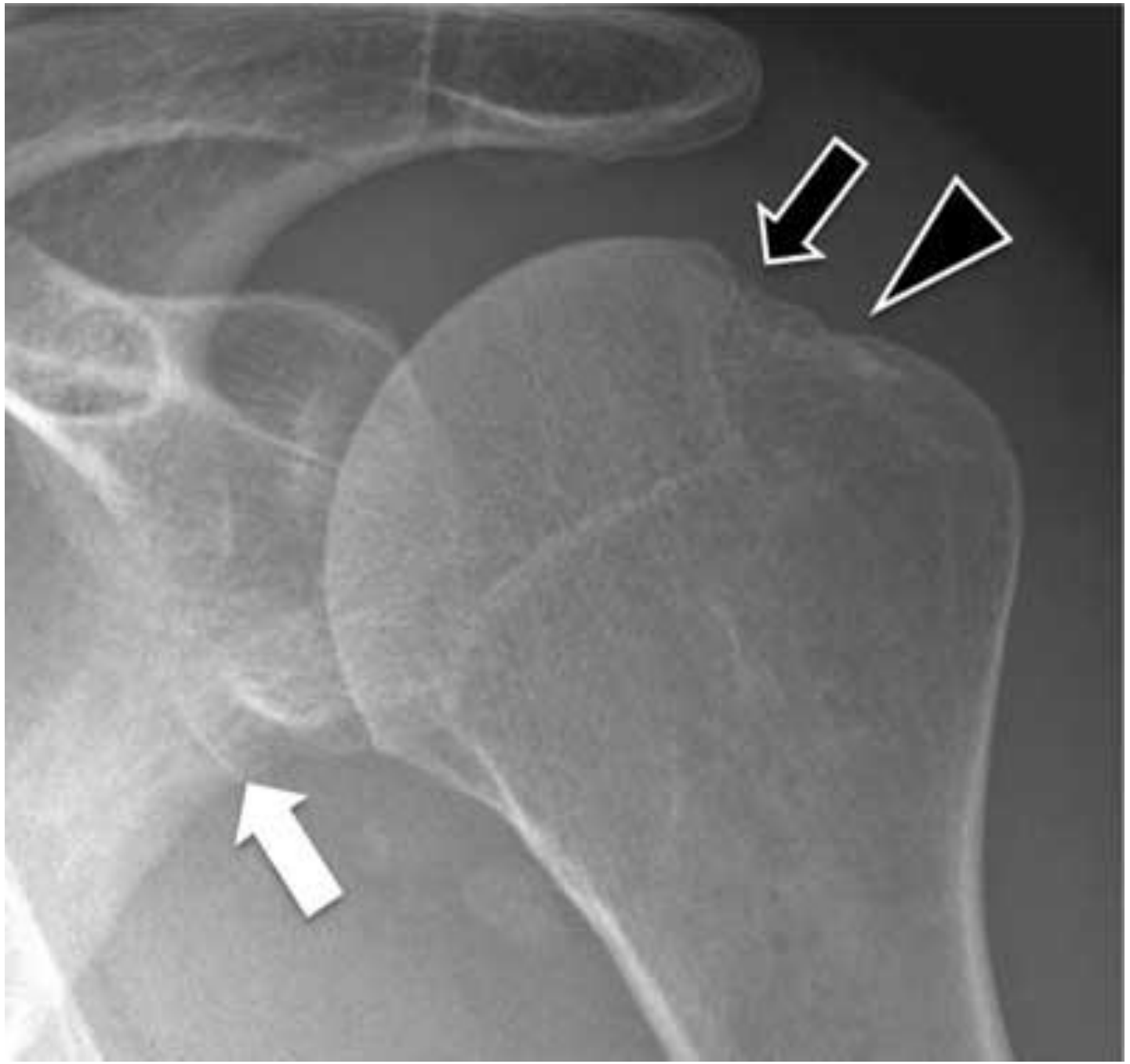
Bankart Lesion
Detachment of the anterior inferior labrum from underlying glenoid

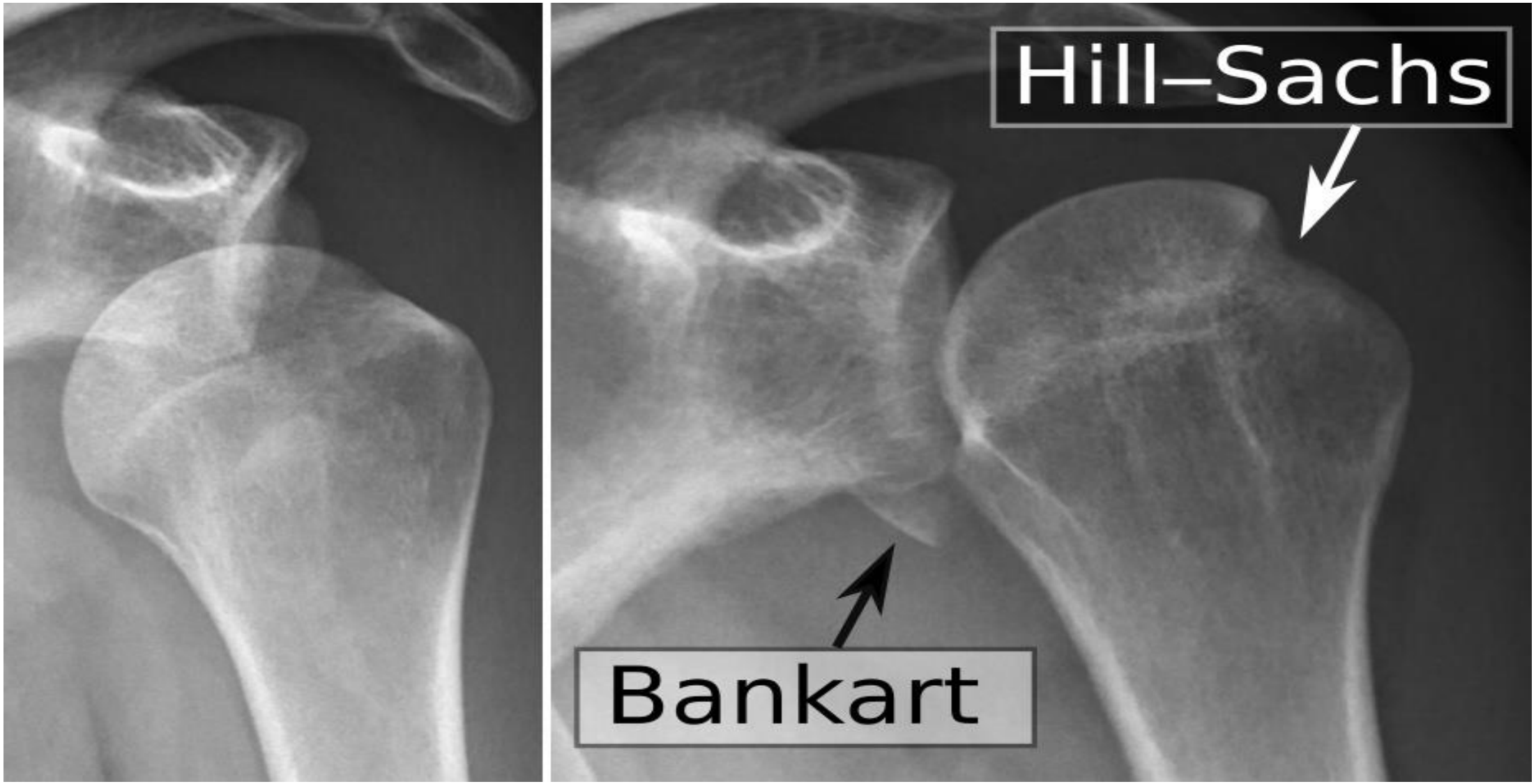


Status after relocated anterior shoulder luxation. The AP image (in endorotation) reveals a defect of the posterolateral humeral head consistent with Hill-Sachs lesion (a). A normal AP image for comparison (b).

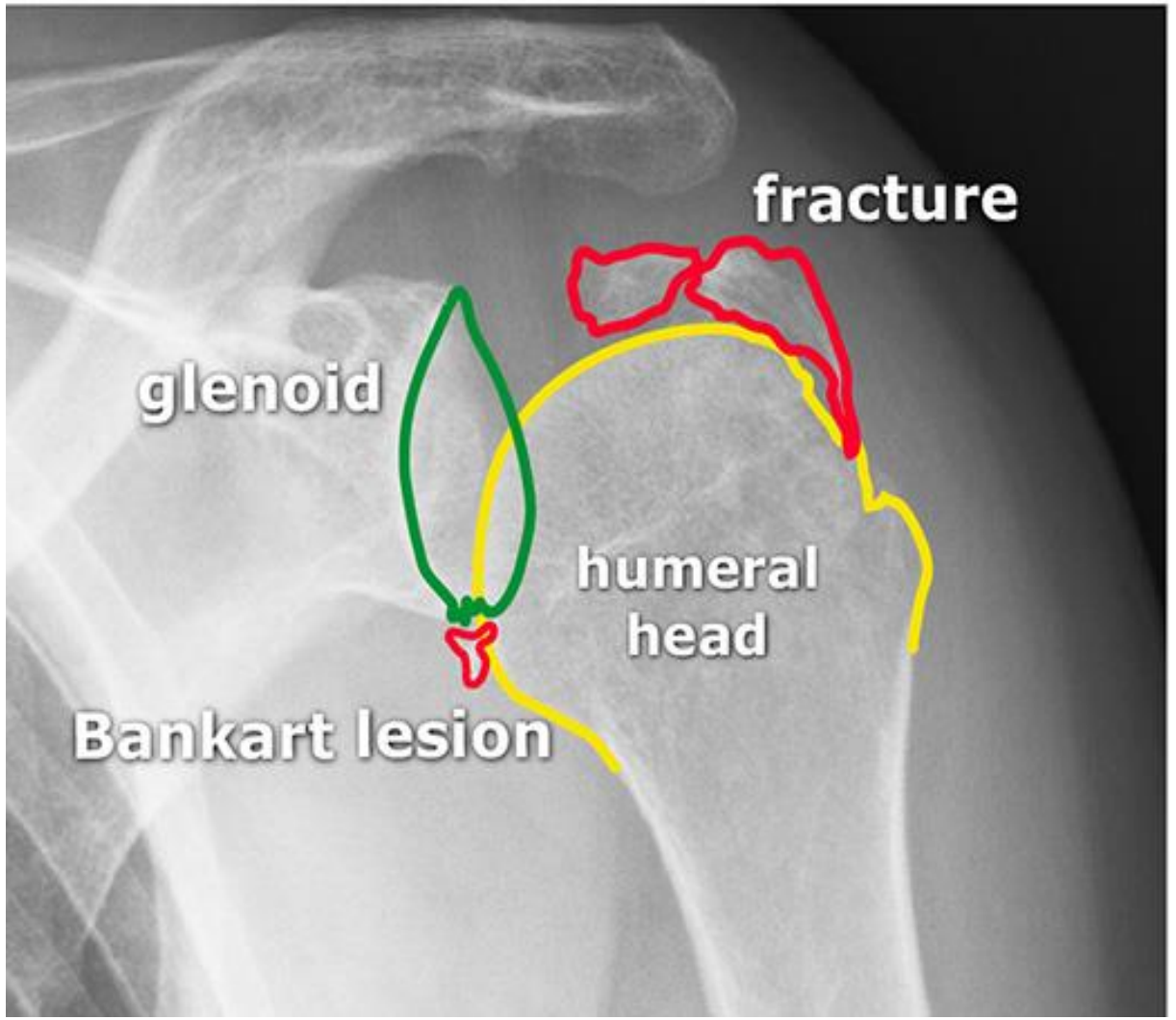


Hill Sachs lesion (black arrow) seen as a cortical depression of the postero-superior aspect of the humeral head. This depression is located medial to the head-neck junction (arrowhead). A Bankart lesion (fracture of the antero-inferior aspect of the glenoid rim) is also seen (white arrow).



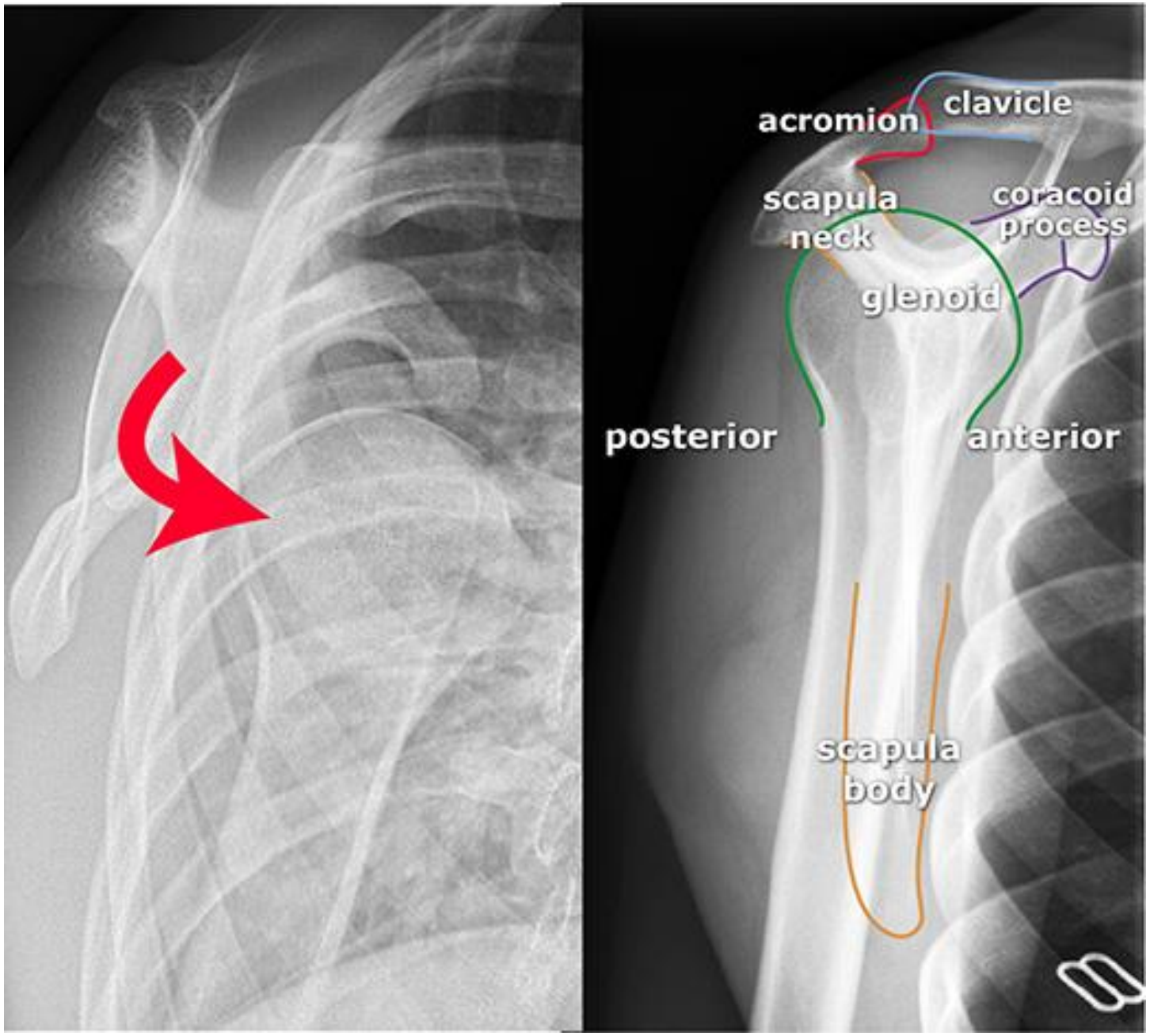


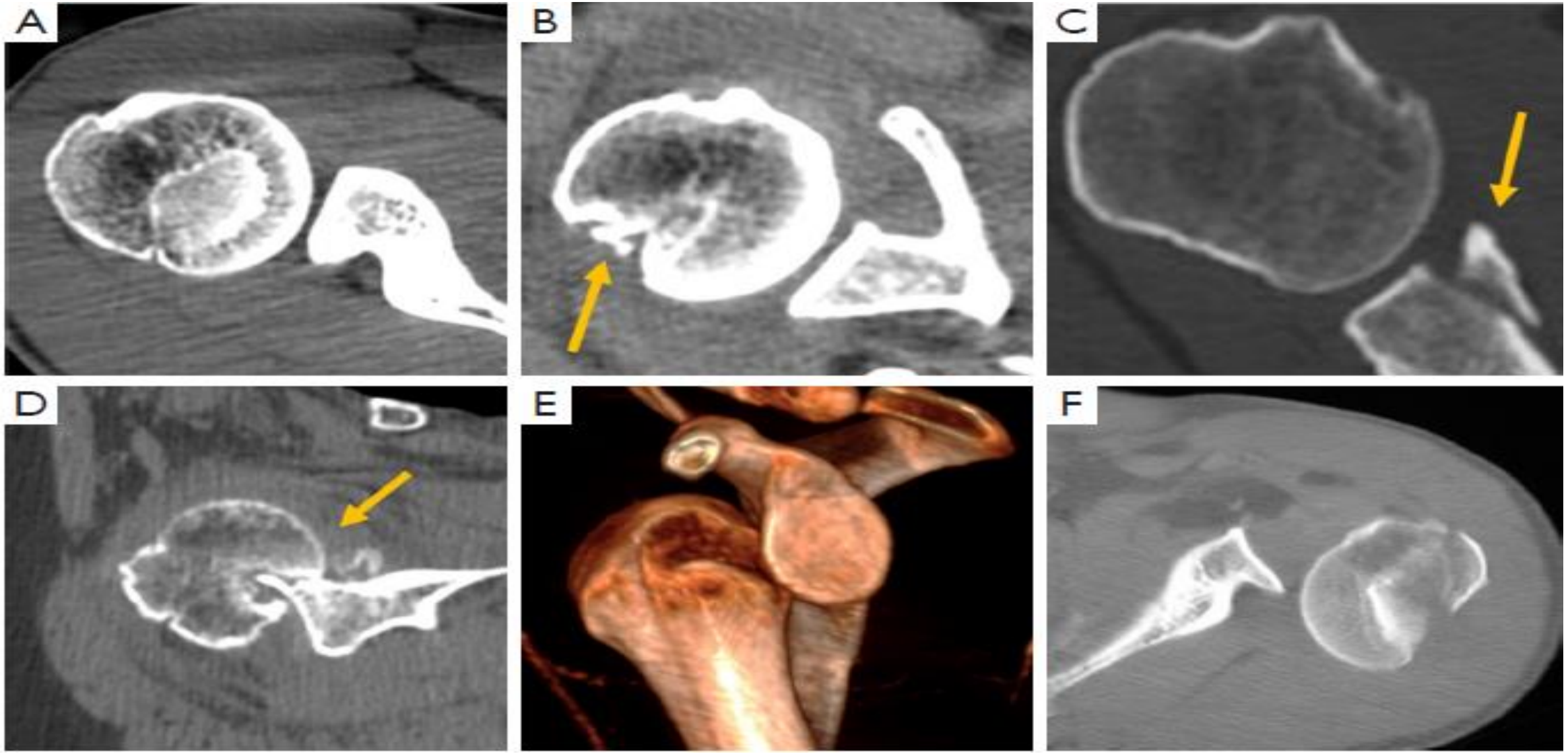
X-ray at left shows anterior dislocation in a young man after trying to get up from his bed. X-ray at right shows same shoulder after reduction and internal rotation, revealing both a Bankart lesion and a Hill-Sachs lesion.



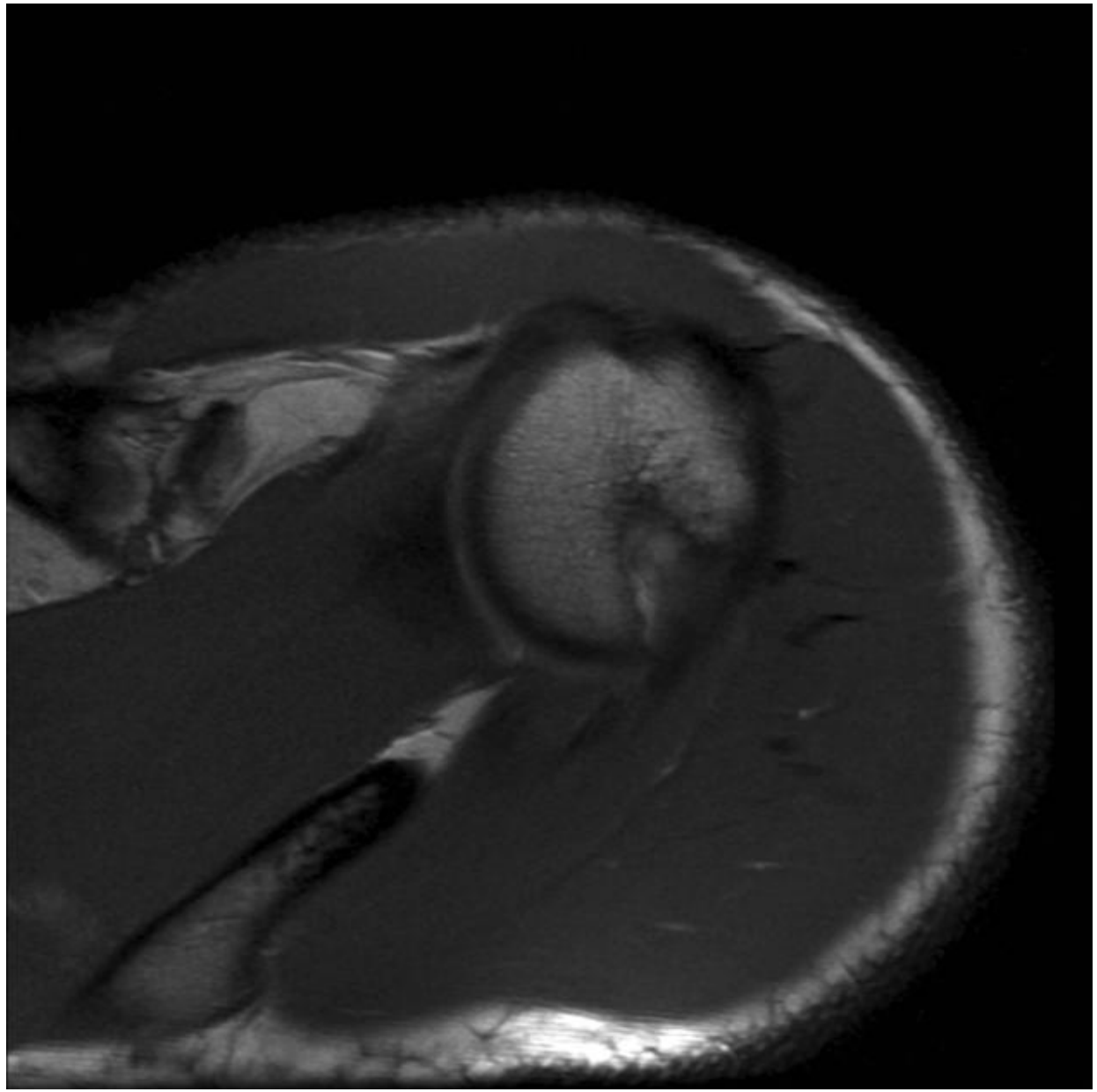
A patient with persistent instability following anterior shoulder luxation. This is a bony Bankart lesion. Also note the fractures and the irregular humeral head.

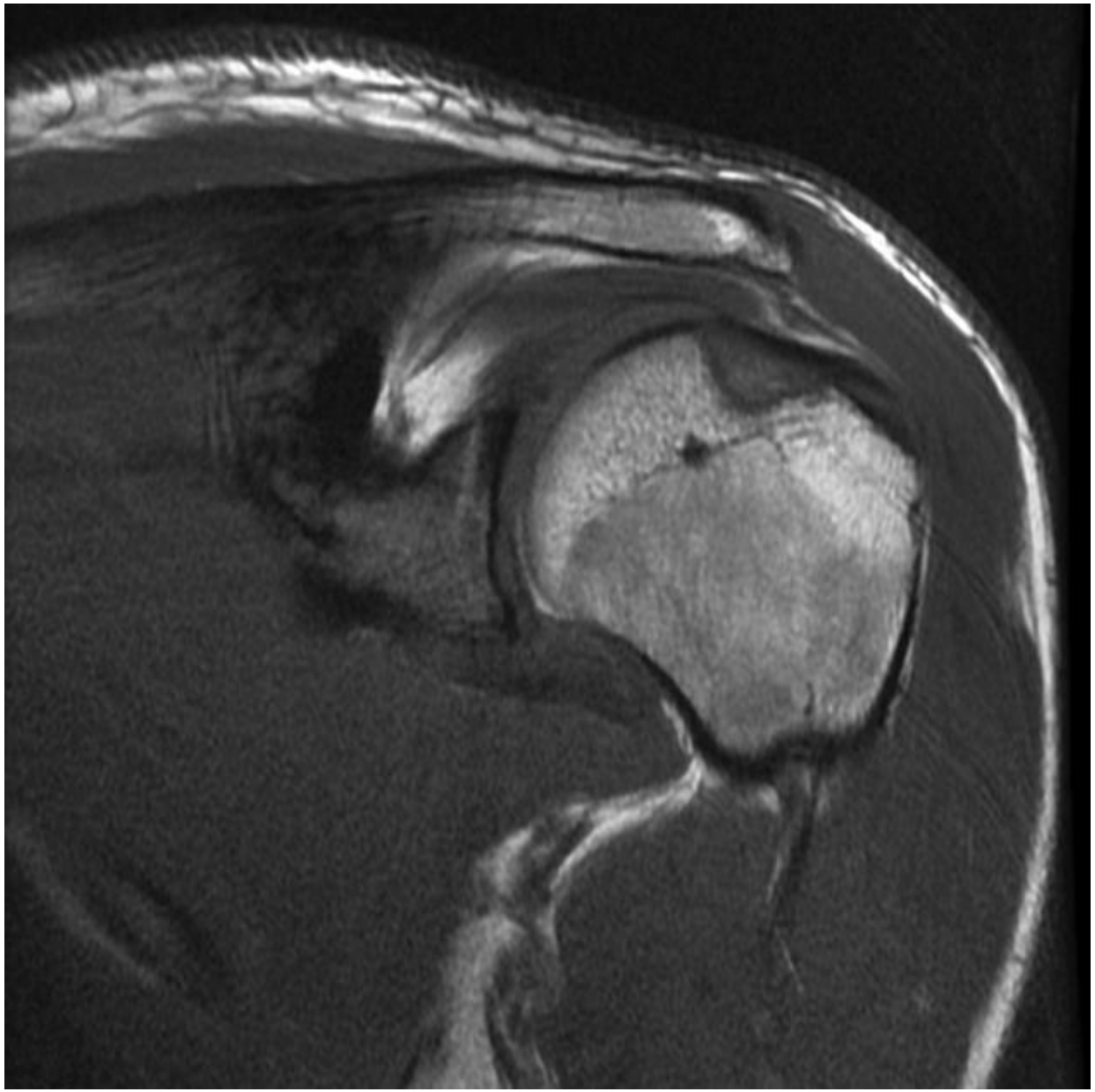
Y image in an anterior shoulder luxation. The shoulder is dislocated to anterior (a). Compare the position with a normal Y image (b).





3 CT in shoulder instability. (A) Small dysplastic glenoid; (B) Hill-Sachs lesion after reduced shoulder dislocation (arrow); (C) anterior osseous Bankart lesion is detected (arrow); (D) engaged Hill-Sachs lesion in the anterior labrum (arrow); (E) 3DCT of anterior shoulder dislocation; (F) posterior fracture dislocation of the shoulder. 3D, three-dimensional; CT, computed tomography.

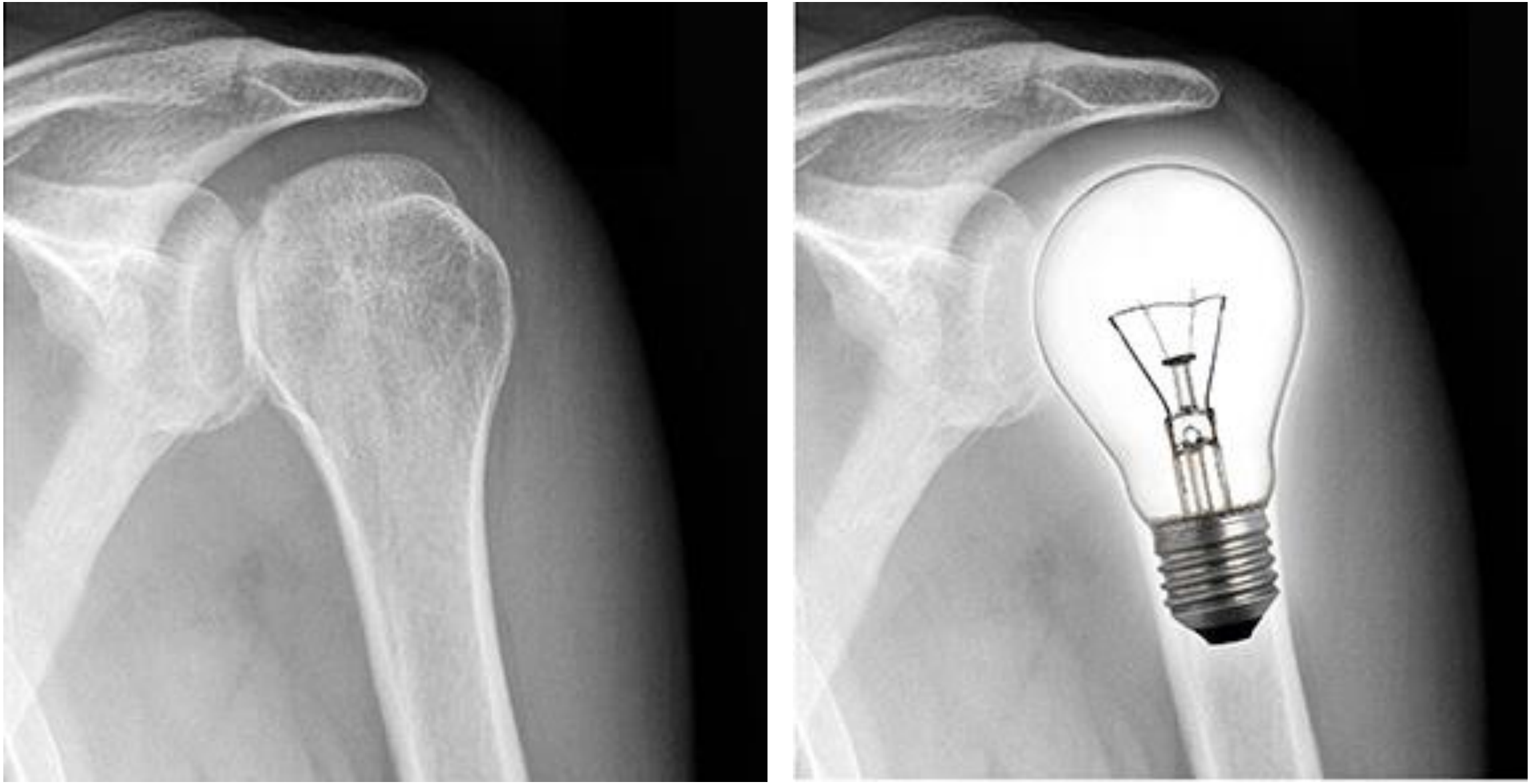




MRI images demonstrate a defect at the posterosuperior aspect of the humeral head consistent with a Hill-Sachs lesion.

Posterior shoulder dislocation (2-4%)

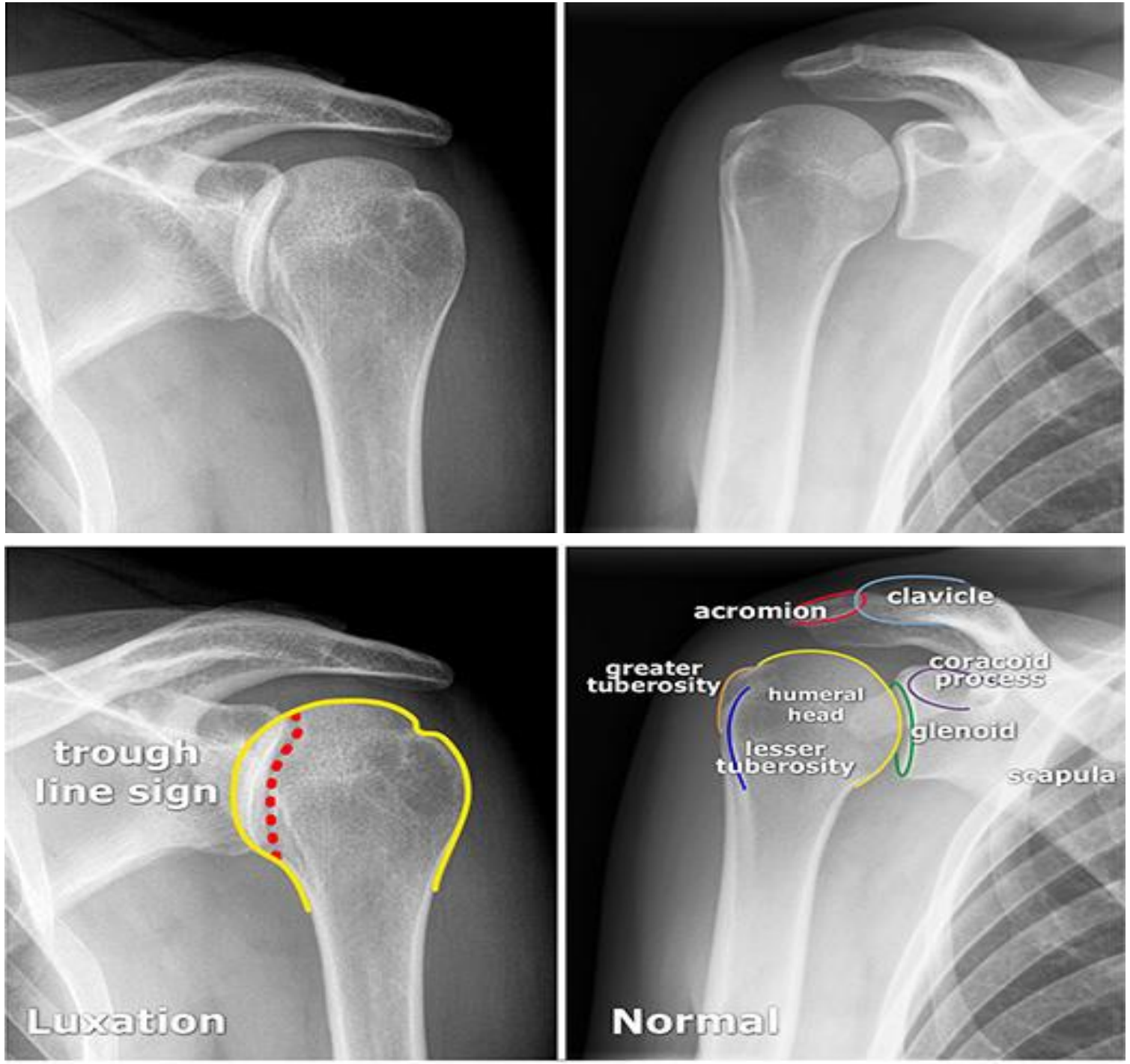
- May be difficult to see on AP radiograph. Scapular Y view or transthoracic lateral of the humeral head.
- **Rim sign** (66%) = distance between medial border of humeral head and anterior glenoid rim >6 mm
- Humeral head is fixed in internal rotation "**lightbulb sign**"
- **Trough sign** (75%) = "reverse Hill-Sachs" = compression fracture of anteromedial humeral head
- Fracture of posterior glenoid rim
- Avulsion fracture of lesser tuberosity



Lightbulb sign in a posterior shoulder luxation on an AP image

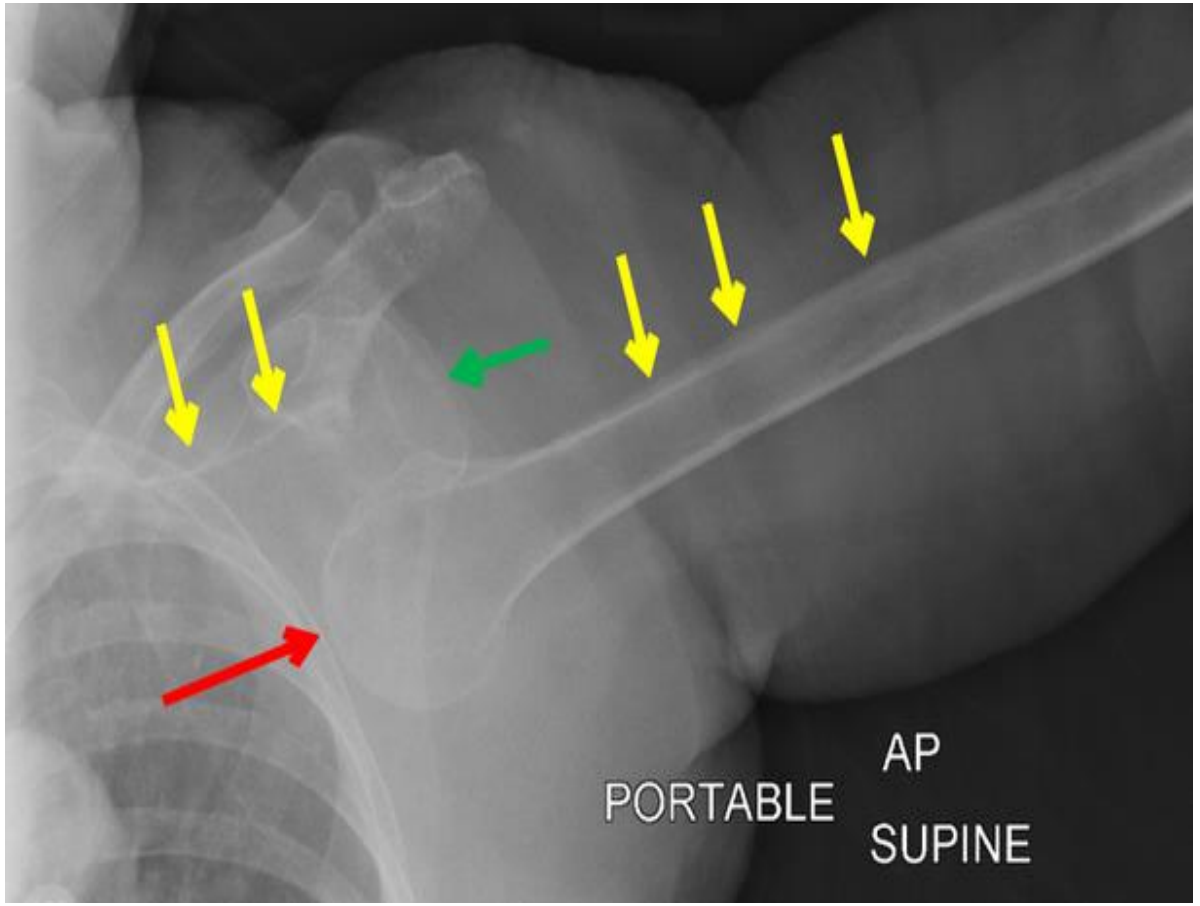
Trough line sign (red dotted line) in posterior shoulder luxation (a).

A normal AP image of the right shoulder for comparison (b).

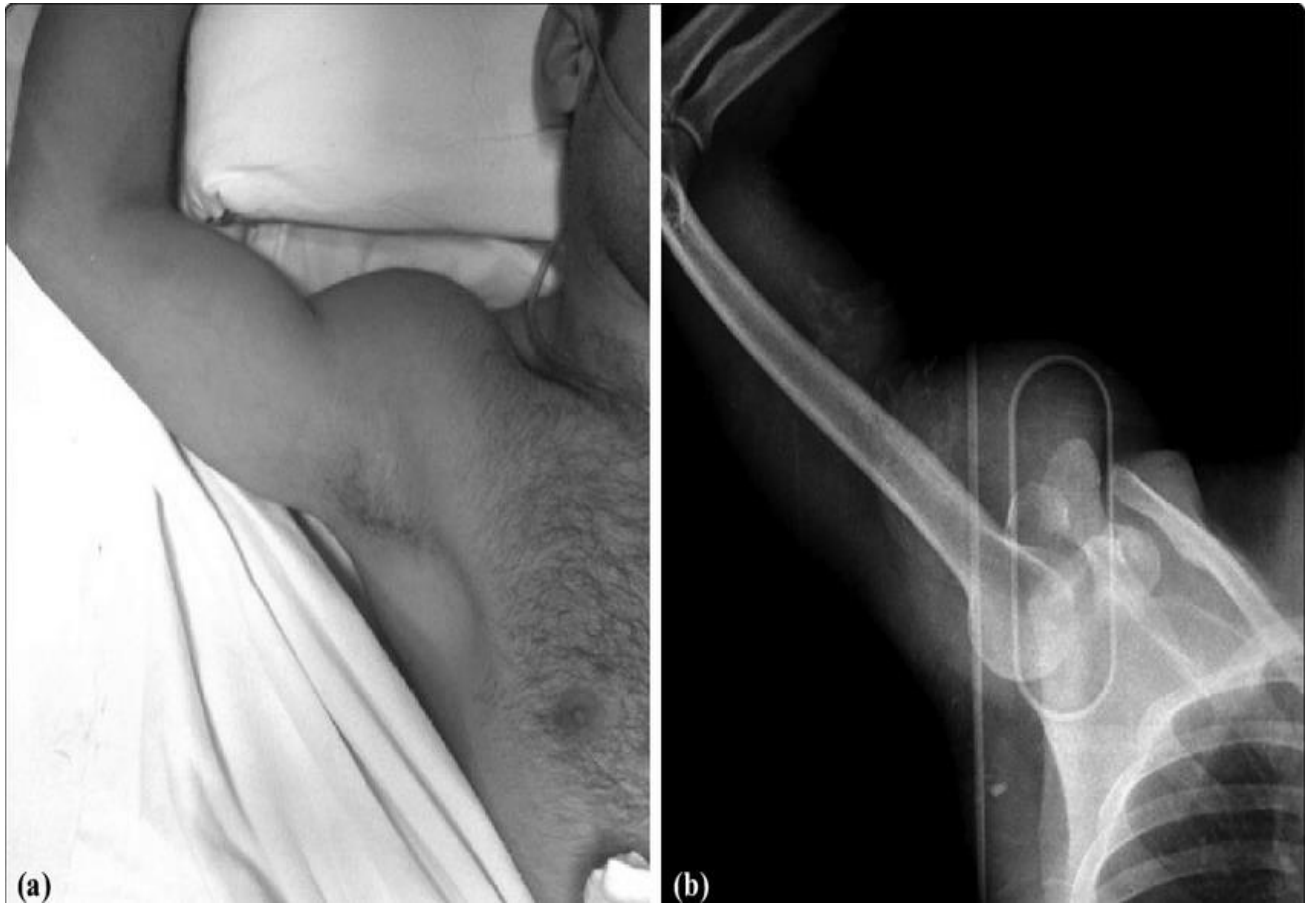


Inferior shoulder dislocation

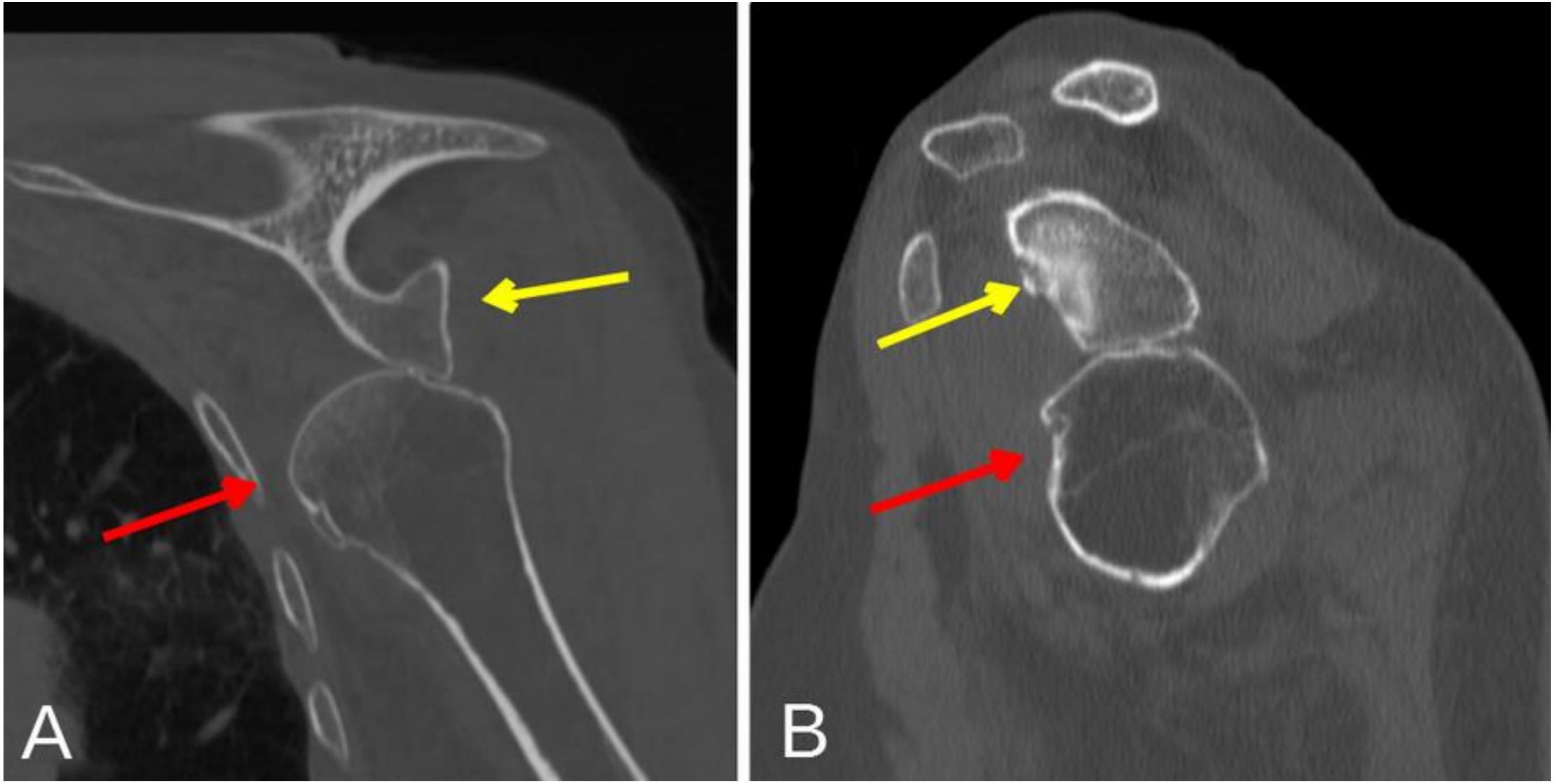
- An inferior shoulder dislocation is the least common form of shoulder dislocation.
- The condition is also called **luxatio erecta** because the arm appears to be permanently held upward, in fixed abduction.
- The patient will often present with their hand placed on the head or near it.
- **Radiographic features**
- Humeral head is displaced directly below and slightly medial to the glenoid fossa, with the arm often in marked abduction (luxatio erecta).
- MRI is performed post-reduction. Common findings include:
- Rotator cuff tears, injuries to the glenoid labrum and both the anterior and posterior band of the inferior glenohumeral ligament (IGHL)



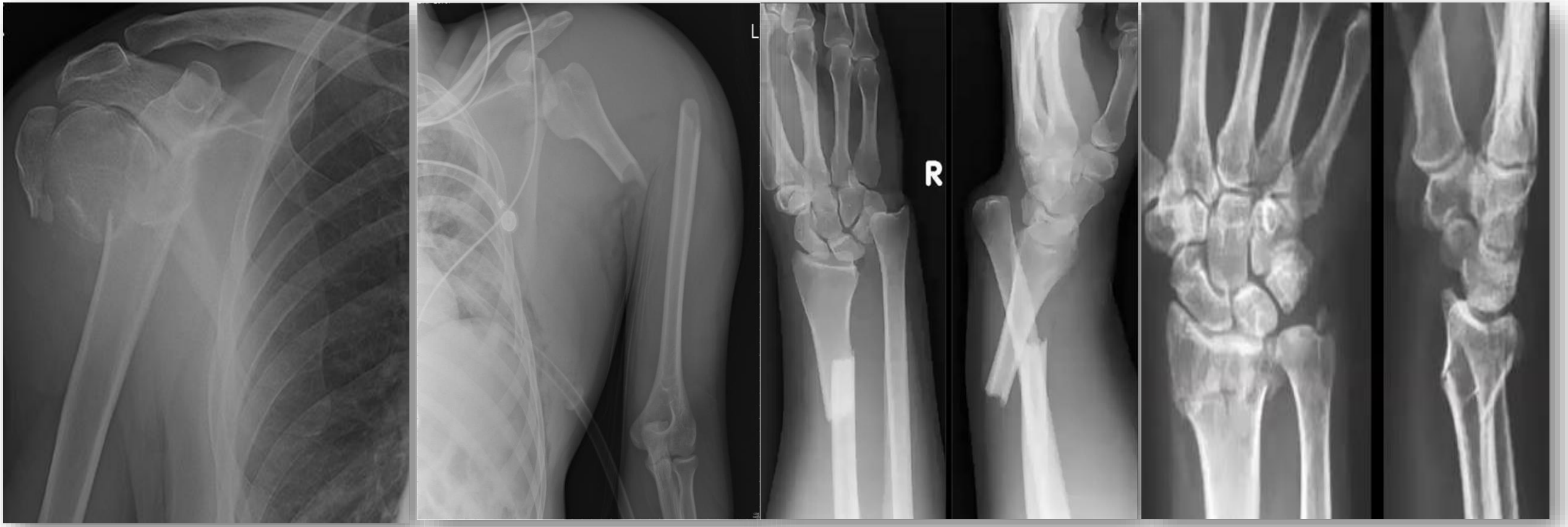
Annotated frontal left shoulder radiograph. The humeral head is dislocated inferiorly (red arrow) and the shaft of the humerus is parallel to the scapular spine (yellow arrows).



a) Patient with right luxatio erecta humeri and (b) radiograph displaying inferior shoulder dislocation with fractured greater tuberosity



CT scan of left shoulder. A. Coronal image. There is inferior dislocation of the humeral head (red arrow) relative to the glenoid (yellow arrow). B. Sagittal image. There is inferior dislocation of the humeral head (red arrow) relative to the glenoid. Also note the well corticated old Bankart fracture of the anterior glenoid (yellow arrow) indicating previous anterior glenohumeral dislocation.



Upper Limb Fractures

Lecturer: Husam Mansour

Master of Public Health

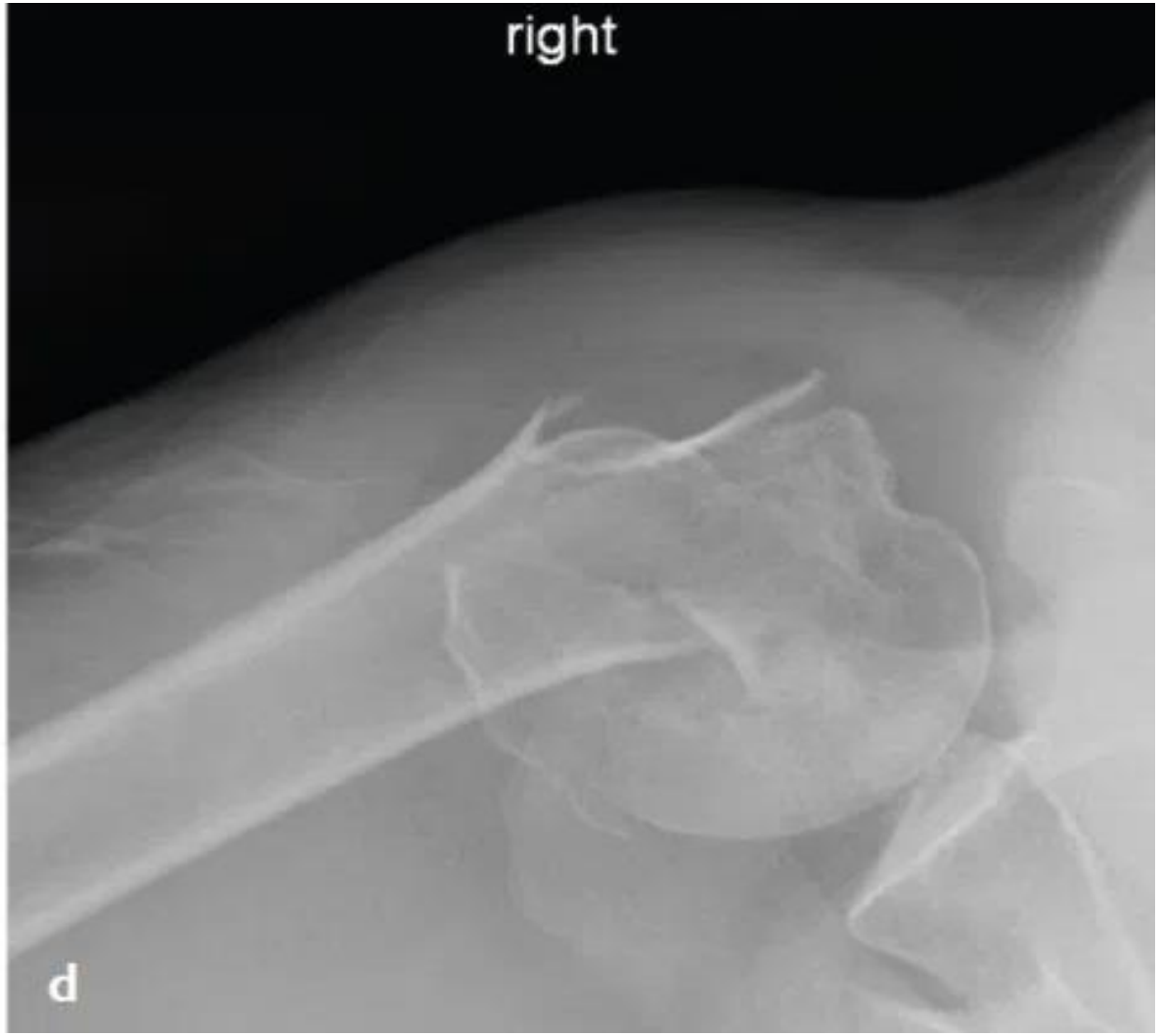
Master of Medical Imaging

Part 2

Proximal humeral fracture

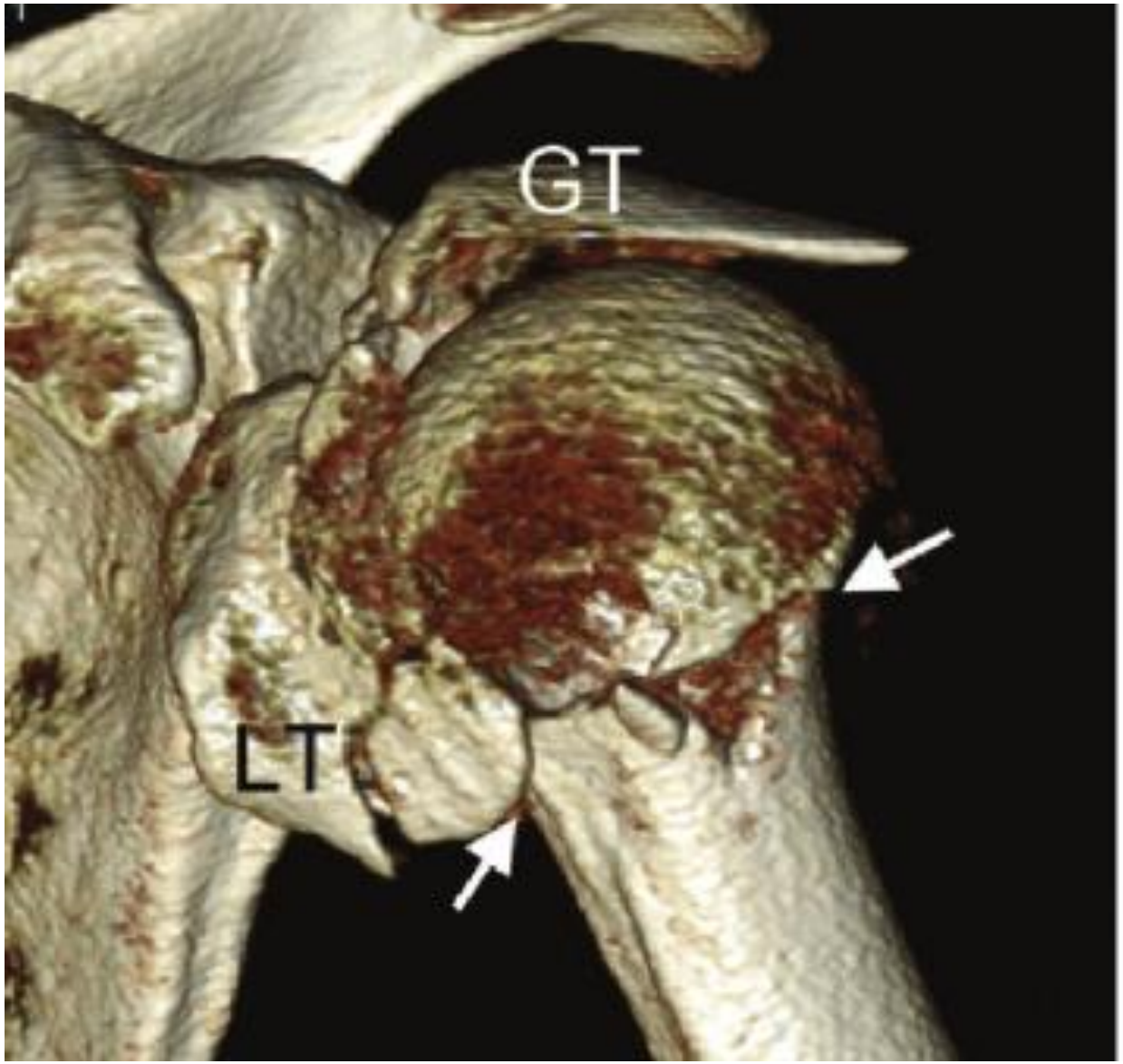
- Proximal humeral fractures are common upper extremity fractures, particularly in older patients, and can result in significant disability.
- They are most common in older populations and especially in those who are osteoporotic.
- Proximal humerus fracture includes the fracture of shaft of the humerus and greater or lesser tuberosity.
- If the fractured segment lies inside the joint then fracture is known as **intra-articular fracture** and if the fracture segment lies outside the joint or joint capsule then fracture is known as **extra-articular fracture.**

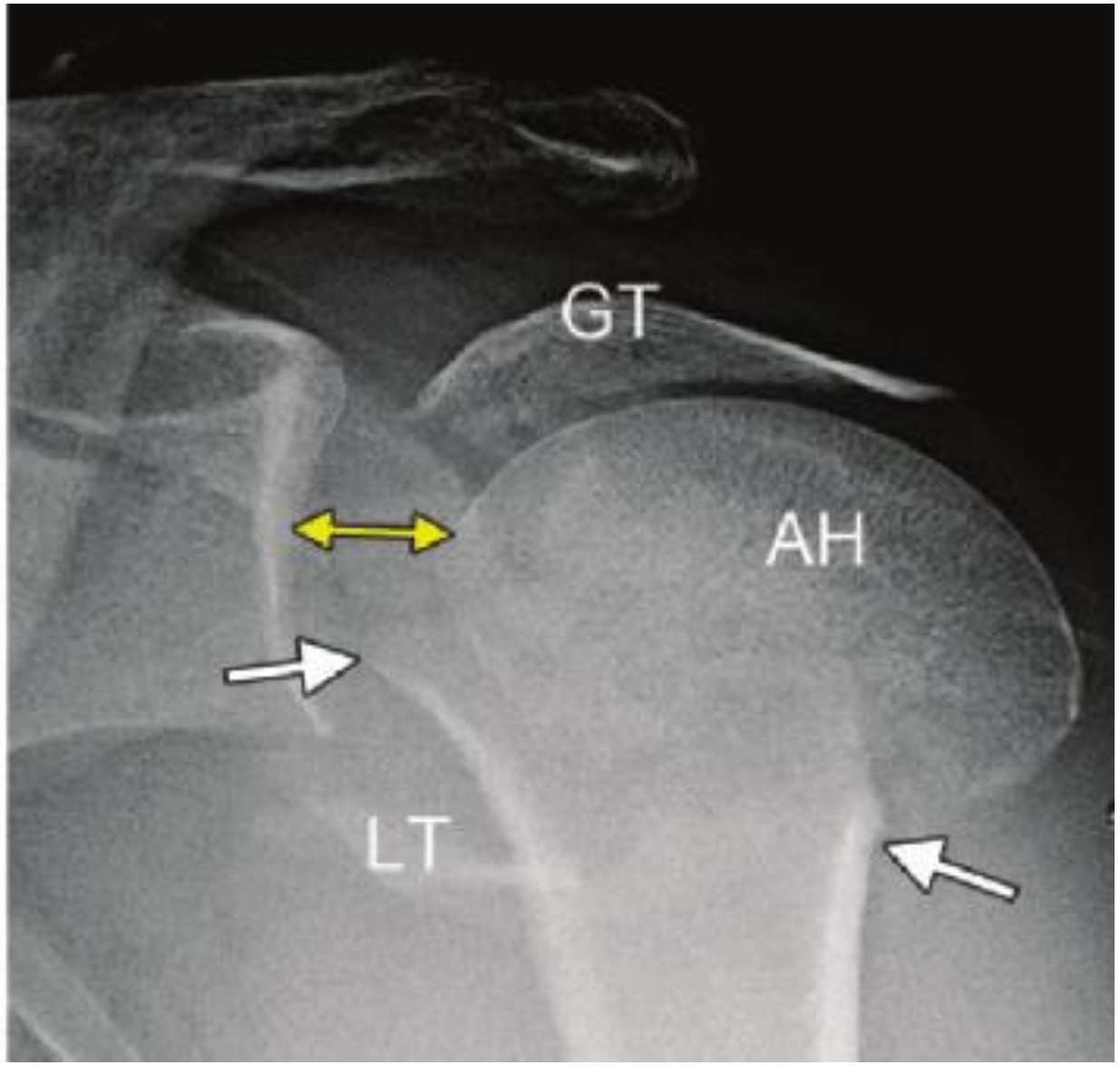
a,b One-part humeral head fracture. Transverse, slightly impacted fracture of the surgical humeral neck with separation and minimal displacement of the greater tuberosity. No fragments are separated by more than 1 cm, nor is there significant angulation of any fragment.



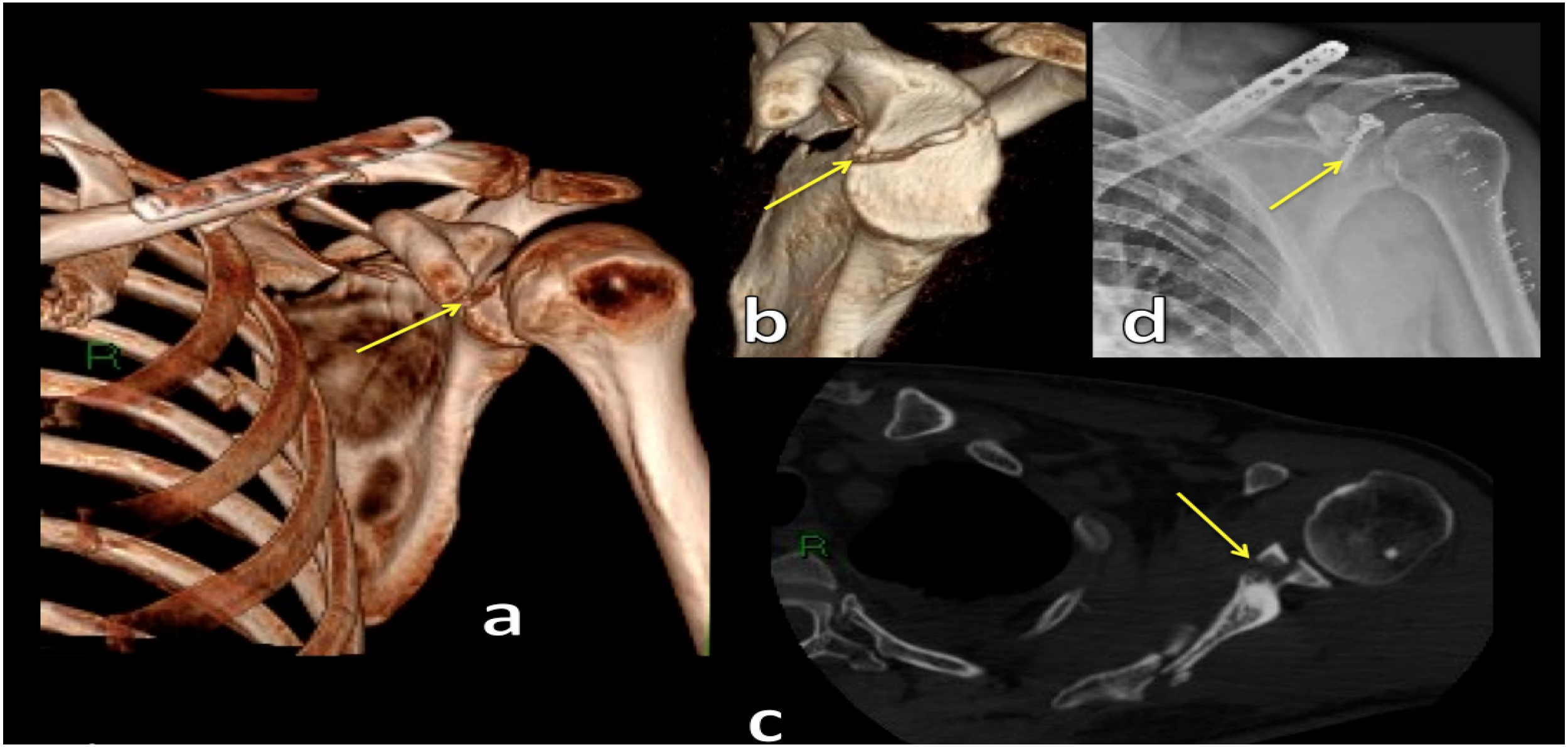
c,d Two-part humeral head fracture. Impacted fracture of the surgical humeral neck with ~ 90° angulation of the humeral head with respect to the shaft.

e Three-part humeral head fracture. Displaced fractures with separation and displacement of the diaphysis, medial humeral head, and greater tuberosity.
f Pseudosubluxation. Nondisplaced (one-part) fractures of the humeral neck and greater tuberosity. The humeral head is inferiorly subluxed

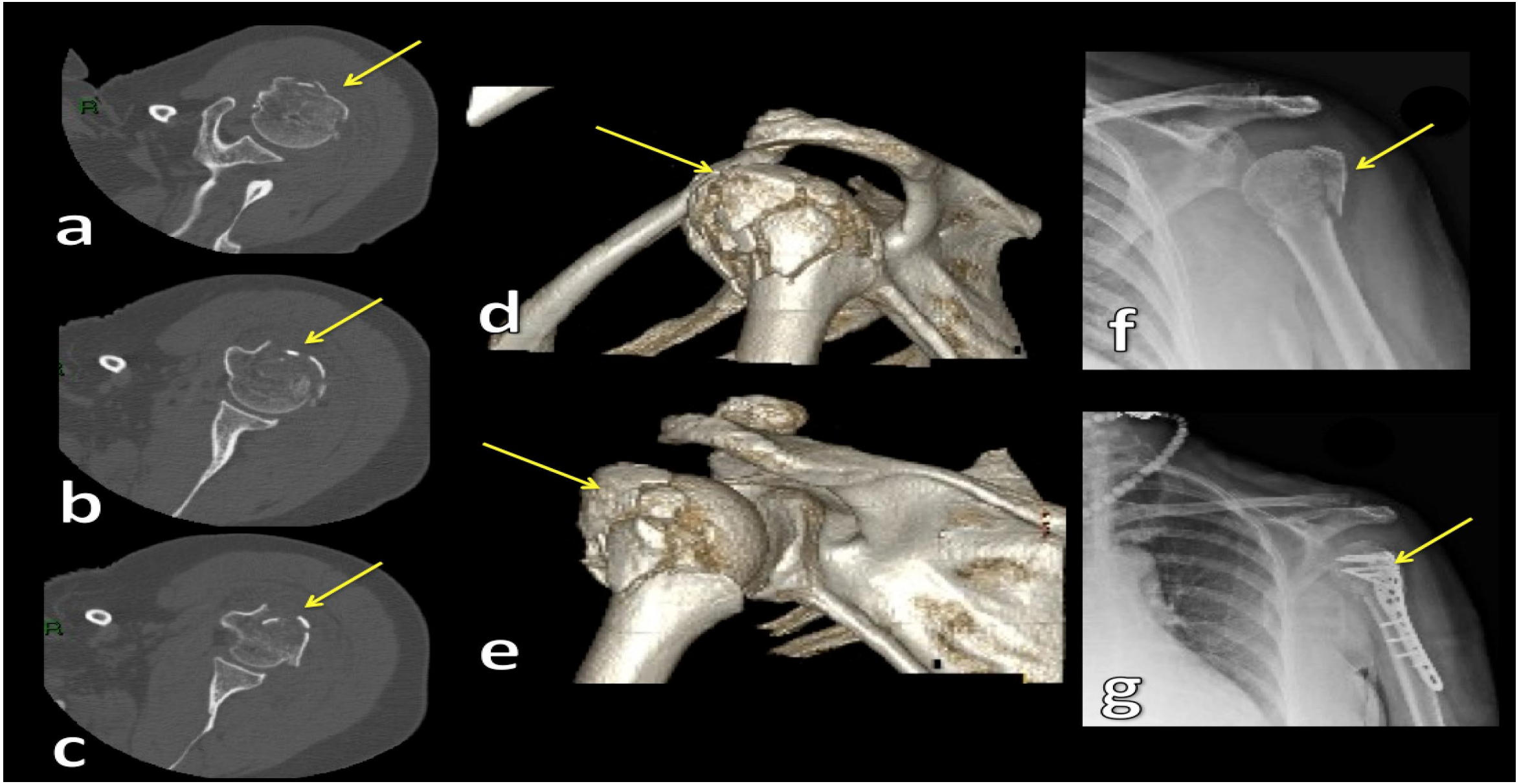




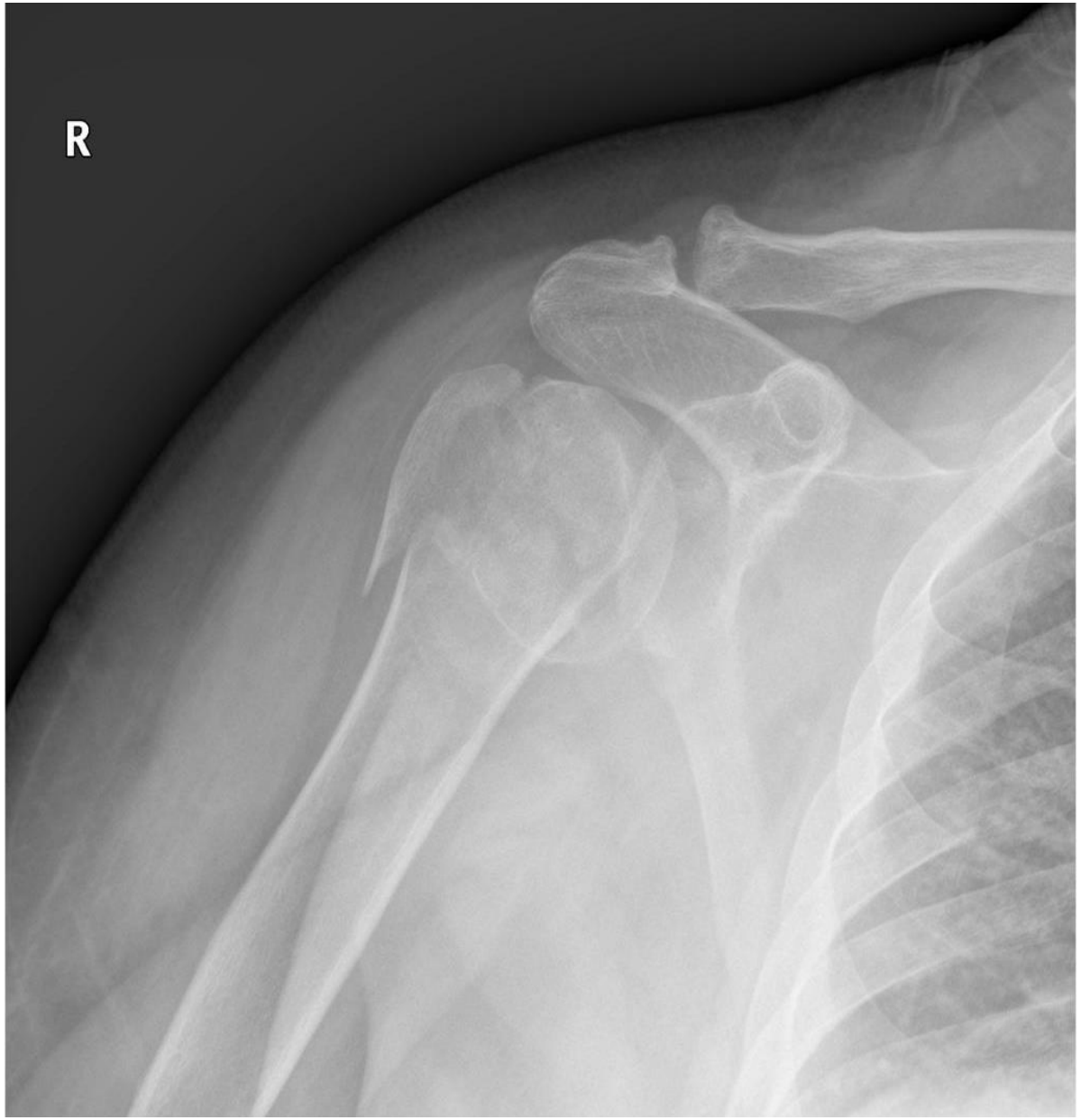
Comminuted proximal humerus fracture with an isolated anatomic head fragment. Impacted fracture of the surgical neck (single arrows) and displaced fracture fragments, one containing the lesser tuberosity (LT) and the other the greater tuberosity (GT)



3D CT images (a,b) and axial CT scan (c) show intraarticular fracture of the glenoid fossa treated with surgical fixation (d).



Axial CT images (a,b,c), 3D-CT images (c,d) and radiograph (f) show multi-fragment fracture of proximal humerus.



Surgical neck of humerus fracture and fracture separating the greater tuberosity.

Multi-fragmented fracture of the proximal humerus with involvement of the greater tuberosity

Greater tuberosity fracture

X-ray and 3D CT Displaced surgical neck of humerus fracture

(A) Right shoulder antero-posterior view shows isolated humerus greater tuberosity fracture of a 36 years old female patient. (B) Immediate postoperative radiograph shows the acceptable reduction and fixation by transosseous sutures. (C) Eight months later, radiograph shows that complete bone union was achieved.

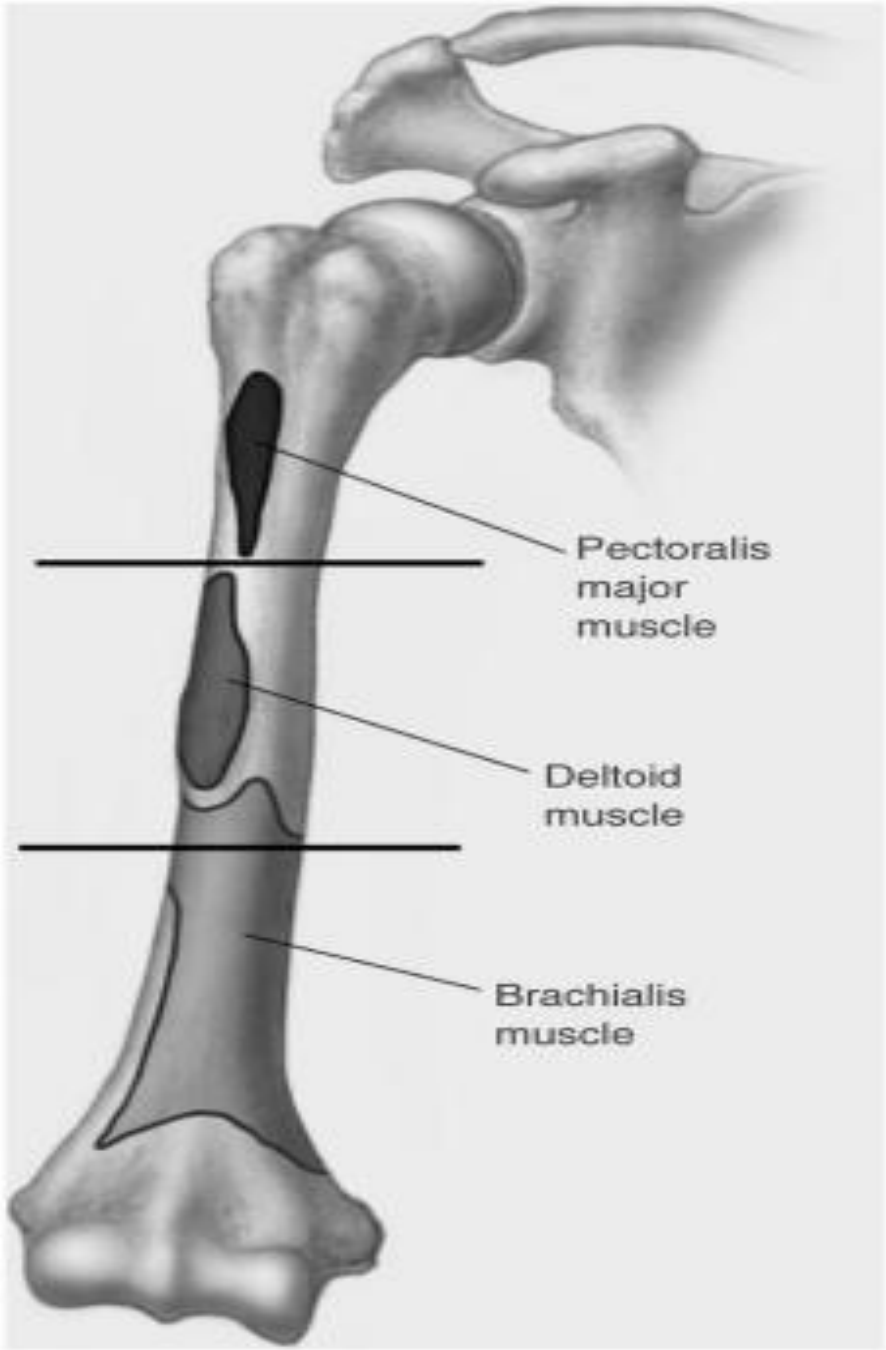
Humeral Shaft Fractures

- Humeral shaft fractures account for 3% to 5% of all skeletal fractures.
- The location of the fracture in relation to the muscle attachments affects the displacement of fragments.
- Fractures of the proximal third (between the pectoralis and deltoid muscle insertions) result in medial displacement of the proximal fragment due to pectoralis major muscle forces.
- When fractures occur distal to the deltoid muscle attachment, the deltoid muscle forces cause abduction of the proximal fragment.
- The brachialis and biceps muscles cause proximal displacement of the distal fragment

- **Radiographic features**
- Fractures are unevenly distributed along the humeral diaphysis :

1. **Proximal third: 30%**
2. **Middle third: 60%**
3. **Distal third: 10%**

- Plain radiography
- The radiographic series comprises an AP and a lateral projection to allow adequate assessment of angulation, displacement and shortening.
- It is essential both the shoulder joint and the elbow joint are included to assess for proximal or distal extension of the fracture.



- Illustration of the three segments of the humerus with muscle insertions affecting displacement. Black lines mark the junction of the proximal and middle thirds and middle and distal thirds.
- Fractures are described by their location as proximal, mid or distal or at the junctions of the proximal and middle thirds, or mid and distal thirds.
- Muscle forces displace the fragments differently depending on the location of the fracture.

Fracture at the deltoid insertion with adduction of the proximal fragment due to pectoralis muscle forces (arrow).

Fracture distal to the deltoid insertion with abduction (arrow) of the proximal fragment by the deltoid muscle and proximal displacement of the distal fragment (arrow) due to brachialis muscle forces.

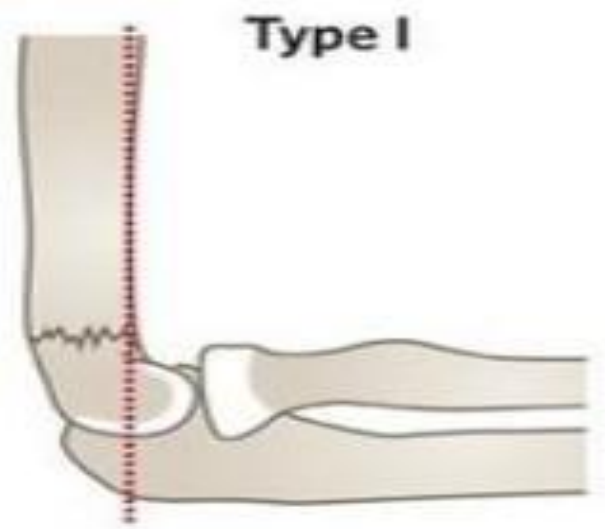
Supracondylar humeral fracture

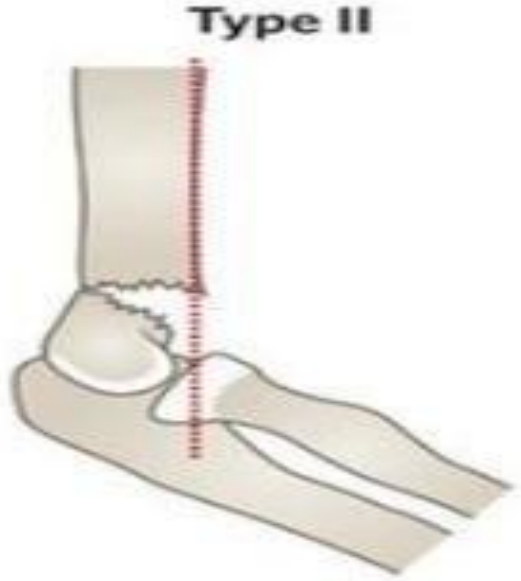
- Supracondylar humeral fractures, often simply referred to as supracondylar fractures.
- Simple supracondylar fractures are typically seen in younger children, and are uncommon in adults; 90% are seen in children younger than 10 years of age.
- Classification of supracondylar fractures
- Type I: un-displaced
- Type II: displaced with intact posterior cortex
- Type IIa: no rotational deformity
- Type IIb: rotational deformity
- Type III: complete displacement

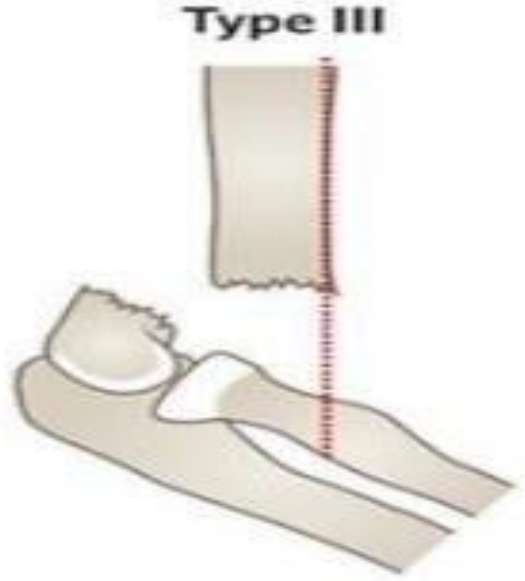
a)

b)

c)

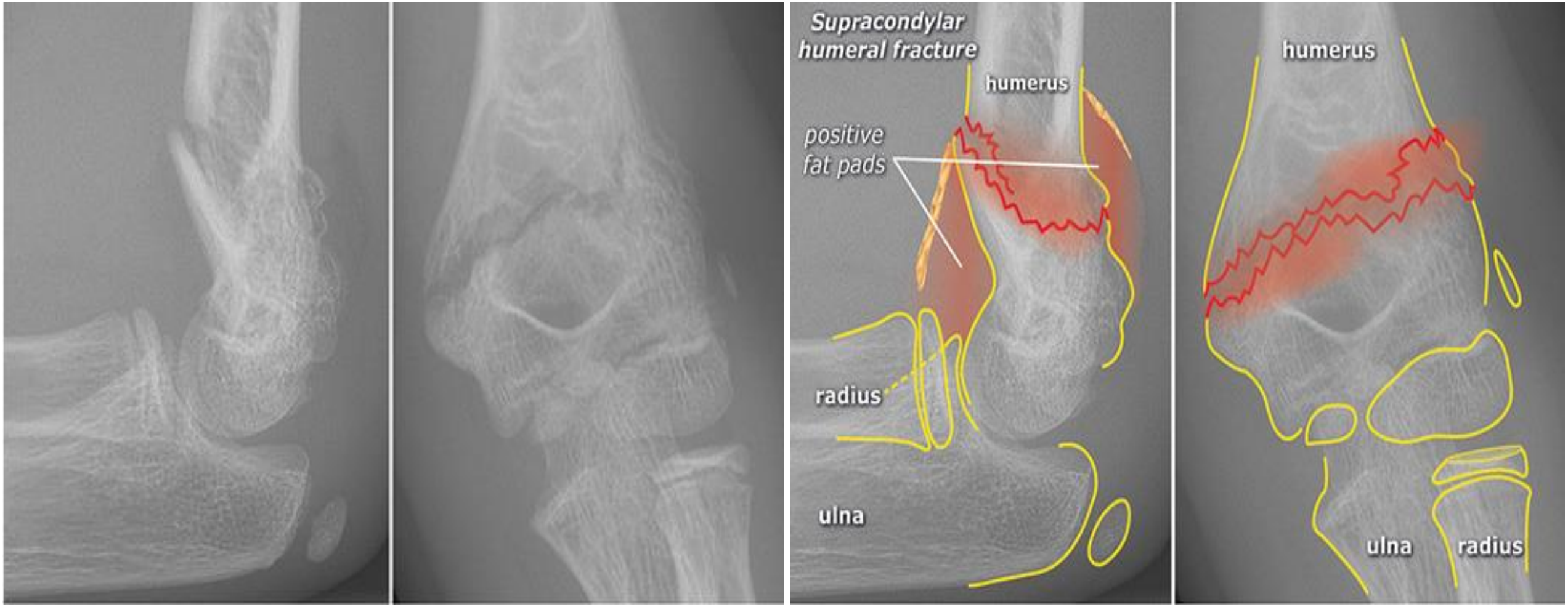

Type I


Type II


Type III

- **Plain radiograph**
- Lateral and AP radiographs demonstrate an obvious fracture.
- Often, however, no fracture line can be identified. In such cases assessing for **indirect signs** is essential:

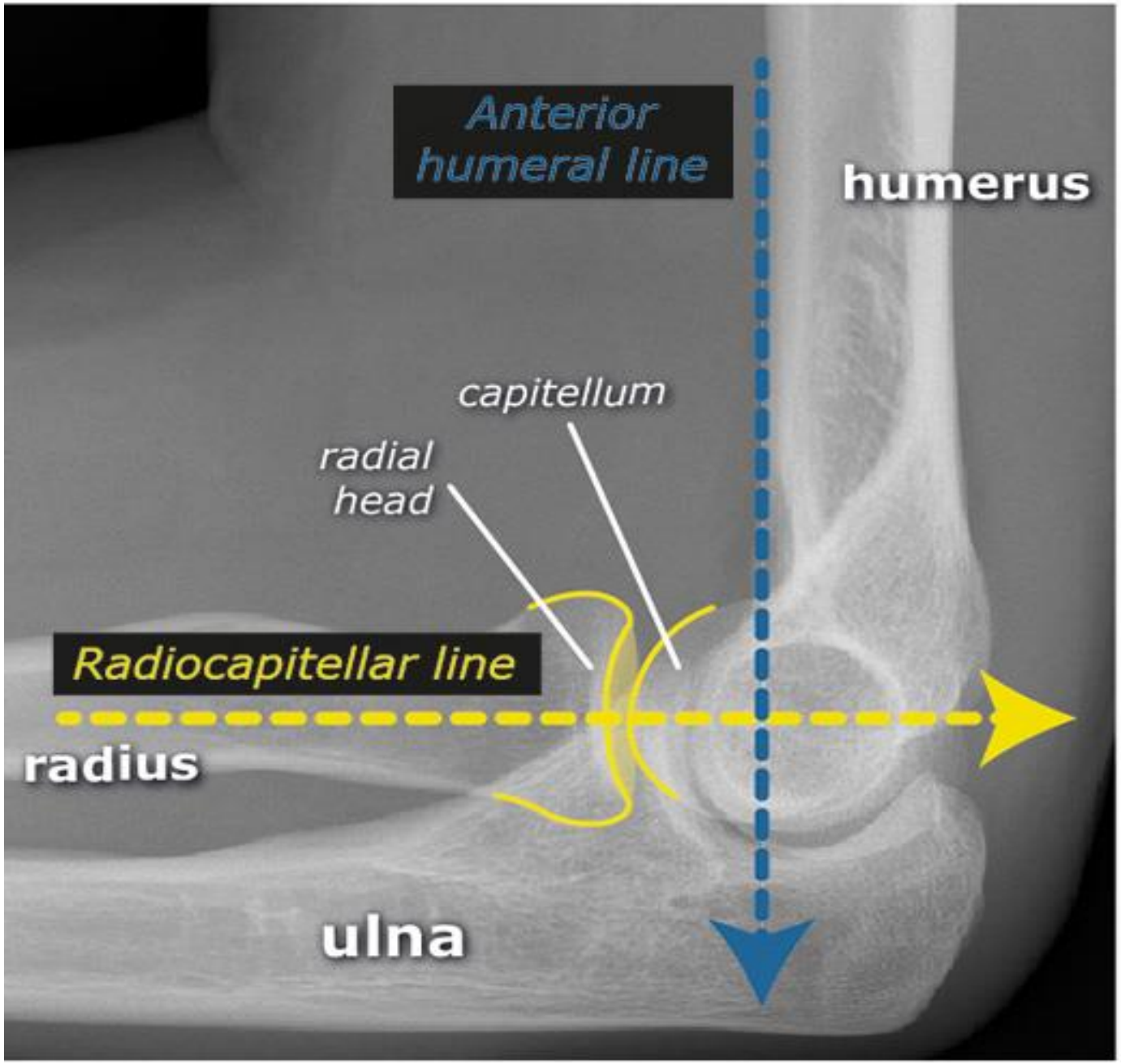
1. **Anterior fat pad sign** (**sail sign**): the anterior fat pad is elevated by a joint effusion and appears as a lucent triangle on the lateral projection.
2. **Posterior fat pad sign**: is the visualization of a lucent crescent of fat located in the olecranon fossa on a true lateral view.
3. **Anterior humeral line**: should intersect the middle third of the capitellum in most children.

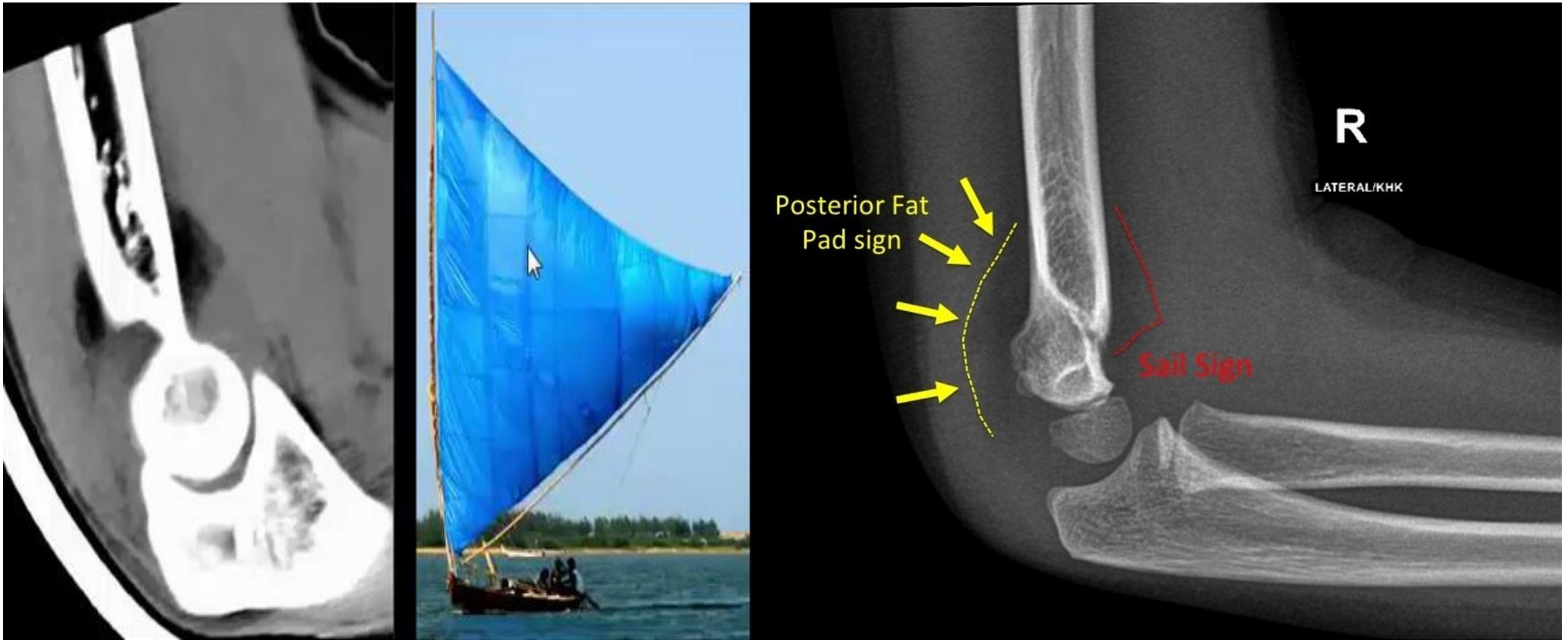


Supracondylar humeral fracture in a 9-year-old child.

Anterior Humeral Line: The anterior humeral line is drawn along the front of the humerus and passes the anterior 1/3 of the capitellum. Less than 1/3 there is likely supracondylar fracture with posterior displacement of the distal fragment.

Radiocapitellar line: a line drawn through the longitudinal axis of the radius should pass through the capitellum. If it does not then there is a dislocation of the radial head.



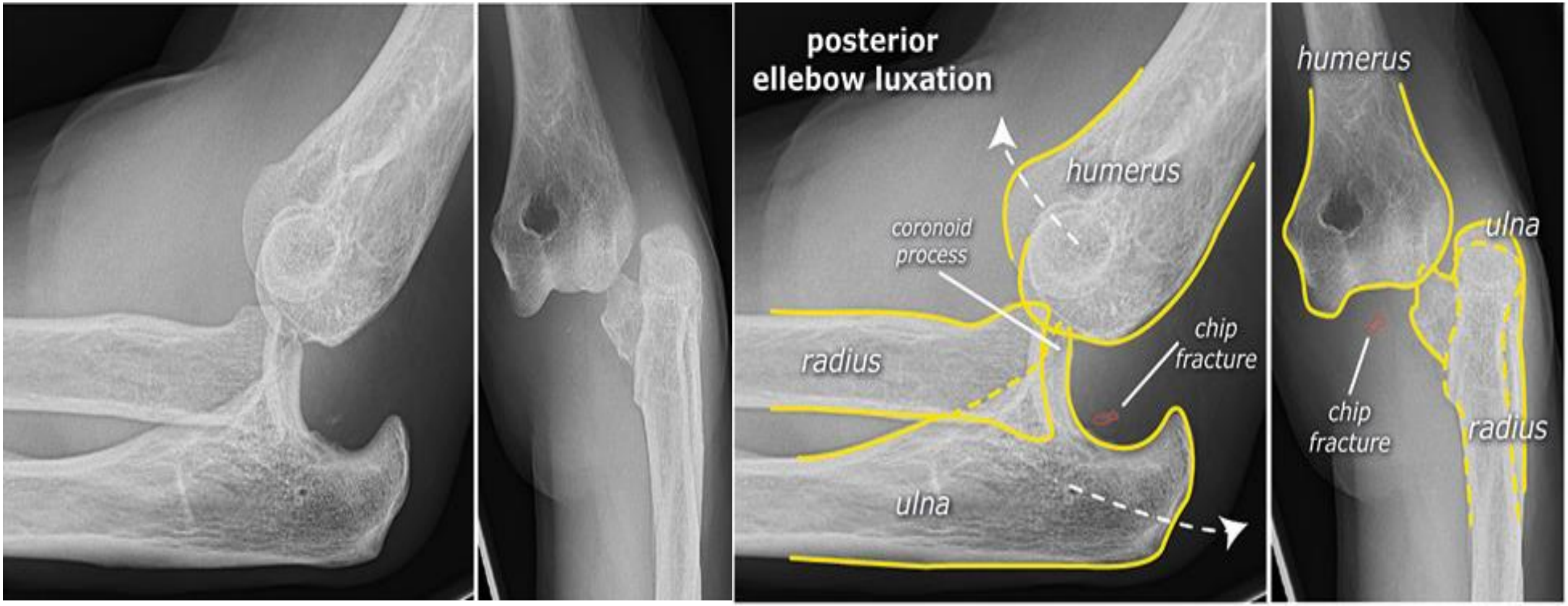


Positive posterior fat pad sign as indicated by yellow line in the olecranon fossa. This is not seen on normal elbow radiographs. The positive sail sign is highlighted in red, the latter represents superior displacement of the anterior fat pad. Both of these findings are caused by joint effusion associated with elbow joint fracture and can help identify an occult fracture (although the fracture can be seen in this case).

Elbow dislocation

- Elbow dislocation is the second most common large joint dislocation in adults and the most common in children.
- If an elbow dislocation is associated with a fracture (fracture-dislocation), it is called "complex." An isolated dislocation without fracture is "simple."
- The most common associated fracture in adults is a radial head fracture, although coronoid process fracture is also common. When all of these occur together in a severe posterior dislocation, it is known as the terrible triad of the elbow.
- The most common associated fracture in children is a medial epicondyle fracture

Two views of the left elbow demonstrate significant deformity with posterolateral dislocation at the elbow joint. No definite evidence of a displaced fracture. Small elbow joint effusion. Significant soft tissue swelling around the elbow with gross deformity.

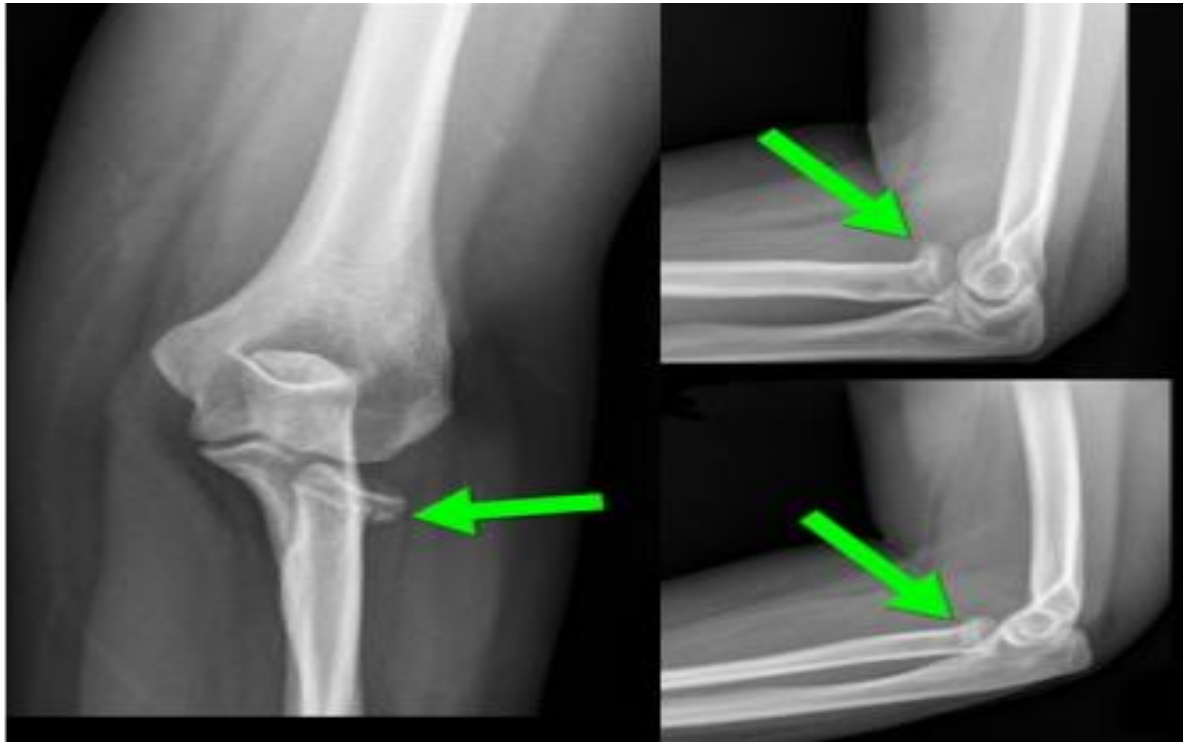


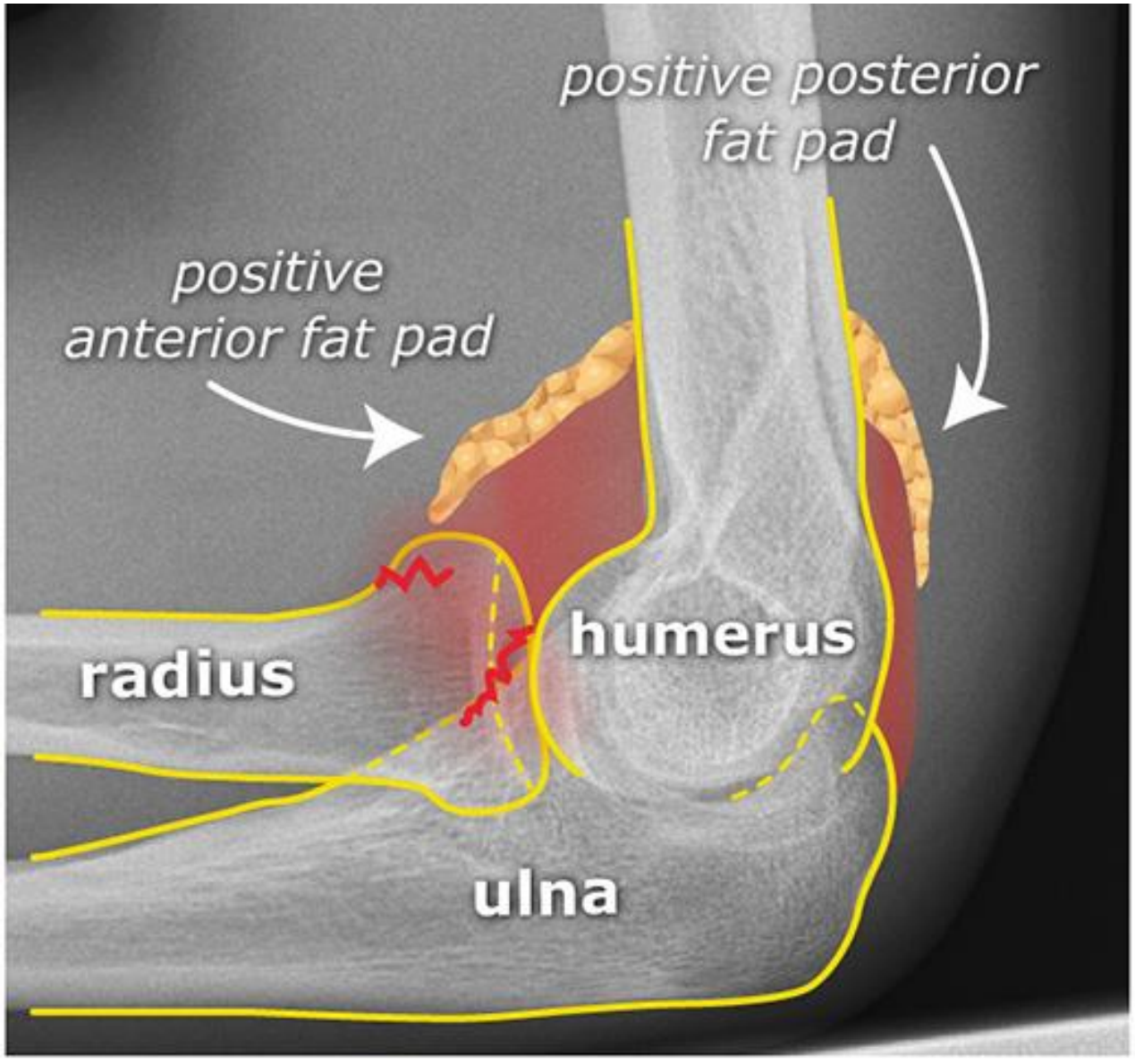
Posterior elbow luxation, associated with a small chip fracture (possibly originating from the coronoid process).

Radial head fractures

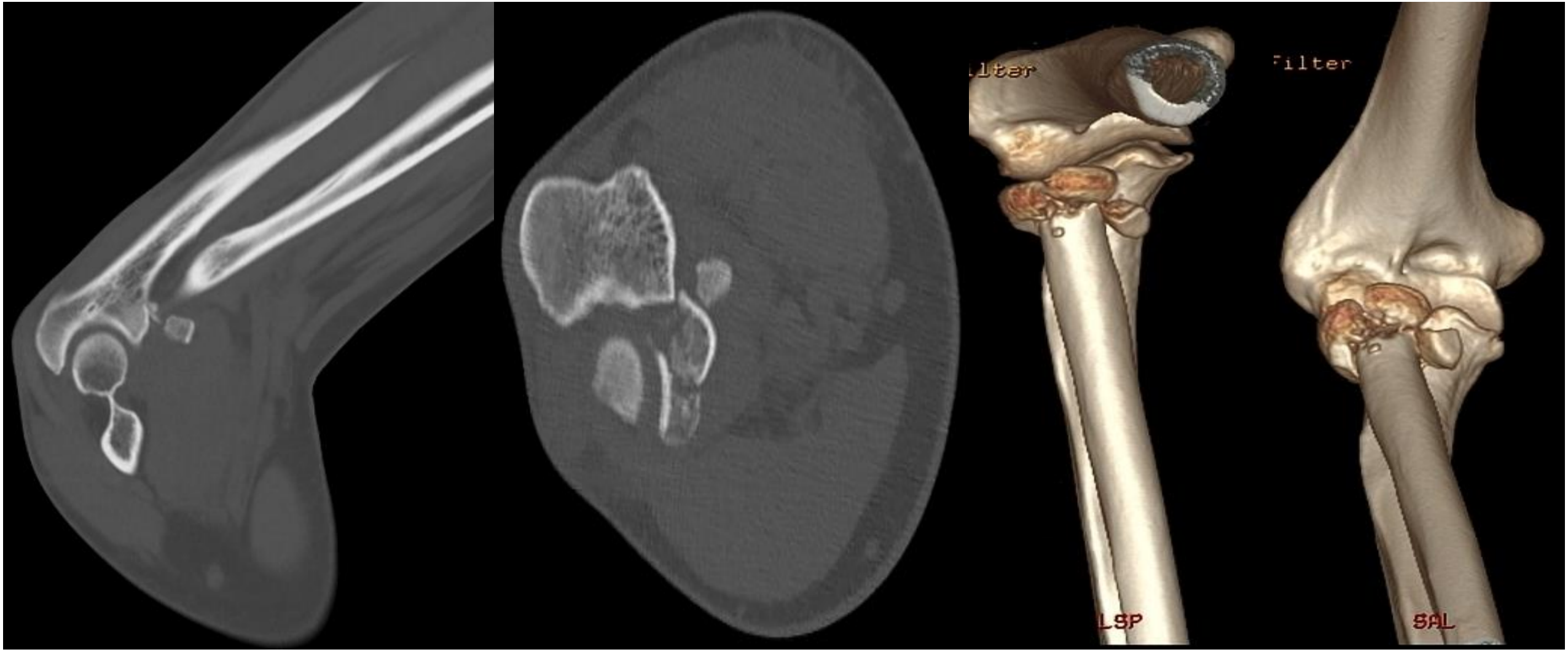
- Relatively common injuries, especially in adults
- **Mason classification of Radial Head fractures**
- Type I: non-displaced radial head fractures (or small marginal fractures), also known as a "chisel" fracture.
- Type II: partial articular fractures with displacement (>2mm).
- Type III: comminuted fractures involving the entire radial head.
- Type IV: fracture of the radial head with dislocation of the elbow joint.

Type I
Type II
Type III
Type IV





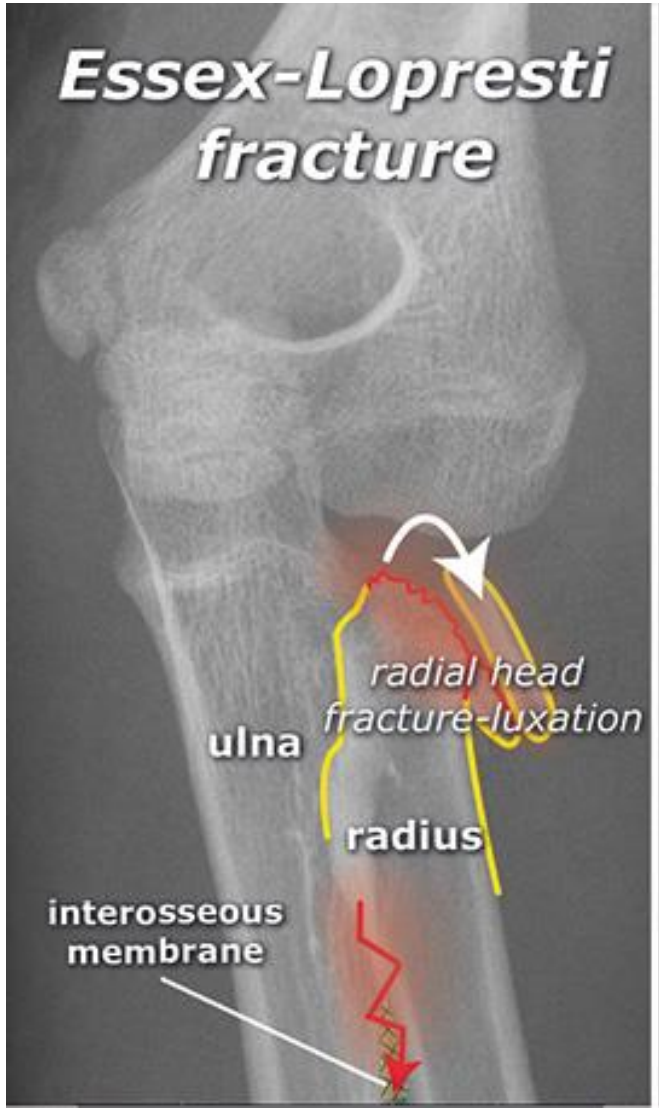
Radial head fracture with positive anterior & posterior fat pad.

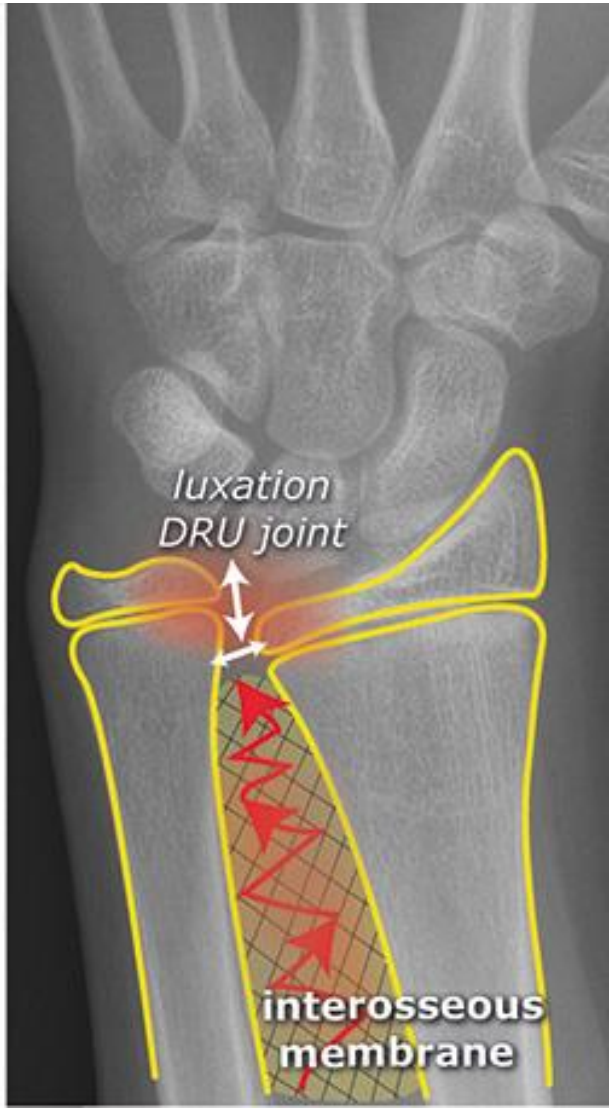


Comminuted intraarticular fracture of the radial head is noted with impaction and tilting consistent with type III radial head fracture with disruption of the radiohumeral articulation. No frank elbow joint dislocation. Mild joint effusion.

Essex-Lopresti fracture

- A comminuted fracture of the radial head by a fall on an outstretched arm may be associated with tearing of the interosseous membrane of the lower arm.
- The force released during the trauma will be transferred distally through the interosseous membrane to the distal radioulnar joint (DRU joint), causing luxation of the DRU joint.
- Learning point: in a comminuted fracture of the radial head, the wrist joint should always be imaged to exclude an Essex-Lopresti fracture.





Essex-Lopresti fracture. The AP image of the elbow reveals a radial head fracture-luxation; the AP image of the wrist shows luxation of the distal radioulnar joint (DRU joint).

Olecranon fracture

- Olecranon fracture is a fracture of the bony portion of the elbow.
- The injury is fairly common and often occurs following a fall or direct trauma to the elbow.
- The olecranon is the proximal extremity of the ulna which is articulated with the humerus bone and constitutes a part of the elbow articulation.
- Imaging: Three radiographic views (anteroposterior, oblique, lateral) are necessary to assist with fracture identification.
- CT can be useful in evaluating high-energy fracture patterns associated with instability.

- **Mayo classification**
- Based on the stability, the displacement and the comminution of the fracture.
- **Type I:** Non-displaced fracture – It can be either non-comminuted ones (Type IA) or comminuted (Type IB).
- **Type II:** Displaced, stable fractures – In this pattern, the proximal fracture fragment is displaced more than 3 mm, but the collateral ligaments are intact. It can be either non-comminuted ones (Type IIA) or comminuted (Type IIB).
- **Type III:** Displaced unstable fracture – In this case, the fracture fragments are displaced and the forearm is unstable in relation to the humerus. It is a fracture dislocation. It also may be either non-comminuted (Type IIIA) or comminuted (Type IIIB).

Mayo type I

Undisplaced

Mayo type II

Displaced

A-Non comminuted

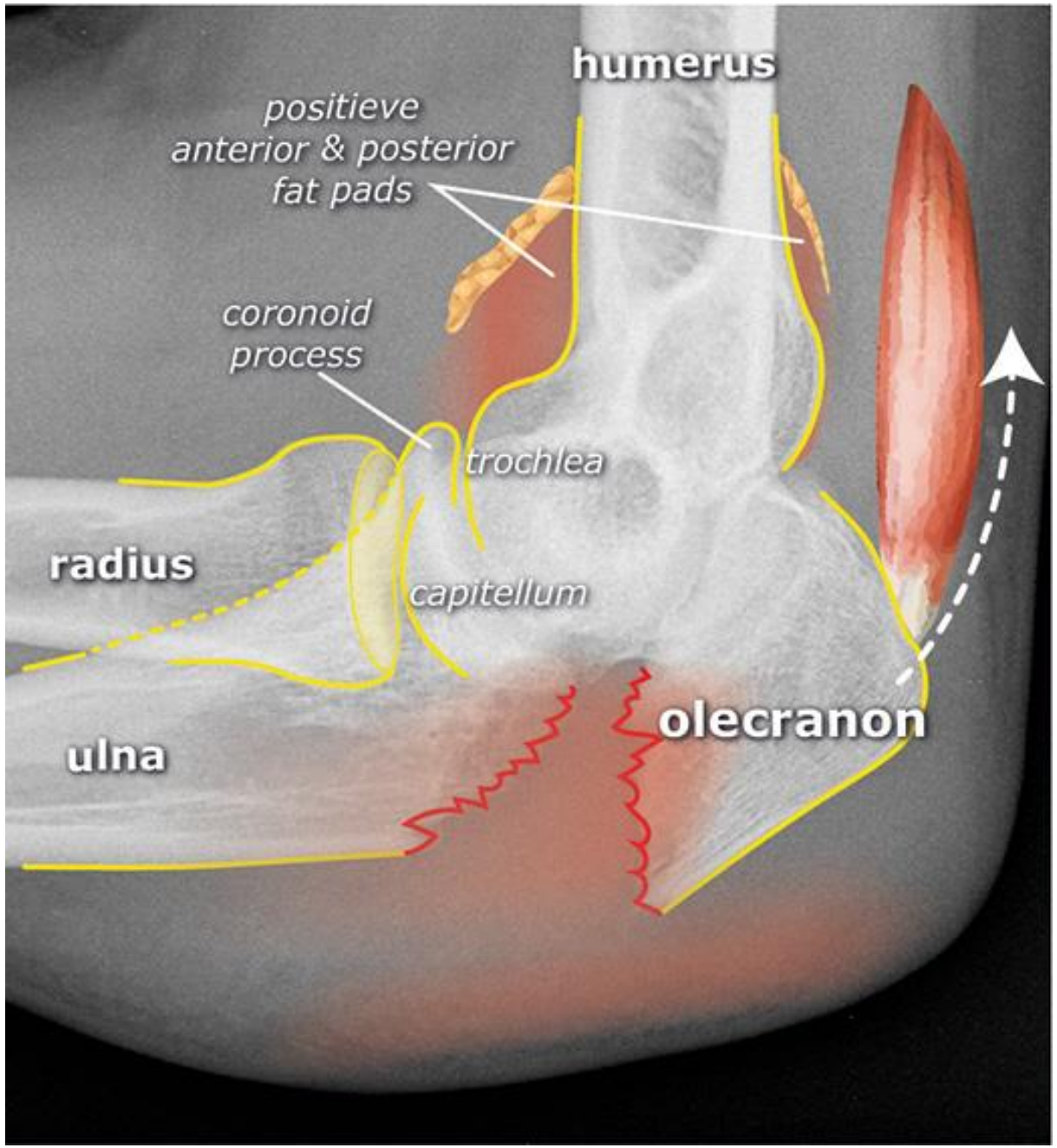
B-Comminuted

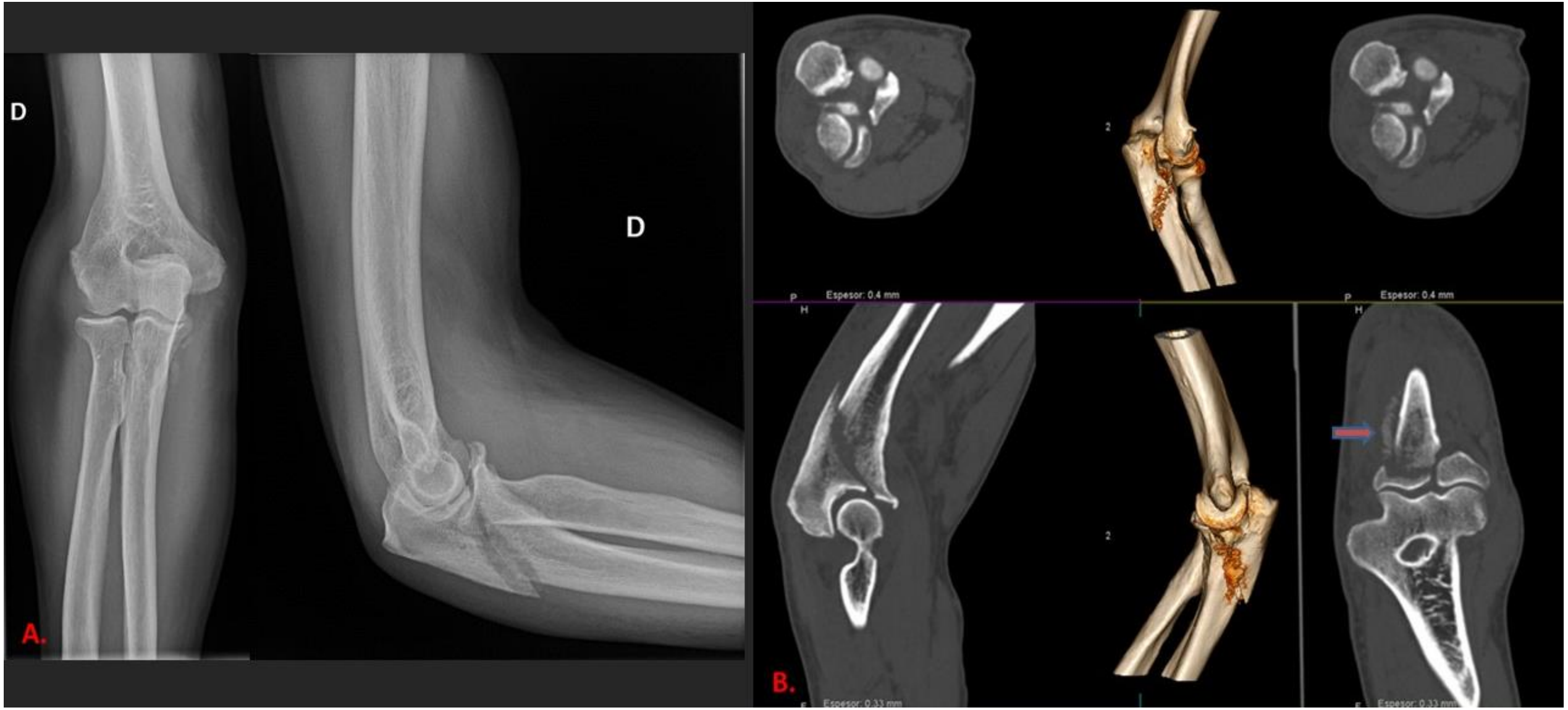
Mayo type III

Accompanying lesions-Instability

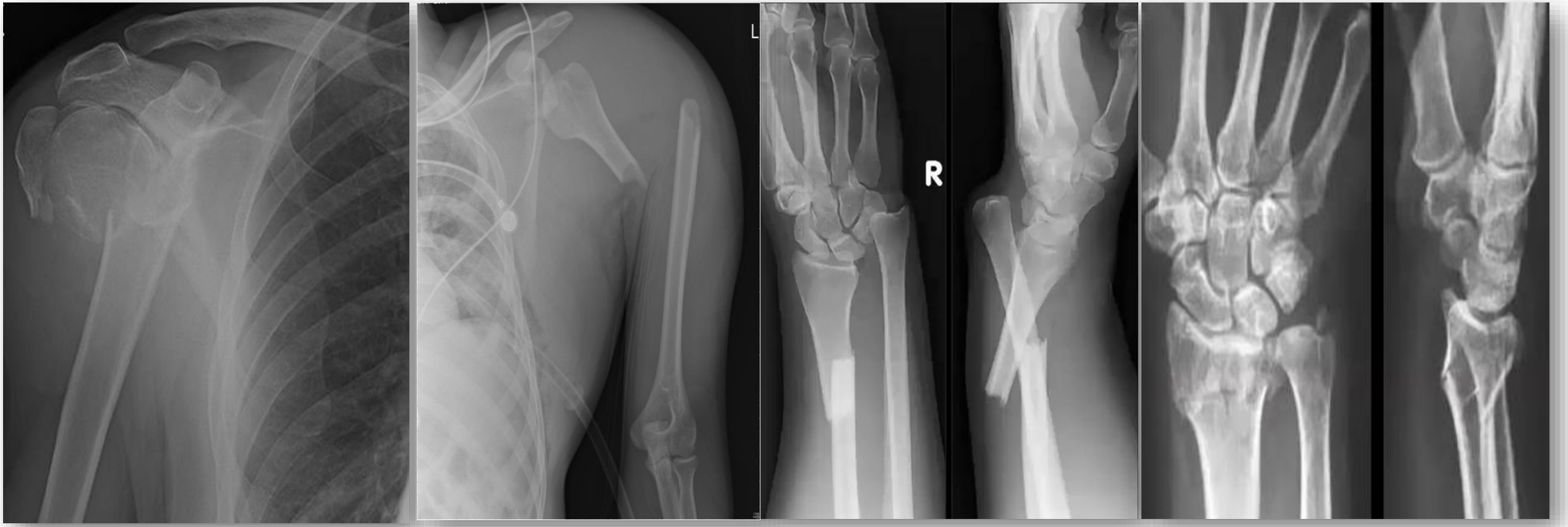
A-Non comminuted

B-Comminuted


humerus
positieve
anterior & posterior
fat pads
coronoid
process
trochlea
radius
capitellum
olecranon
ulna



Olecranon fracture. 43-year-old man. AP and lateral right elbow x-rays (A.) and CT with multiplanar and 3D reconstructions (B.): Ulna metaphyseal-epiphyseal fracture with intraarticular component. It is an oblique fracture and presents distraction of its fragments (more than 3 mm) and an incipient callus formation (red arrow). It is a type II A of Mayo classification.



Upper Limb Fractures

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Part 3

Galeazzi fracture-dislocation

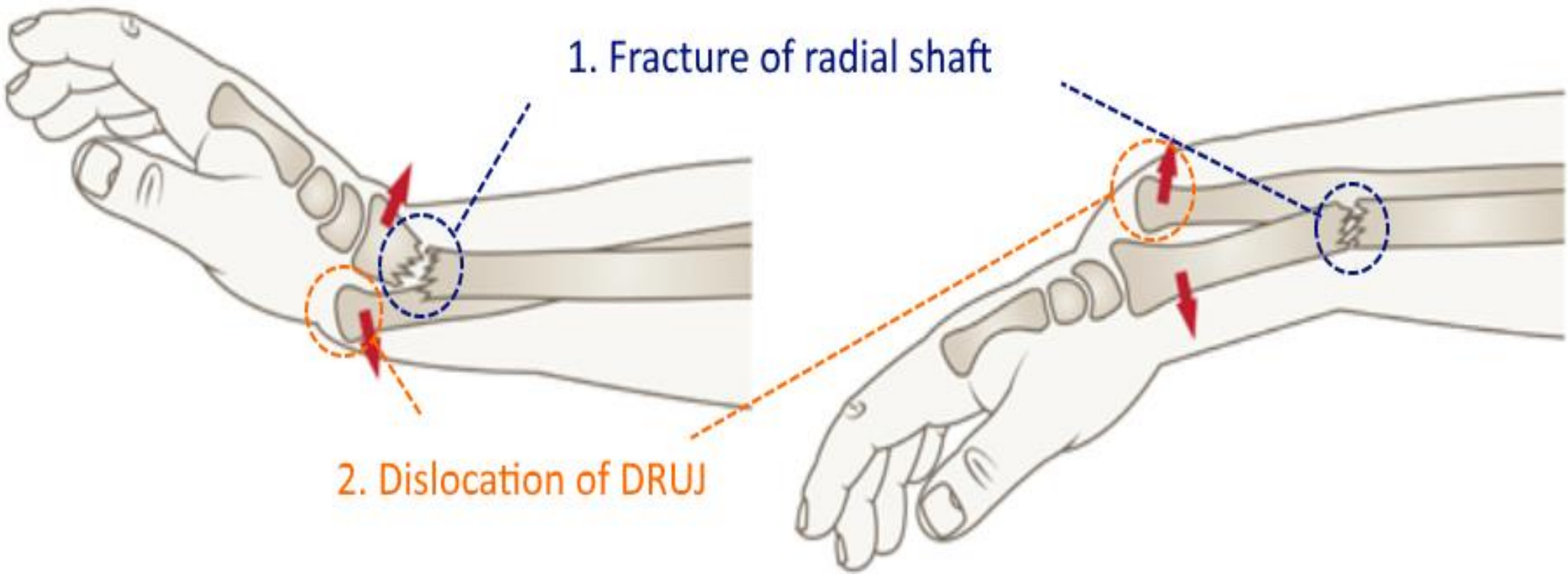
- **Galeazzi fracture-dislocations** consist of fracture of the distal part of the radius (distal and middle 1/3 rd) with dislocation of distal radioulnar joint and an intact ulna.
- **Galeazzi-equivalent fracture** is a distal radial fracture with a distal ulnar physeal fracture
- Galeazzi fractures are classified according to the position of the distal radius:

  type I: dorsal displacement

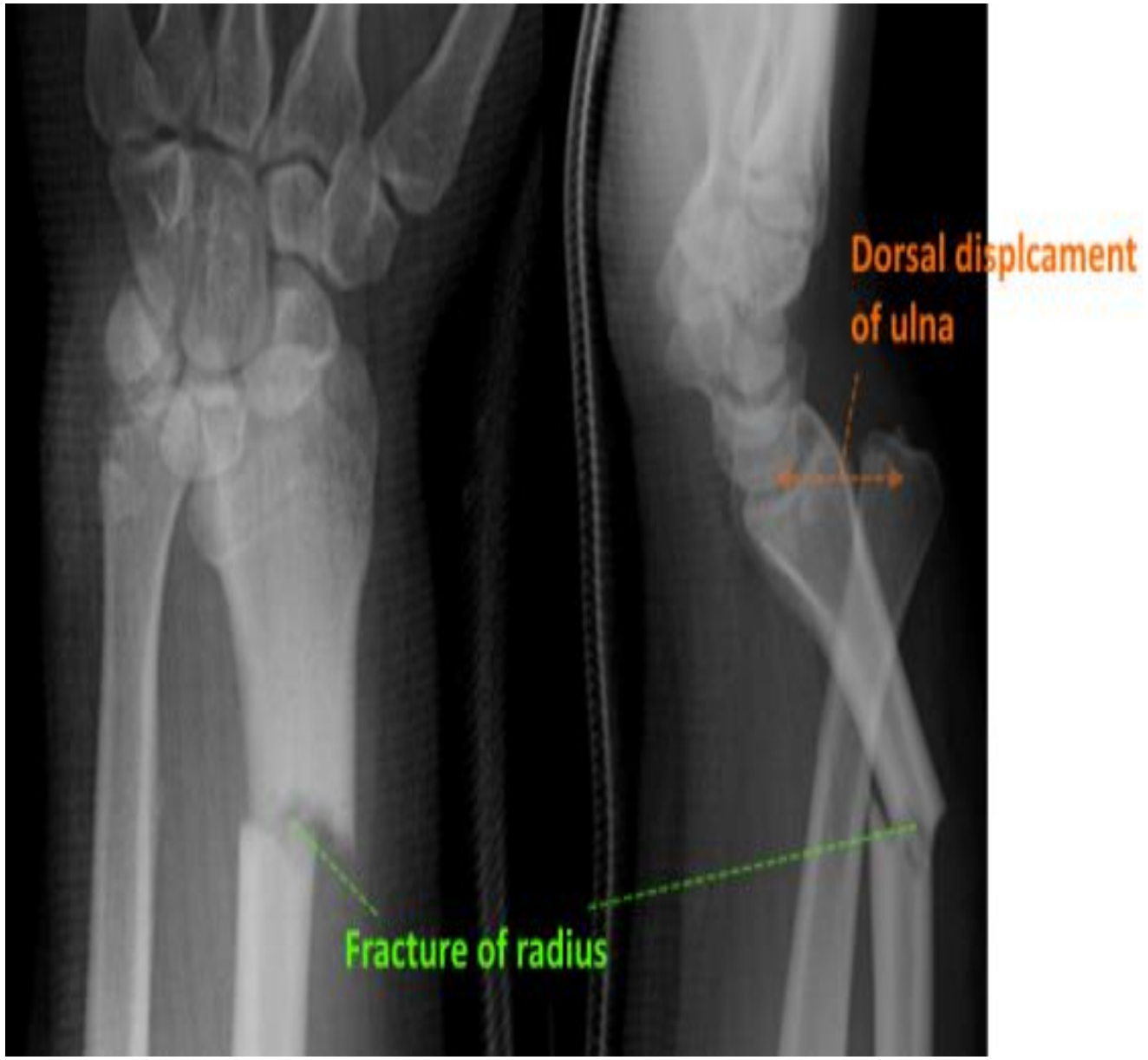
  type II: volar displacement

Volar Galeazzi Fracture

Dorsal Galeazzi Fracture



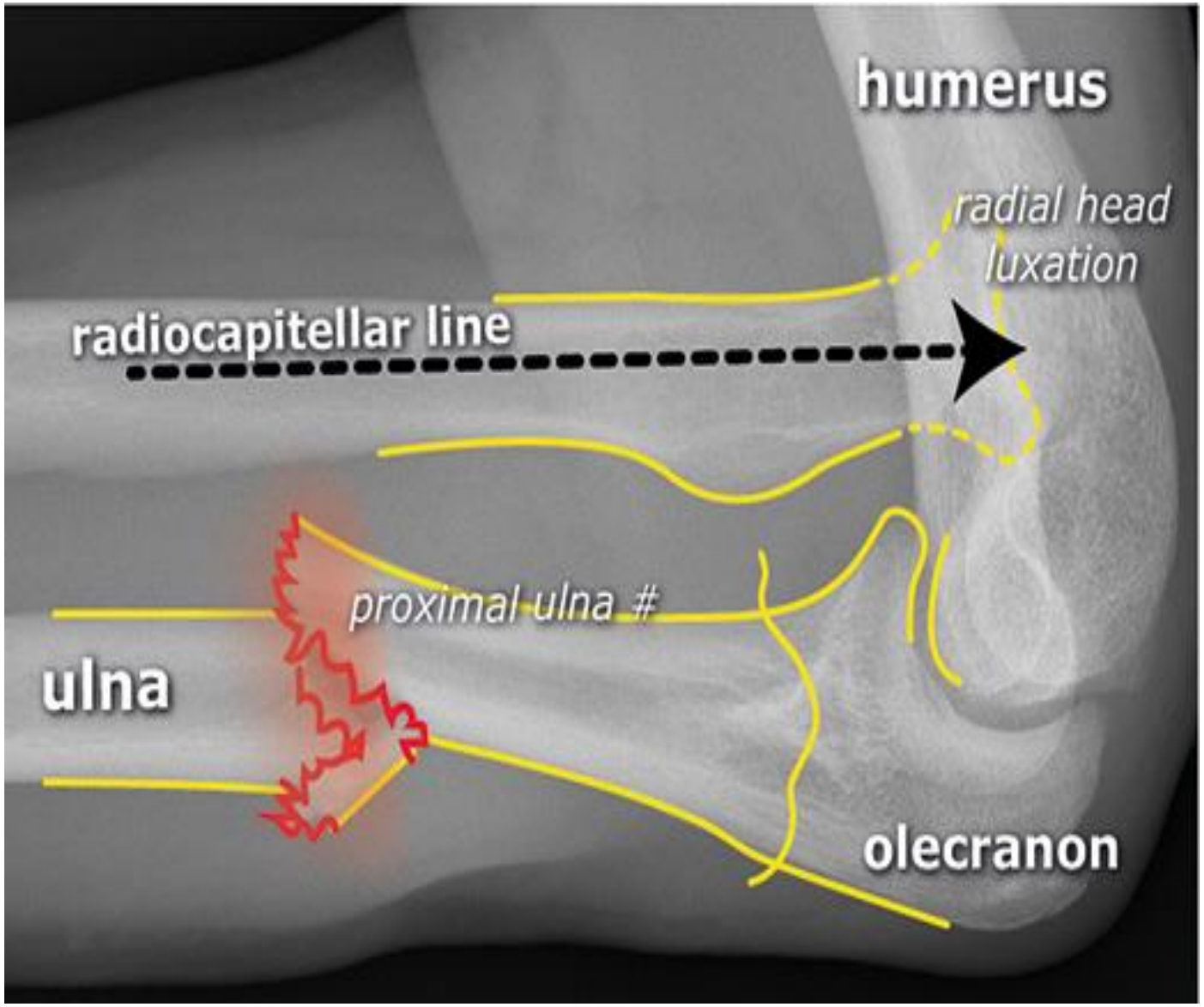
Volar displacement

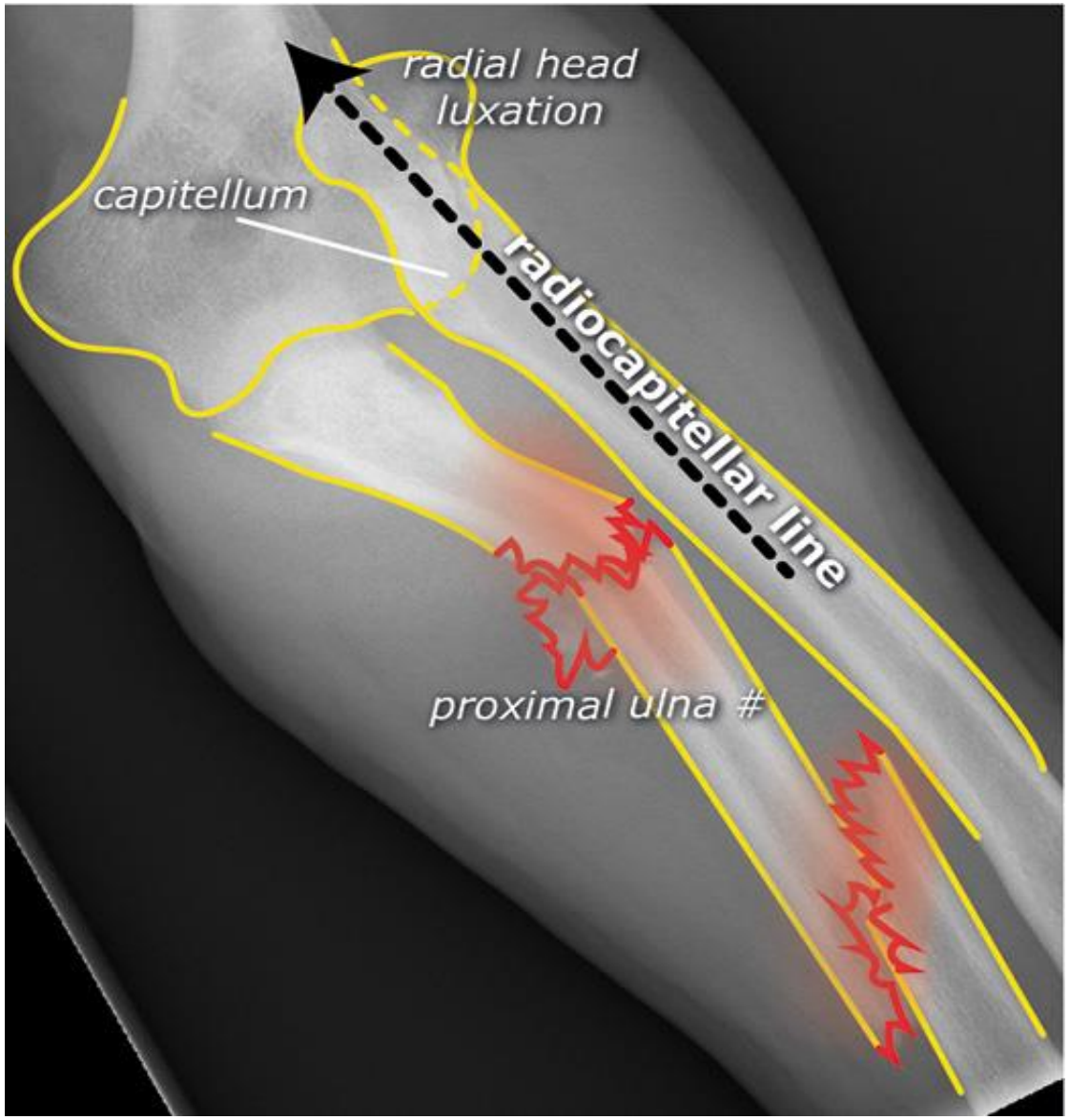


Dorsal displacement

Monteggia fracture-dislocation

Monteggia fracture-dislocations consist of a fracture of the ulnar shaft with concomitant dislocation of the radial head.

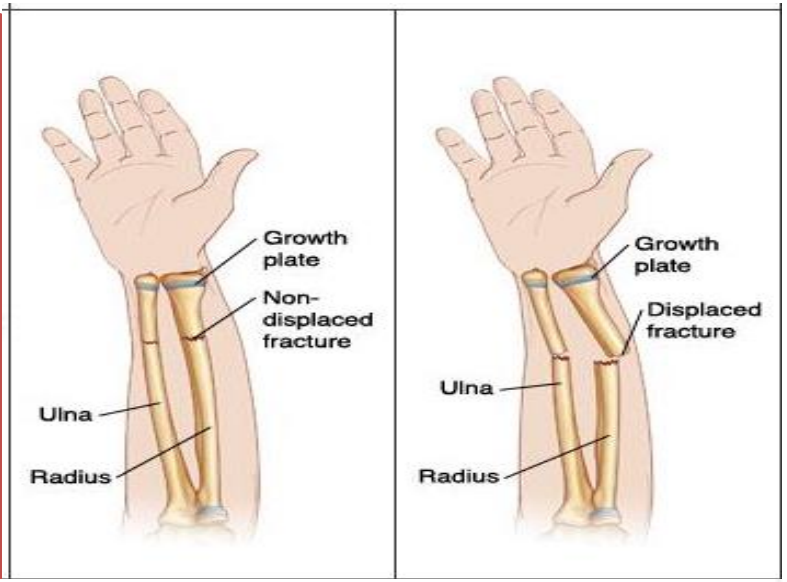




Monteggia fracture; fracture of the proximal ulna and radial head luxation. Note the abnormal radiocapitellar line.

Radius and Ulnar Shaft Fractures

- When they both break, it is called **a both-bone forearm fracture.**
- one of the most common pediatric fractures estimated around 40% of all pediatric fractures.

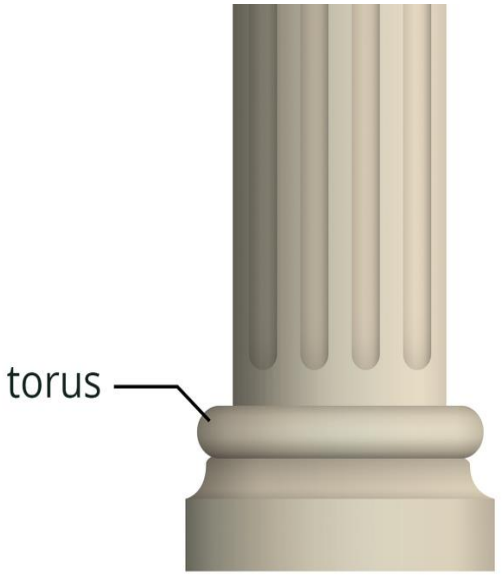


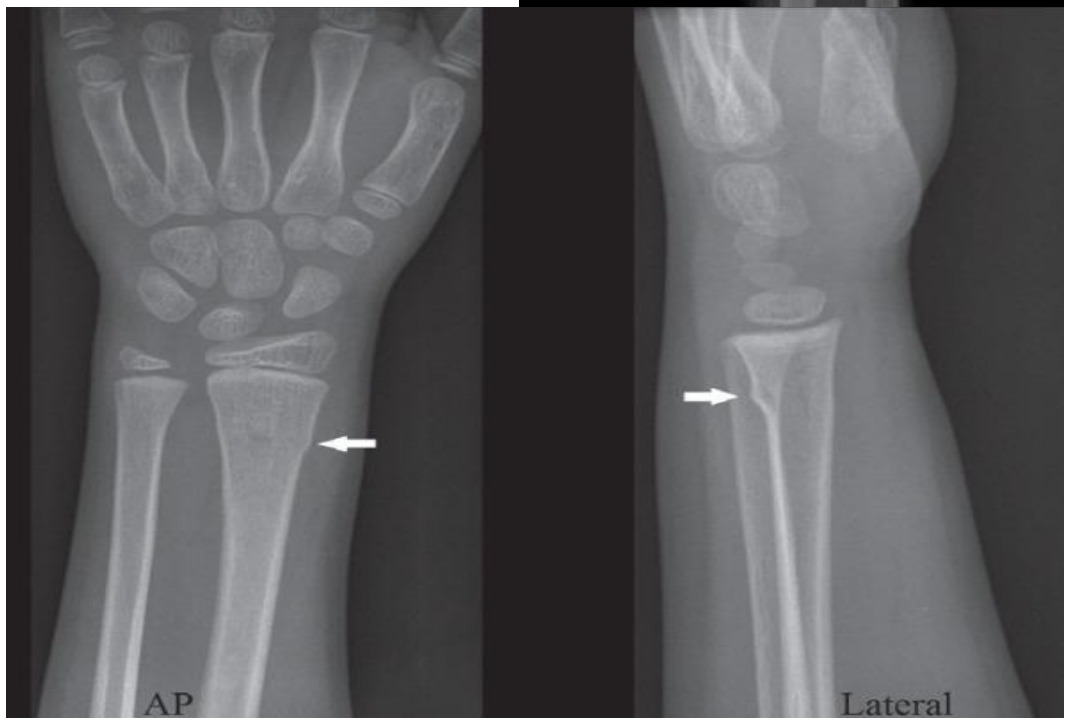
Greenstick fracture in children

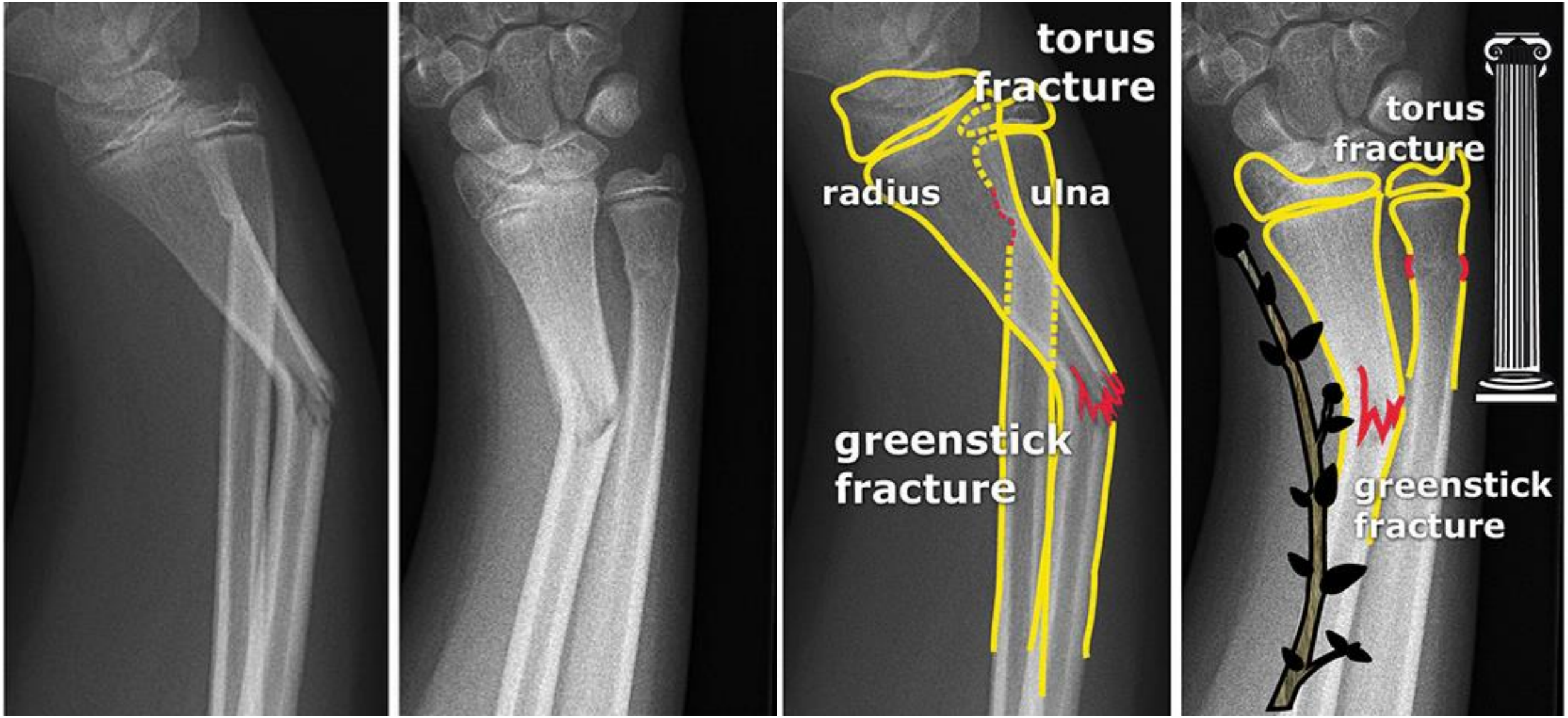
- A greenstick fracture occurs when a bone bends and breaks, but doesn't break into two separate pieces.
- It's called by this name because it looks similar to what happens when you try to break a "green" branch from a tree.
- Greenstick fractures are incomplete fractures of long bones and are usually seen in young children, more commonly less than 10 years of age.

Torus buckle fracture in children

- Buckle (torus) fractures occur when the bony cortex is compressed and bulges.
- They result from trabecular compression due to an axial loading force along the long axis of the bone.
- They are usually seen in children, frequently involving the distal radial metaphysis.





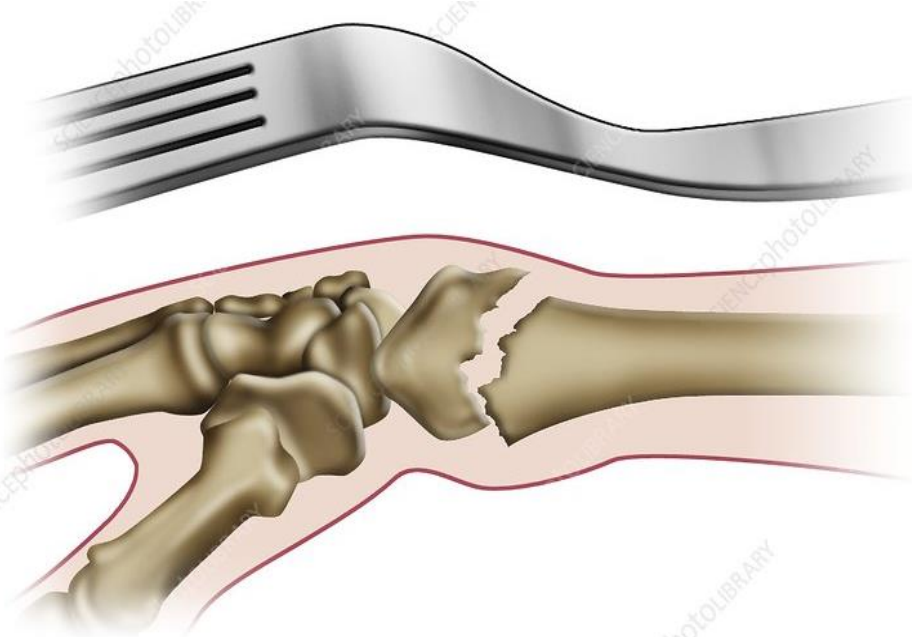


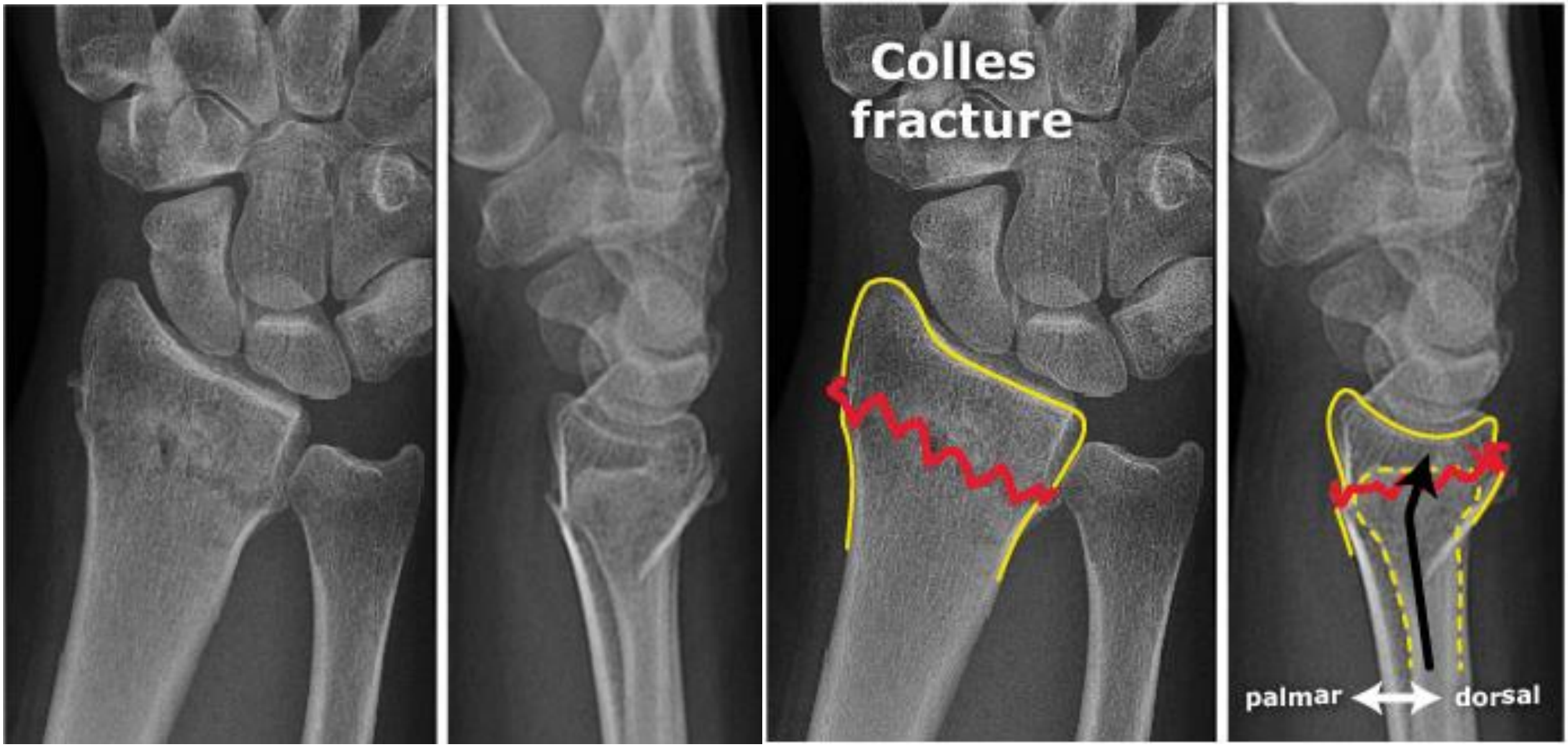
Lateral image (a) and PA image (b) of a radial greenstick fracture and ulnar torus fracture.

Colles' fracture

- Colles' fracture of the distal radius with fracture of the ulnar styloid process.
- Usually seen in elderly osteoporotic patients.
- Results from fall on the outstretches hand.
- The classic deformities are:

1. Dorsal displacement of the distal fragment leading to a 'silver fork deformity.
2. Radial shortening.





PA and lateral image of a Colles' fracture.

Smiths fracture

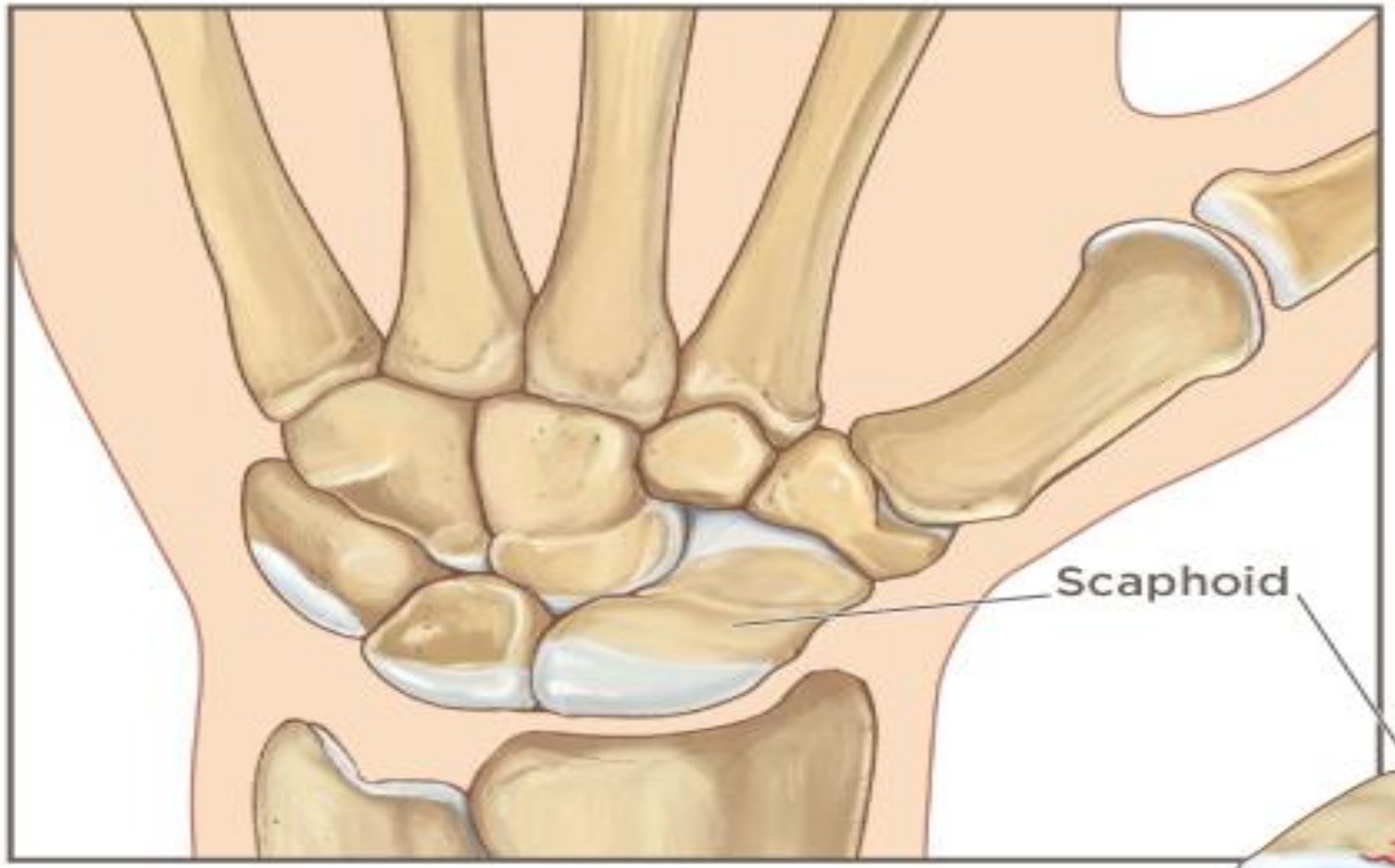
- The mechanism of this injury is the wrist flexed when the patients hand hits the ground, resulting in volar (anterior) displacement of the distal fragment of the radius.
- Smith's fracture is palmar displacement of the wrist that results in a "garden-spade deformity.

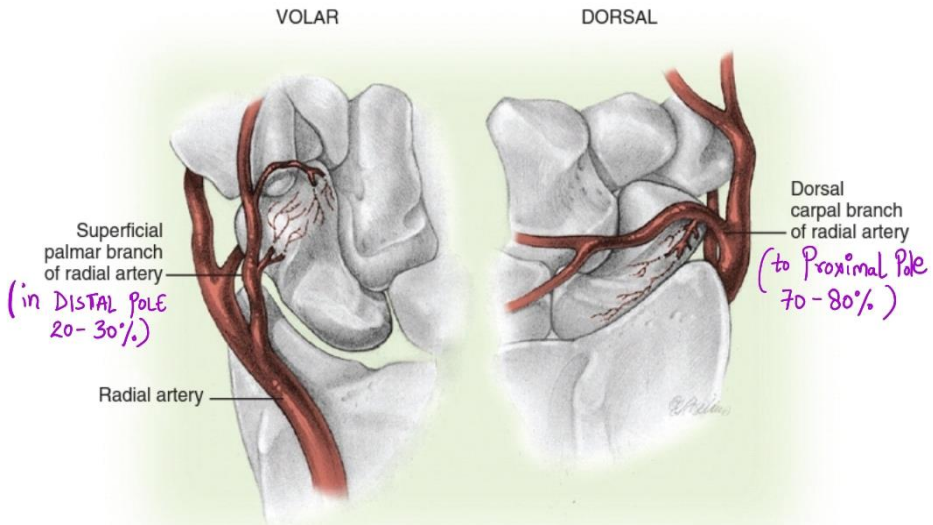
Smith's "garden spade"

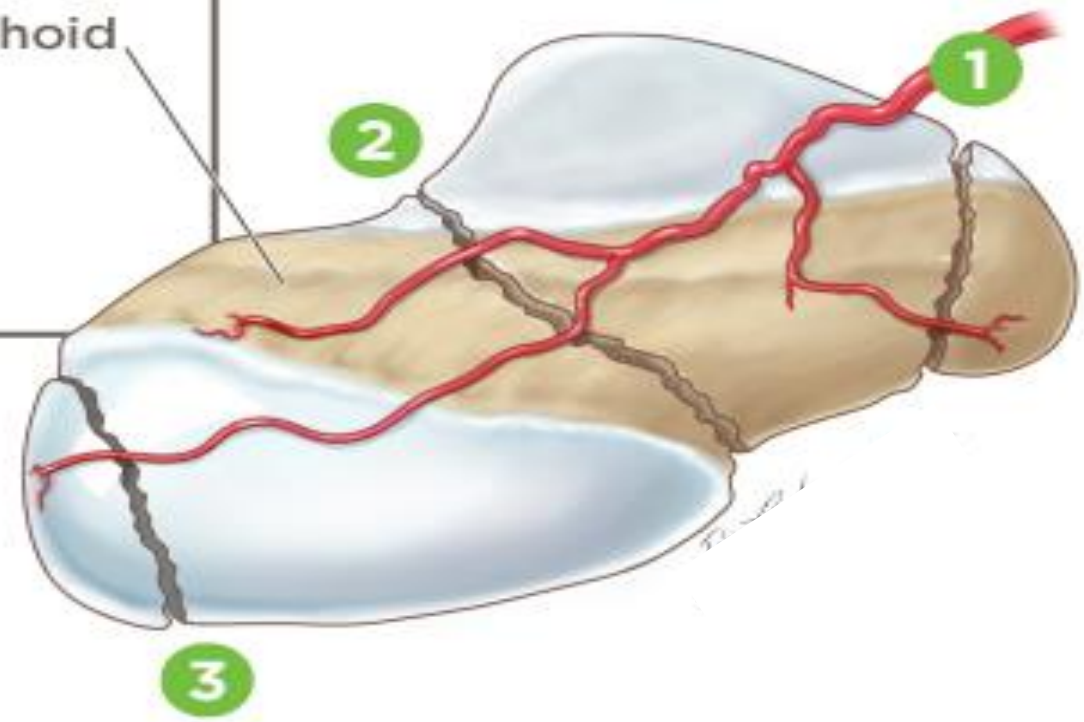
Volar
Dorsal

Scaphoid fracture

- The most common fracture of the carpal bones.
- Occurs during extreme dorsiflexion of the wrist (fall on the out stretched hands). Tenderness in the anatomic snuffbox is considered a sign of scaphoid fracture.
- If this fracture is missed, Avascular Necrosis (AVN) of the proximal pole may occur.
- Radiographs taken soon after the injury frequently fail to reveal the fracture. CT or MRI maybe needed.

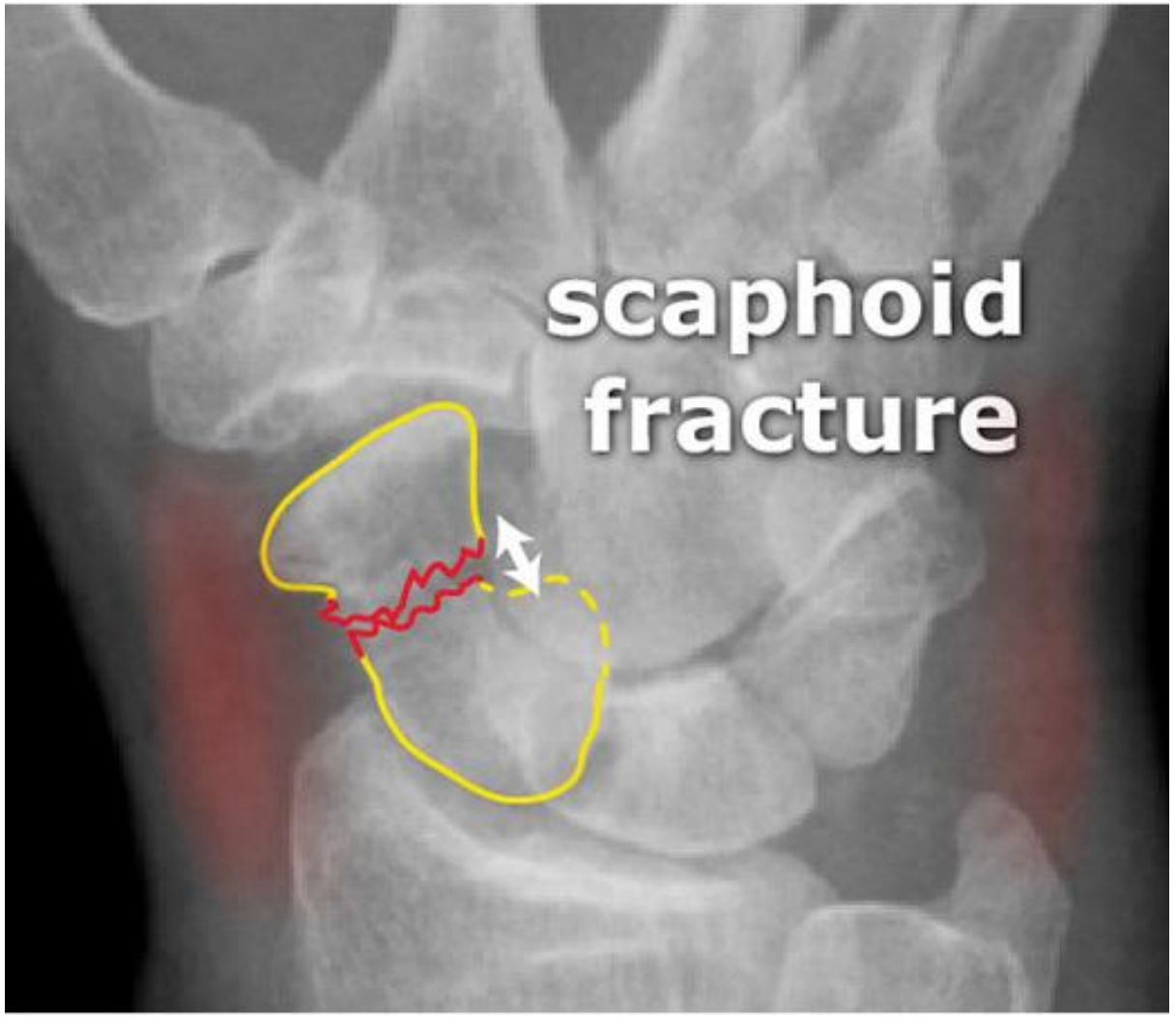




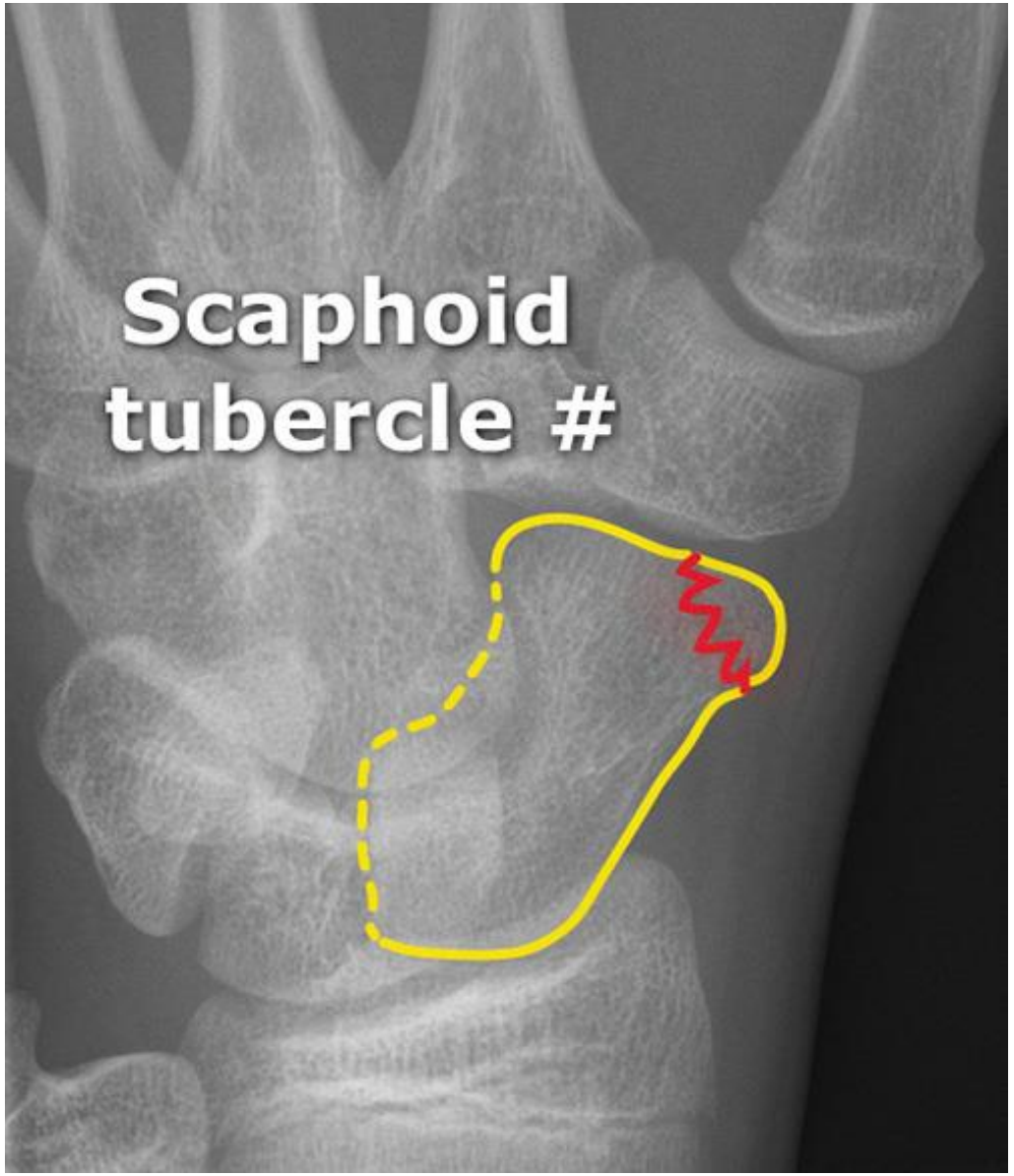


Classifiction of scaphoid fractures

1. Distal fracture
2. Waist fracture
3. Proximal fracture

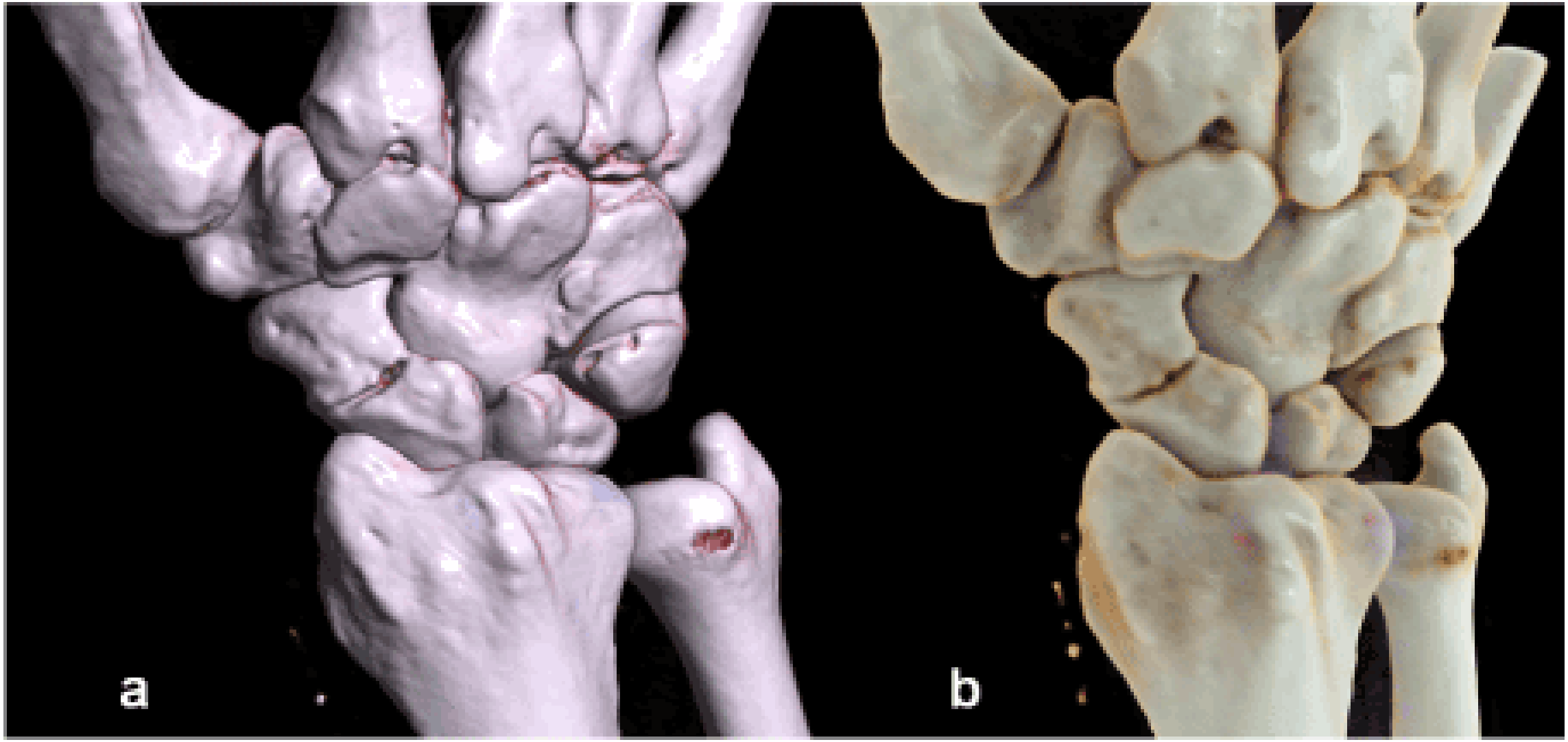


Transversal scaphoid fracture with cortical interruption > 1mm.
Note also marked soft tissue swelling.



Tubercle scaphoid fracture.

Computed tomography, multiplanar reconstructions: a) coronal orientation, b) adequate angulated sagittal reconstruction visualizing the transverse scaphoid fracture.



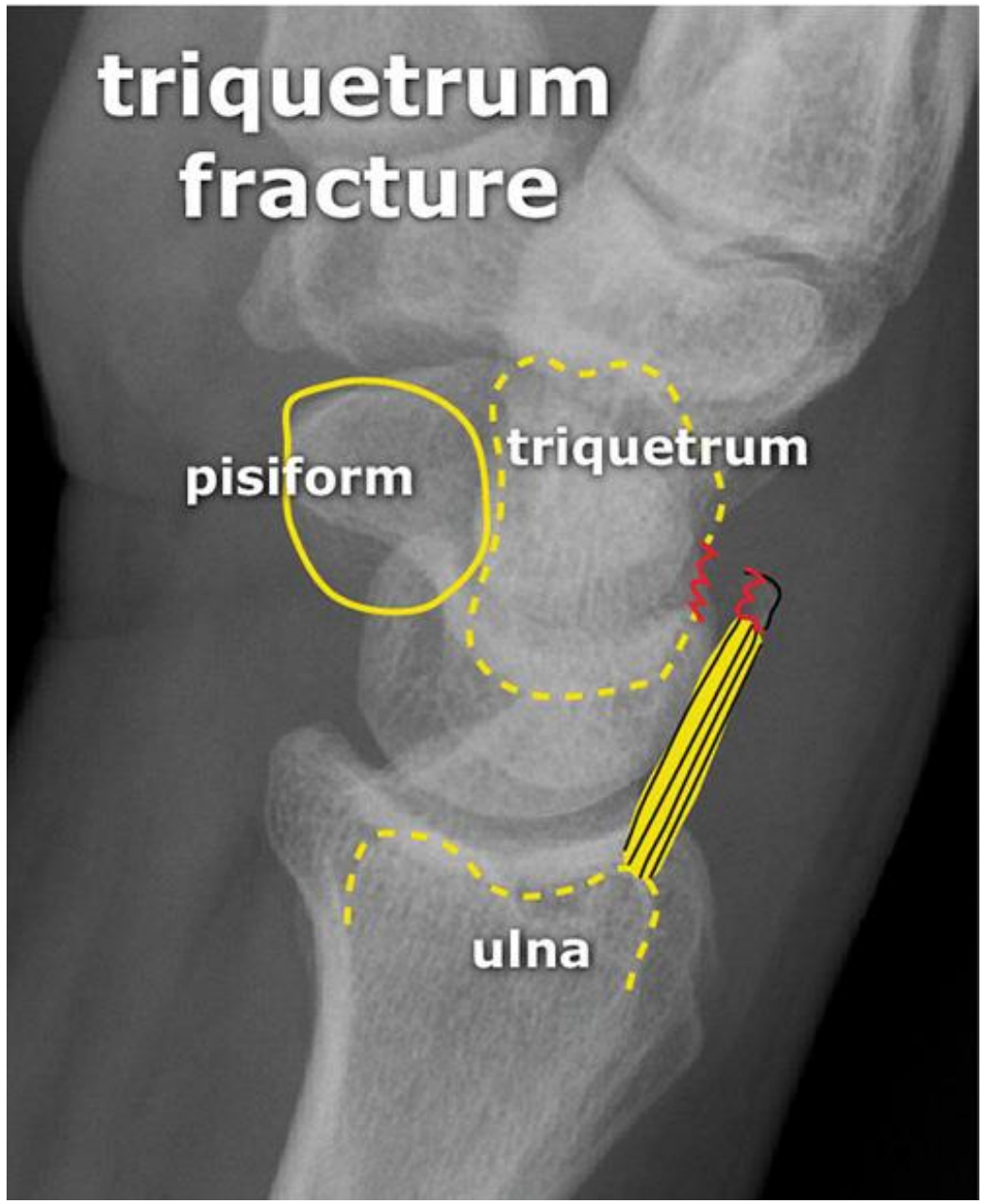
Three-dimensional visualization of the CT data set with Volume Rendering (a) and Cinematic Rendering (b). In comparison to Volume Rendering the display using Cinematic Rendering appears more realistic.

3T MR examination, T2 fat suppressed (STIR) images demonstrate significant bone marrow edema as it is found in recent fractures.

Post gadolinium image shows no enhancement of the proximal scaphoid pole in keeping with establish avascular necrosis.

Triquetral fracture

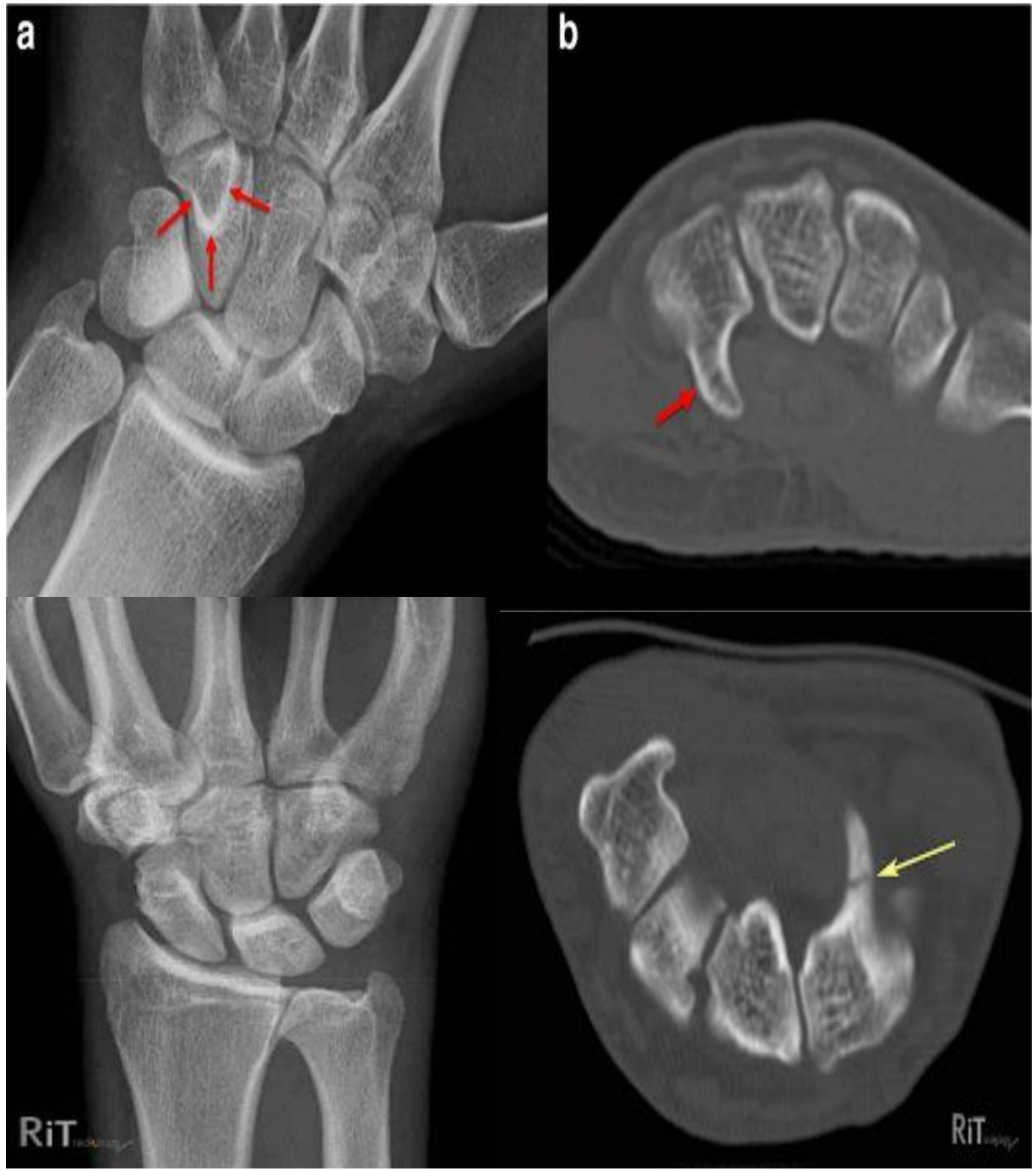
- The triquetrum is the 2nd most commonly fractured carpal.
- Fractures of the triquetrum usually occur from forced hyperflexion or as an avulsion injury.
- They are frequently seen as dorsal chip fractures only on the lateral projection since the pisiform usually overlies the triquetrum on the frontal projection of the wrist.

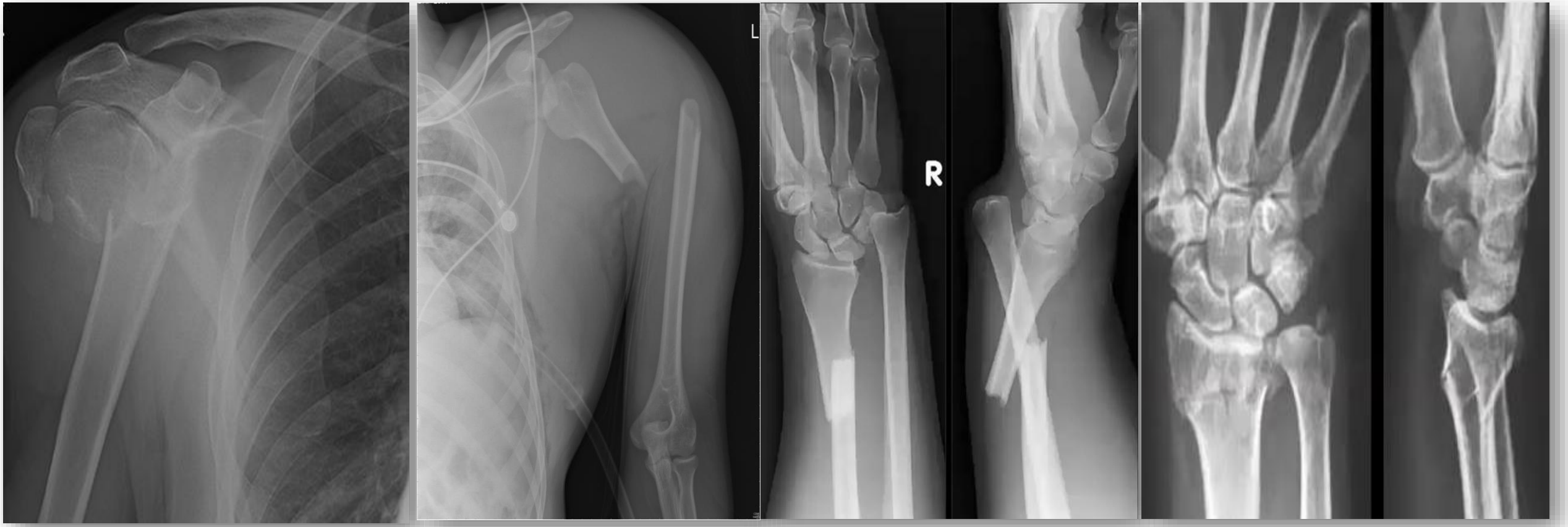


Avulsion fracture of the triquetral bone (not visible on a PA image).

Hook of hamate fracture

- Uncommon fracture that is easily missed on radiography
- Hook of hamate fracture is more common than fracture of the hamate body.
- On frontal radiograph, there is absence or indistinctness of the "eye" of hamate (oval, dense cortical ring shadow over the hamate)
- On CT scan, the fracture line is apparent at the hook best seen on axial images.





Upper Limb Fractures

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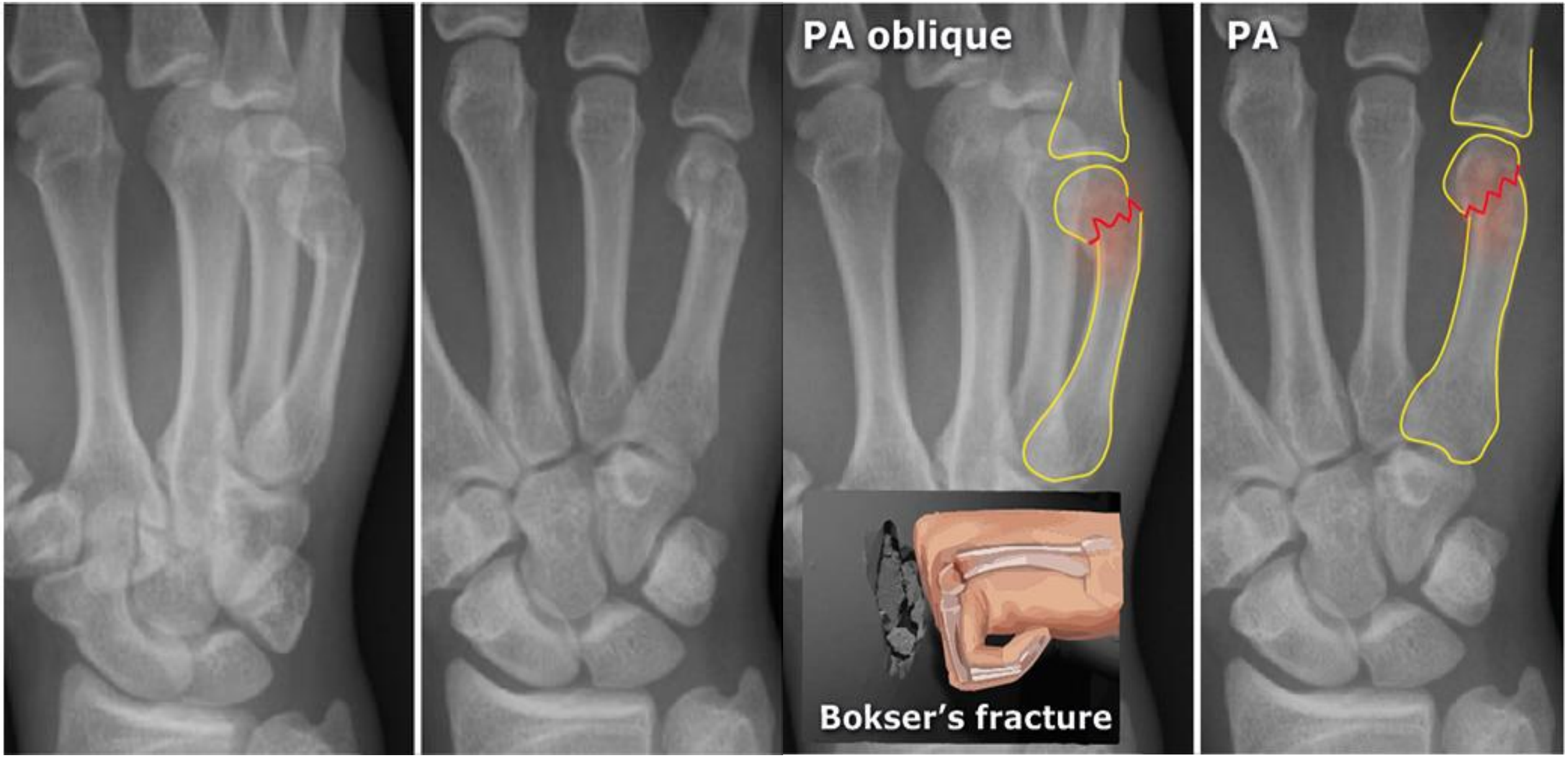
Part 4

Boxers fracture

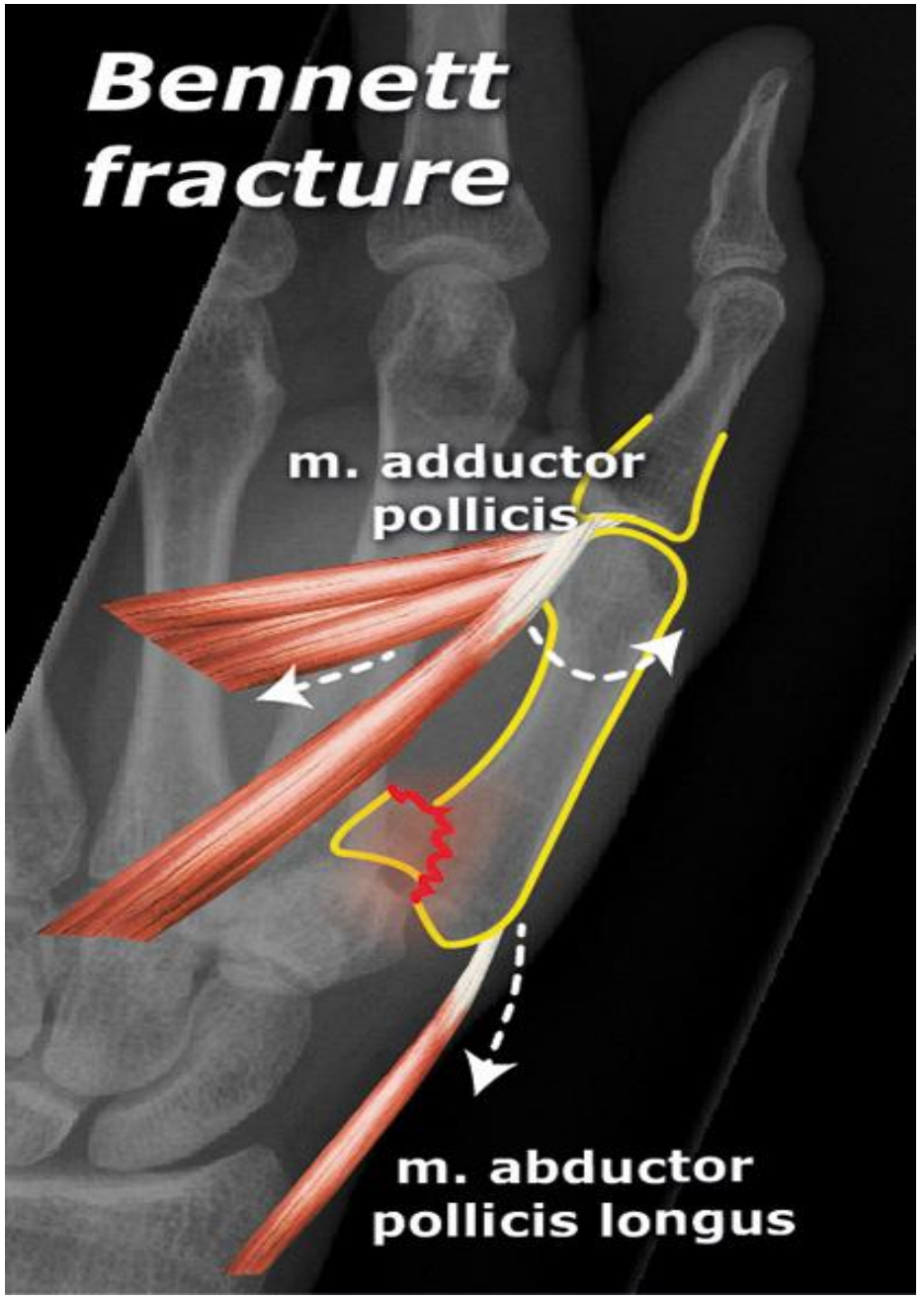
- Occurs during a punch with a closed fist.
- Usually involves the fifth metacarpal bone.
- The fourth metacarpal may also be involved.



PA oblique image and PA image of the left hand. Subcapital fracture of MC-V (= boxer's fracture).

Bennett fracture

- An intra-articular fracture dislocation of the base of the first metacarpal bone.
- Affects the base of the thumb.
- Two- piece fracture dislocation of the base of the thumb.
- Small fragment of 1st metacarpal continues to articular with trapezium.
- Intra-articular dorso lateral dislocation.



Bennett fracture.

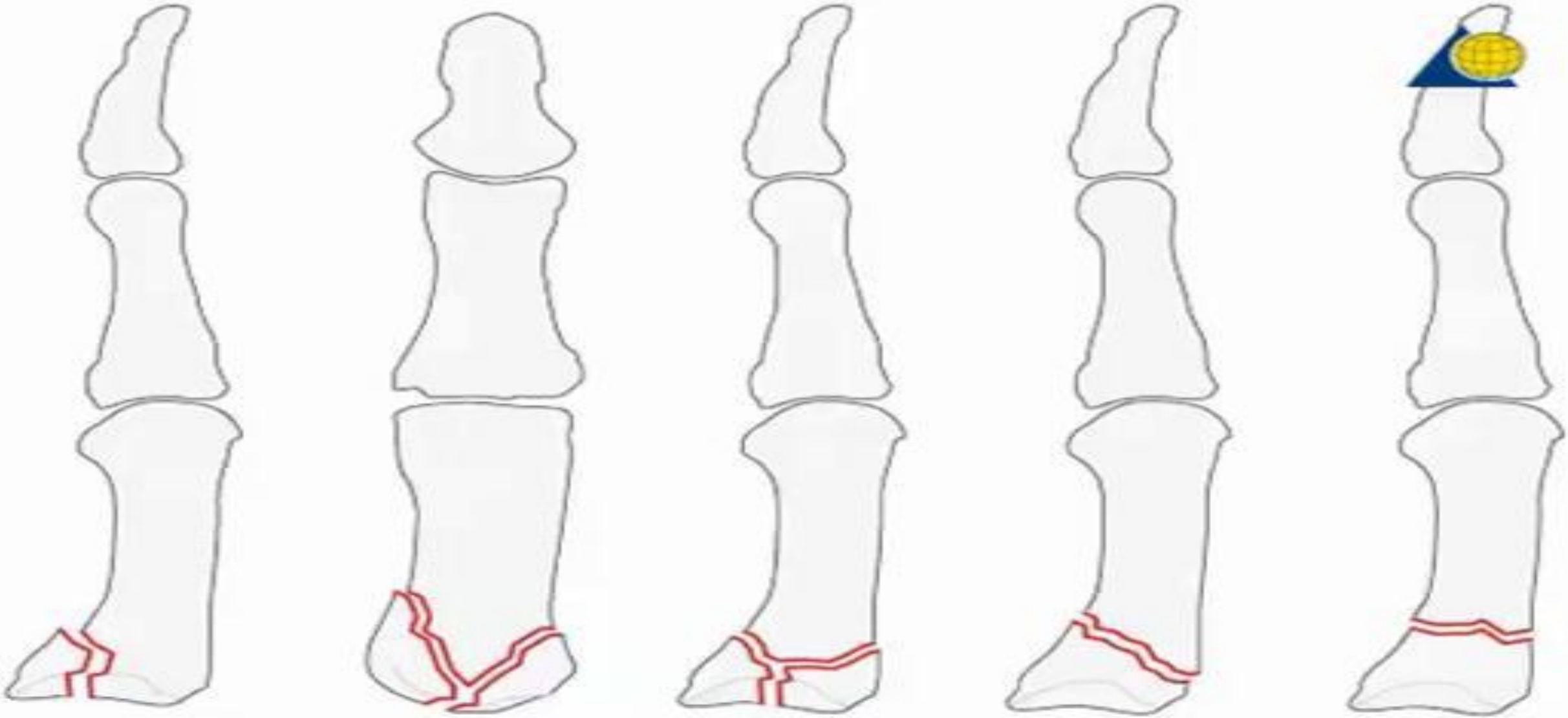
Rolando fracture

- **Rolando fracture** is a comminuted intraarticular fracture through the base of the first metacarpal bone (the first bone forming the thumb).
- This is a fracture consisting of **3 distinct fragments**.
- It is typically **T- or Y-shaped**.

AP image (a) and CT scan (b) of the right hand (a) Rolando fracture, type Y.

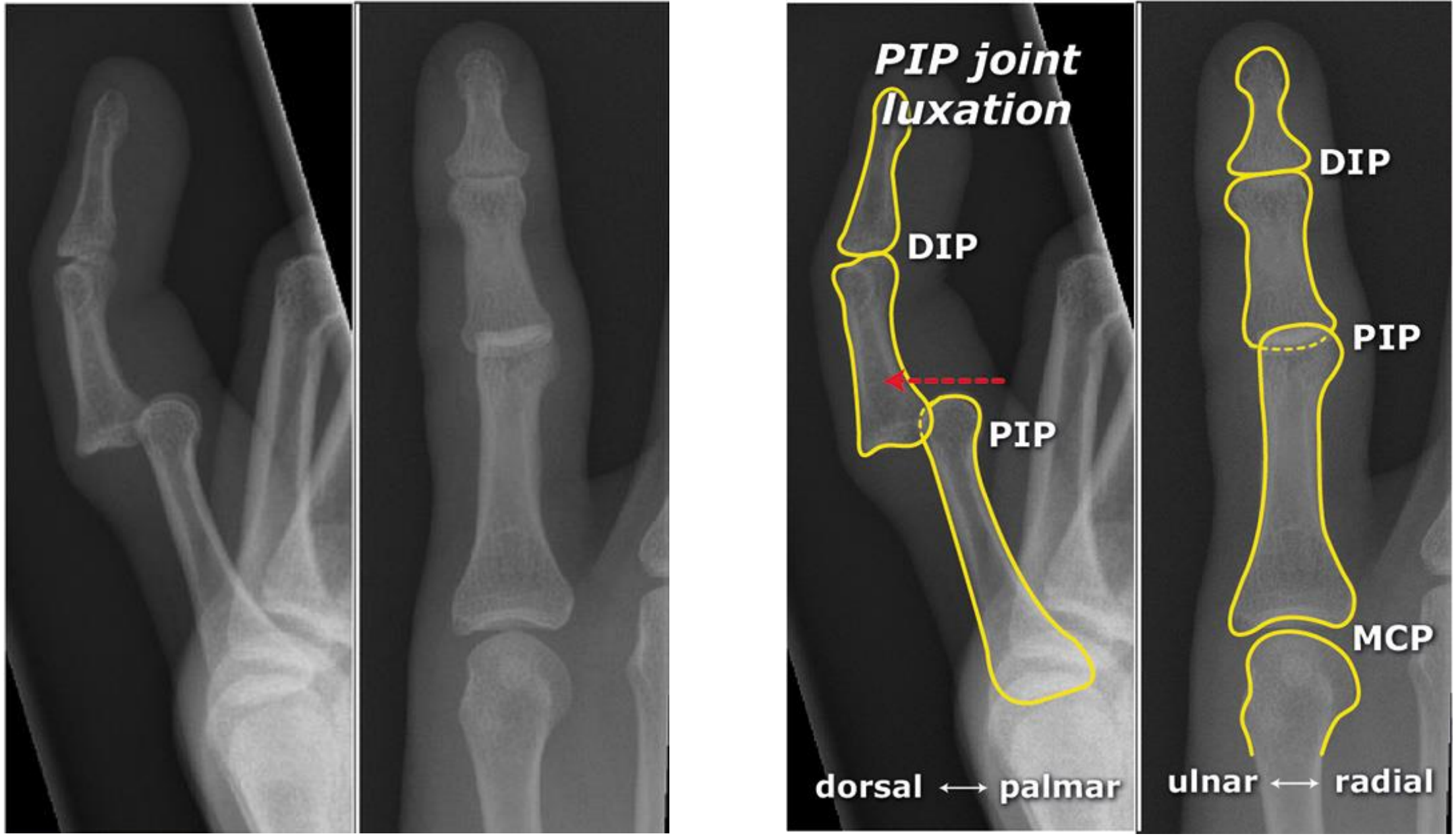
Epibasal thumb fracture

- This two part fracture of the first metacarple base does not extend to the articular surface allowing distinction from a true Bennetts fracture.
- Called pseudo Bennentts fracture.


Bennett
Rolando
Epibasal

Interphalangeal joint dislocations

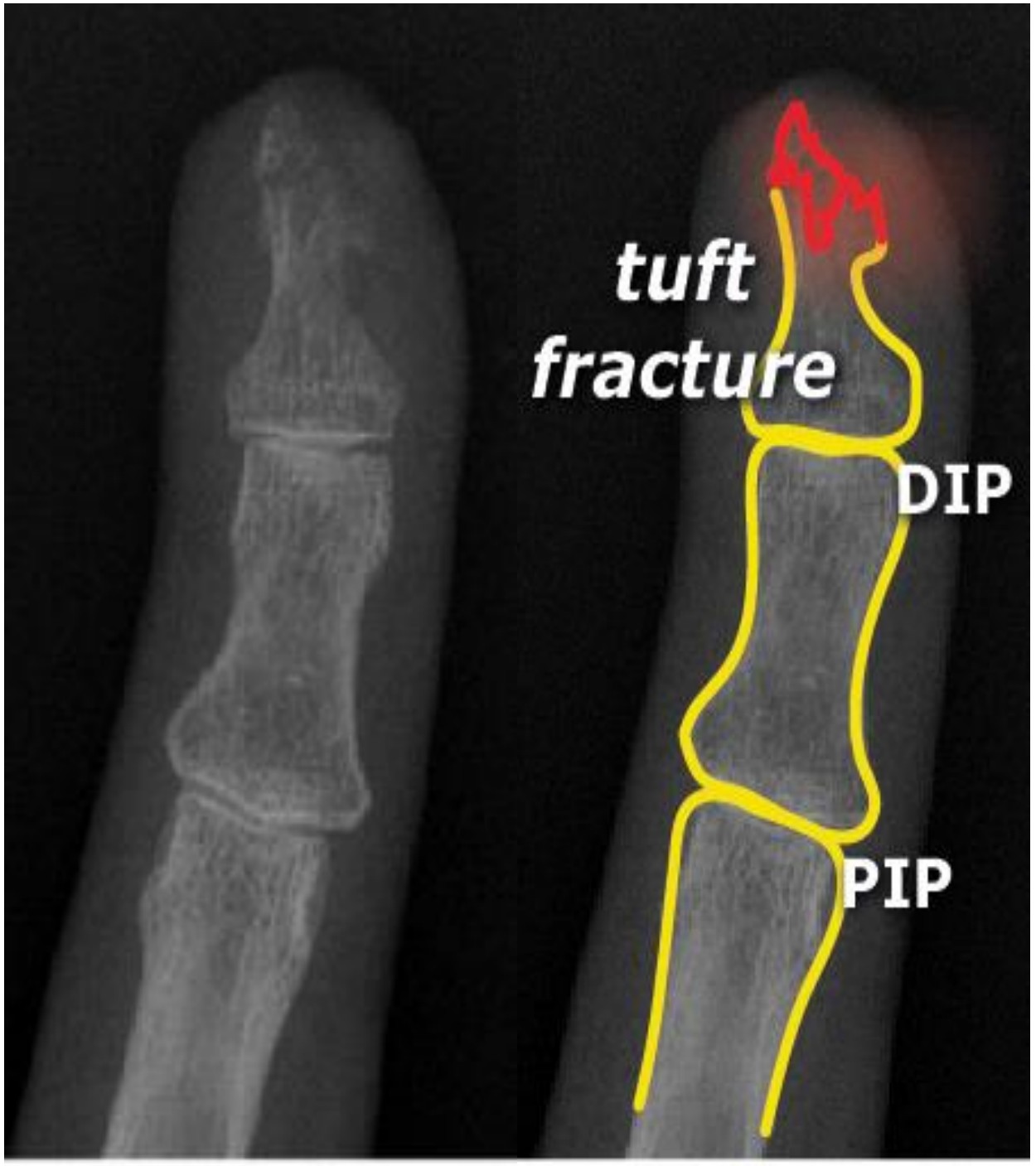
- Interphalangeal joint dislocations are common upper extremity dislocations.
- The typical mechanism is a hyperextension injury.
- The proximal interphalangeal joints are the most commonly.
- The dislocation is dorsal.
- Collateral ligaments most commonly tear.
- Avulsion fracture occasion also result.



Lateral image and PA image of dig-V in the left hand. Dorsal luxation of the PIP joint. Lateral image; note the overlap of the proximal phalanx and the middle phalanx.

Tuft fracture

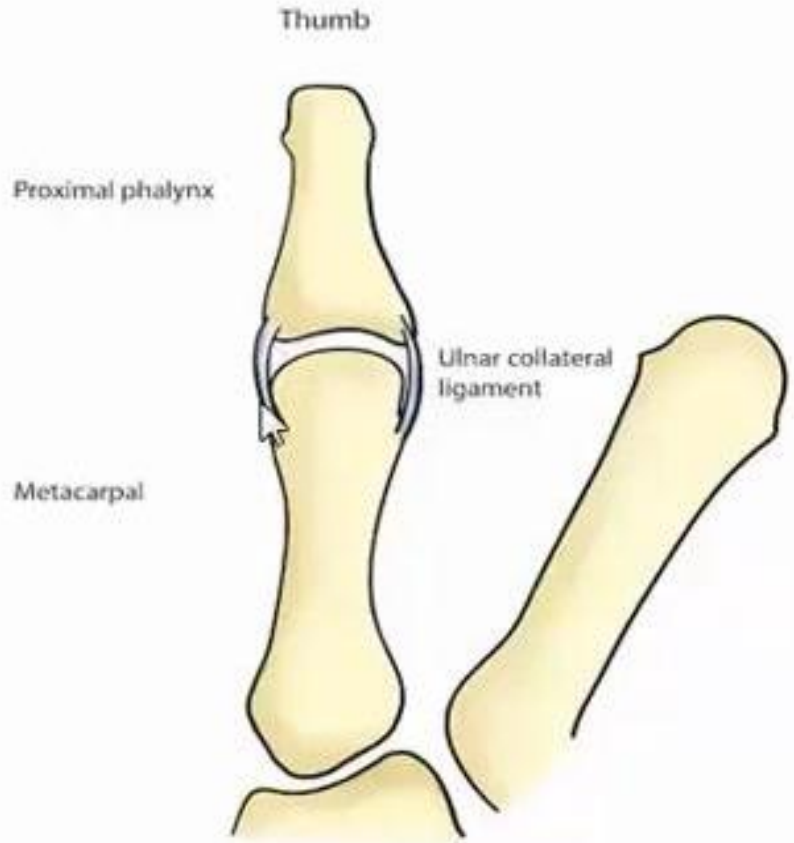
- A tuft fracture is a comminuted fracture of the distal phalanx and usually results from a crush injury (e.g. finger caught between the door).
- Tuft fractures are associated with Subungual hematoma (hematoma under the nail bed). Additionally, be alert for damage of the flexor/extensor tendons.

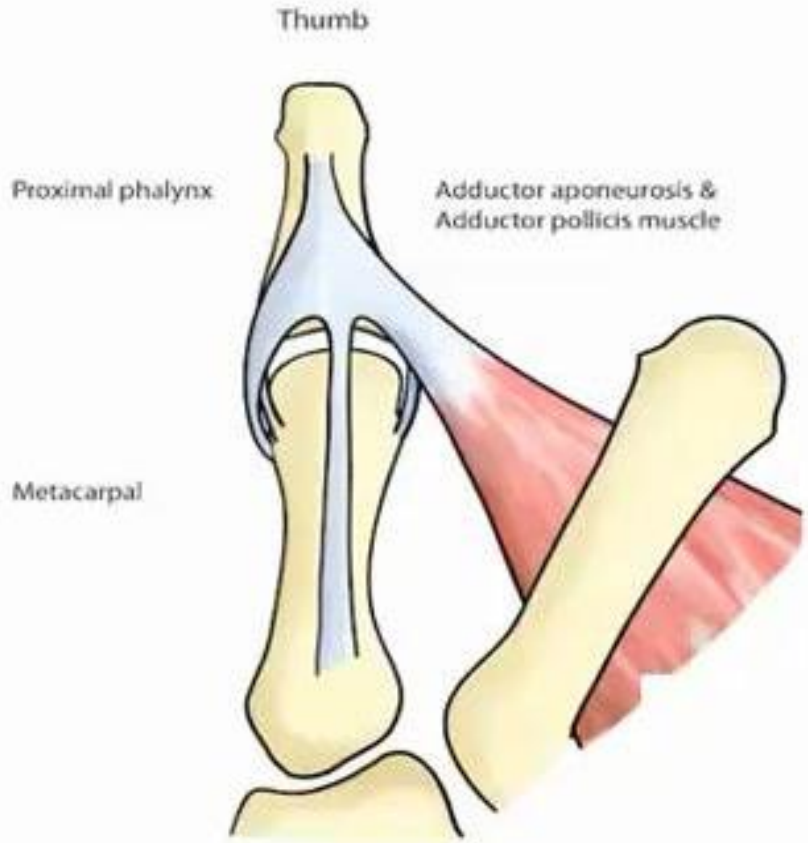


Gamekeepers/Skiers thumb

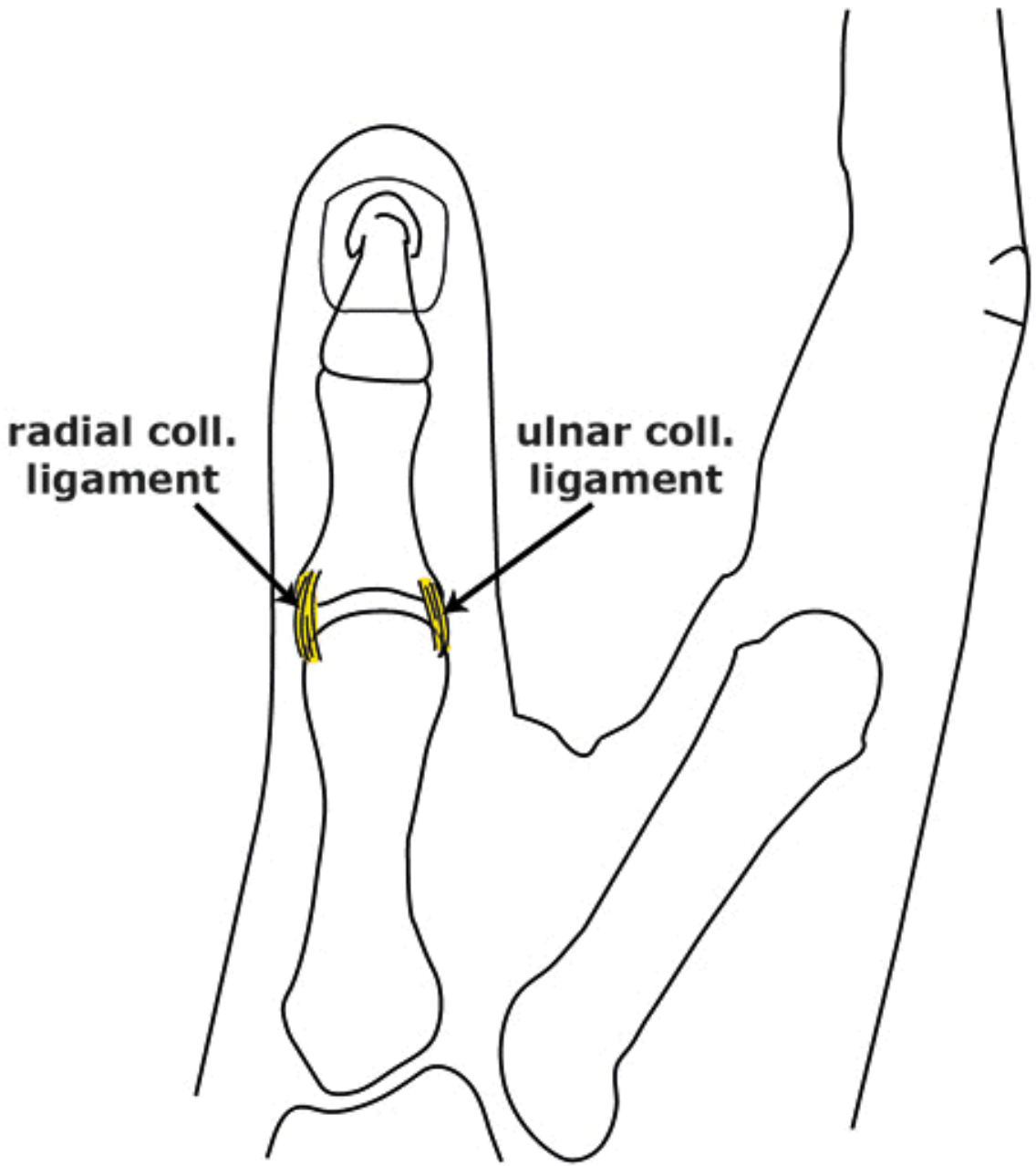
- It is an injury to the ulnar collateral ligament (UCL), which is located in the metacarpophalangeal (MCP) joint where the thumb meets the hand.
- Occur during a fall with a valgus stress and severe abduction are applied to the thumb.

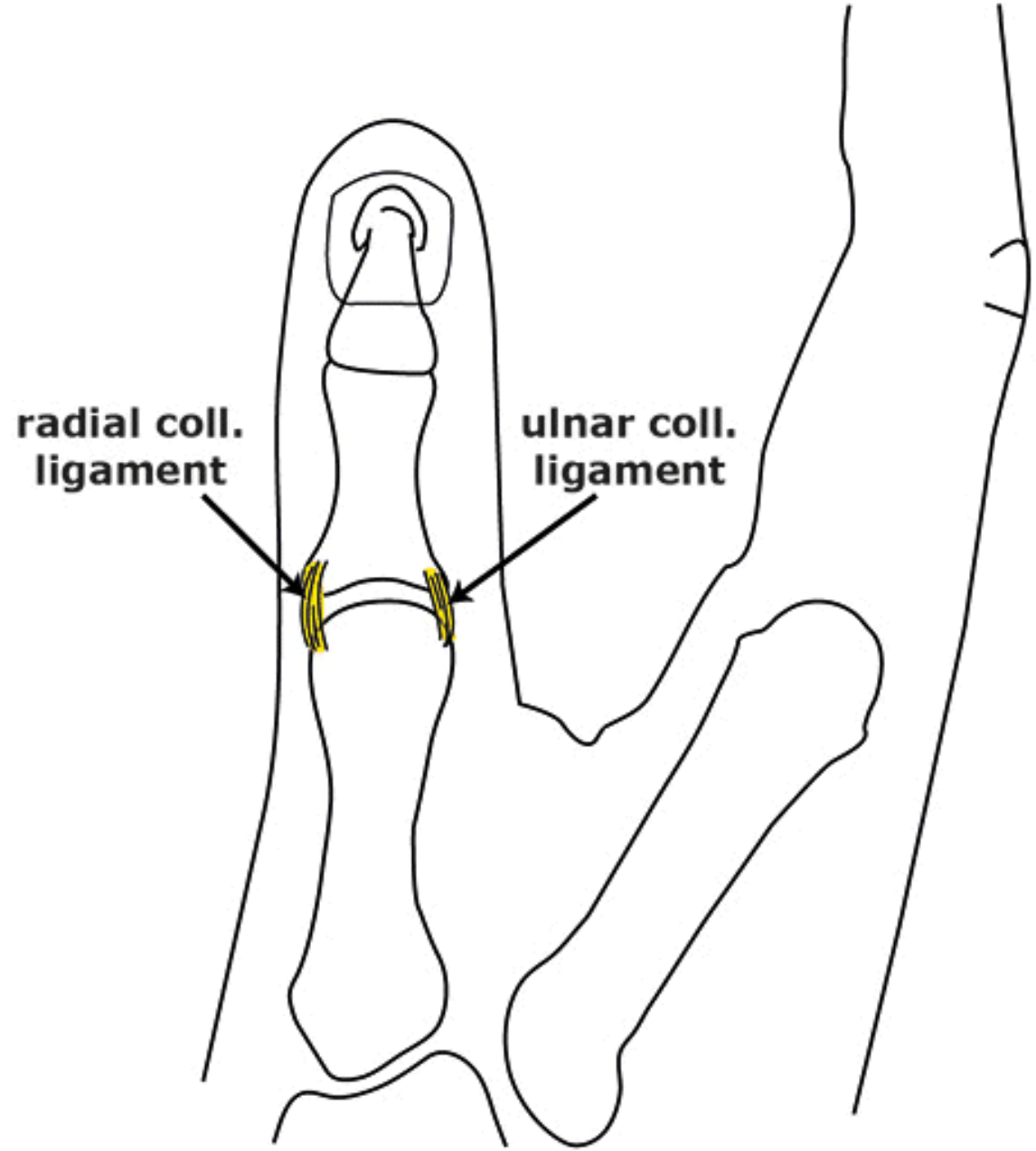
Gamekeeper's / Skier's Thumb





Skier thumb, is an acute injury . It is an avulsion or rupture of the ulnar collateral ligament (UCL) of the thumb


radial coll.
ligament
ulnar coll.
ligament


radial coll.
ligament
ulnar coll.
ligament

Plain radiograph

- This is almost always the first examination and is often able to give the diagnosis.
- If a small avulsion fracture is present, then this will be seen at the ulnar corner of the base of the proximal phalanx.

MRI

- MRI is increasingly used to asses x-ray occult injuries to the ulnar collateral ligament, bone marrow edema and fracture.

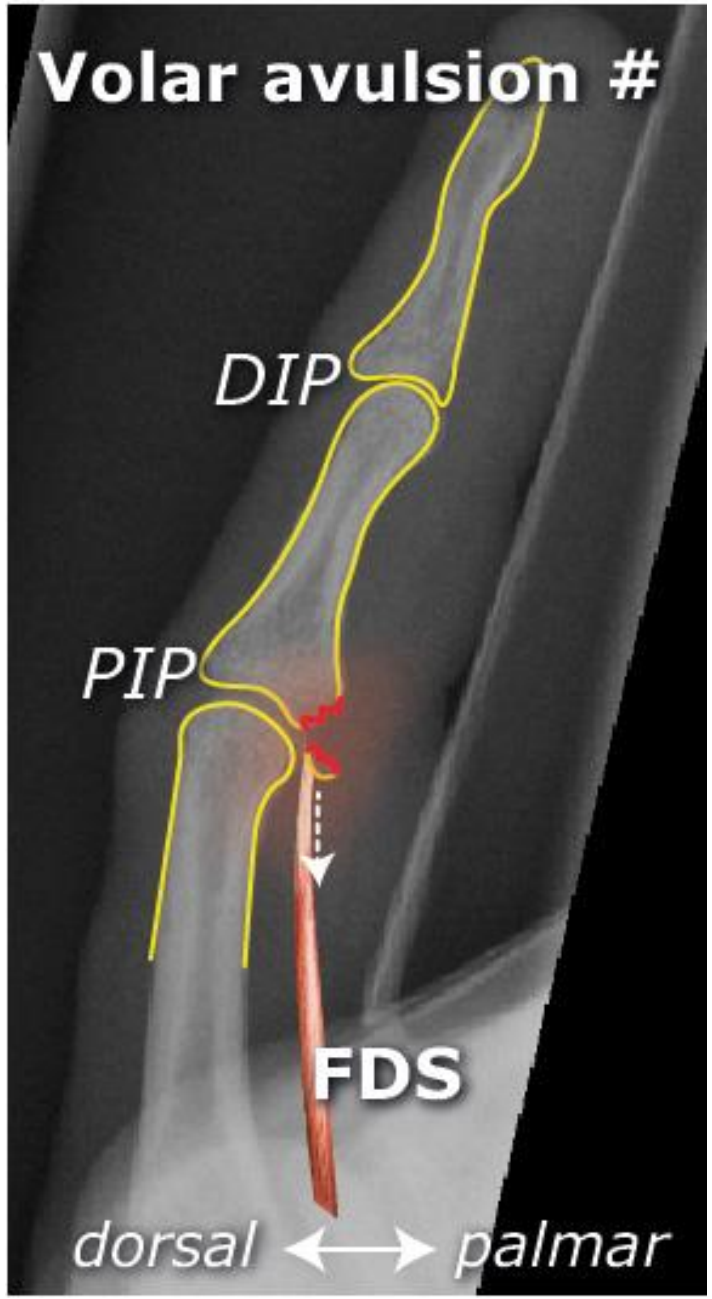
Volar avulsion fracture

- An **avulsion fracture** is a fracture at the level of a tendon insertion. The bone of the insertion site is ripped loose by the tendon/muscle (excessive traction on the bone).
- A **volar avulsion fracture** develops as a result of forced hyperextension of the finger.
- The loose fragment is generally visible on the lateral image only and frequently involves the PIP joint.
- The joint has lost its stability. Eventually, the extensor tendon (without the resistance of the flexor tendon) may cause a hyperextension deformity of the finger.

Lateral image of dig-V of the left hand.

Volar avulsion fracture of the PIP joint with marked proximal dislocation.

FDS = flexor digitorum superficialis or superficial digital flexor.

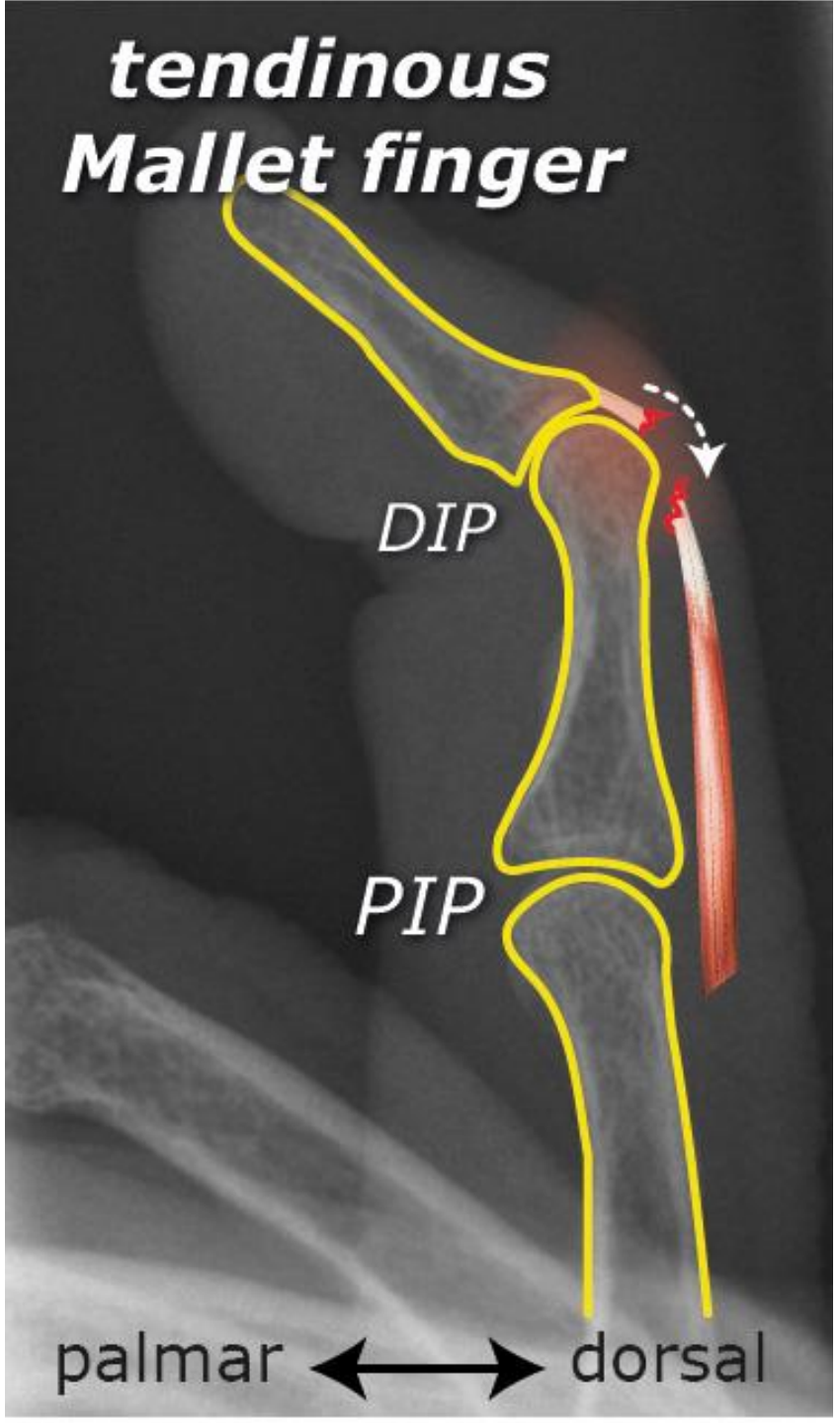


Extensor tendon injuries

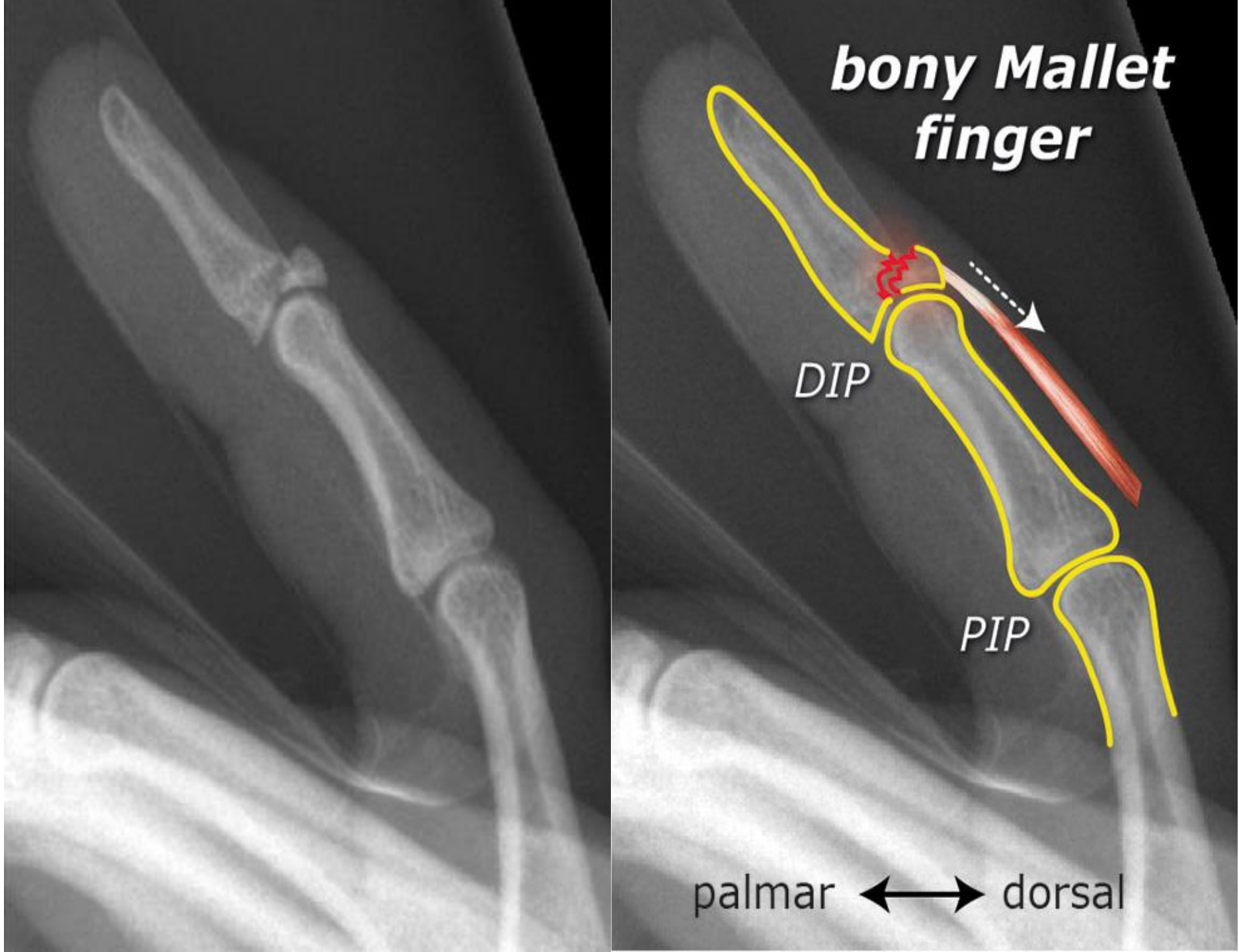
Mallet finger

- A **Mallet finger** involves an avulsion of the extensor tendon on the distal phalanx. The tendon rupture prevents active extension, possibly causing the distal phalanx to assume a position of flexion.
- This may be a purely **ligamentous avulsion** (**tendinous Mallet**), which may or may not be combined with an **ossal avulsion** (**ossal Mallet**).
- Note: the absence of a fracture therefore does not exclude a Mallet finger. The extension function of the DIP joint requires clinical evaluation in this setting.

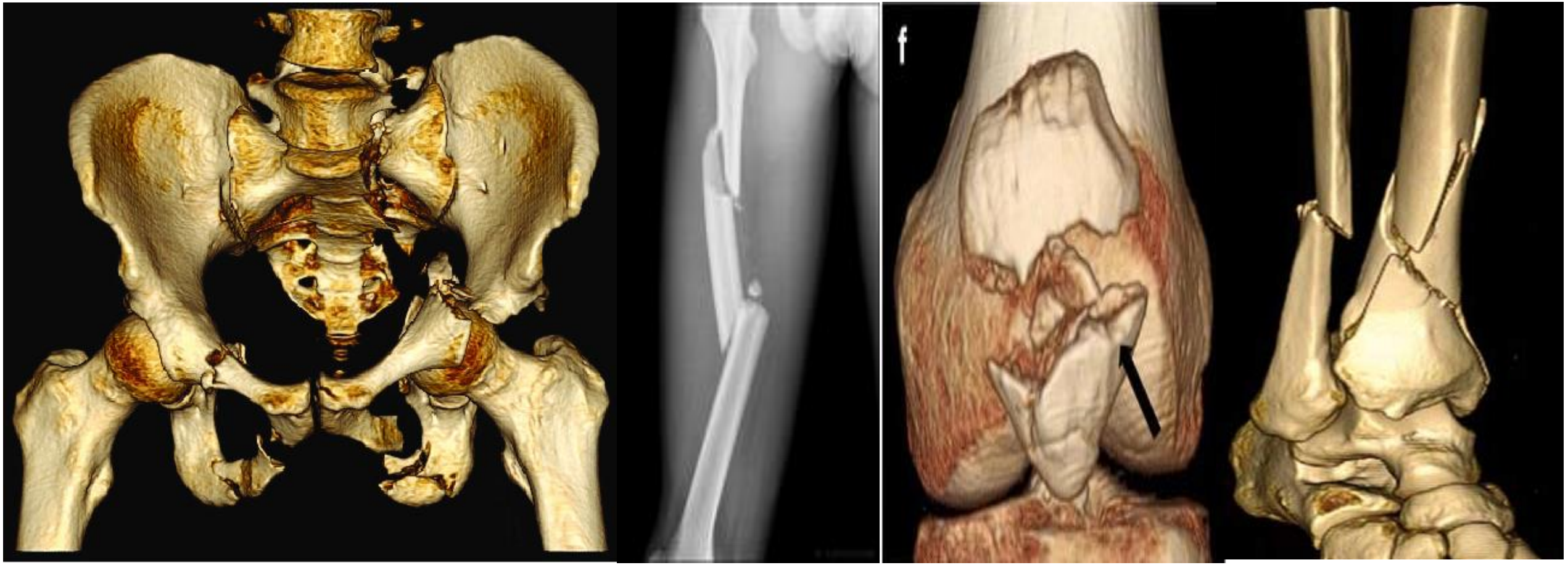
Lateral image of dig-V of the right hand. The position of flexion of the DIP joint and the inability to induce extension are signs of a tendinous Mallet finger.



Lateral image of dig-V of the right hand. Ossal Mallet finger with dislocation to dorsal.



Thank
you



Lower Limb Fractures

Lecturer: Husam Mansour
Master of Public Health
Master of Medical Imaging

Part I

Pelvic fractures

- Pelvic fractures can be simple or complex and can involve any part of the bony pelvis.
- Pelvic fractures can be fatal due to pelvic hemorrhage, and an unstable pelvis requires immediate management.
- **Classification:**

1. Low- energy blunt force trauma resulting in stable fractures.

A. Acetabular fracture.

B. Pubic ramus fracture.

C. Iliac wing fracture.

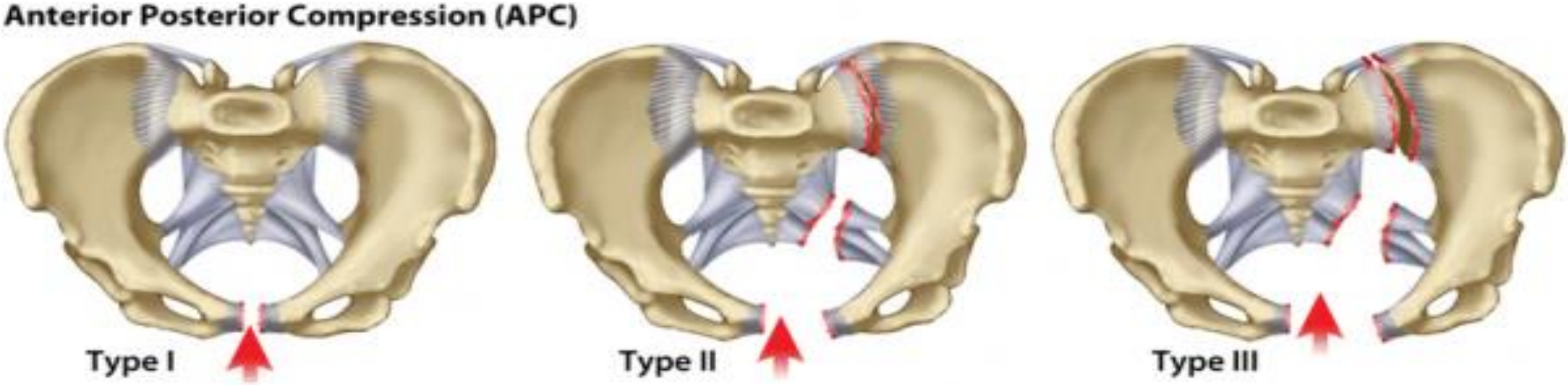
D. Avulsion fractures.

- **High energy blunt force trauma resulting in unstable fractures.**
- **Anteroposterior compression:** result in **open book** or **sprung pelvis fractures.**
- **Lateral compression:** result in a **windswept pelvis**.
- **Vertical shear:** result in **malgaigne fracture** or **bucket handle fracture.**
- **Combined mechanical:** two different forces resulting in a **complex fracture** pattern.

1. Anteroposterior compression:

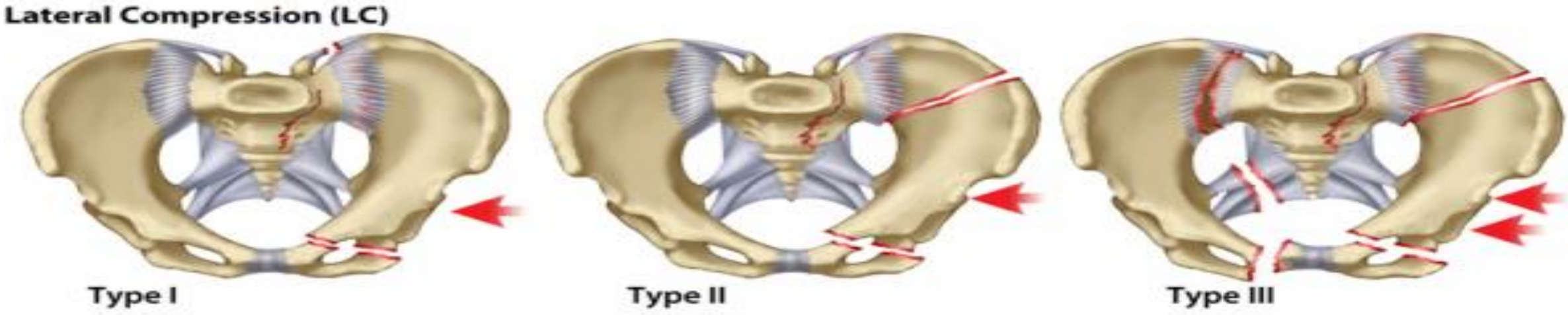
- Produce external rotation of one or both hemi-pelvises, with the rotation at the SI joints leading to an "open-book" or "spring pelvis" type of injury.
- External rotation of the hemipelvis causes the iliac wing to appear broader than normal on AP radiographs.
- Substantial hemorrhage and arterial bleeding are most prevalent with AP compression injuries.
- The arteries that are most frequently severed are the superior gluteal (in posterior pelvic injuries) and the internal pudendal (in anterior pelvic arch injuries) arteries.

- **APC I:** Pubic symphyseal diastasis, <2.5 cm, no significant posterior ring injury (stable)
- **APC II:** Pubic symphyseal diastasis >2.5 cm, tearing of anterior sacral ligaments (rotationally unstable, vertically stable)
- **APC III:** Hemipelvis separation with complete disruption of pubic symphysis and posterior ligament complexes (completely unstable)



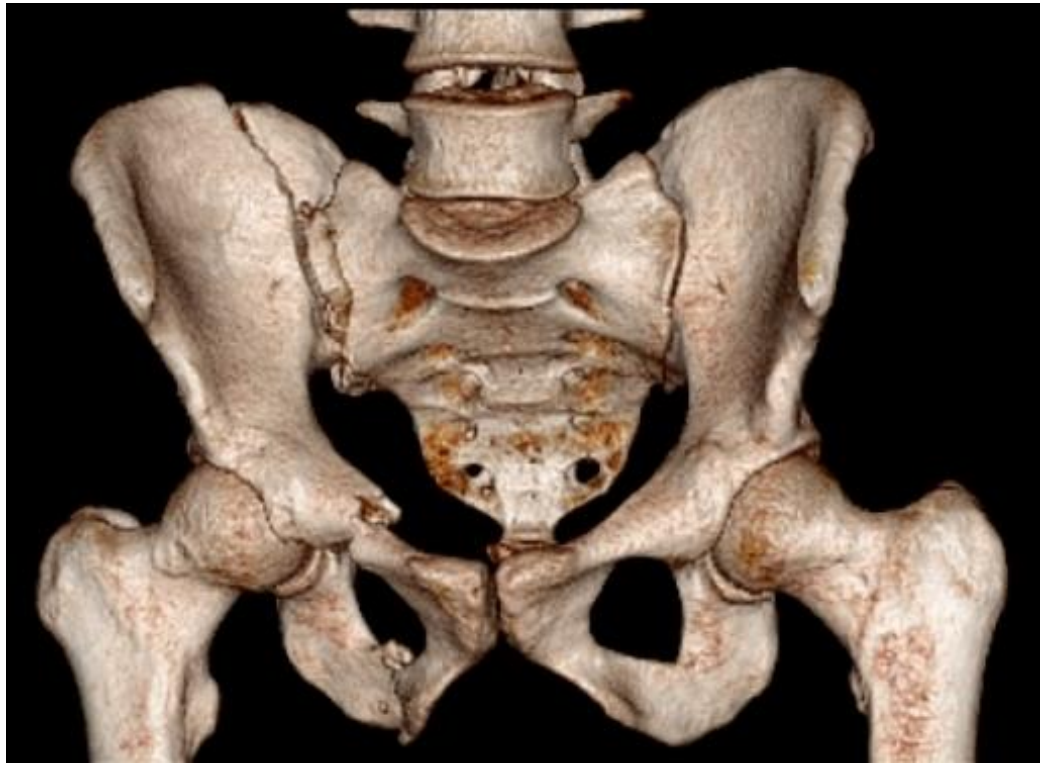
2. Lateral compression:

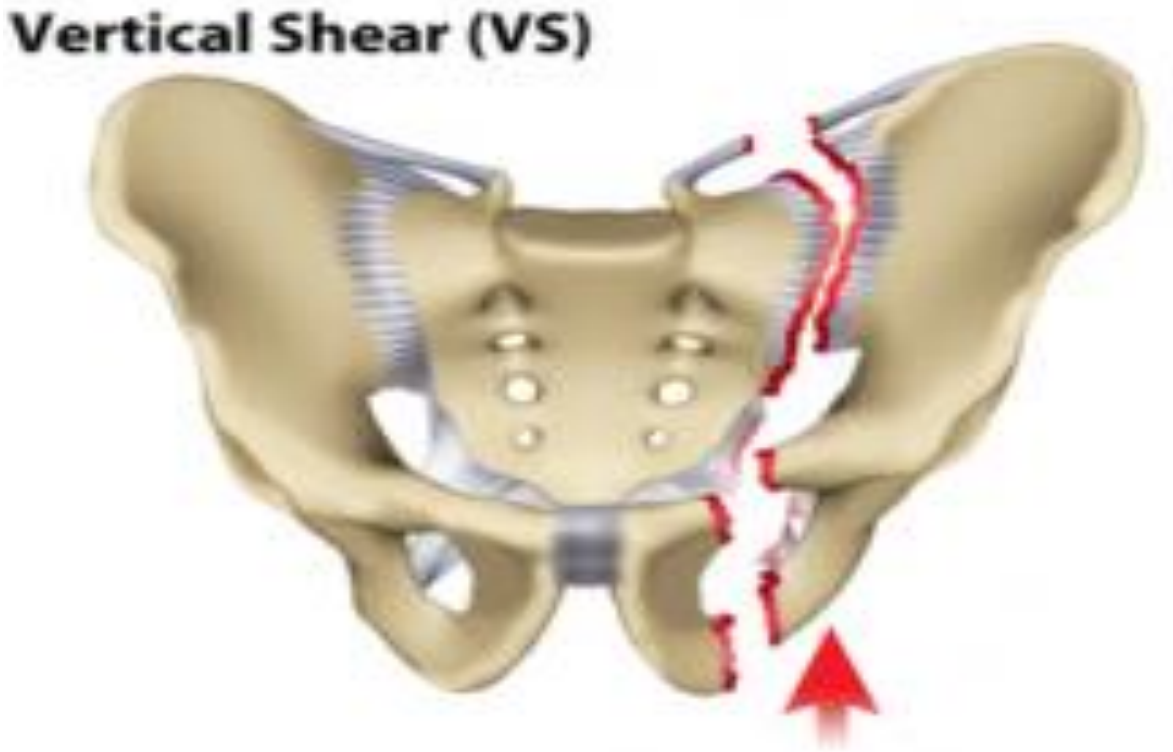
- Common feature is a transverse fracture of the pubic rami
- **LC I:** Posterior compression of sacroiliac (SI) joint without ligament disruption (stable)
- **LC II:** Posterior SI ligament rupture, sacral crush injury or iliac wing fracture (rotationally unstable, vertically stable)
- **LC III:** LC II, with open book (APC) injury to contralateral pelvis (completely unstable)



3. Vertical shear:

- Unstable ipsilateral anterior and posterior fractures of the pelvic ring, with resultant superior displacement of one hemipelvis.
- Also known as the Malgaigne fracture.





Avulsion Iliac Apophysis

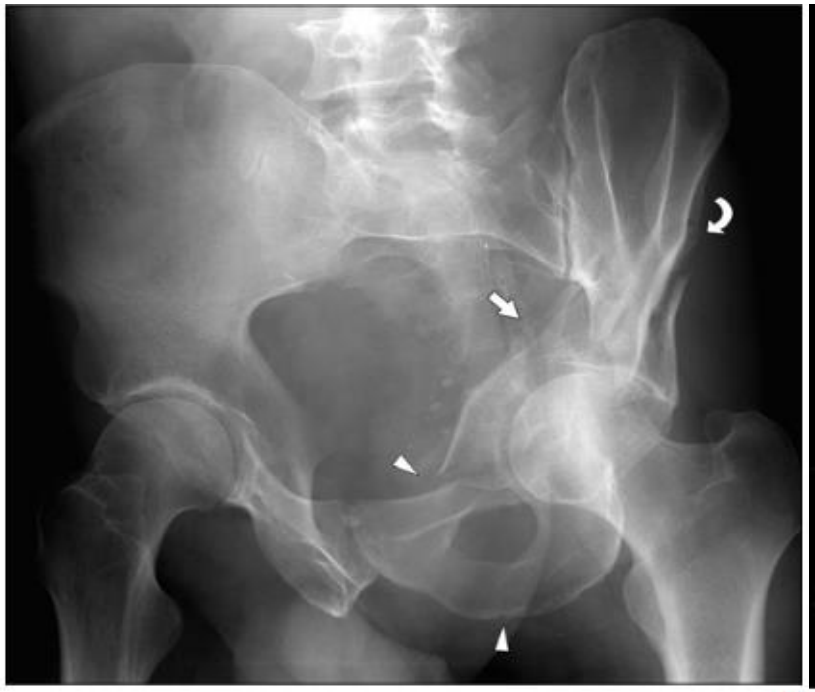
Avulsion Anterior Superior Iliac Spine (ASIS)

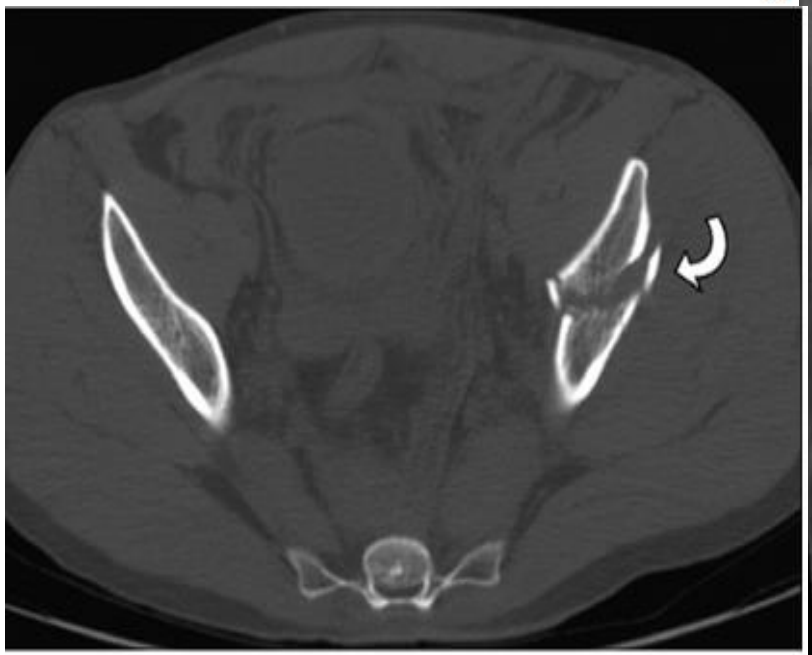
Avulsion Anterior Inferior Iliac Spine (AIIS)

Avulsion Ischial Tuberosity

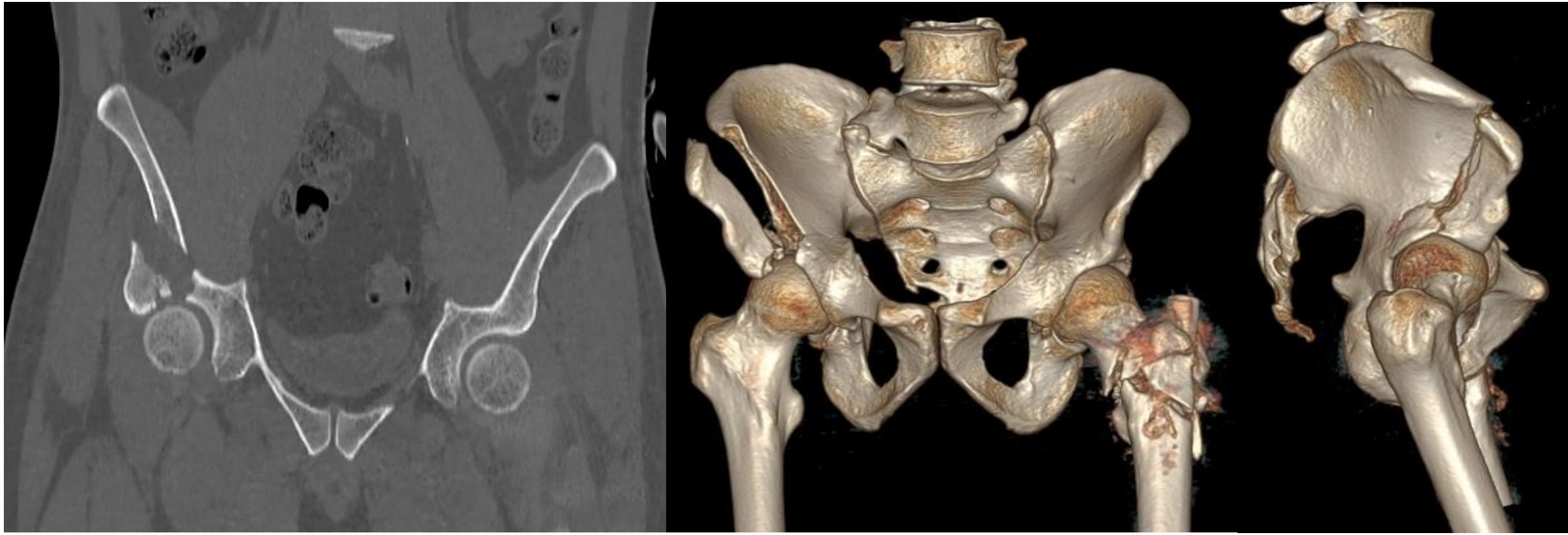
Avulsion Greater Trochanter

Avulsion Lesser Trochanter

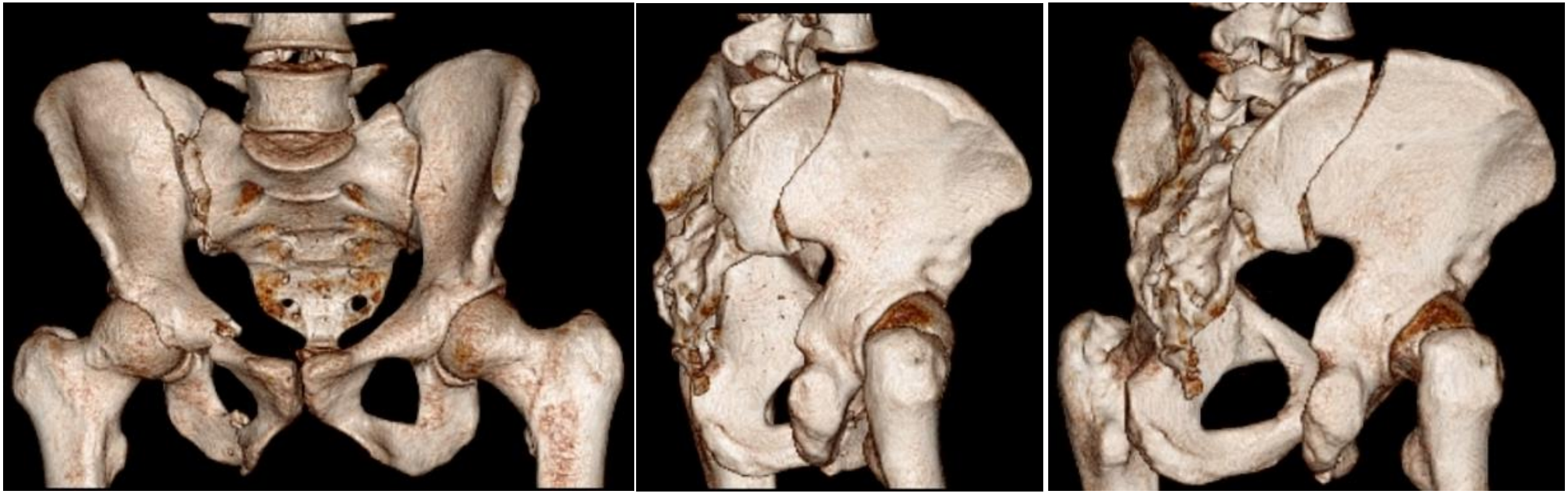




45-year-old man with both-column acetabular fracture. A–E, Anteroposterior pelvic radiograph (A), bilateral oblique pelvic radiographs (B, C), axial CT scan (D), and sagittal reconstruction CT scan (E) show acetabular fracture (straight arrows, A–C), with break in obturator ring (arrowheads, A–C) and extension into iliac wing (curved arrows). Note coronal plane of fracture on CT and superior pubic ramus fractured at puboacetabular junction.



10 cm long displaced fracture of right ilium along long axis of bone with a fracture gap exceeding 17 mm. There is caudal extension to roof of acetabulum which is disintegrated into a minimum of three parts. Non-displaced small fracture of inferior lip of acetabulum. Mild cranial migration of right femur with no femur fracture. Comminuted fracture of left femur neck, post fixation. Rest of the pelvic bones are normal.

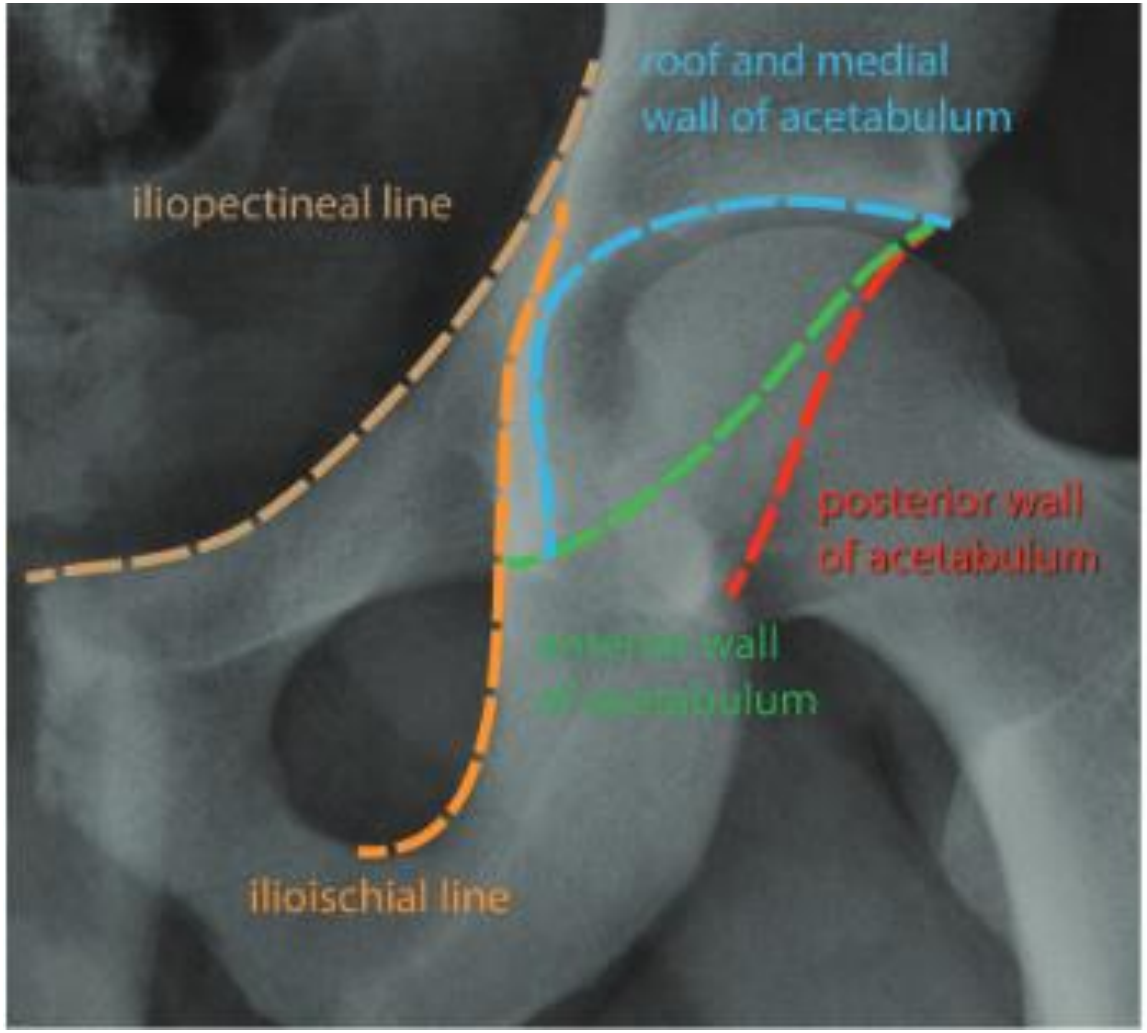


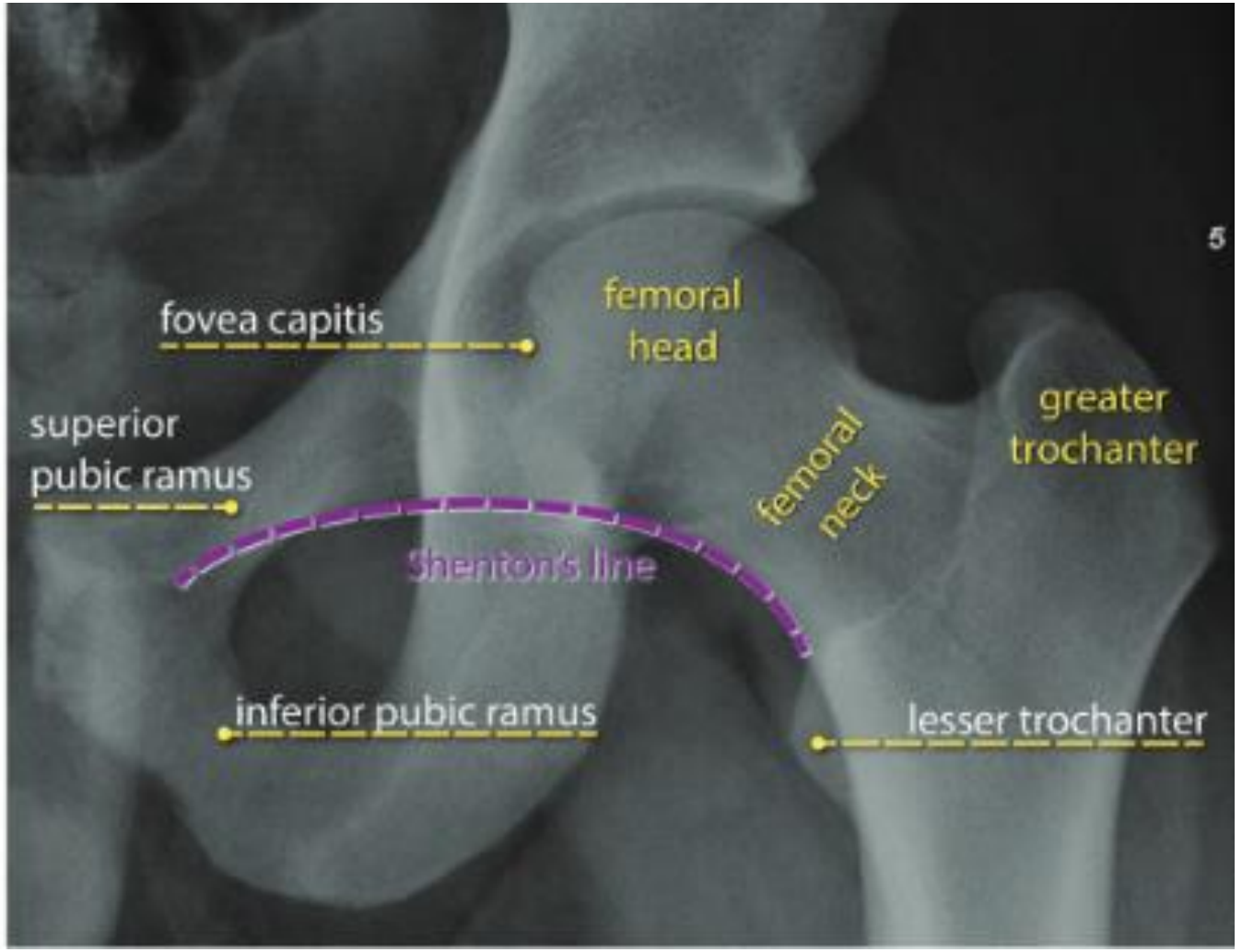
Vertical fractures involving the right side of pelvic ring. Above the acetabulum, the oblique fracture in right iliac bone involving its posterior third and start from the iliac crest and exit into the lower one-third of ipsilateral SI joint with involvement of articular surface without its significant diastasis. Below the acetabulum, there are comminuted fractures of ipsilateral superior and inferior pubic rami; however, there is no evidence of fracture of acetabular walls. There is fracture of ipsilateral ischial spine as well. Contralateral hemipelvis is intact without fracture.

Hip joint dislocation

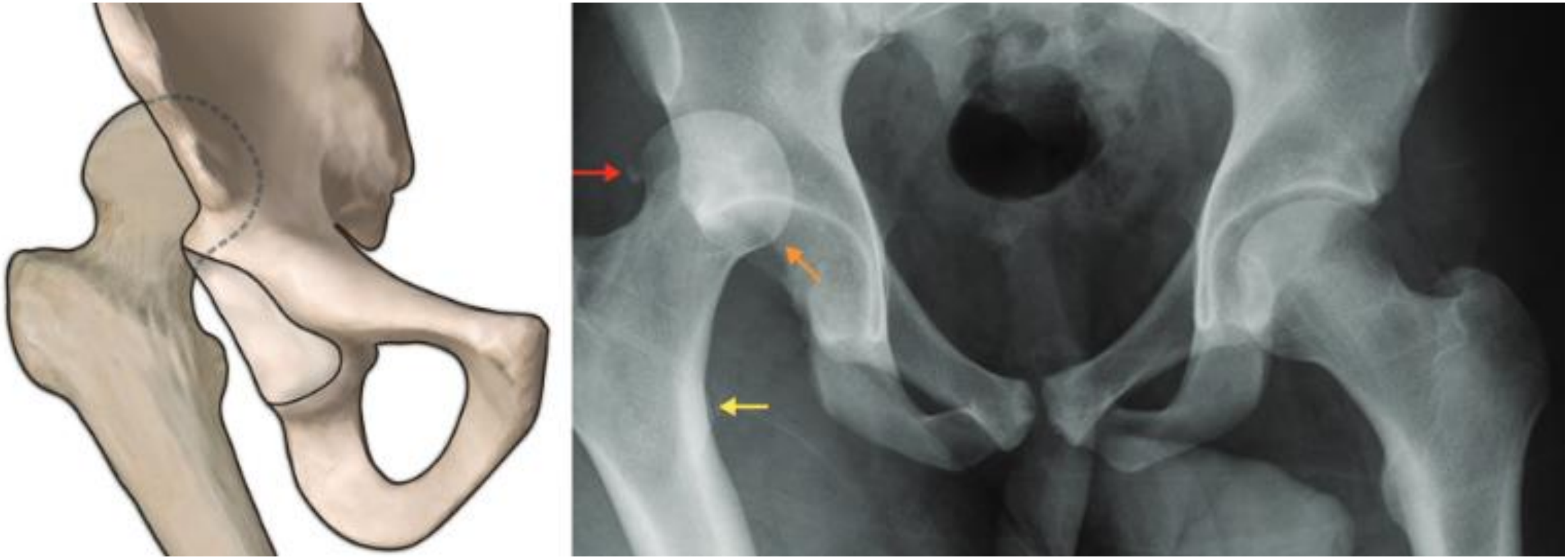
- Hip dislocation may be congenital or acquired.
- Congenital hip dislocation is now considered part of the spectrum of developmental dysplasia of the hip.
- Acquired hip dislocation (Traumatic Hip Dislocation) is normally associated with high-speed trauma.
- **Posterior dislocations** of the hip are the most common outnumbering anterior dislocations 9:1.
- Posterior dislocation may have associated fractures of the posterior lip of the acetabulum and/or injury to the acetabular labrum.
- AP pelvic x-rays will usually be sufficient for the diagnosis, although associated acetabular fractures will require CT to fully characterize.

- In a posterior dislocation, the femoral head is usually displaced posterior, superior, and slightly lateral to the acetabulum and also internally rotated hence the lesser trochanter is often obscured on the AP view.
- In AP the posteriorly dislocated femoral head will appear smaller than the contralateral hip, and vice versa.
- Radiographic signs of anterior hip dislocation are the lesser trochanter being more visible due to external rotation.
- The hip is abducted, and the femur head is usually inferior to the acetabulum. Shenton's line is also broken (is an imaginary curved line drawn along the inferior border of the superior pubic ramus (superior border of the obturator foramen) and along the inferomedial border of the neck of femur).

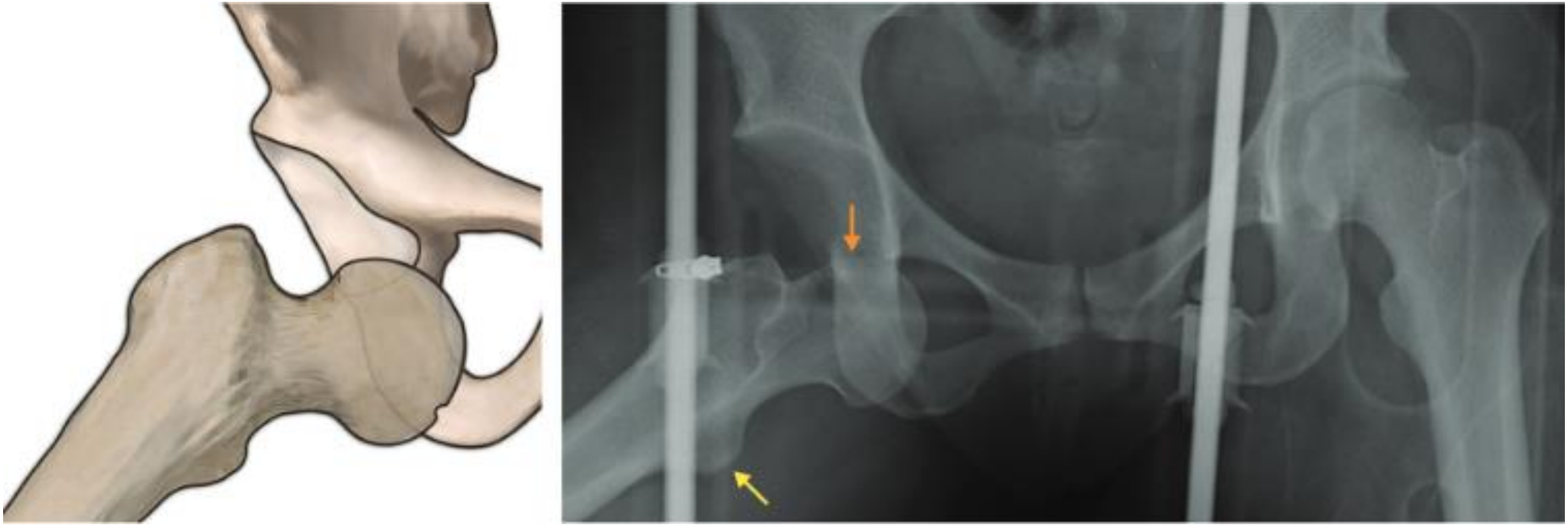




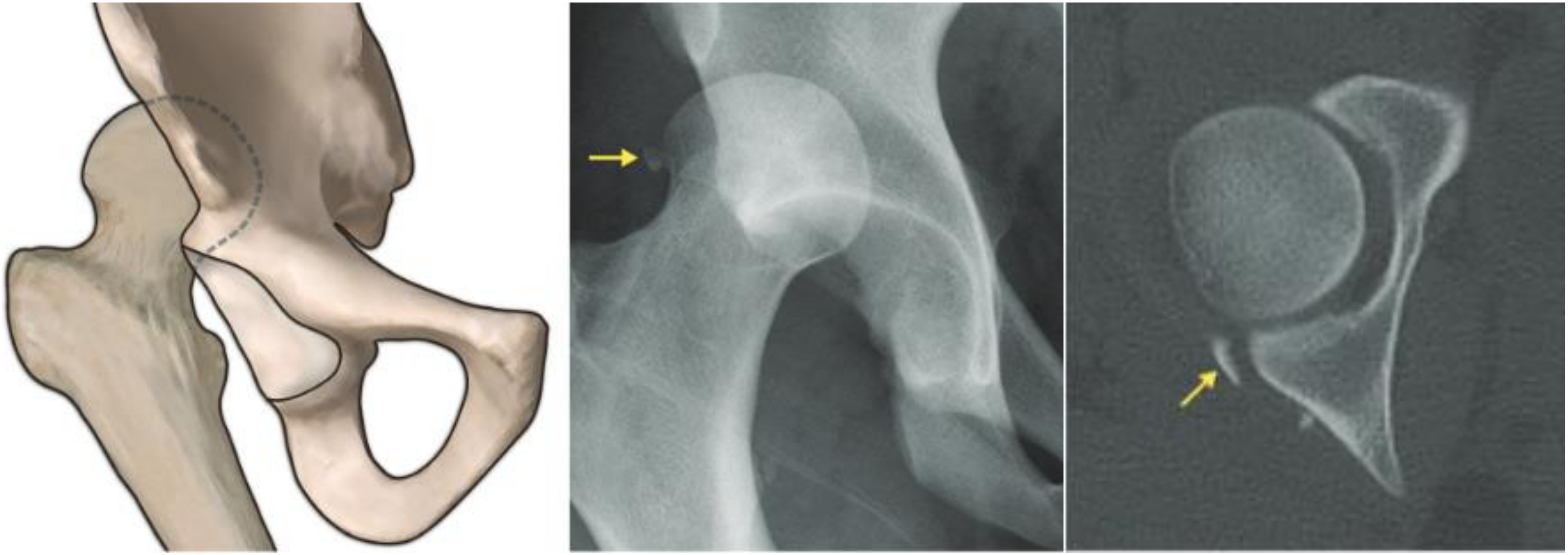
Annotated radiographs show the normal anatomy of the hip. (a) The iliopectineal line comprises the anterior column of the acetabulum, and the ilio-ischial line is the posterior column. The anterior and posterior walls of the acetabulum can be visualized through the femoral head. The medial wall of the acetabulum is contiguous with the roof. (b) Of particular importance to hip dislocation is the fovea capitis, which is an important landmark to classify femoral head fractures



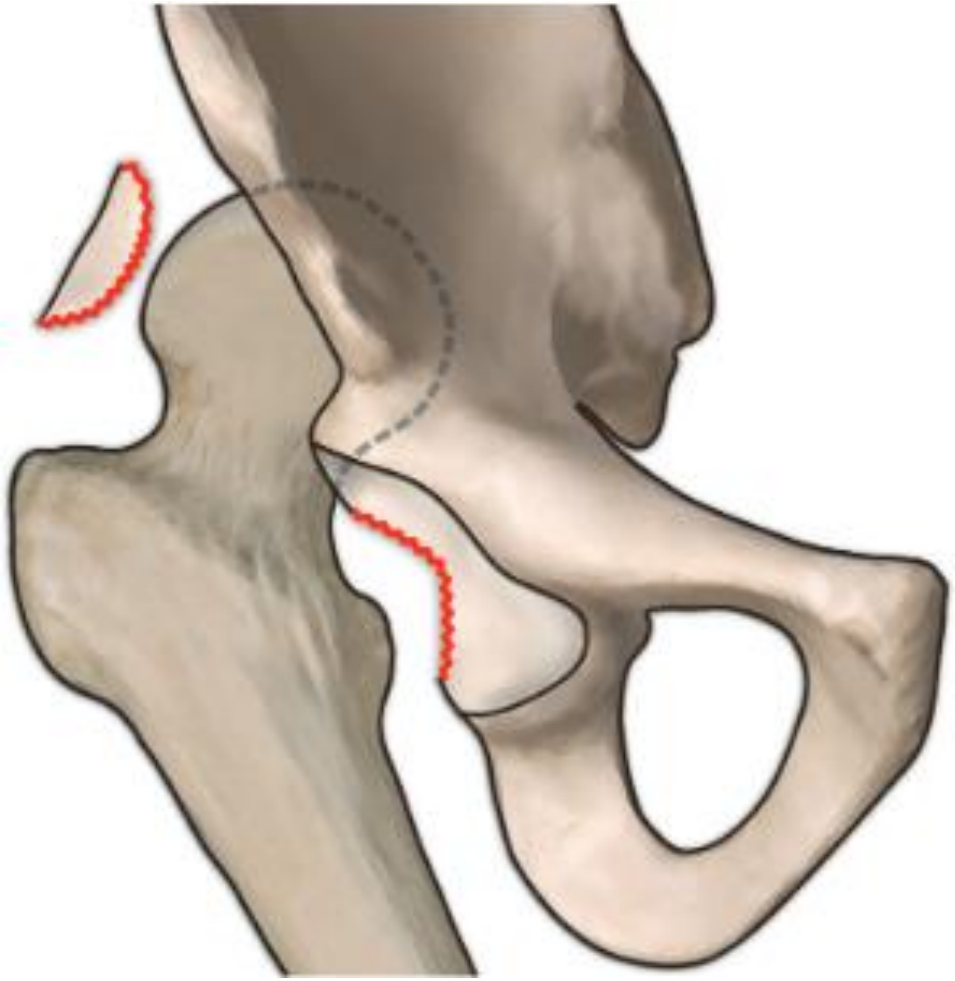
(a) Drawing demonstrates the typical position of the femur in a posterior dislocation. (b) Posterior hip dislocation in a 19-year-old man with a basketball injury who struck the foot of another player with the knee in extension. Frontal radiograph of the pelvis shows that the femoral head (orange arrow) is typically positioned superolateral to the acetabulum, with the hip in flexion, internal rotation, and adduction. Internal rotation of the femur causes decreased conspicuity of the lesser trochanter (yellow arrow). The dislocated femoral head may appear smaller than the normally positioned femoral head owing to AP radiographic technique and magnification effect. Note the tiny posterior acetabular wall fracture (red arrow) in this case.

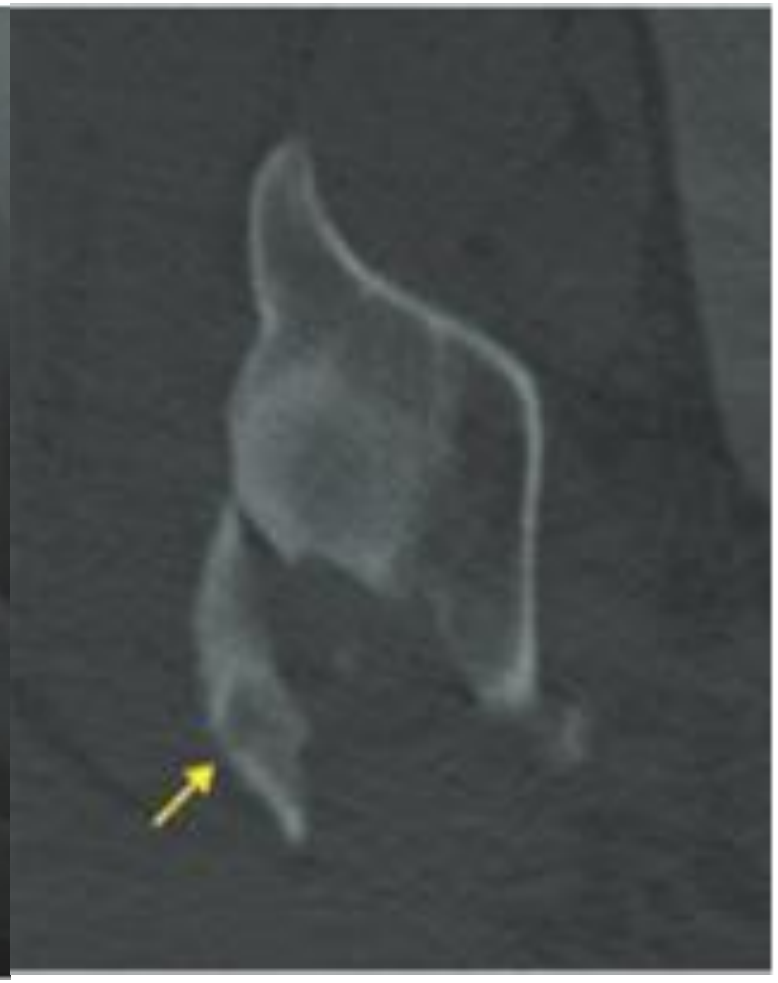


(a) Drawing demonstrates the typical position of the femur in an anterior dislocation. (b) Anterior dislocation in a 25-year-old woman who was on a sled that crashed into a tree. Frontal radiograph of the pelvis shows the femoral head (orange arrow) typically positioned inferomedial to the acetabulum, with the hip in marked external rotation, with milder flexion and abduction. Anterior superior dislocation of the hip is rare in comparison. External rotation of the femur causes increased conspicuity of the lesser trochanter (yellow arrow). The dislocated femoral head may appear larger than the normally positioned femoral head owing to AP technique and magnification effect.

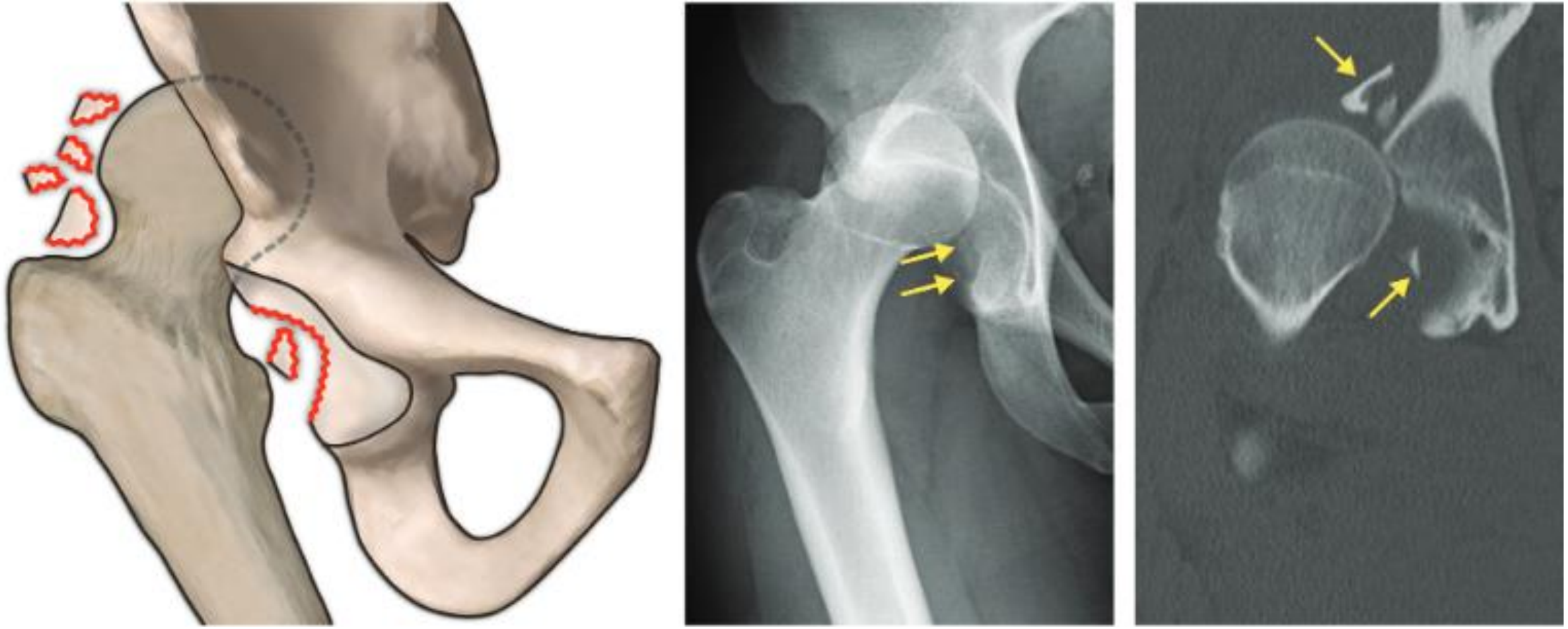


a) Drawing shows the typical appearance of posterior hip dislocation without acetabular fracture. (b, c) Posterior dislocation with insignificant acetabular fracture in a 19-year-old man with a basketball injury who struck the foot of another player with the knee in extension. Frontal radiograph of the right hip (b) demonstrates a posterior dislocation. There is a tiny posterior acetabular fracture (arrow) seen on the radiograph and on the subsequent post reduction CT image (c), which does not require fixation and is considered an insignificant acetabular injury.

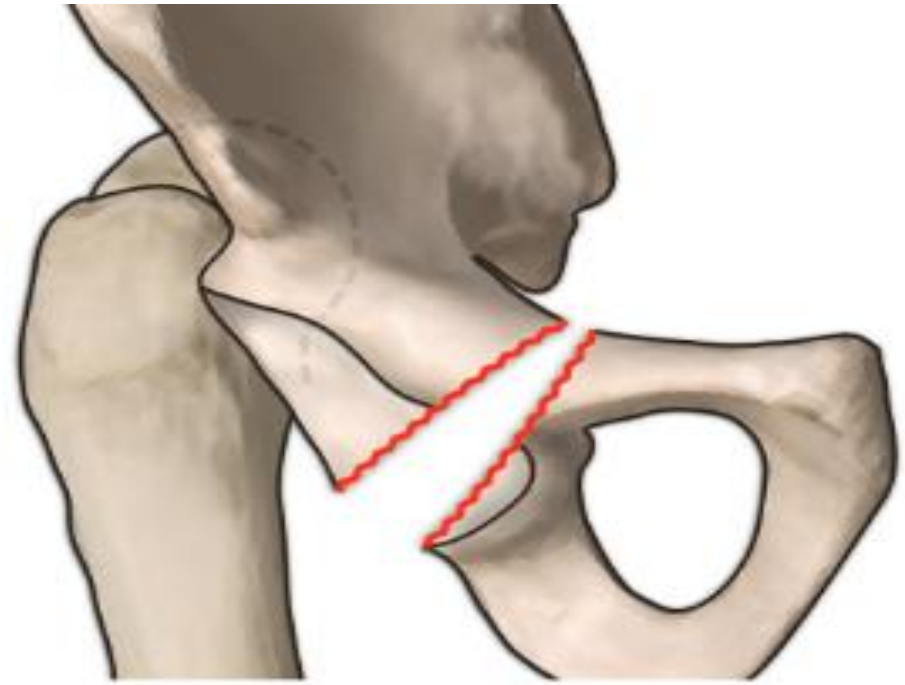




(a) Drawing shows a posterior dislocation and a single dominant posterior acetabular wall fracture fragment requiring fixation. (b, c) Initial pelvic radiograph (b) and post reduction axial CT image (c) in a 27-year-old man in a car-versus-tree motor vehicle crash show a posterior dislocation with a dominant posterior acetabular fracture fragment (arrow).

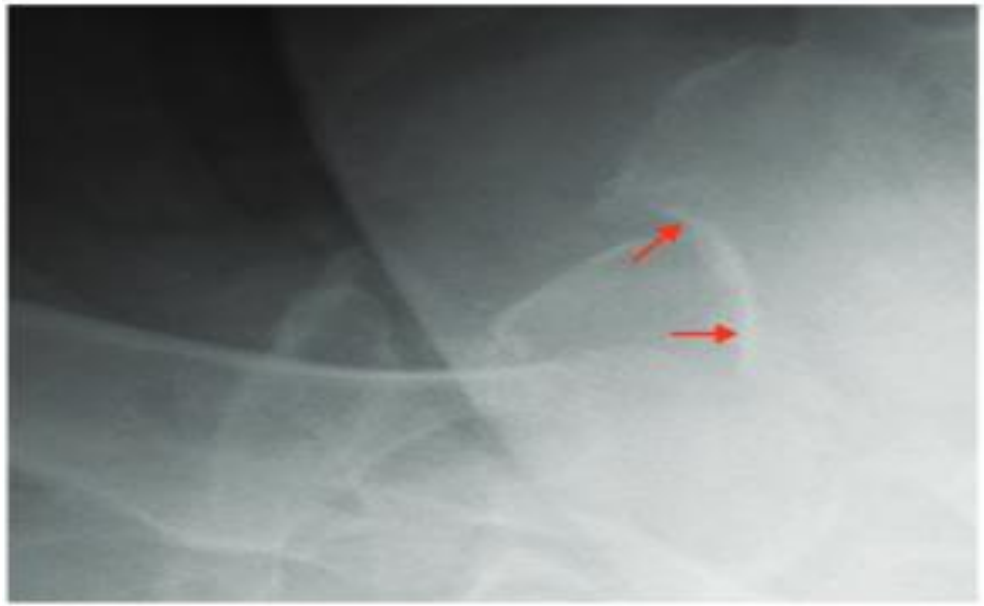


(a) Drawing shows a posterior dislocation and a comminuted posterior acetabular wall fracture. (b, c) Posterior dislocation and fracture of the left hip in a 24-year-old man who was tackled while playing football. (b) Initial pelvic radiograph demonstrates a cortical irregularity (arrows) of the acetabulum. (c) Coronal CT image after attempted reduction shows a comminuted posterior acetabular wall fracture with fracture fragments (arrows) superior to and within the joint.

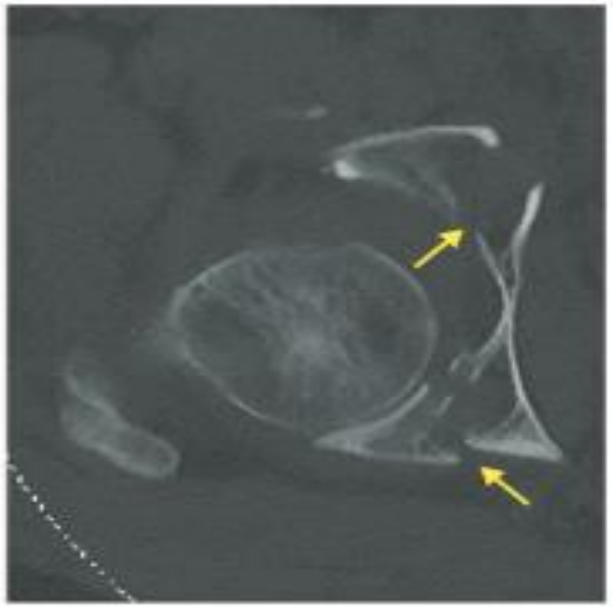


a.

b.



c.

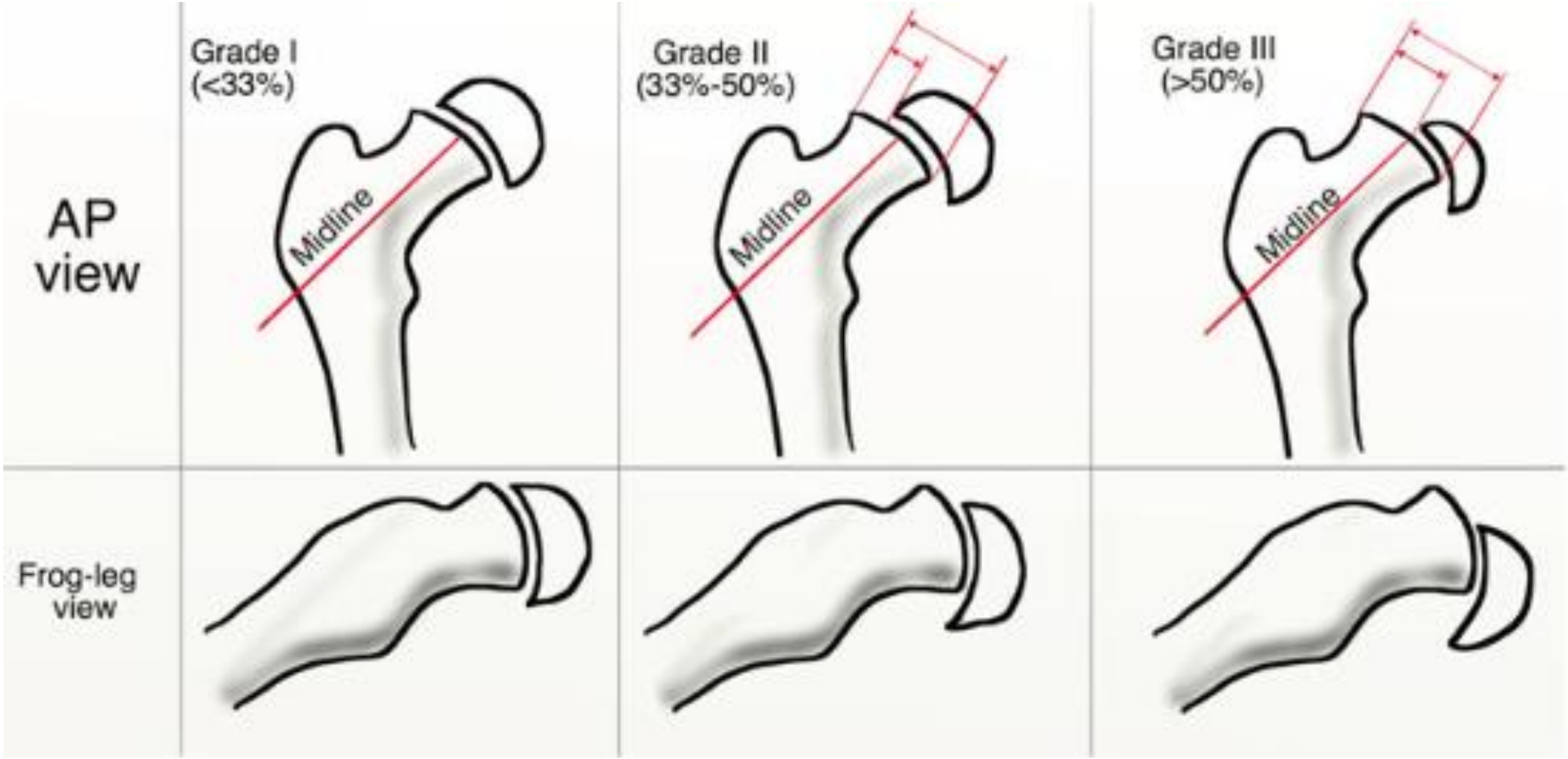


d.

(a) Drawing shows a posterior dislocation and a medial acetabular wall and floor fracture. (b) Initial pelvic radiograph in a 59-year-old man who tripped on wires and fell from standing, landing on his right side, demonstrates a fracture through the entire medial acetabular wall (arrow). (c) Posterior dislocation is more clearly evident on a lateral radiograph, which demonstrates an empty acetabular fossa (arrows) and posterior dislocation of the femoral head. (d) Axial post reduction CT image demonstrates comminution of the medial acetabulum (arrows).

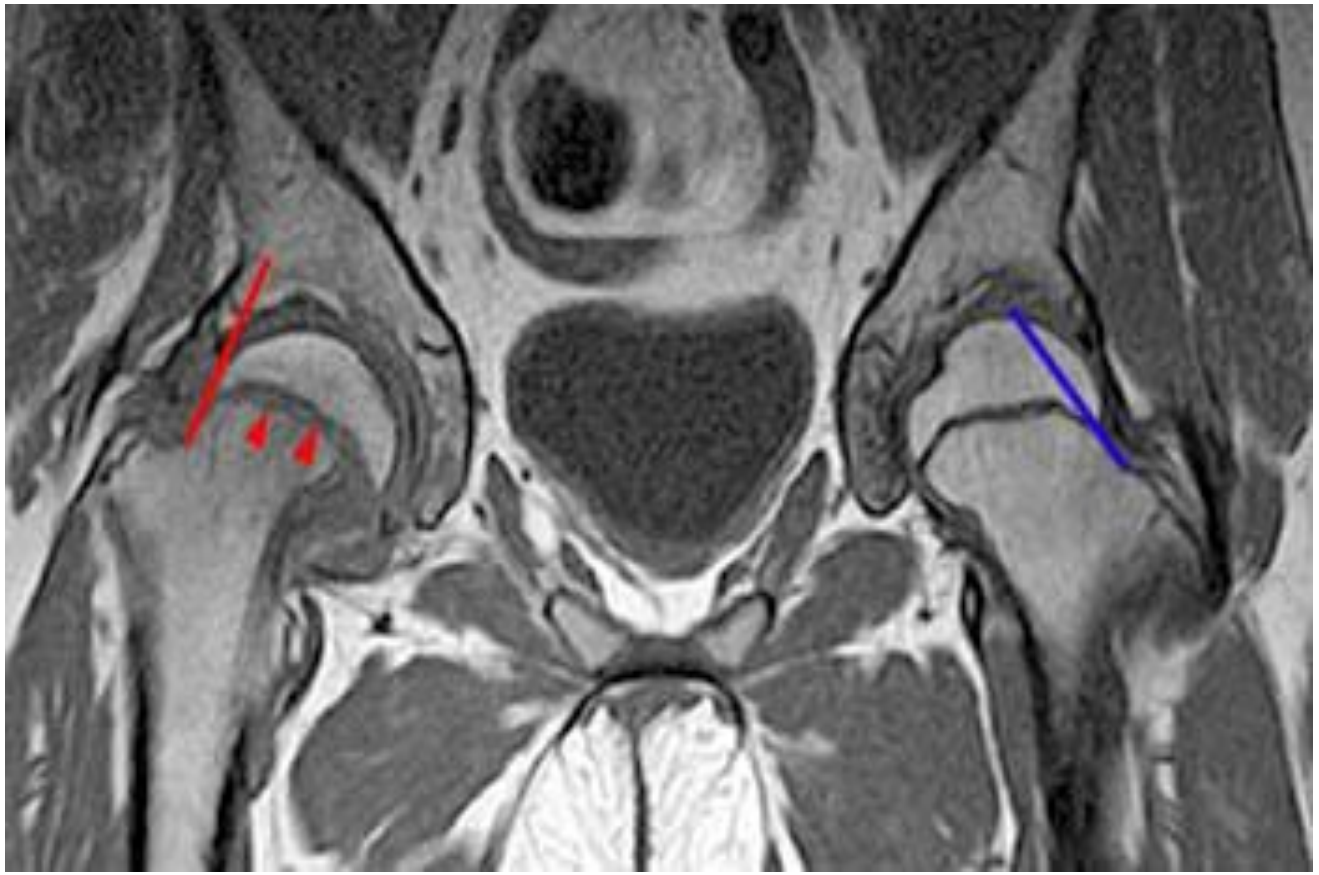
Slipped Upper Femoral Epiphysis SUFE

- Slipped upper femoral epiphysis (SUFE), also known as a slipped capital femoral epiphysis (SCFE), (plural: epiphyses) is a relatively common condition affecting the epiphysis of the proximal femur in adolescents.
- It is one of the commonest hip abnormalities in adolescence and is bilateral in ~20% of cases.
- Slipped upper femoral epiphysis is more common in boys than girls
- The age of presentation is somewhat dependent on gender with boys presenting later (10-17 years) than girls (8-15 years)

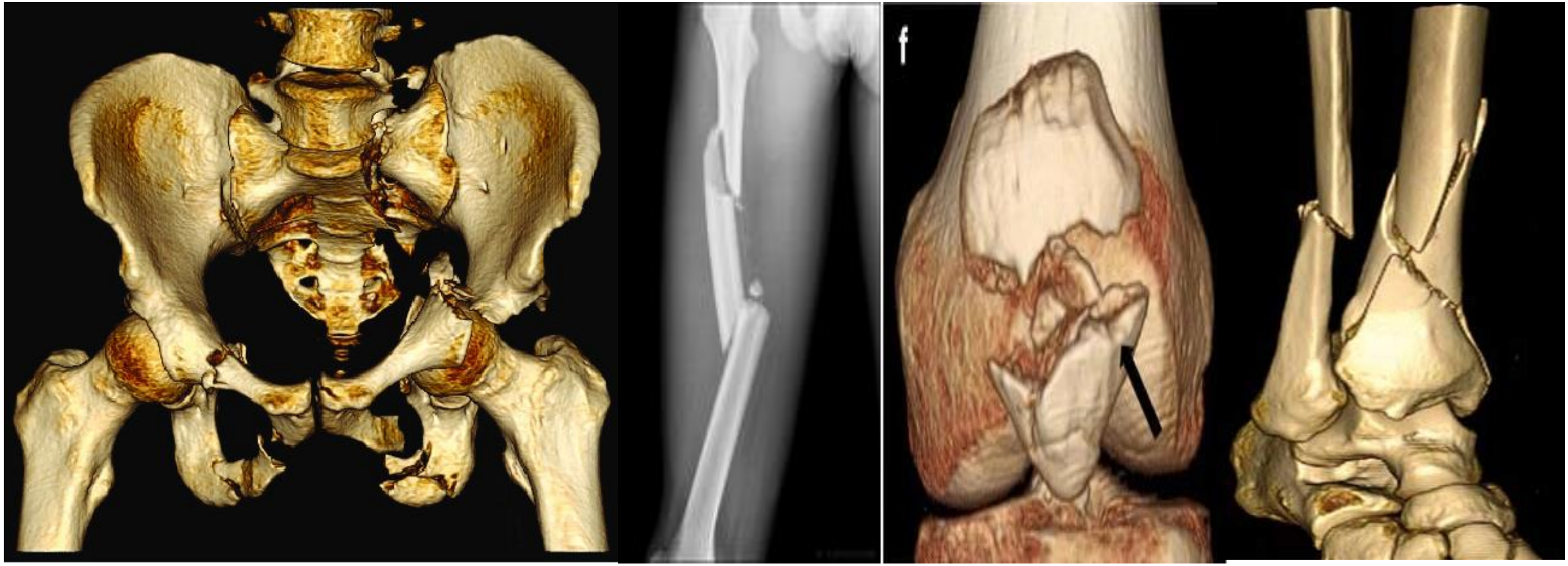


The three types of the slipped capital femoral epiphysis based on radiographic findings: type I displacement <33 %; type II displacement 33–50 %; displacement >50 %

A pelvis x-ray with left slipped upper femoral epiphysis. The x-ray shows a severe slipped upper epiphysis



The coronal T1-weighted image shows widening of the right femoral physis (arrowheads) and a medial slip of the femoral head on the right.



Lower Limb Fractures

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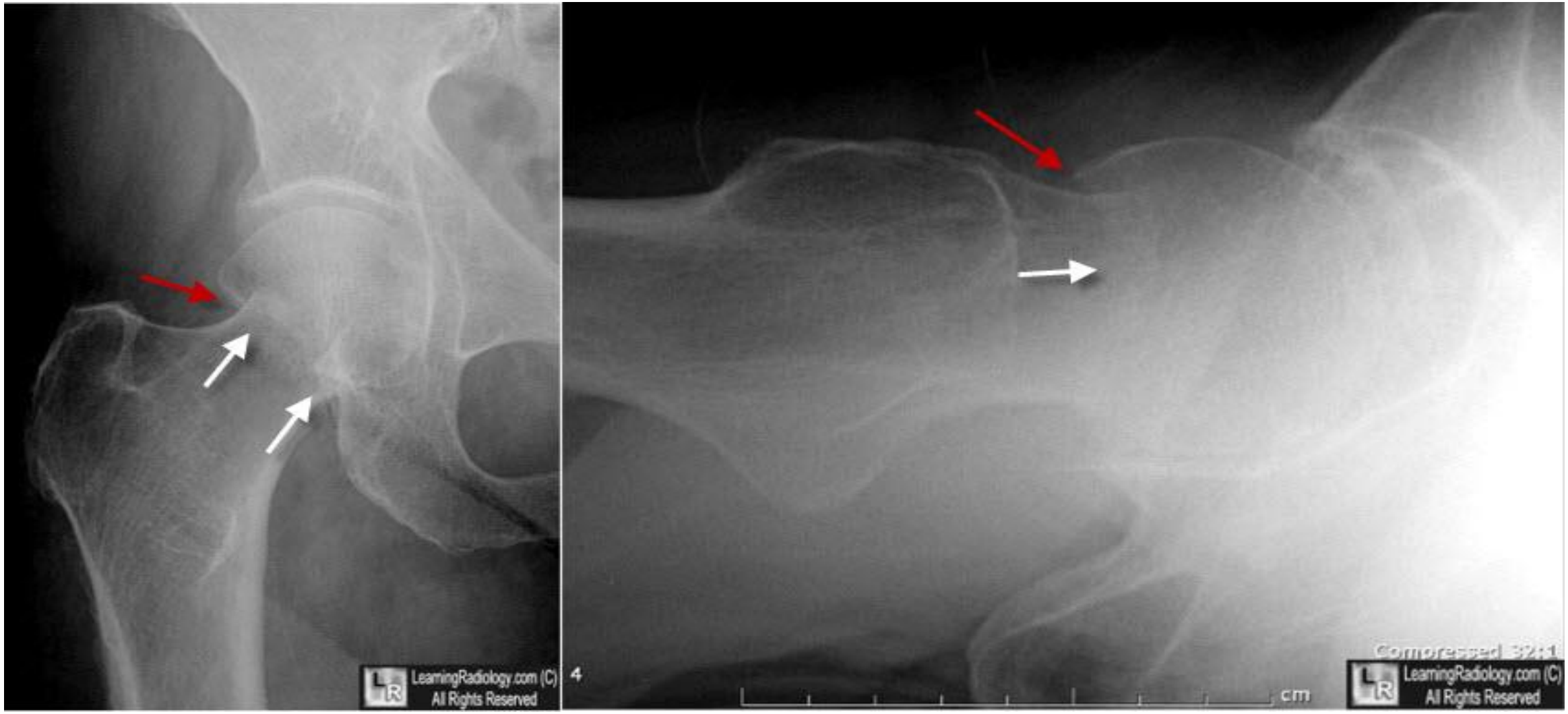
Part 2

Proximal femoral fracture

- Proximal femoral fractures are a subset of fractures that occur in the hip region.
- Proximal femoral fractures are therefore divided into groups based on their location with regard to the capsule, i.e. whether they are intracapsular or extracapsular.
- **Intracapsular fractures**
- Intracapsular fractures are important because of their propensity to damage the small intracapsular vessels that provide the majority of the blood supply to the femoral head.
- Neck of femur (NOF) fractures are common injuries sustained by older patients who are more likely to have reduced bone mineral density.

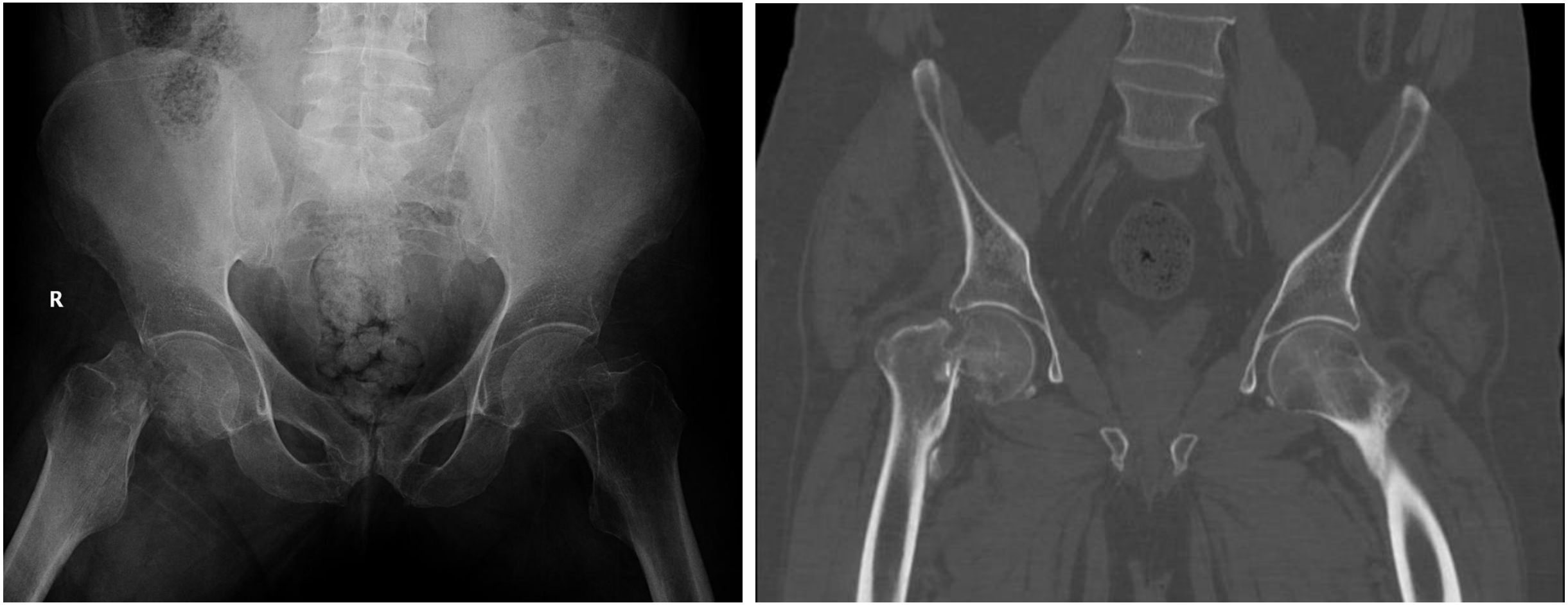
- There are three types:
- **Subcapital:** femoral head/neck junction
- **Transcervical:** midportion of femoral neck
- **Basicervical:** base of femoral neck
- **Extracapsular fractures**
- Fractures outside the capsule do not cause the same degree of vascular damage as intracapsular fractures and therefore can be treated differently.
- The trochanteric fractures are extracapsular injuries.



Sub-capital hip fracture. On the frontal view, there is a step-off in the cortex superiorly (red arrow) while there is abnormal overlapping of the femoral head and neck (white arrows) due to impaction. On the lateral view, the same step-off can be seen (red arrow) as well as the impaction (white arrow).

Base-of-right-femoral-neck-fracture

Displaced-right-femoral-neck-fracture



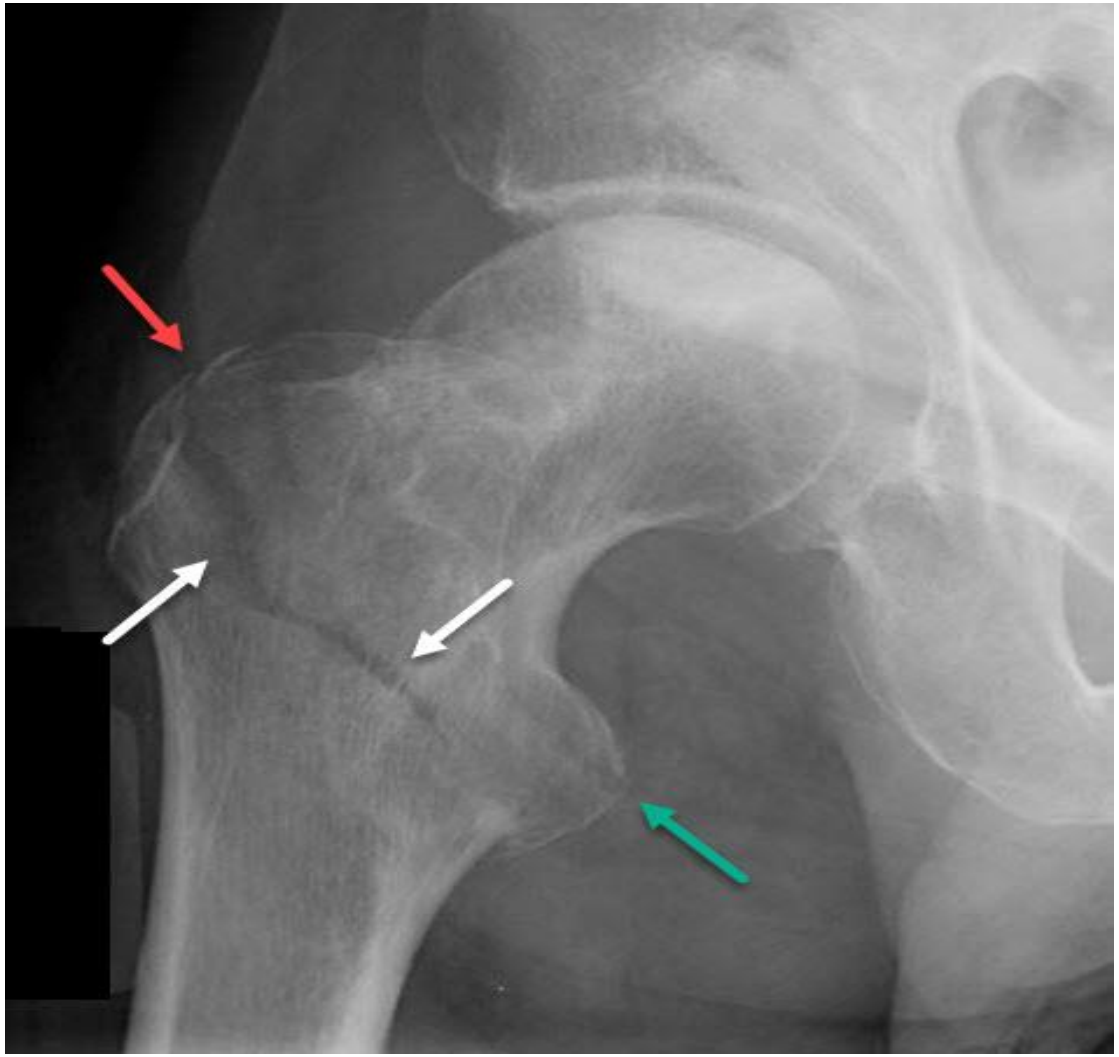
Subcapital femural neck fracture.

- **Intertrochanteric fractures**

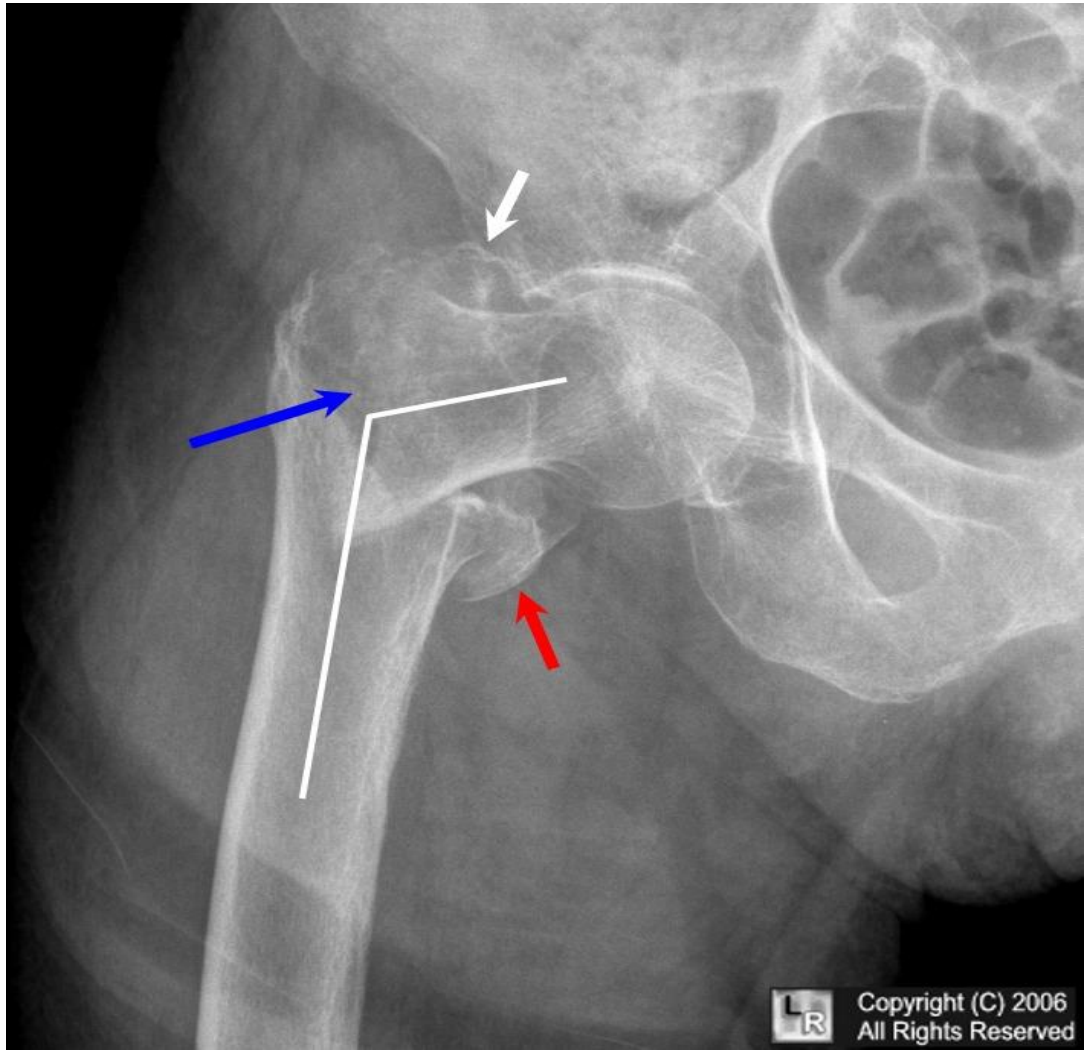
- Most common of the extracapsular hip fractures

- Comminuted fractures may also manifest separate fragments of either or both of the lesser and greater trochanters

- Intertrochanteric fractures associated with a separate fragment of lesser trochanter may also include a portion of the posterior cortex of femoral neck and are considered unstable



Intertrochanteric fracture. There is a fracture from the greater (red arrow) to the lesser (green arrow) trochanter. A lucency, representing the fracture, runs between the two trochanters (white arrows).



Comminuted intertrochanteric fracture. There is a fracture from the greater to the lesser trochanter (blue arrow). There are separate fragments of the greater trochanter (white arrow) and lesser trochanter (red arrow). There is varus deformity (white line) of the femoral shaft.

Subtrochanteric fracture

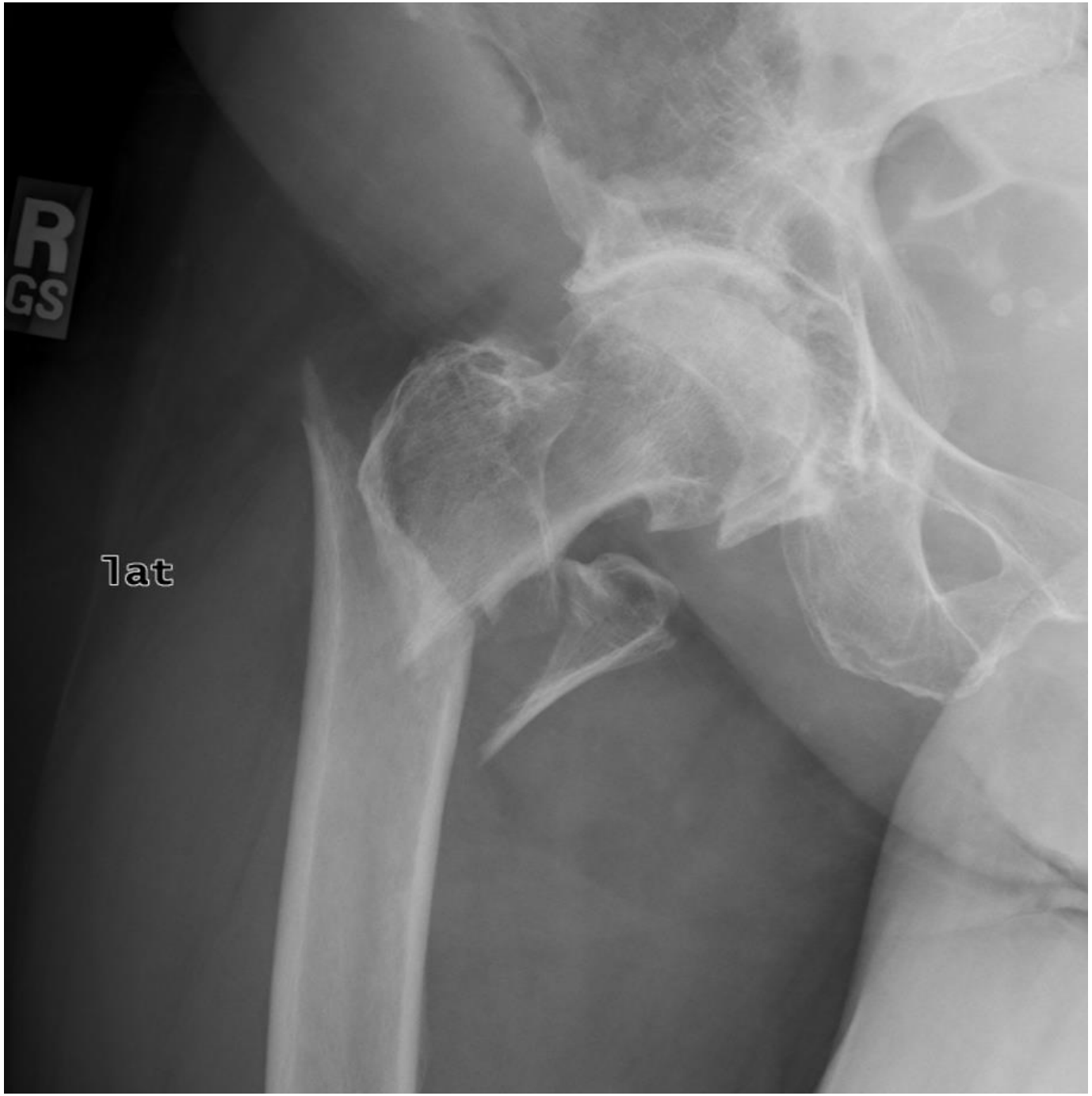
- These fractures occur at the level distal to the lesser trochanter.
- **Proximal fragment**

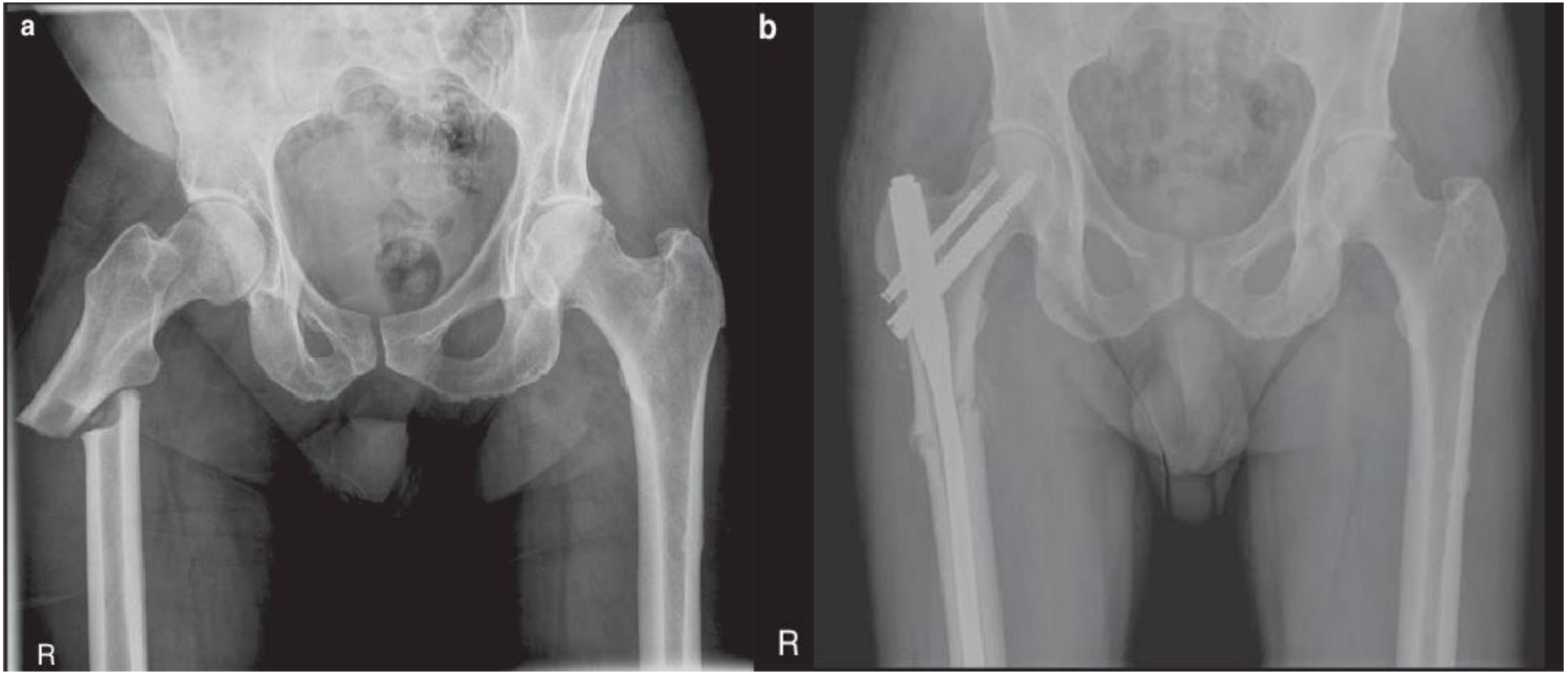
flexed and abducted

- **Distal fragment**

adducted and shortened

- Oblique view of the right pelvis demonstrates a comminuted subtrochanteric fracture (inferior to the greater and lesser trochanters).
- The proximal fragment appears abducted.
- The lesser trochanter is itself fractured and separated from the femur.





(a) Radiograph of subtrochanteric fracture of the right femur.
(b) Radiograph after intramedullary fixation.

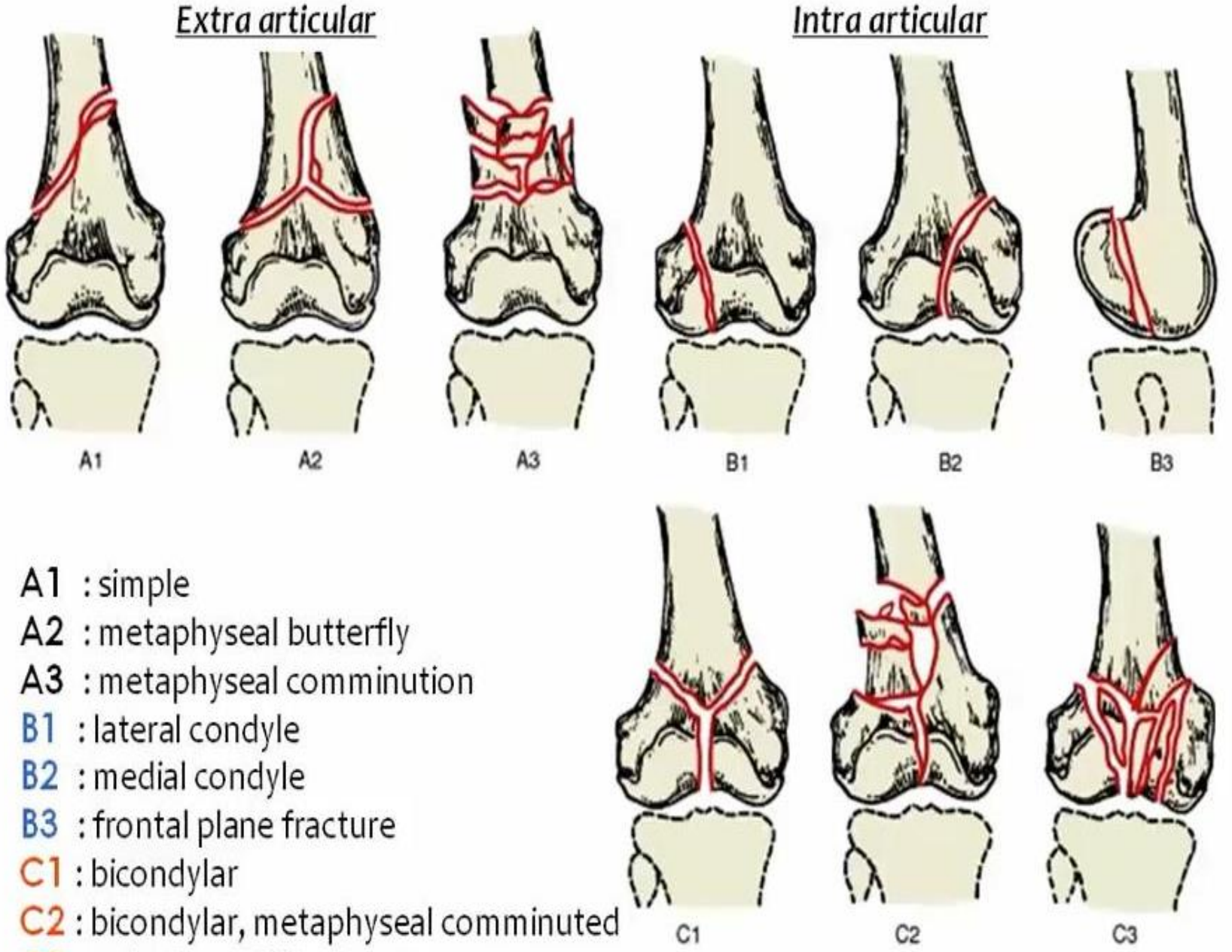
Lateral x ray shows a transverse fracture of the femur.

(front view x-ray, the fracture has been treated with intramedullary nailing.

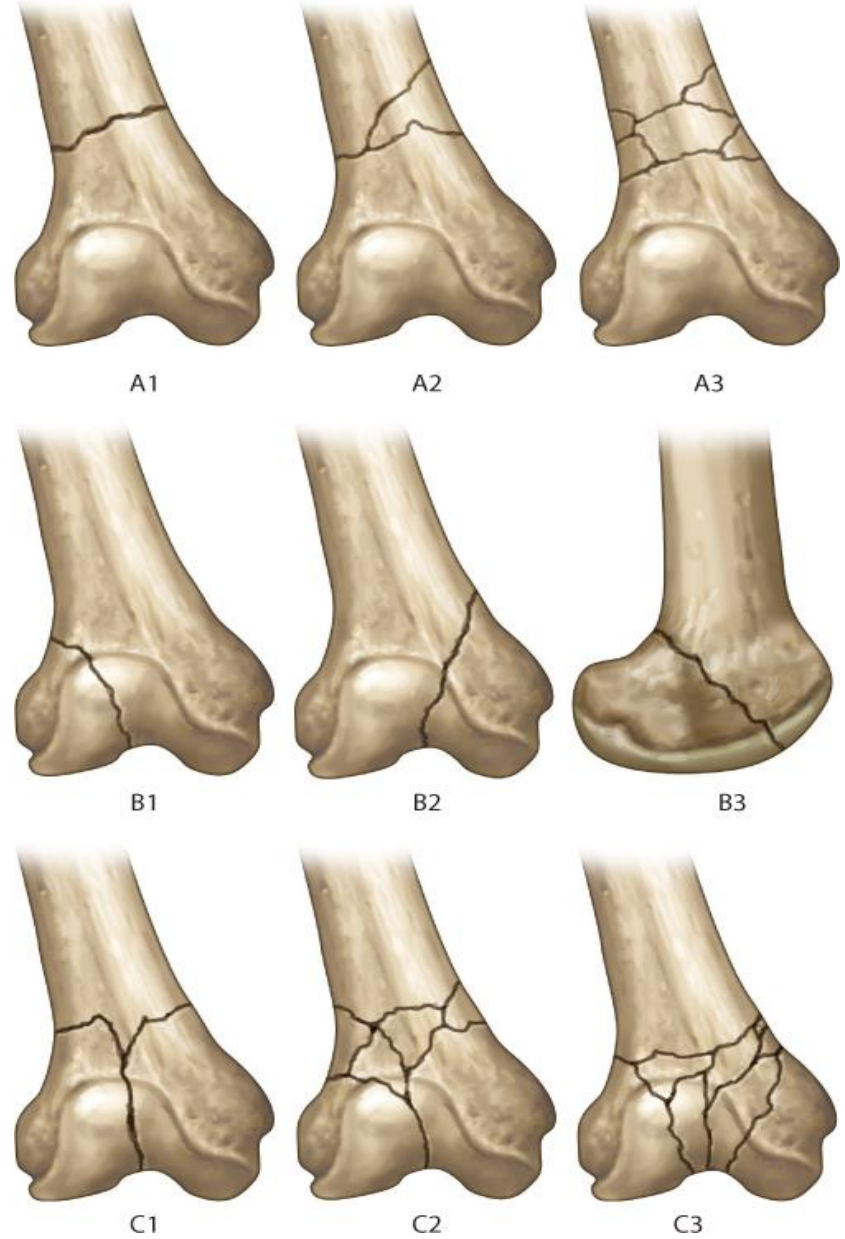
X-ray shows a transverse fracture of the femur. The break is a straight horizontal line across the shaft

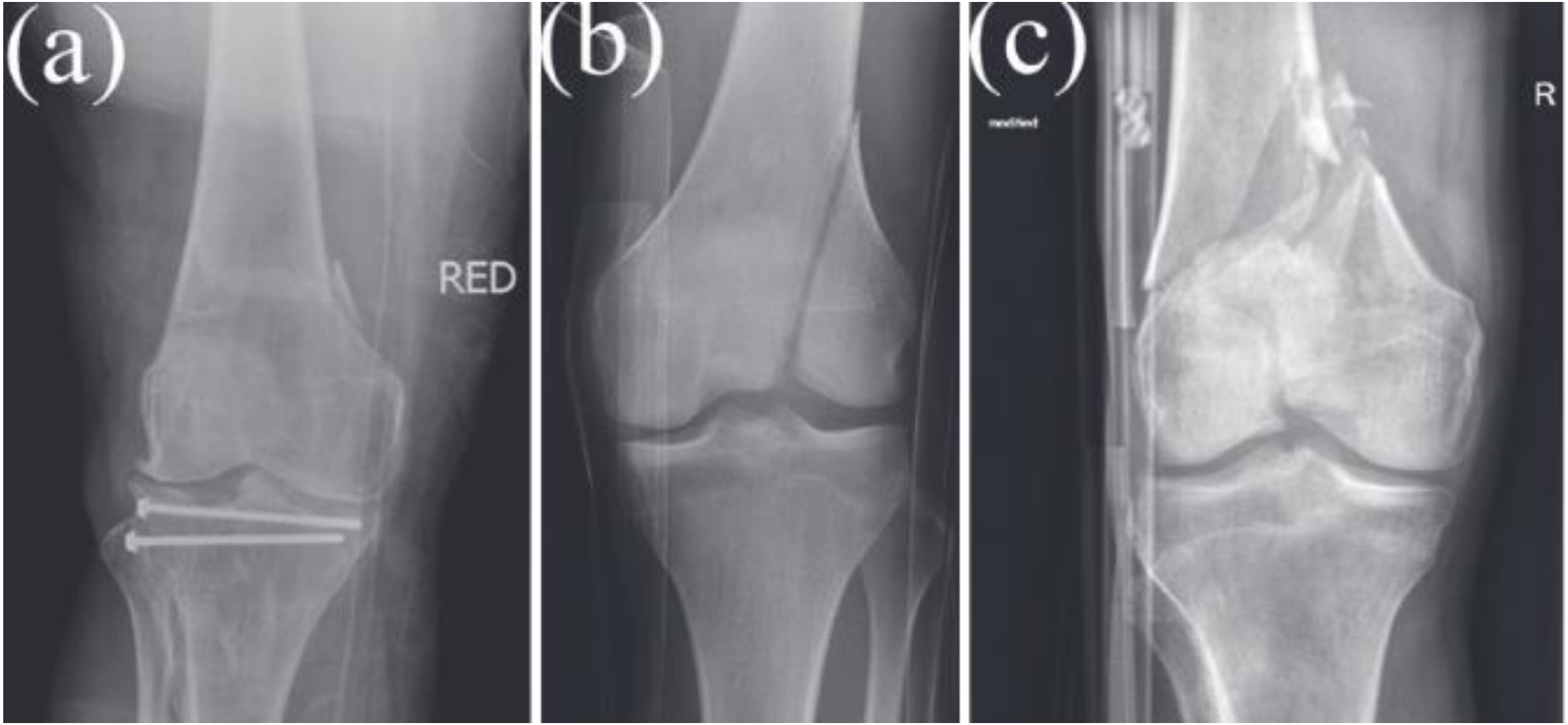
Distal femoral fracture

- Distal femoral fractures involve the femoral condyles and the metaphyseal region.
- Result from high energy trauma such as motor vehicle accidents or a fall from a height.
- The most commonly followed classification system is AO/Orthopedic Trauma Association (OTA) system.
- This system classifies fracture into A, B and C corresponding to extra-articular and intra-articular injuries.
- Further sub-classified from 1to 3 according to degree of comminution and fracture pattern.

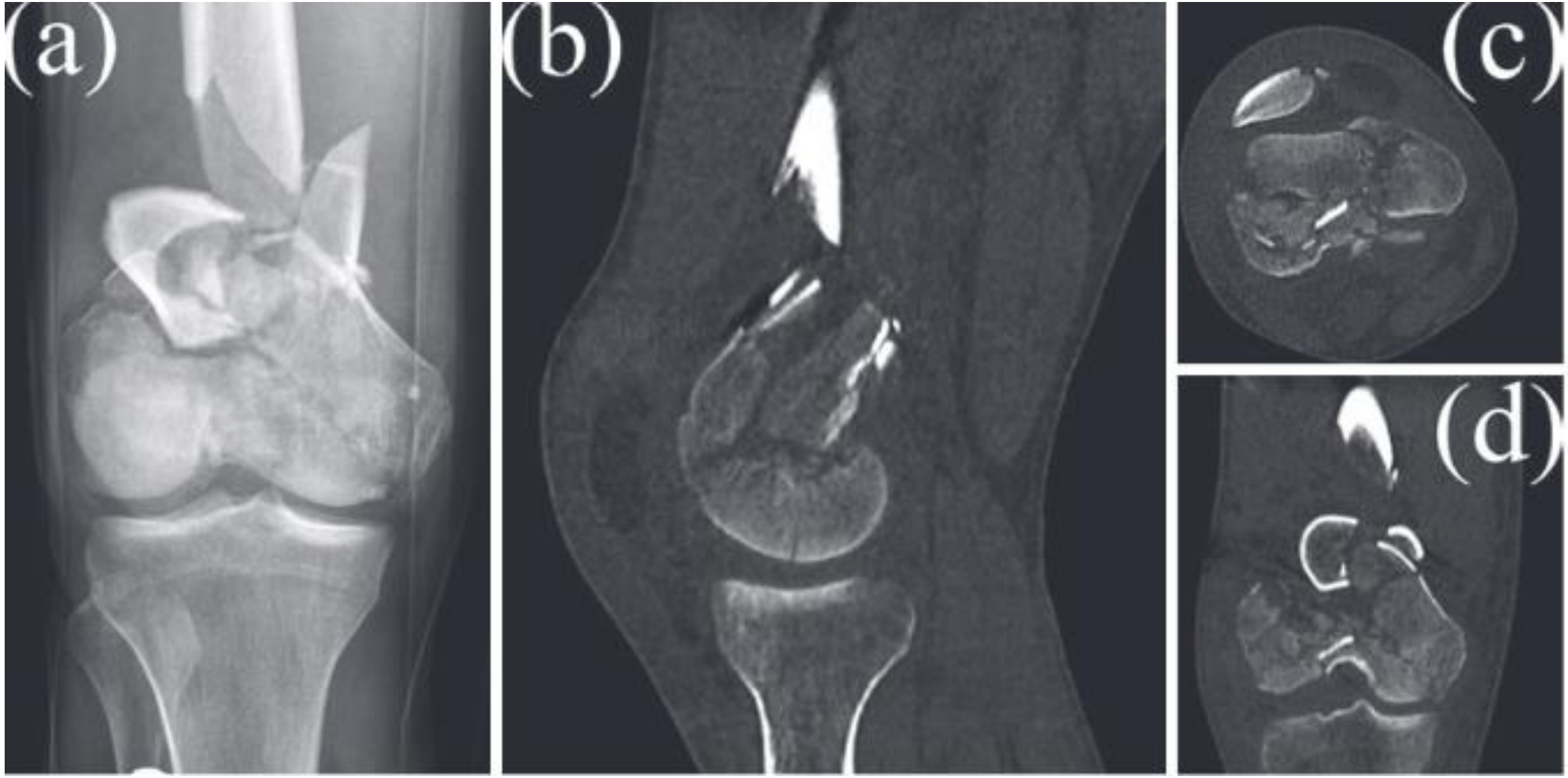


A1 : simple
A2 : metaphyseal butterfly
A3 : metaphyseal comminution
B1 : lateral condyle
B2 : medial condyle
B3 : frontal plane fracture
C1 : bicondylar
C2 : bicondylar, metaphyseal comminuted
C3 : articular multifragmentary

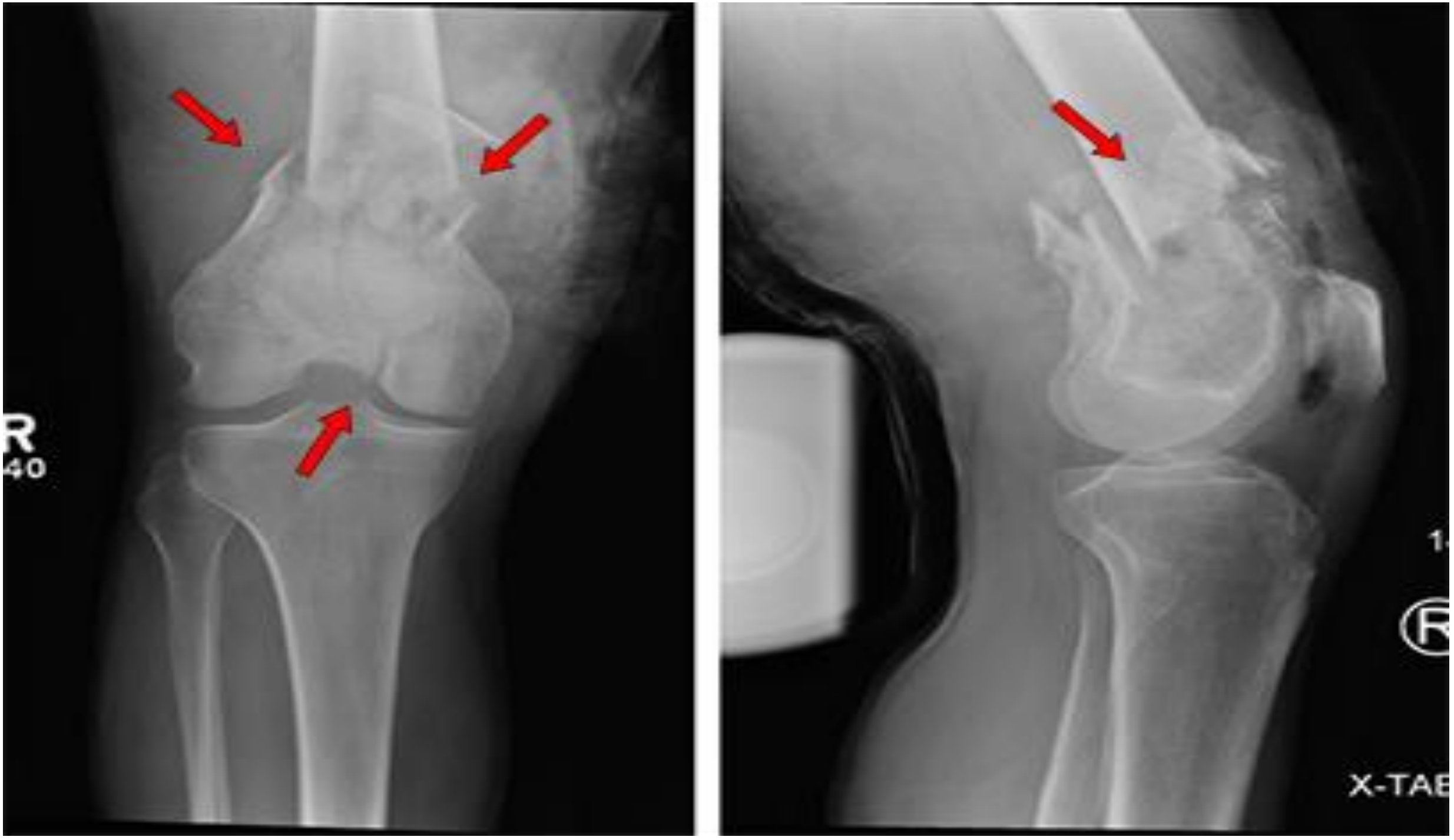




AP radiographs of (a) supracondylar fracture in a 65-year-old following a fall from standing height having had previous tibial plateau fixation, (b) Unicondylar fracture in a 45-year-old who fell from a horse and (c) Multi-fragmentary total articular) fracture from a dashboard impact in a high-speed motor vehicle collision



Preoperative planning of articular reconstruction and fixation using plain radiograph (a) and sagittal (b), axial (c) and coronal (d) and CT reconstructions to visualize the alignment and orientation of the individual fracture lines and bone fragments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(A) Anteroposterior view, (B) lateral view and (C) oblique view demonstrate ipsilateral femoral neck, subtrochanteric, comminuted femoral shaft, and supracondylar fracture of left femur.

Salter–Harris fracture

- A Salter–Harris fracture is a fracture that involves the epiphyseal plate or growth plate of a bone, specifically the zone of provisional calcification.
- It is thus a form of child bone fracture. It is a common injury found in children, occurring in 15% of childhood long bone fractures.
- Salter- Harris type II fracture is the most common and fracture passes across most of the growth plate and up through the metaphysis

Salter- Harris II fracture

Knee joint dislocation

- Knee dislocations are rare, but a significant number have a serious associated neurovascular injury.
- ***Kennedy classification based on the direction of displacement of the tibia***

1. Anterior (30-50%)

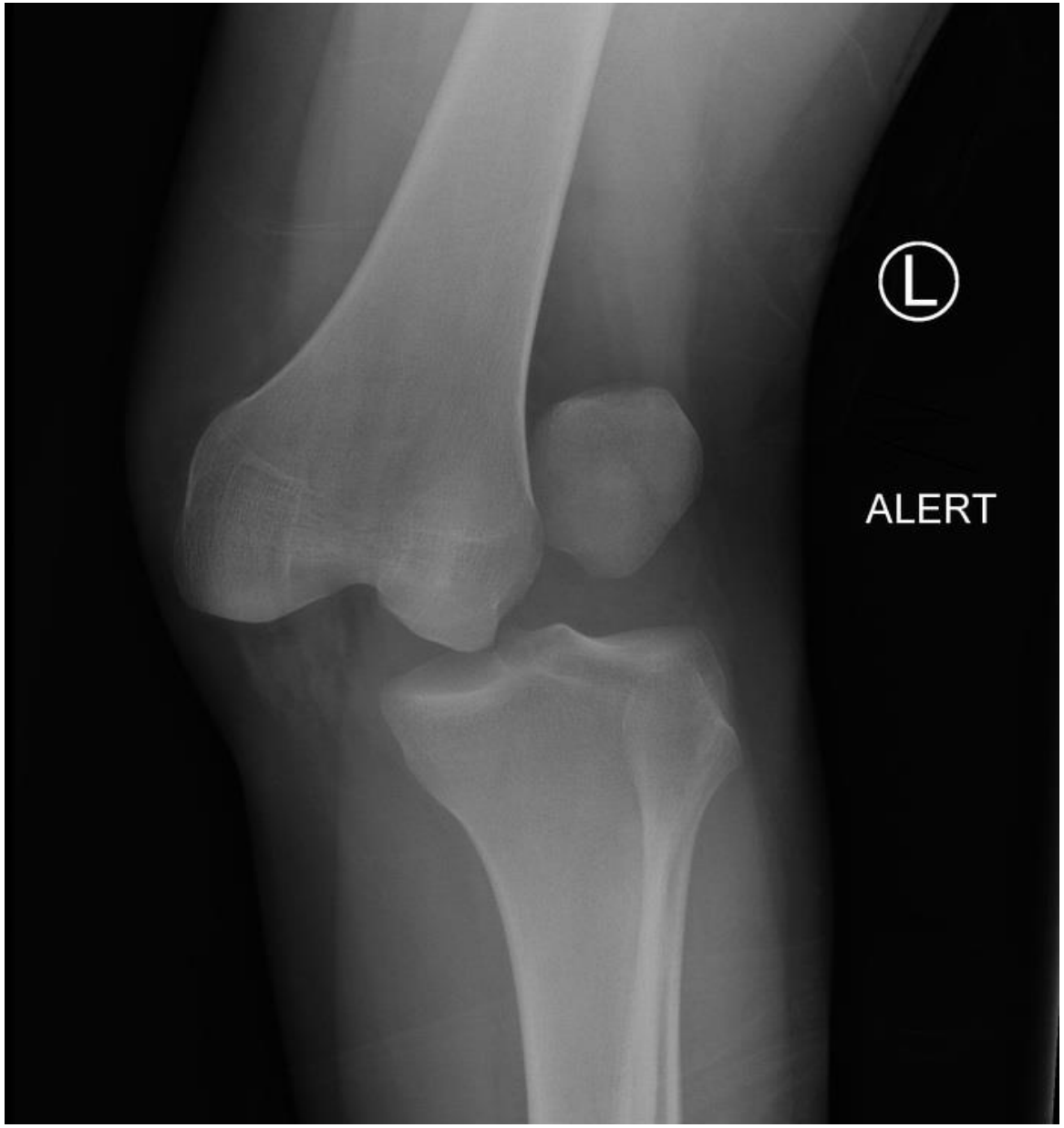
- Most common
- Due to hyperextension injury
- Usually involves tear of posterior cruciate ligament (PCL)
- An arterial injury is generally an intimal tear due to traction
- The highest rate of peroneal nerve injury

2. Posterior (30-40%)

- 2nd most common
- Due to axial load to the flexed knee (dashboard injury)
- The highest rate of vascular injury (25%) based on Kennedy classification has highest incidence of a complete tear of the popliteal artery

Plain film lateral view of left knee showing anterior knee dislocation

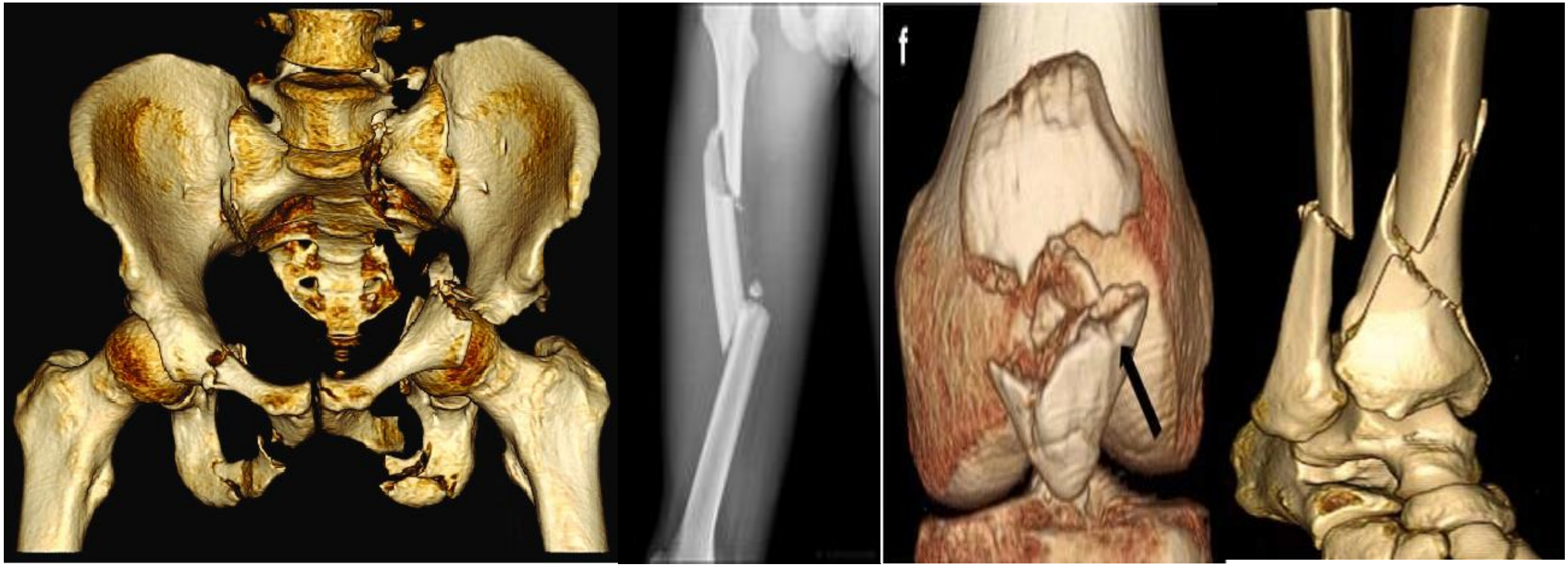
Plain film lateral view of left knee showing posterior knee dislocation



Lateral dislocation of the knee

Medial dislocation of the knee

- Angio CT-scan reconstruction. 1a : posterior view. 1b : lateral view.
- Initial angio CT-scan reconstruction in an obese female patient (fall from height). Complete interruption of popliteal flow associated with antero-lateral knee dislocation.
- Resumption of the flow in the infra-articular portion with visualization of the three artery lines of the leg.
- No associated fracture.



Lower Limb Fractures

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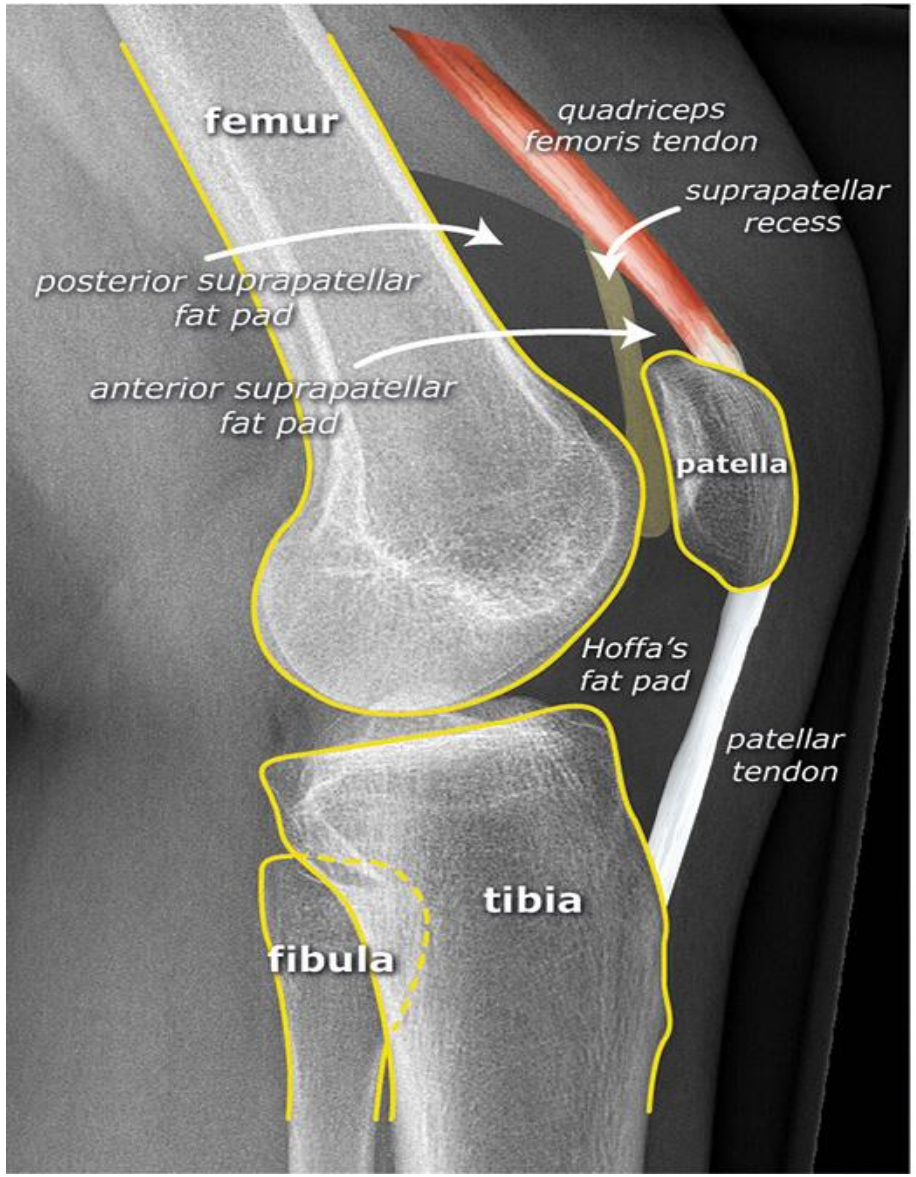
Part 3

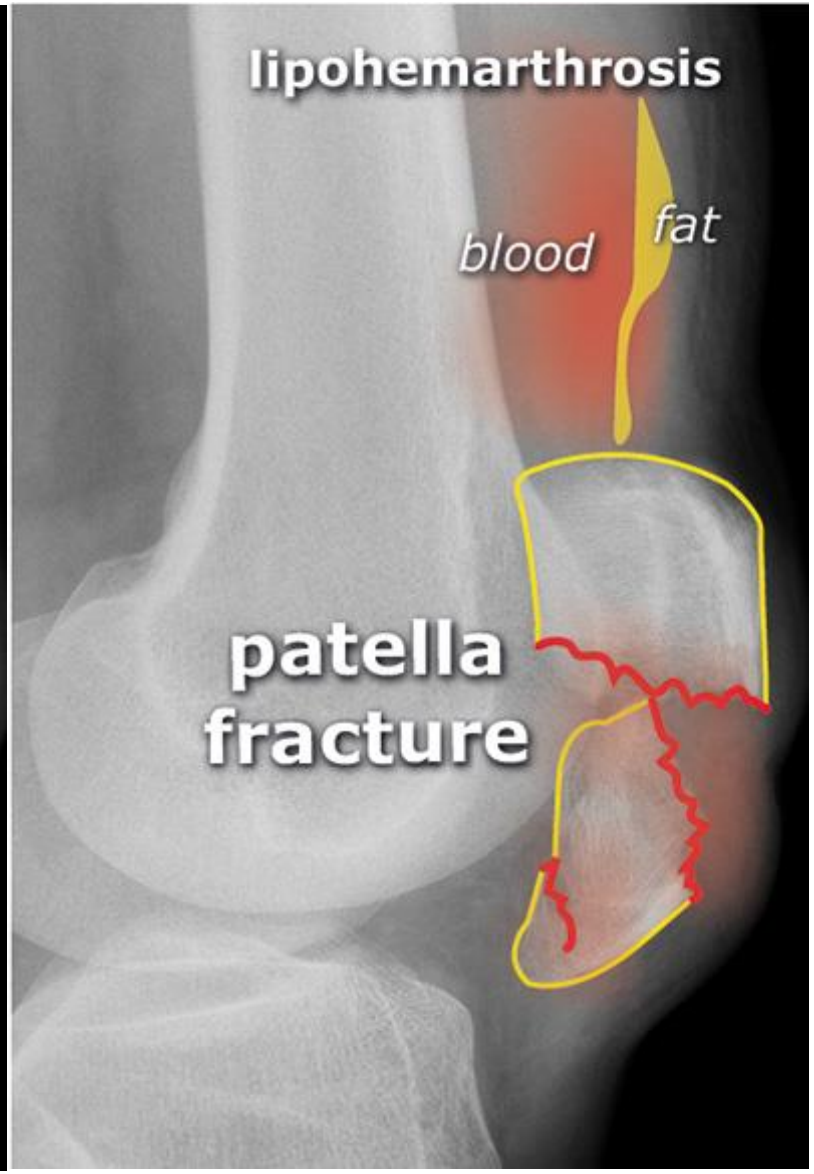
Patellar fracture

- Patellar fracture is one of the common knee injuries usually post direct trauma to the patella.
- Patients present with marked swelling and pain over the patella.
- Usually, there is a large joint effusion or hemarthrosis.
- Types of Patellar fracture

1. Transverse fracture of mid patella (most common)
2. Comminuted fracture
3. Vertical fracture (least common)
4. Osteochondral defect usually from medial facet
5. Patellar sleeve fracture in children



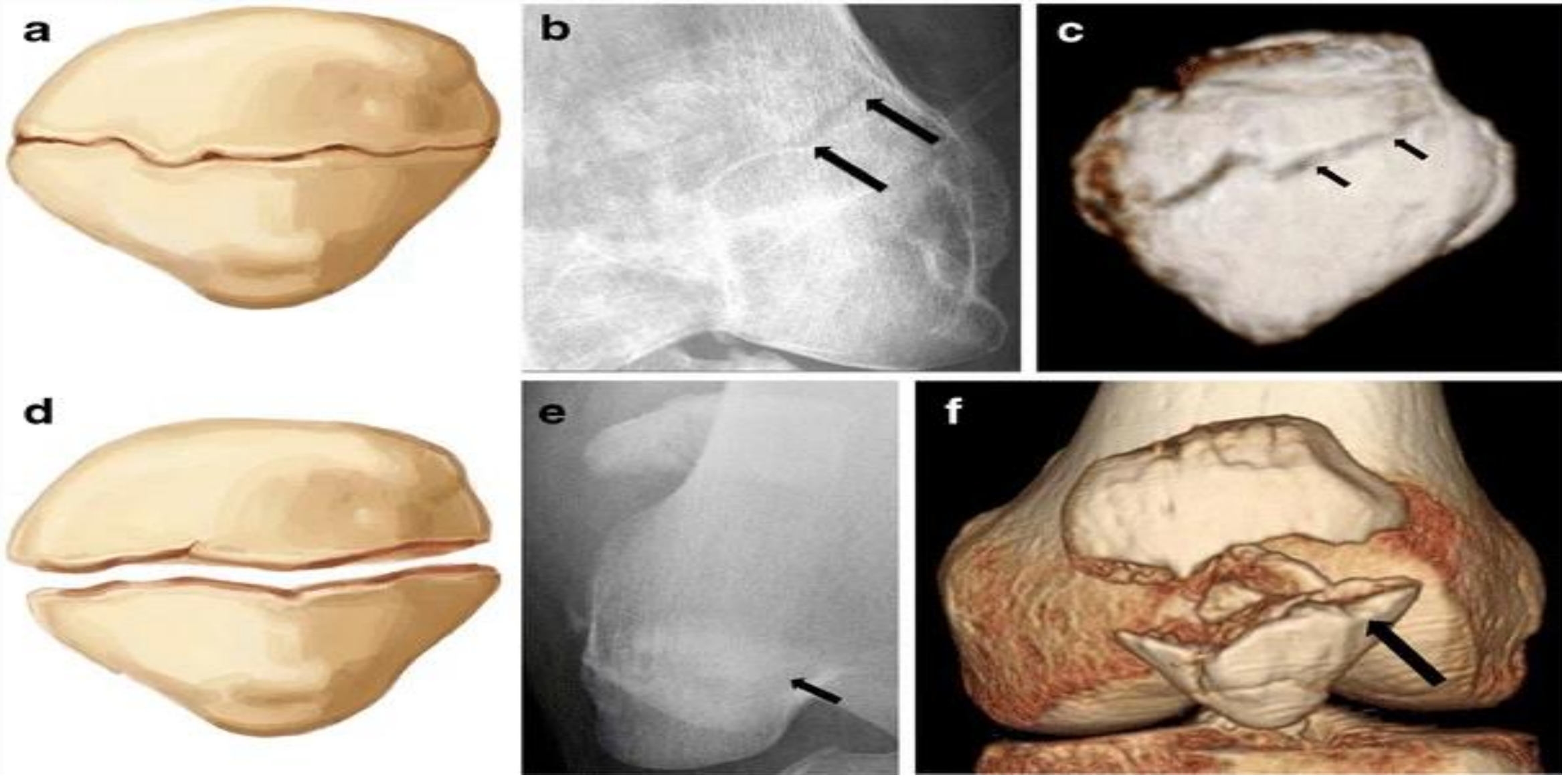
Lateral image (mediolateral projection) of the left knee. Normal anatomy.



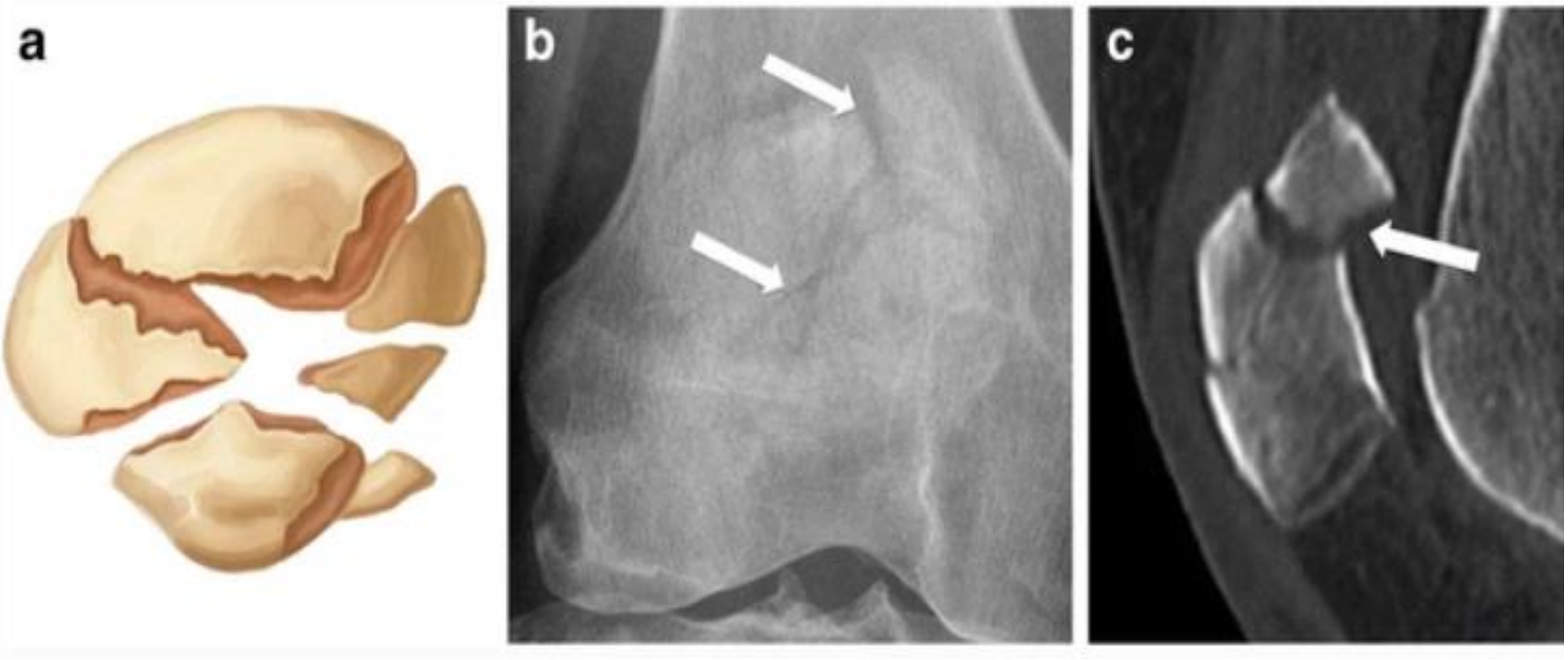
Transversal patellar fracture with significant dislocation. Also note the lipohemarthrosis in the suprapatellar recess.

Lateral x-ray of a patellar fracture shows significant displacement between the broken pieces of bone.

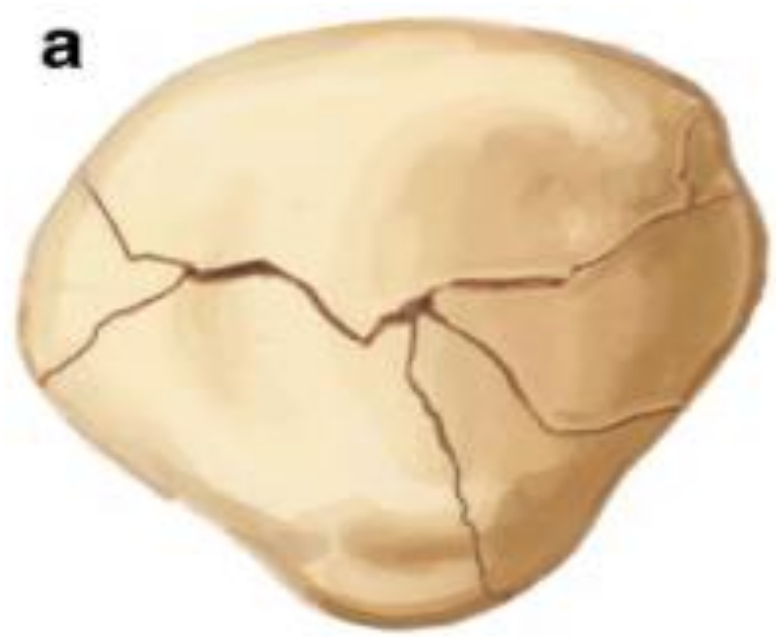
Lateral x-ray shows transverse fracture across the patella with slight displacement between the broken pieces of bone.

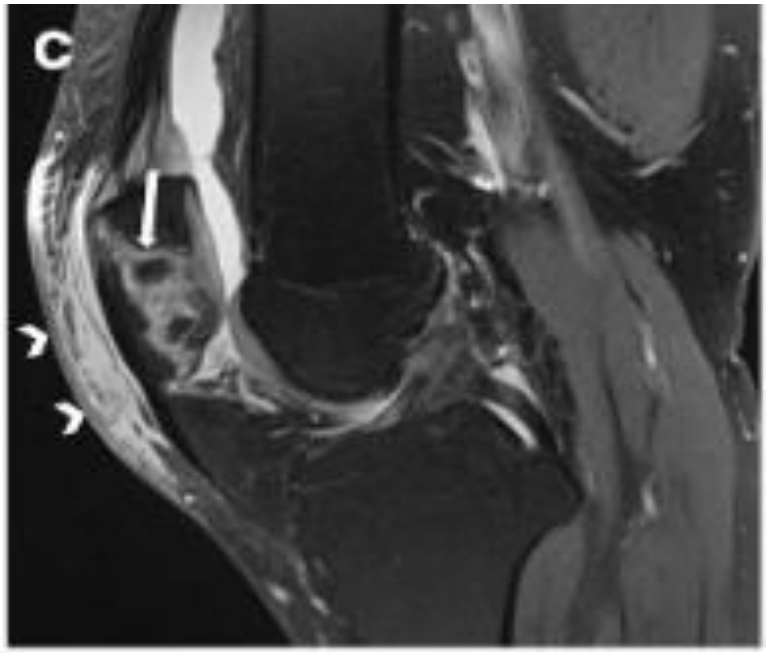


Two examples of transverse fractures in young adults. (a, d) Drawing, (b, e) anteroposterior radiograph, and (c, f) 3D CT surface reconstruction. (a–c) Non-displaced and (d–f) displaced transverse fractures through the body of the patella.



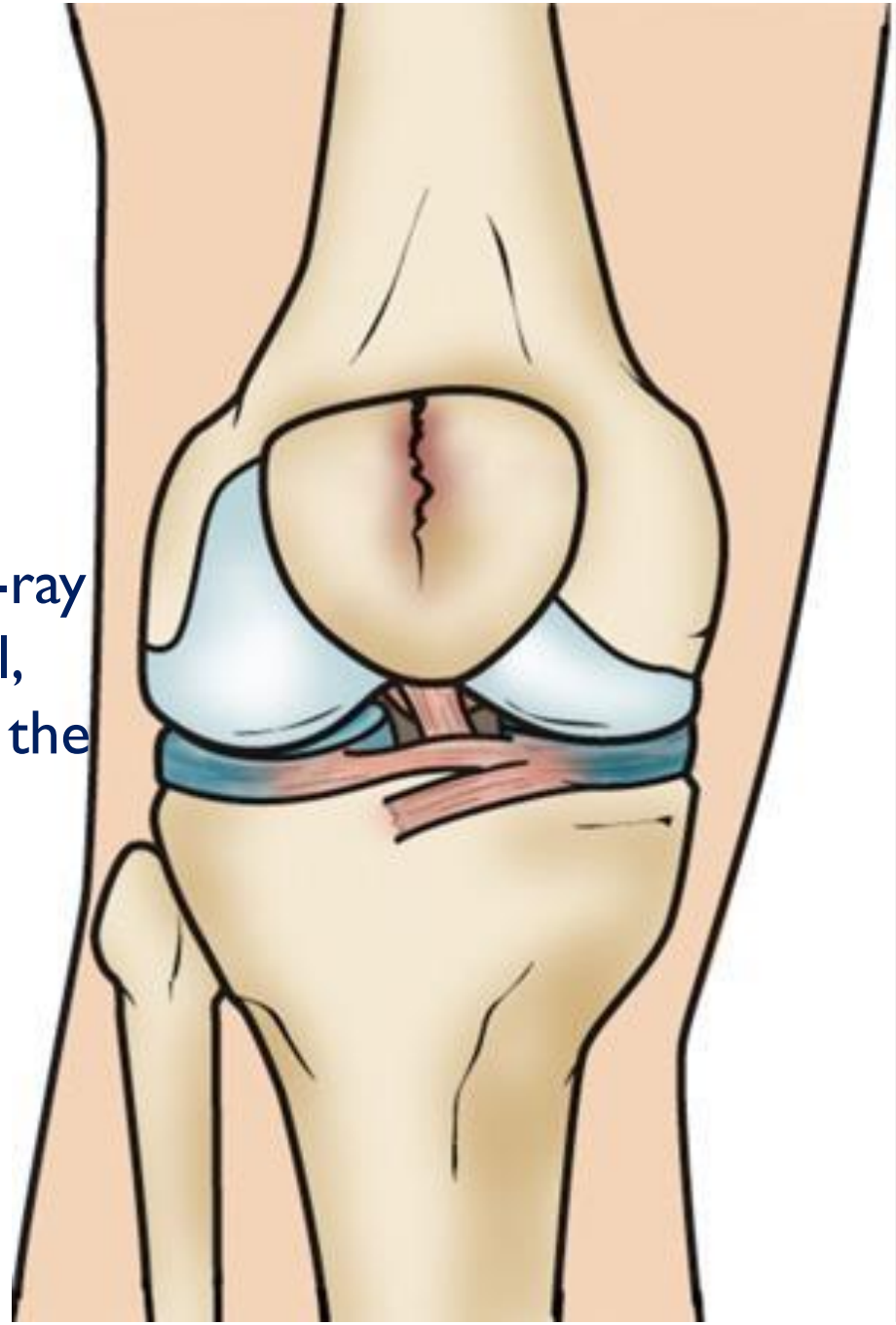
CT reformatted image showing displaced comminuted fracture. (a) Drawing, (b) anteroposterior radiograph, and (c) sagittal computed tomography reformat in a middle-aged man following direct blow to the knee in a car accident, showing displaced comminuted fracture. Note that the degree of displacement is better appreciated on the reformatted CT image than on conventional radiograph

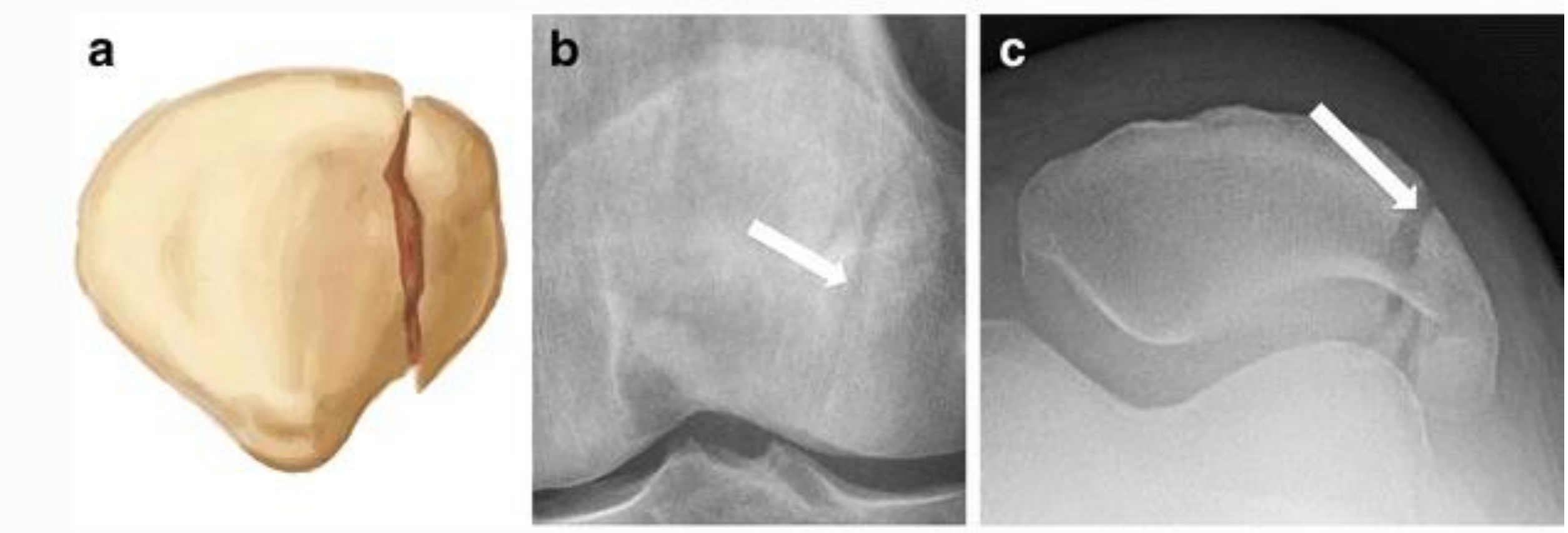




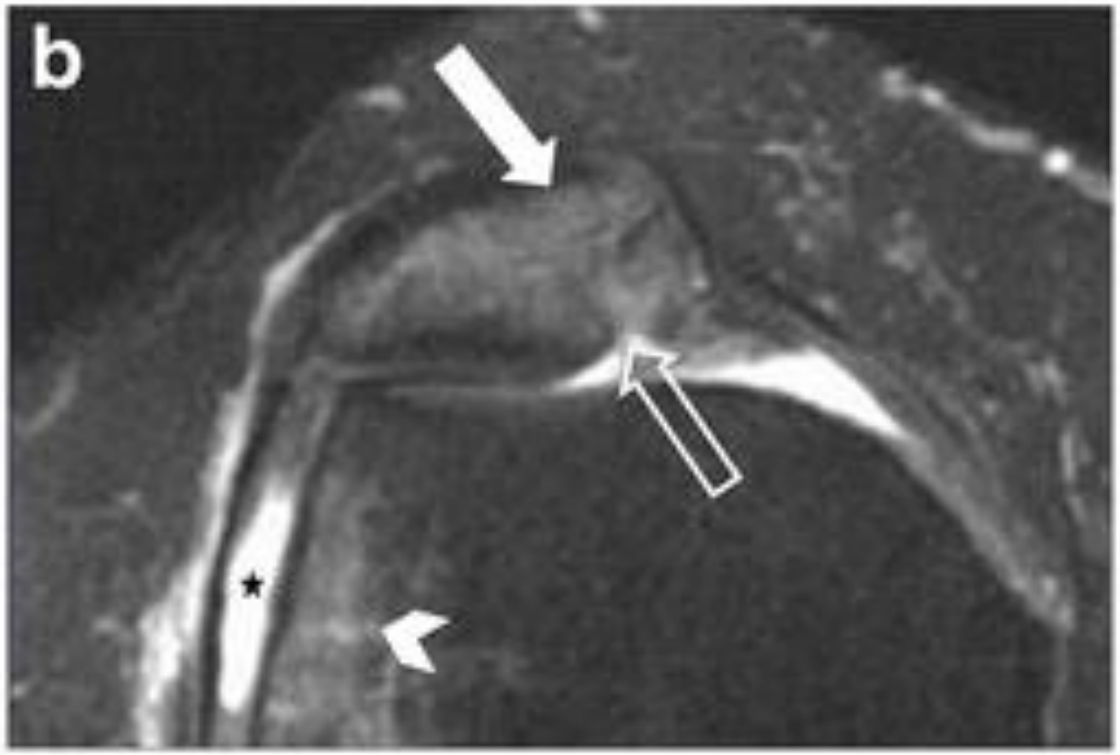
(a)Drawing,(b)anteroposterior radiograph, and(c)sagittal and(d) coronal fat-suppressed proton density MR images showing non-displaced stellate fracture (solid arrows)ina36-year-old man. Note the subcutaneous prepatellar edema related to direct trauma mechanism (arrowheads)

Illustration and x-ray show a vertical, stable fracture of the patella.





Vertical fracture of the right patella in a 34-year-old man. (a) Drawing and (b) anteroposterior and (c) skyline views of the right knee show a vertical fracture of the lateral facet (red arrows). Note that the fracture line is more conspicuous on the tangential than the anteroposterior view



Osteochondral fracture from lateral patellar dislocation in a 27-year-old woman. (a) Drawing shows an incomplete osteochondral fracture. (b) Axial fat-suppressed proton density MR image shows an osteochondral injury to the articular surface (open arrow) with bone marrow edema of the medial patella (solid arrow) and lateral aspect of the lateral femoral condyle (arrowhead), in keeping with transient lateral patellar dislocation. Note also a moderate amount of joint effusion (star)

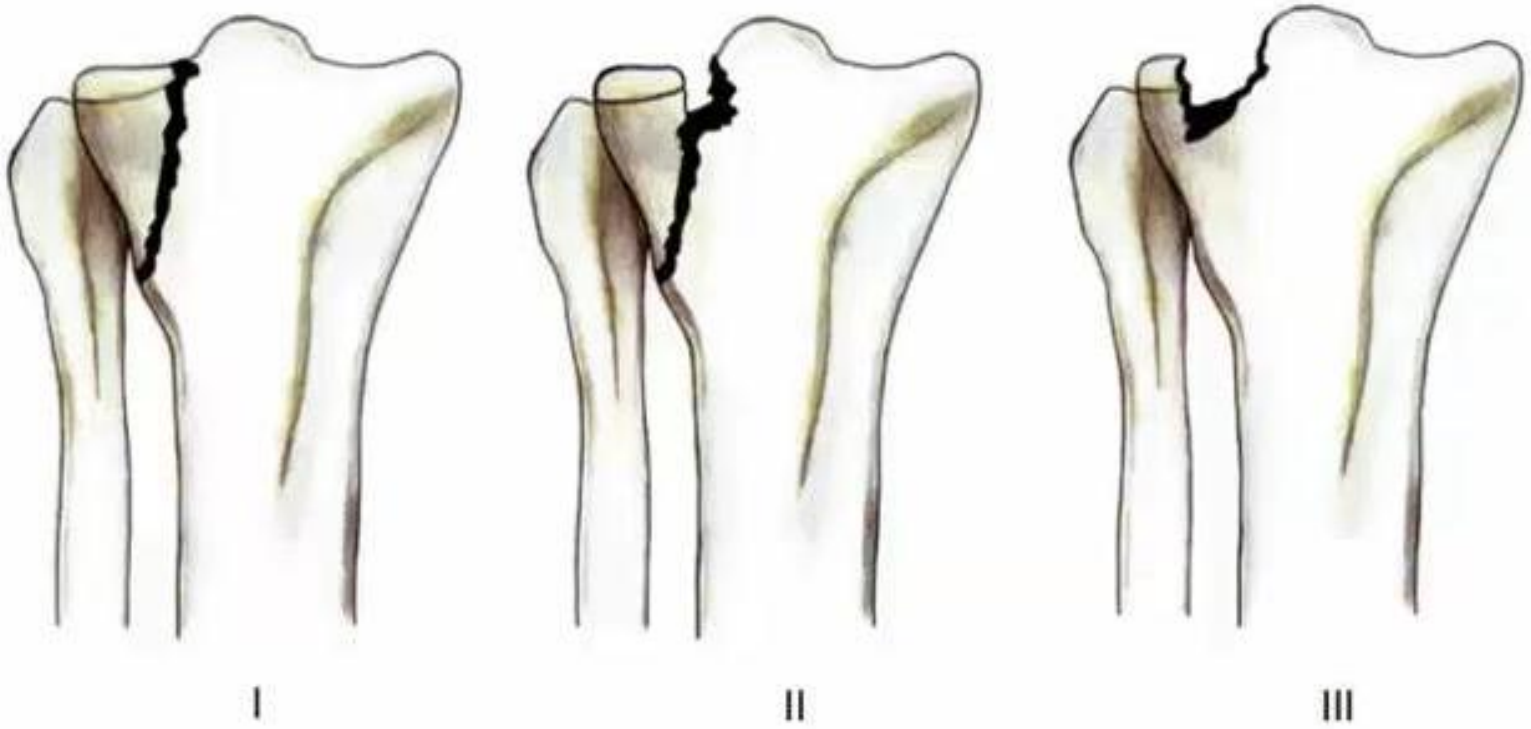
A 14-year-old boy was evaluated for knee pain that occurred while running. On examination, tenderness to palpation was noted along the patella and extensor mechanism. The lateral radiograph demonstrates prepatellar soft-tissue swelling with separation of the anterior and inferior calcified fibrocartilage of the patella (Fig. 1, lines). MRI confirms a patellar sleeve fracture with laminar tearing of the extensor mechanism (Fig. 2, arrows).

Tibial plateau fracture

- A tibial plateau fracture is a break of the upper part of the tibia (shinbone) that involves the knee joint
- Fractures of the lateral plateau are much more common than the medial plateau.
- To injure the medial plateau, a large amount of force is required; fractures of the medial plateau are usually seen in conjunction with fractures of the lateral plateau and other bones around the knee joint.

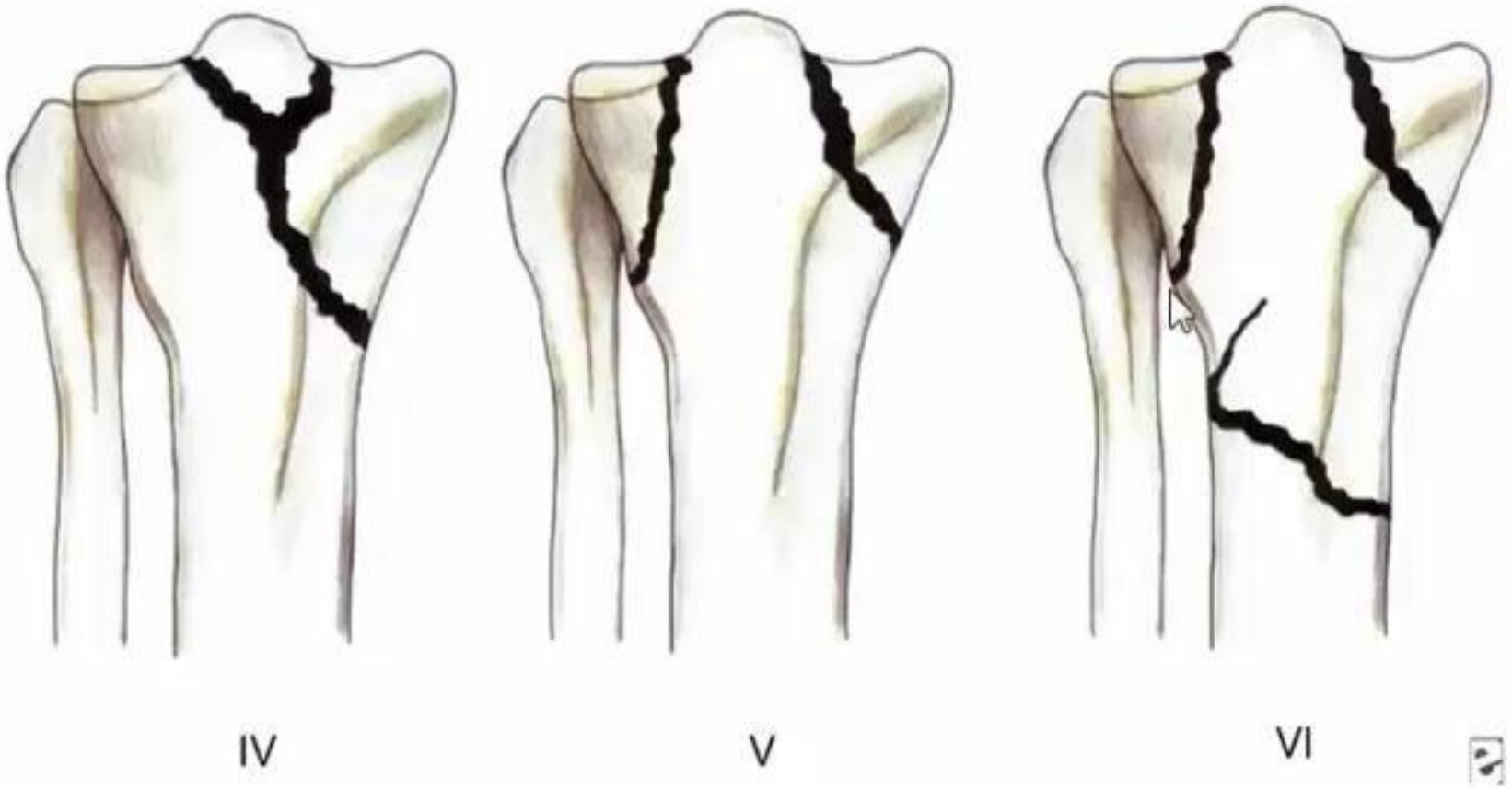
- **Radiographic features**
- Plain radiography often underestimates the severity of the injury. Lipohemarthrosis should be present.
- Lipohemarthrosis results from an intra-articular fracture with escape of fat and blood from the bone marrow into the joint, and is most frequently seen in the knee, associated with a tibial plateau fracture
- CT is very helpful in accurately defining the extent of the bony injury and facilitates orthopedic intervention.
- MRI is very helpful in the assessment of soft tissue injury around the joint.

Schatzker classification of tibial plateau fractures

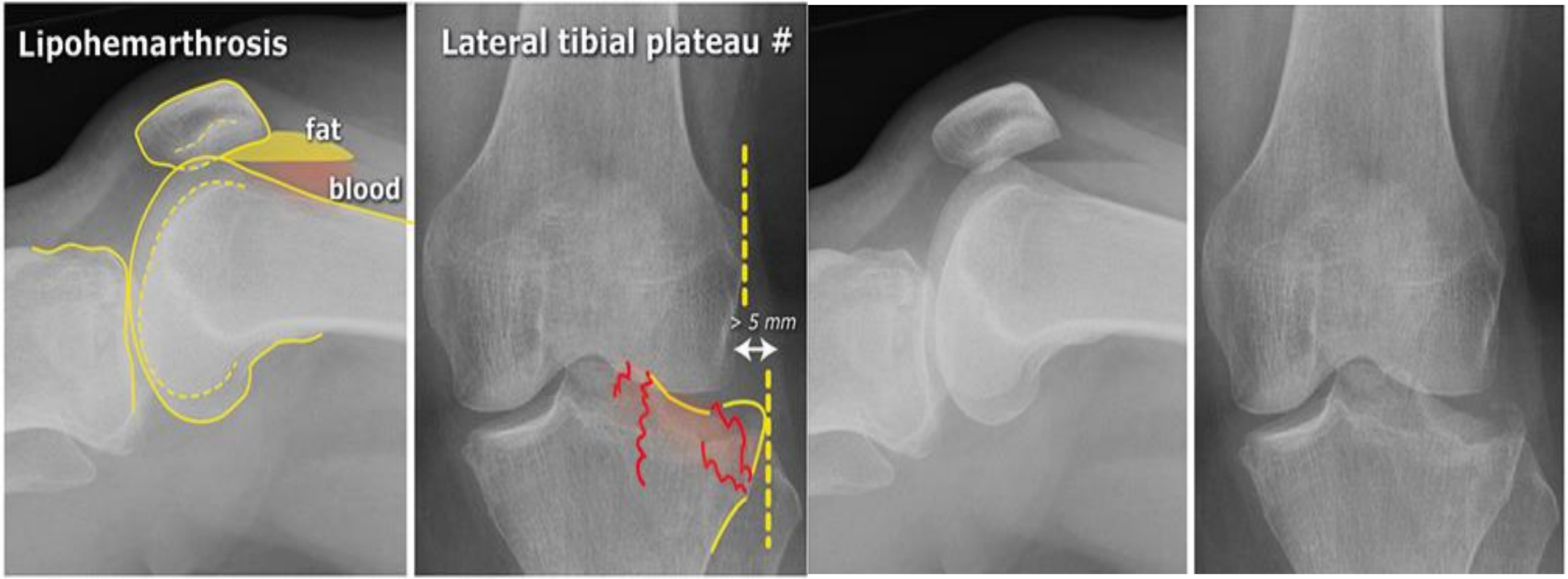


- I : A wedge-shaped pure cleavage fracture of the lateral tibial plateau,
- Originally defined as having less than 4 mm of depression or displacement
- II : Split and depression , type I fracture with a depressed component
- III : Pure depression of the lateral tibial plateau ,divided into two subtypes
  - IIIa: those with lateral depression
  - IIIb: those with central depression

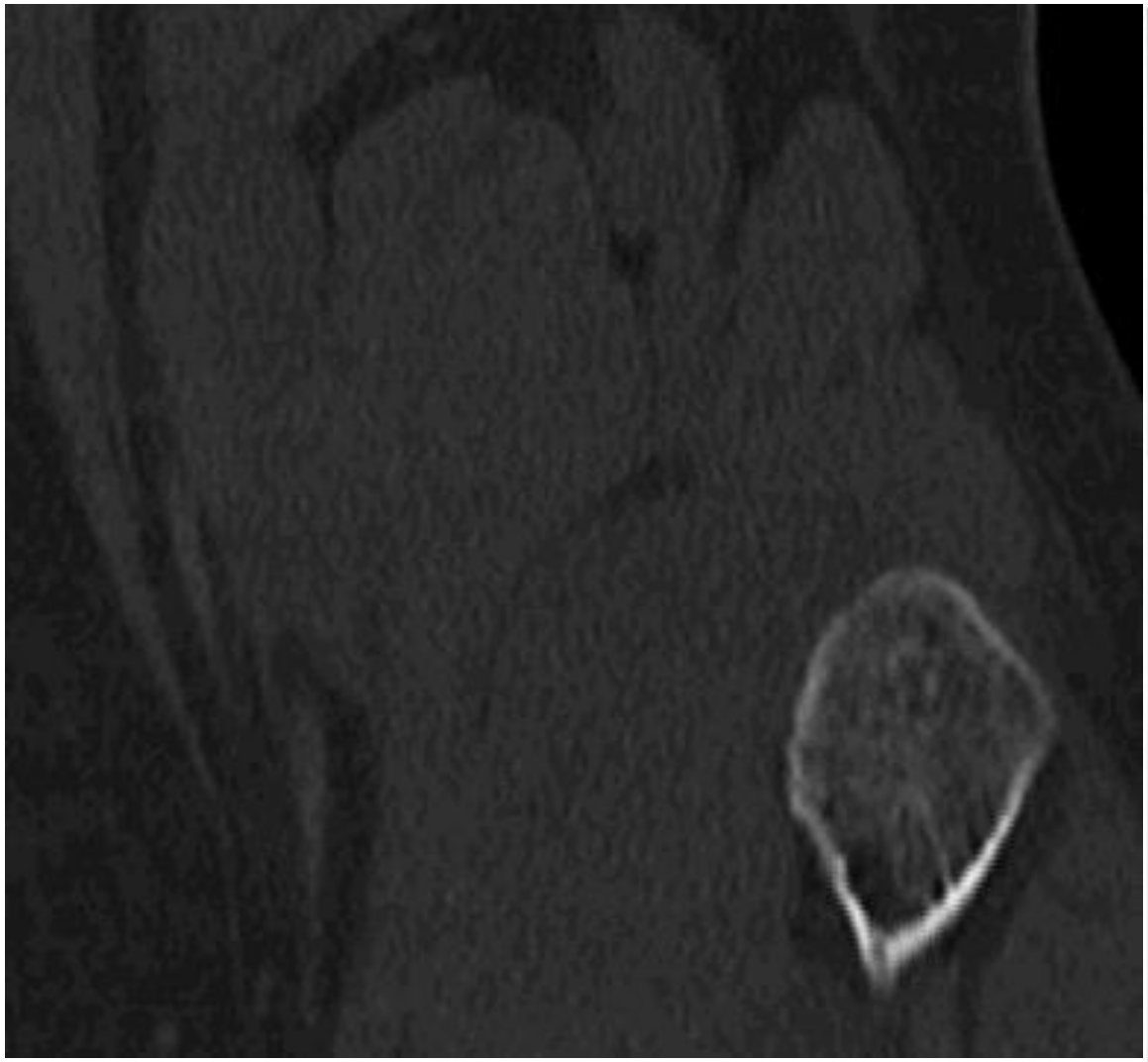
Schatzker classification of tibial plateau fractures

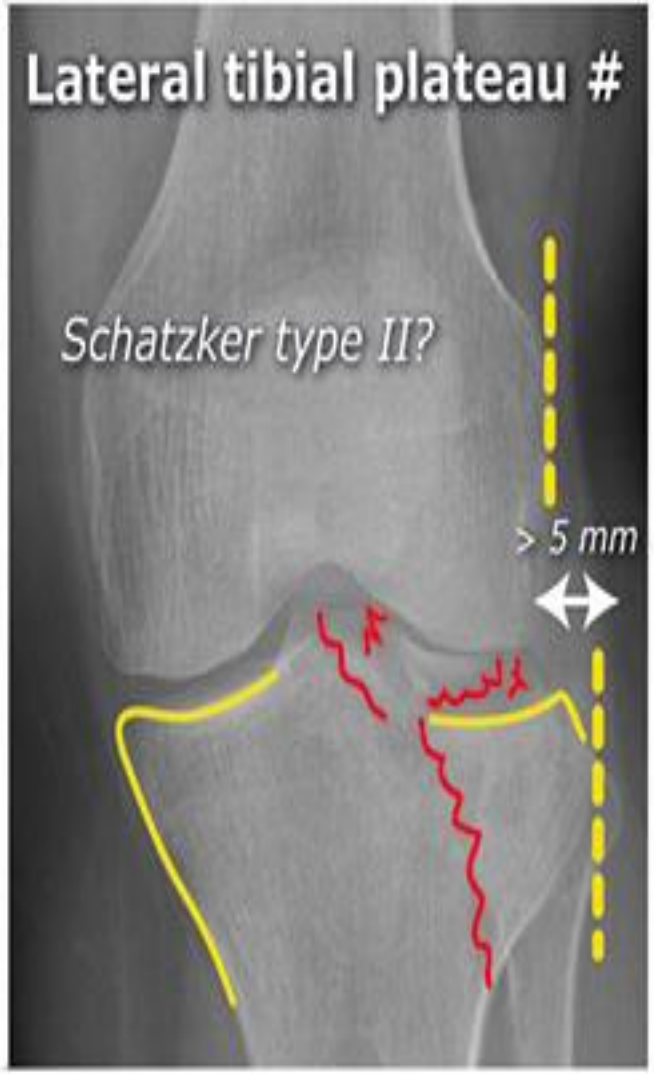


- IV : Comminuted split fracture of the medial tibial plateau and tibial spine
- V : Involving both tibial plateau regions
- VI : Fracture through the metadiaphysis of the tibia



Lateral knee X-ray with horizontal beam & AP image of the left knee. The lateral image shows lipohemarthrosis and a lateral tibial plateau fracture. Note the abnormal tibiofemoral alignment (> 5 mm).



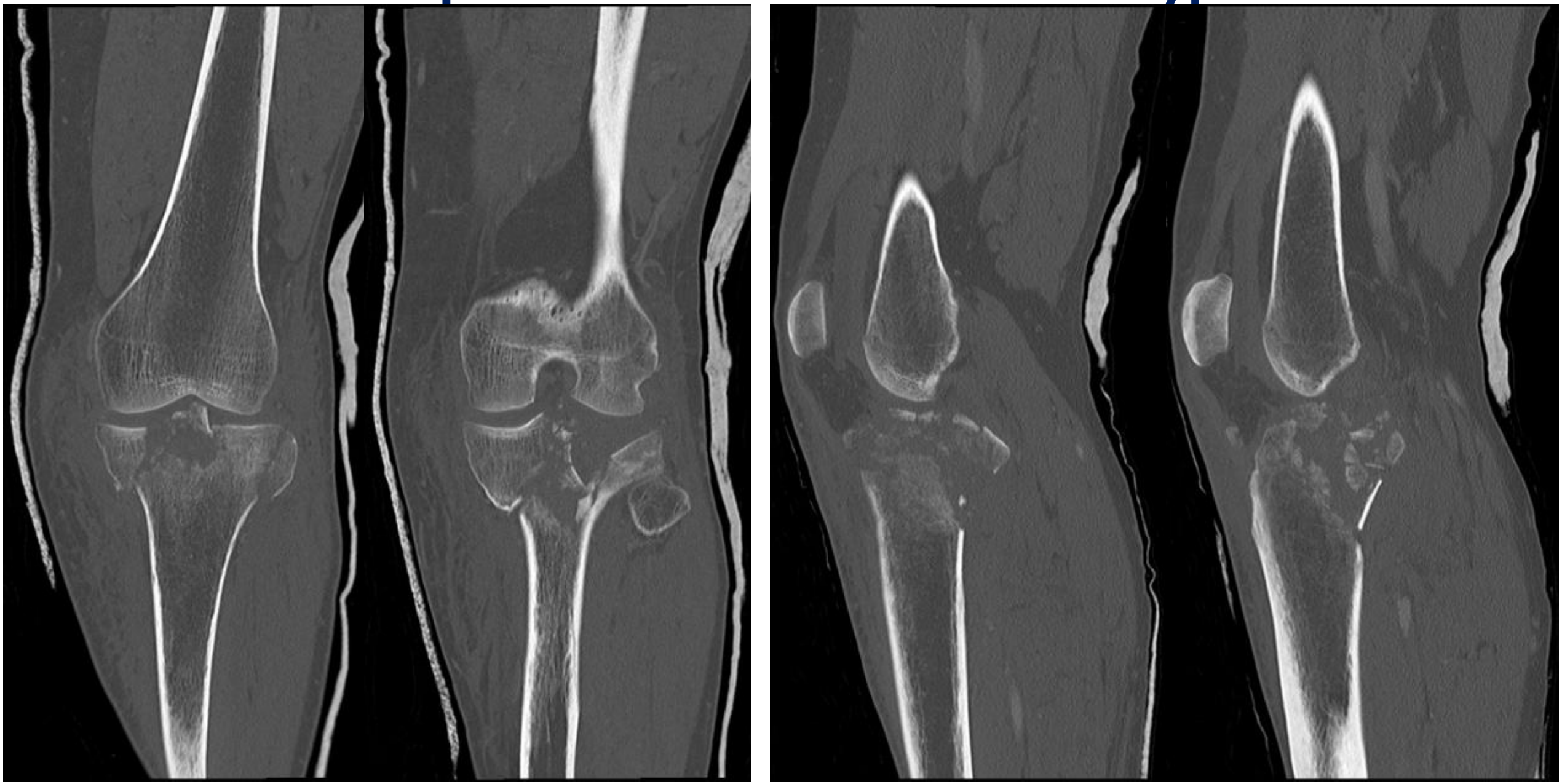


Lateral knee X-ray with a horizontal beam & AP of the left knee reveal lipohemarthrosis with a lateral tibial plateau fracture. There is depression (> 4 mm) of the lateral tibial plateau, therefore consistent with type II Schatzker. However, CT scan also reveals a fracture of the medial tibial plateau. Therefore, this is actually a type V Schatzker tibial plateau fracture.

A severe tibial plateau fracture with an associated fibular head fracture

A fracture of the lateral tibial plateau (arrows) in an osteopenic patient who had a recent trauma.

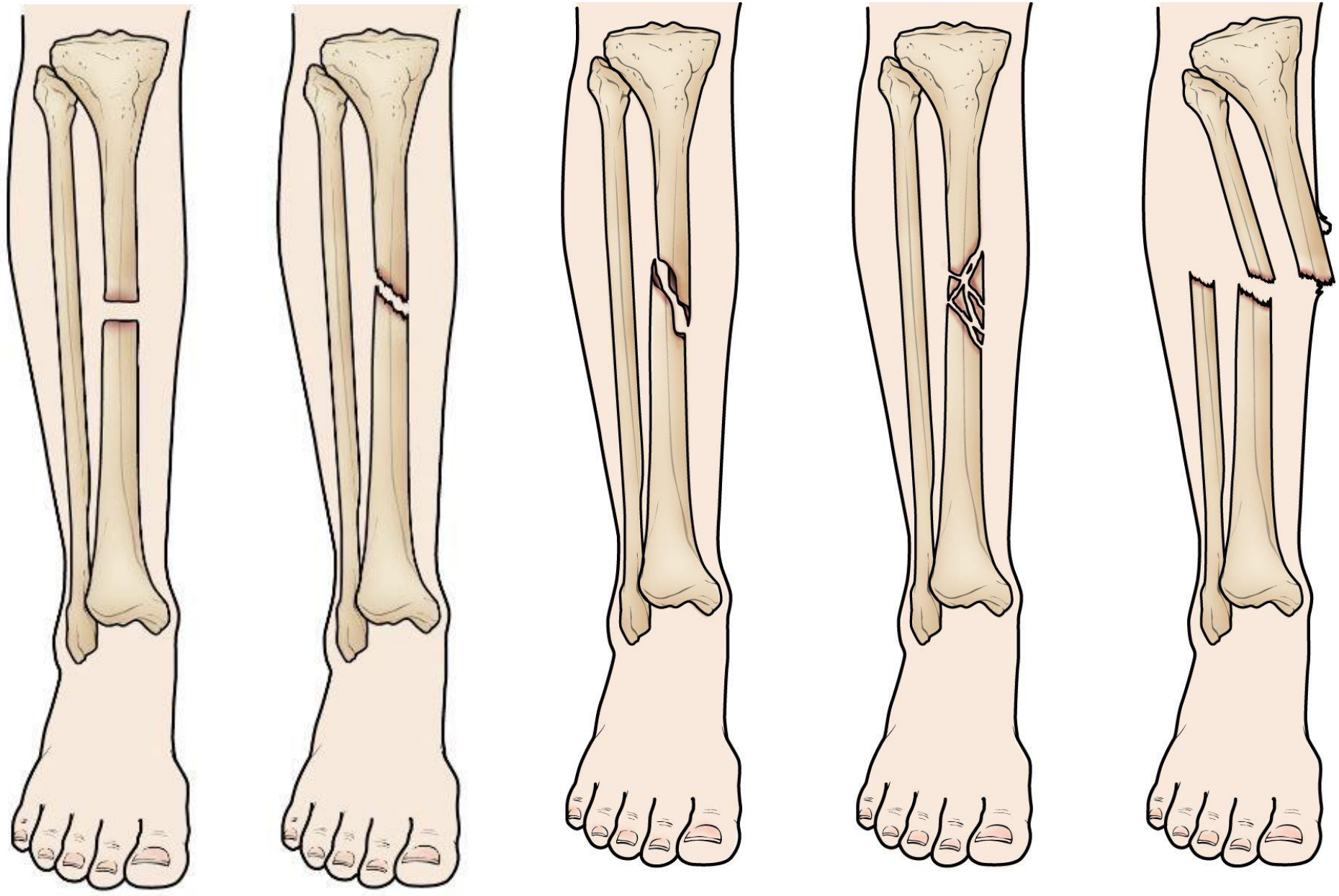
CT: Tibial plateau fracture: Schatzker type VI

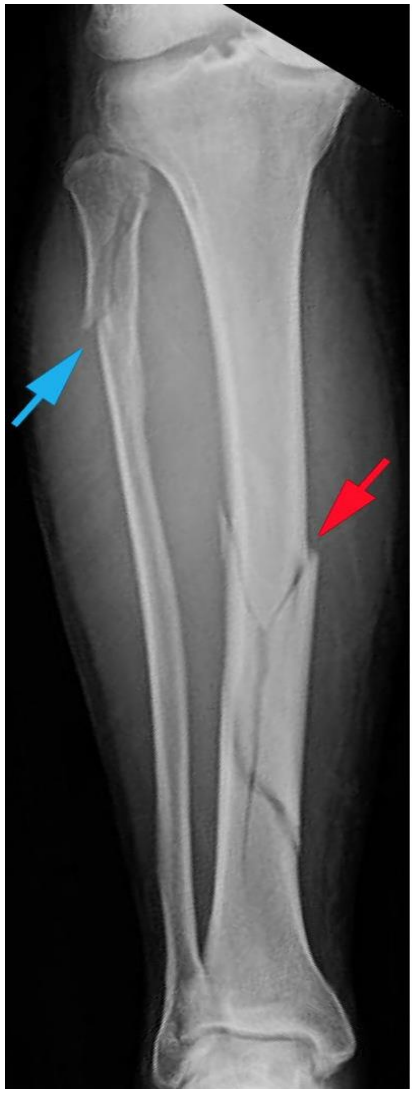


Tibial Shaft Fractures

- The tibia, or shinbone, is the most commonly fractured long bone in the body. A tibial shaft fracture occurs along the length of the bone, below the knee and above the ankle.
- ***The most common types of tibial shaft fractures include:***

1. **Transverse fracture:** horizontal line going across the tibial shaft.
2. **Oblique fracture:** an angled line across the shaft.
3. **Spiral fracture:** line encircles shaft like the stripes on a candy cane.
4. **Comminuted fracture:** the bone breaks into three or more pieces.
5. **Open fracture:** If a bone breaks in such a way that bone fragments stick out through the skin or a wound penetrates down to the broken bone, the fracture is called an open or compound fracture.


Transverse
Oblique
Spiral
Comminuted
Open



X-ray shows a tibial shaft fracture (red arrow) and a fibula fracture (blue arrow). The tibial shaft fracture has been treated with intramedullary nailing.

X-ray shows a fibula fracture (blue arrow) and a tibial shaft fracture (red arrows) that extends into the ankle joint. Both fractures have been treated with plates and screws.

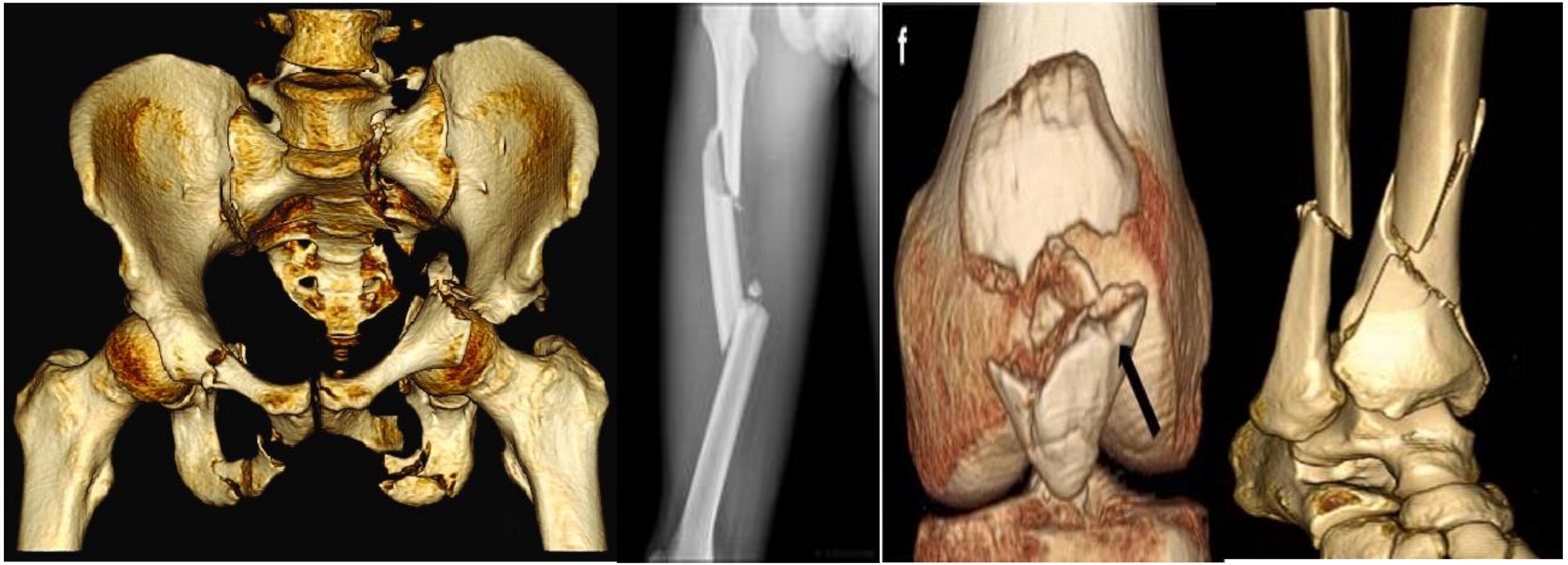
Minimally displaced spiral fracture of the distal tibia due to a twisting injury in soccer. Anteroposterior (AP) (A) and lateral (B) radiographs demonstrate the tibial fracture (arrow) with no associated fibular fracture.

Anteroposterior (AP) (A) and lateral (B) radiographs of minimally displaced transverse fractures of the mid tibia and fibula with cast immobilization.

Toddler fracture

- A toddler fracture is a minimally or un displaced spiral fracture, usually of the tibia, typically encountered in toddlers.
- Conventional radiographs are all that are required, however, multiple views may be necessary as the fracture is often inapparent on one or more views.
- Internal oblique projection can better demonstrate the fracture in some cases. If despite multiple views, no fracture can be identified, follow-up radiographs usually will demonstrate slight sclerosis and periosteal reaction.
- Nondisplaced oblique fracture will be noted in the middle or distal diaphysis of the tibia

At Injury
3 Weeks Later



Lower Limb Fractures

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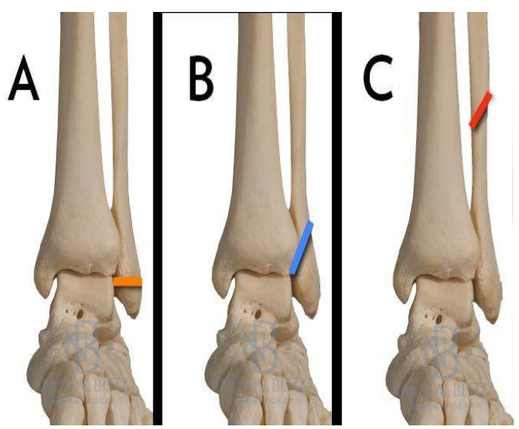
Part 4

Fractures around ankle

Weber classification of ankle fractures

- The Weber ankle fracture classification (or Danis-Weber classification) is a simple system for classification of lateral malleolar fractures, relating to the level of the fracture in relation to the ankle joint, specifically the distal tibiofibular syndesmosis.



❑ Type A

- Below the level of the syndesmosis (infra-syndesmotic).
- Abduction or adduction force.
- Tibiofibular syndesmosis and deltoid ligament intact.
- Medial malleolus occasionally fractured.
- Usually transverse and stable if medial malleolus intact.

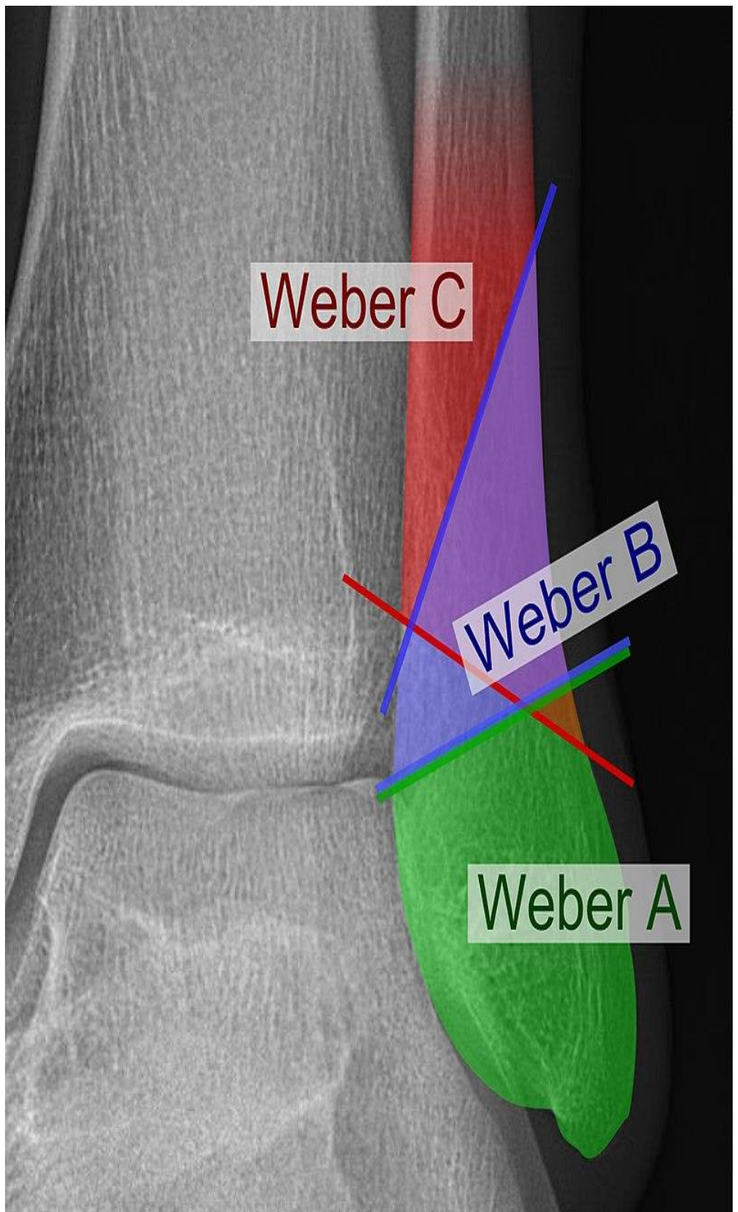
❑ Type B

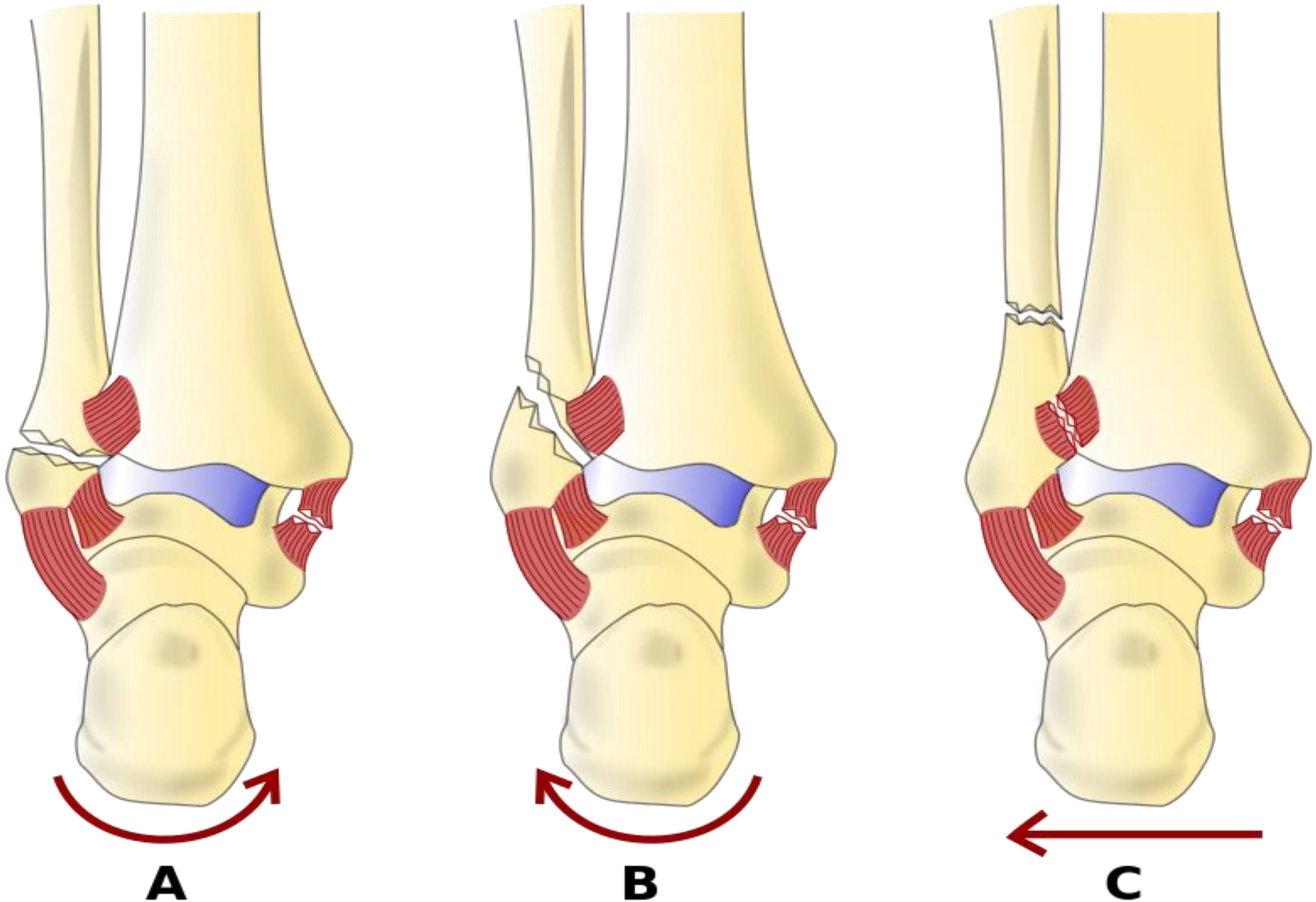
- Fracture at the level of the syndesmosis (trans-syndesmotic).
- Spiral or oblique fibular fracture.
- Tibiofibular syndesmosis usually intact, but widening of the distal tibiofibular joint.
- External rotation force
- Medial malleolus may be fractured
- Deltoid ligament may be torn, indicated by widening of the space between the medial malleolus and talar dome.

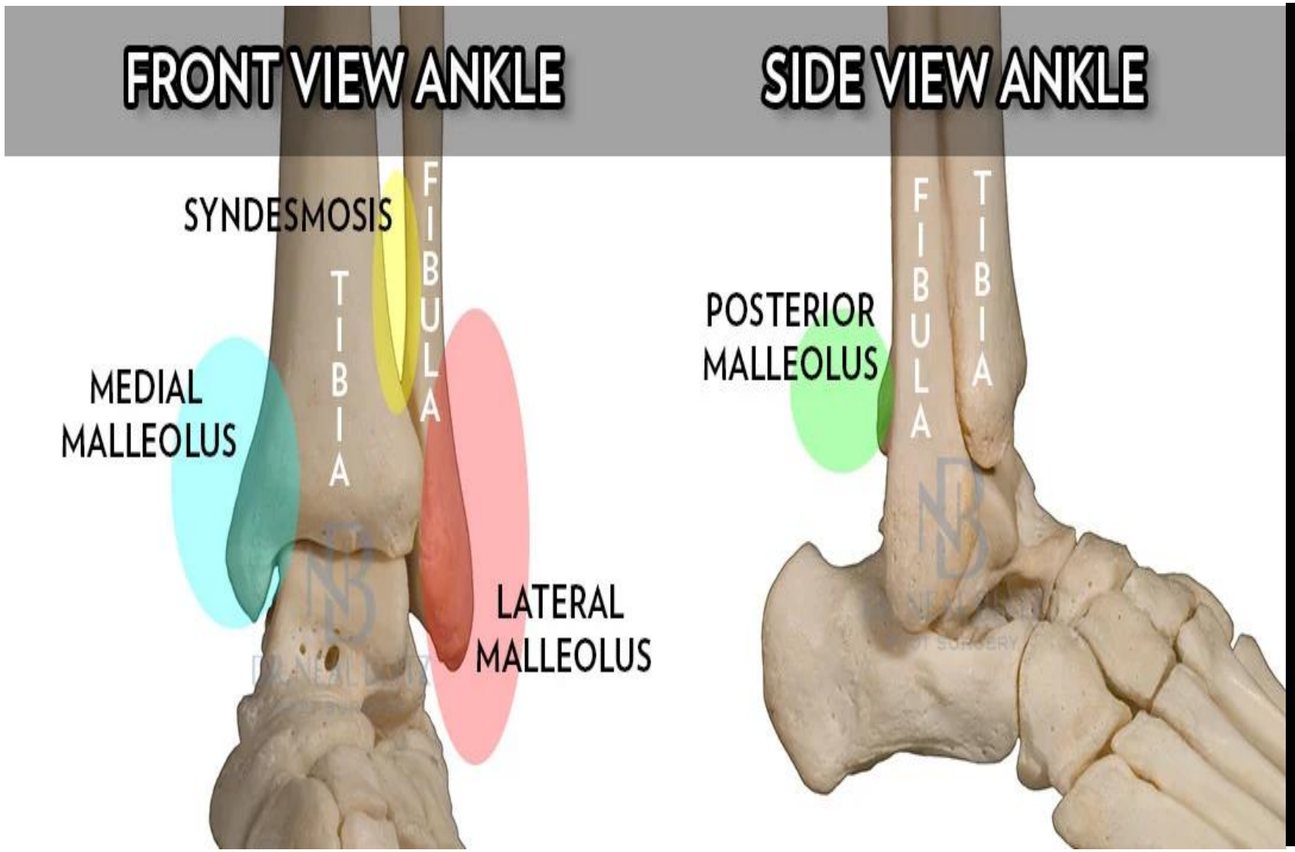
❑ Type C

- Fracture above the level of the syndesmosis (supra-syndesmotic).
- Tibiofibular syndesmosis disruption with widening of the distal tibiofibular articulation.
- Medial malleolus fracture or deltoid ligament injury often present.

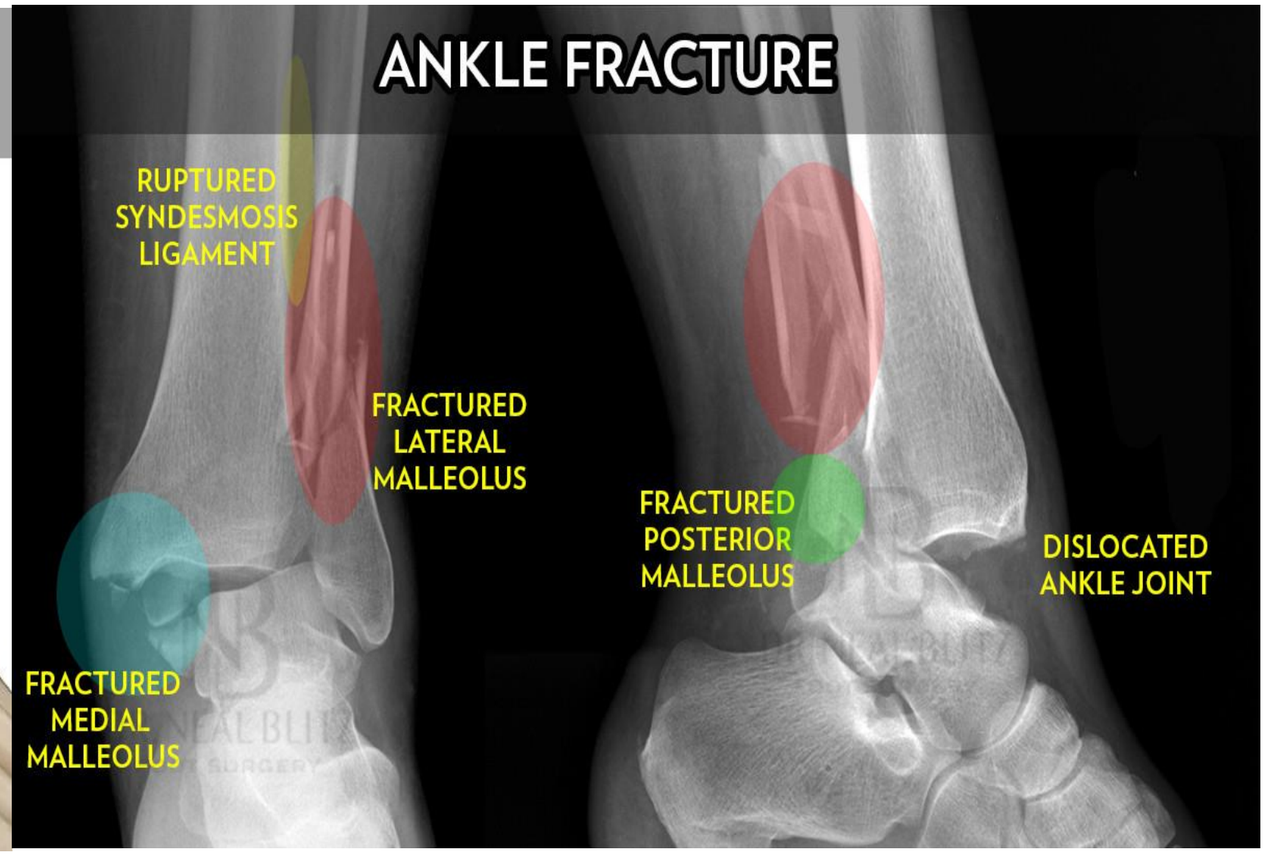
Weber classification of ankle fractures

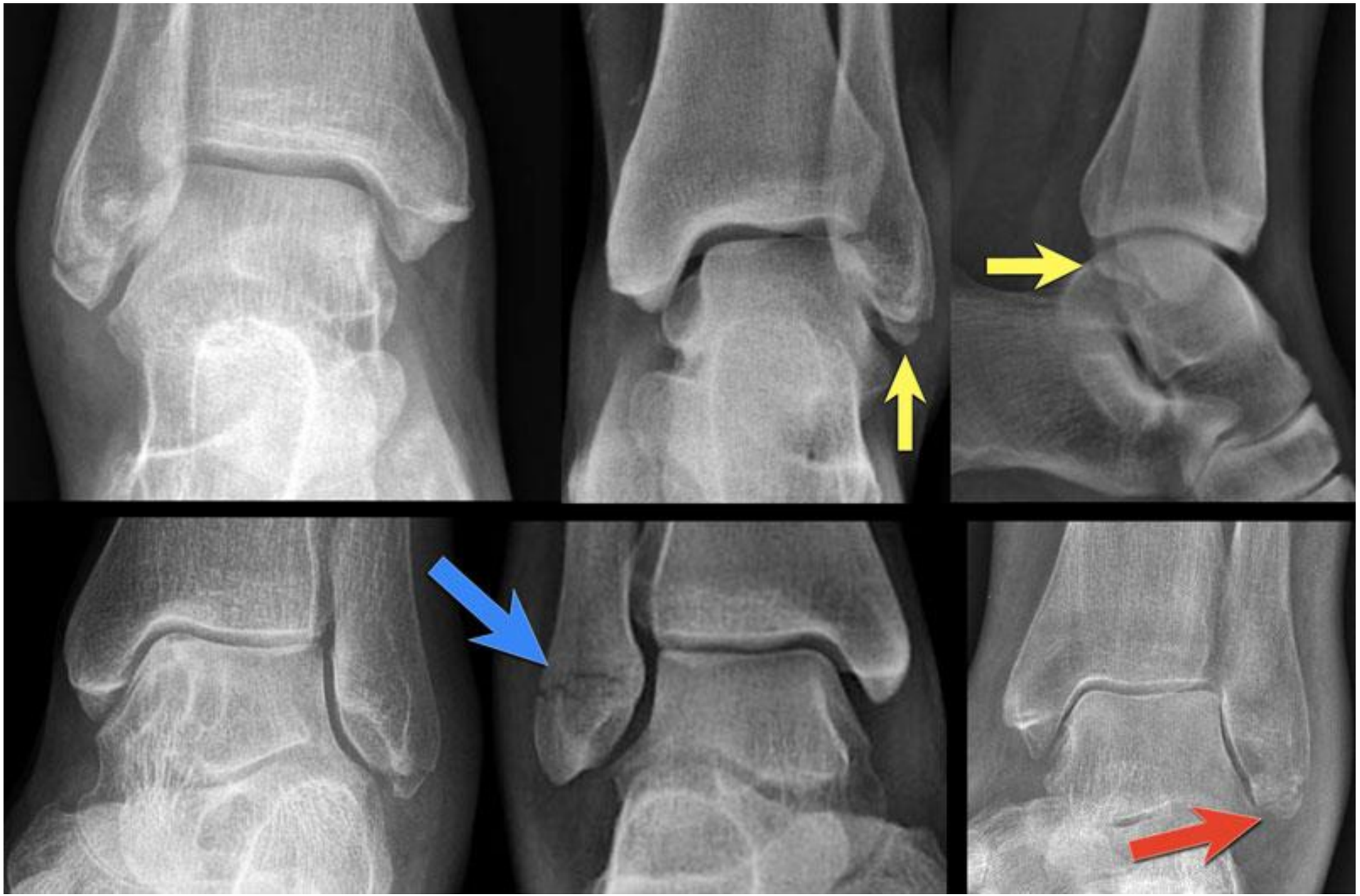





FRONT VIEW ANKLE
SYNDESMOSIS
TIBIA
FIBULA
MEDIAL MALLEOLUS
LATERAL MALLEOLUS

SIDE VIEW ANKLE
POSTERIOR MALLEOLUS
FIBULA
TIBIA


ANKLE FRACTURE
RUPTURED SYNDESMOSIS LIGAMENT
FRACTURED LATERAL MALLEOLUS
FRACTURED MEDIAL MALLEOLUS
FRACTURED POSTERIOR MALLEOLUS
DISLOCATED ANKLE JOINT



Weber A fracture

Weber B fracture

- Oblique fibula fracture at the level of the syndesmosis - i.e. Weber B fracture.
- Fracture - seen on AP view (red arrow) and on lateral view (yellow arrow).
- Rupture of the medial collateral ligaments - seen as widening of the space between the medial malleolus and the talus (medial clear space)

Weber B fracture

- There is an oblique fracture of the fibula.
- There is an avulsion of the posterior malleolus and an avulsion of the medial malleolus.

Weber C fracture

- There is an avulsion fracture of the medial malleolus and a fibula fracture above the level of the syndesmosis.

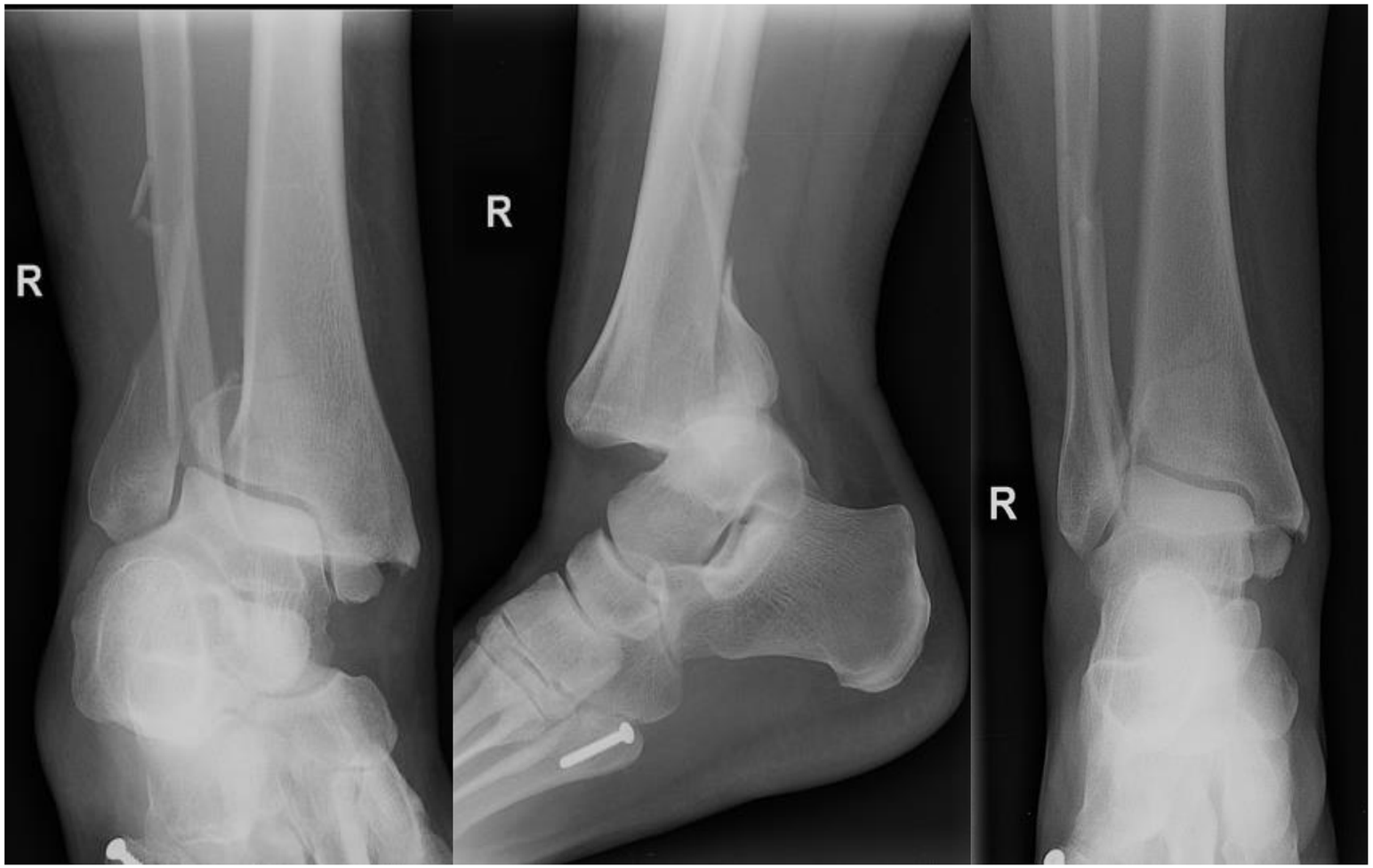
Weber C fracture

Weber C fracture

- Radiograph at the time of injury. A Weber type C ankle fracture is observed. Closed reduction was not possible.

Trimalleolar fracture

- Trimalleolar fractures refer to a three-part fracture of the ankle.
- The fractures involve the medial malleolus, the posterior aspect of the tibial plafond (referred to as the posterior malleolus) and the lateral malleolus.
- Having three parts, this is a more unstable fracture and may be associated with ligamentous injury.
- A trimalleolar fracture is a serious injury that will dramatically reduce a person's mobility and quality of life while the injury heals.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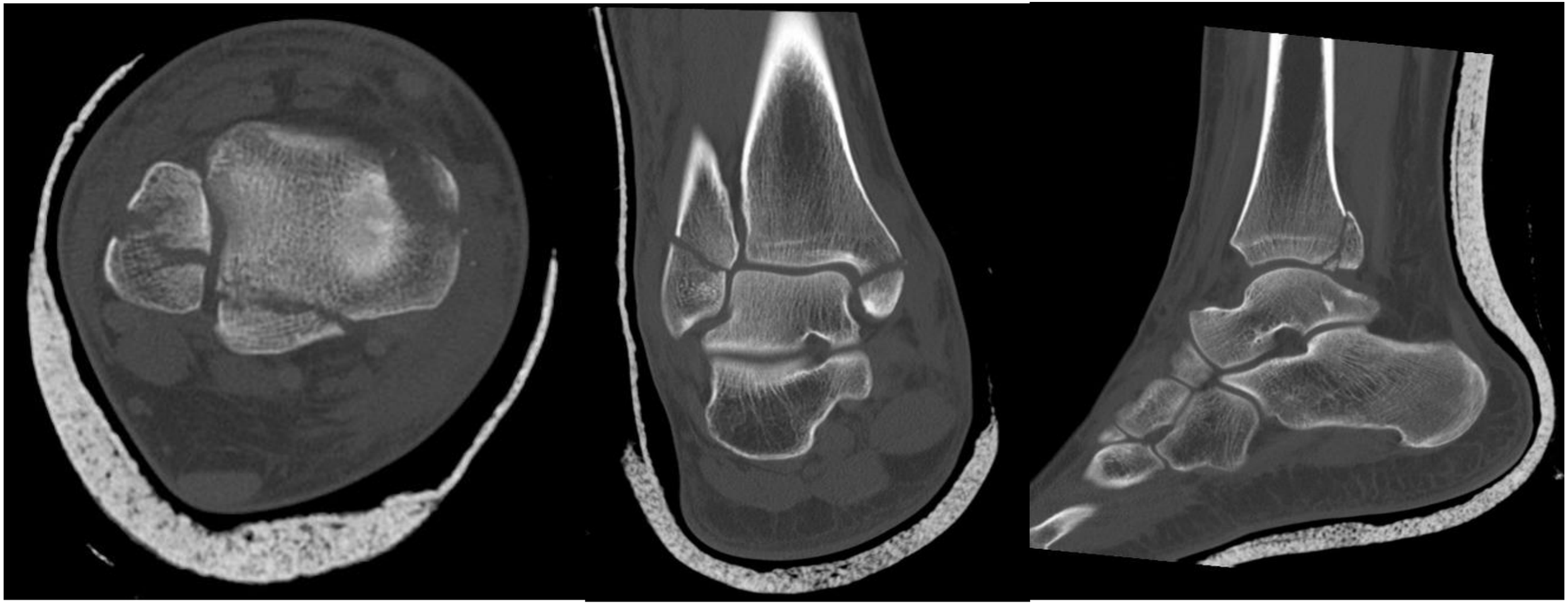


Oblique fracture through the distal fibular metaphysis extending into the ankle joint. Distal tibiofibular articulation is not widened. Fracture through the medial tibial malleolus associated with tiny bone loose bodies. Comminuted fracture involving the posterior aspect of the distal tibial plafond, also with intraarticular extension. No other fractures seen within the mid and hindfoot. Small sclerotic foci within the talus and medial cuneiform bones are consistent with enostoses. Soft tissue swelling around the ankle joint.

Foot fractures

1. **Lisfranc injury**
2. **Chopart injury**
3. **Snowboarders fracture**
4. **Calcaneal fracture**
5. **Fractures of 5th metatarsal bone**

A. Stress fracture

B. Jones fracture

C. Avulsion fracture

- **Lisfranc injury**
- The Lisfranc joint is the articulation of the tarsus with the metatarsal bases, whereby the first three metatarsals articulate respectively with the three cuneiforms, and the 4th and 5th metatarsals with the cuboid.
- The Lisfranc ligament is a strong band attaching the medial cuneiform to the 2nd metatarsal base on the plantar aspect of the foot.
- Lisfranc injuries, also called Lisfranc fracture-dislocations, are the most common type of dislocation involving the foot and correspond to the dislocation of the articulation of the tarsus with the metatarsal bases.

Types of Lisfranc injury

1. **Homolateral:** a homolateral injury is a lateral displacement of the 1st to 5th metatarsals, or of 2nd to 5th metatarsals where the 1st MTP joint remains congruent.
2. **Isolated:** this involves one or two metatarsals that dislocate dorsally in isolation.
3. **Divergent:** a divergent injury is a lateral dislocation of the 2nd to 5th metatarsals with medial dislocation of the 1st metatarsal.

- **Radiographic features**
- **Plain radiograph/CT**
- These injuries are well demonstrated on the standard views of the foot but subtle injuries may be missed and require further imaging such as CT, MRI.
- CT is however favored as it will also demonstrate unsuspected associated fractures.
- The key finding is malalignment of the second tarsometatarsal joint, such as lateral displacement of the second metatarsal base on AP view and/or dorsal step-off sign on lateral view.
- Associated fractures most often occur at the base of the second metatarsal, seen as the fleck sign.

Homolateral Lisfranc fracture-dislocation.

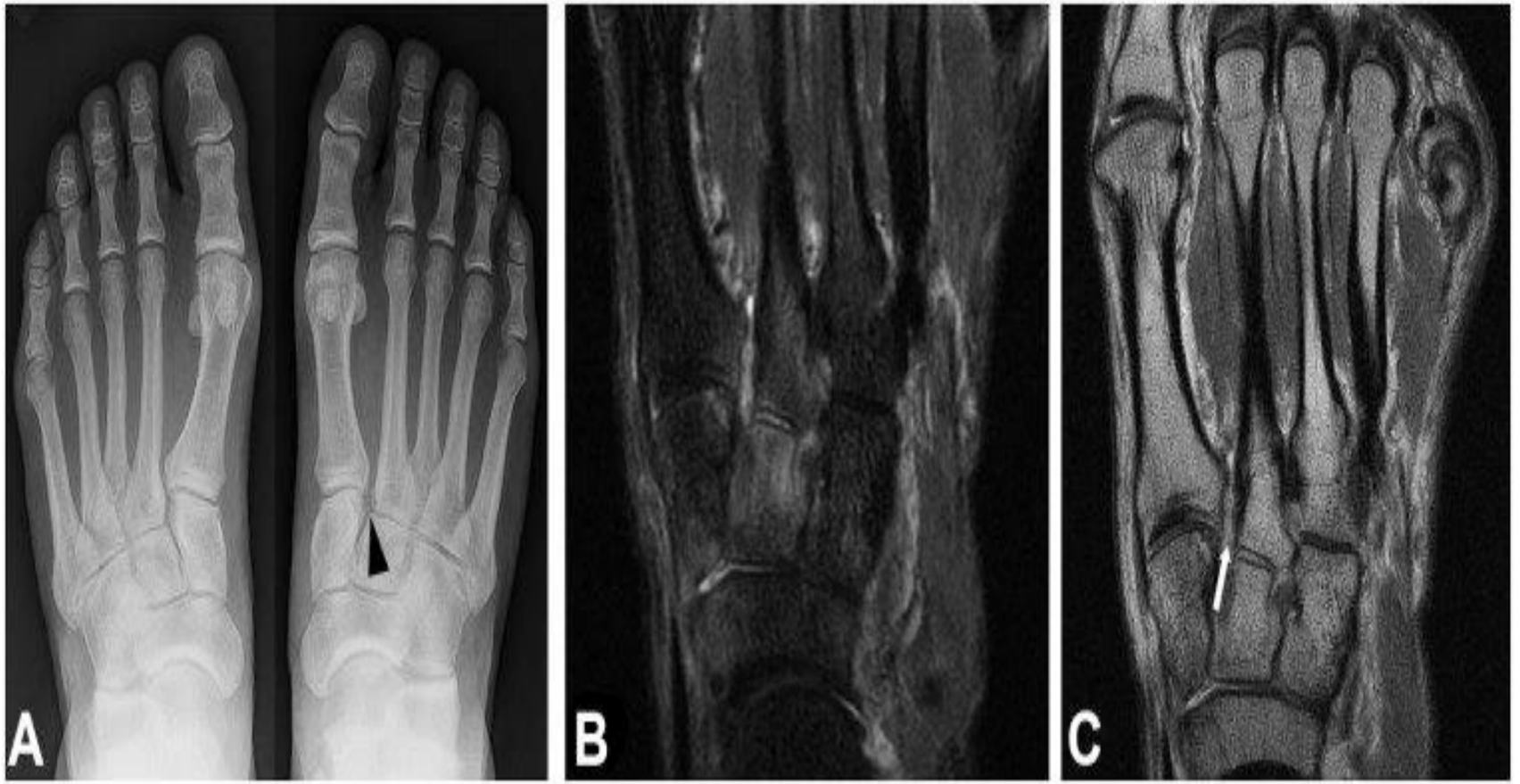
The step-off sign is the radiographic appearance of Lisfranc injury on lateral projections of the foot when there is a dorsal displacement of the second metatarsal base.

Lateral displacement of the lesser metatarsals with respect to the first metatarsal with widening of the space between the 1st and 2nd metatarsal base, with an intra-articular fracture from the medial margin of the base of the 2nd metatarsal. Homolateral Lisfranc fracture dislocation.

Fleck sign indicating avulsion of the Lisfranc ligament.

Multiplanar reformation axial CT image demonstrates the ‘Fleck sign’, a fracture fragment in the interspace between the first and second metatarsal base indicated avulsion of the Lisfranc ligament.

AP radiograph showing a divergent type Lisfranc injury

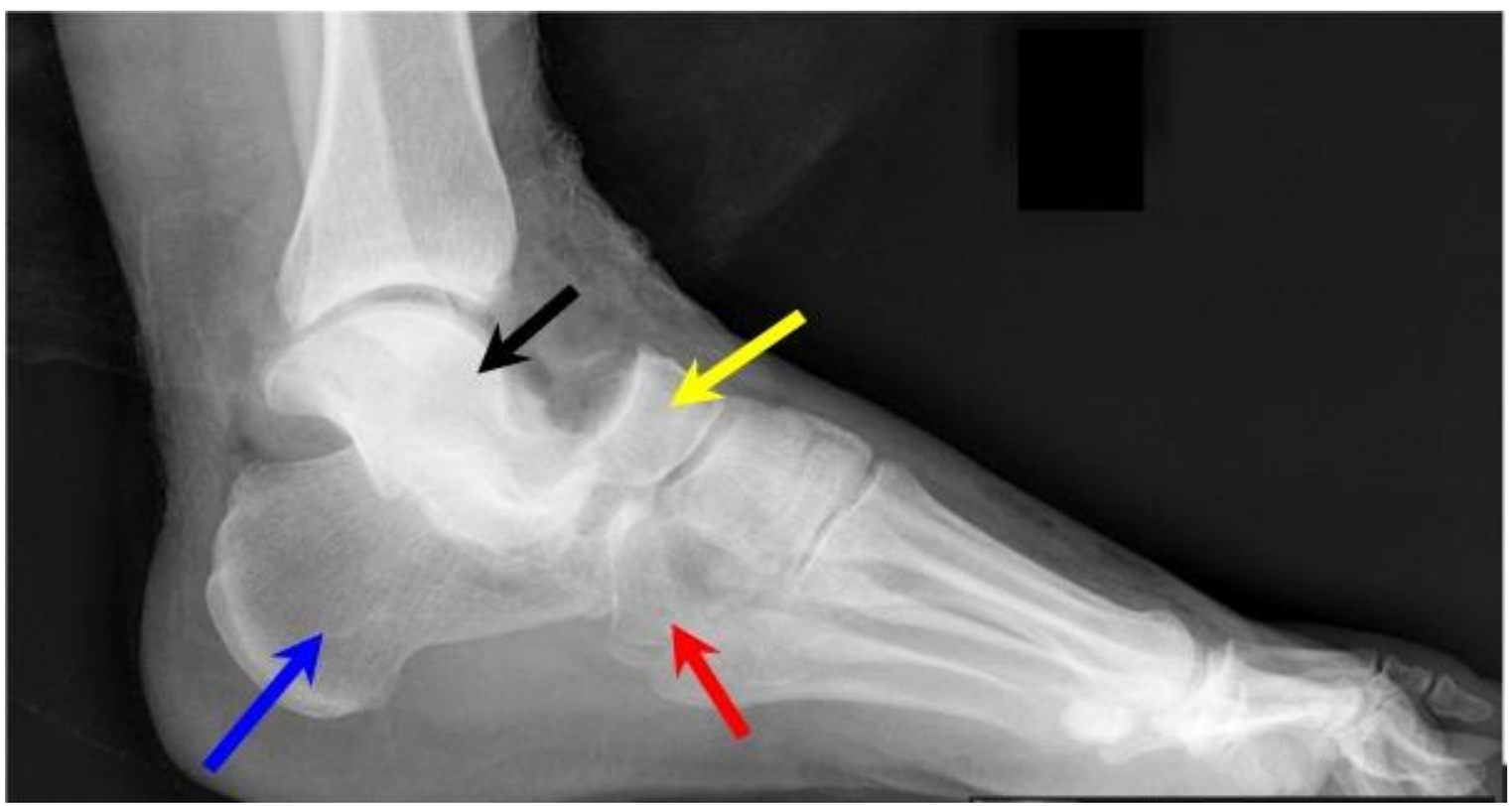


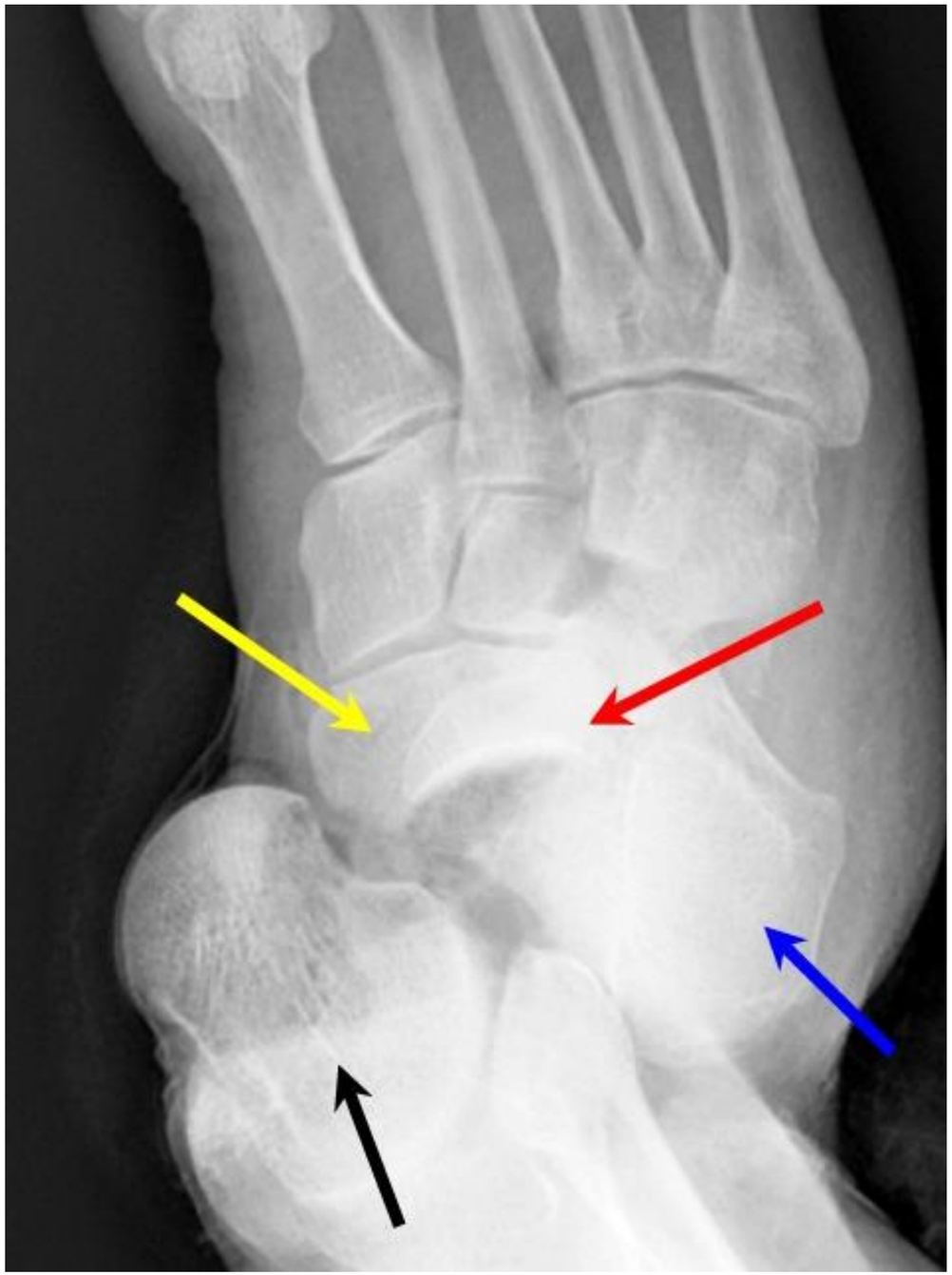
Isolated Lisfranc ligament injury. Radiographs of the bilateral feet (A) demonstrate widening of the right Lisfranc interval (black arrowhead). MRI (B) demonstrates marrow edema signal within the medial and middle cuneiforms and first and second metatarsal bases, (C) demonstrates disruption of the dorsal fibers of the Lisfranc ligament (white arrow).

- **Chopart injury**
- Chopart fracture-dislocations occur at the midtarsal (Chopart) joint in the foot, i.e. talonavicular and calcaneocuboid joints which separate the hindfoot from the midfoot.
- There are frequently associated fractures of the calcaneus, cuboid and navicular.
- The foot is usually dislocated medially and superiorly as it is plantarflexed and inverted, usually as a result of a high-energy impact, e.g. fall from height or road traffic collision.

Chopart's fracture dislocation.

- Black arrow points to talus which is dislocated from navicular (yellow arrow) at talonavicular joint.
- Calcaneus (blue arrow) is dislocated from the cuboid (red arrow), which is also fractured. The dislocation is at the calcaneocuboid joint.
- The forefoot is usually displaced medially rather than laterally as in this case.





- **Snowboarders fracture**

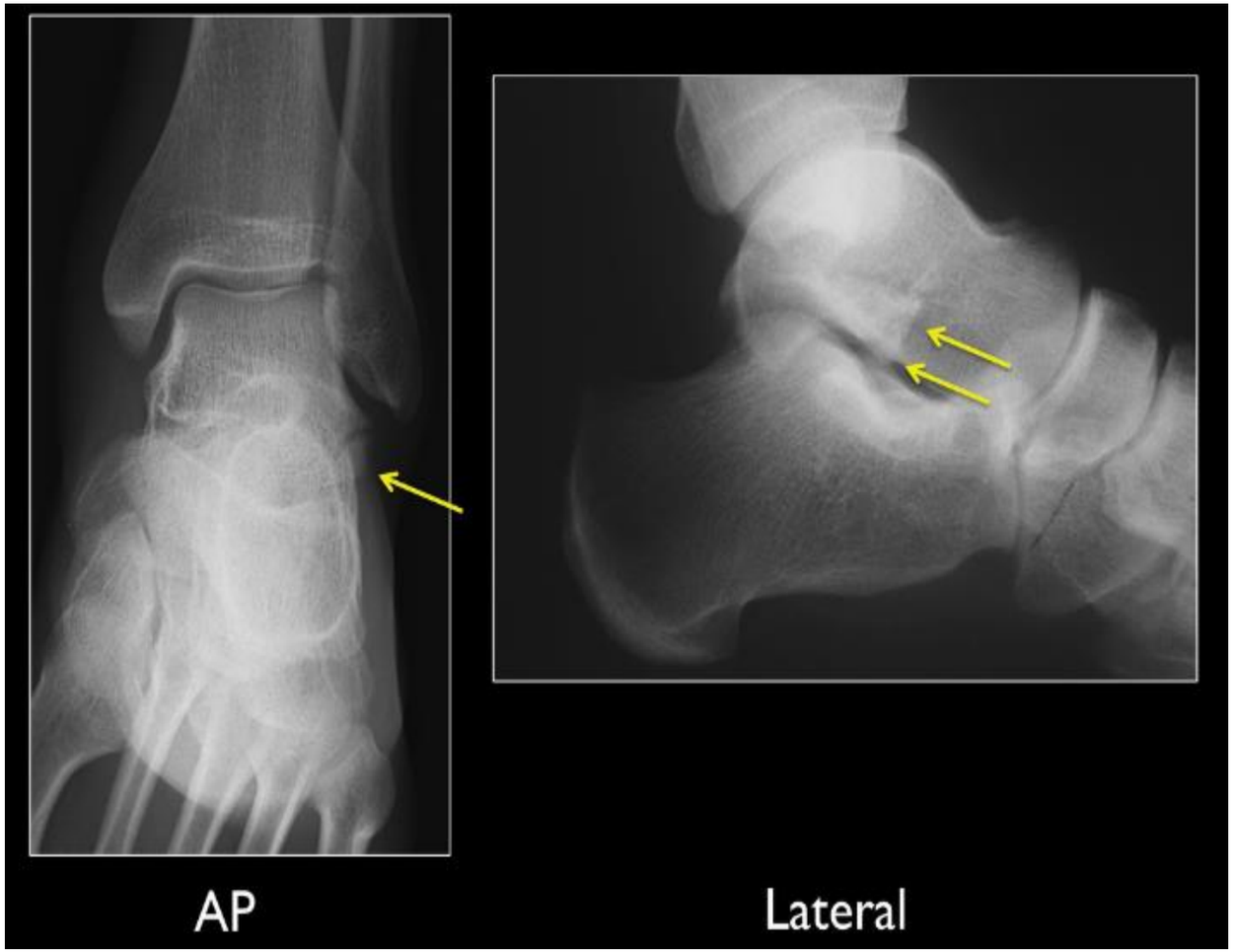
(Lateral talar process fractures)

- Fractures of the lateral process of the talus.
- It may be an isolated fracture or occur as a component of more complex ankle fractures.
- The fracture occurs when the foot is dorsiflexed and inverted, as can happen with snowboarding (hence the term "snowboarder fracture").
- A fracture should be suspected when there is soft tissue swelling inferior to the lateral malleolus.

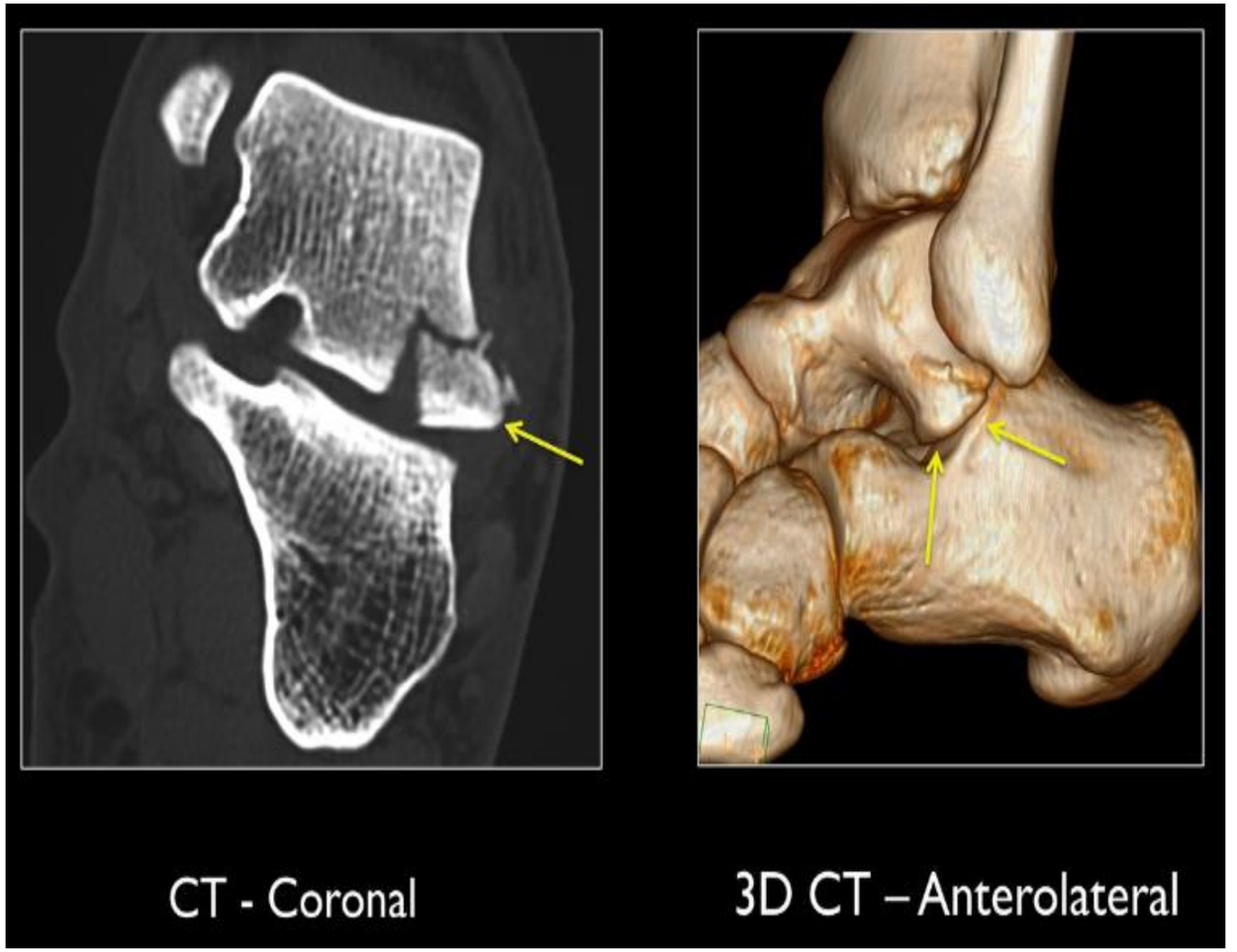
X-ray image demonstrating a lateral process fracture. Notice that the fracture is difficult to identify leading many of these injuries to be initially missed

CT scan demonstrating the same fracture. Note the small fragments in the joint and the displacement between the joint surface

Front and side views of the same lateral process fracture treated with 2 screws for fixation



AP and lateral radiograph of the left ankle. Yellow arrows = small bony fragment at the lateral process of the talus



Coronal CT (bone window) and anterolateral 3D-CT showing a bony fragment of 12mm originating from the lateral process of the talus (yellow arrow). Lateral displacement of 3mm was measured.

- **Calcaneal fracture**
- Calcaneal fractures are the most common tarsal fracture and accounts for about 2% of all fractures and ~60% of all tarsal fractures.
- Calcaneal fractures can be divided broadly into two types depending on whether there is articular involvement of the subtalar joint

1. **Intra-articular:** 70-75%: intra-articular body fracture
2. **Extra-articular:** 25-30%: anterior calcaneal process fracture, calcaneal tuberosity avulsion fracture, extra-articular body fracture.

- **Imaging Findings**
- Conventional radiographs are the study of first choice.

❑ Lateral and axial views.

❑ Radiographs of the thoraco-lumbar spine may also be indicated.

- **CT** is the examination of choice to demonstrate all of the fragments and their relationships.
- **Boehler's angle** is measured at the intersection of a line drawn between the posterior superior aspect of the calcaneal tuberosity to the highest point of the posterior articular facet and another line drawn to the anterior process of the calcaneus.
- It is normally 25-40°.

- As the calcaneous compresses secondary to a fracture, there will be a decrease in Boehler's angle to less than 20° (frequently to 0°)

P

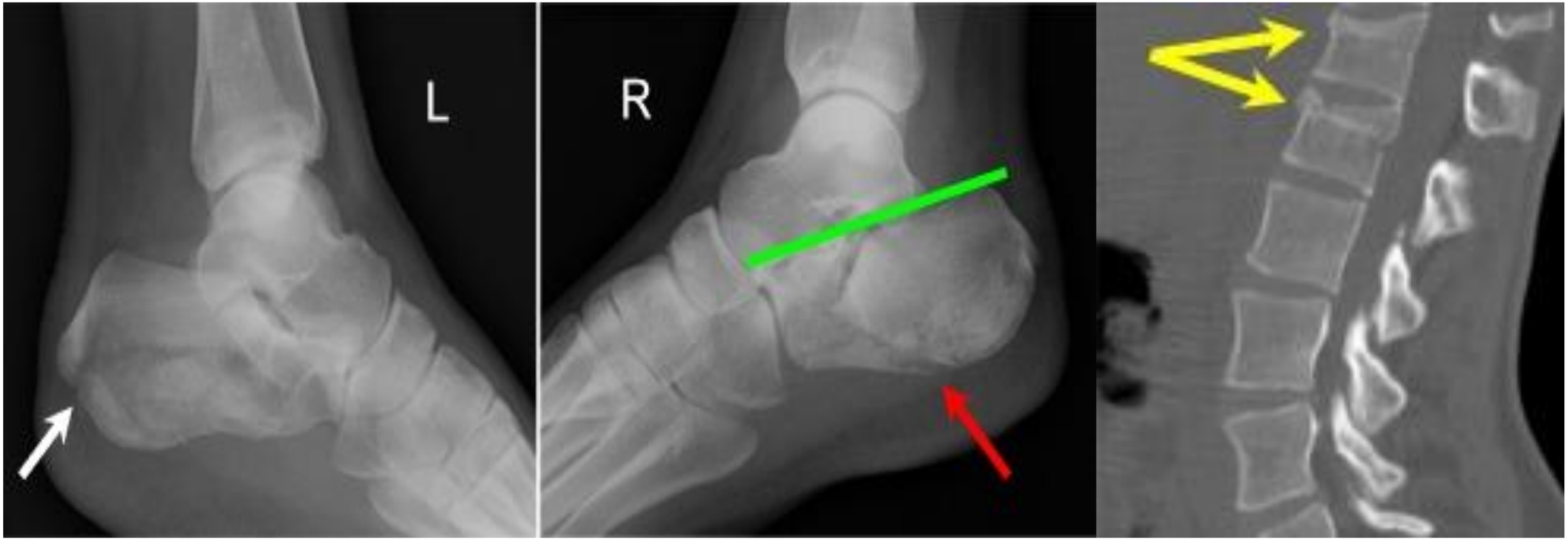
I

A
B
C
D

MRI

Normal

Stress fractures of the calcaneus



Bilateral calcaneal fractures and fractures of spine. There is a comminuted fracture of the left calcaneus (white arrow); there is a comminuted fracture of the left calcaneus (red arrow) with flattening of Boehler's Angle to 0 degrees. A sagittal reconstructed CT scan of the spine shows compression fractures of the superior endplates of T12 and L1 (yellow arrows) .

- **Fractures of 5th metatarsal bone**

1. Stress fracture

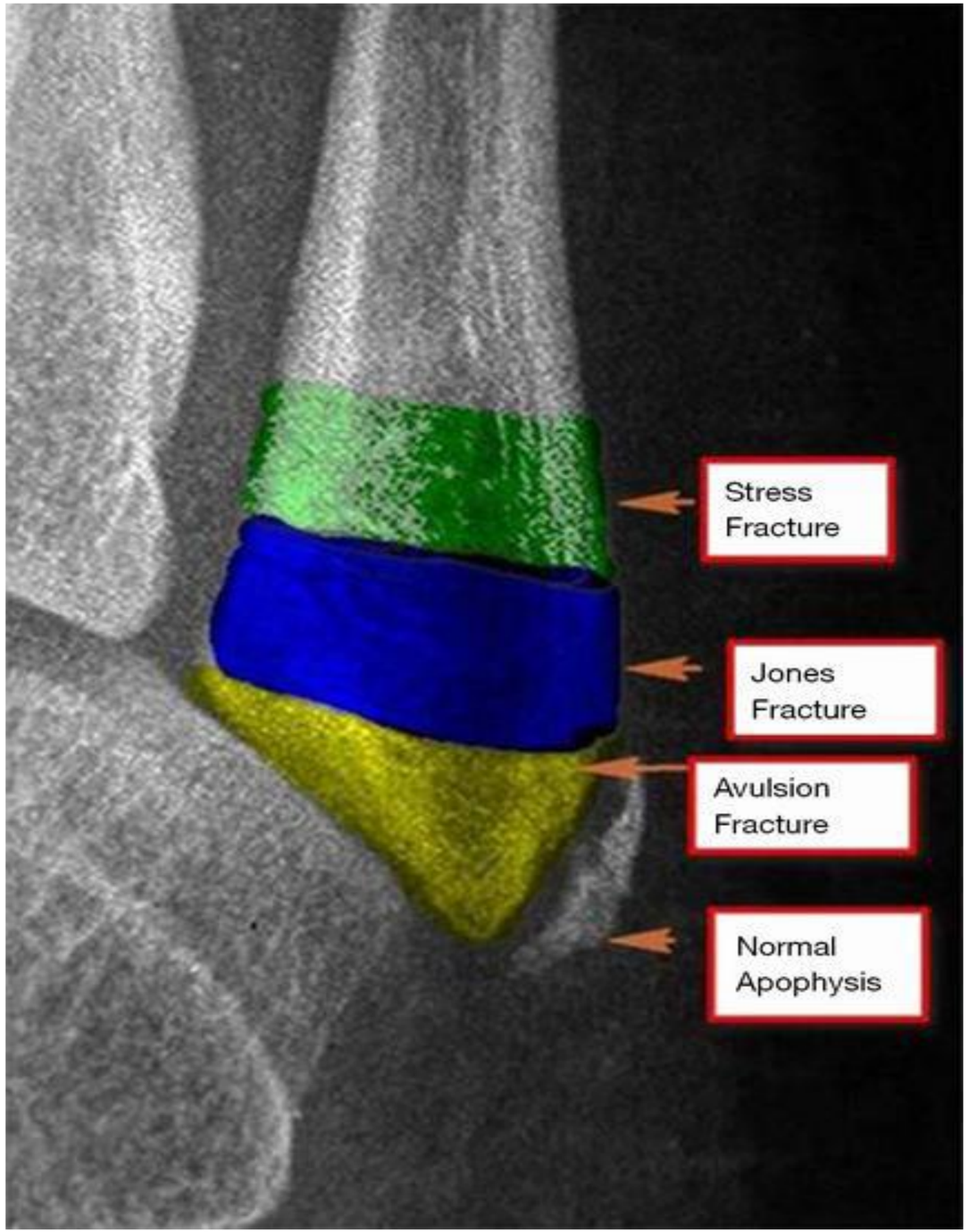
- Stress fractures typically occur within 1.5 cm of the metadiaphyseal junction, slightly distal to the site of a Jones fracture. At the proximal diaphysis, distal to the level of the fourth-fifth intermetatarsal articulation.

2. Jones fracture

- A Jones fracture is an extra-articular fracture at the base of the fifth metatarsal bone. At the metaphysis-diaphysis junction, at the level of the fourth-fifth intermetatarsal articulation

3. Avulsion fracture

- Avulsion fracture of the 5th metatarsal styloid, also known as a pseudo-Jones fracture or a dancer fracture, is one of the more common foot avulsion injuries and accounts for over 90% of fractures of the base of the 5th metatarsal.



Proximal fractures of the 5th metatarsal

Location	Associated with
Proximal diaphysis	Stress fracture
Metaphysis	Jones fracture
Tuberosity	Avulsion fracture
Apophysis	Normal at 10 - 16 years
Os vesalianum	Present in 0.1–1%

Jones fracture. Fracture of the metaphysis of the fifth metatarsal

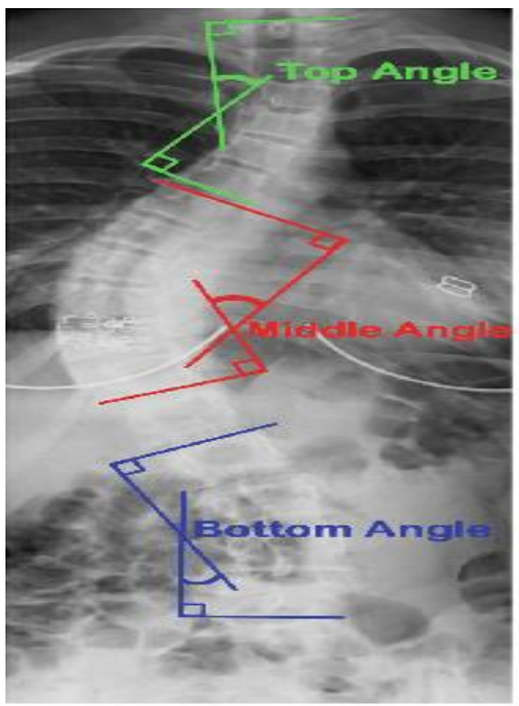
Fracture line run transversely. Apophysis lies longitudinal to long axis of metatarsal

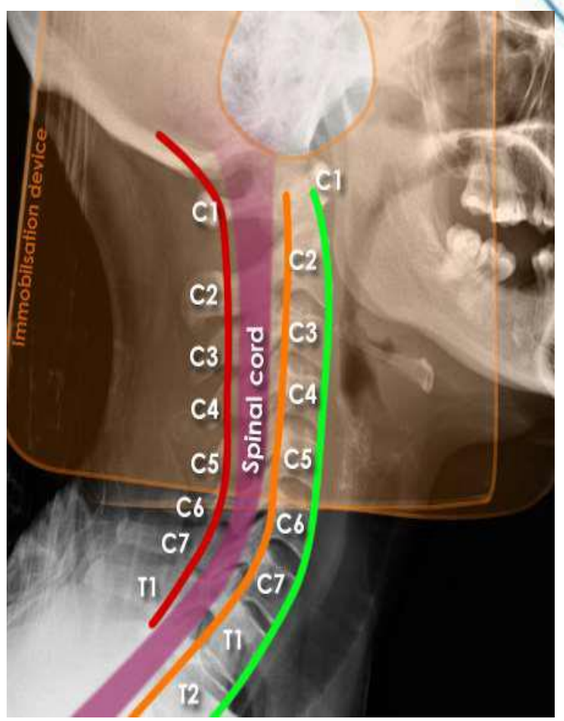
Avulsion the 5th metatarsal with proximal retraction of the fragment.

Avulsion the 5th metatarsal: Oblique radiograph of the foot shows a transverse avulsion fracture (arrow) of the 5th metatarsal bone

Thank you

Al-Azhar University – Gaza
Faculty of Applied Medical Sciences





Diagnostic Radiology of Spine

Lecturer: Husam Mansour
Master of Public Health. Health Management
Master of Medical Imaging. Computed Tomography

Radiological Assessment of Cervical Spine

Clinical considerations are particularly important because

1. Normal C-spine X-rays cannot exclude significant injury.
2. A missed C-spine fracture can lead to death.
3. Life long neurological deficit.
4. Clinico-radiological assessment of spinal injuries.
5. Should be managed by experienced clinicians .

CT or MRI is often appropriate in the context of

1. High risk injury,
2. Neurological deficit,
3. Limited clinical examination, or
4. Where there are unclear X-ray findings.

Cervical Spine Systemic Approach: Normal

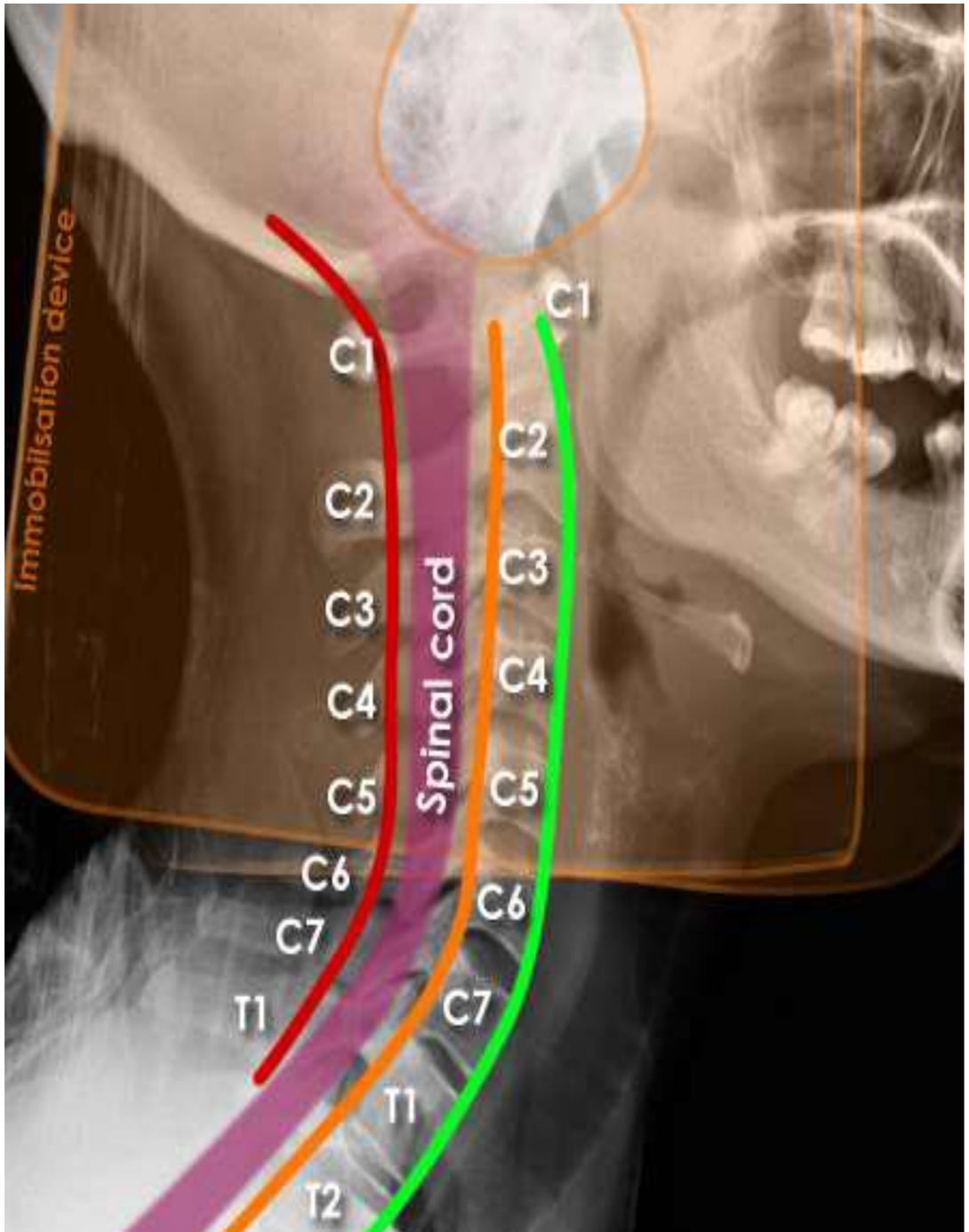
- **Coverage:** All vertebrae are visible (from the skull base to the top of T1).
- **Alignment:**

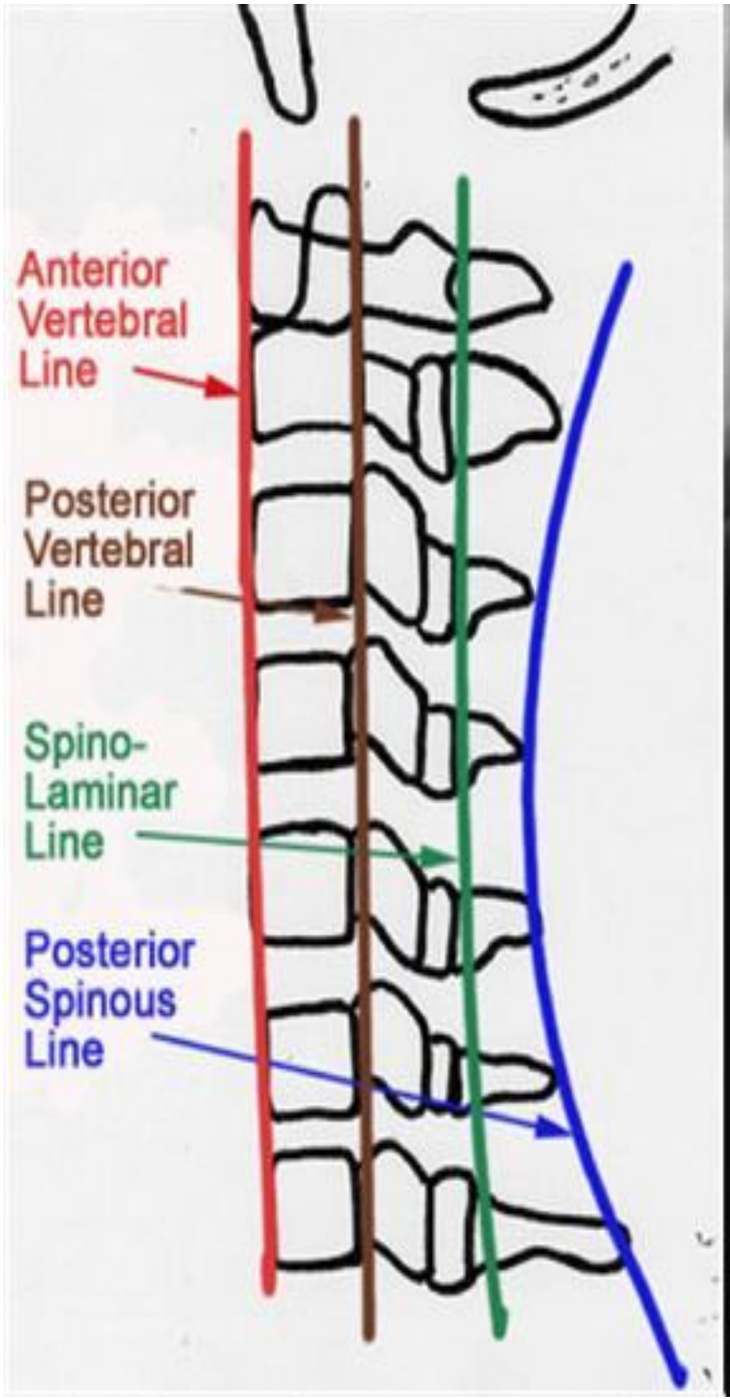
1. **Anterior line** (the line of the anterior longitudinal ligament).
2. **Posterior line** (the line of the posterior longitudinal ligament).
3. **Spinolaminar line** (the line formed by the anterior edge of the spinous processes - extends from inner edge of skull).

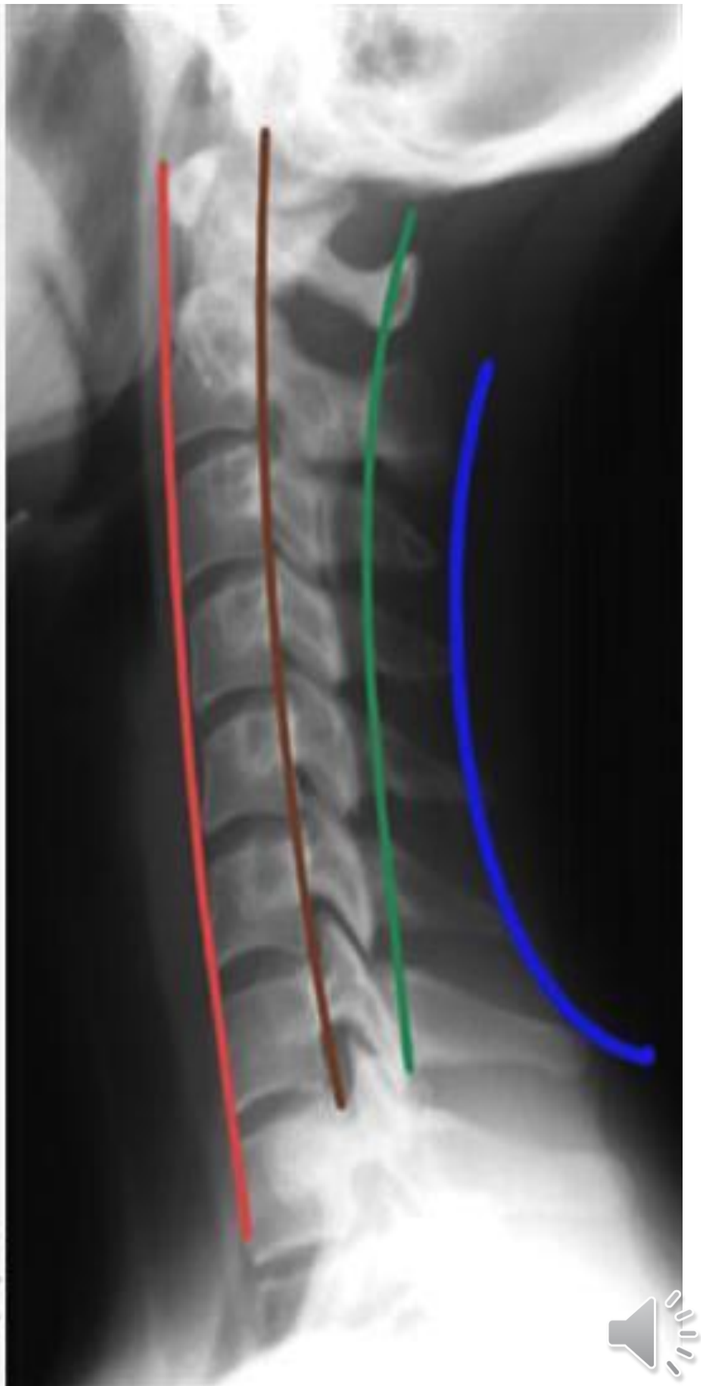
- **C2 Bone Ring** - At C2 (Axis) the lateral masses viewed side on form a ring of corticated bone.

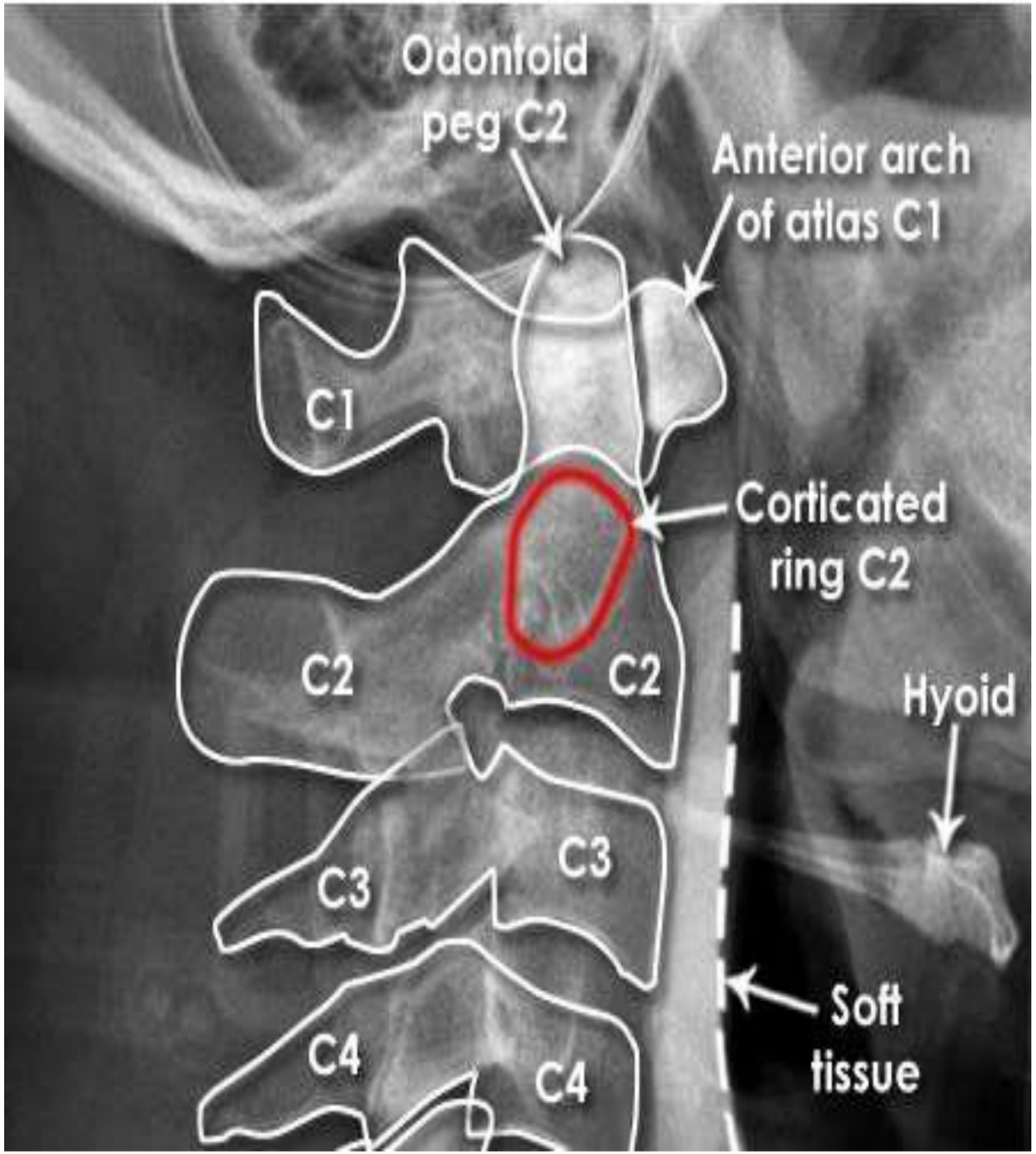
Note: The spinal cord (not visible) lies between the posterior and spinolaminar lines.

- **Disc spaces:** The vertebral bodies are spaced apart by the intervertebral discs - not directly visible with X-rays. These spaces should be approximately equal in height.
- **Prevertebral soft tissue:** Some fractures cause widening of the prevertebral soft tissue due to prevertebral haematoma:
  - **Normal prevertebral soft tissue – *narrow down to C4 and wider below***
    - *Above C4 ≤ 1/3rd vertebral body width*
    - *Below C4 ≤ 100% vertebral body width*

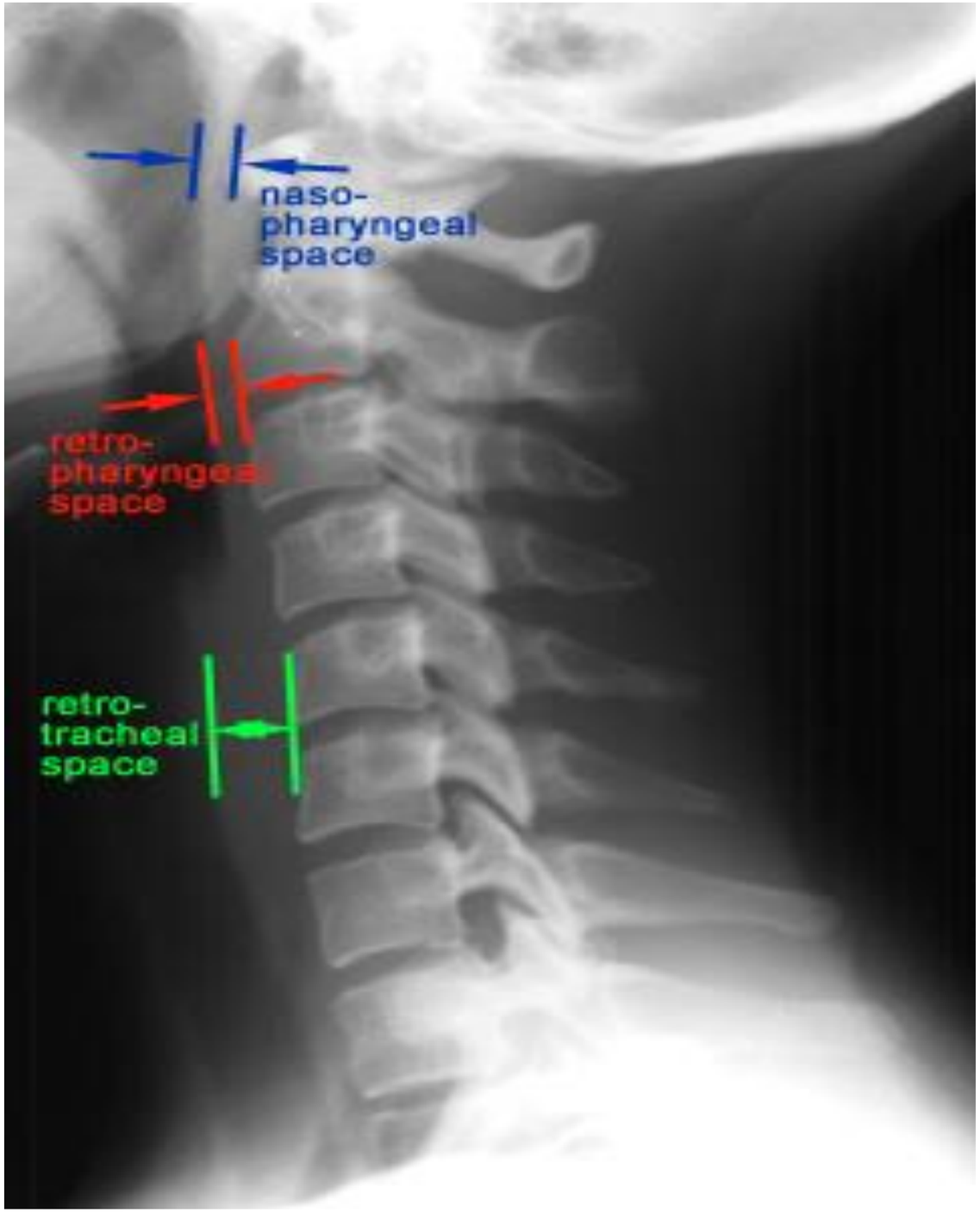

Immobilisation device
Spinal cord
C1
C2
C3
C4
C5
C6
C7
T1
T2

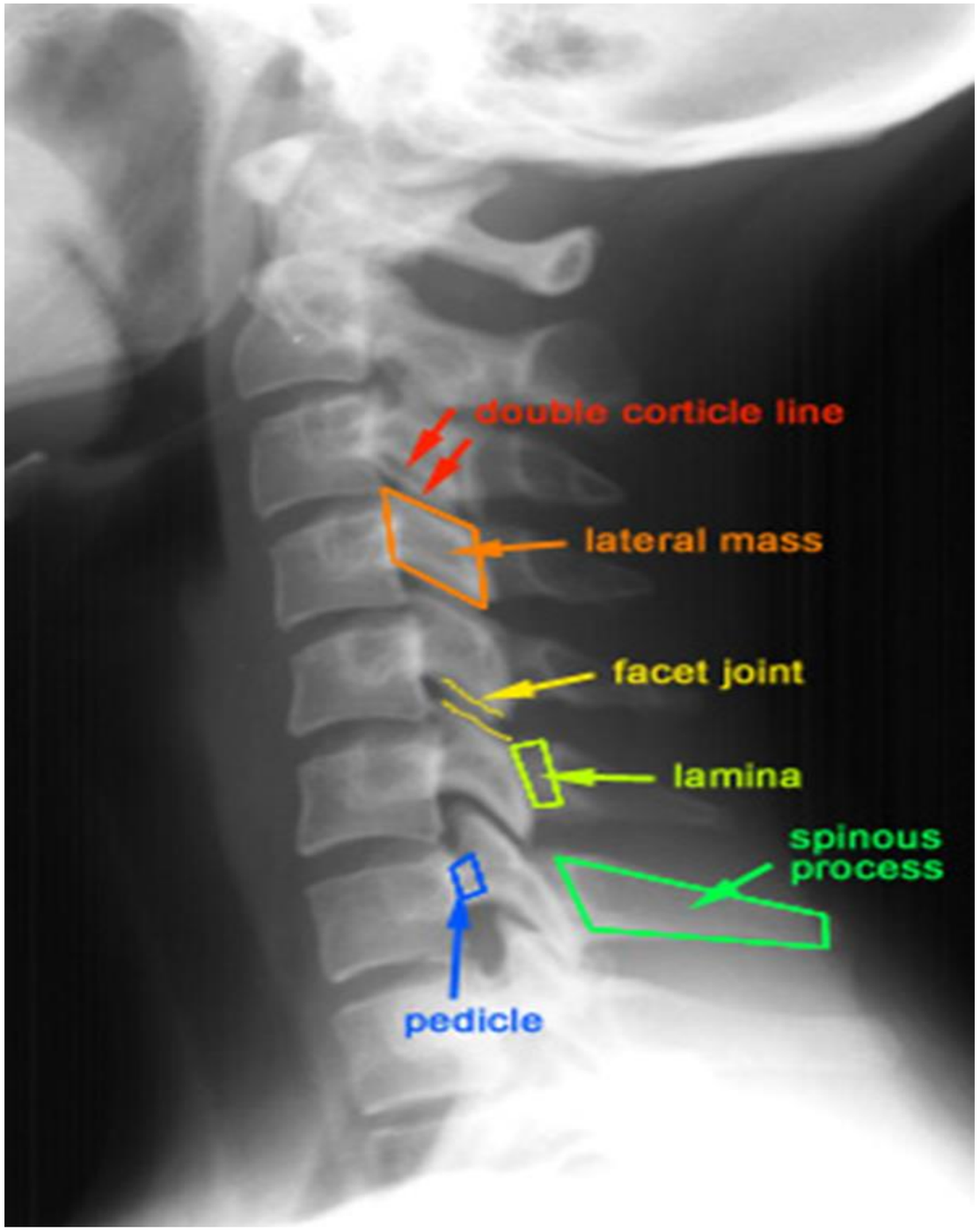

Anterior Vertebral Line
Posterior Vertebral Line
Spino-Laminar Line
Posterior Spinous Line




Odontoid peg C2
Anterior arch of atlas C1
C1
Corticated ring C2
C2
C2
Hyoid
C3
C3
Soft tissue
C4
C4

Disc spaces
C4
Trachea

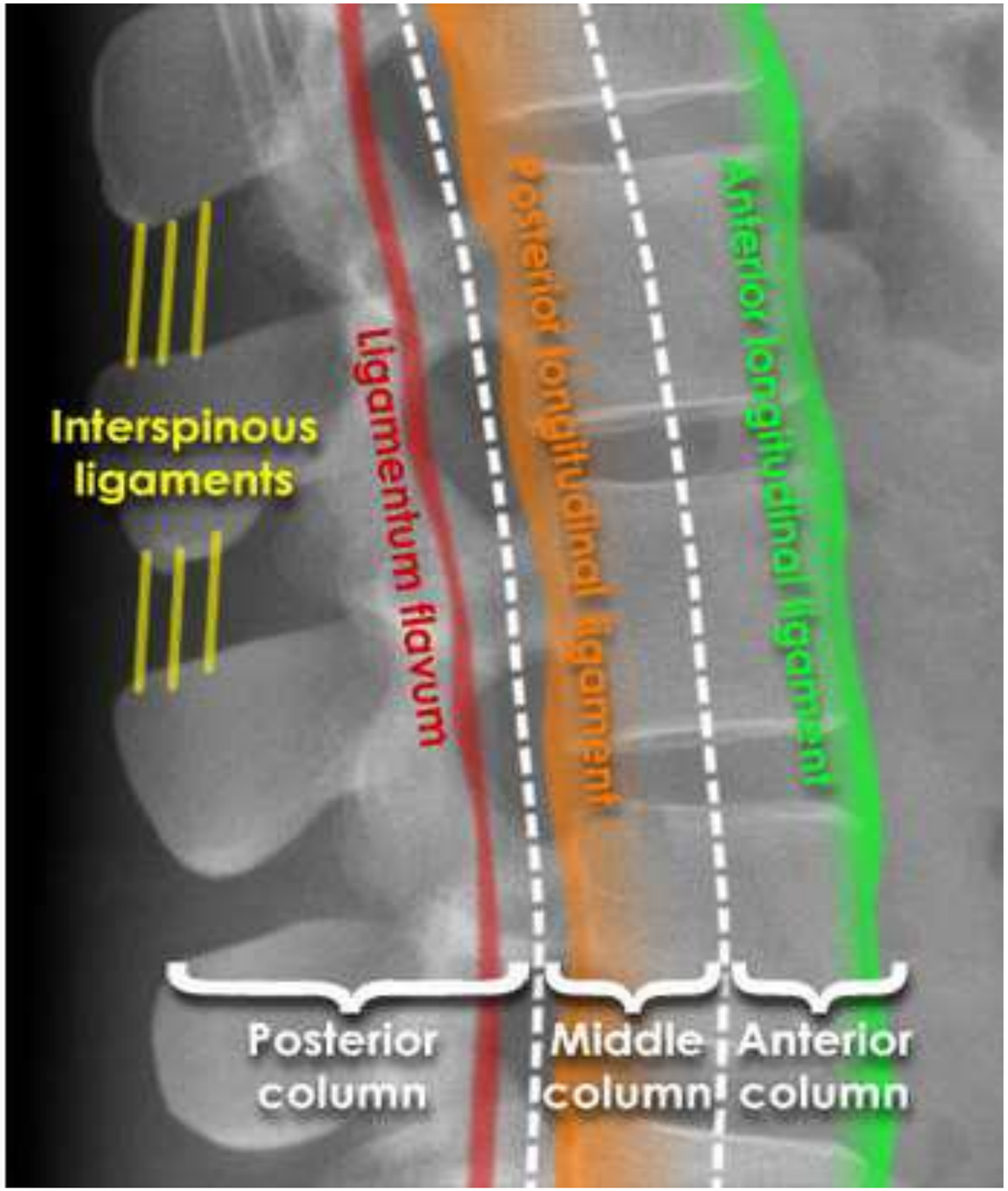

naso-
pharyngeal
space
retro-
pharyngeal
space
retro-
tracheal
space

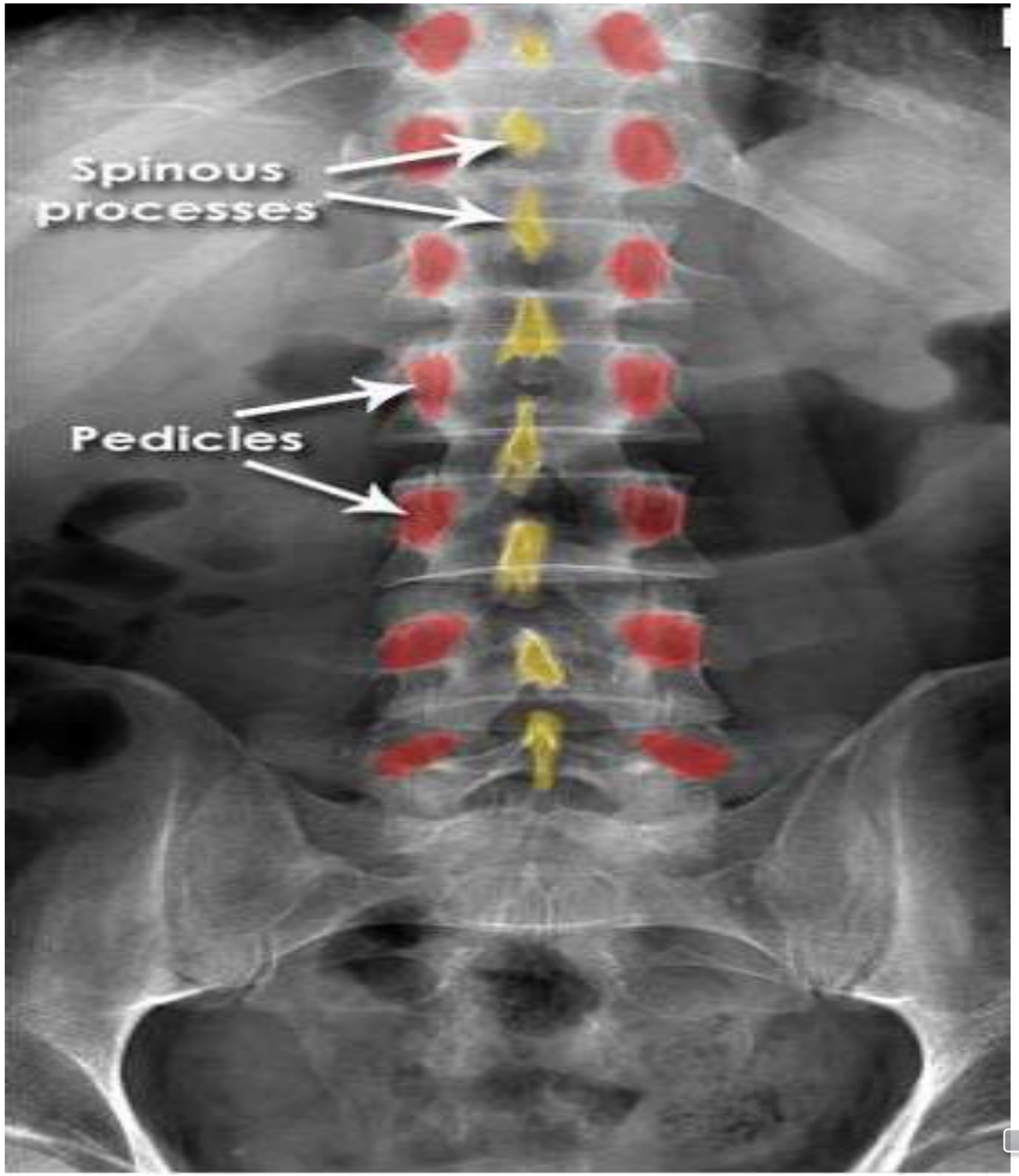

double corticle line
lateral mass
facet joint
lamina
spinous
process
pedicle

Lumber Spine –Systemic Approach

- **Alignment**: The vertebral bodies and spinous processes are aligned.
- **Bones:** The vertebral bodies and pedicles are Intact.
- **Spacing:** Gradually increasing disc height from superior to inferior.
- **Pedicles** gradually become wider apart from superior to inferior
- **Three column model of Lumber Spine**

1. **Anterior column:** Anterior half of the vertebral bodies and soft tissues.
2. **Middle column:** Posterior half of the vertebral bodies and soft tissues.
3. **Posterior column:** Posterior elements and soft tissues.


Interspinous ligaments
Ligamentum flavum
Posterior longitudinal ligament
Anterior longitudinal ligament
Posterior column
Middle column
Anterior column


Spinous processes
Pedicles

Spondylolisthesis

- Spondylolisthesis refers to a slip of one vertebra on another, usually forwards but it may occasionally be backwards.
- **Spondylolytic spondylolisthesis:**
- Is the most common type and results from bilateral defects in the pars interarticularis.
- Posterior body remains fixed in position while the anterior part is free to slip forward.
- Fractures of the pars may heal with a pseudarthrosis or fibrous ankylosis.

Spondylolisthesis

- **Degenerative spondylolisthesis:**
- Most frequent at the L4-L5 level.
- Is not the result of a pars defect but a complex interaction between the disk, facets joints and the ligamentous structures.
- There is usually narrowing, sclerosis of the facet joints from osteoarthritis.
- Retrolisthesis (backward slippage of a vertebral body on the body below it) may occur with osteoarthritis of the facets joints.

Spondylolisthesis

- **Dysplastic spondylolisthesis:**
- Results from congenital abnormalities of the body and/or facets in the lumbar region such that the alignment of the facets allows spondylolisthesis to occur.
- **Traumatic spondylolisthesis:**
- Trauma can lead to an acute fracture through a normal pars interarticularis

Spondylolisthesis

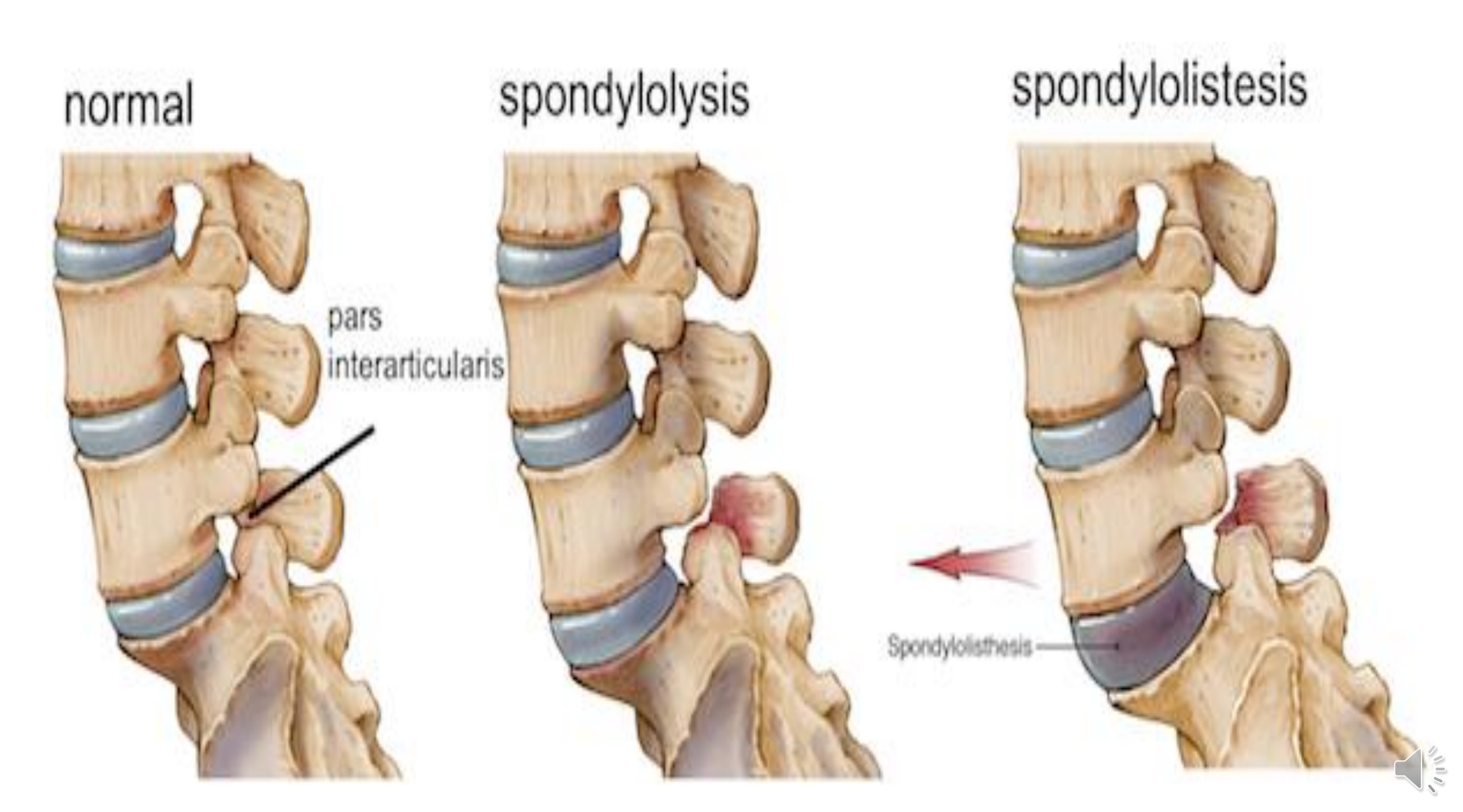
Grading of Spondylolisthesis

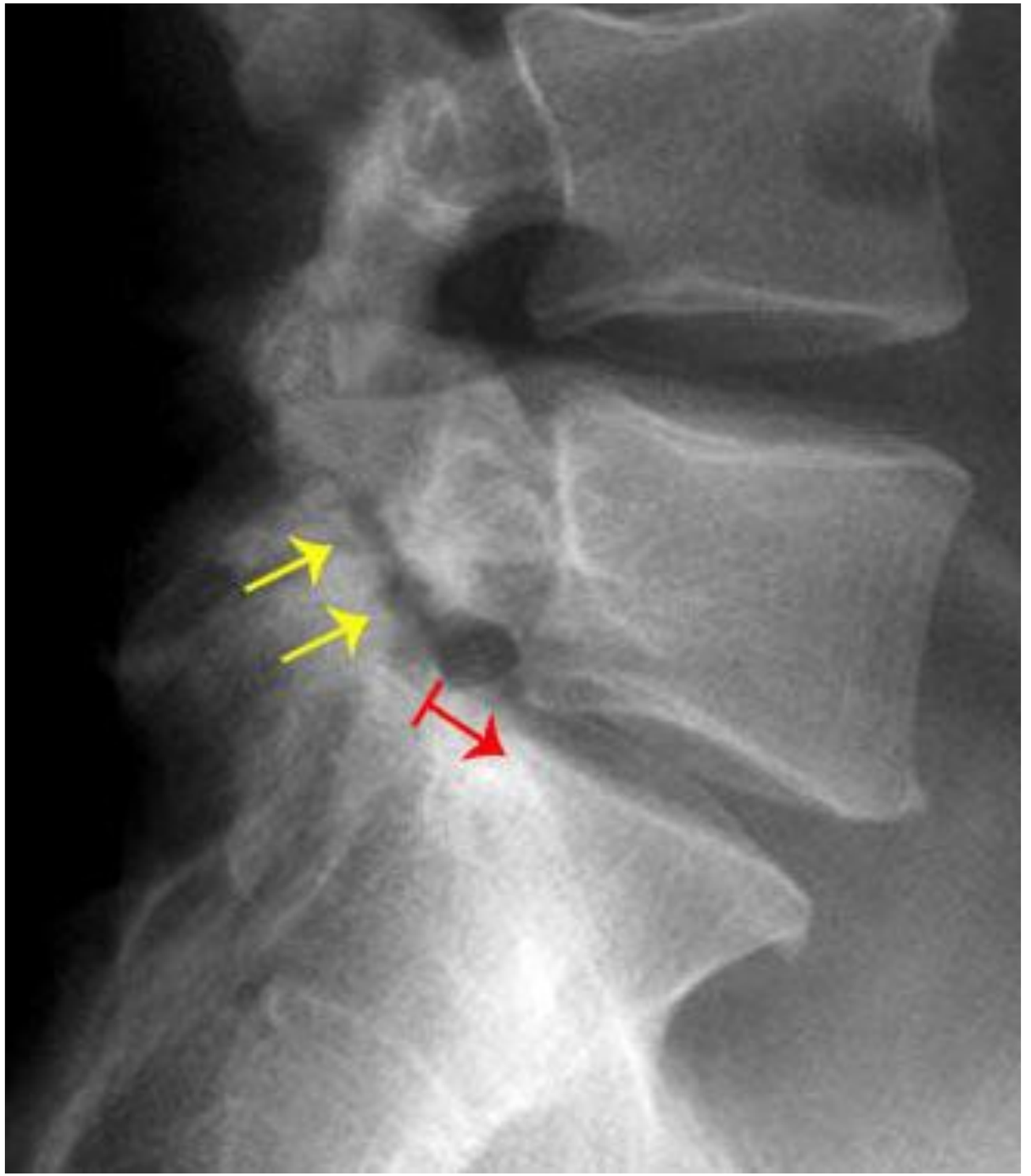
- **Grade 1**-vertebral body above subtends 1/4 of the AP diameter of the vertebral body below
- **Grade 2**- vertebral body above subtends 1/2 of the AP diameter of the vertebral body below
- **Grade 3**- vertebral body above subtends 3/4 of the AP diameter of the vertebral body below
- **Grade 4**- vertebral body above subtends the full AP diameter of the vertebral body below

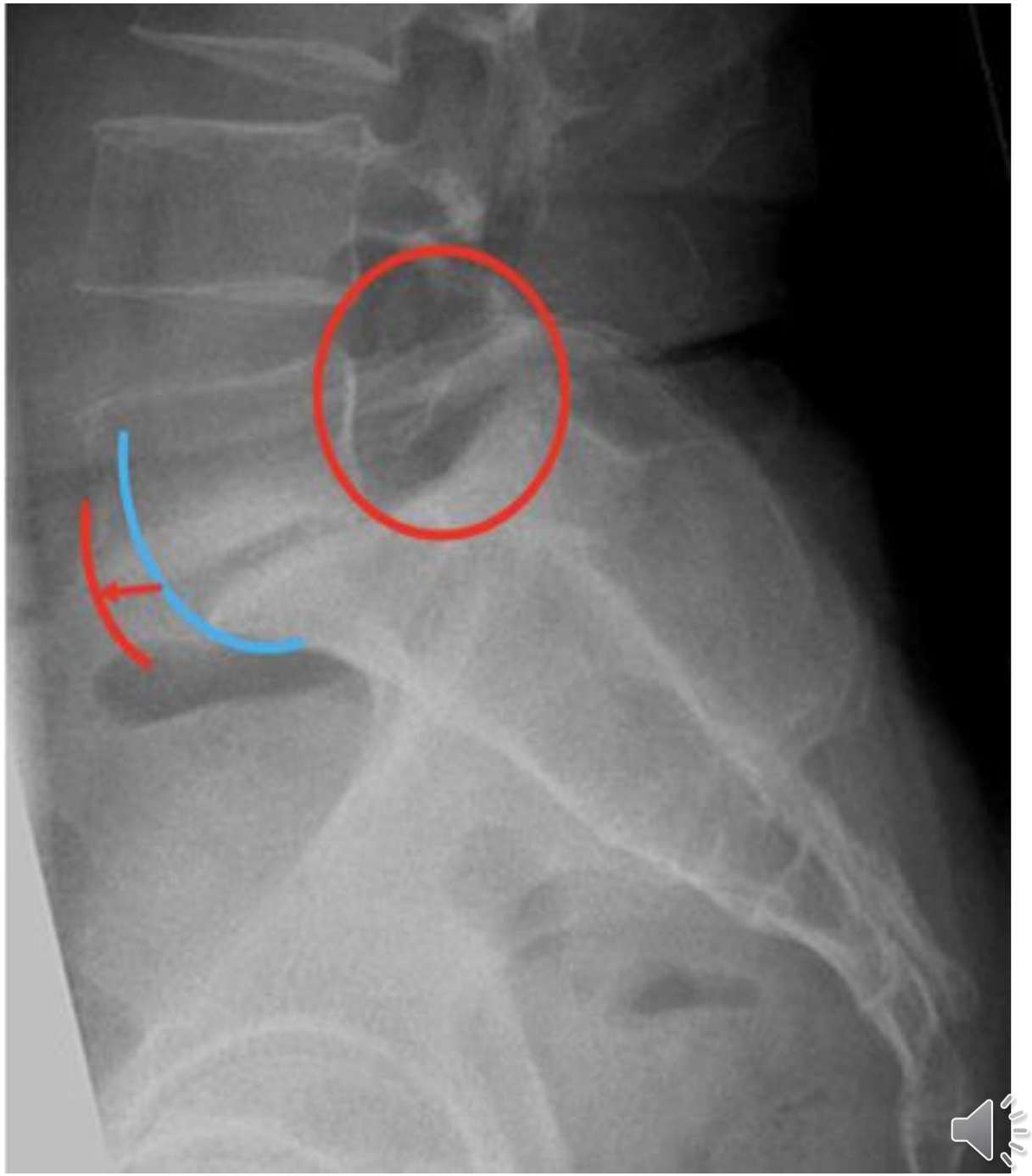
Spondylolisthesis

Imaging findings

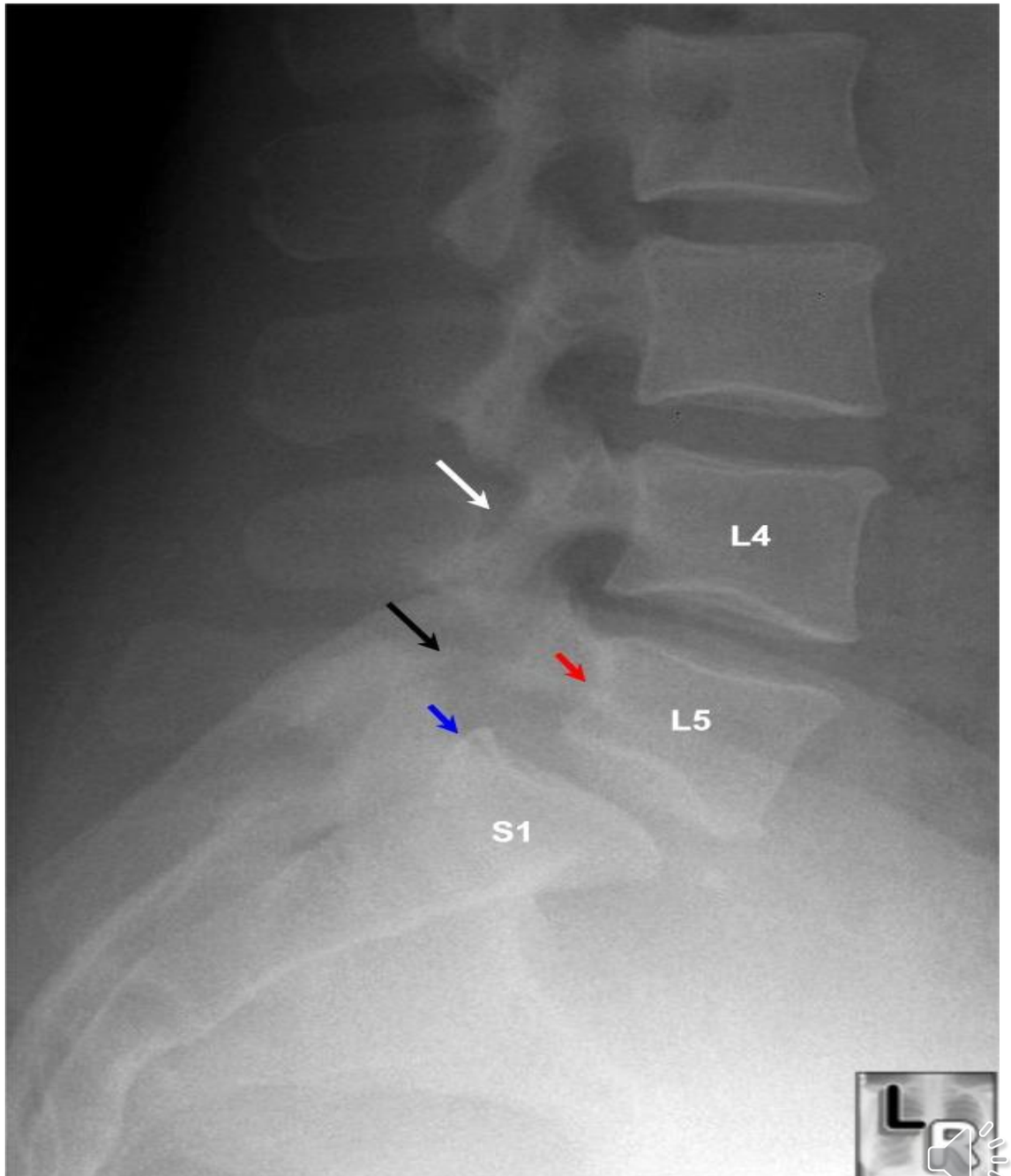
- Conventional radiography in the anteroposterior, lateral and both oblique projections is usually adequate to demonstrate both spondylolysis and spondylolisthesis
- Spondylolysis appears as a break in the “neck” of the Scottie Dog on the oblique view
- Bilateral spondylolysis is visible on the lateral view
- Forward slippage of one vertebral body on the other is observed and graded on the lateral view

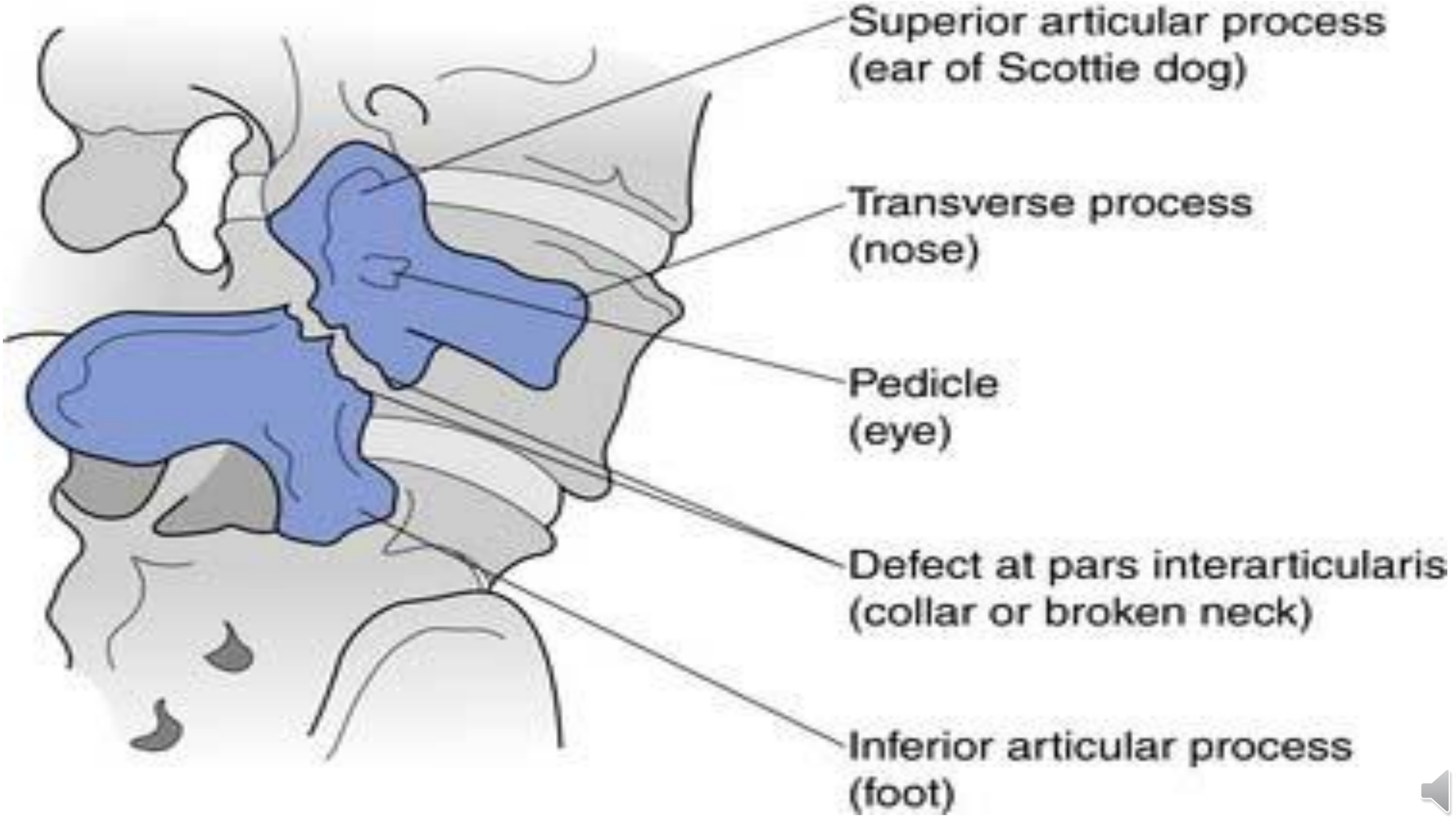

normal
pars
interarticularis
spondylolysis
spondylolistesis
Spondylolisthesis

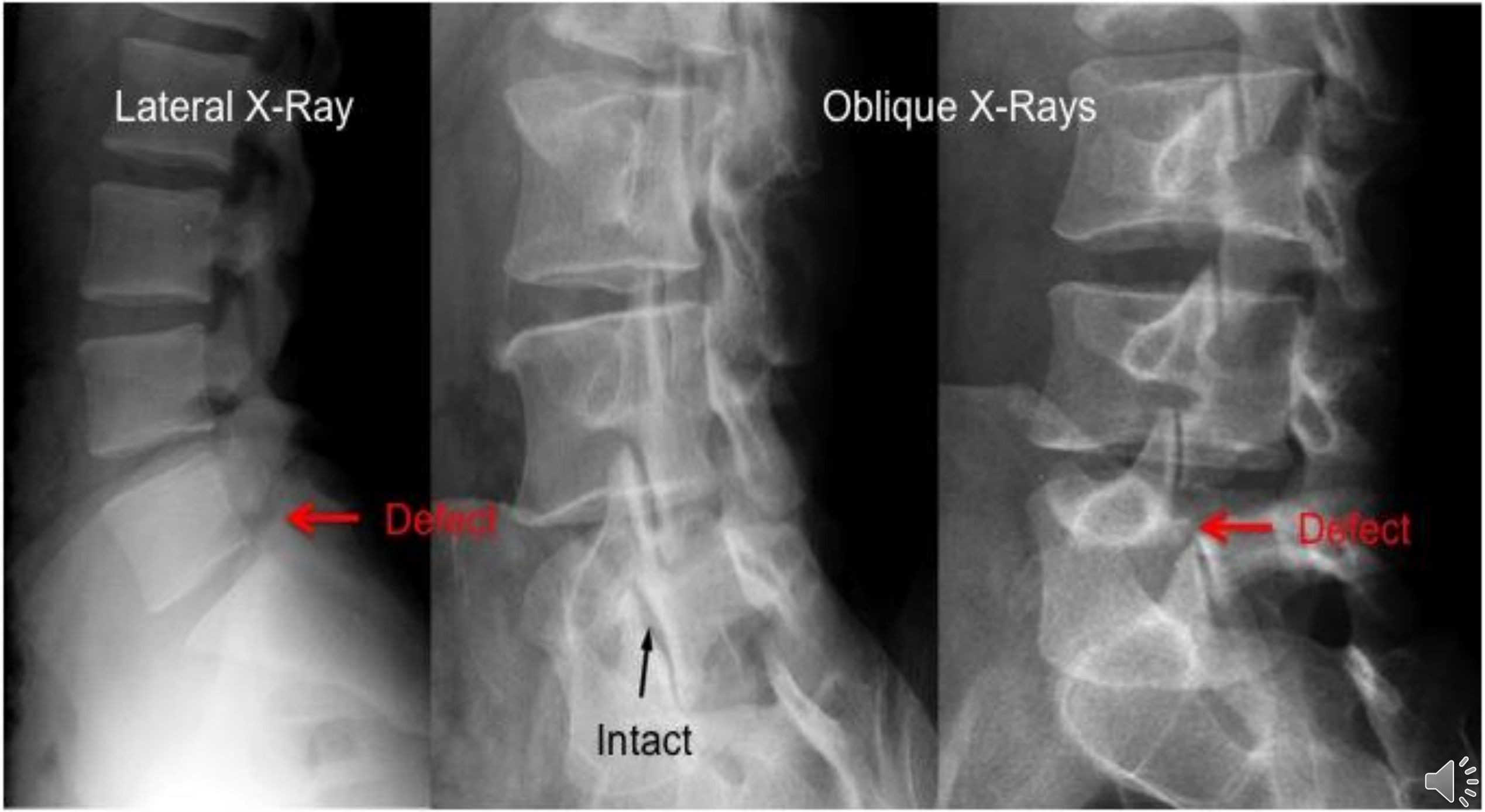


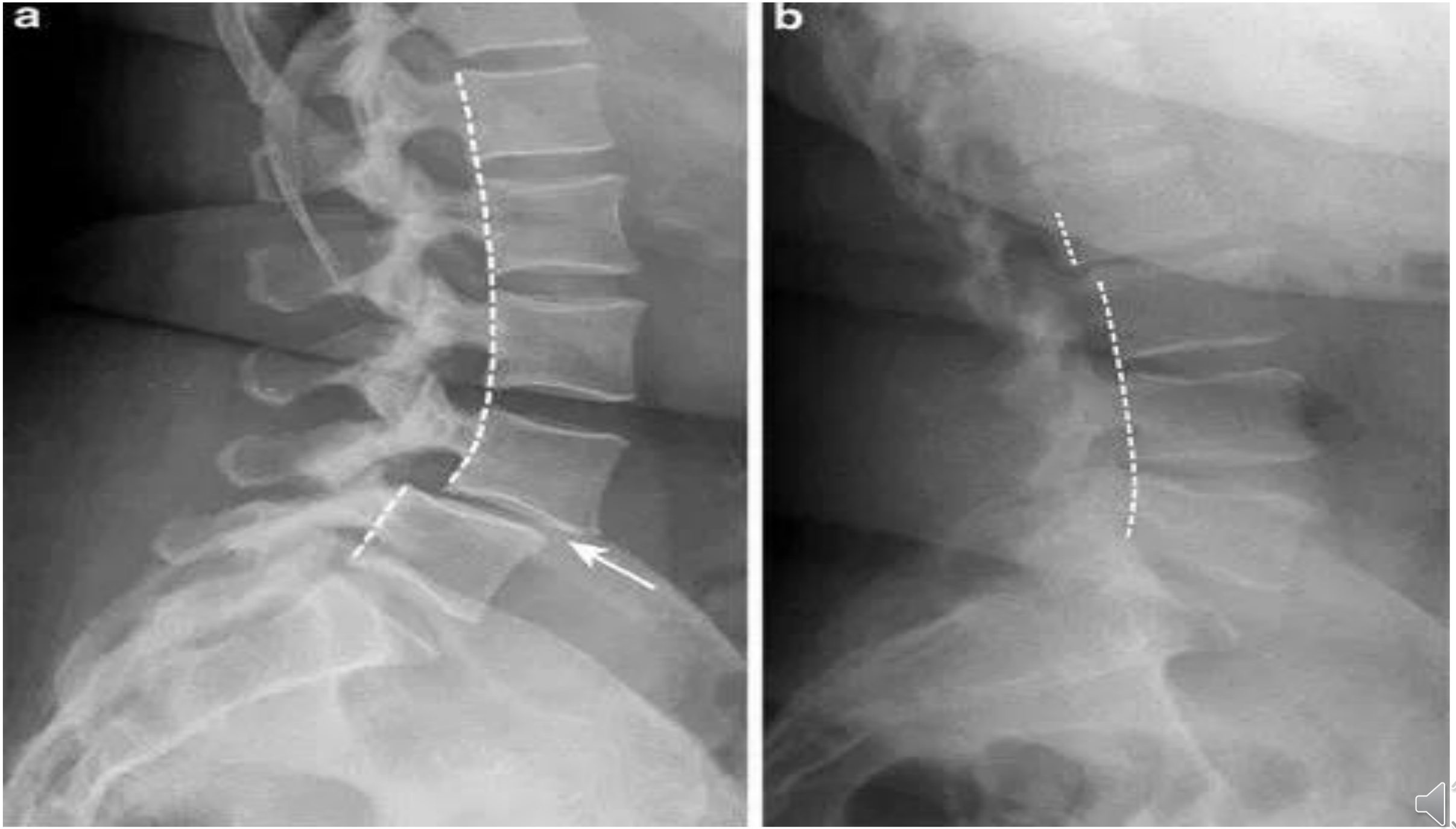


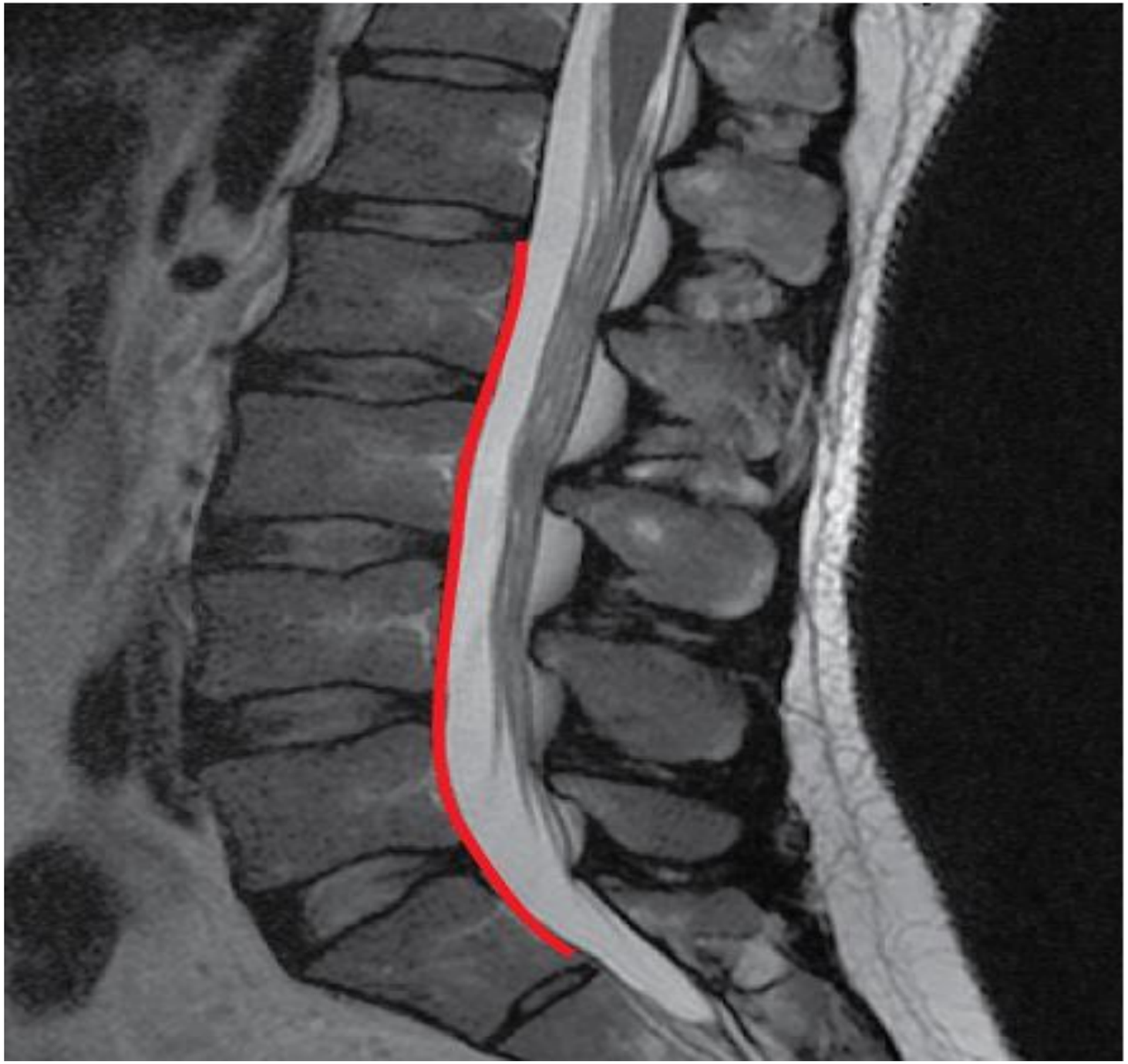
Spondylolytic Spondylolisthesis. Lateral view of the lumbar spine demonstrates a bilateral break in the pars interarticularis or spondylolysis (lucency shown by black arrow) that allows the L5 vertebral body (red arrow) to slip forward on the S1 vertebral body (blue arrow). The forward slippage is called spondylolisthesis. The normal pars interarticularis is shown by the white arrow. The degree of forward slippage is equal to about 1/4 to 1/2 of the AP diameter of S1 so this is a Grade1-Grade 2 spondylolisthesis.

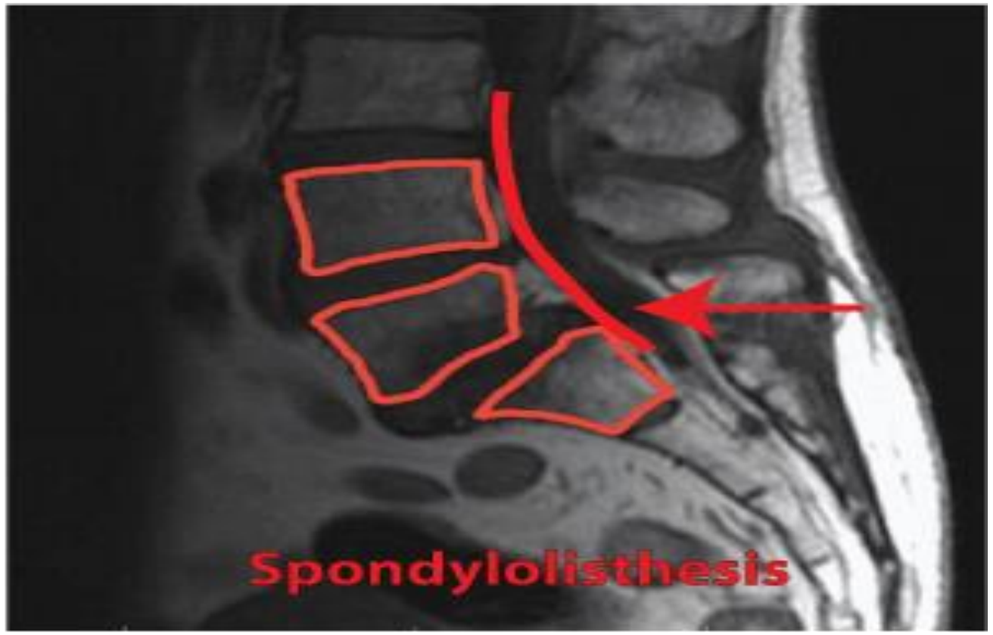


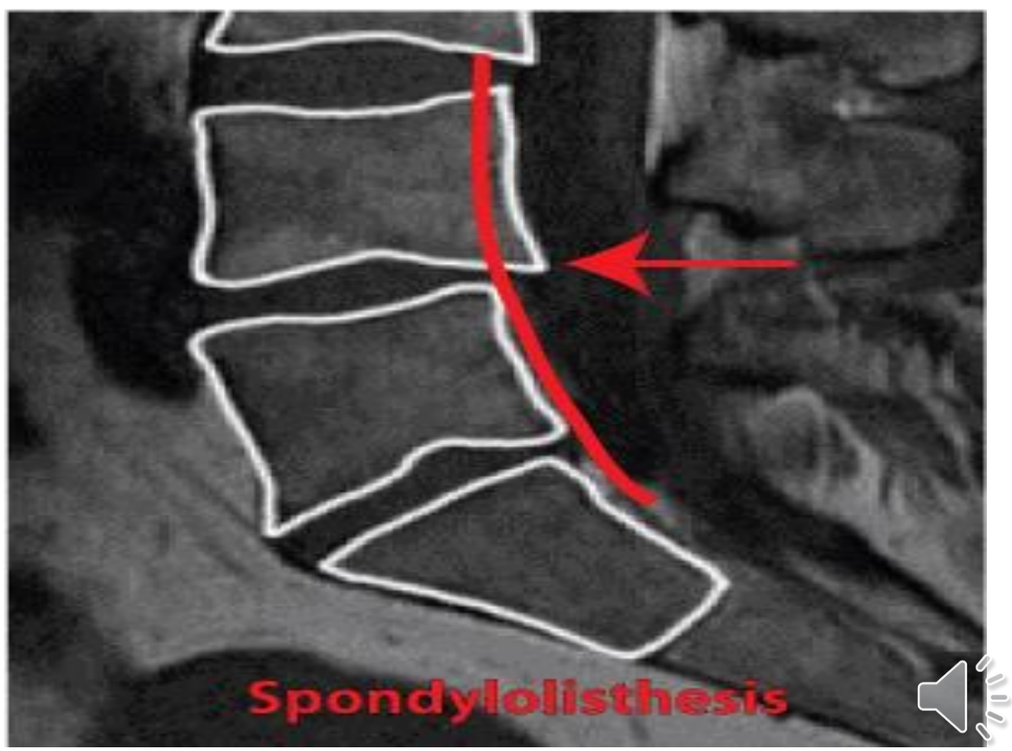

Superior articular process
(ear of Scottie dog)
Transverse process
(nose)
Pedicle
(eye)
Defect at pars interarticularis
(collar or broken neck)
Inferior articular process
(foot)


Lateral X-Ray
Oblique X-Rays
Defect
Intact
Defect


a
b

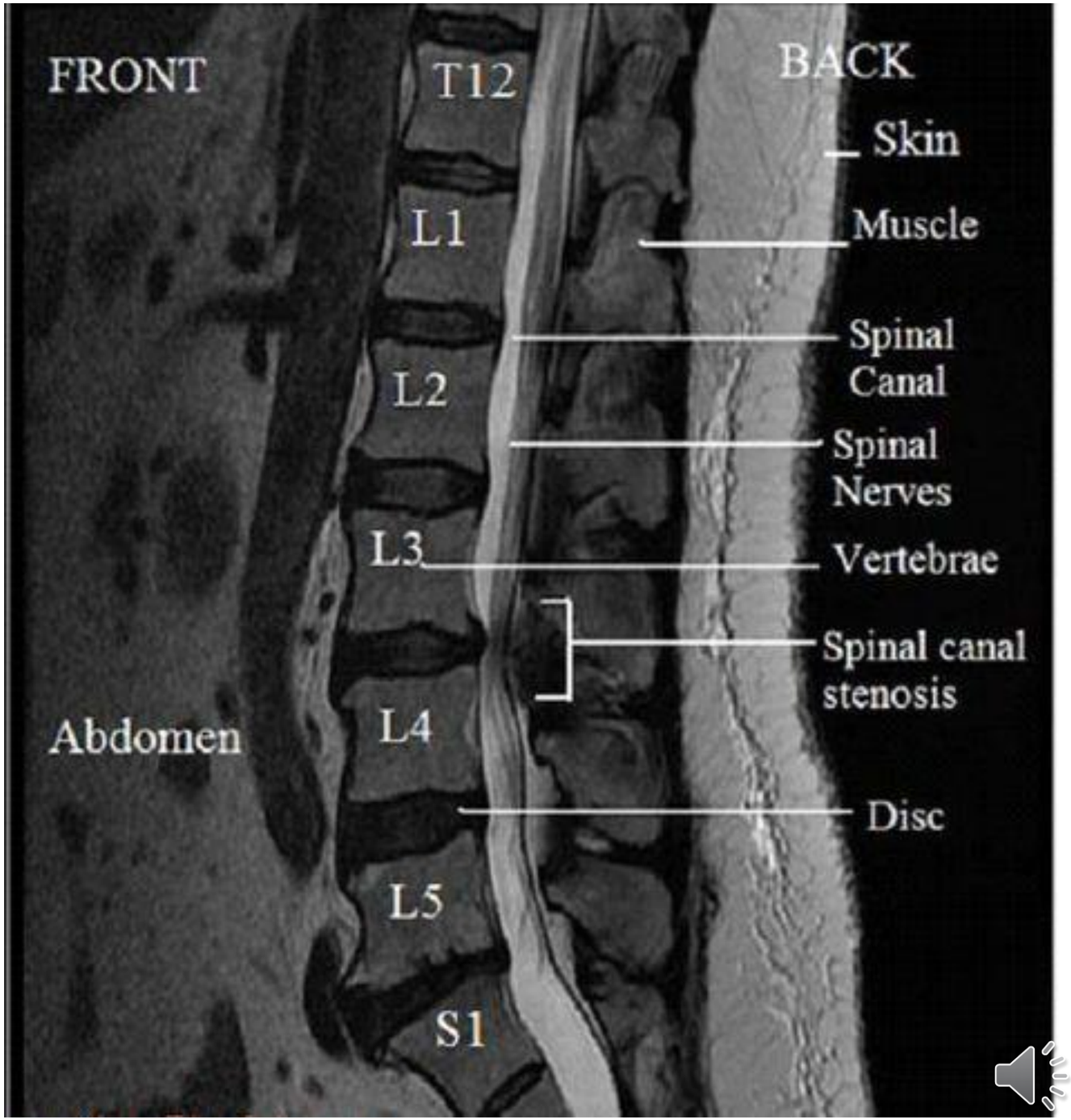



Spondylolisthesis


Spondylolisthesis

Spinal Stenosis

- Spinal stenosis refers to narrowing of the spinal canal or the neural foramina secondary to soft tissue or bony abnormalities.
- Soft tissue abnormalities that can lead to spinal stenosis include hypertrophy of the ligamentum flavum, bulging disk(s), and ossification of the posterior longitudinal ligament (OPLL).
- Bony abnormalities that can lead to spinal stenosis include a congenitally narrow spinal canal, osteophytes, facet osteoarthritis, or spondylolisthesis.

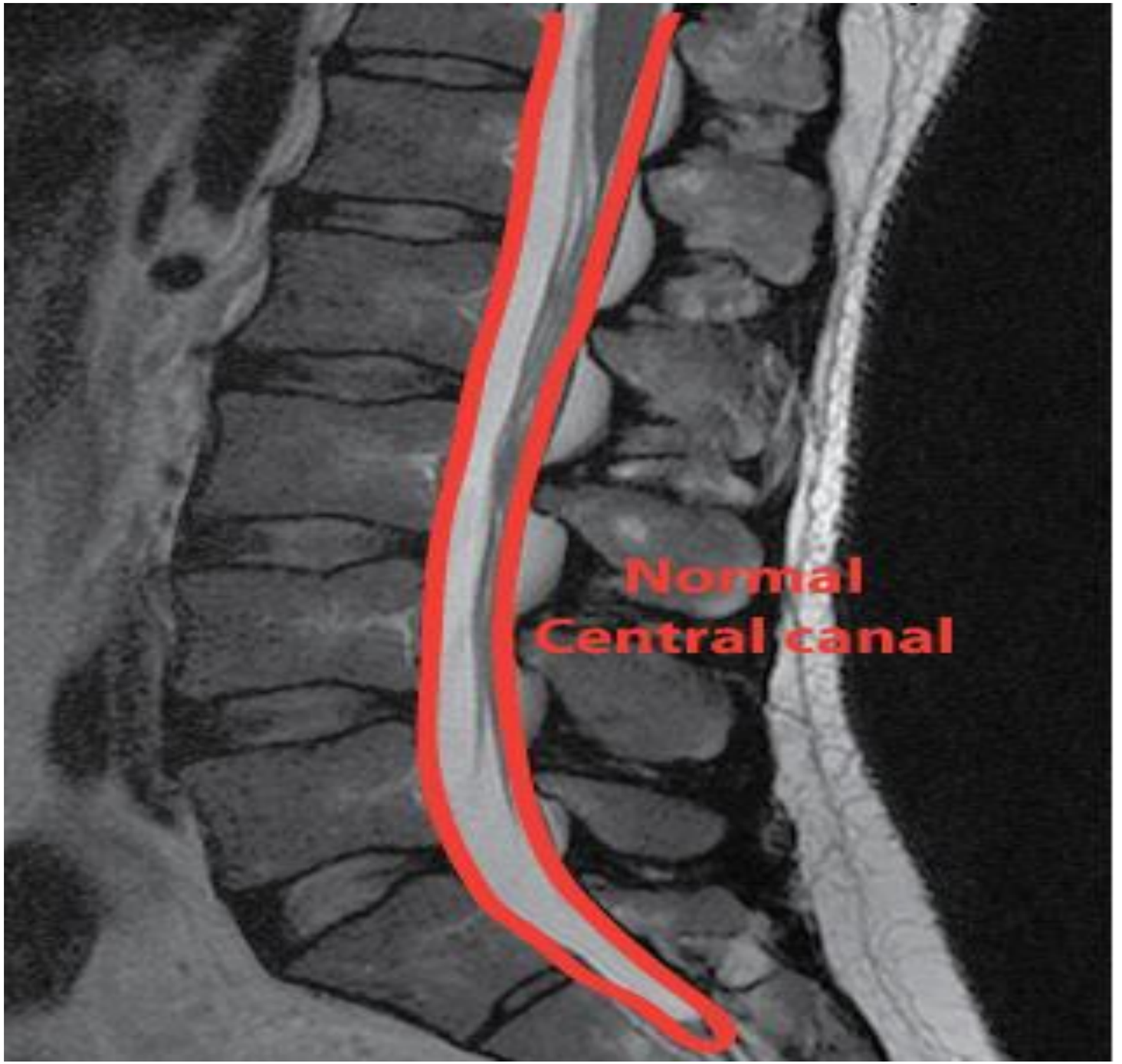

FRONT
T12
BACK
Skin
L1
Muscle
Spinal Canal
L2
Spinal Nerves
L3
Vertebrae
Spinal canal stenosis
Abdomen
L4
Disc
L5
S1

Normal

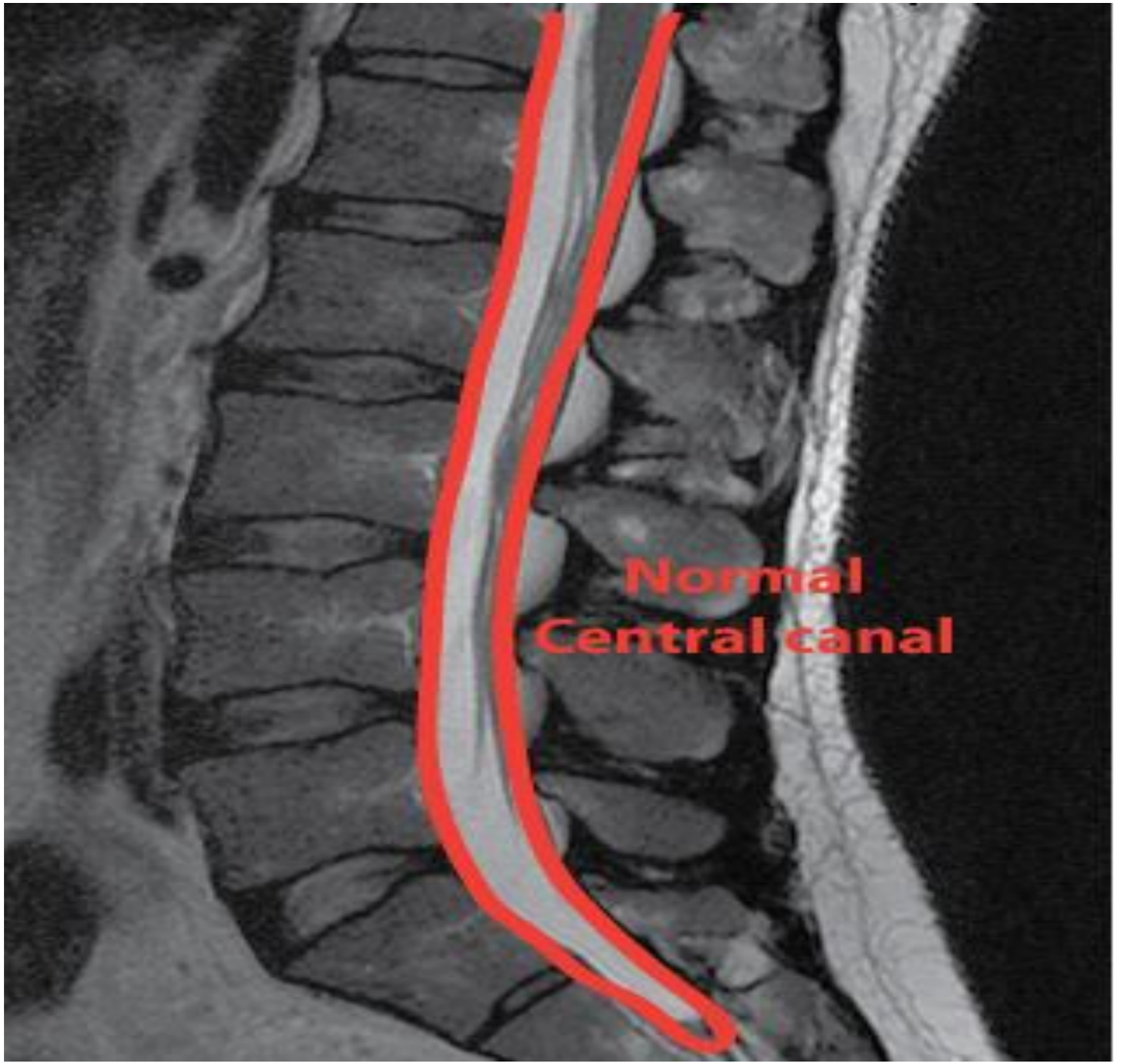
Mild

Moderate

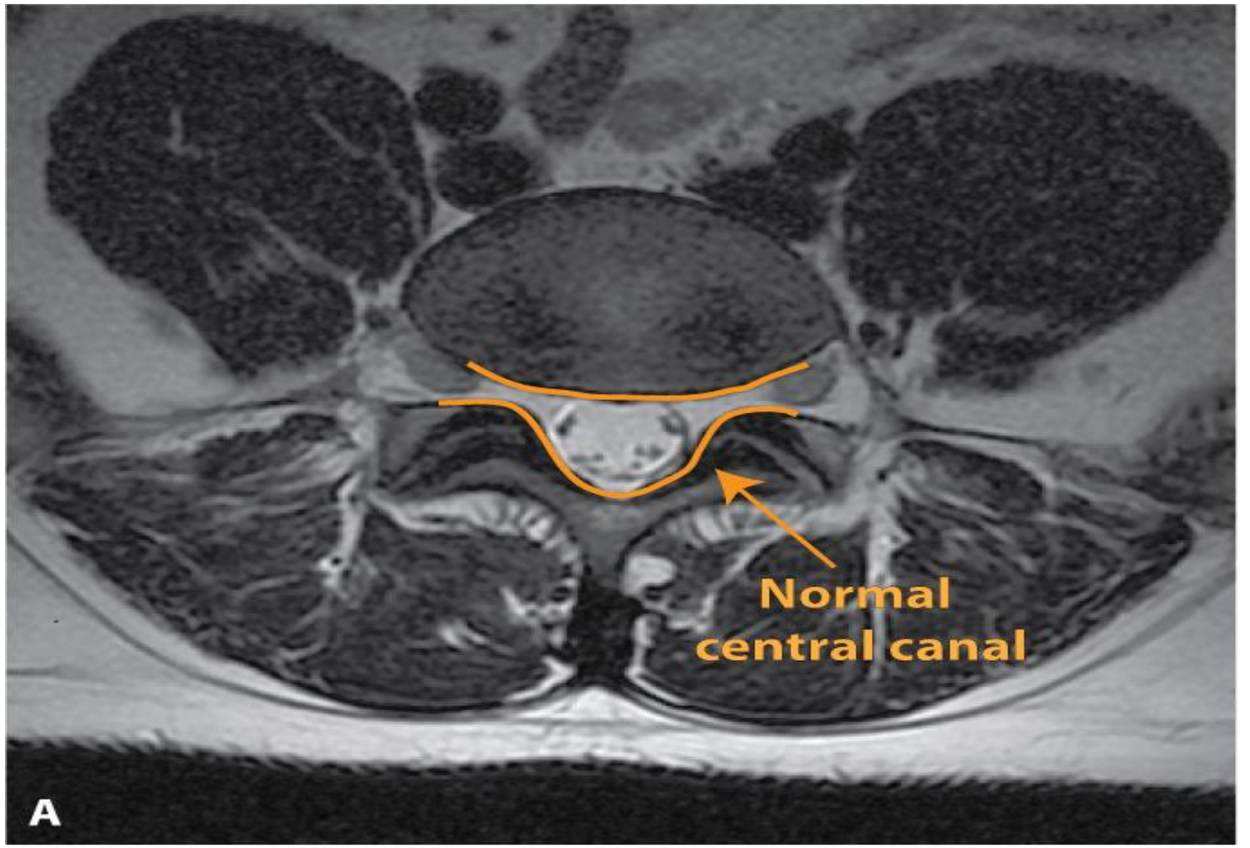
Sever

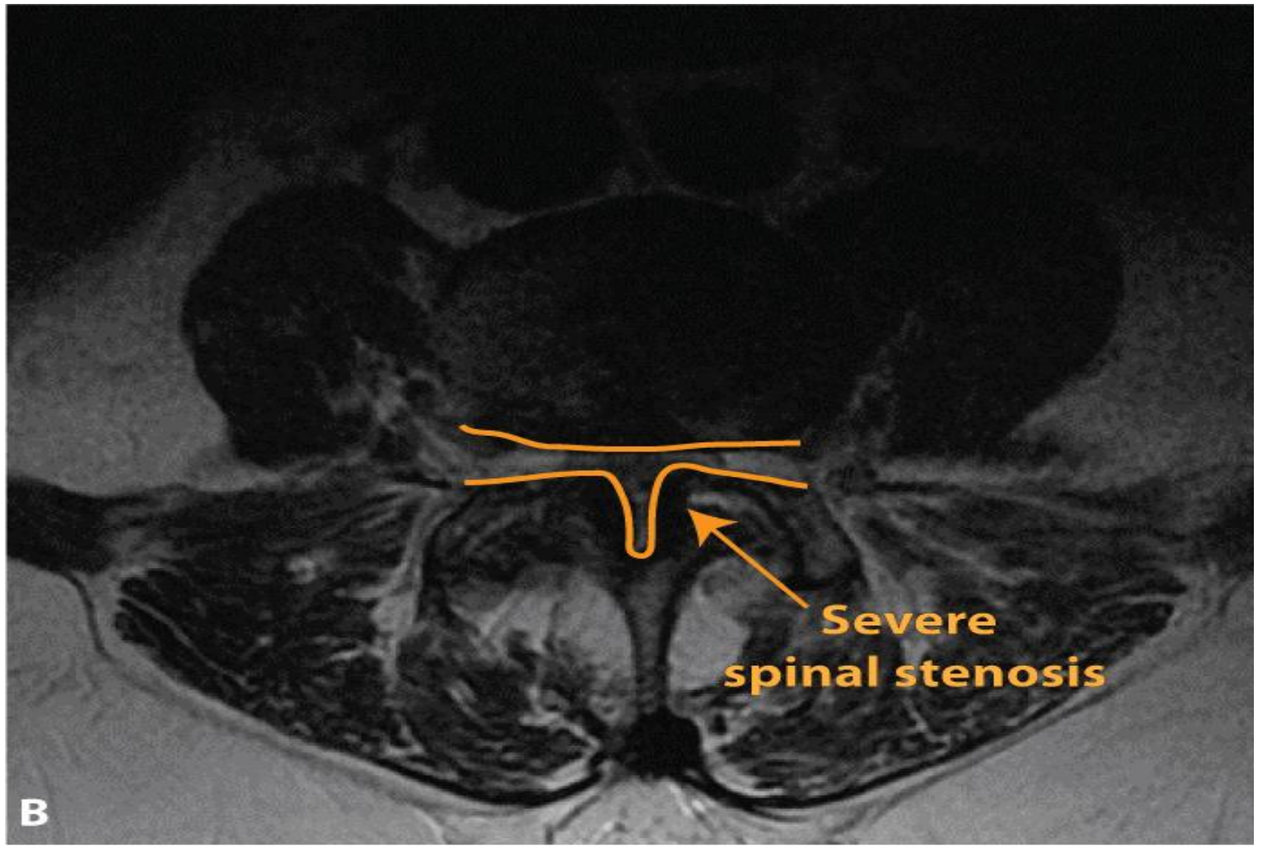

Normal
Central canal

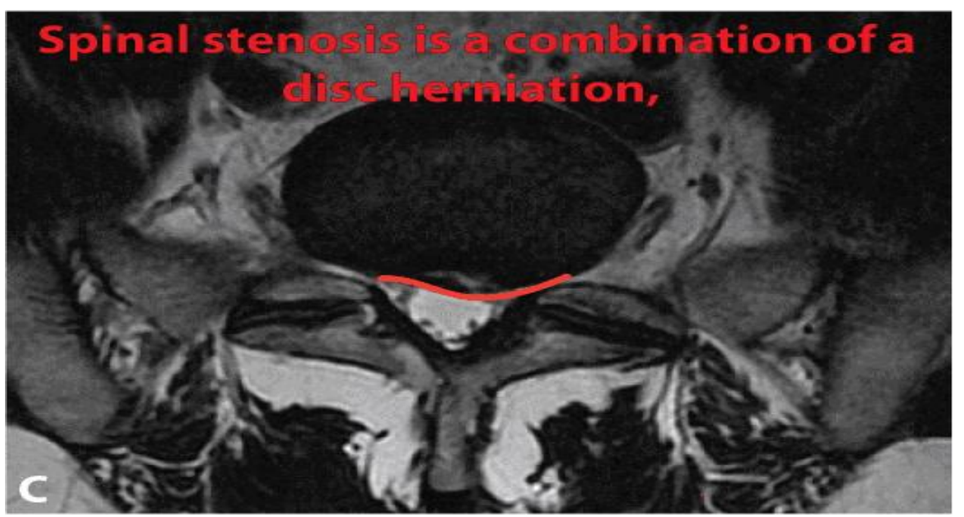
Severe
spinal stenosis

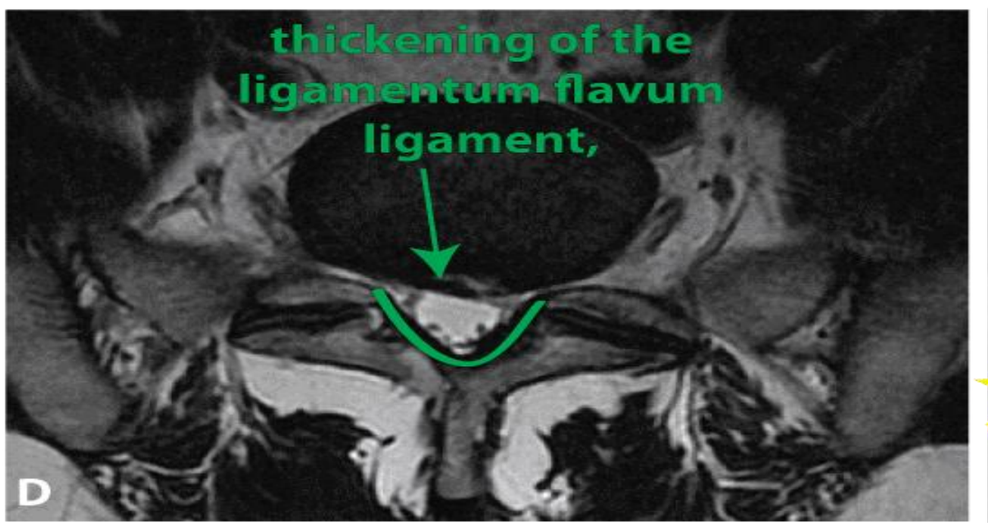

Normal
Central canal

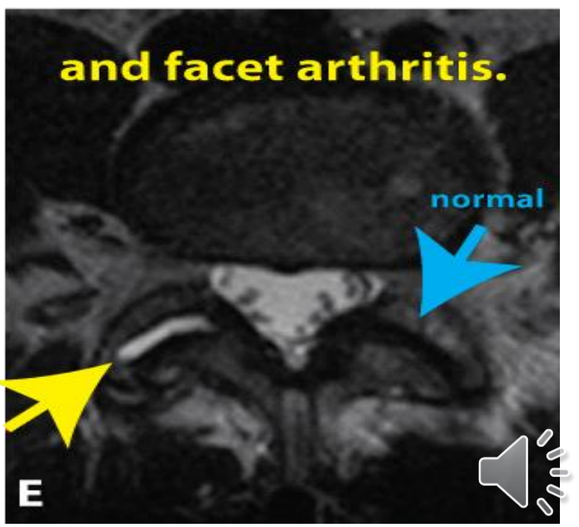
Severe
spinal stenosis


Normal
central canal
A


Severe
spinal stenosis
B

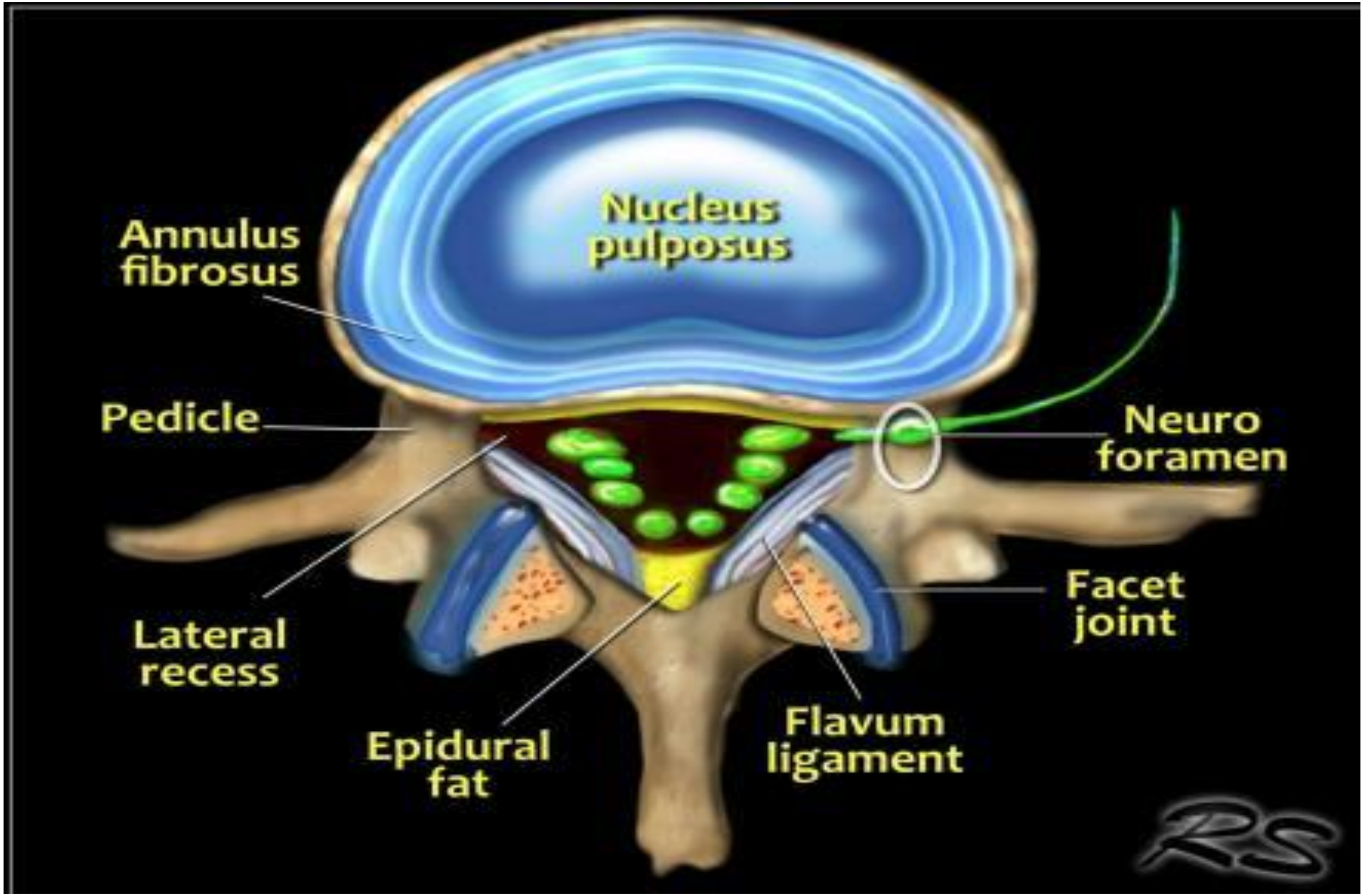

Spinal stenosis is a combination of a
disc herniation,
C

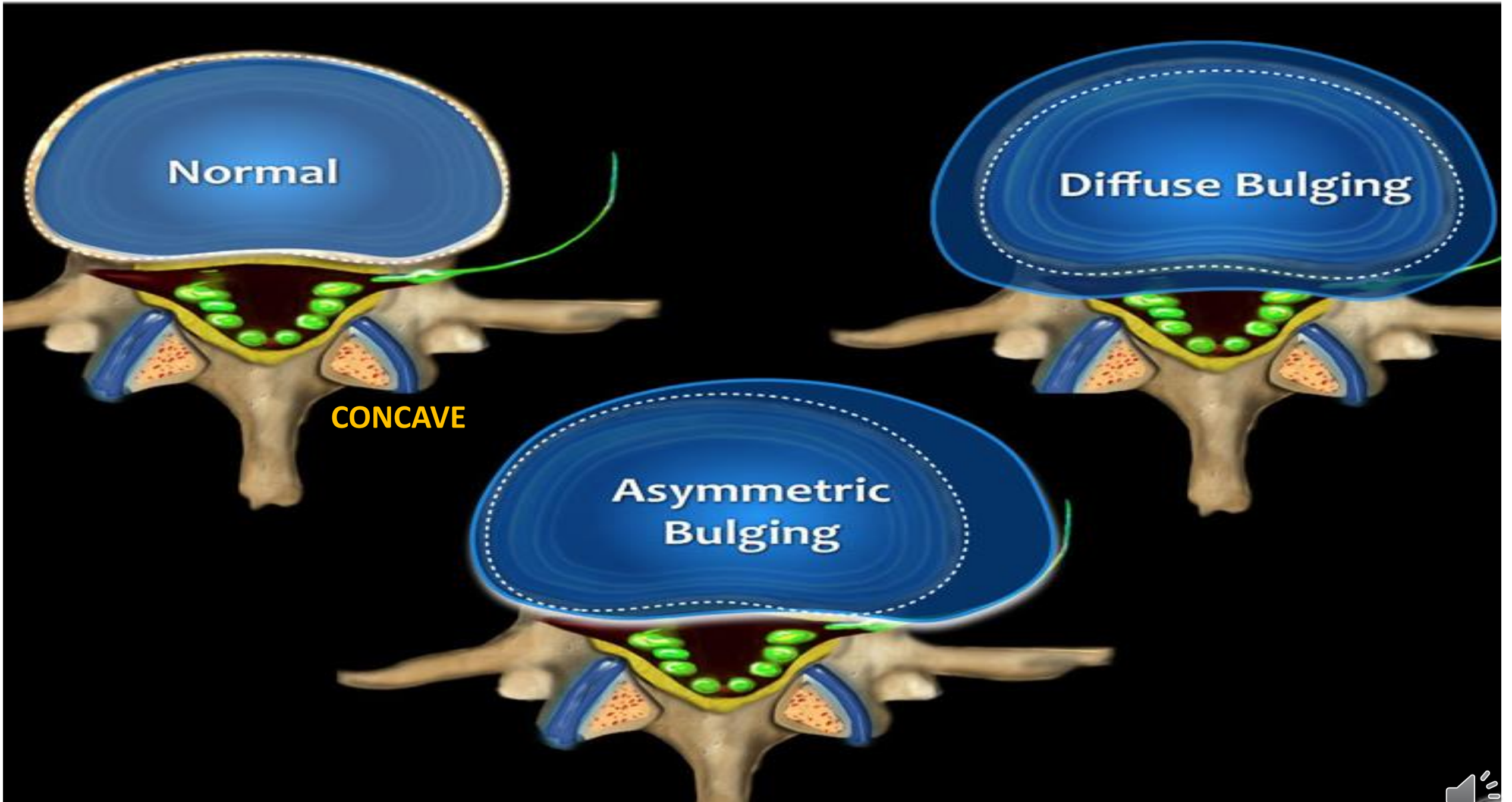

thickening of the
ligamentum flavum
ligament,
D

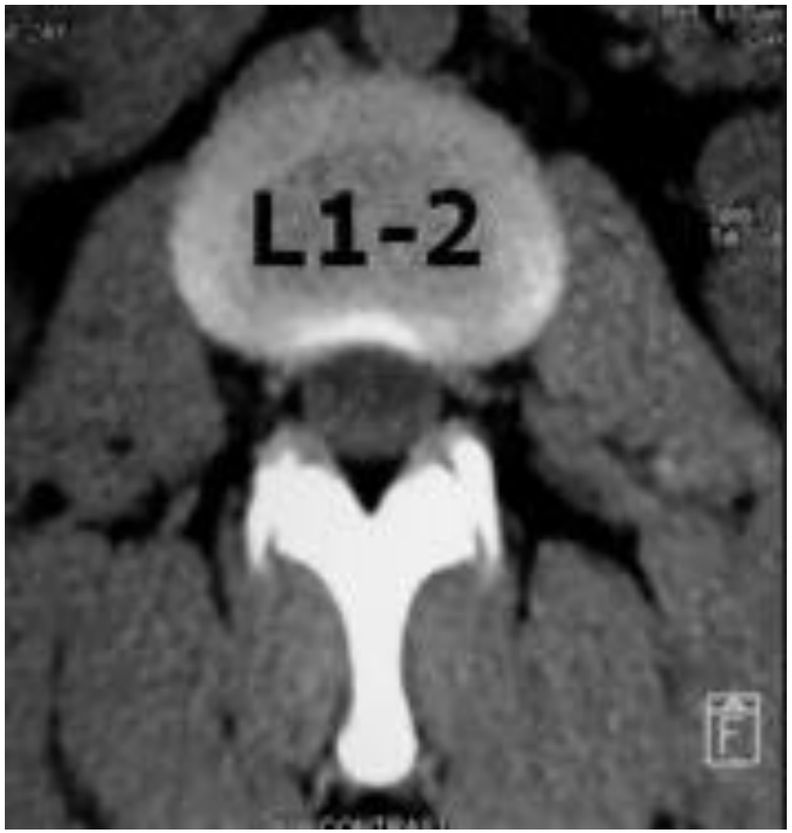

and facet arthritis.
normal
E

Bulging disc

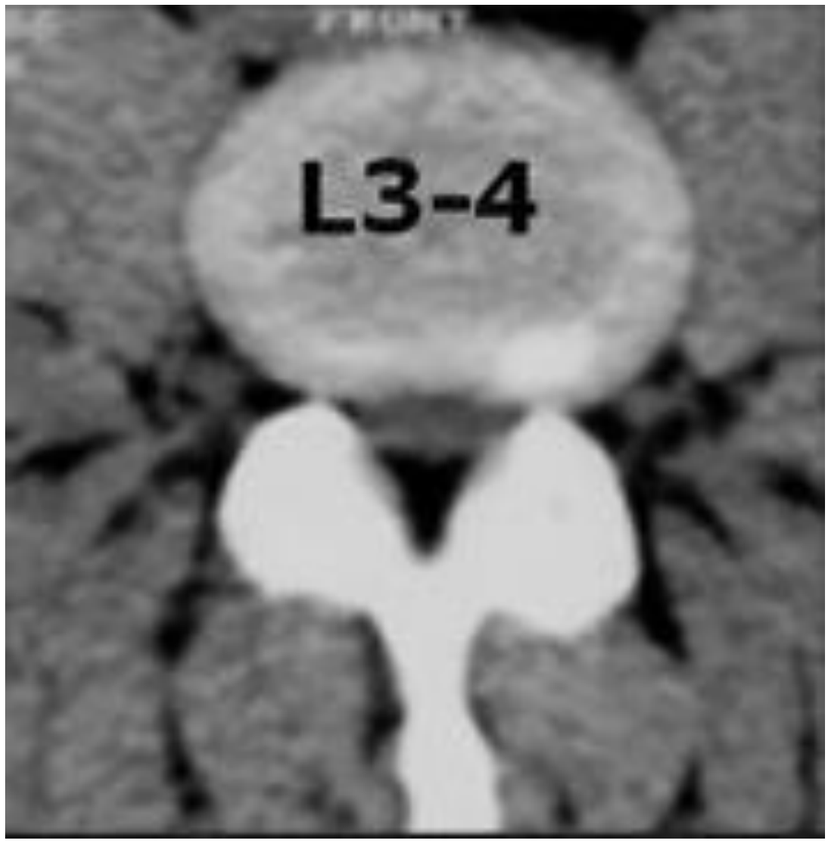
- The presence of disc tissue extending beyond the edges of the ring apophyses, throughout the circumference of the disc, is called "bulging" and is not considered a form of herniation.
- Posterior border of the disc is more important as it has relation to the disc.
- The normal posterior border of the disc is **CONCAVE**.
- The abnormal is **STRAIGHT OR CONVEX**


Annulus fibrosus
Nucleus pulposus
Pedicle
Neuro foramen
Lateral recess
Facet joint
Epidural fat
Flavum ligament
RS

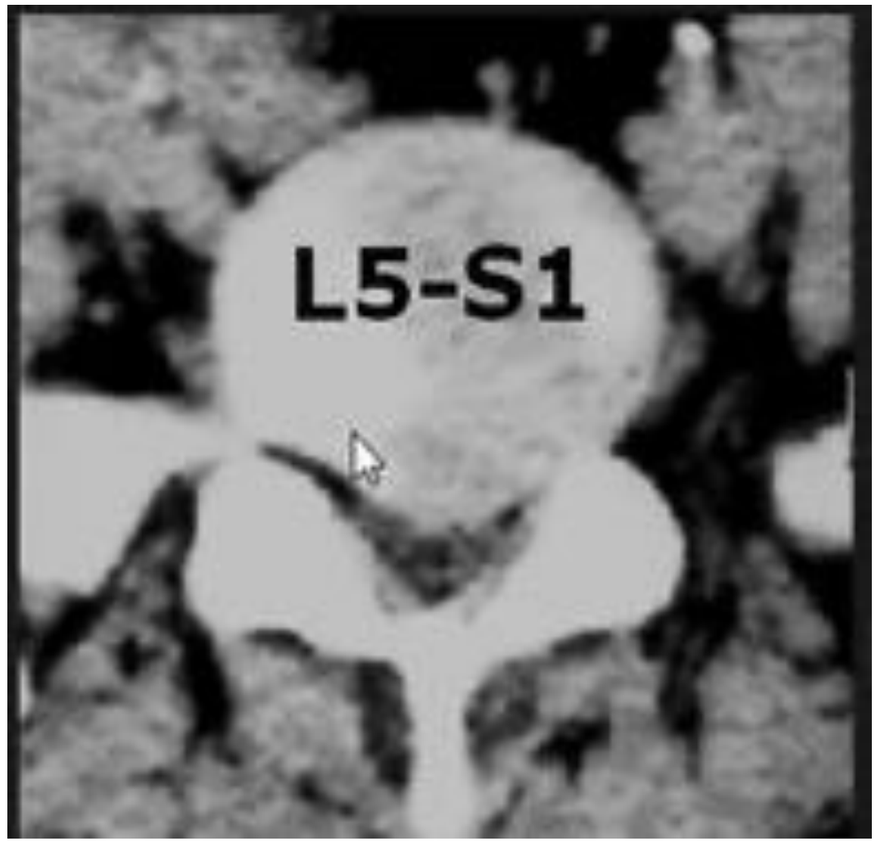

Normal
Diffuse Bulging
CONCAVE
Asymmetric
Bulging



Normal disc



Mild disc bulge

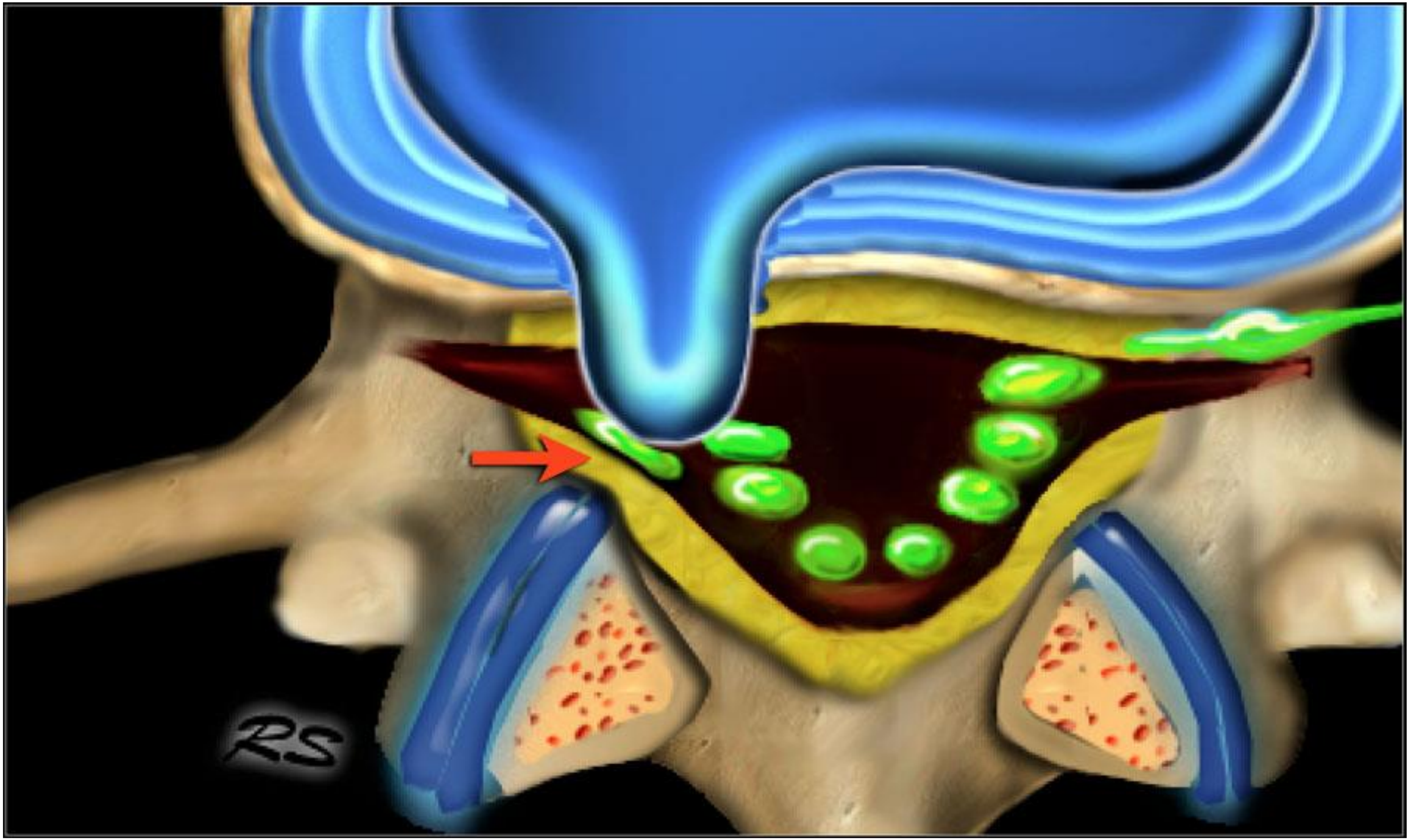


Moderate disc bulge

Herniated Disks

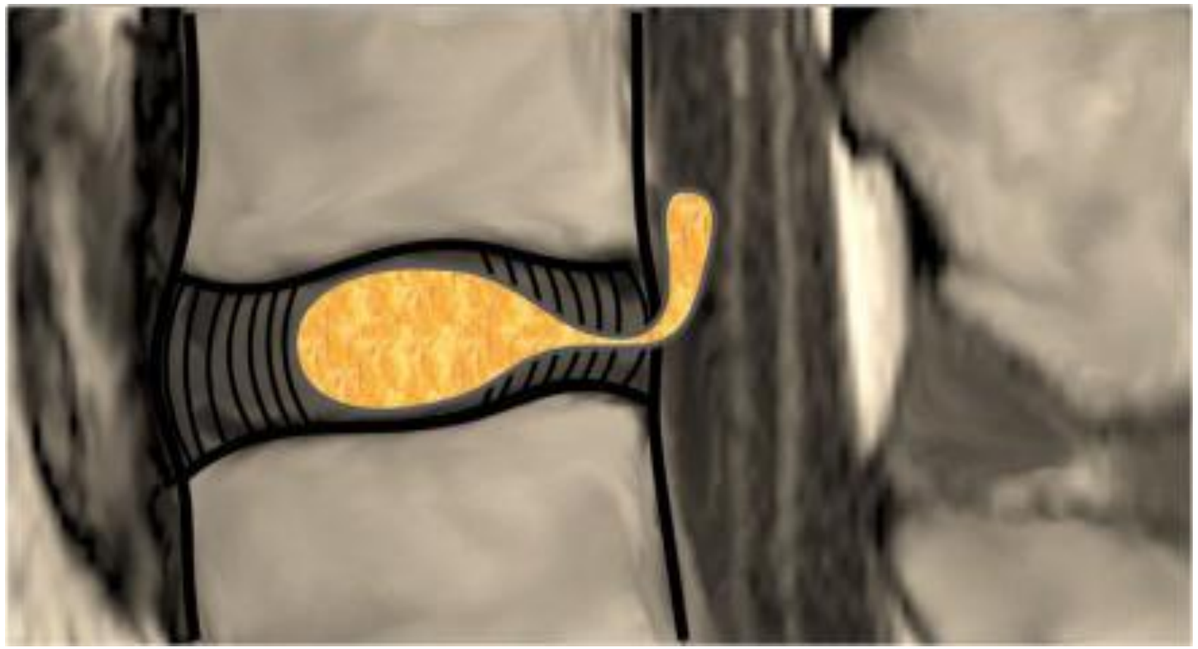
- Only about 2% of patients with acute low back pain have a herniated disk.
- In the lumbar region, disk herniation may lead to back pain and sciatica.
- Herniation of a cervical disk may produce radiculopathy and myelopathy.
- MRI is the study of choice for evaluating herniated disks.

- In the cervical spine, disk herniation's occur most frequently at C4-C5, C5-C6, and C6-C7.
- The majority of disk herniations occur at the lower three lumbar disk levels, L3-L4, L4-L5 and L5 to S1.
- More than 60% of disk herniation's occur posterolaterally.


RS

Migration

Migration indicates displacement of disc material away from the site of extrusion, regardless of whether sequestrated or not.



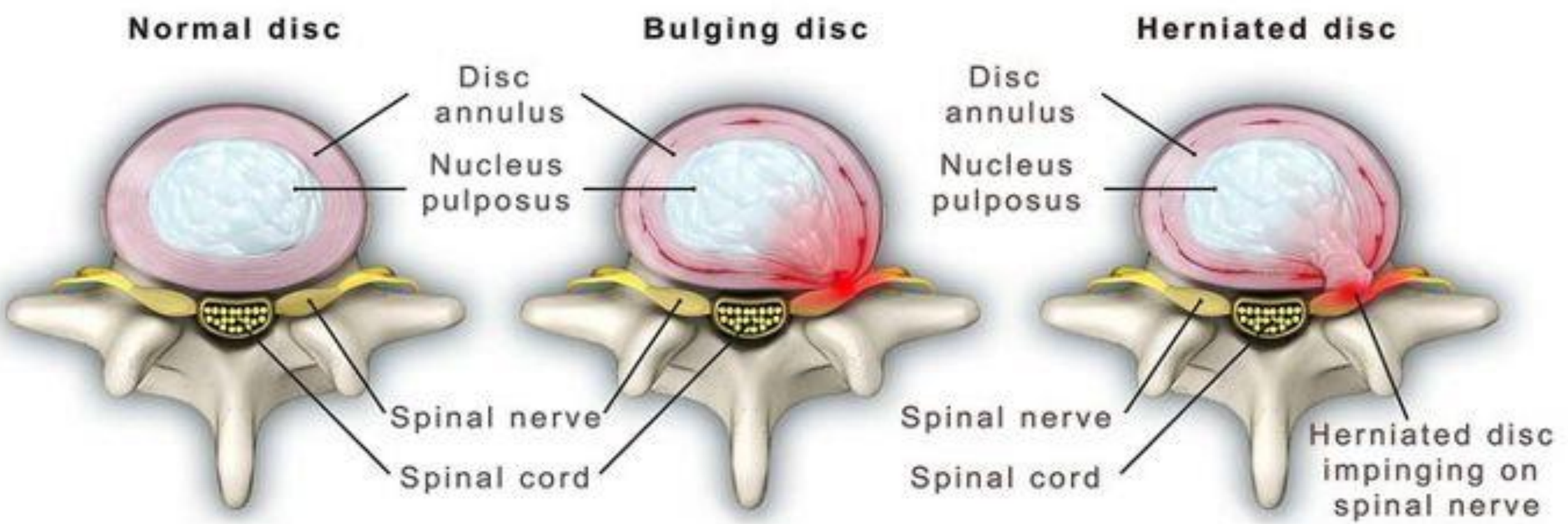
Sequestration

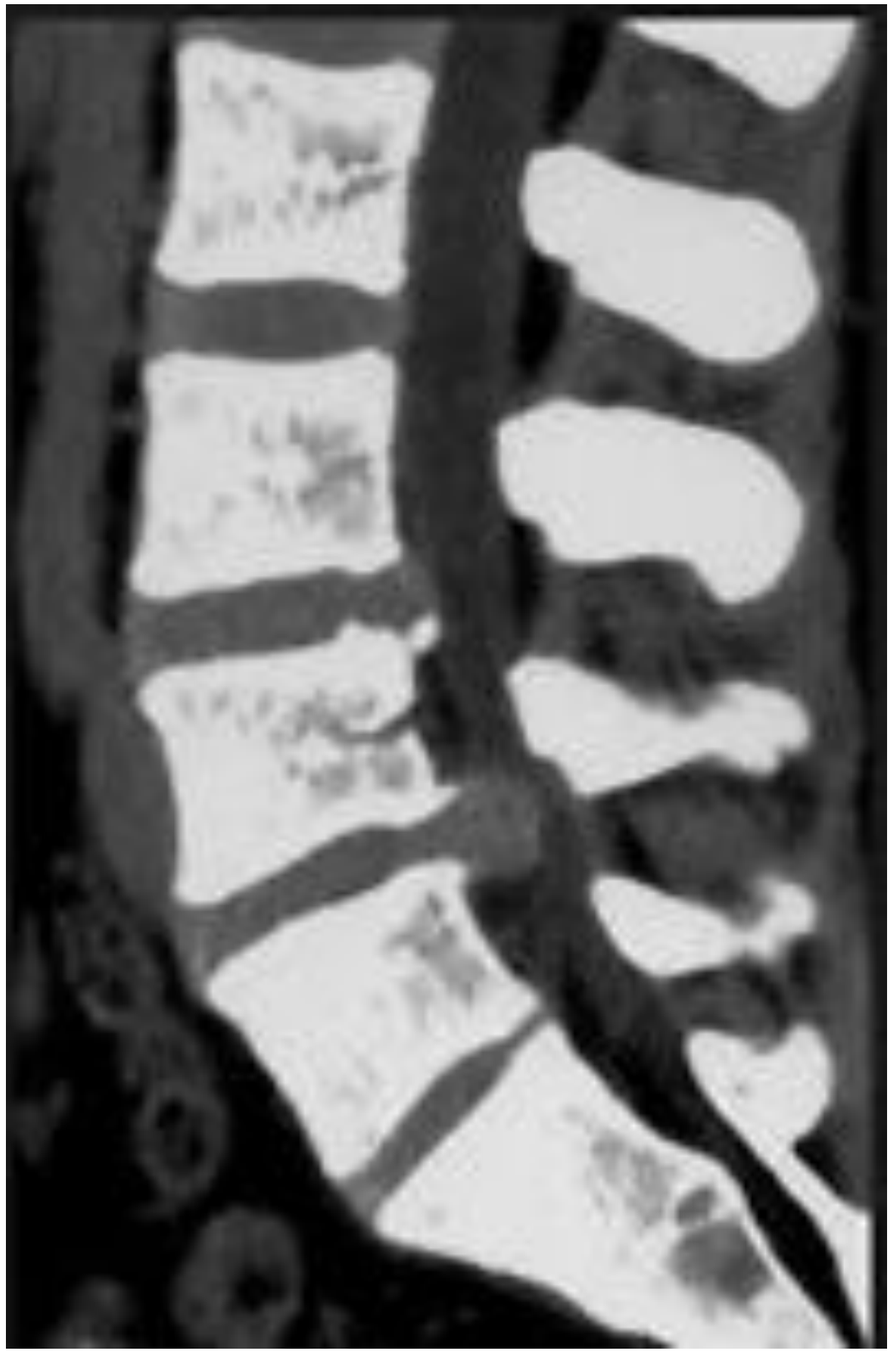
The term sequestration is used to indicate that the displaced disc material has lost continuity with the parent disc.

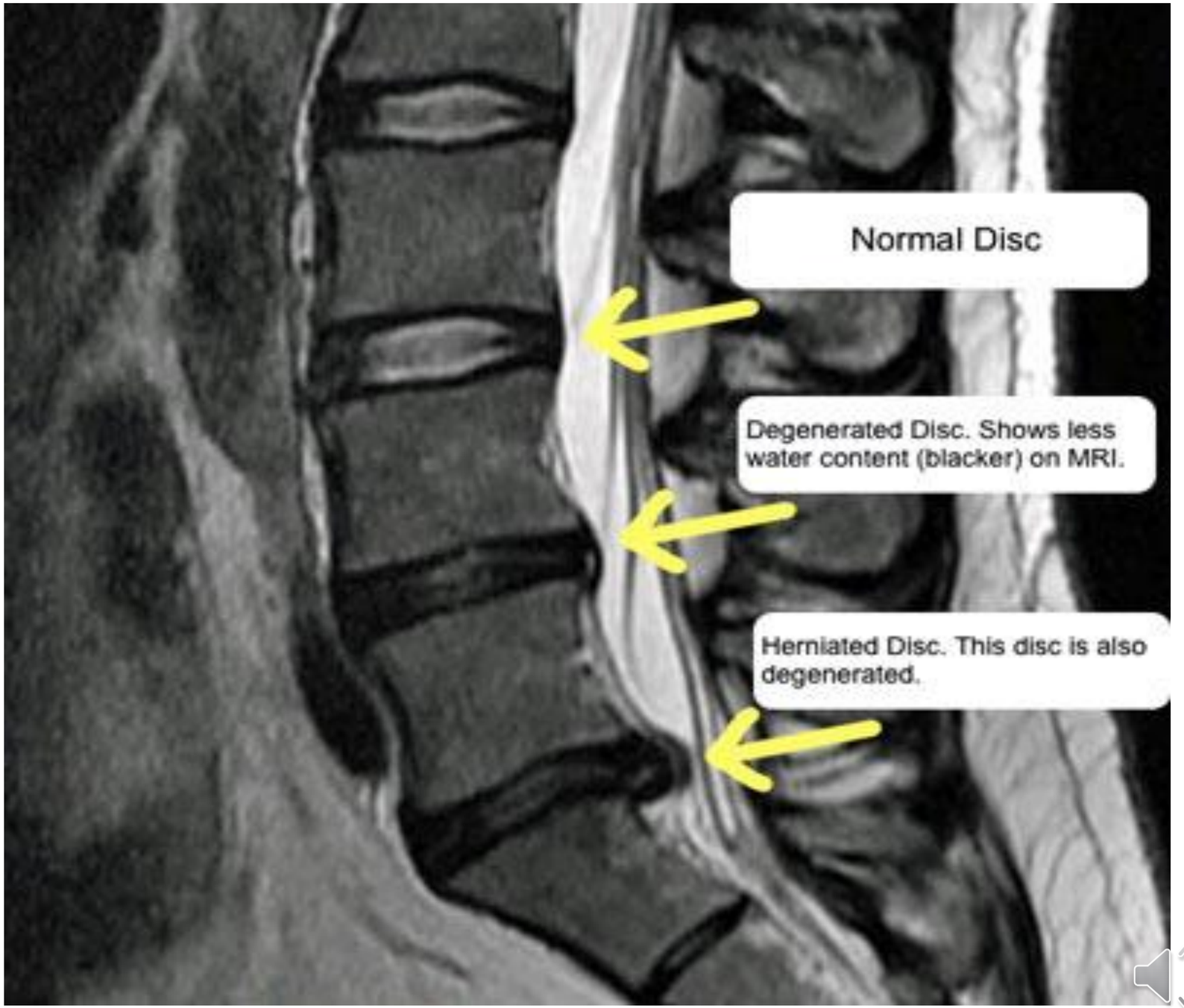
Disc bulge / Herniation

Disc bulge= intact annulus = diffuse pathology

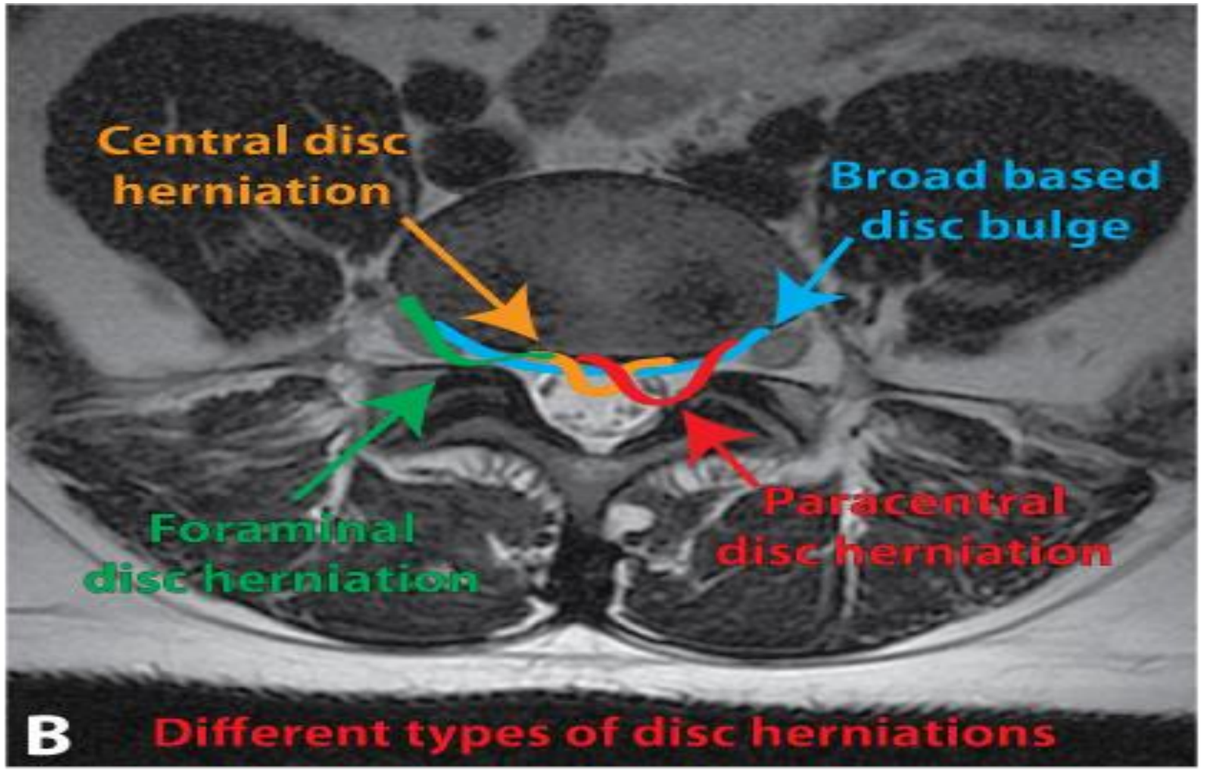
Disc Herniation= torn annulus = focal pathology

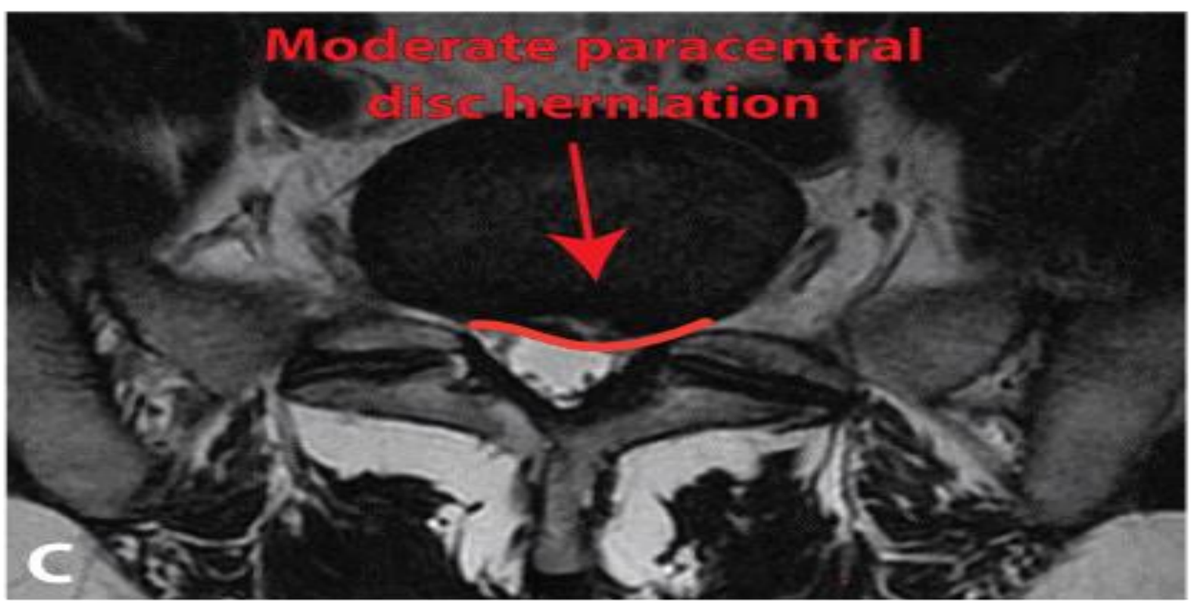


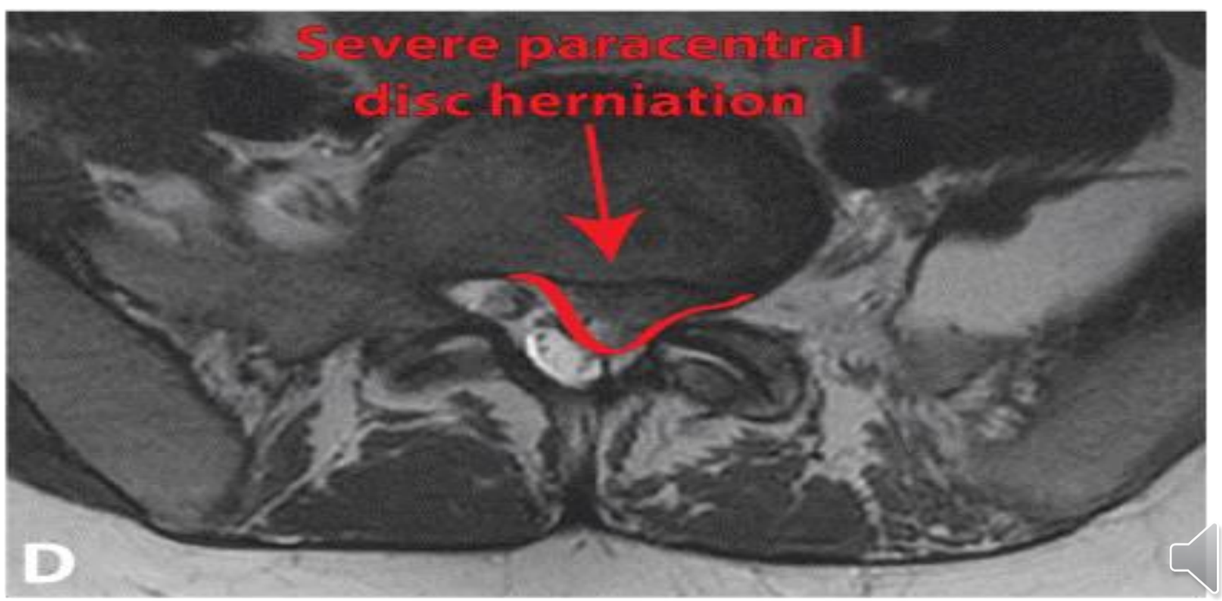



Normal Disc
Degenerated Disc. Shows less water content (blacker) on MRI.
Herniated Disc. This disc is also degenerated.

A
Normal disc


Central disc herniation
Broad based disc bulge
Foraminal disc herniation
Paracentral disc herniation
B
Different types of disc herniations


Moderate paracentral disc herniation
C


Severe paracentral disc herniation
D

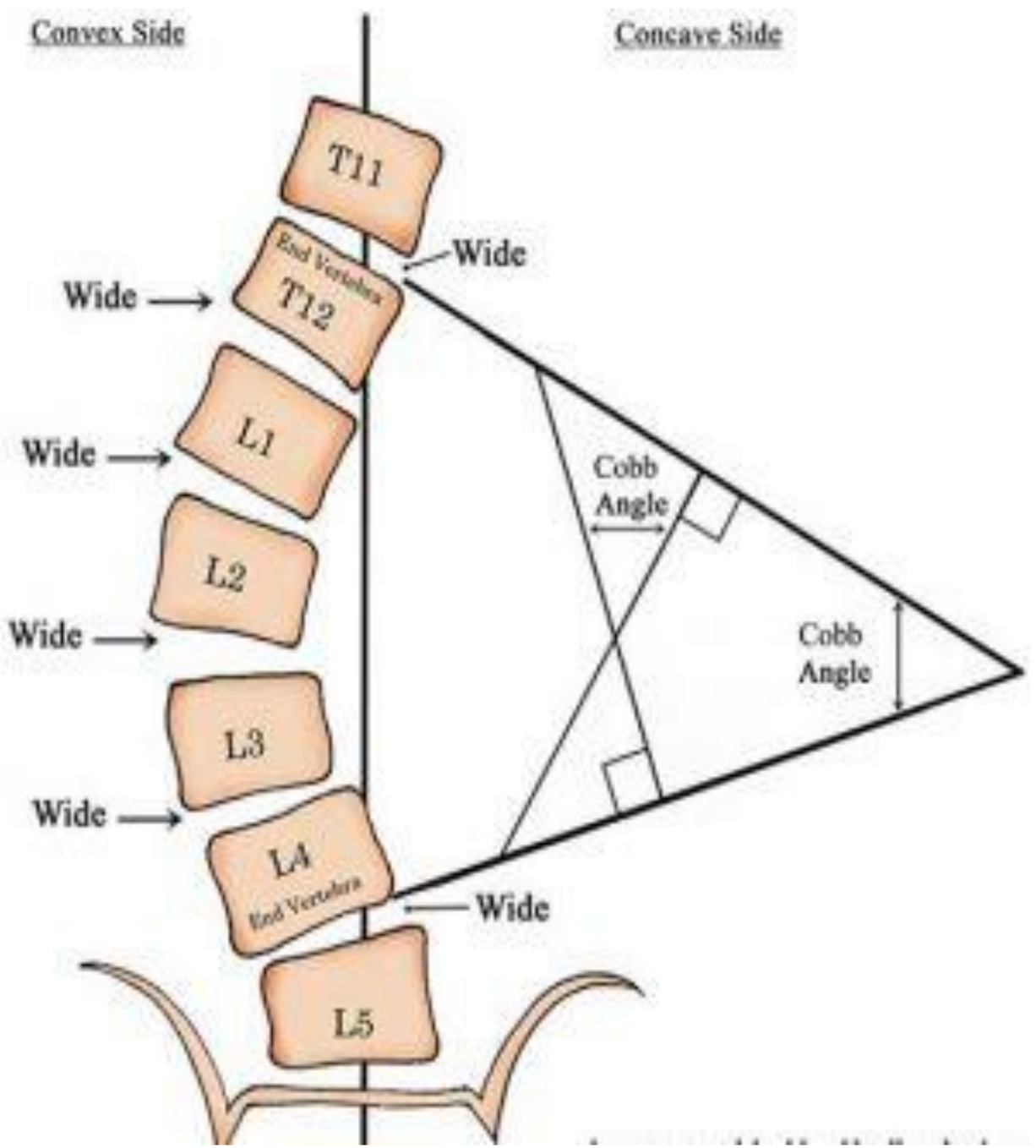
Scoliosis

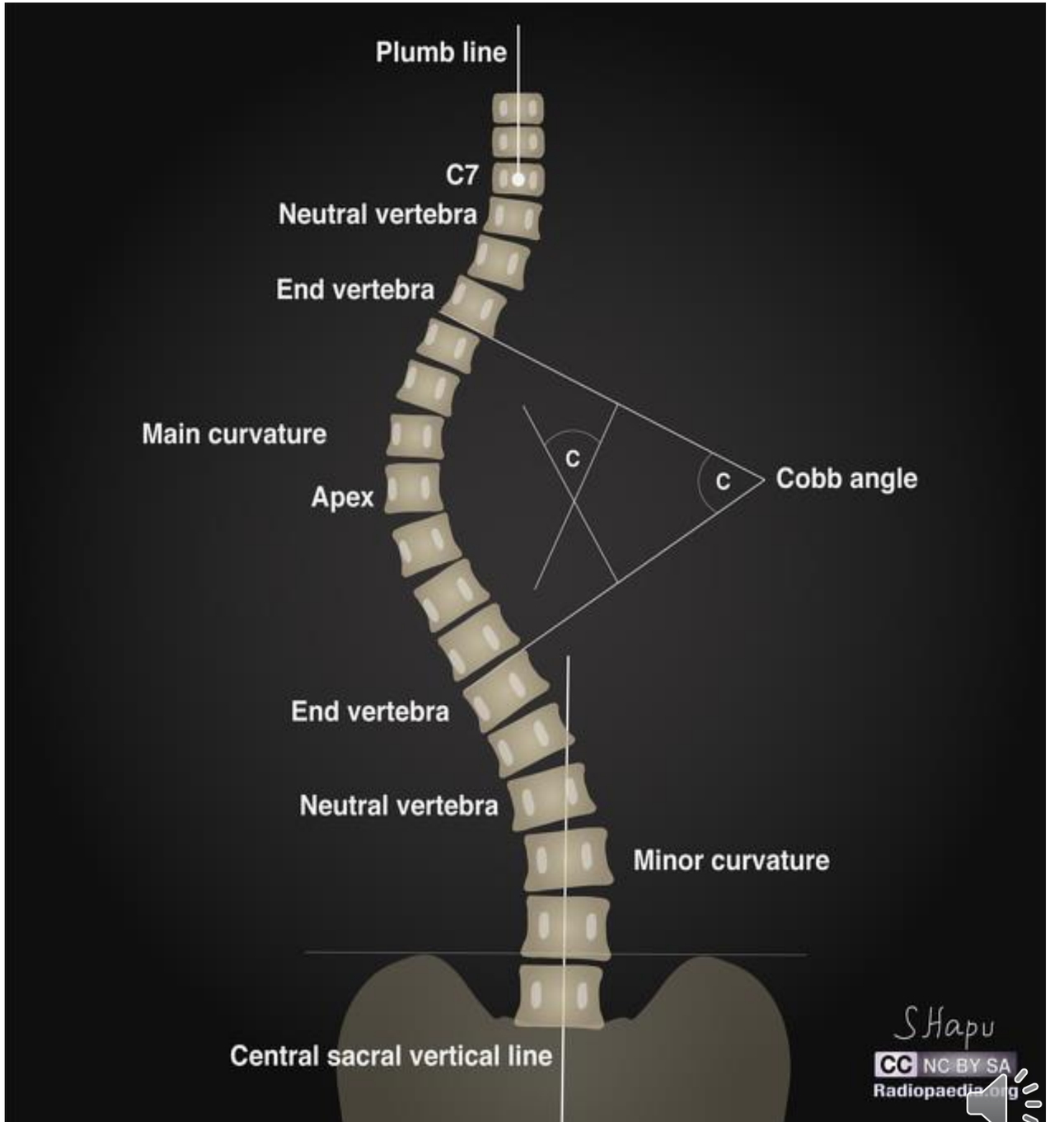
- Scoliosis is defined as a lateral spinal curvature with a Cobb angle of 10° or more.
- The Cobb angle is measured between the superior endplate of the proximal end vertebra and the inferior endplate of the distal end vertebra.
- An increase in the Cobb angle by 5° or more per year indicates progression of scoliosis.

Radiology Interpretation

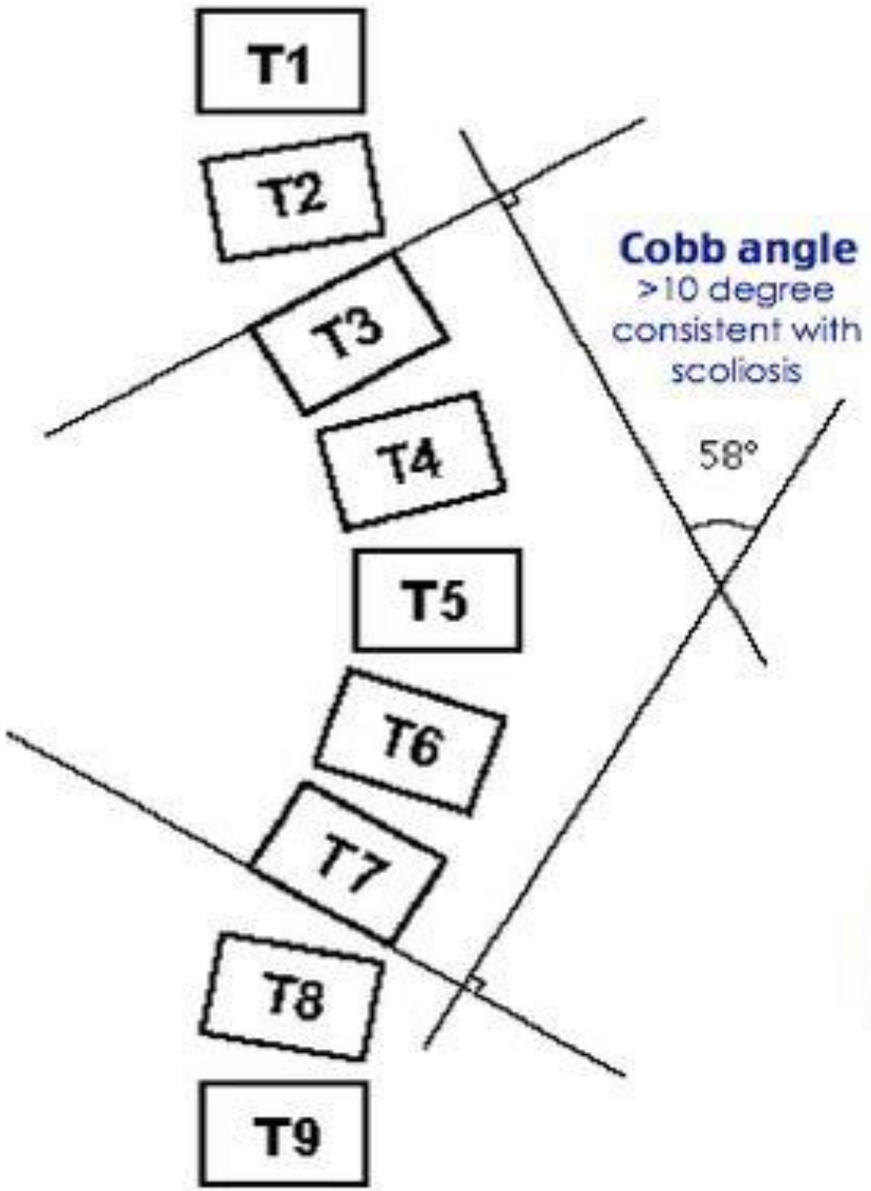
Curve Interpretation

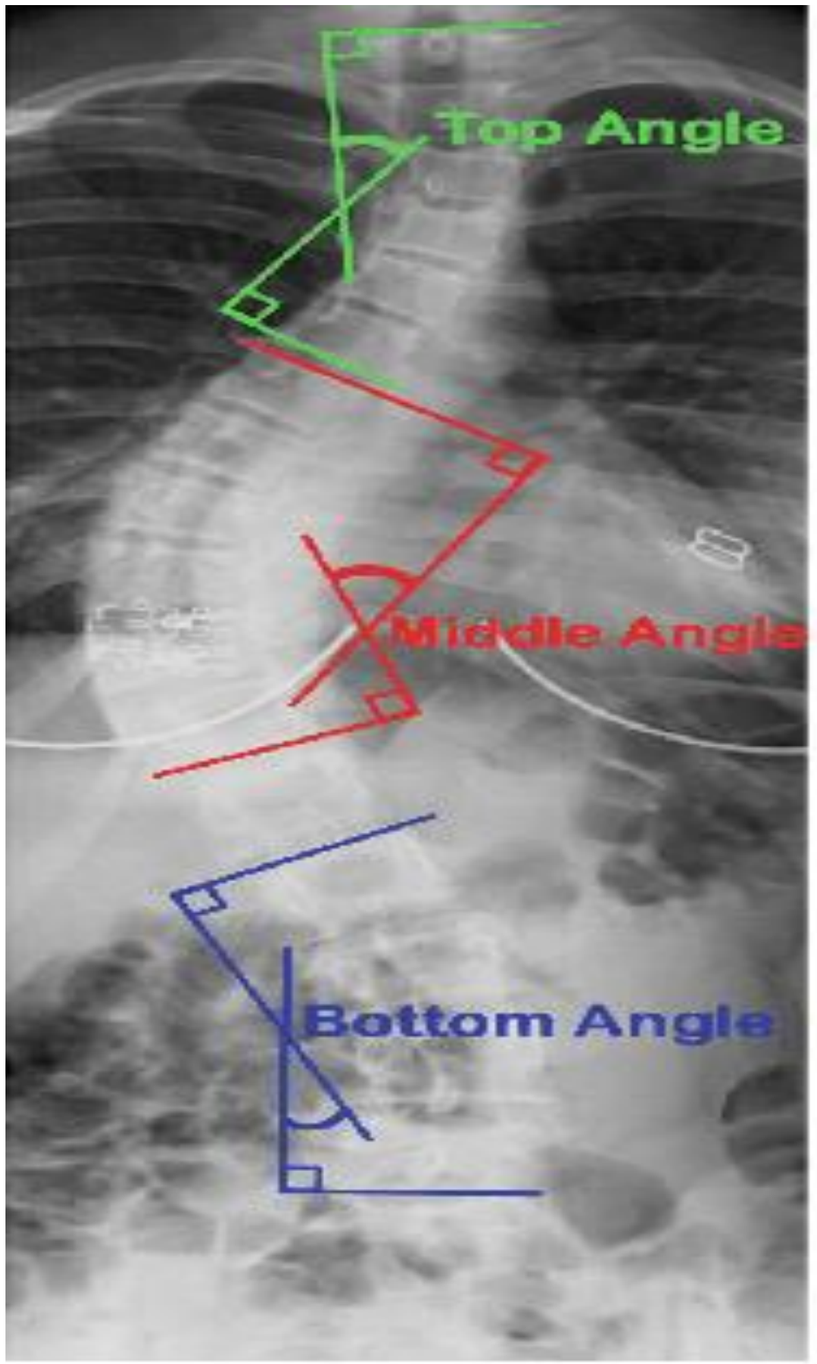
- **Mild**: Curve <10-15 degrees
- **Moderate**: Curve 20-45 degrees
- **Severe**: Curve >45-50 degrees
- **Respiratory compromise**: Curve >90 degrees

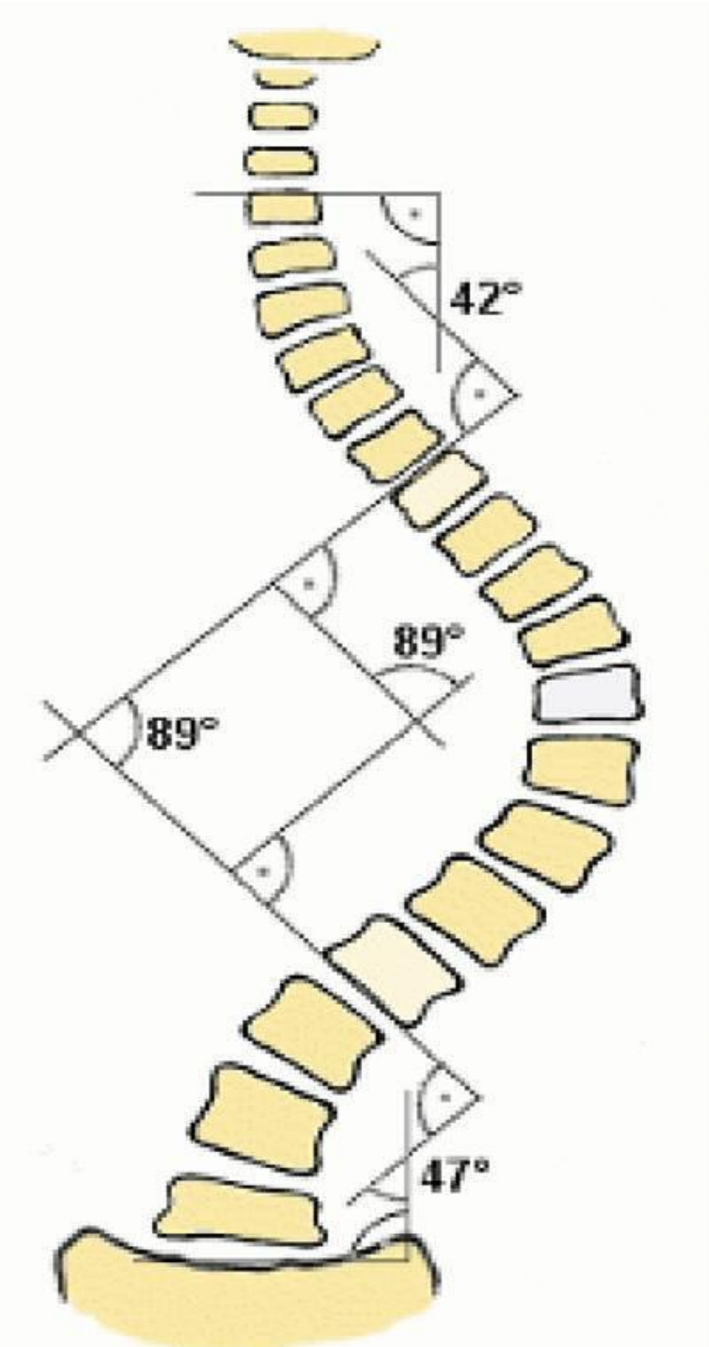

Convex Side
Concave Side
T11
End Vertebra
T12
Wide
Wide
L1
Wide
L2
Wide
L3
Wide
L4
End Vertebra
Wide
L5
Cobb Angle
Cobb Angle

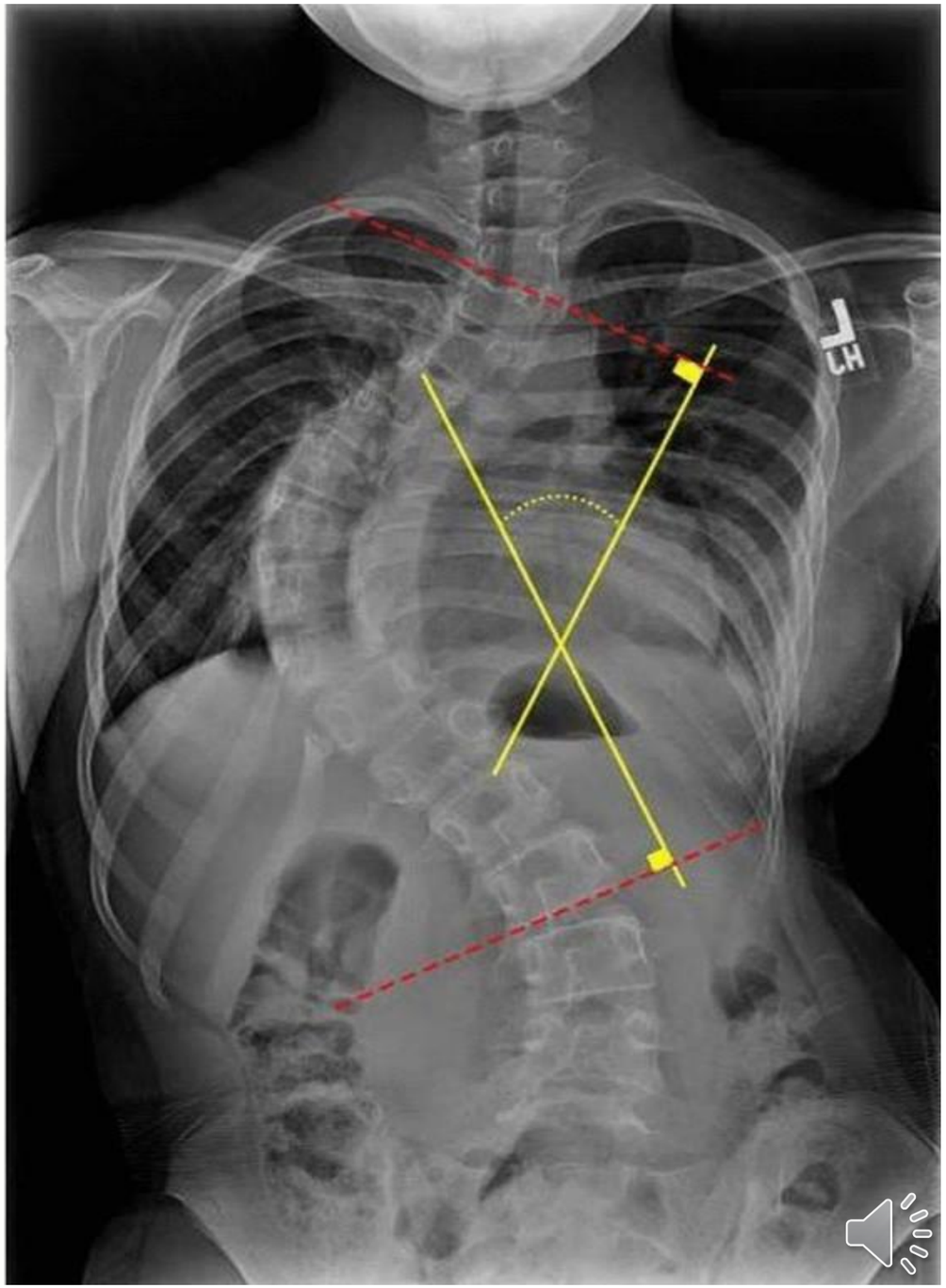

Plumb line
C7
Neutral vertebra
End vertebra
Main curvature
Apex
C
C
Cobb angle
End vertebra
Neutral vertebra
Minor curvature
Central sacral vertical line
S Hapu
CC NC BY SA
Radiopaedia.org

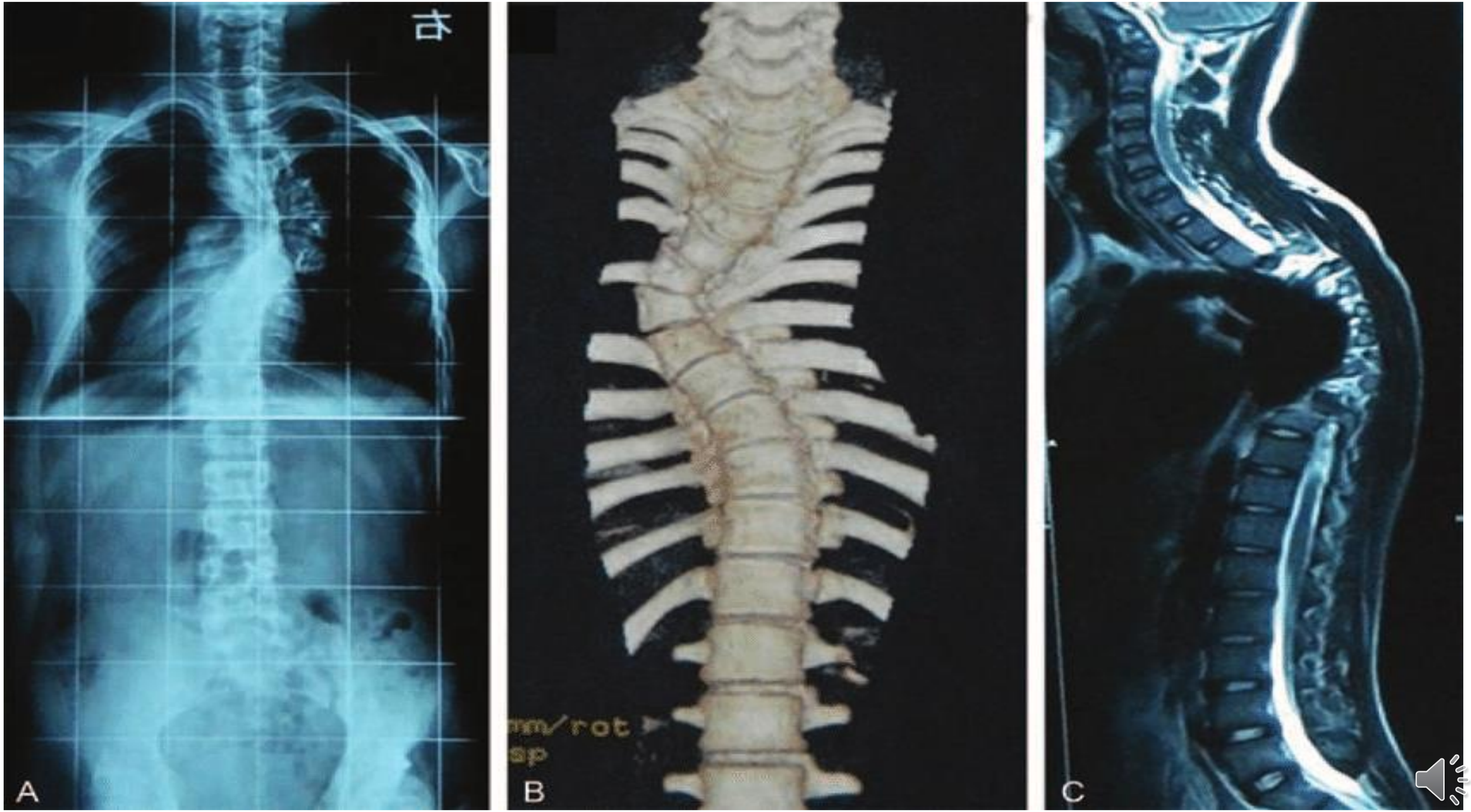
Scoliosis and Cobb Angle




Top Angle
Middle Angle
Bottom Angle


42°
89°
89°
47°


CH

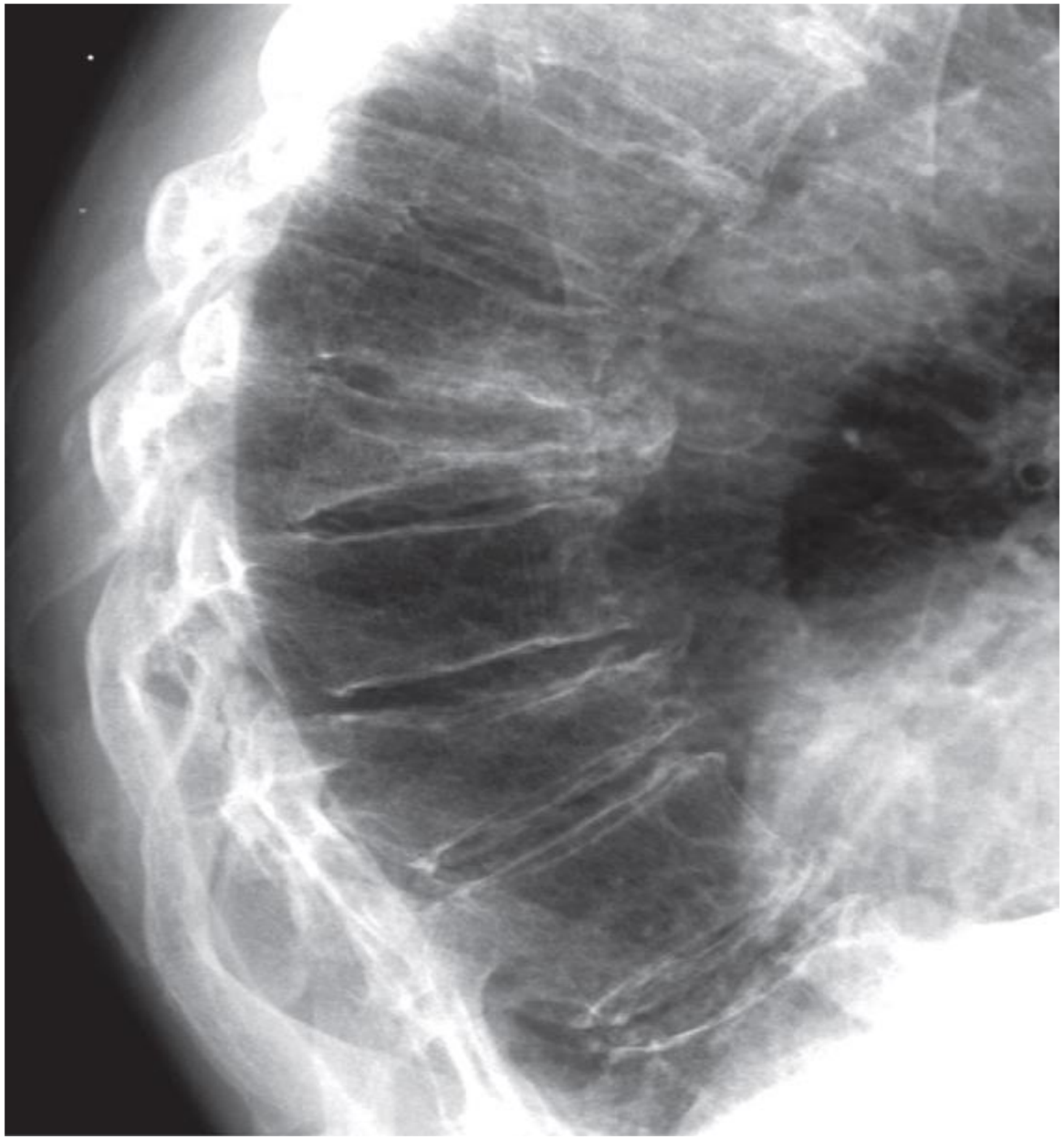

A
B
C

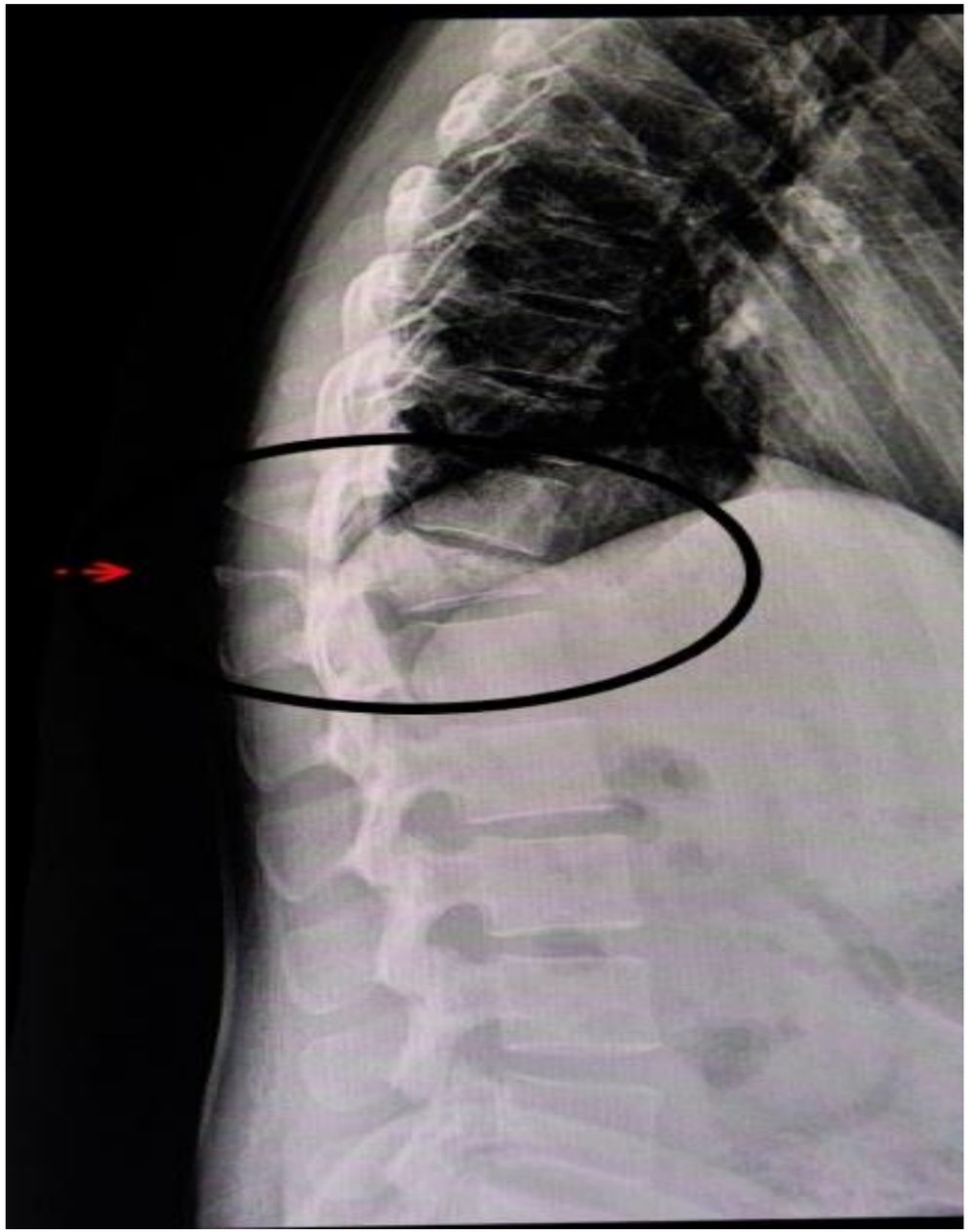
Spinal Trauma

- Fractures of the spine are infrequent compared with fractures of other parts of the skeleton.
- When they occur, they have particular importance because of the implications for associated spinal cord injury.
- The most commonly fractured vertebrae are L1, L2, and T12, accounting for more than half of all thoracolumbar spine fractures.
- In the thoracic spine, compression fractures are the most common type of spinal fracture.

Compression Fractures of the Spine

- Vertebral compression fractures are common, affecting women more than men, and typically secondary to osteoporosis.
- They may be asymptomatic, or they may produce pain in the midthoracic or upper lumbar area that typically disappears in 4 to 6 weeks. Sometimes they are first noticed because of increasing kyphosis.
- Conventional spine radiographs are usually the study of first choice. MRI can be utilized for differentiating osteoporotic compression fractures from malignancy.





S

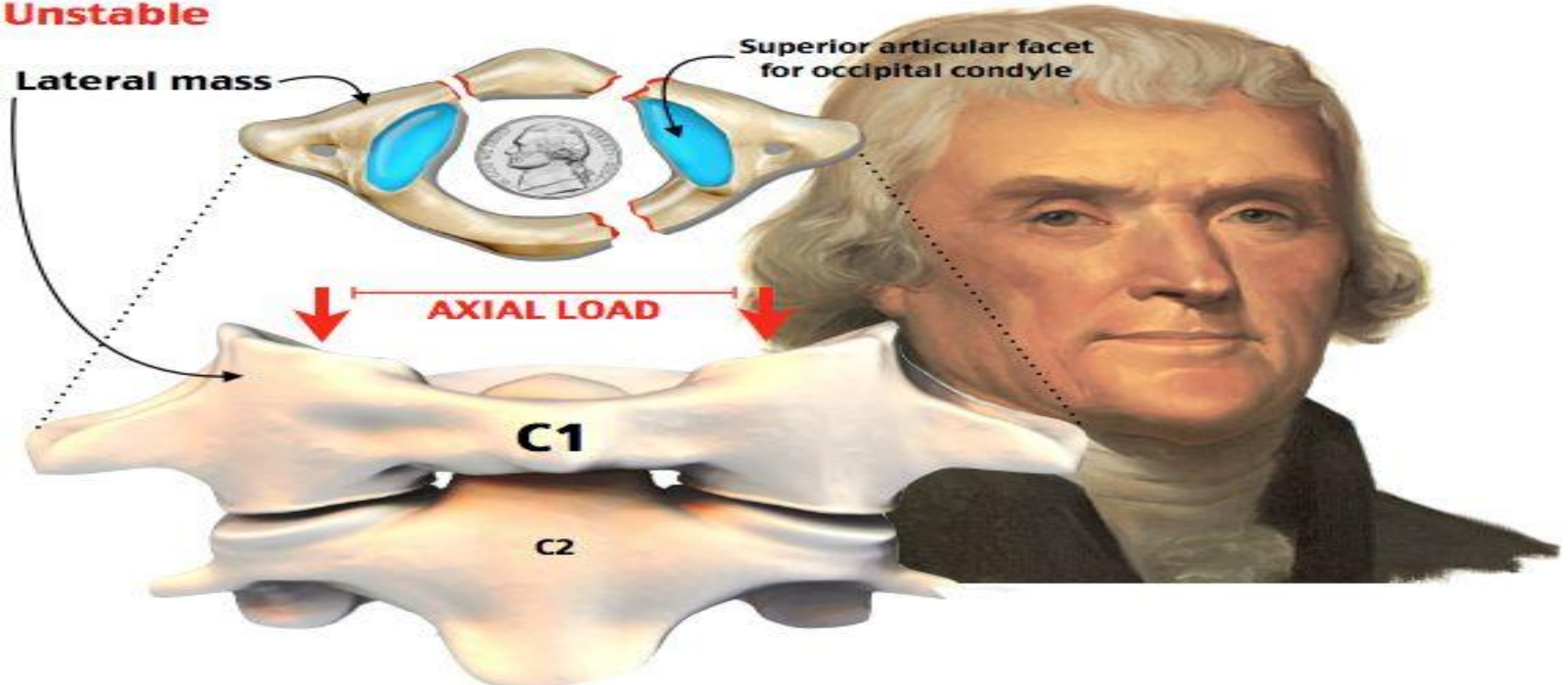
Jefferson Fracture

- A Jefferson fracture is a fracture of C1 usually involving both the anterior and posterior arches.
- In its classical presentation, there are bilateral fractures of both the anterior and posterior arches of C1, producing four fractures in all.
- It is caused by an axial loading injury (such as diving into a swimming pool and hitting one's head on the bottom).

- **On conventional radiographs**, the hallmark of a Jefferson fracture is bilateral, lateral offset of the lateral masses of C1 relative to C2 as seen on the open mouth view (atlantoaxial view) of the cervical spine.
- The fracture is confirmed utilizing CT .
- A Jefferson fracture is a "***self-decompressing***" fracture in that the spinal canal at the level of the fracture is wide enough to accommodate any swelling of the cord.
- There is usually ***no neurologic deficit*** associated with this type of fracture.

C1 Jefferson Burst Fracture

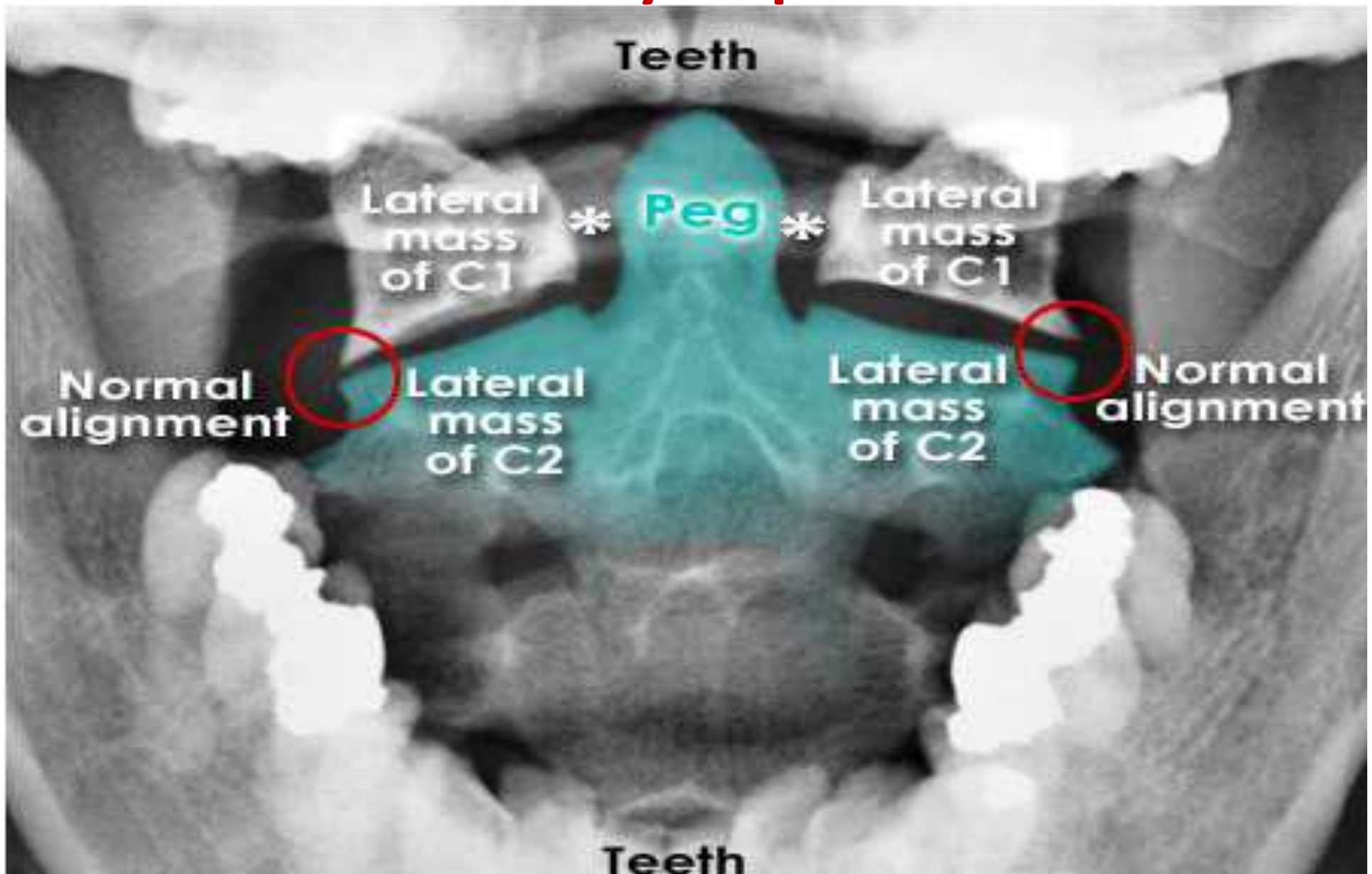
Unstable



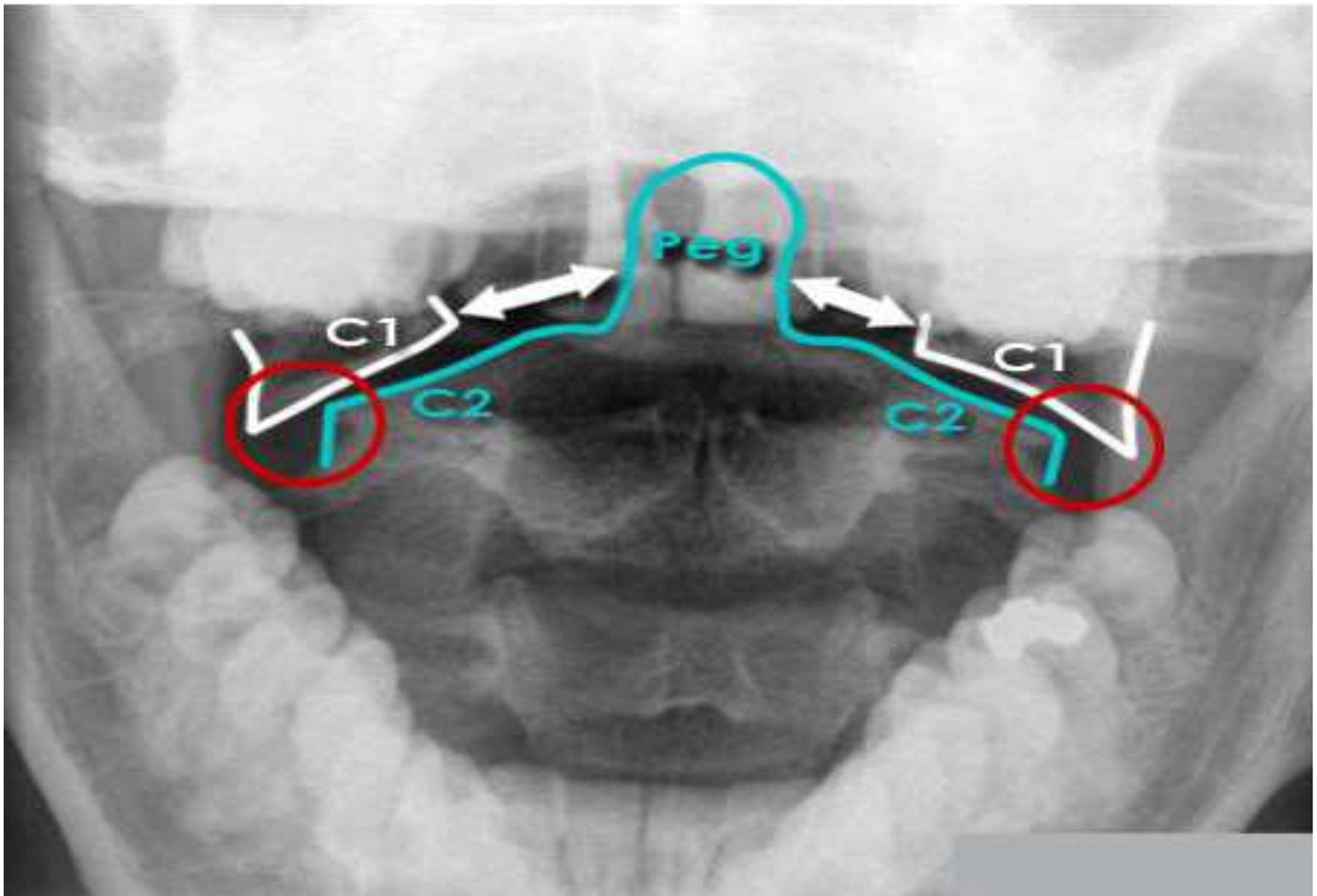
Football player
spearing another player

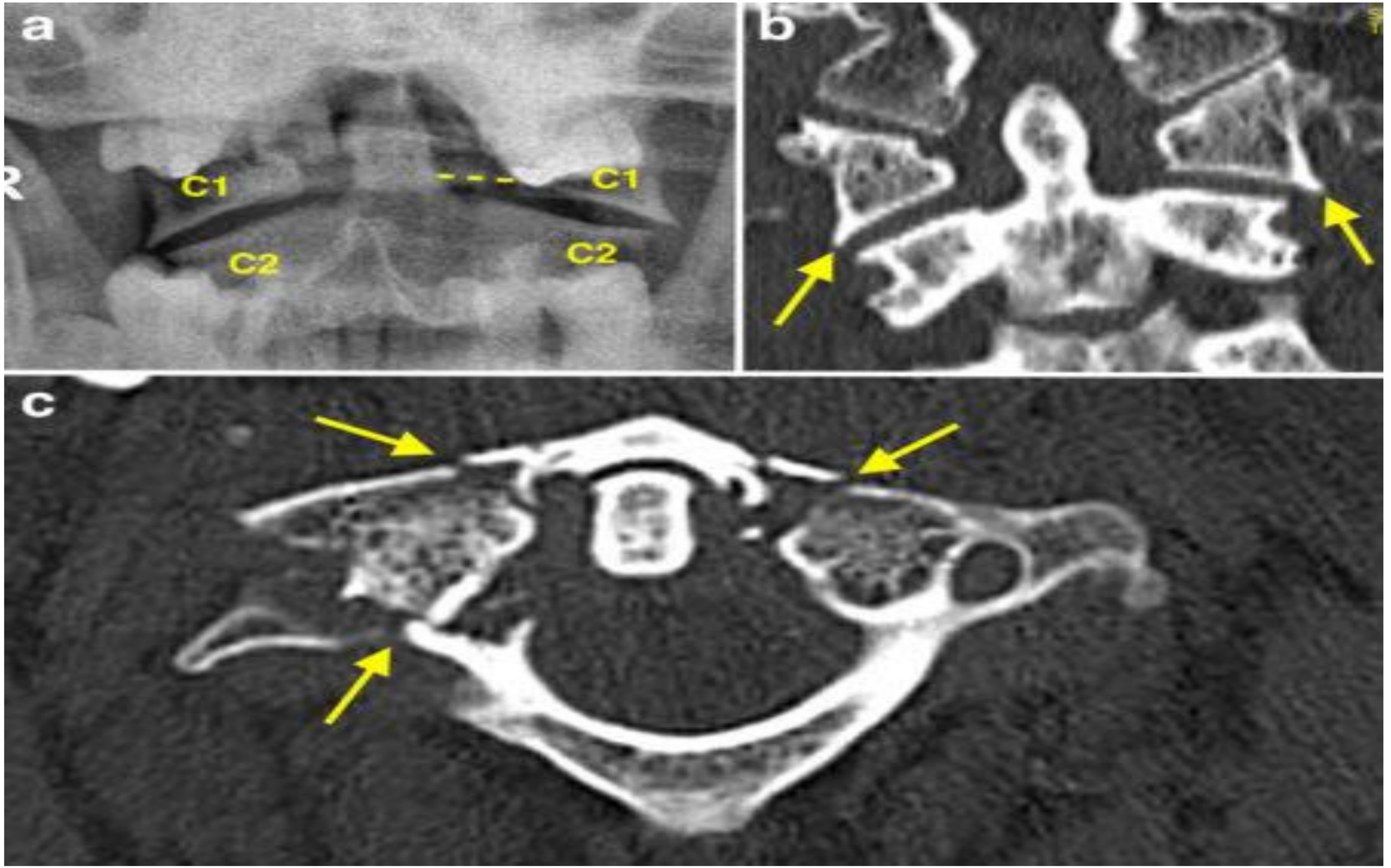
Diving into shallow pool

Normal anatomy - Open mouth view



C1 'Jefferson' fracture - Open mouth view

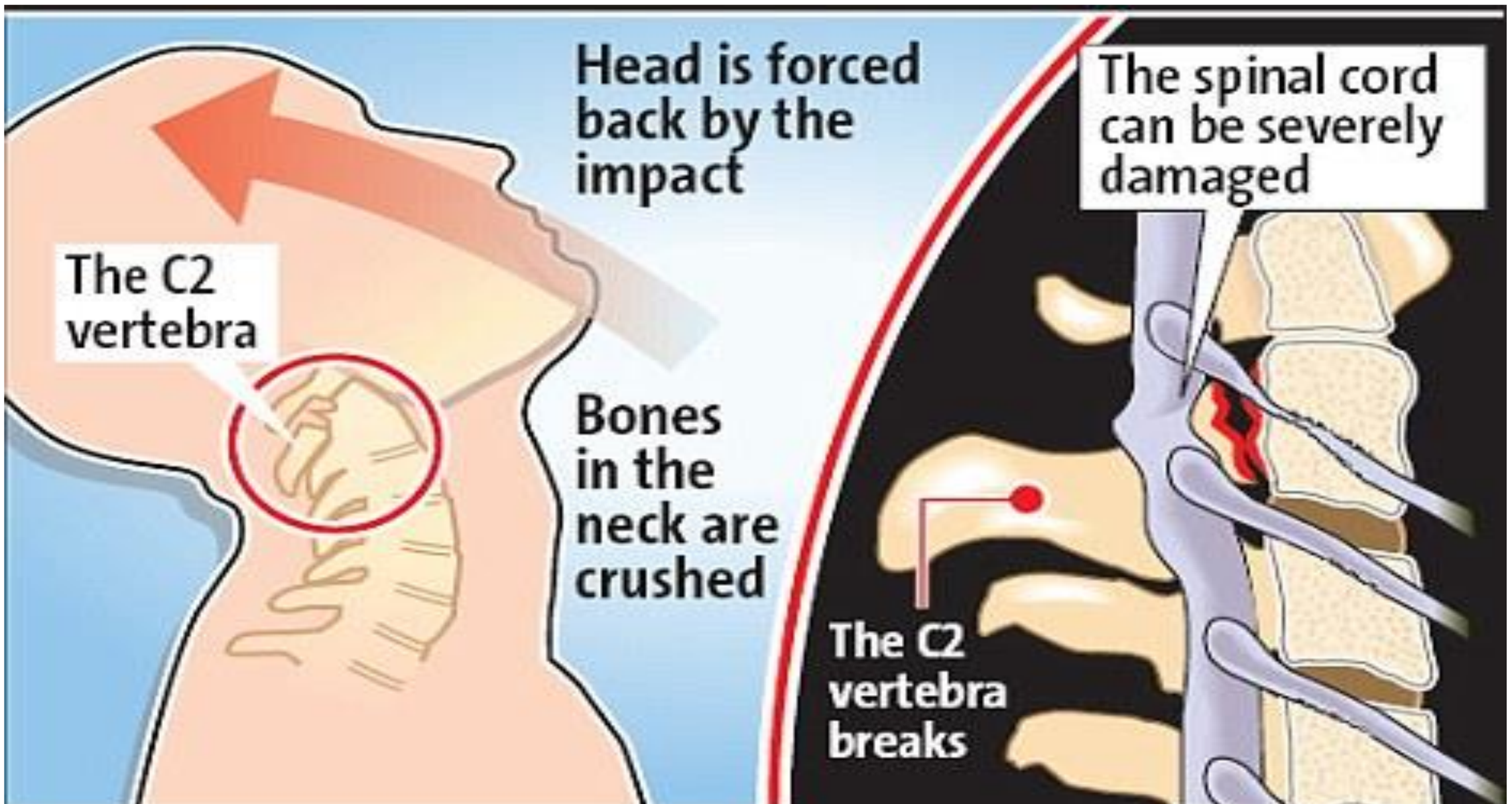



a
C1
C1
C2
C2
b
c

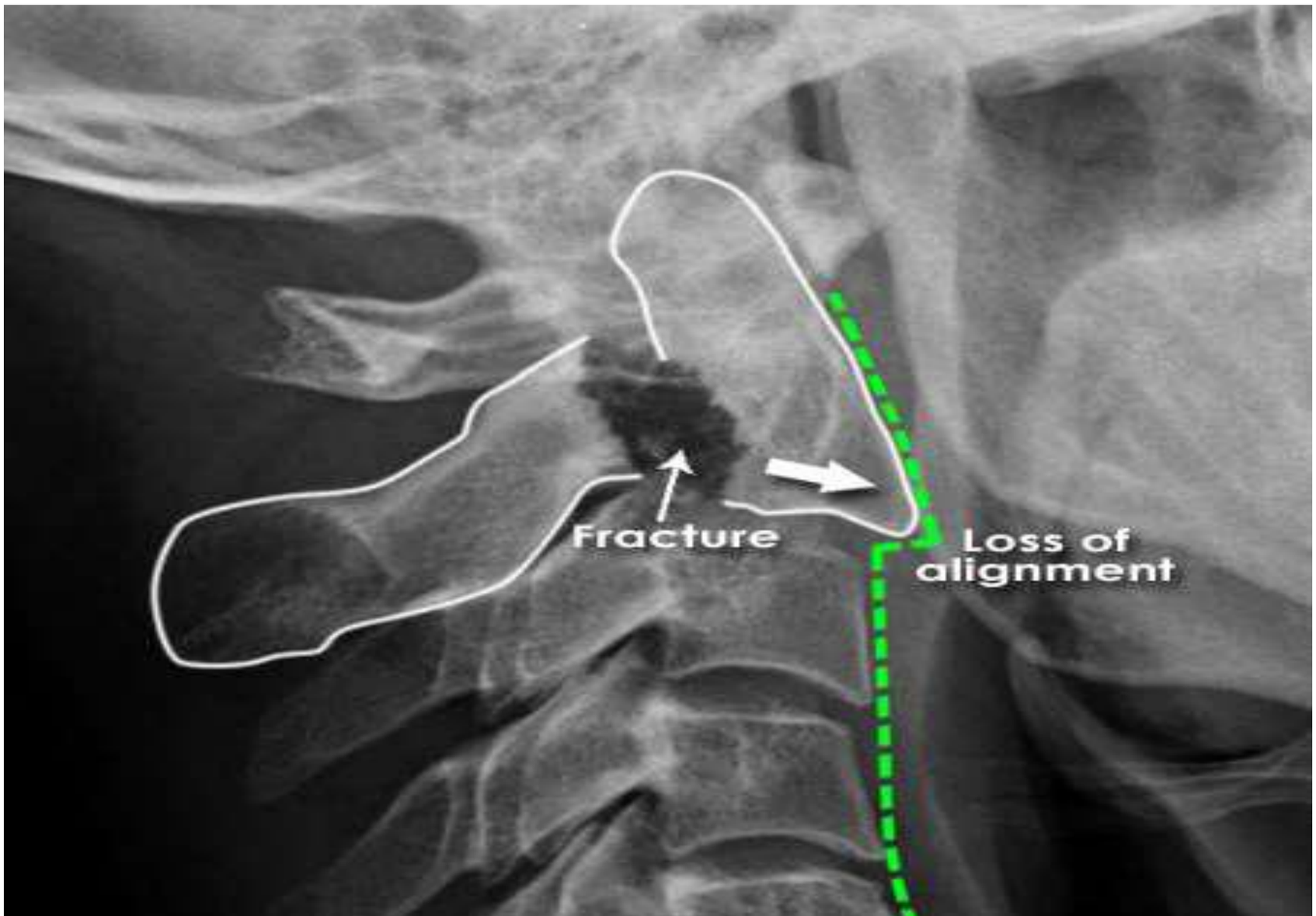
Hangman’s Fracture

- A hangman’s fracture is a fracture of the posterior elements of C2.
- Hangman’s fractures result from a hyperextension-compression injury typically occurring in an unrestrained occupant in a motor vehicle accident who strikes his forehead on the windshield.
- Hangman’s fractures are best evaluated on the lateral view of the cervical spine on conventional radiography and the sagittal view on CT.
- The fracture effectively separates the posterior aspect of the C2 vertebral body from the anterior aspect of C2, allowing the anterior aspect of C2 to sublux forward on the body of C3.

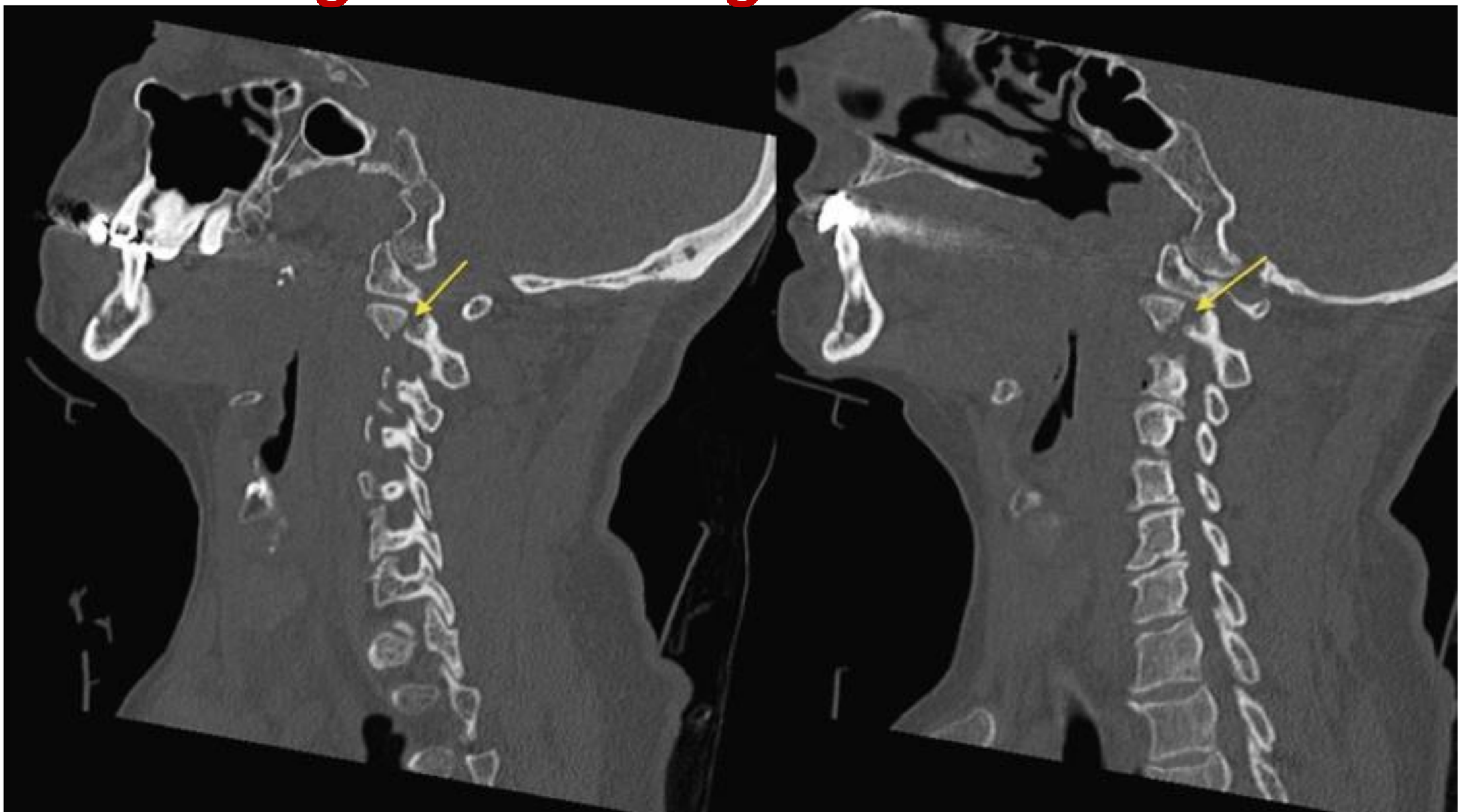
Hangman's Fracture



Lateral; X-ray: Hangman's Fracture

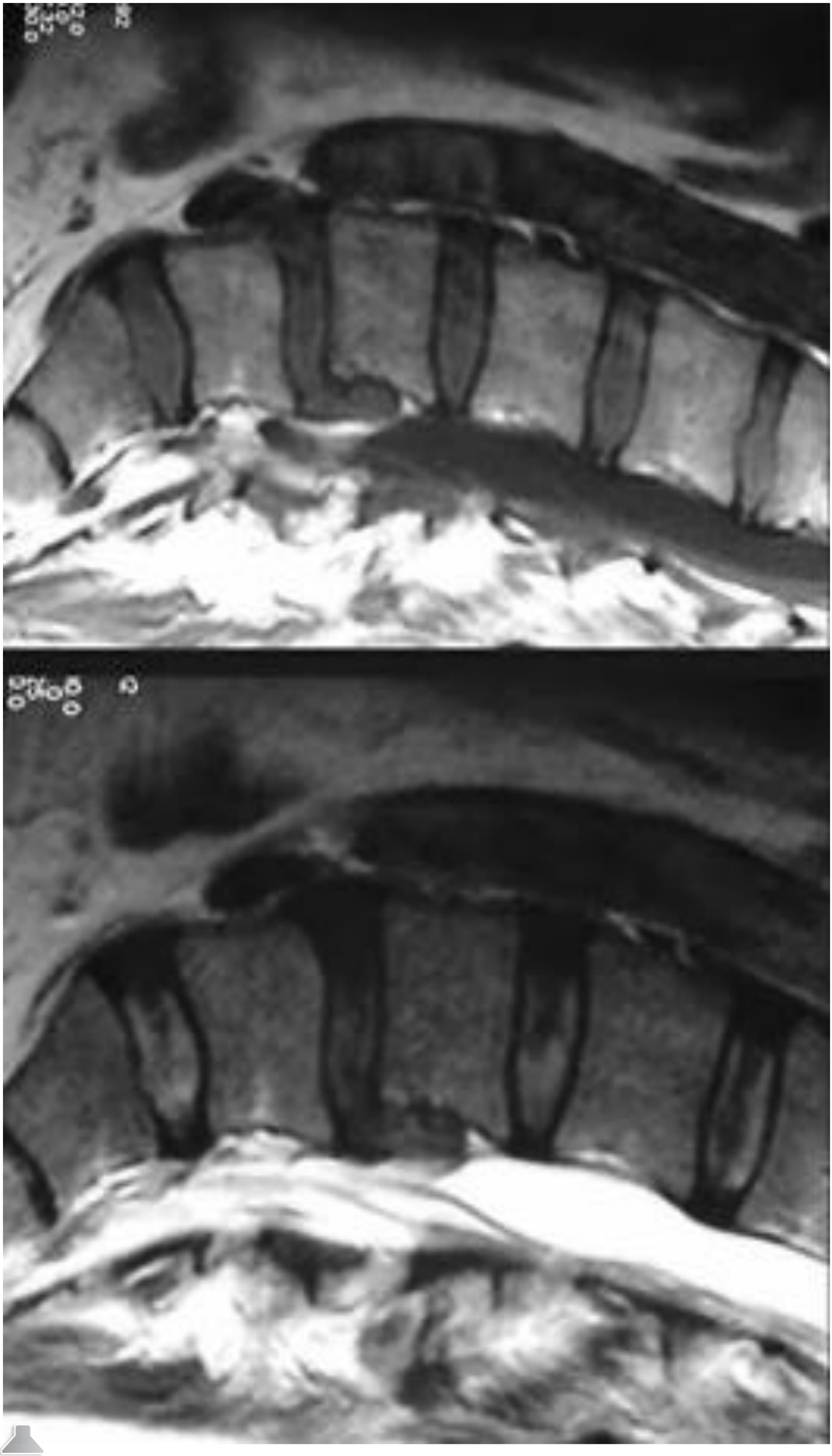


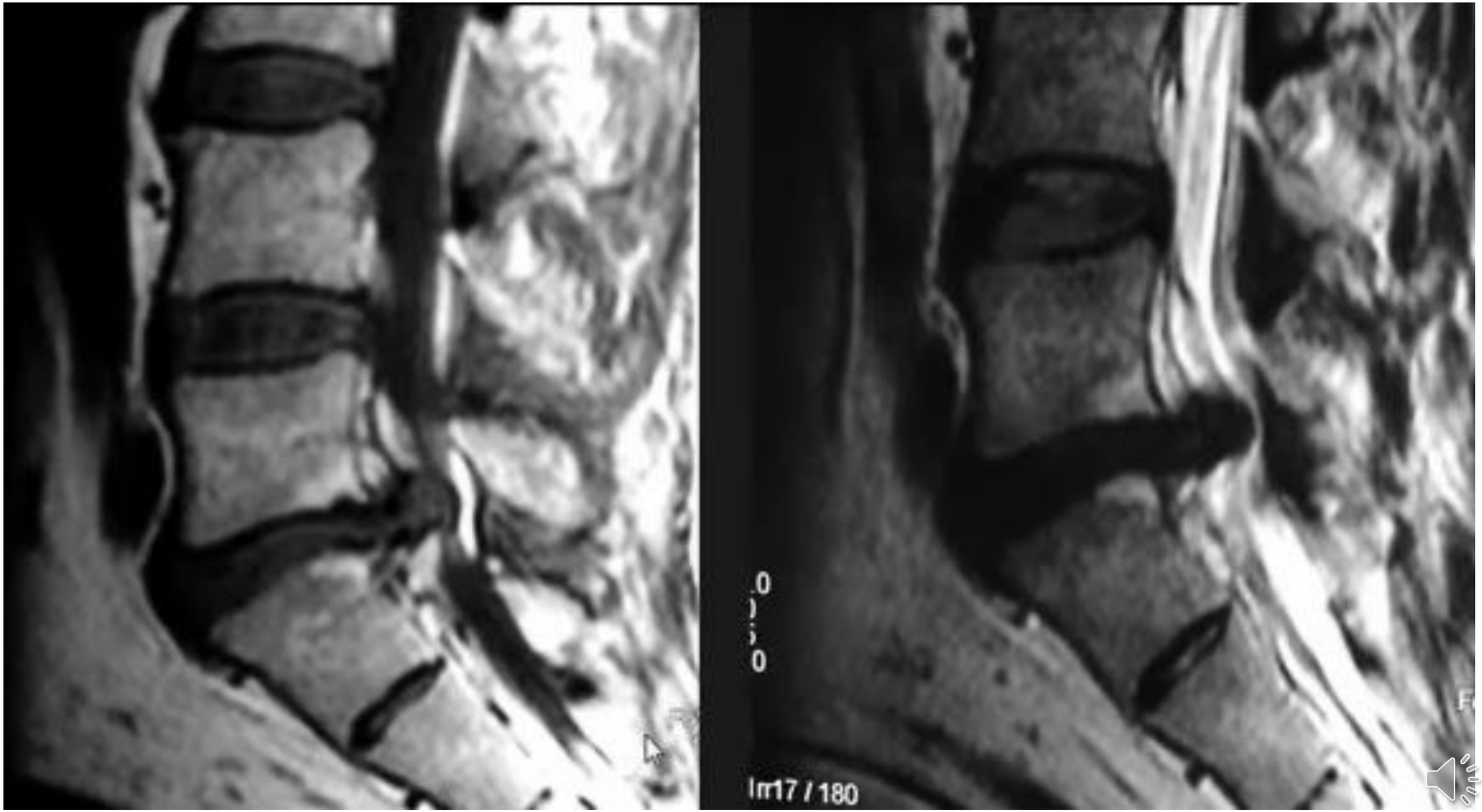
Sagittal CT: Hangman's Fracture



Group Discussion

Give spot diagnosis for the following Images

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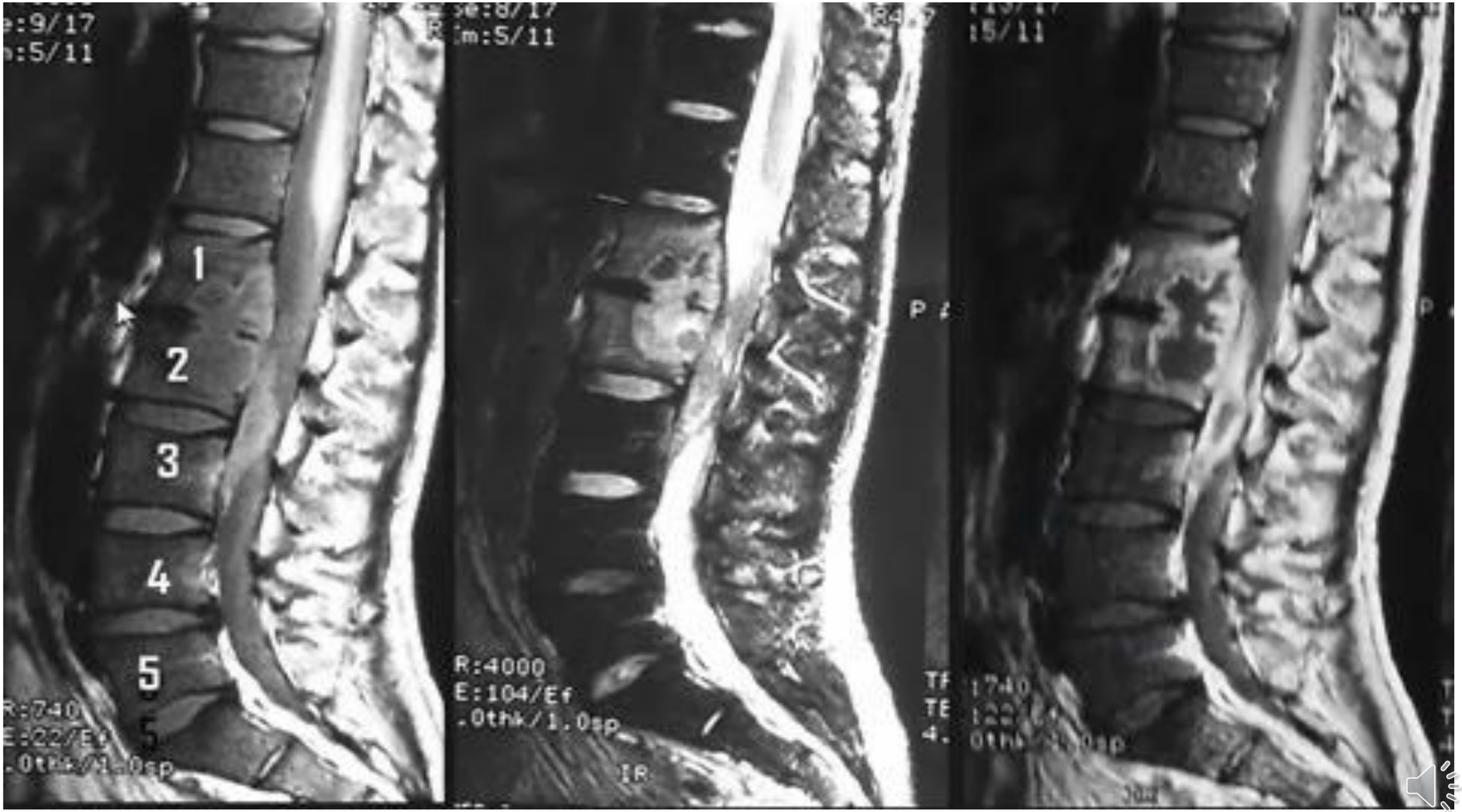

1
2
3
4
5
5
R:4000
E:104/Ef
.0thk/1.0sp
IR

Thank you

Al-Azhar University – Gaza
Faculty of Applied Medical Sciences

Musculoskeletal System Pathology

Osteoarthritis

Husam Mansour

Osteoarthritis

Osteoarthritis is a non-inflammatory characterized by degeneration of articular cartilage and is part of the normal ageing process due to wear and tear of the articular surface.

Primary osteoarthritis is more common than secondary osteoarthritis, common in elderly and its mainly due to wear and tear changes occurring in old ages mainly in weight bearing joints.

Secondary osteoarthritis results from previous trauma with mal-alignment of articular surfaces, joint infection and rheumatoid arthritis.

General radiological features

1. **Osteophyte formation:** osteophytes are spurs of compact bone that form at joint margins.

2. **Joint space narrowing:** cartilage loss eventually leads to non - uniform joint space narrowing.

3. **Loose bodies:** result from separation of cartilage and osteophytes.

4. **Subchondral cysts and sclerosis:** increased bone density around joints with degenerative cyst formation.

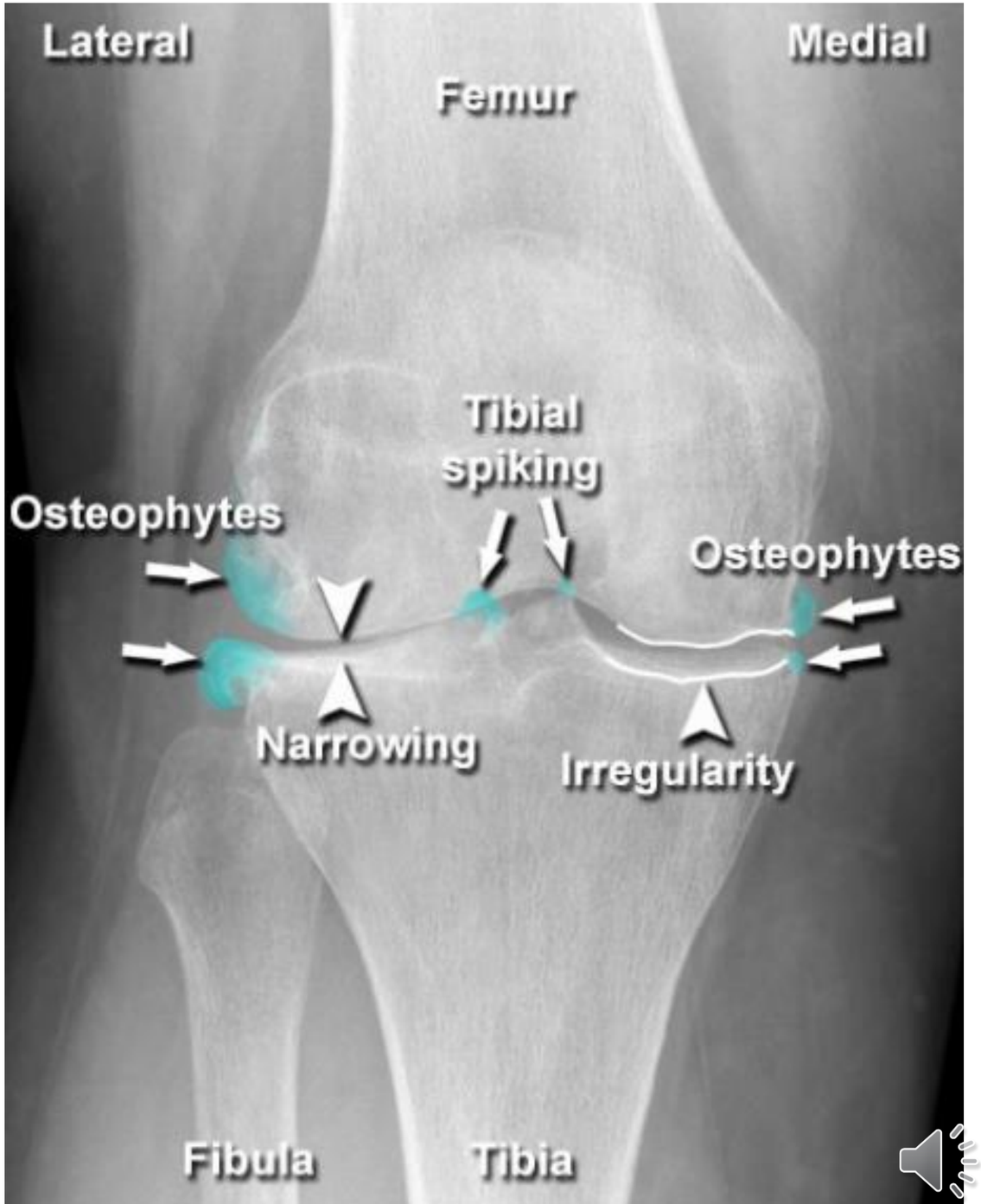
Common sites of involvement

Knee

- The most common joint involved, with femorotibial compartment loss of joint space. The medial compartment is the weight-bearing part under greatest stress, and so almost always shows the earliest narrowing.
- The joint may become stiff and swollen, making it difficult to bend and straighten the knee.
- Severe changes may require total knee joint replacement.

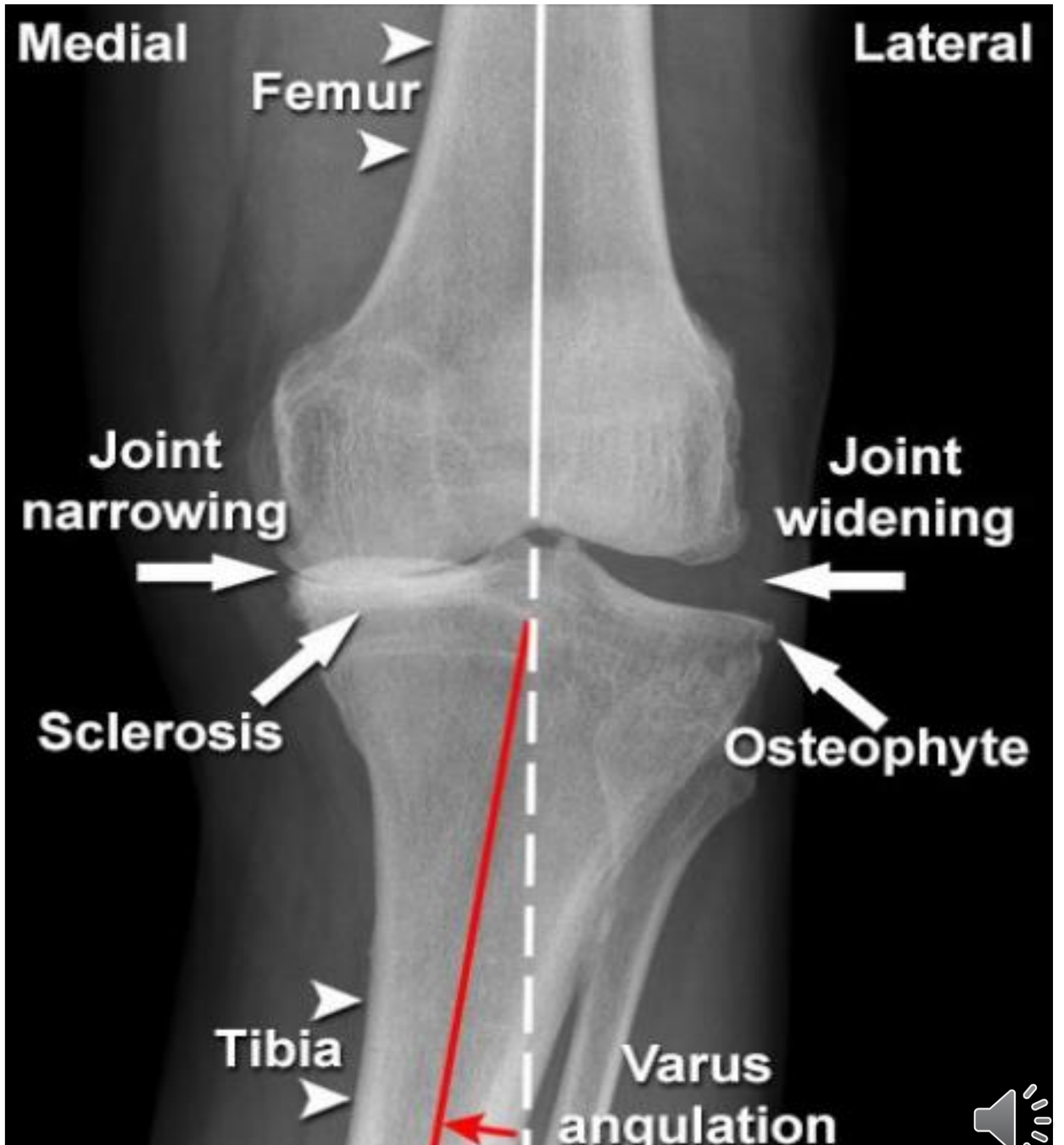
Osteophytes - Knee

- The formation of osteophytes is a common feature of osteoarthritis
- This AP view of the knee shows osteophytes of the tibia and femur
- Small osteophytes have also formed at the tibial spines (tibial spiking)
- The knee joint spaces are narrowed and irregular



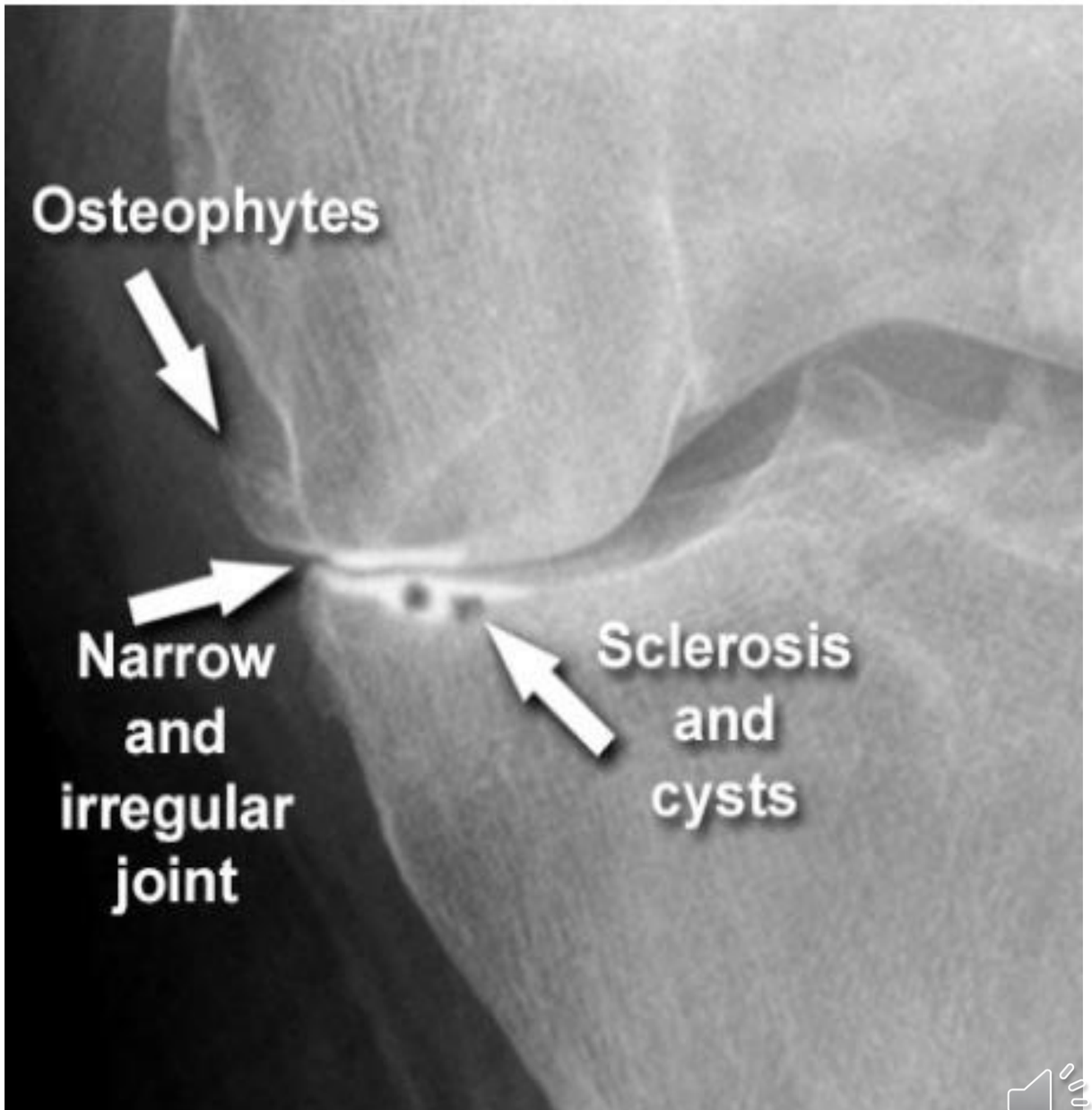
Joint space narrowing

- The medial side of the knee joint is severely narrowed such that the femur and tibia are touching
- The lateral side of the joint is widened and there is abnormal (varus) angulation of the tibia
- Increased density indicates articular surface sclerosis



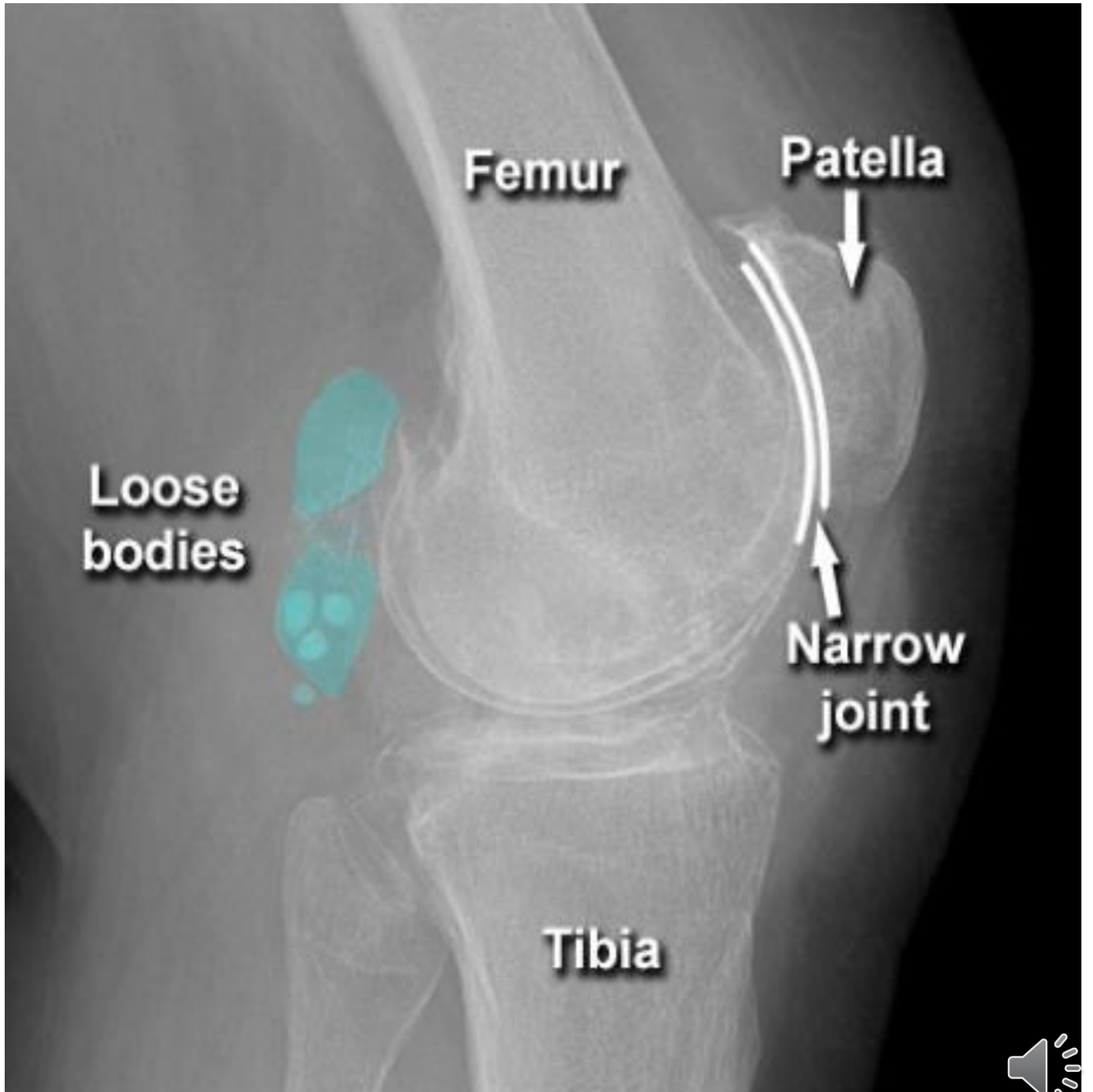
Articular surface cortical irregularity and/or sclerosis

- The joint space is narrowed and the cortical surface is irregular
- Increased density of the articular surface indicates sclerosis
- Small sub-cortical cysts have formed



Intra-articular loose bodies

- In some patients calcified loose bodies may be seen within an osteoarthritic joint
- These are fragments of bone or cartilage which have detached from the joint surfaces and remain in the joint capsule
- This lateral view of the knee also shows severe osteoarthritis of the patellofemoral joint which is very narrow



Hips

- Joint space narrowing is seen initially at the superior maximum weight - bearing aspect, with femoral and acetabular osteophytes. Other findings may include sclerosis and subchondral cyst formation.
- Factors that may make you more likely to develop osteoarthritis

1. Increasing age
2. Family history of osteoarthritis
3. Previous injury to the hip joint
4. Improper formation of the hip joint at birth, DDH

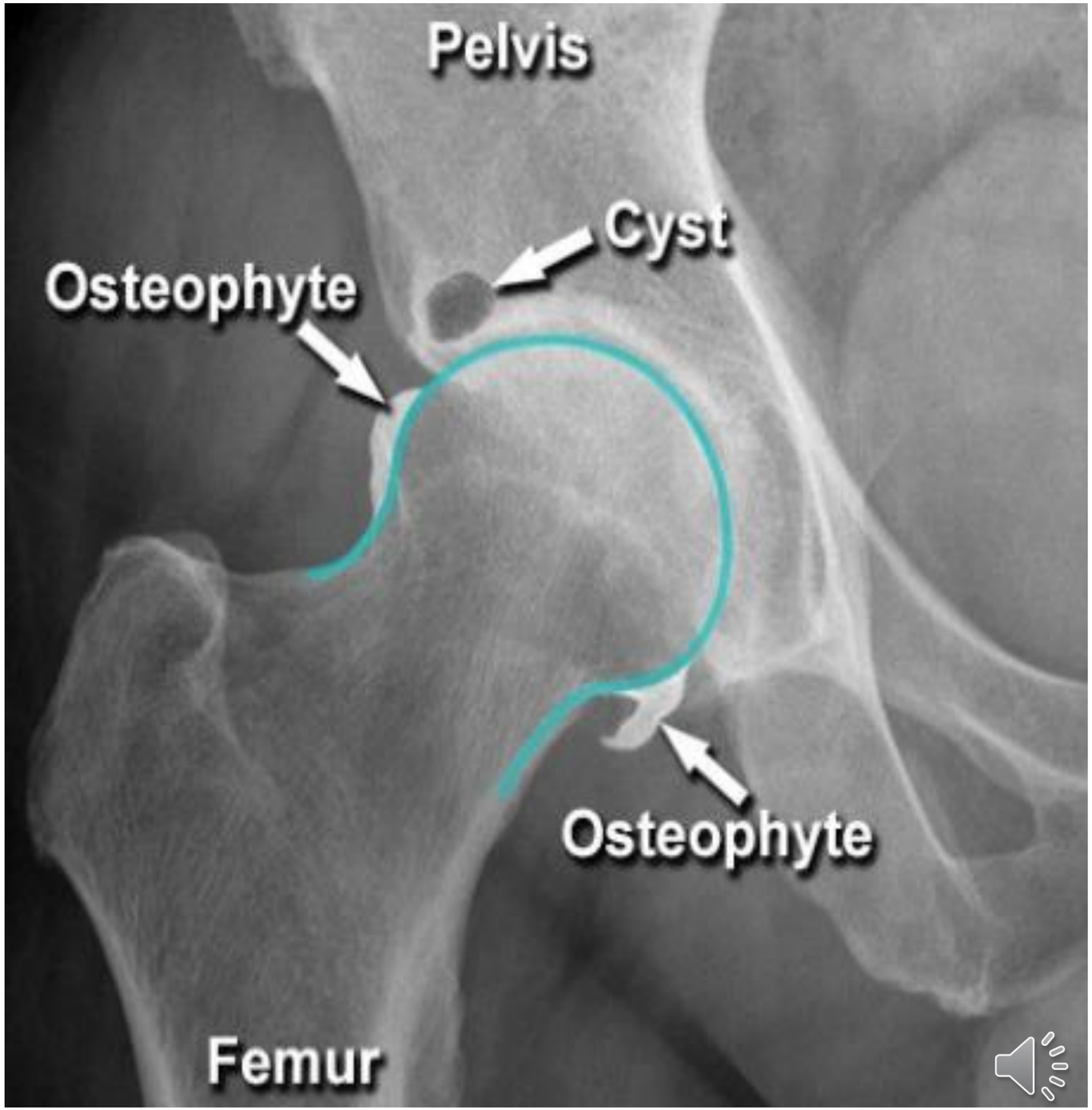
(Left) In this x-ray of a normal hip, the space between the ball and socket indicates healthy cartilage. (Right) This x-ray of an arthritic hip shows severe loss of joint space and bone spurs.

Sub-cortical cysts (geodes)

Although sub-cortical bone cysts are a characteristic finding of osteoarthritis, they are only visible in approximately one third of X-rays of osteoarthritic joints

This hip joint is narrowed and large osteophytes have formed

Note: These cysts may be called ‘sub-chondral cysts’, ‘sub-cortical cysts’, or 'geodes' – any of these terms are acceptable



Spine

- In the spine, OA can affect the facet joints, the intervertebral discs, and the ligaments supporting the spine.
- Degenerative changes are present in nearly all elderly patients. Features include:

1. Narrowing of disc space;

2. New bone formation (spurring) between adjacent vertebrae may cause nerve root

3. Spinal cord compression;

4. Sclerosis and osteophytes at intervertebral apophyseal joints.

Degenerative changes in the cervical spine with mid and lower cervical loss of disc space

R
Osteophyte
Narrowed
disc space
Osteophytes
Sclerosis

Multiple osteoarthritic changes

- Large osteophytes (**arrowheads**) have formed at the joints of the fingers
- These osteophytes correspond to the clinical finding of bony swelling at the distal interphalangeal joints and the proximal interphalangeal joints
- The joints are severely narrowed (**arrows**)
- Large sub-cortical cysts have also formed in this patient with severe osteoarthritis of the finger joints
- The ring could no longer be removed

Thank you

Al-Azhar University – Gaza
Faculty of Applied Medical Sciences

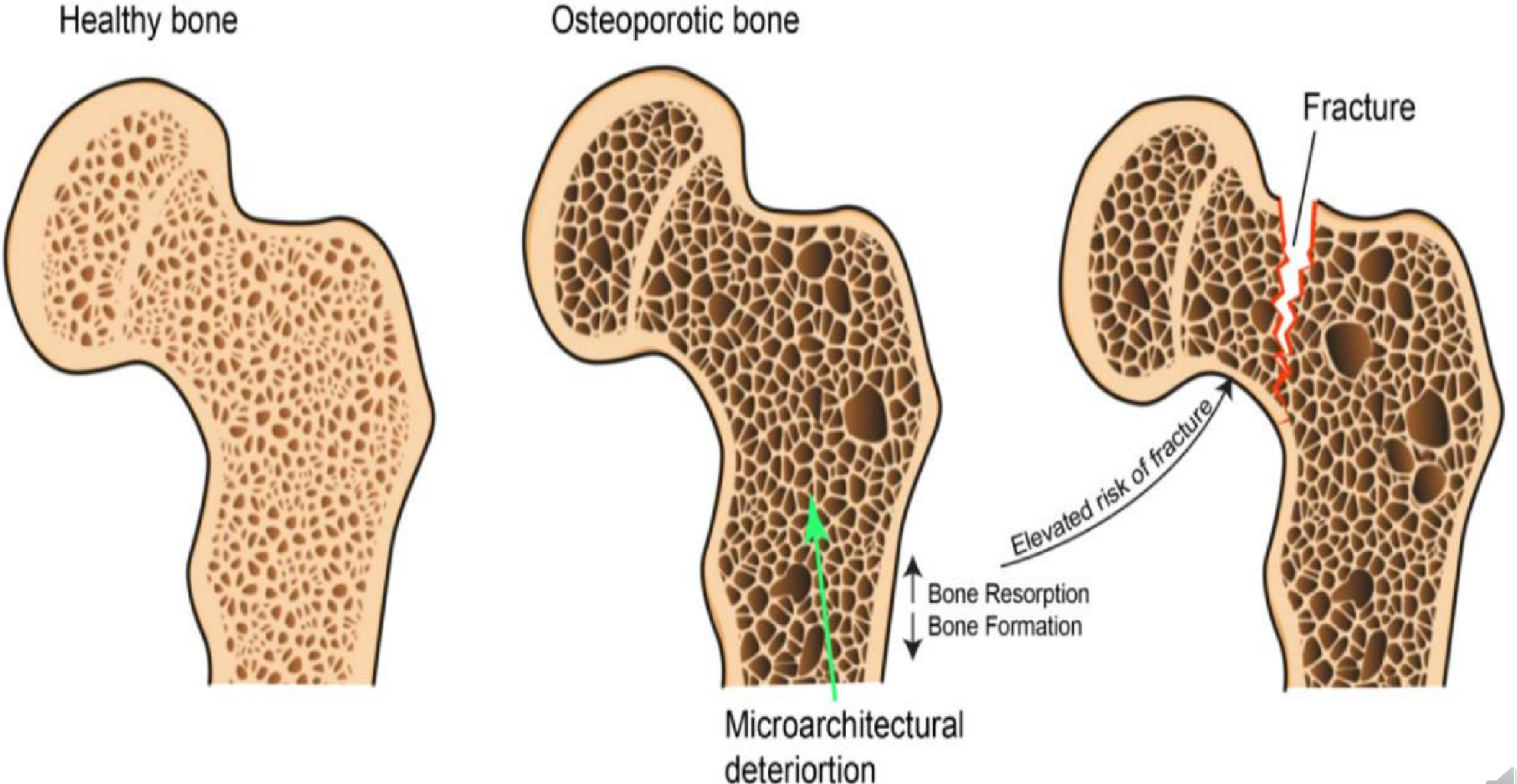
Musculoskeletal System Pathology

Osteoporosis

Husam Mansour

Osteoporosis

- Osteoporosis, the most common of all metabolic bone disorders, is defined by the World Health Organization (WHO) as “a skeletal disease, characterized by low bone mass (reduction of bone mass) and micro-architectural deterioration of bone tissue, with a consequent increase in bone fragility and susceptibility to fracture.


Healthy bone
Osteoporotic bone
Fracture
Elevated risk of fracture
Bone Resorption
Bone Formation
Microarchitectural deteriortion

- **Role of Diagnostic Imaging**

- Diagnostic imaging of osteoporosis has two principal aims:

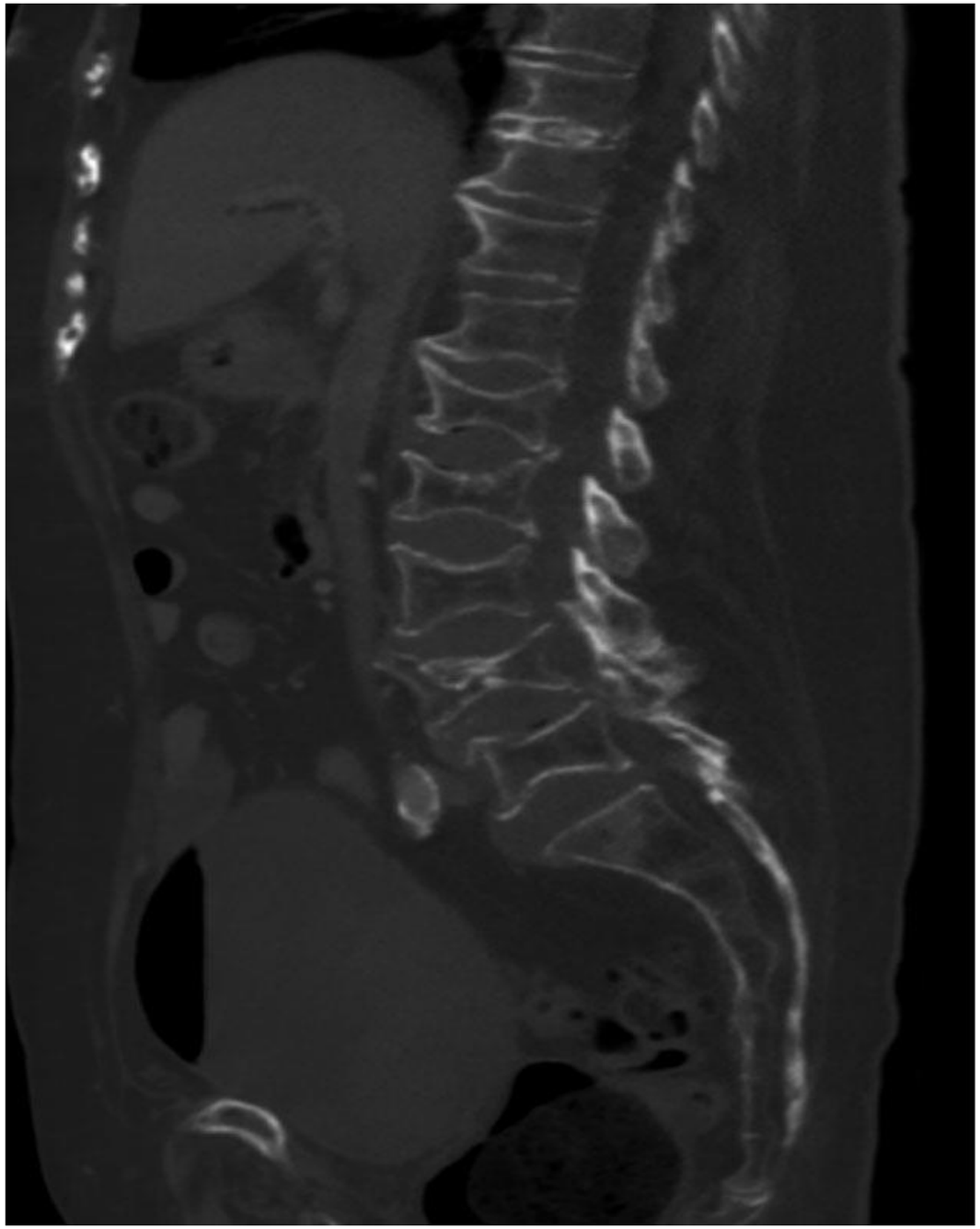
1. To identify the presence of osteoporosis.

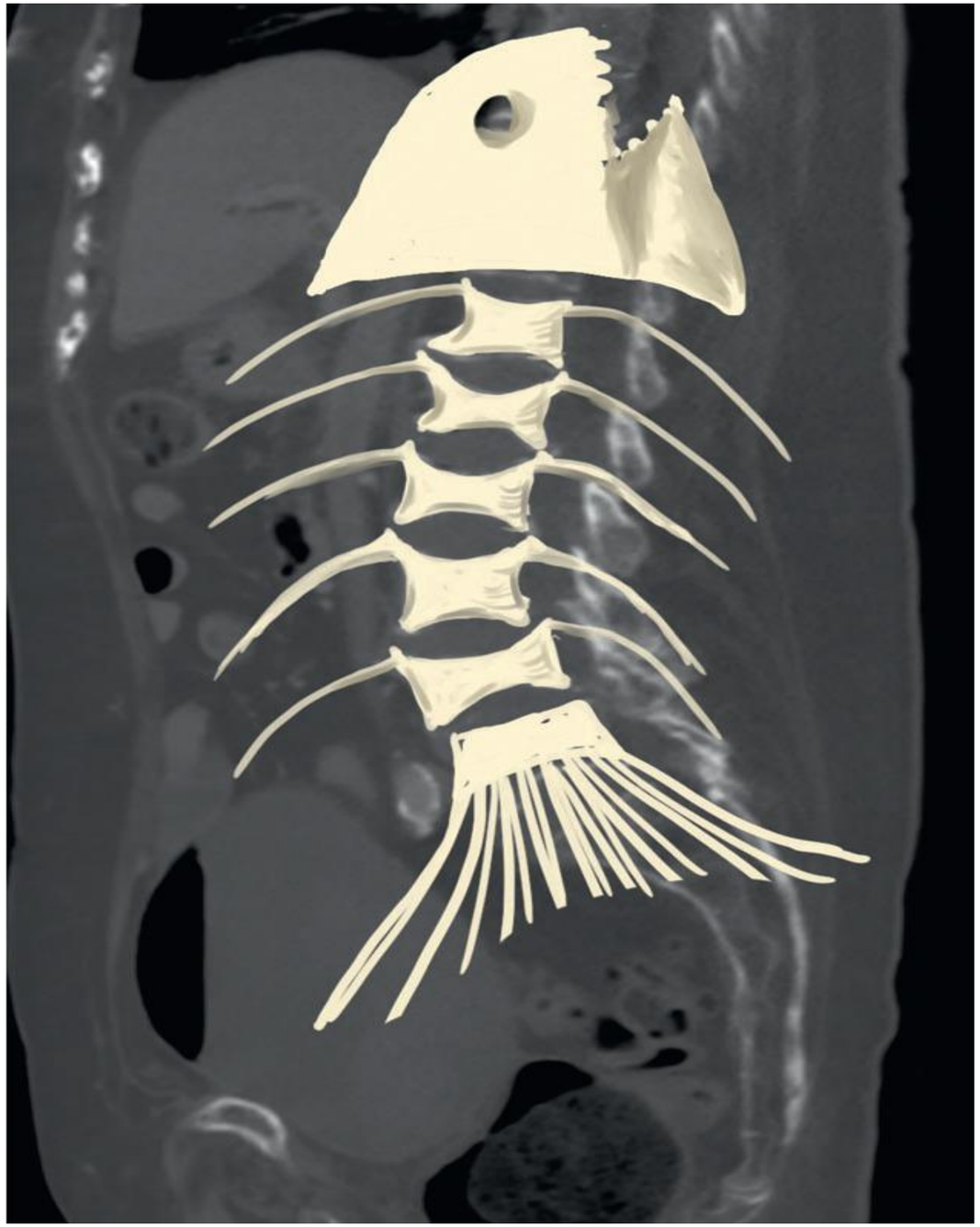
2. To quantify bone mass with use of semiquantitative (conventional radiography) or quantitative (Dual-Energy X-Ray Absorptiometry (DEXA)) methods.

Conventional Radiography (X-Ray image)

- Conventional Radiography is able to help detect bone loss only when a substantial amount (approximately 30%) occurs.
- The main radiographic features of generalized osteoporosis are increased radiolucency and cortical thinning .
- The term osteopenia (“poverty of bone”) is used as a generic designation for radiographic signs of decreased bone density.

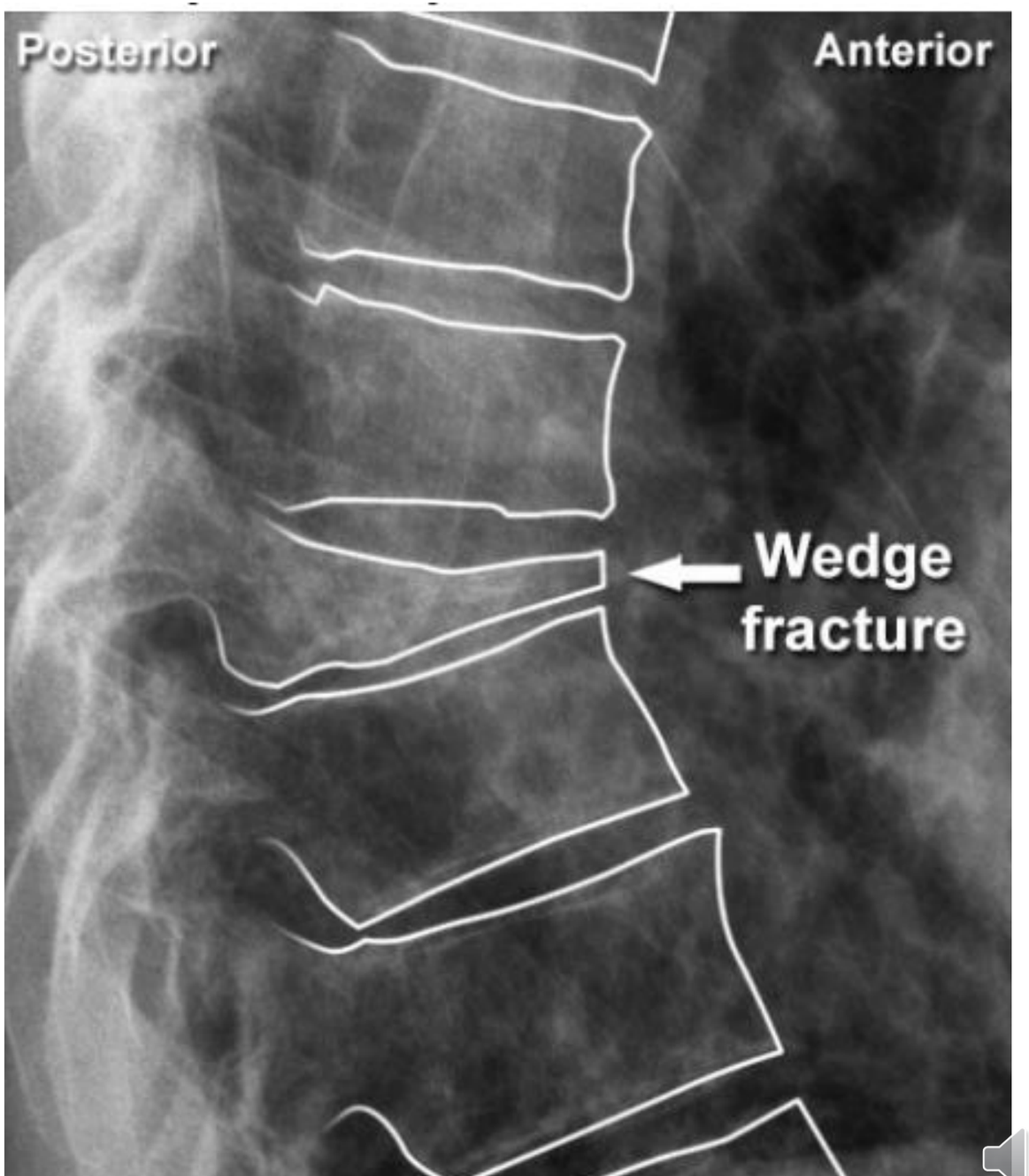
- **Radiological features**
- Osteoporosis results in a loss of bone density. The condition manifests itself most prominently in the spine. The vertebral bodies appear lucent with thin cortical lines, often with a biconcave appearance (‘**cod fish’ vertebrae**), vertebral wedging and collapse; this subsequently leads to a kyphosis.
- Fractures of the peripheral skeleton, including femoral neck fractures, commonly occur even after minor trauma.

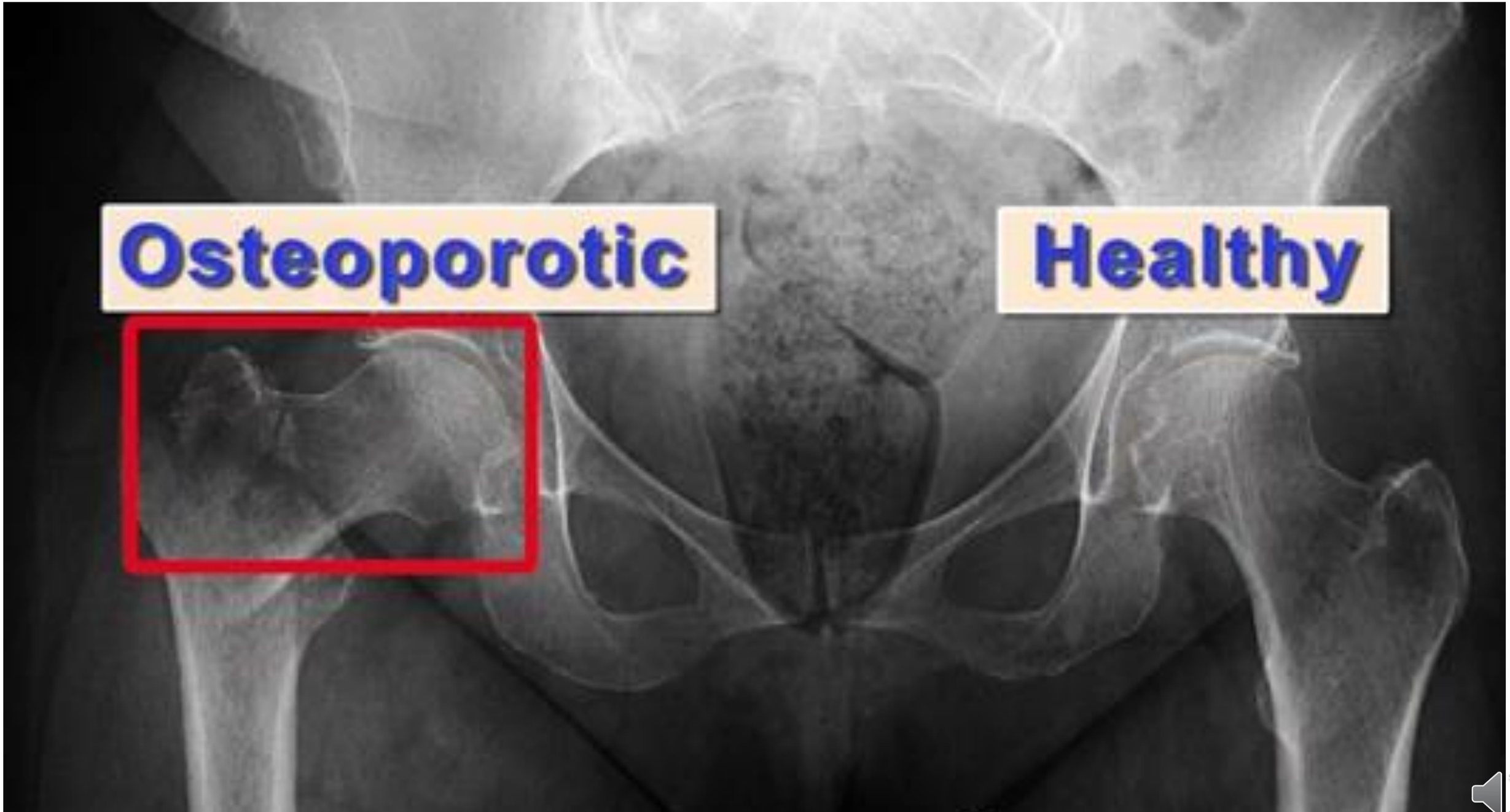




Osteoporotic wedge fracture – T-spine lateral view

- All the bones appear of reduced density (blacker than normal)
- One of the vertebral bodies of the T-spine has collapsed
- This fracture is often described as a wedge fracture

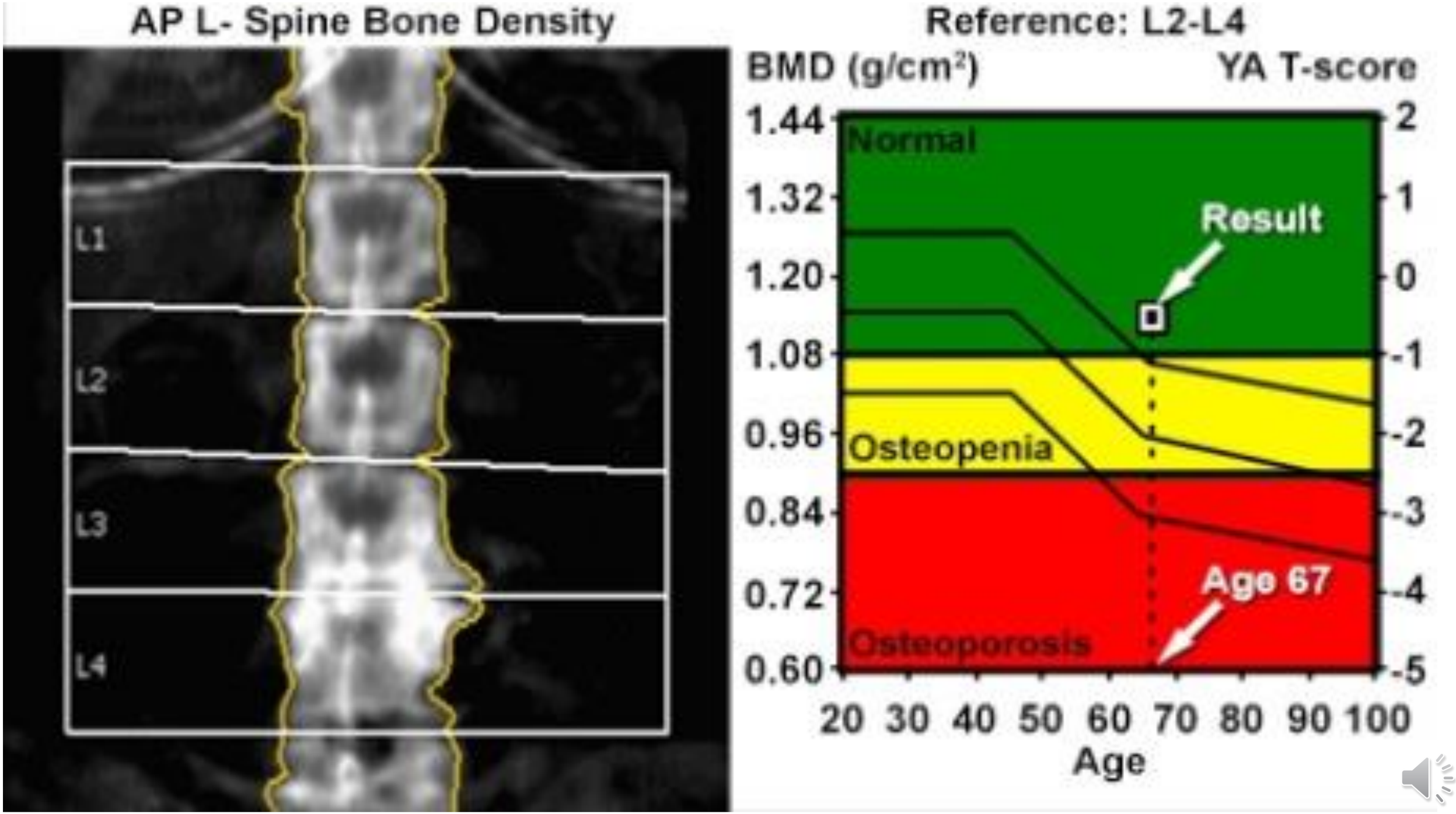


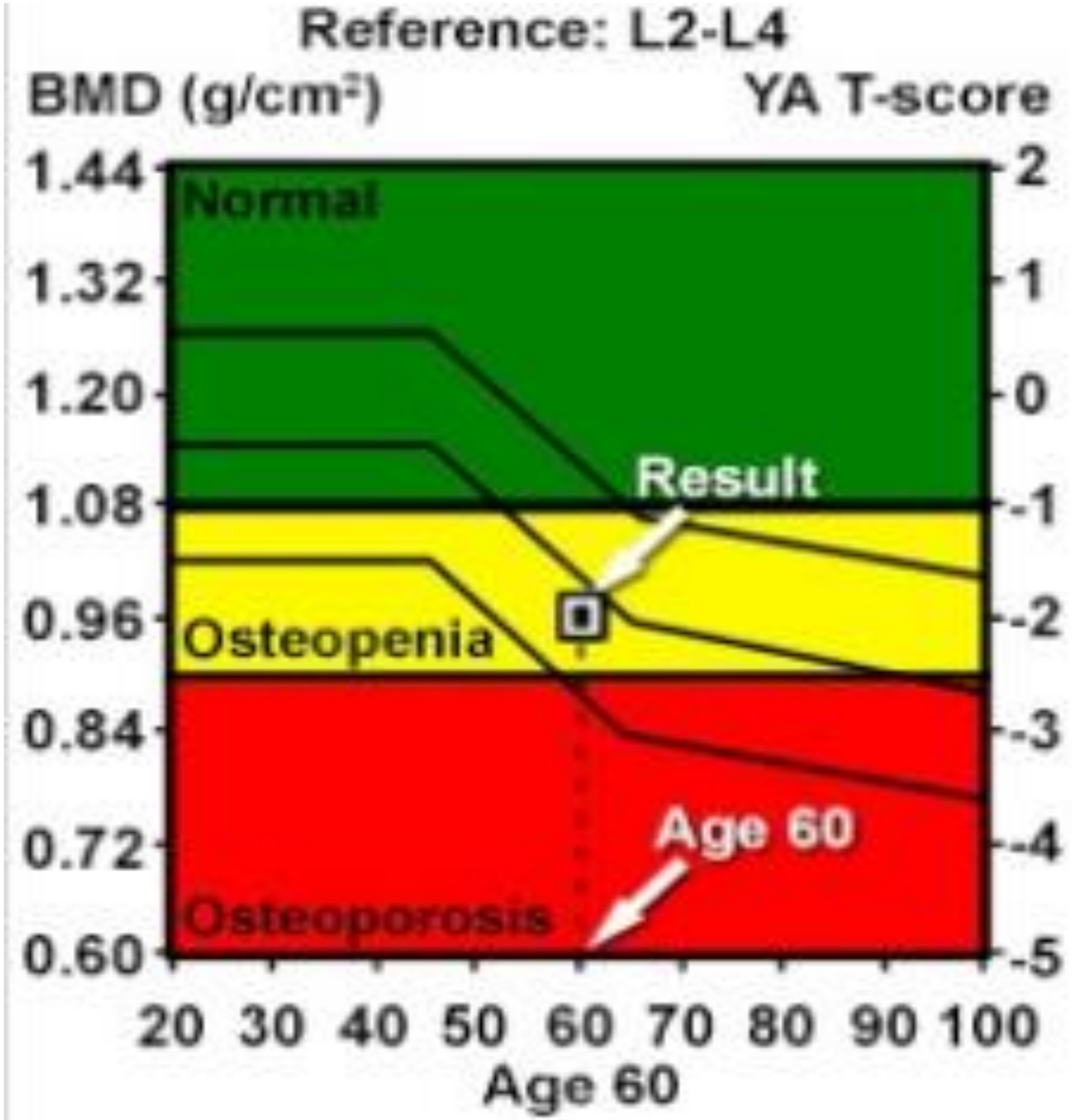

Osteoporotic
Healthy

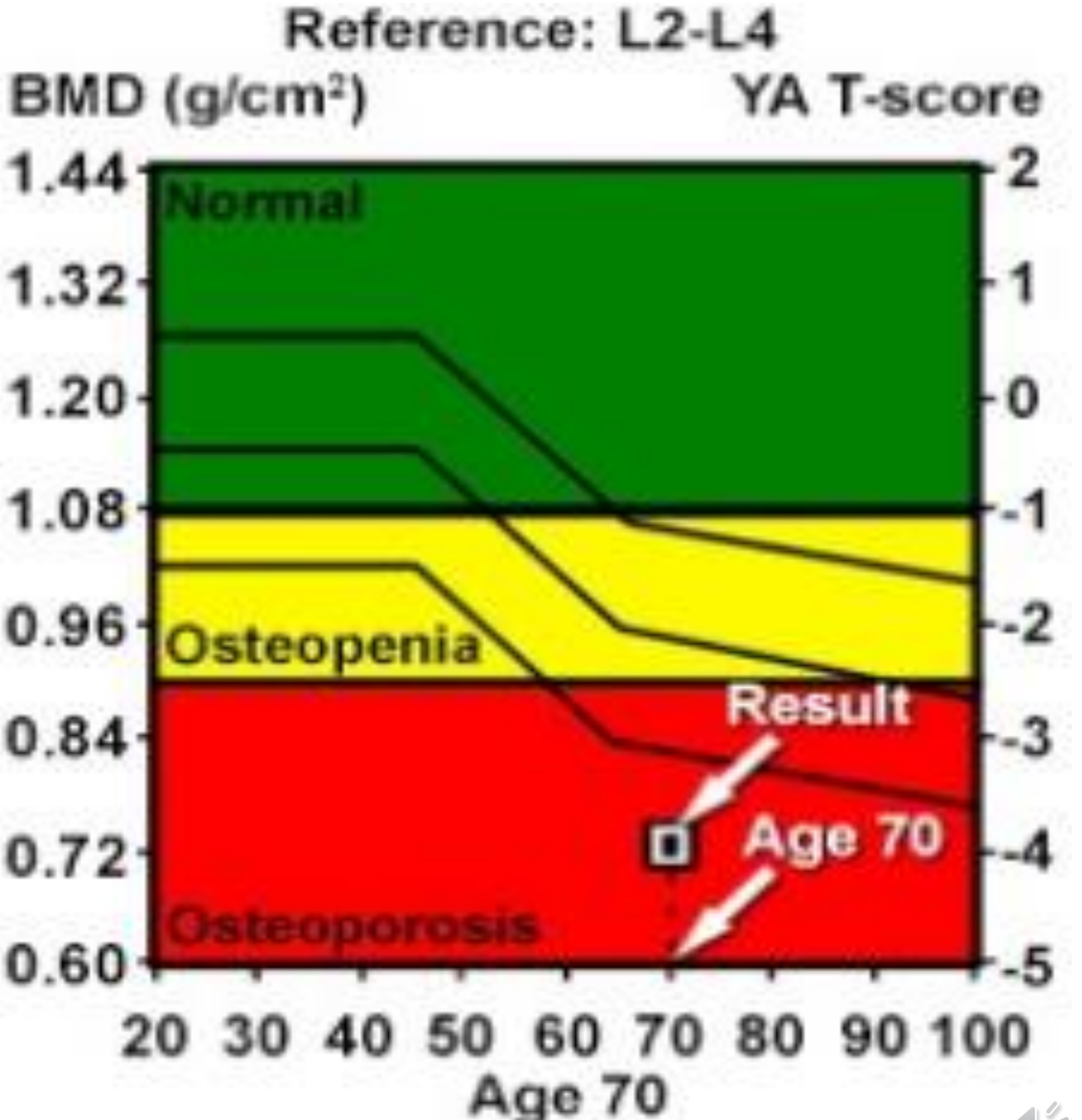
- **Dual-Energy X-Ray Absorptiometry (DEXA)**
- The measurements provided by DEXA are bone mineral content (in grams) and the projected area of the measured site (in square centimeters).
- Dividing bone mineral content by the area yields bone mineral density BMD (in grams per square centimeter).
- BMD is expressed in terms of standard deviation (SD) as a T score and a Z score.

- The T score describes the difference between the BMD of the patient being examined and the mean BMD of a standard young adult population (20–30 years of age), and refers to the peak of bone mass.
- The Z score shows the difference between the patient's BMD and the mean BMD of age- and gender-matched controls.

- **WHO Diagnostic Criteria for Osteoporosis**
- Diagnosis T-Score
- **Normal** –1.0 or higher
- **Osteopenia** Less than –1.0 but greater than –2.5
- **Osteoporosis** Less than –2.5 but without fractures
- **Established osteoporosis** Less than –2.5 with fractures


AP L- Spine Bone Density
L1
L2
L3
L4
Reference: L2-L4
BMD (g/cm²)
YA T-score
1.44
1.32
1.20
1.08
0.96
0.84
0.72
0.60
2
1
0
-1
-2
-3
-4
-5
Normal
Result
Osteopenia
Age 67
Osteoporosis
20 30 40 50 60 70 80 90 100
Age

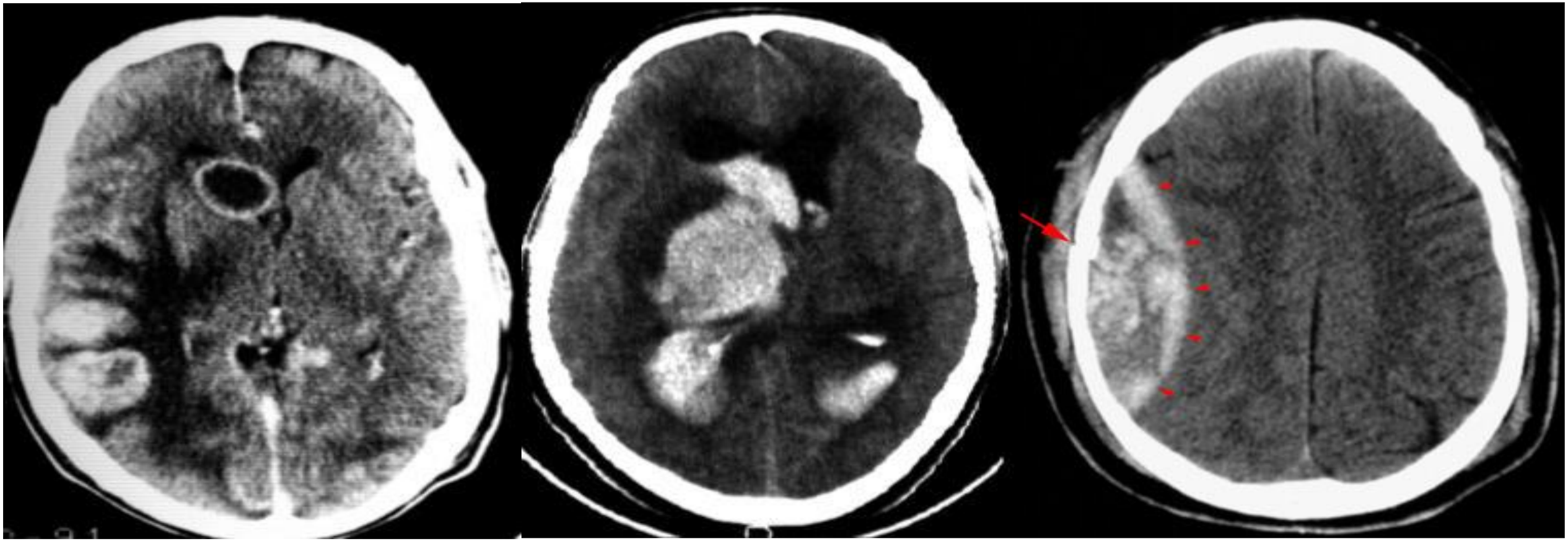

Reference: L2-L4
BMD (g/cm²)
YA T-score
1.44
1.32
1.20
1.08
0.96
0.84
0.72
0.60
2
1
0
-1
-2
-3
-4
-5
Normal
Result
Osteopenia
Age 60
Osteoporosis
20 30 40 50 60 70 80 90 100
Age 60


Reference: L2-L4
BMD (g/cm²)
YA T-score
1.44
1.32
1.20
1.08
0.96
0.84
0.72
0.60
2
1
0
-1
-2
-3
-4
-5
Normal
Osteopenia
Result
Age 70
Osteoporosis
20 30 40 50 60 70 80 90 100
Age 70

Case Study

- 56 female patient
- Image (b) after image (a) one year.
- Progress of osteoporosis with wedge fractures.

Thank you



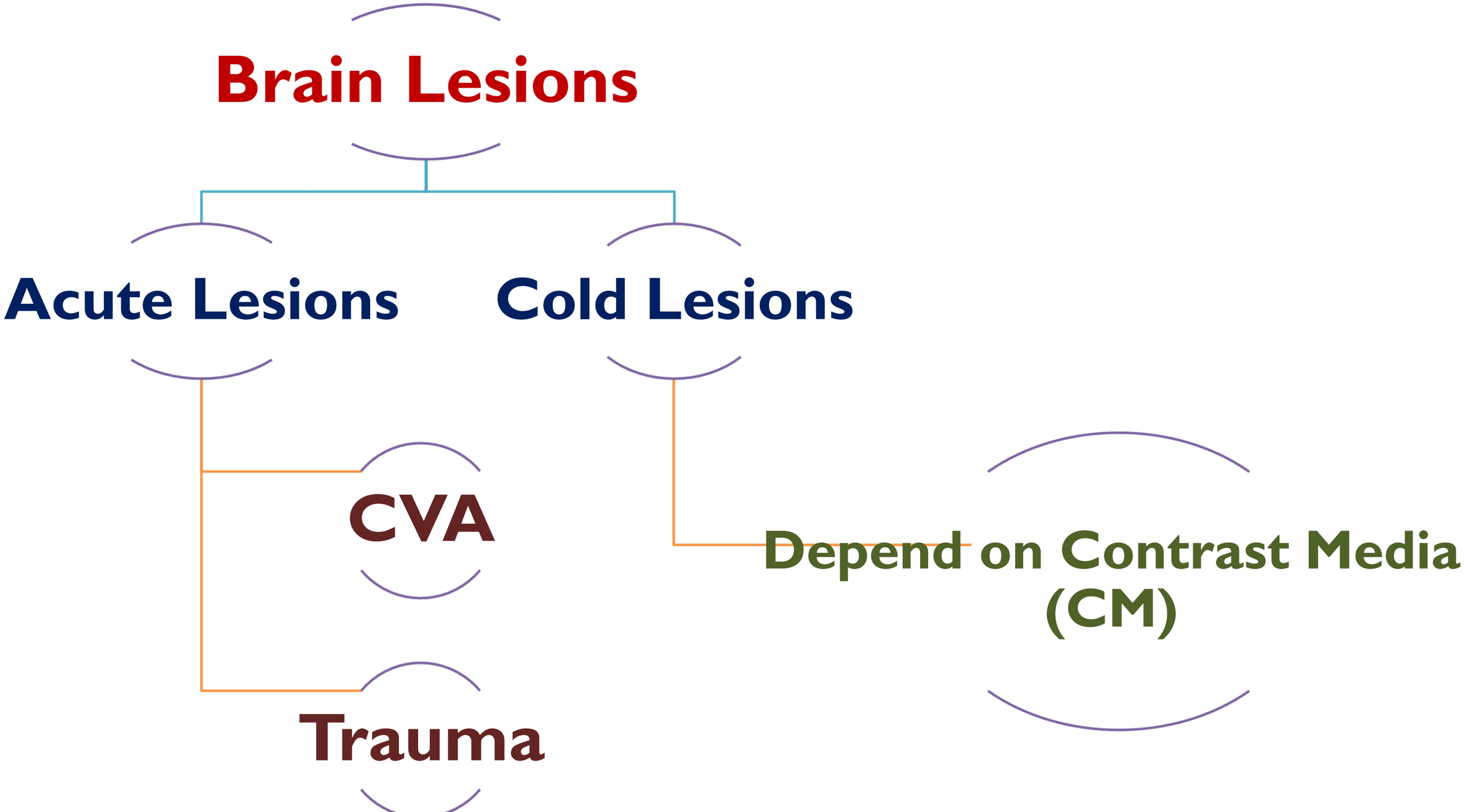
Diagnostic Radiology of Brain CT

Recognizing Some Common Causes of Intracranial Pathology

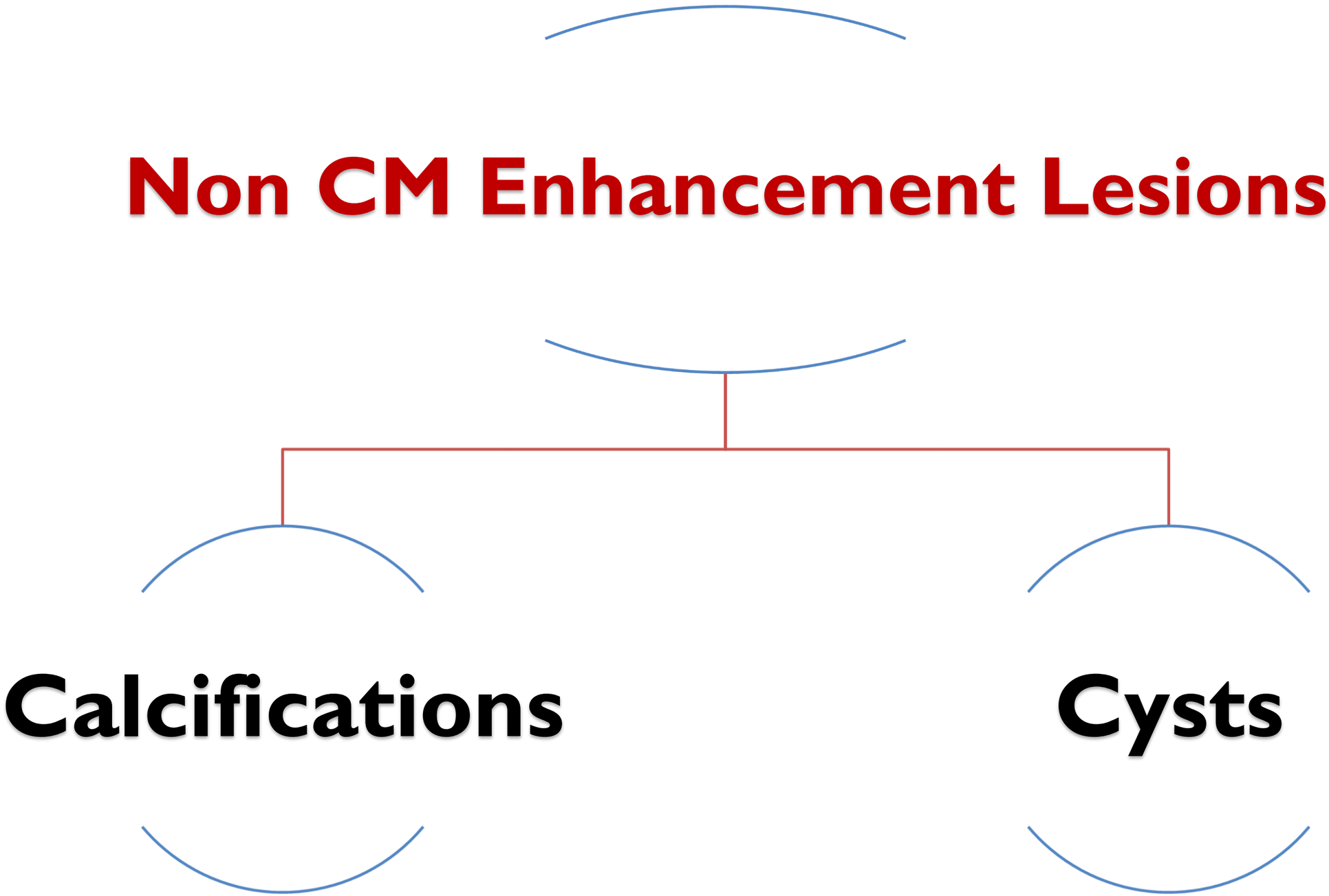
Lecturer: Husam Mansour

Master of Public Health

Master of Medical Imaging


Brain Lesions
Acute Lesions
Cold Lesions
CVA
Trauma
Depend on Contrast Media (CM)

Non CM Enhancement Lesions



CM Enhancement Lesions

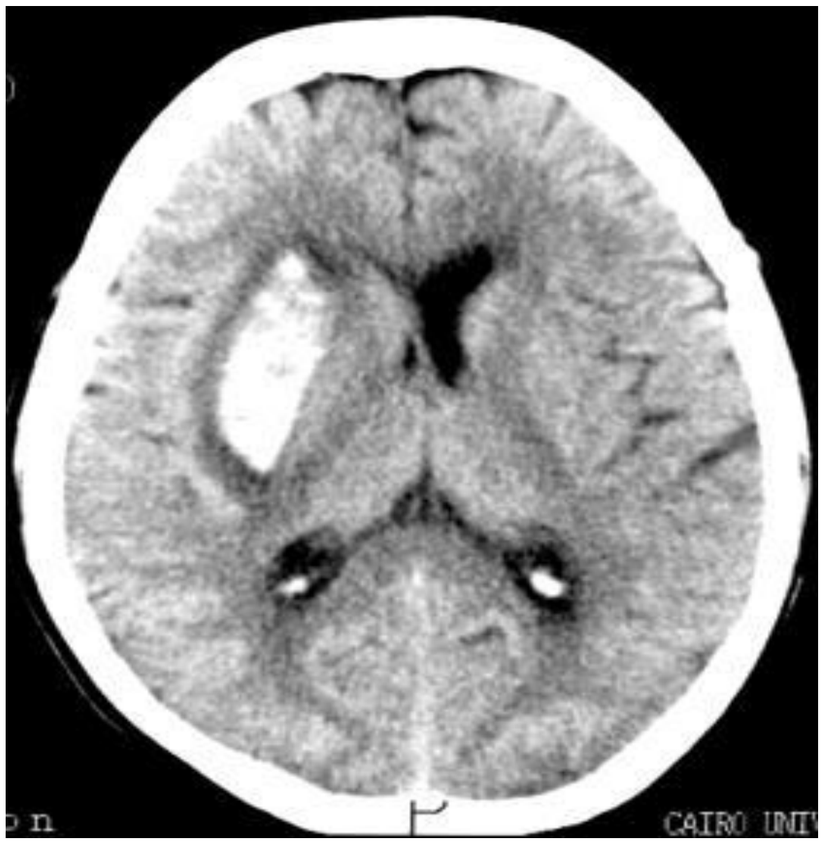
- Homogenous Lesions
- Heterogenous Lesions
- Ring CM Enhancement

Cerebro-vascular stroke

CT Pathology Terminology

- **Hypodense lesion** = lesion of lower density than the normal brain tissue
- **Hyperdense lesion** = lesion of higher density than normal brain tissue
- **Isodense lesion** = lesion of similar density to the normal brain tissue. The site of this lesion is identified by its mass effect on the adjacent ventricle and sulci

Hypo dense



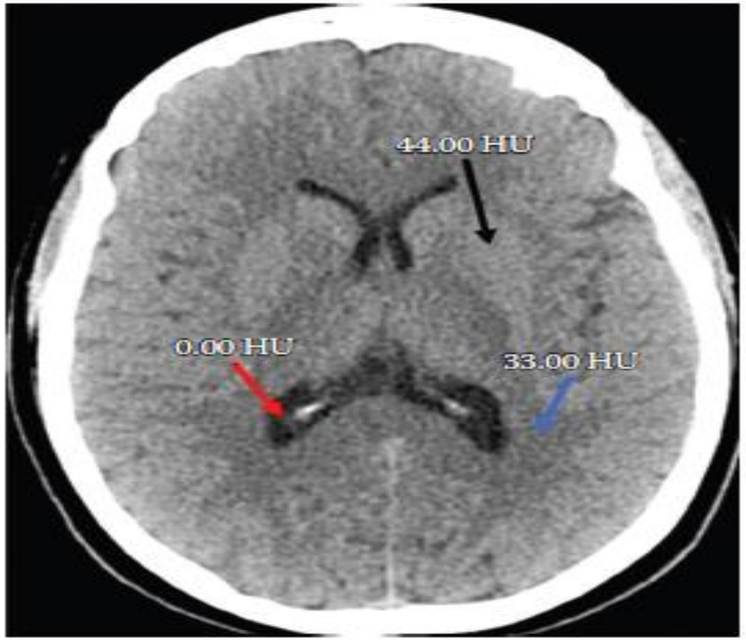
Hyper dense

Isodense

CT density [HU]

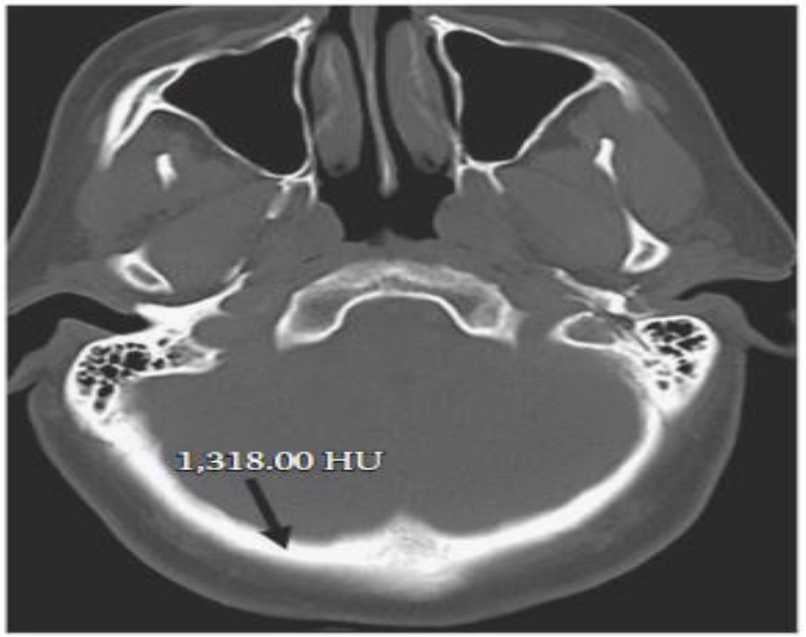
Structure	Attenuation value in HU
Air	From -500 To -1000
Fat	From -10 To - 200
Water	From 0 To 15
Brain edema and infarction	around 20
Normal Brain tissue	From 30 To 40
Recent hematoma	From 60 To 90
Calcifications	More than 100
Bone	From 200 and above

- The normal brain density ranger from 30- 40 HU
- Any lesion with CT density lower than 30HU will appear hypodense
- Any lesion with CT density higher than 40 HU will appear hyperdense

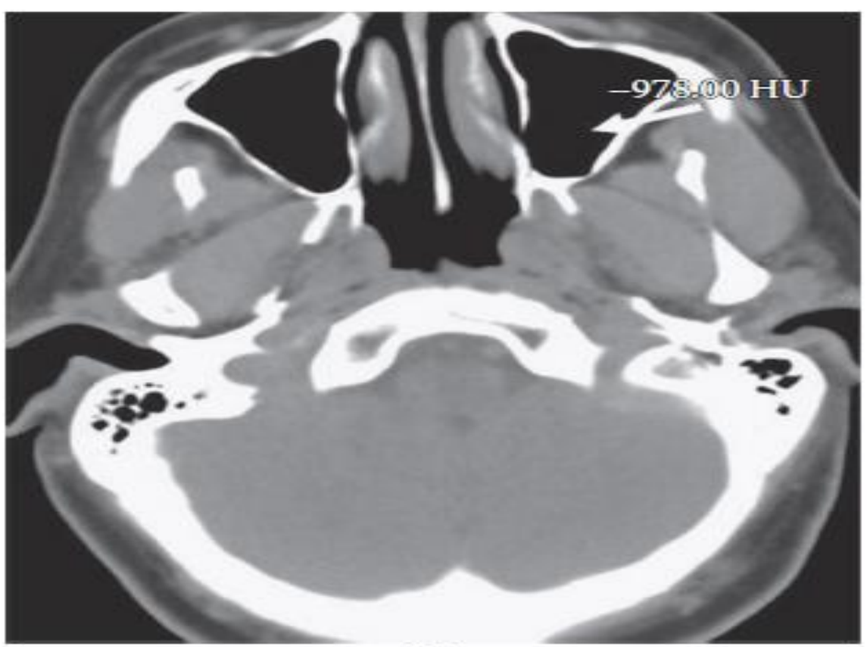


(a)

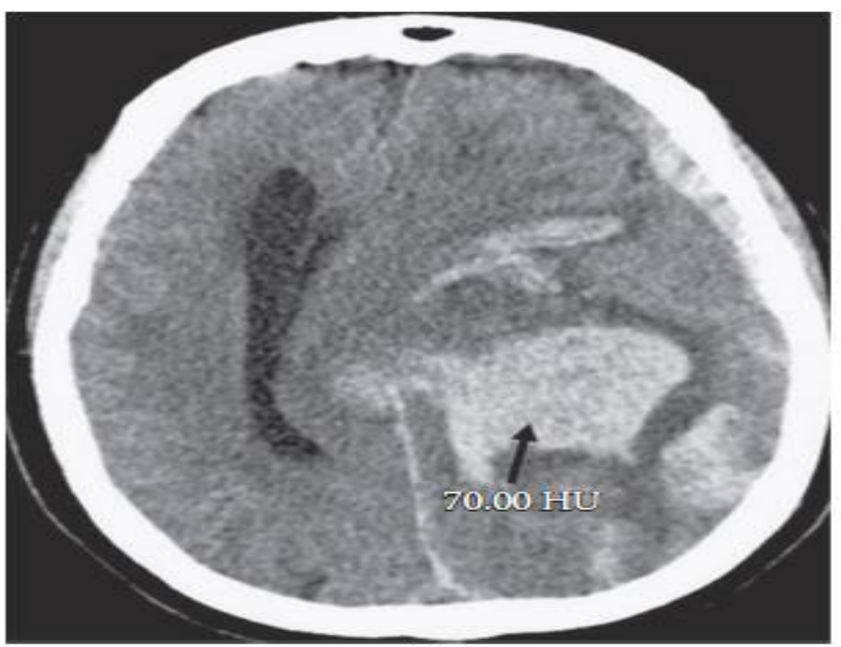
(b)



(c)



(d)



(e)

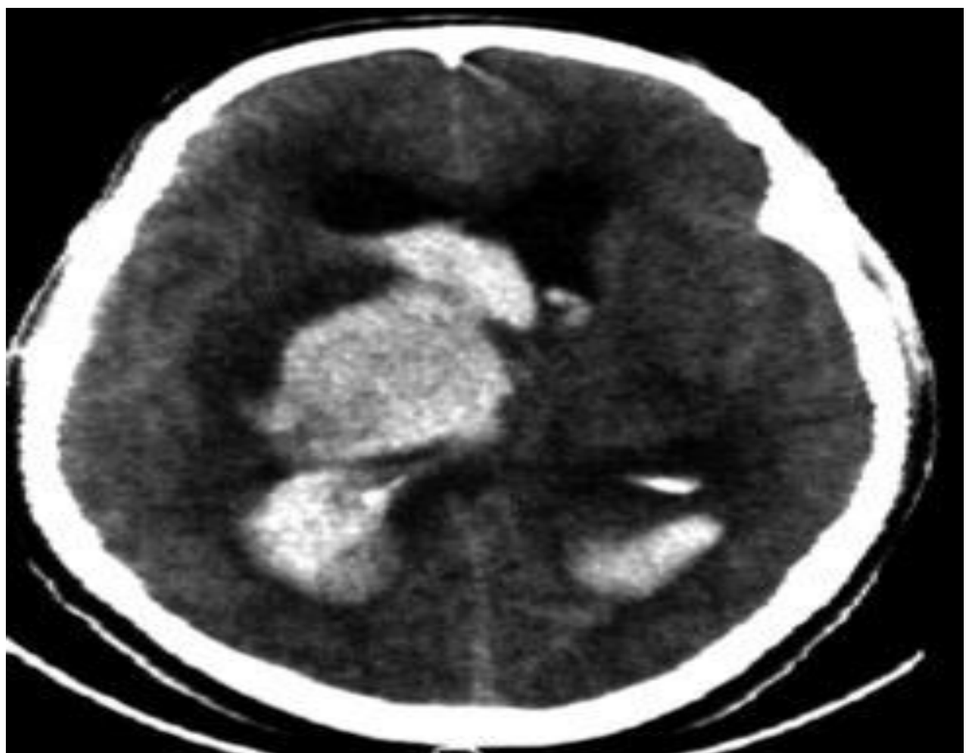
Mass effect: 3 grades

- Effacement of the cortical sulci
- Compression of the ventricle
- Contralateral shift of the midline structures

NCCT of a large right sided subacute subdural hematoma showing all grades of mass effect in the form of effacement of the cortical sulci, marked compression of the right lateral ventricle with contralateral shift of the midline structures to the left side

Edema

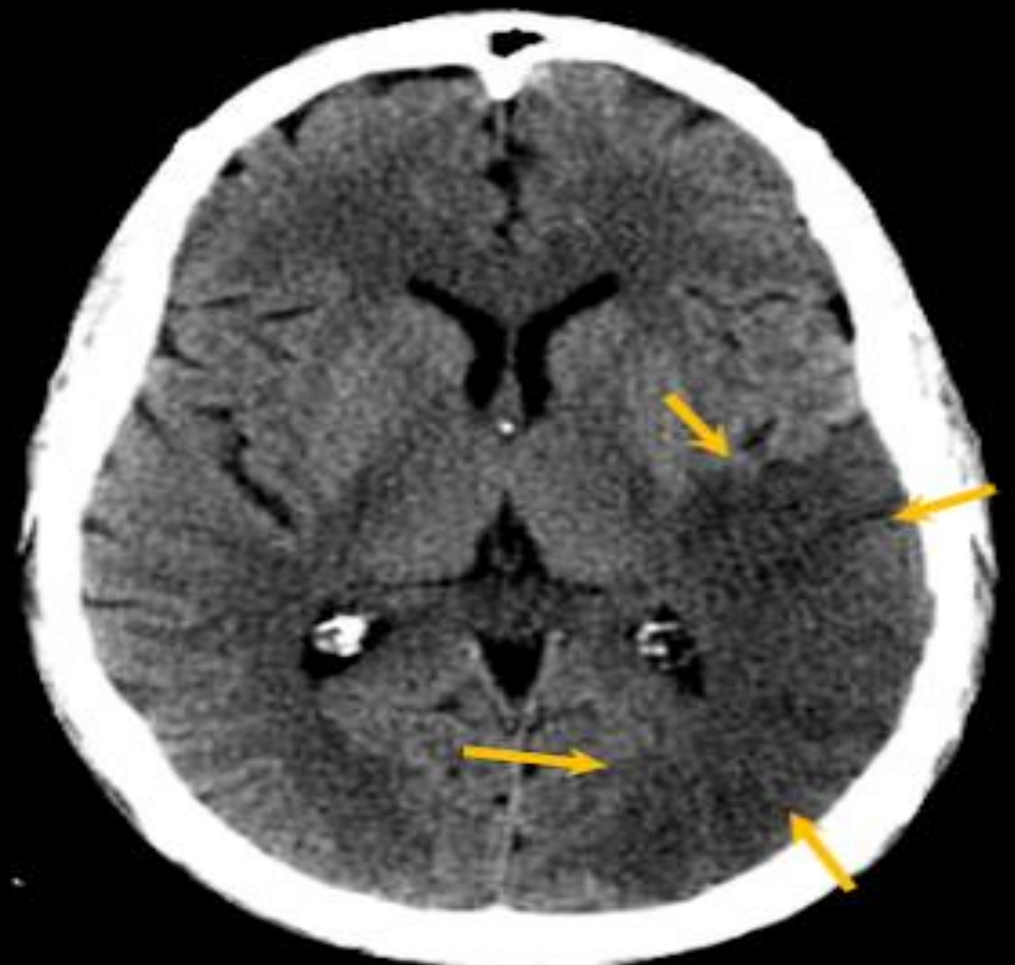
1. **Cytotoxic** : around acute infarction and hemorrhage
2. **Vasogenic** : around SOL (tumors, abscesses,…)



Large right thalamic recent hematoma surrounded by a hypodense rim of edema. Intraventricula extension of the hematoma is seen

Multiple brain deposits in the right cerebral hemisphere surrounded by vasogenic finger like brain edema hypodensity

Stroke vs. Tumor



Cytotoxic Edema

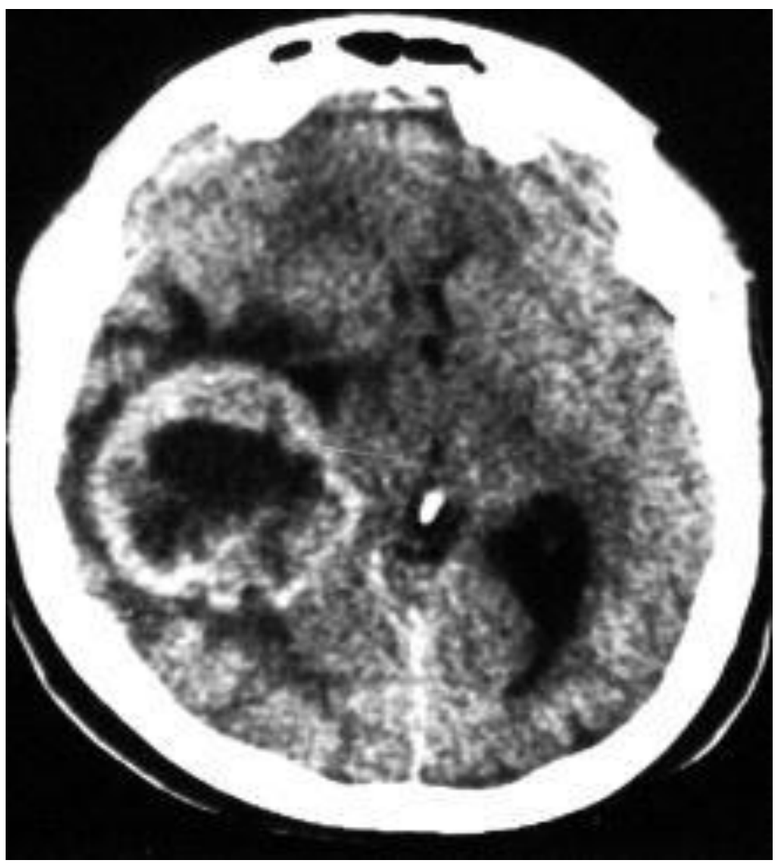
Cellular swelling
Gray-white margin lost

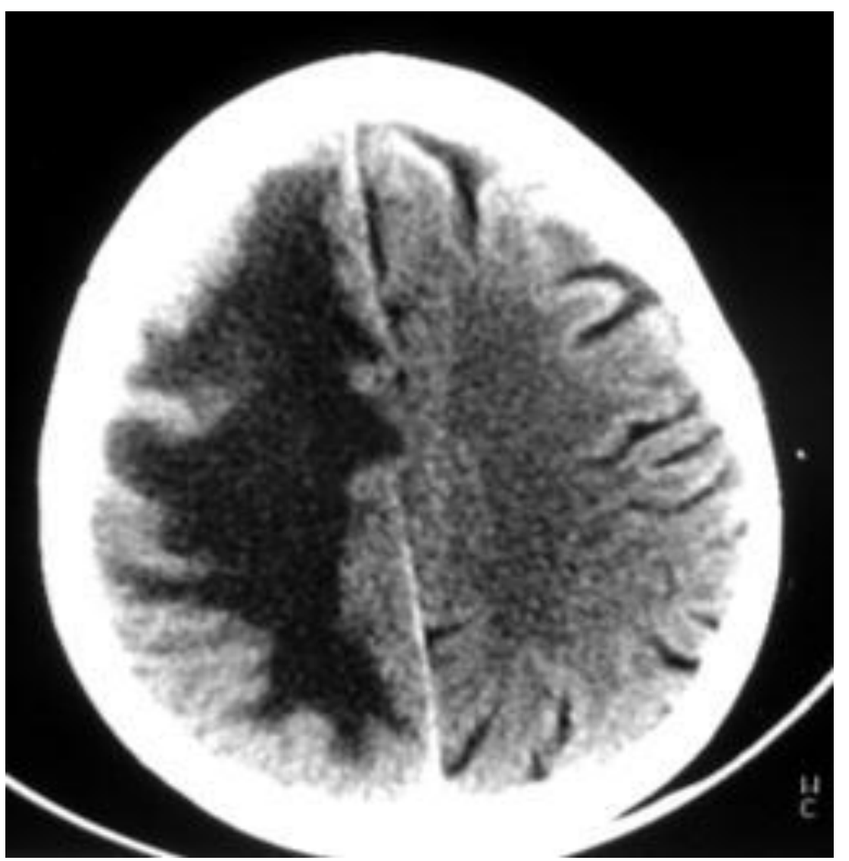
Vasogenic Edema

Leaky capillaries
Gray matter is spared

Vasogenic edema: 3 grades

1. Grade I: 2 cm around the lesion
2. Grade II: more than 2 cm but less than ½ the cerebral hemisphere
3. Grade III: more than ½ the cerebral hemisphere



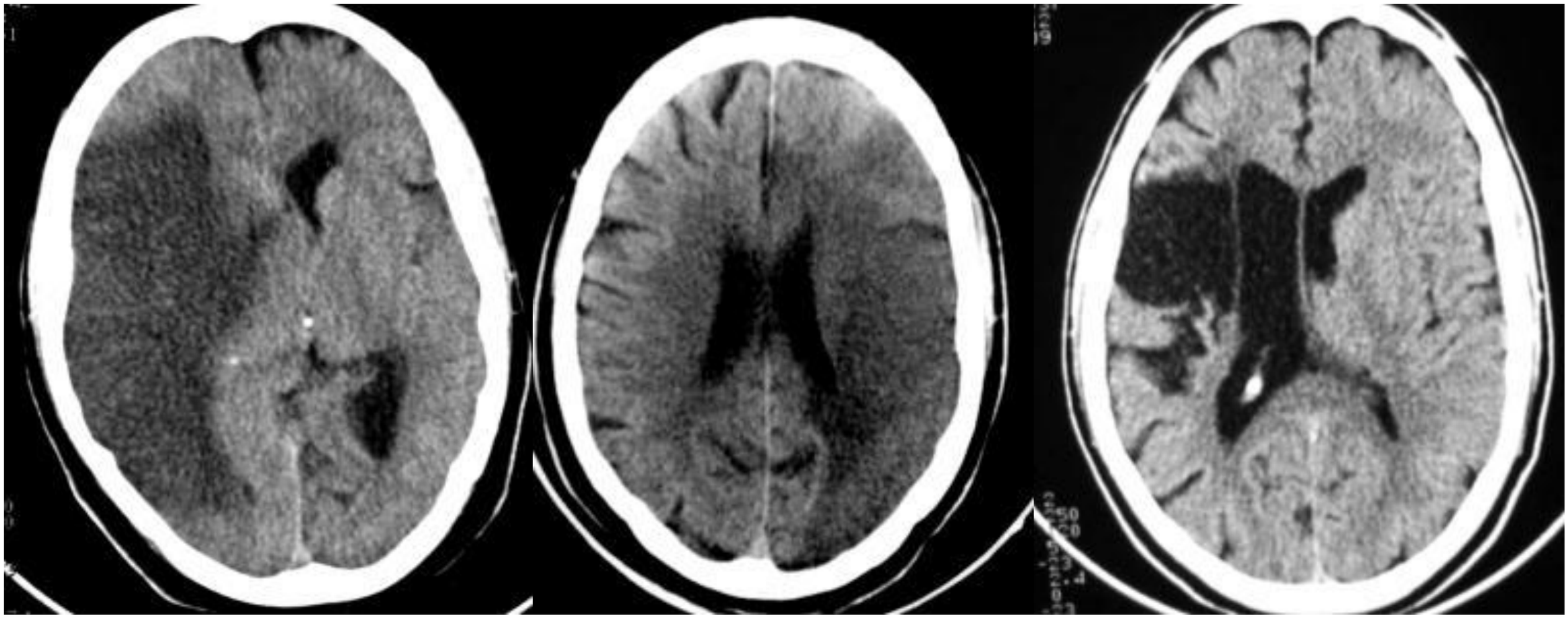


Cerebral Infarction

1. Infarction of the brain results from a deficient cerebral circulation from thrombosis or an embolic event and clinically presents as a stroke.
2. Transient ischaemic attack (TIA) produces a focal neurological deficit in which complete recovery of function occurs within 24 hours.

How to know infarction phase ?!

- **Acute infarction** is associated with edema causing mass effect on the ventricle and sulci
- **Chronic infarction** is associated with volume loss with compensatory dilatation of the adjacent ventricle and sulci
- **Subacute infarction** is not associated with edema or volume loss, so it has no effect on the adjacent ventricle

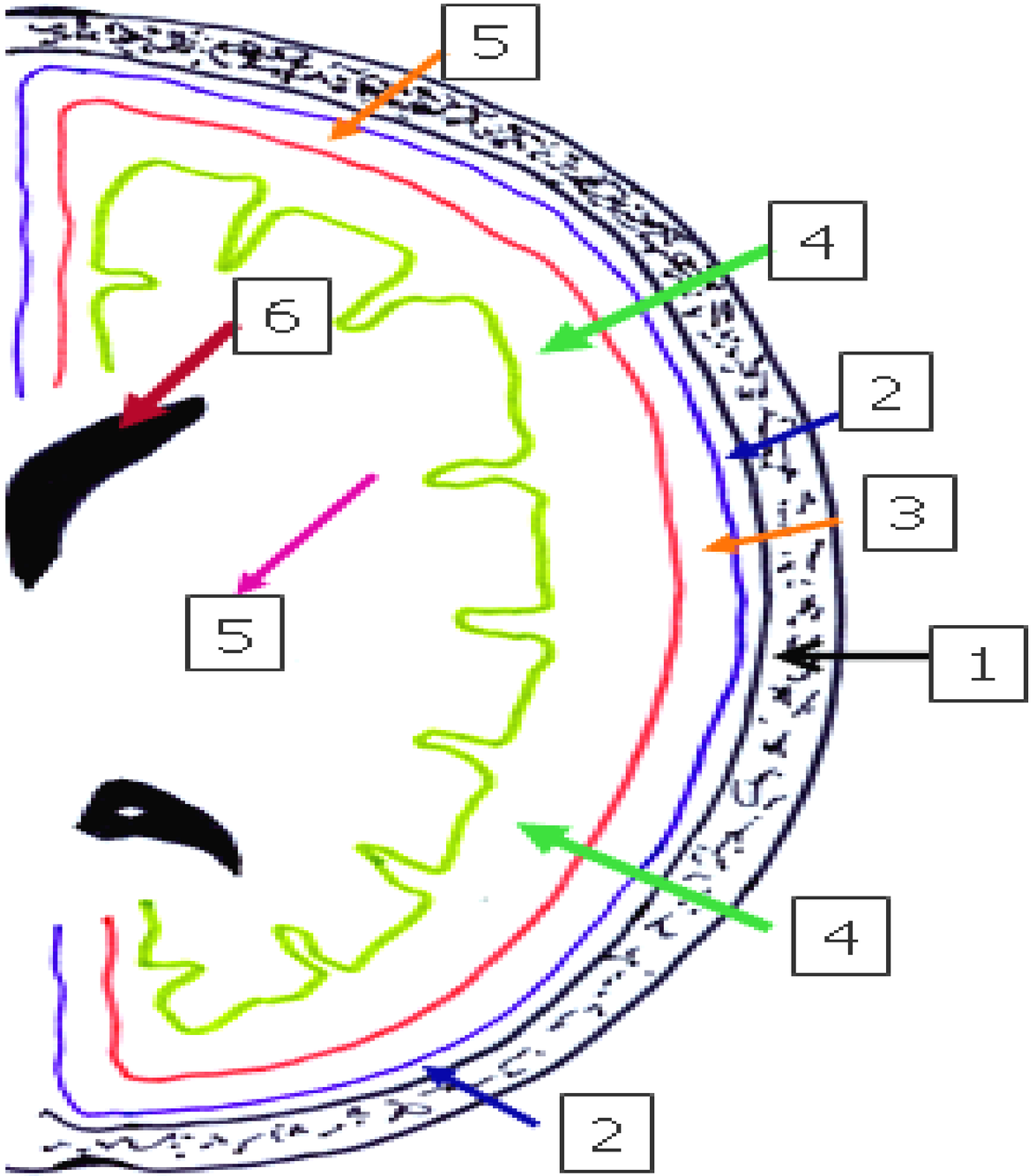


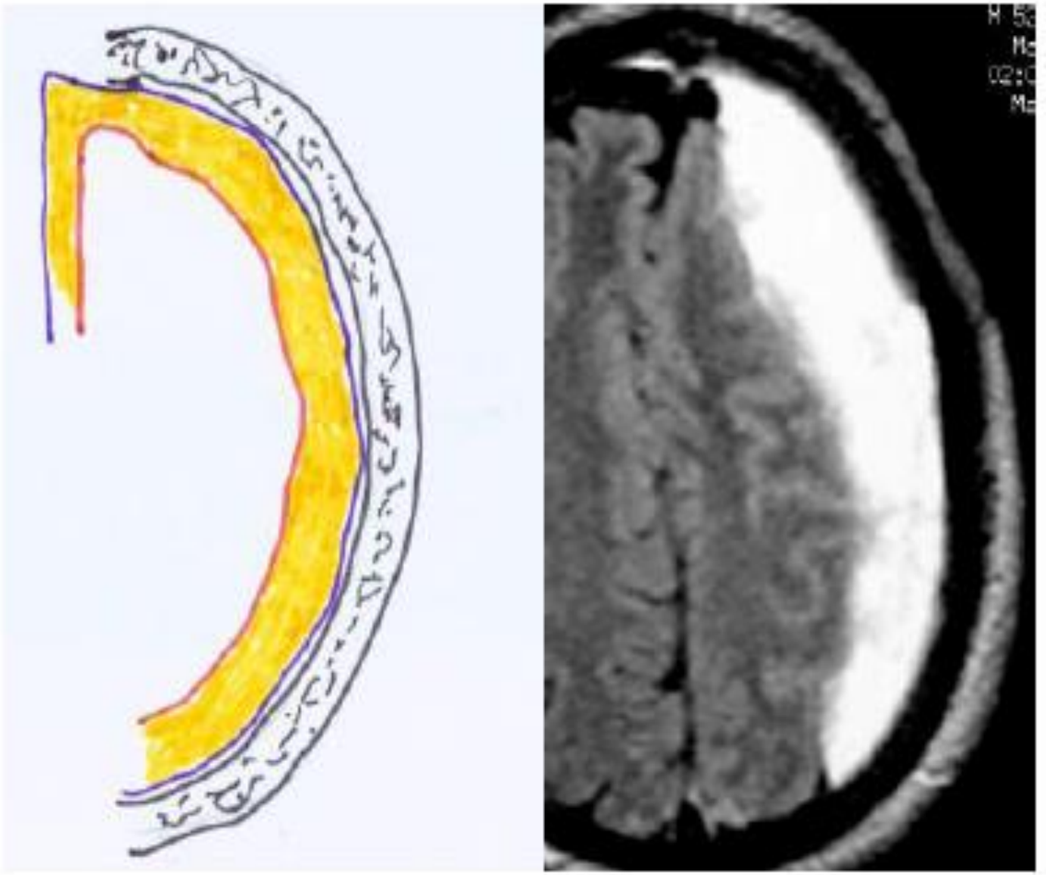
Sites of intracranial hemorrhage

- Calverial bones [1]
- Epidural [2]
- Subdural [3]
- Subarachnoid [4]
- Intra-cerebral [5]
- Intra-ventricular [6]

As a general rule

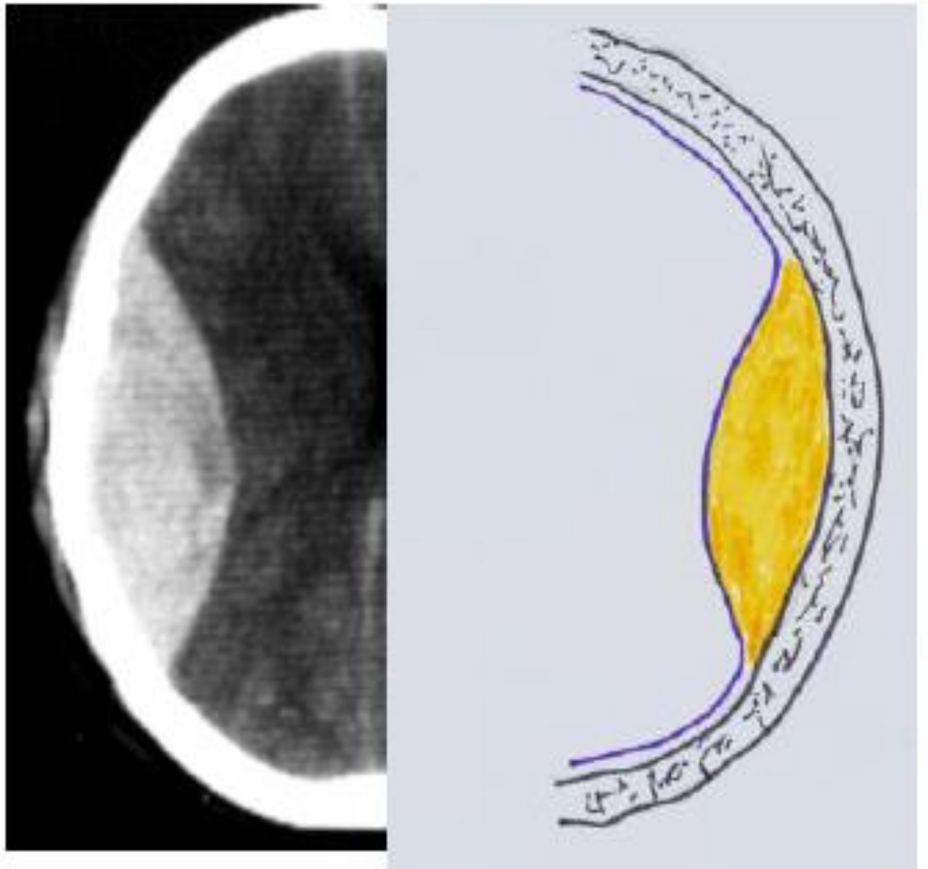
- Acute hematoma ***(Hyperdense)***
- Subacute hematoma ***(Isodense)***
- Chronic hematoma ***(Hypodense)***





Subdural hematoma

- Concave inner margin
- Do not cross midline
- Can be acute, sub acute or chronic

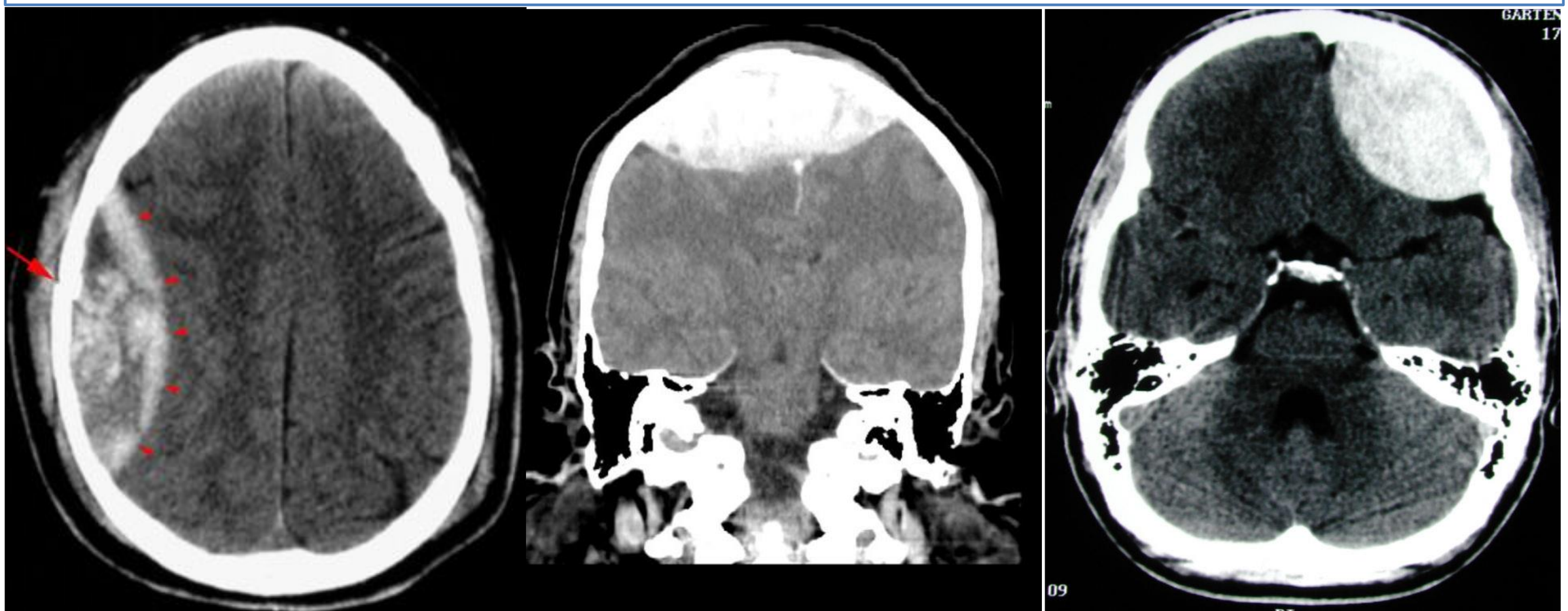


Epidural hematoma

- Convex inner margin
- Can cross midline
- Almost always acute

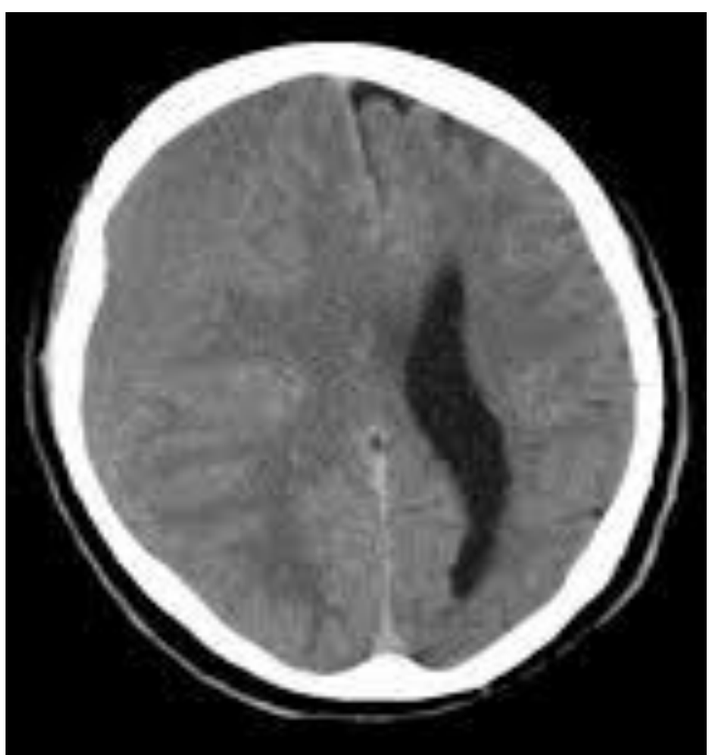
Epidural hematoma

- Convex inner margin.
- Post traumatic.
- Almost always acute due to severe symptoms as it occur in tight space.

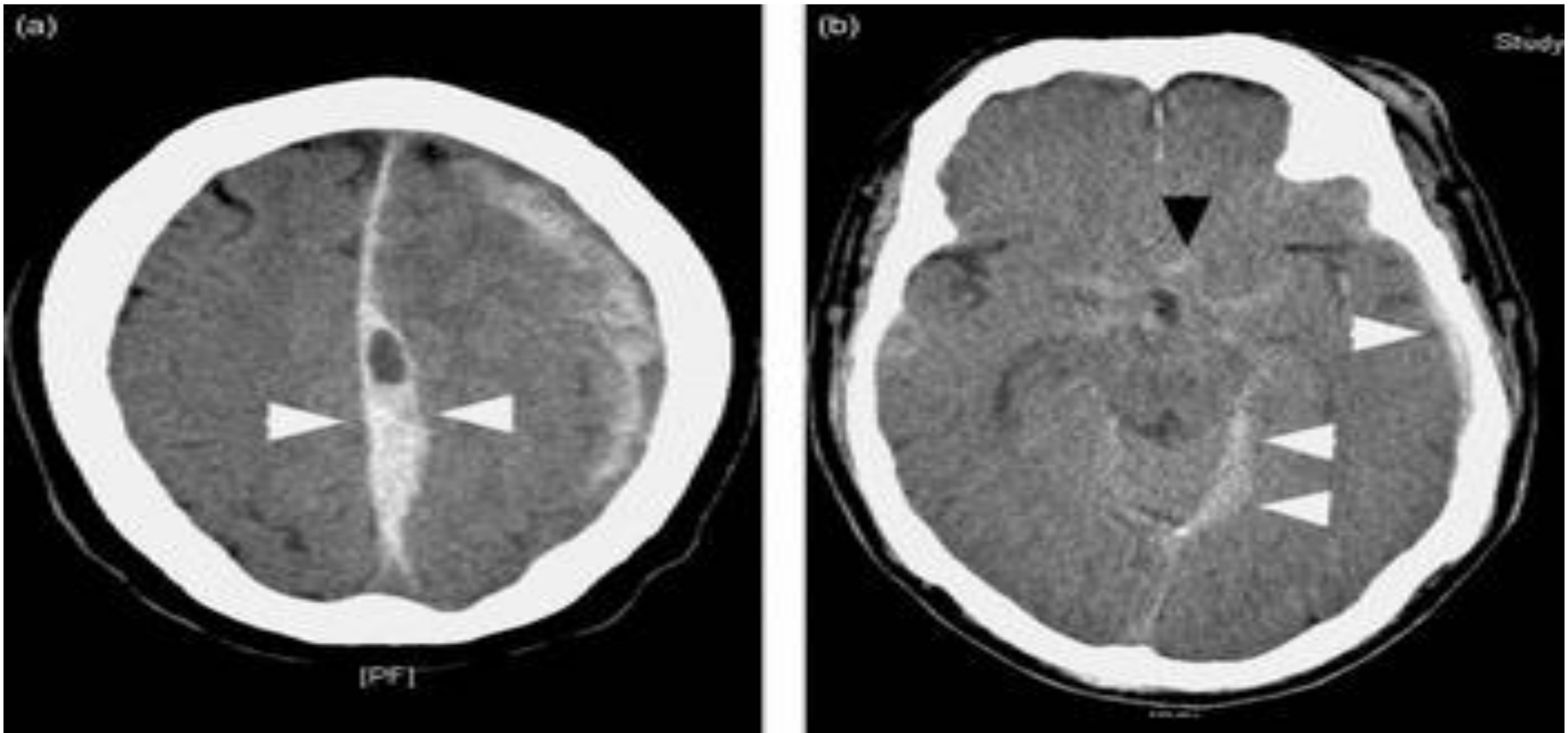


Subdural hematoma

- Concave inner margin In all phases.
- Can be acute ,subacute or chronic
- Usually seen in elderly patient
- History of minor trauma
- Can occur in the interhemispheric fissure

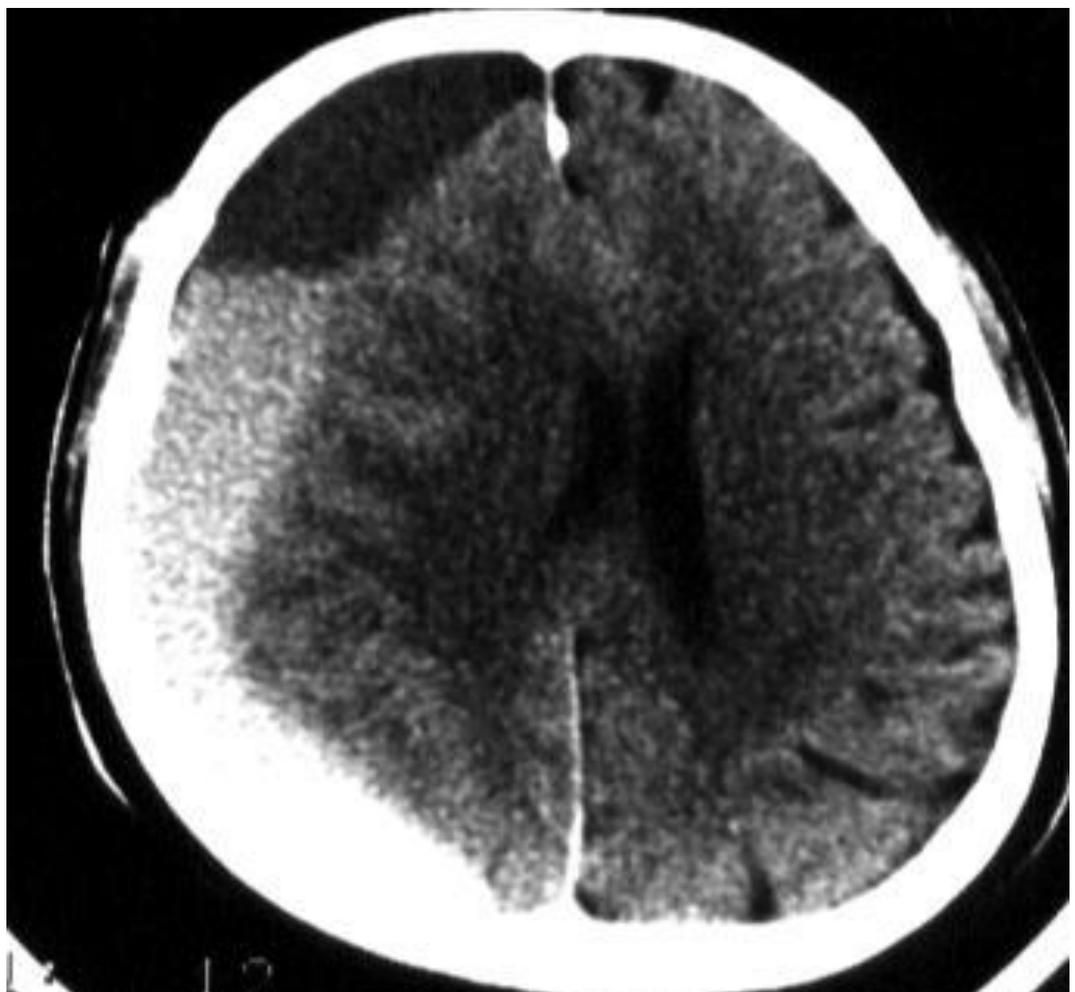


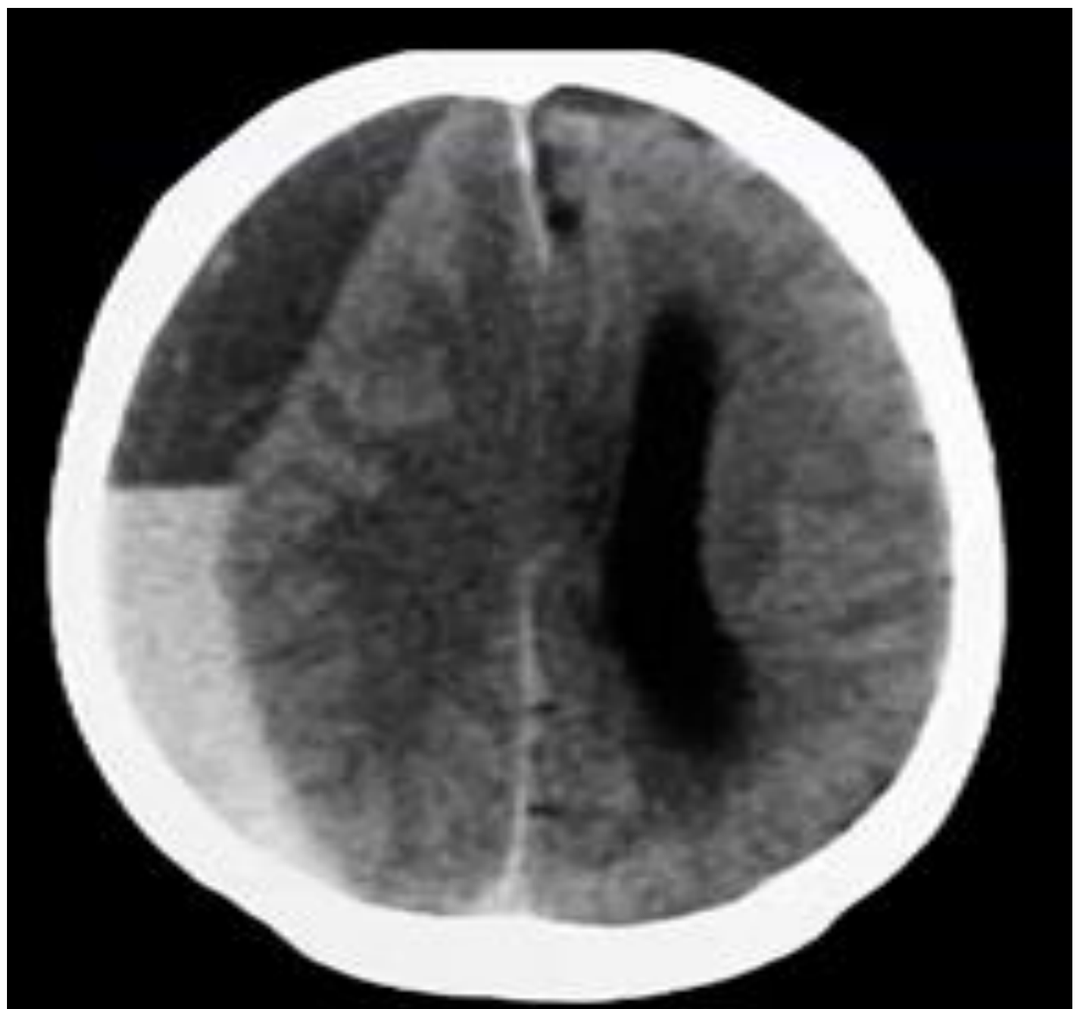
Subdural hematoma in the intrahemispheric fissure



Subacute subdural hematoma

Sedimentation level with upper hypodensity and lower hyperdensity





Subacute subdural hematoma

Re bleeding mixed hyper- and hypodensities of acute and chronic blood

Intraventricular hemorrhage has to main causes

- Intracerebral hematoma extending into the ventricles
- Subarachnoid hemorrhage refluxing into the ventricles

Intracerebral hematoma

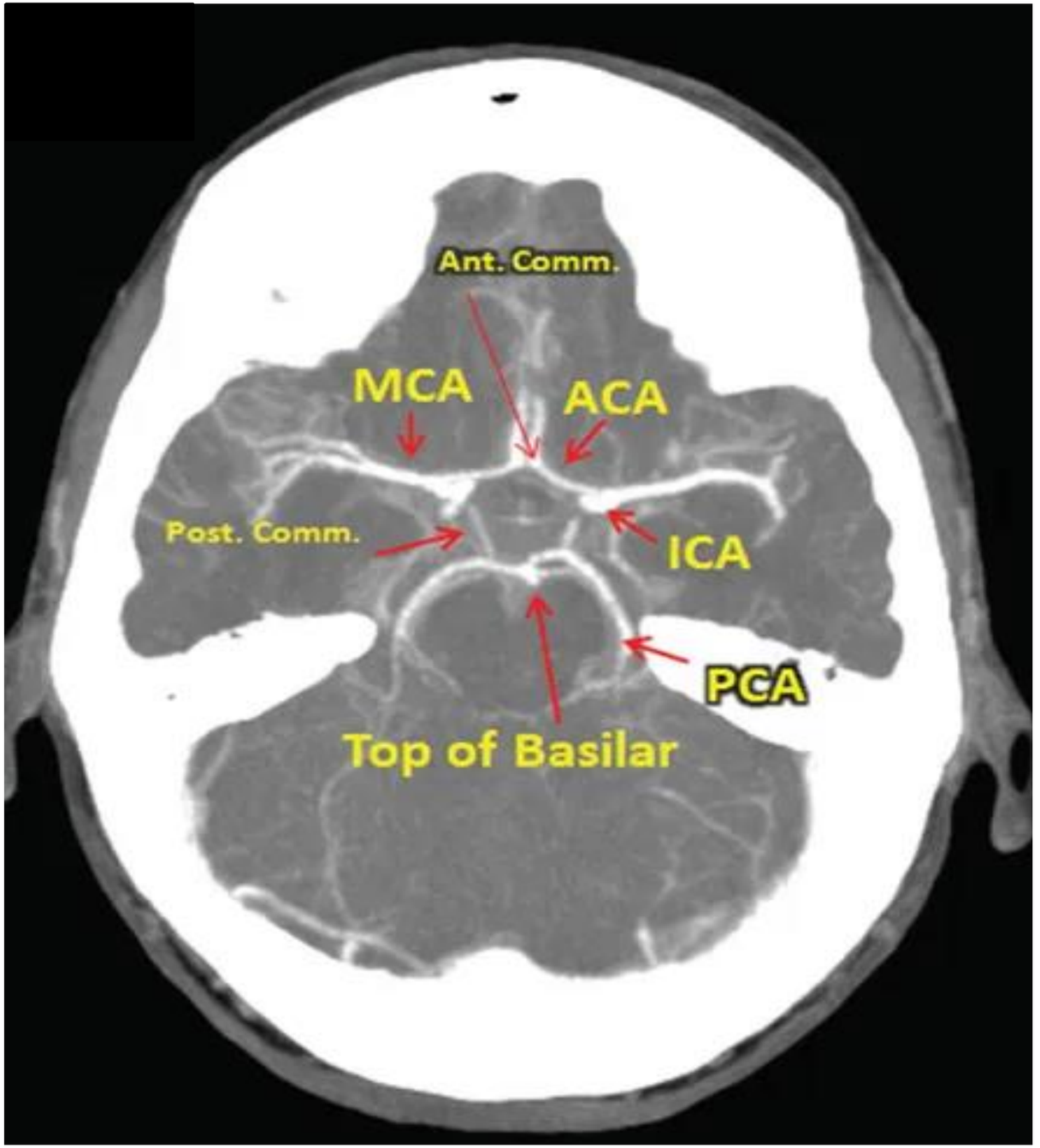
Post traumatic

Non traumatic

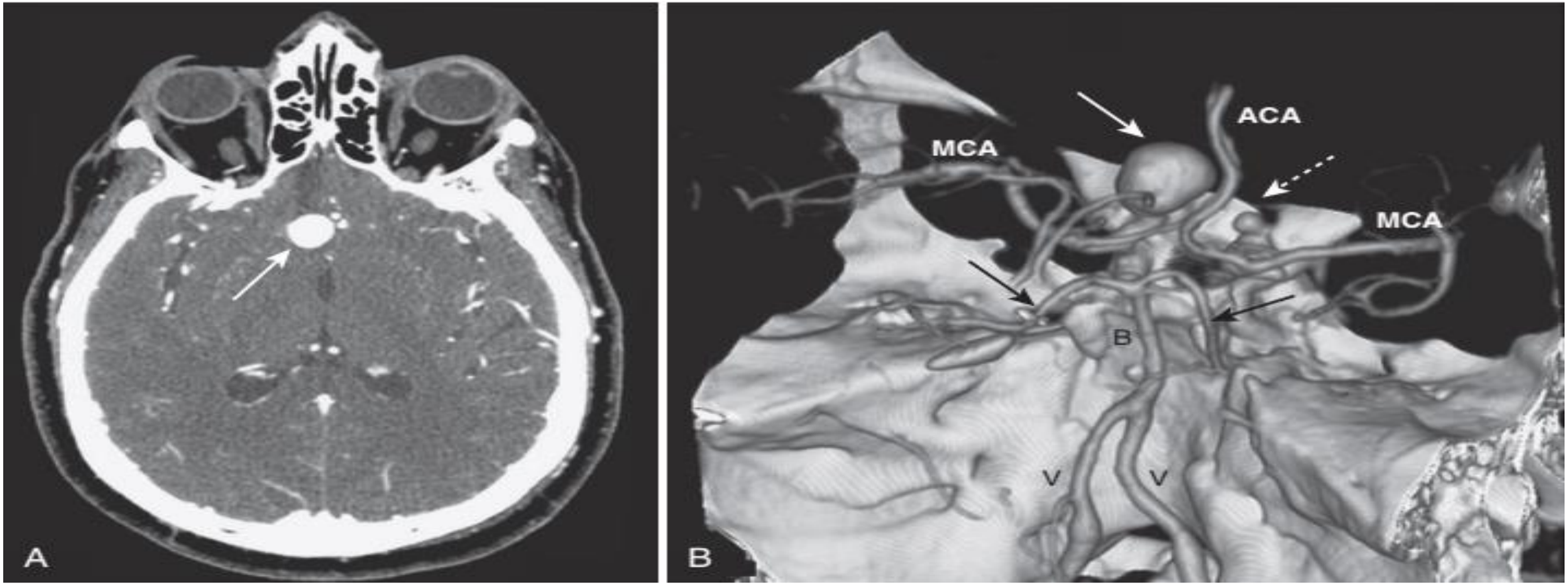
- **Ruptured Aneurysms**
- The most frequent central nervous system aneurysm is the berry aneurysm, which develops from a congenital weakening in the arterial wall, usually at the sites of vessel branching in the circle of Willis at the base of the brain.
- Hypertension and aging play a role in the growth of aneurysms. Larger aneurysms bleed more frequently than do smaller ones.

- The aim is to discover and treat the aneurysm before it has undergone a major bleed. In some studies, 10 mm was found to be the critical size for rupture.
- When aneurysms rupture, the blood usually enters the subarachnoid space.
- Rupture of an aneurysm is the most common nontraumatic cause of a subarachnoid hemorrhage (80%) but not the only cause as trauma, arteriovenous malformations can also produce subarachnoid hemorrhage.

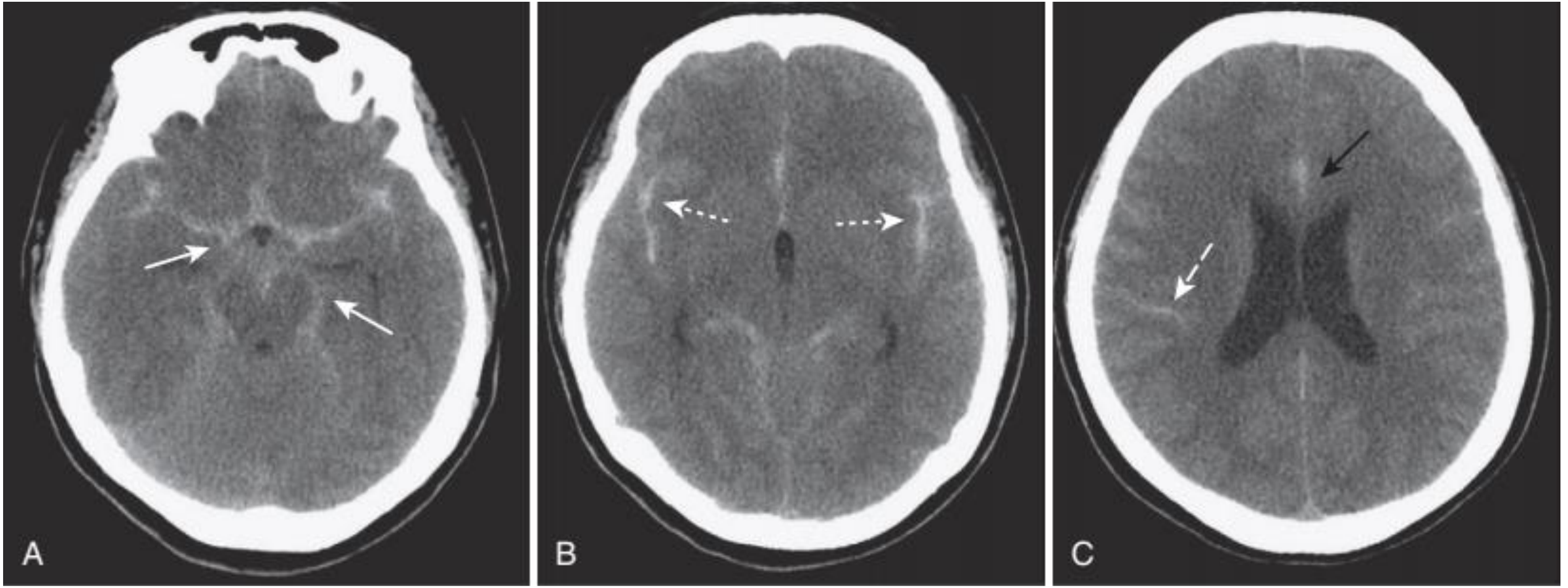
- Today, most aneurysms are detected by either CT angiography or MR angiography. CT angiography is most common performed.
- On CT, acute blood is hyperdense and may be visualized within the sulci and basal cisterns.
- The region of the falx may become hyperdense, widened, and irregularly marginated.
- Generally, the greatest concentration of blood indicates the most likely site of the ruptured aneurysm.


Ant. Comm.
MCA
ACA
Post. Comm.
ICA
PCA
Top of Basilar

A2
A1
Acom
M1
Pcom
P1



Berry aneurysm, axial computed tomography (CT), and 3D reconstruction. A, There is a 2-cm focal outpouching of contrast in the region of the right internal carotid artery (ICA) on this contrast-enhanced CT of the brain (white arrow). This is consistent with an aneurysm. B, A 3D reconstruction of the circle of Willis from a CT-angiogram demonstrates the aneurysm arising from the supraclinoid segment of the right ICA (solid white arrow) and another smaller aneurysm arising from the supraclinoid segment of the left ICA (dotted white arrow). Black arrows point to posterior cerebral arteries. ACA, Anterior cerebral arteries; B, basilar artery; MCA, middle cerebral arteries; V, vertebral artery.

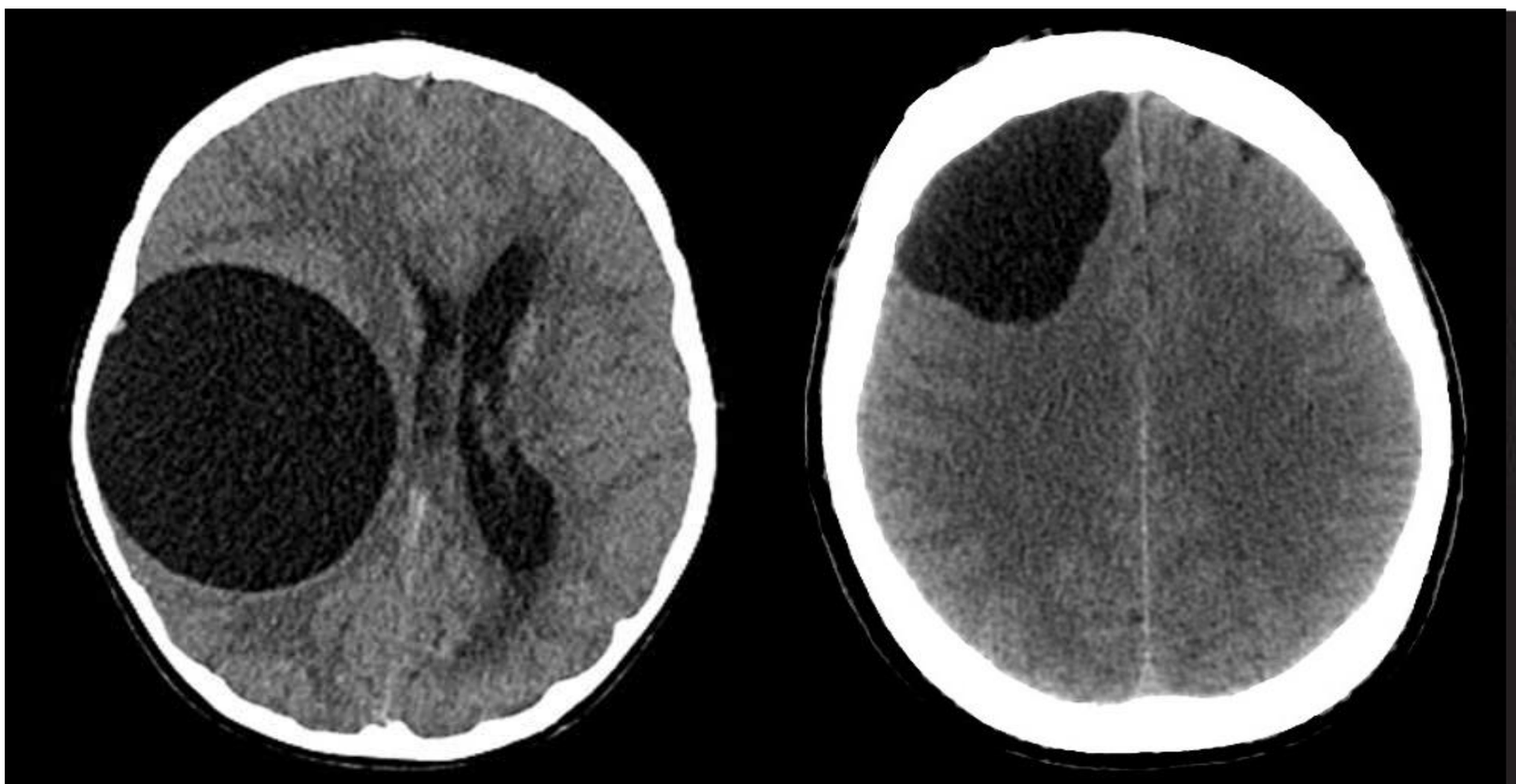


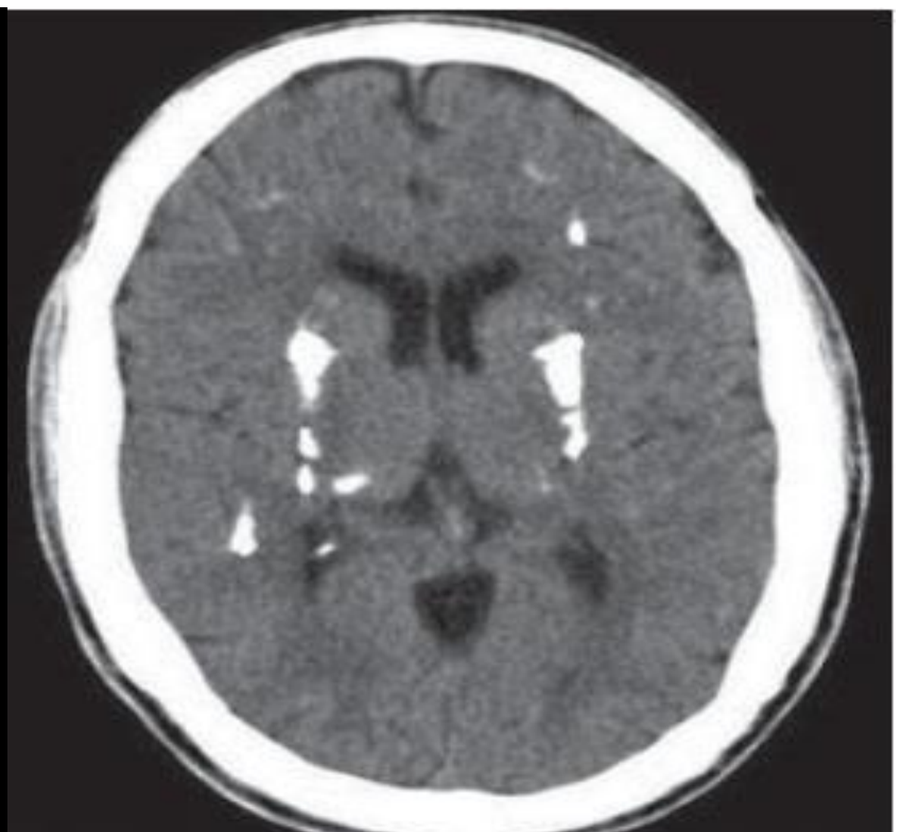
Subarachnoid hemorrhage, unenhanced computed tomography (CT) scans. Subarachnoid hemorrhage is frequently the result of a ruptured aneurysm. Blood may be most easily visualized within the basal cisterns (white arrows) (A), in the fissures (dotted white arrows) (B), and interdigitated in the subarachnoid spaces of the sulci (dashed white arrow) (C). The region of the falx may become hyperdense, widened, and irregularly marginated (solid black arrow).

None CM Enhancement lesions

Cysts

Calcification





Homogenous CM enhancement

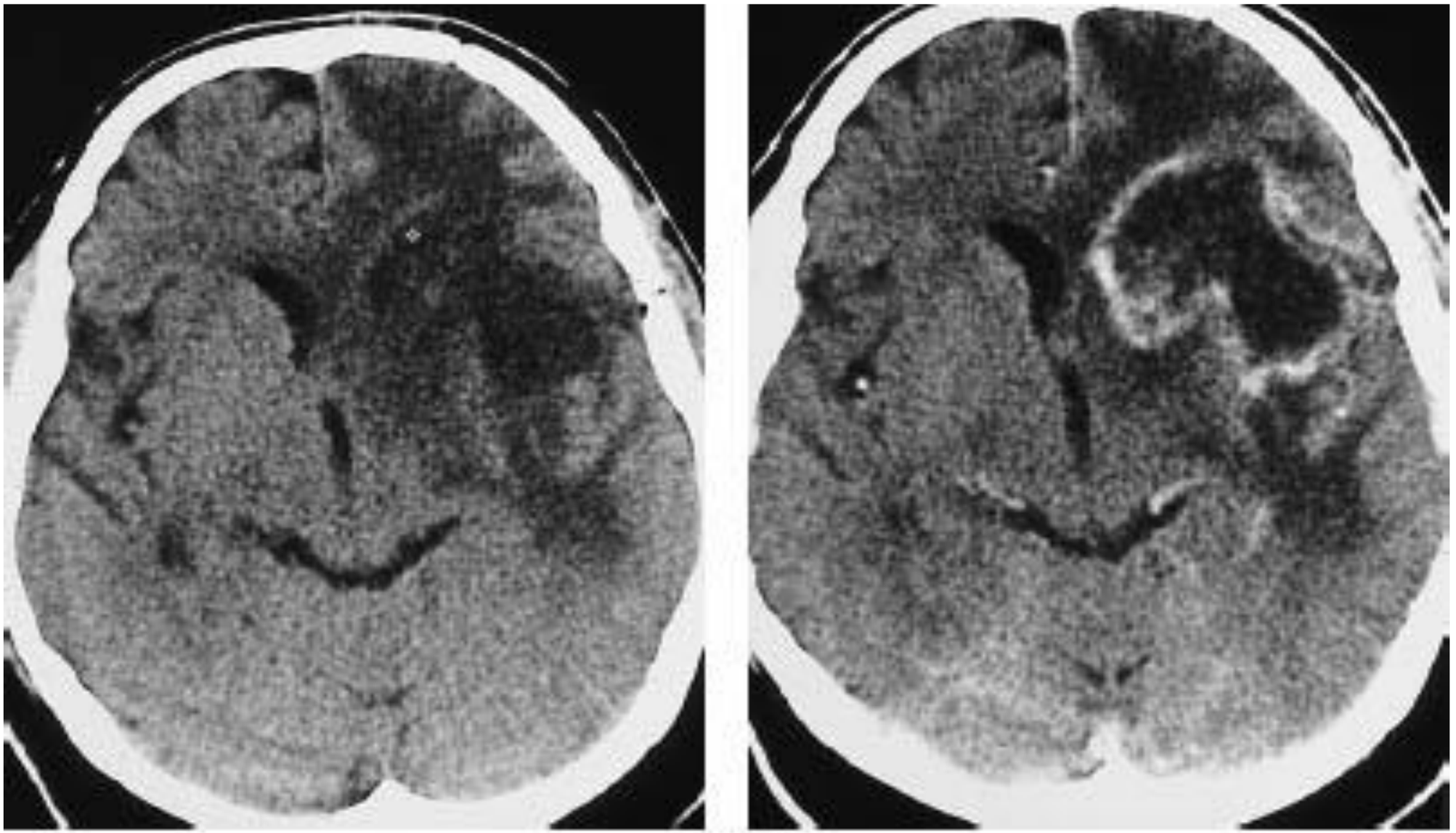
Meningioma

- ❖ Homogenous enhancement.
- ❖ Dural base.
- ❖ Other features:

1. Calcification 10-20%.
2. Bone sclerosis.
3. Brain edema 40%.

Heterogeneous CM enhancement

Gliomas:



a

b

Glioblastoma Multiform

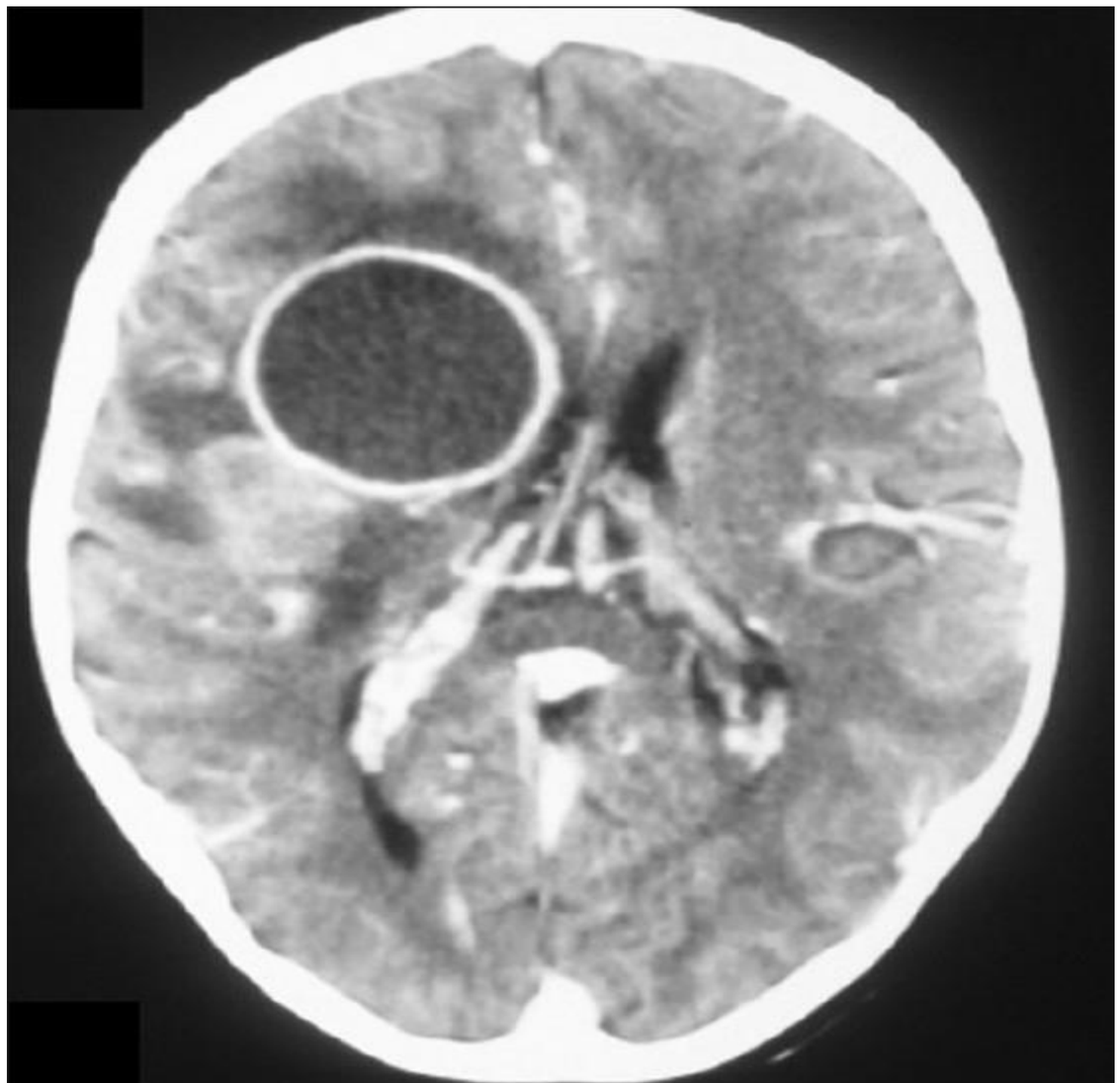
- **Gliomas of the Brain**
- Gliomas are the most common primary, supratentorial, intra-axial mass in an adult. They account for 30% of brain tumors of all types and 80% of all primary malignant brain tumors.
- Glioblastoma multiforme accounts for more than half of all gliomas, astrocytomas about 20%, with the remainder split among ependymoma, oligodendroglioma, and mixed glioma (e.g., oligoastrocytoma).

- Glioblastoma multiforme occurs more commonly in males between 65 to 75 years of age, especially in the frontal and temporal lobes. It has the worst prognosis of all gliomas.
- It infiltrates adjacent areas of the brain along white matter tracts, making it difficult to resect, but like most brain tumors, it does not produce extracerebral metastases.

- **Recognizing glioblastoma multiforme:**
- As the most aggressive of tumors, glioblastoma multiforme frequently demonstrates necrosis within the tumor.
- The tumor infiltrates the surrounding brain tissue, frequently crossing the white matter tracts of the corpus callosum to the opposite cerebral hemisphere, producing a pattern called **a butterfly glioma**.

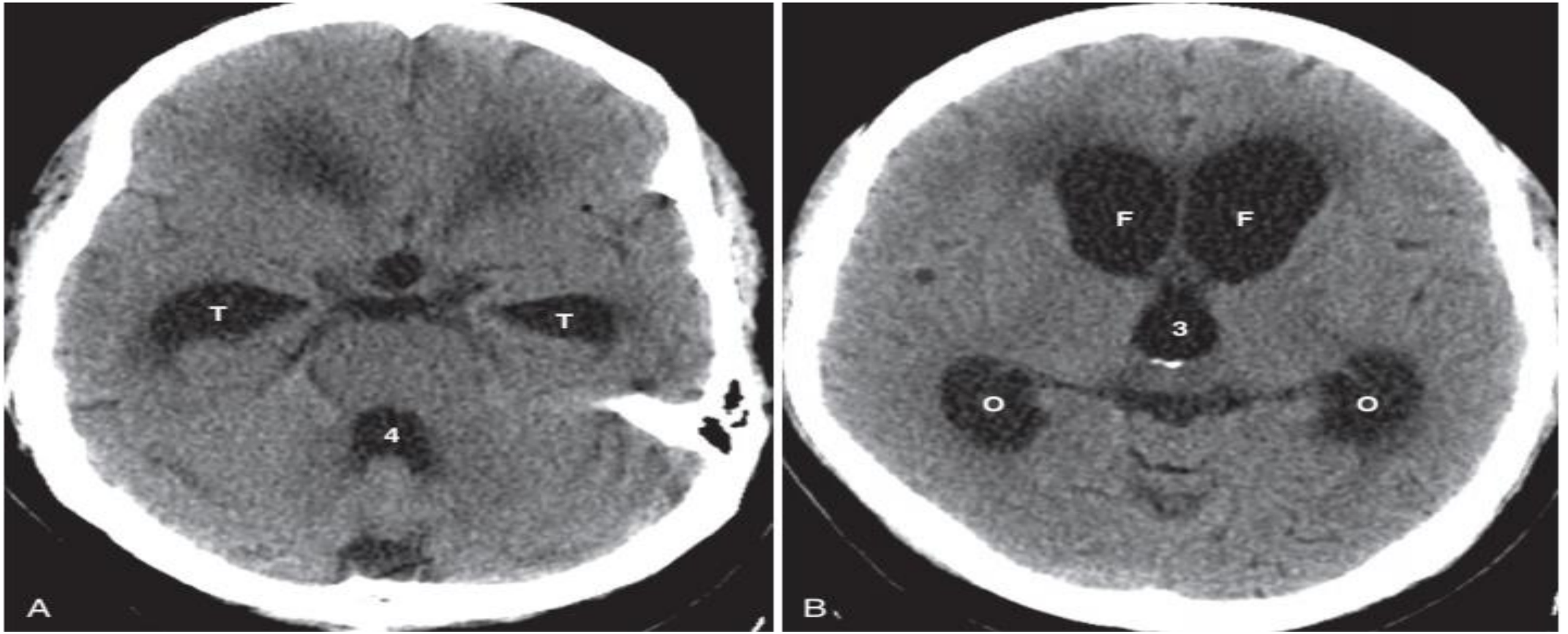
Glioblastoma multiforme. Unenhanced CT image shows heterogeneously hypodense mass with surrounding vasogenic edema. This involves both frontal lobes and crosses genu (G) of corpus callosum, producing butterfly appearance. There is compression and distortion of frontal horns of both lateral ventricles

Marginal CM enhancement
Brain Abscess

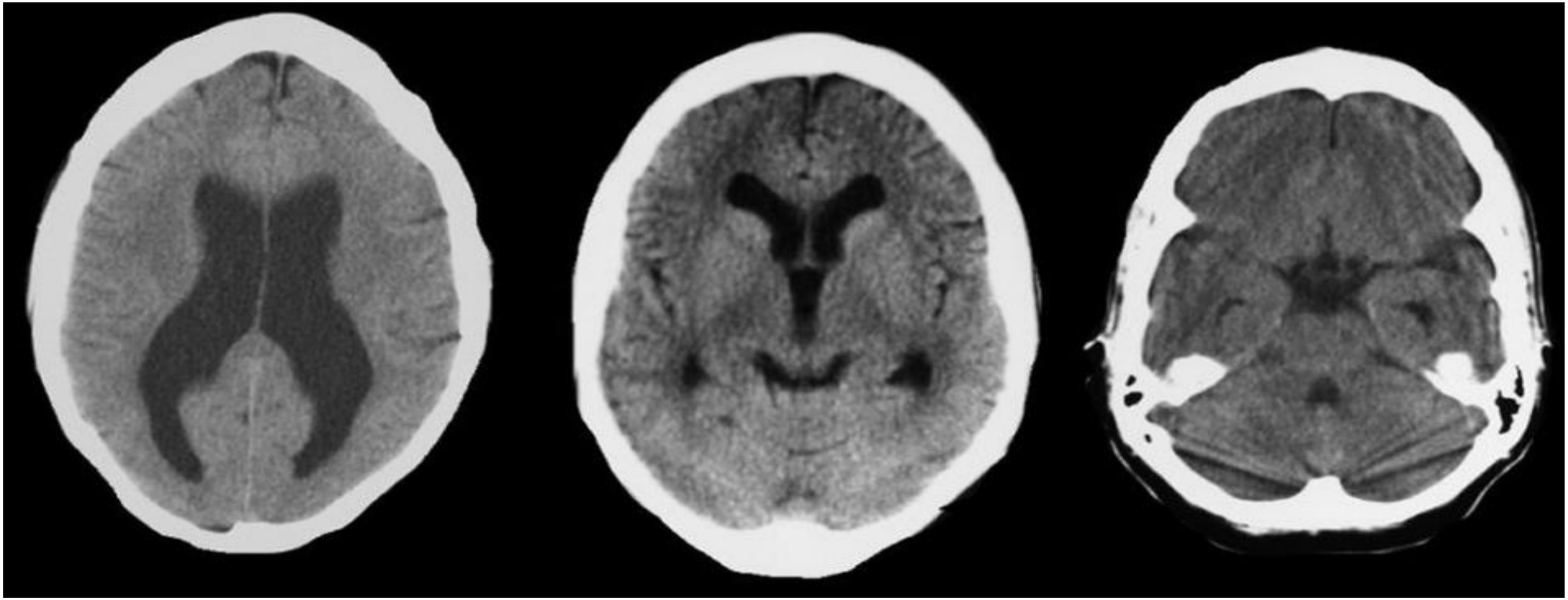


- **Hydrocephalus**
- Hydrocephalus is a condition in which an accumulation of cerebrospinal fluid (CSF) occurs within the brain.
- Hydrocephalus can be classified into communicating and noncommunicating (obstructive). Both forms can be either congenital or acquired.
- **Communicating Hydrocephalus**
- Communicating hydrocephalus, also known as nonobstructive hydrocephalus, is caused by impaired CSF reabsorption in the absence of any obstruction of CSF flow between the ventricles and subarachnoid space.

- **Noncommunicating Hydrocephalus**
- **Foramen of Monro** obstruction may lead to dilation of one, both lateral ventricles.
- **Aqueduct of Sylvius**, normally narrow, may be obstructed by a number of genetic or acquired lesions and lead to dilation of both lateral ventricles, as well as the third ventricle.
- **Fourth ventricle obstruction** leads to dilatation of the aqueduct, as well as the lateral and third ventricles.
- **Foramina of Luschka and foramen of Magendie** may be obstructed due to congenital malformation.



Communicating hydrocephalus. Communicating hydrocephalus is caused by abnormalities that inhibit the resorption of cerebrospinal fluid. A, Classically, the 4th ventricle is dilated in communicating hydrocephalus (4) but normal in size in noncommunicating hydrocephalus. The temporal horns (T) are particularly sensitive to increases in intraventricular volume or pressure and are dilated here. B, The frontal horns (F), occipital horns (O), and 3rd ventricle (3) are markedly dilated.

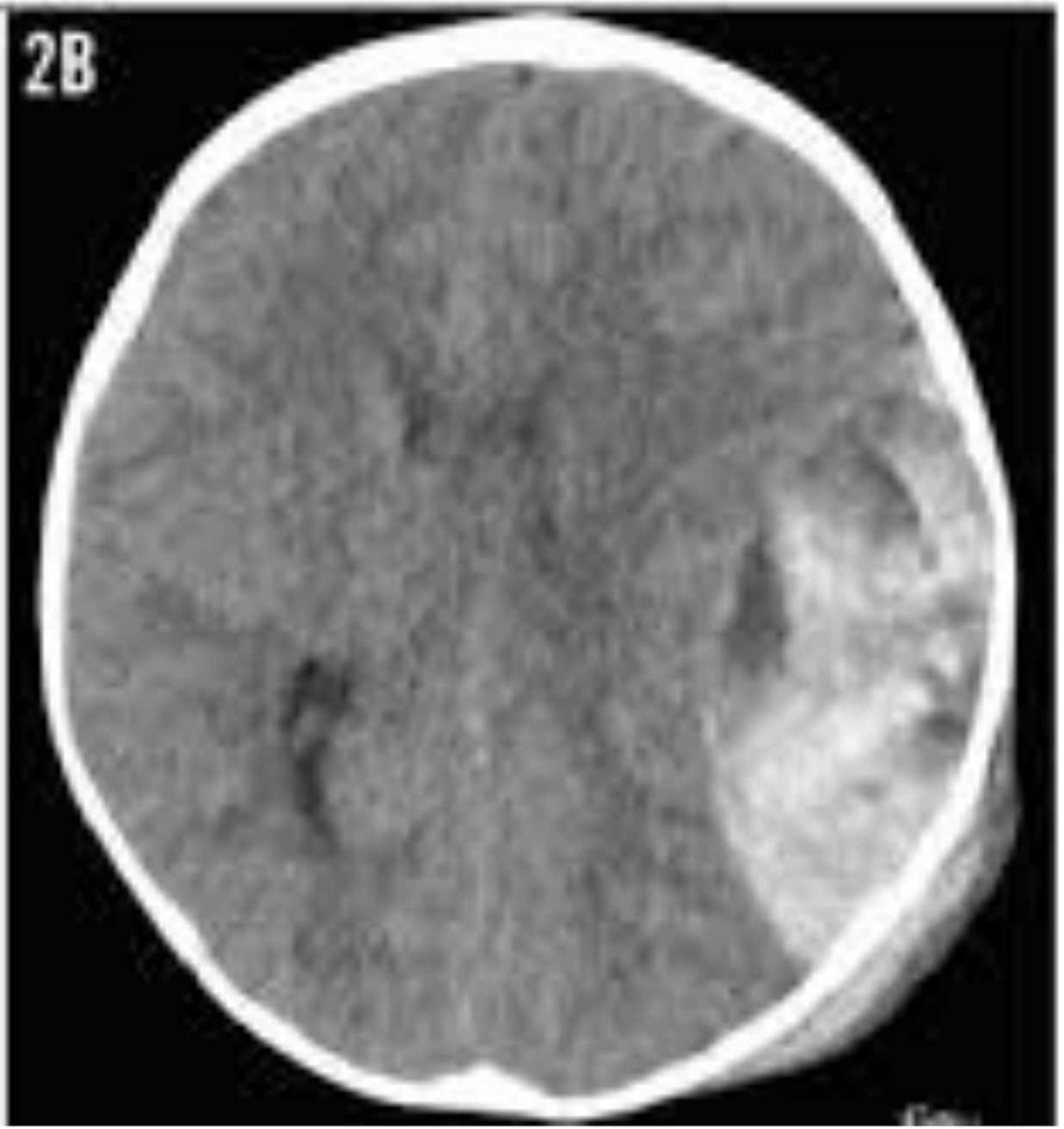


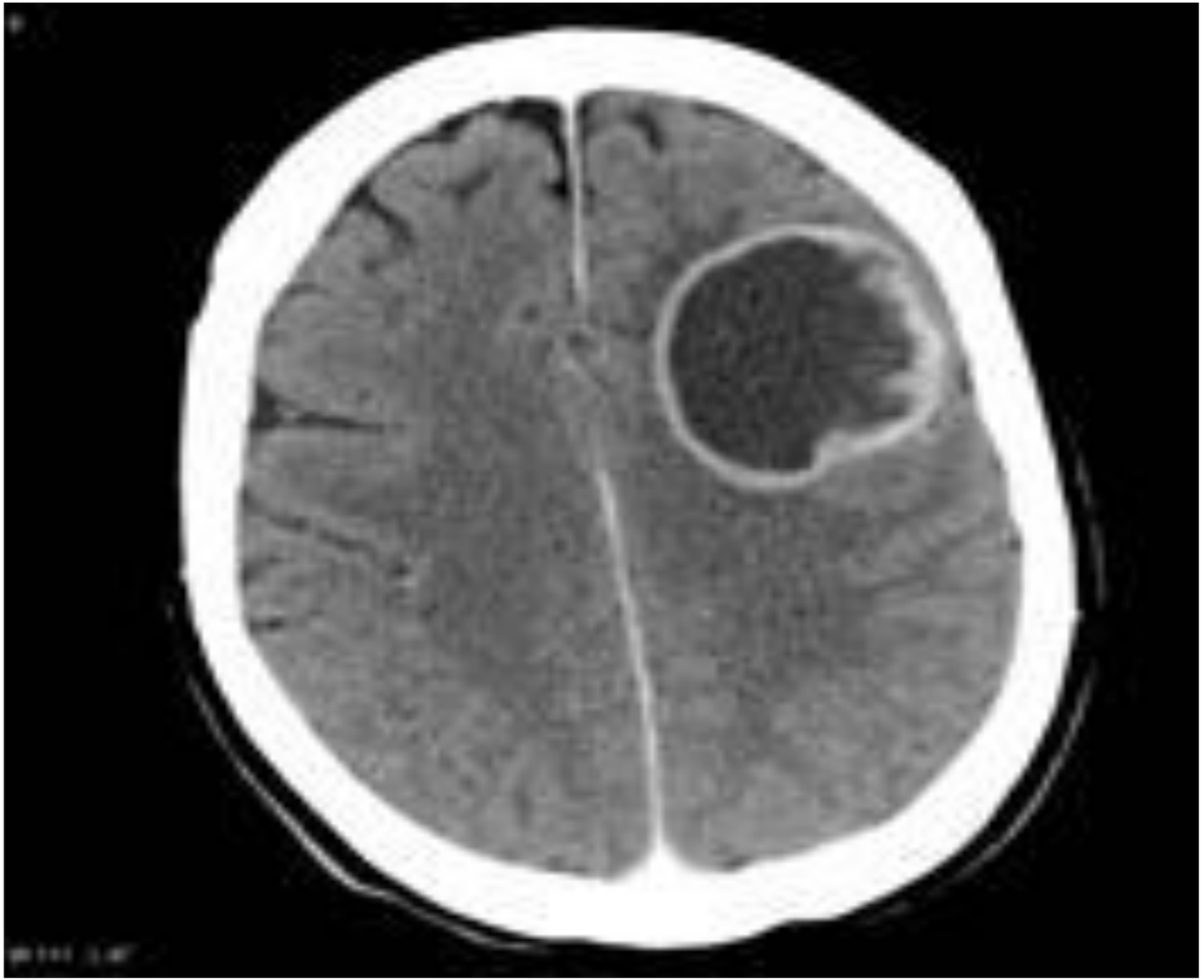
Non-Communicating Hydrocephalus: Axial CT scans. Note the prominent enlargement of the lateral and third ventricles in the setting of a normal sized fourth ventricle. This pattern is one of non-communicating (obstructive) hydrocephalus, which occurs from impaired drainage through the cerebral aqueduct which connects the third and fourth ventricles. This picture differs from communicating hydrocephalus wherein all the ventricles are enlarged. Note also that the frontal horns are enlarged - they extend beyond the head of the caudate nucleus and form somewhat of a picture of "mickey mouse" ears.

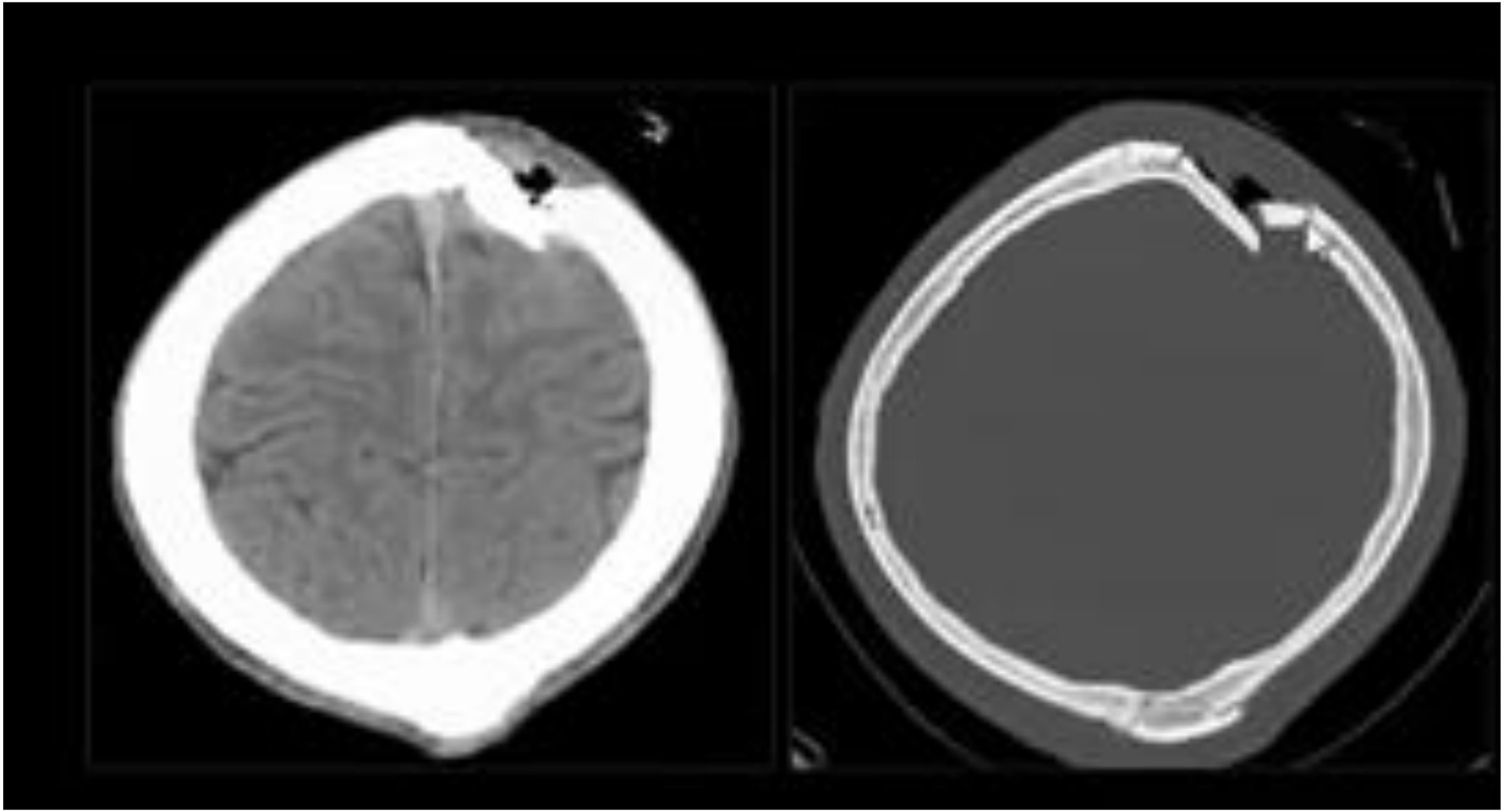
Group Discussion

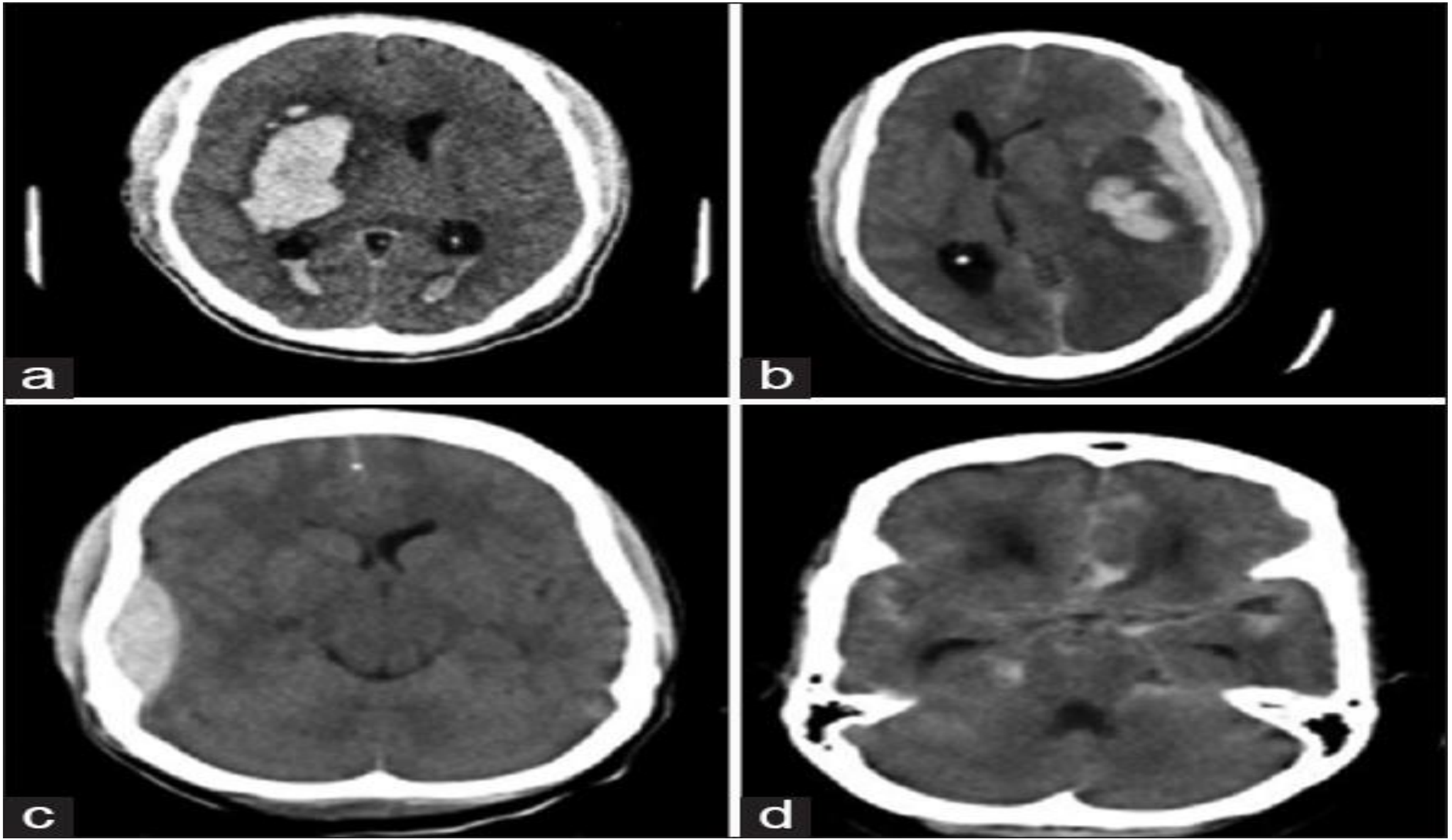
Give spot diagnosis for the following Images

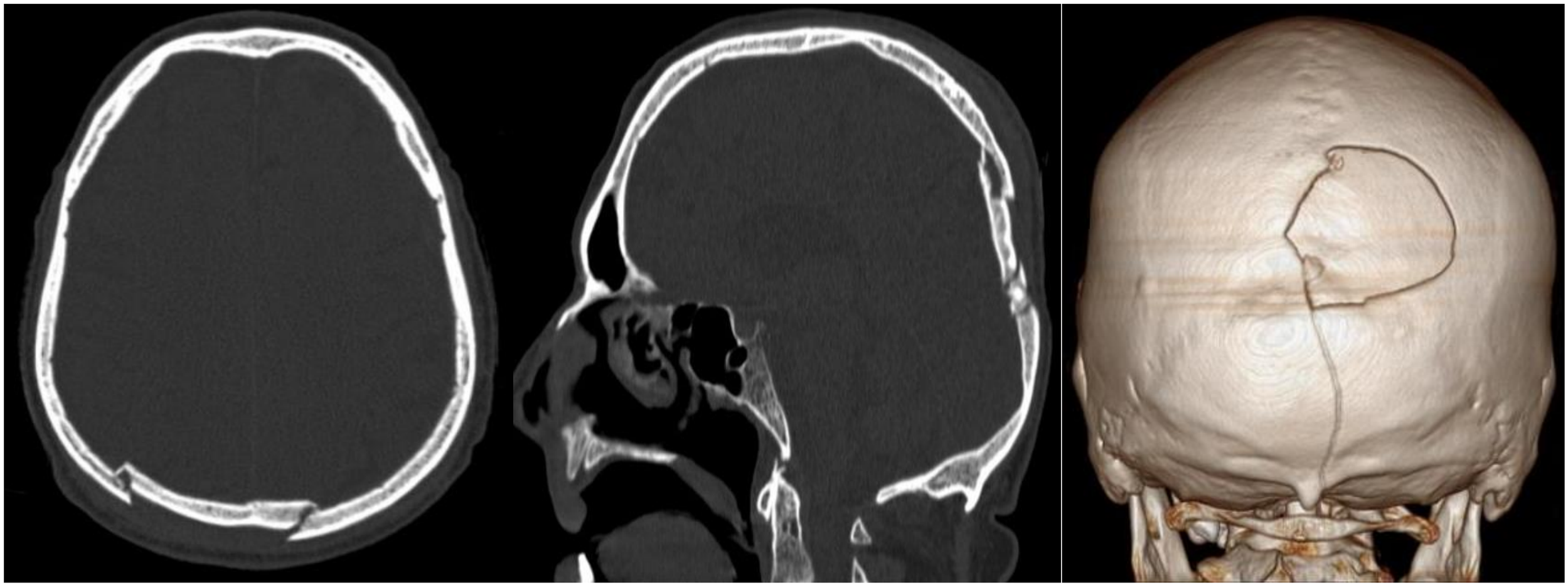
2A


2B

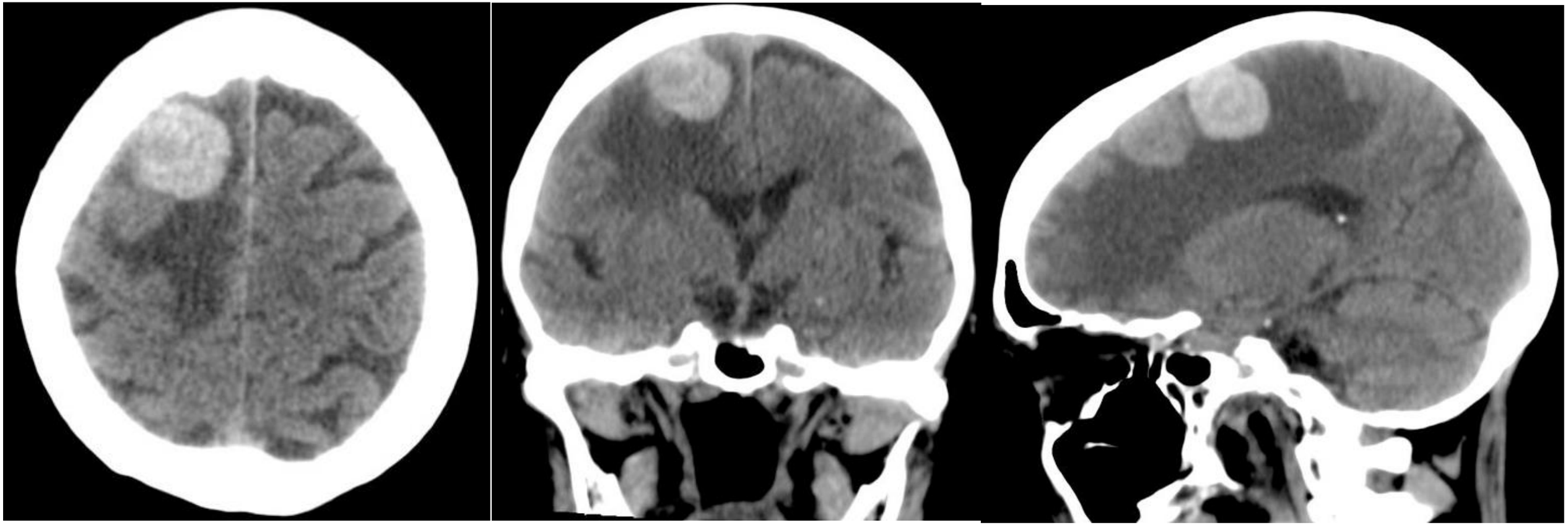





a
b
c
d

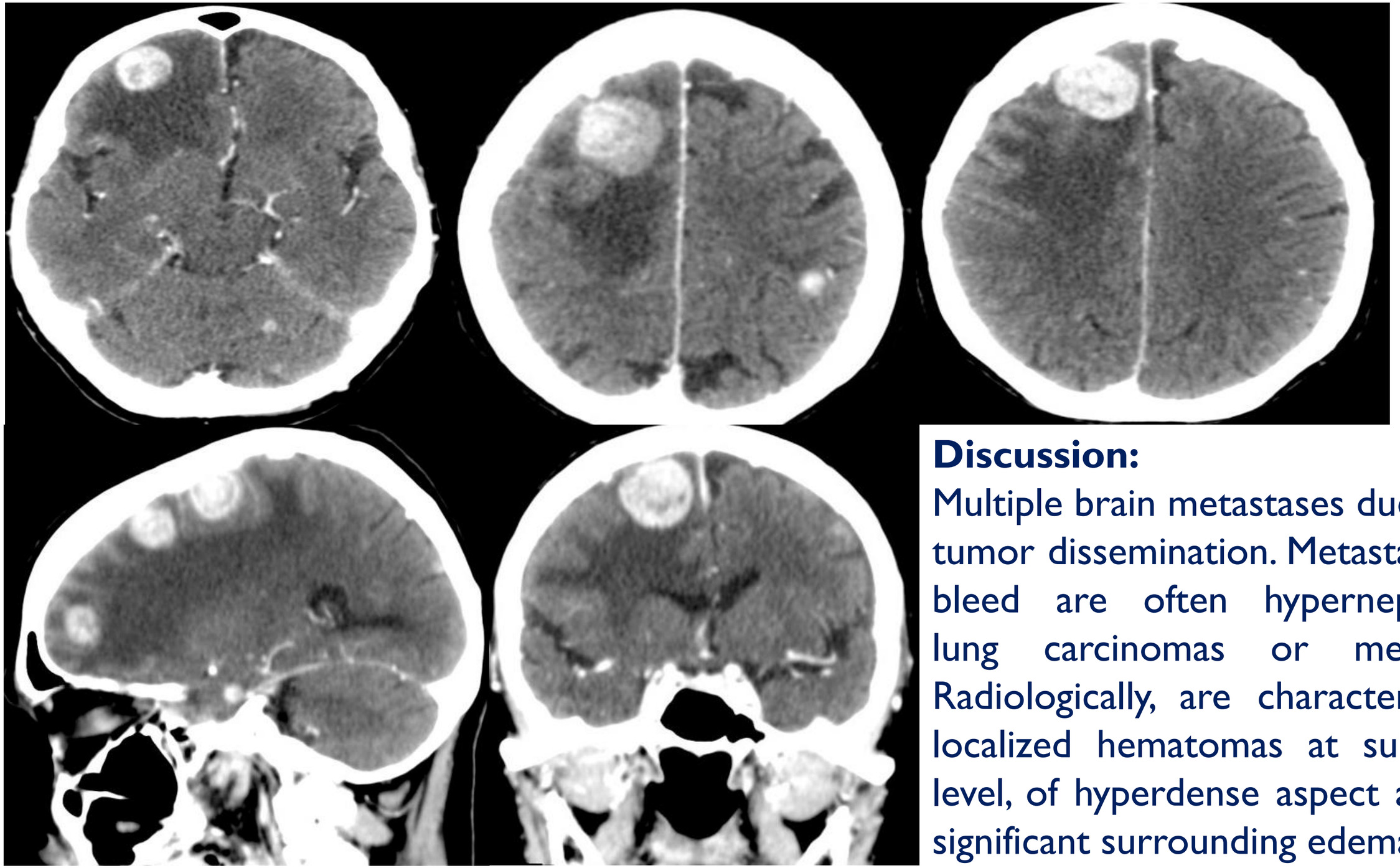


This case shows an depressed skull fracture which crosses the midline.



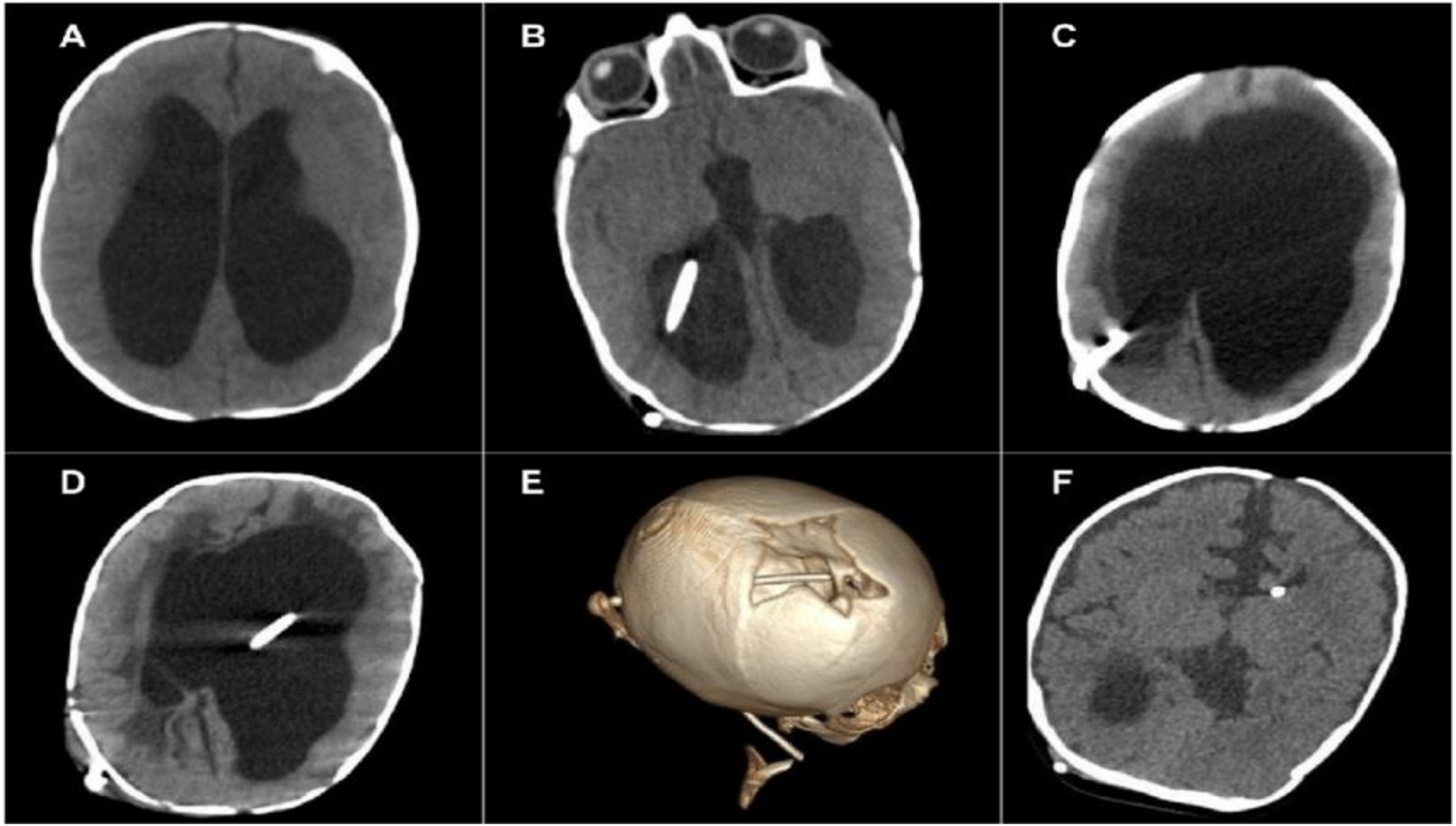
Presentation

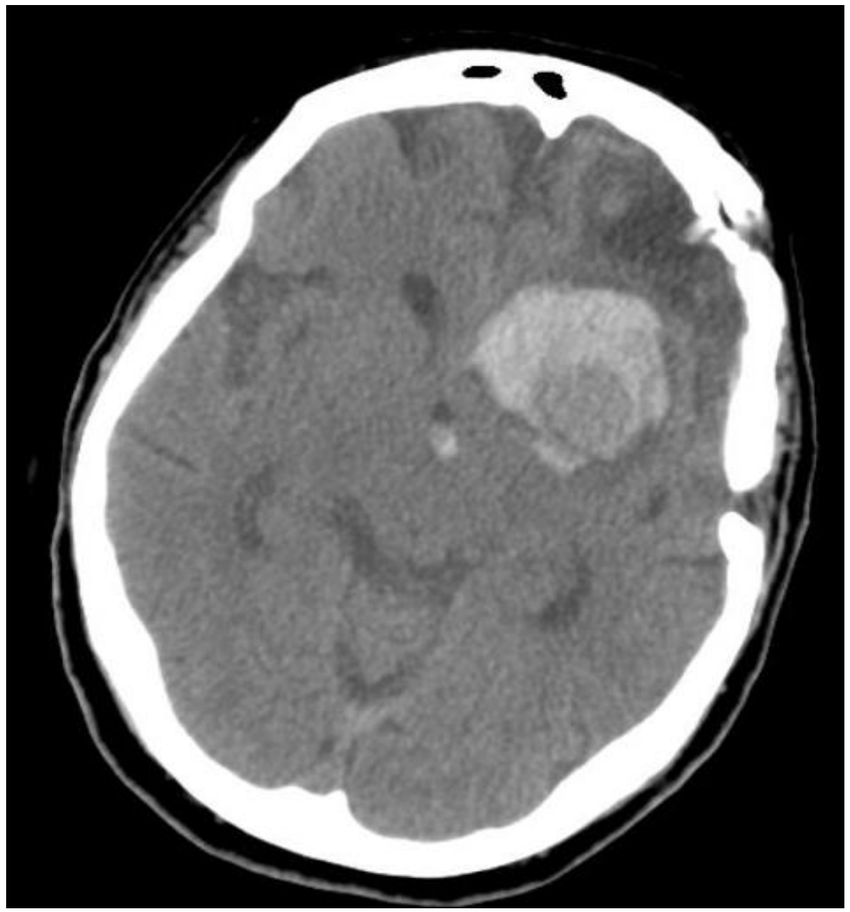
Female patient, 80 years, Impairment of consciousness, history of lung cancer

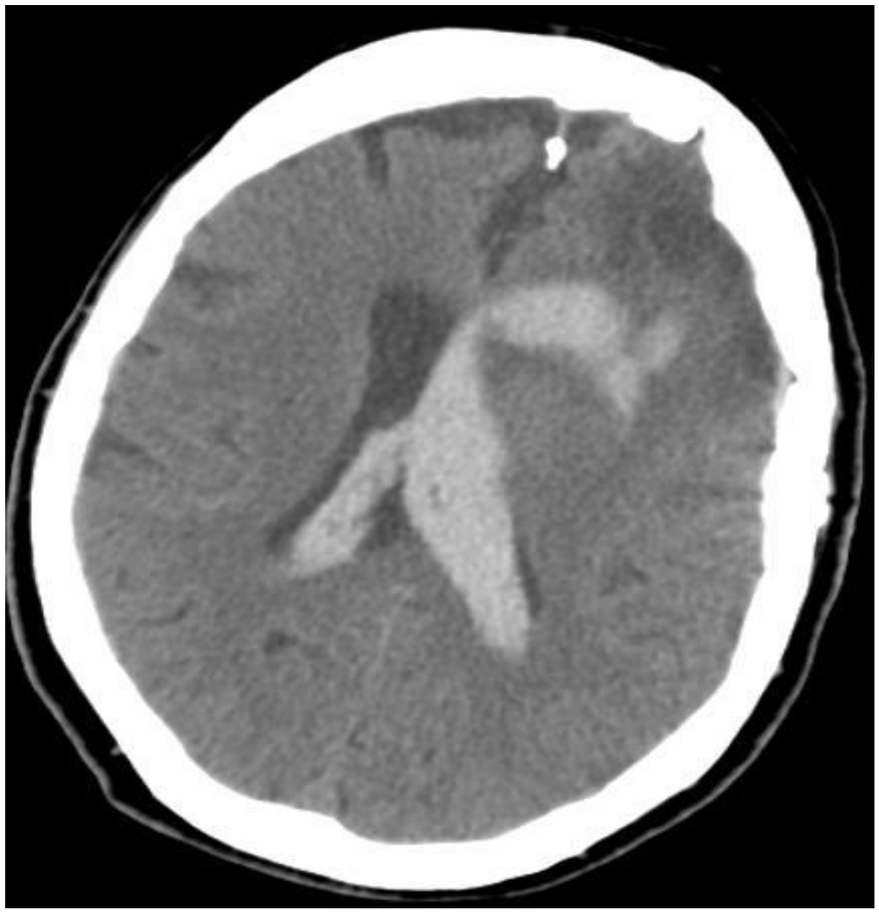


Discussion:

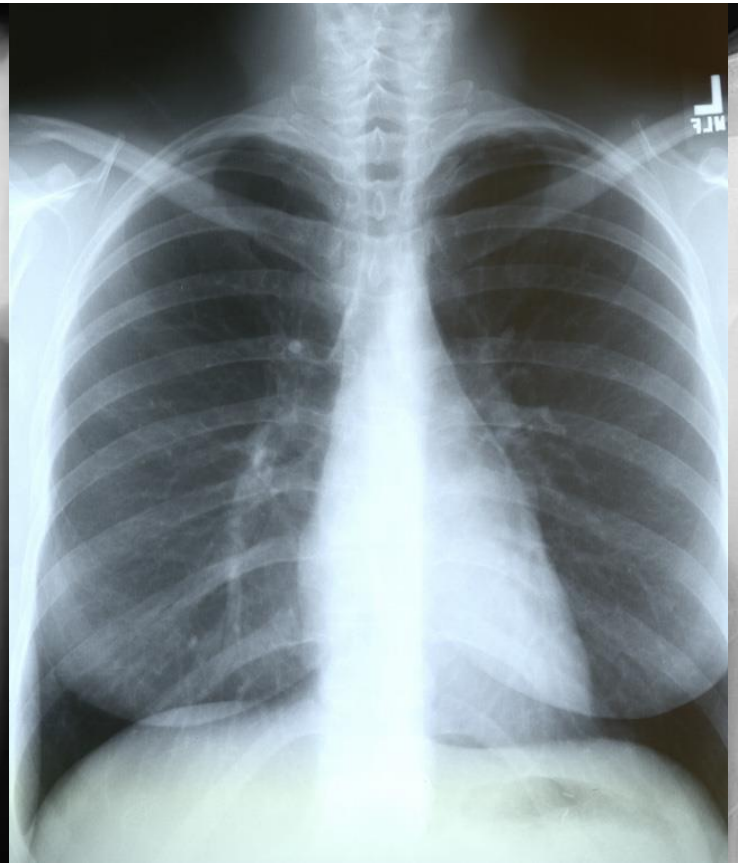
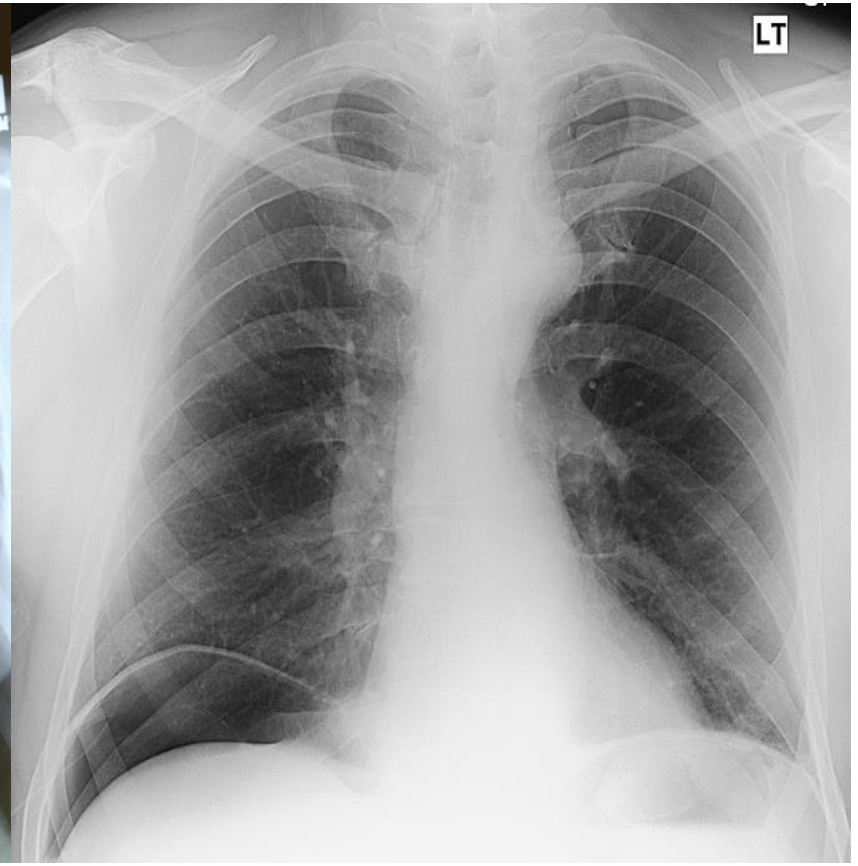
Multiple brain metastases due to lung tumor dissemination. Metastases that bleed are often hypernephromas, lung carcinomas or melanomas. Radiologically, are characterized by localized hematomas at subcortical level, of hyperdense aspect and with significant surrounding edema.


A
B
C
D
E
F



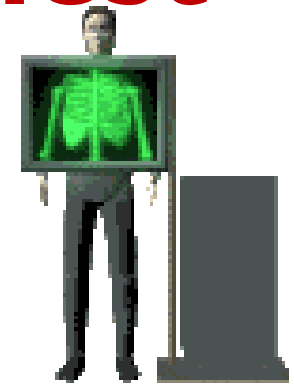


Thank
you

Diagnostic Radiology of Chest

Lecturer: Husam Mansour
Master of Public Health
Master of Medical Imaging



Part I

Criteria of evaluation of chest X-ray 12-Step

1: Name
2: Date
3: Old films
4: What type of view(s)

} **Pre-read**

5: Penetration
6: Inspiration
7: Rotation
8: Angulation

} **Quality Control**

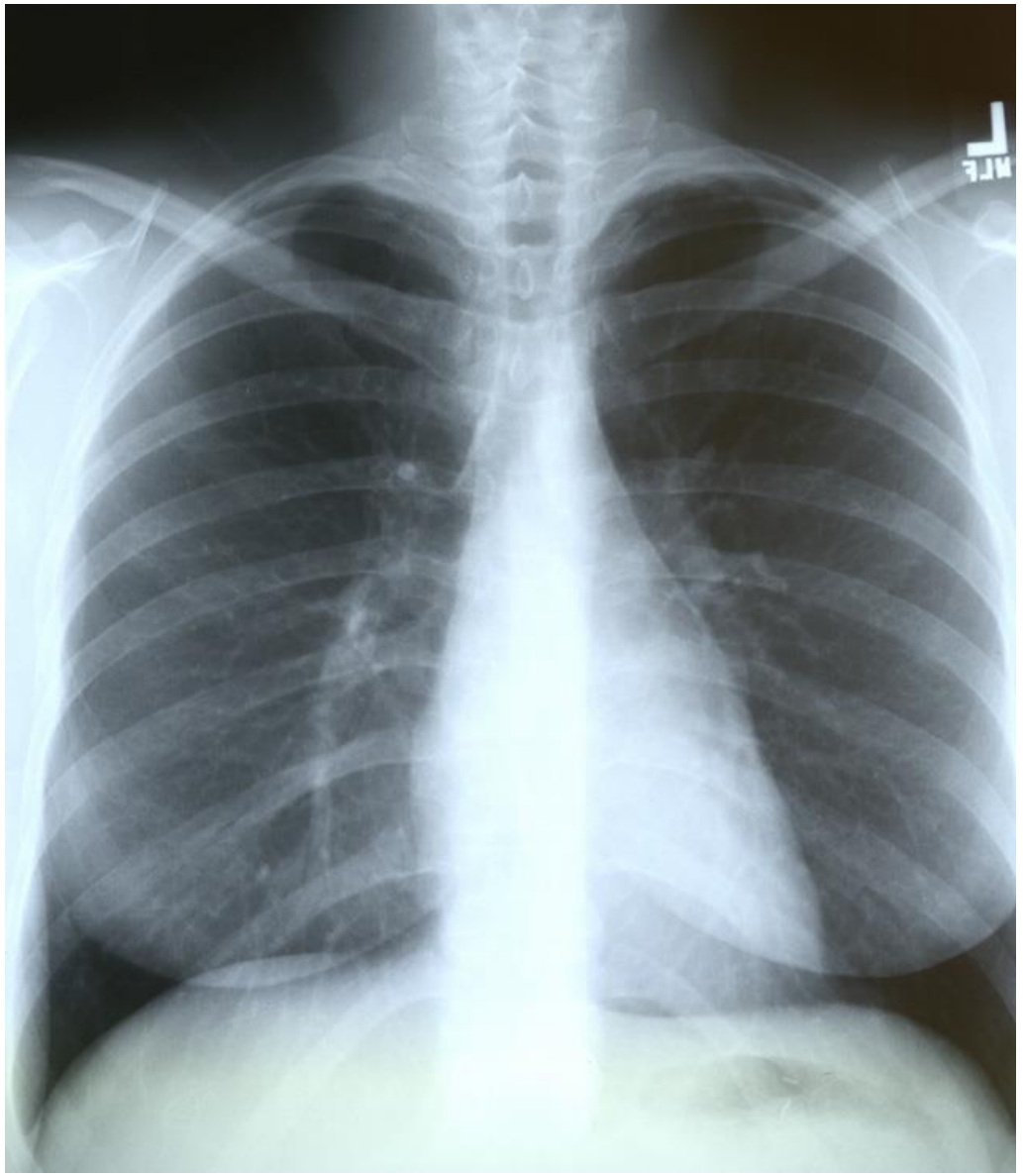
9: Soft tissues / bony structures
10: Mediastinum
11: Diaphragms
12: Lung Fields

} **Findings**

5. Penetration

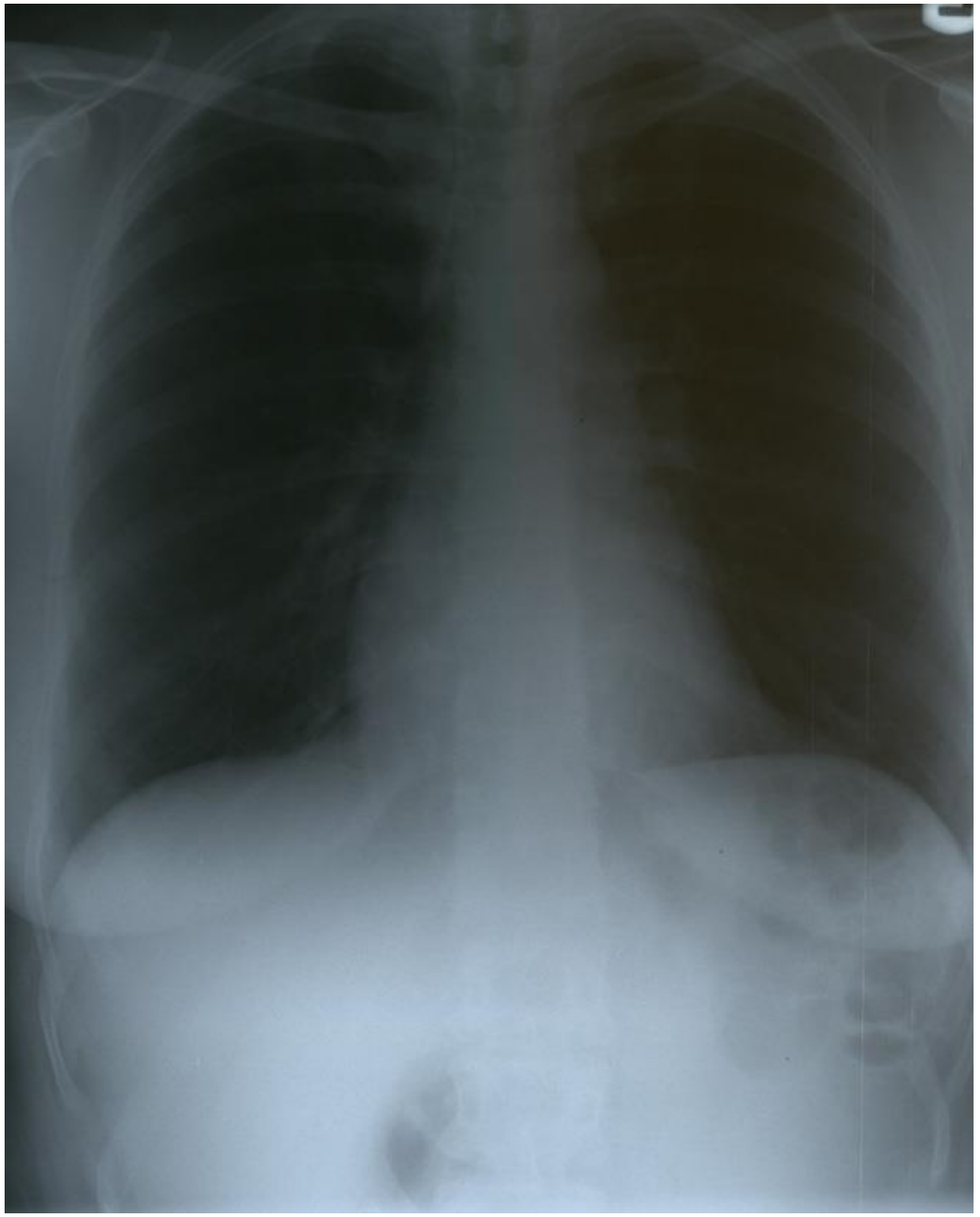
- Refers to adequate photons traversing the patient to expose the radiograph.
- This is often limited in patients of large size such that there is poor visualization of structures in the lower lung fields and in a retro-cardiac location.
- The lack of penetration renders the area "whiter" than with an adequate film and can simulate pneumonia or effusion.
- In an ideal radiograph the thoracic spine should be barely perceptual viewing through the cardiac silhouette.
- The soft tissues at the shoulder can also give an estimate of the relative degree of penetration of the film.

- Should see ribs through the heart
- Barely see the spine through the heart
- Should see pulmonary vessels nearly to the edges of the lungs



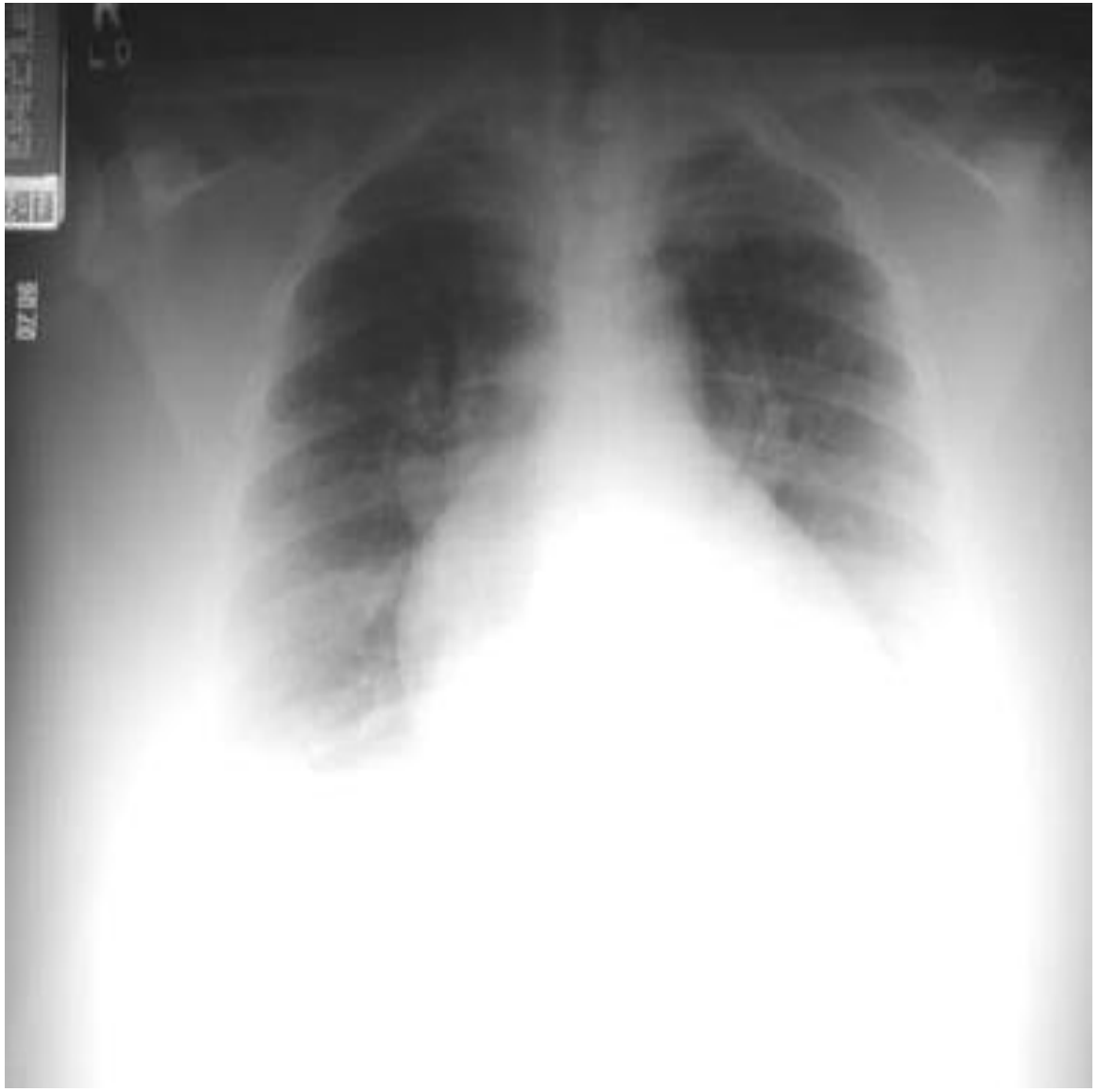
Over penetrated Film

- Lung fields darker than normal—may obscure subtle pathologies
- See spine well beyond the diaphragms
- Inadequate lung detail



Underpenetrated Film

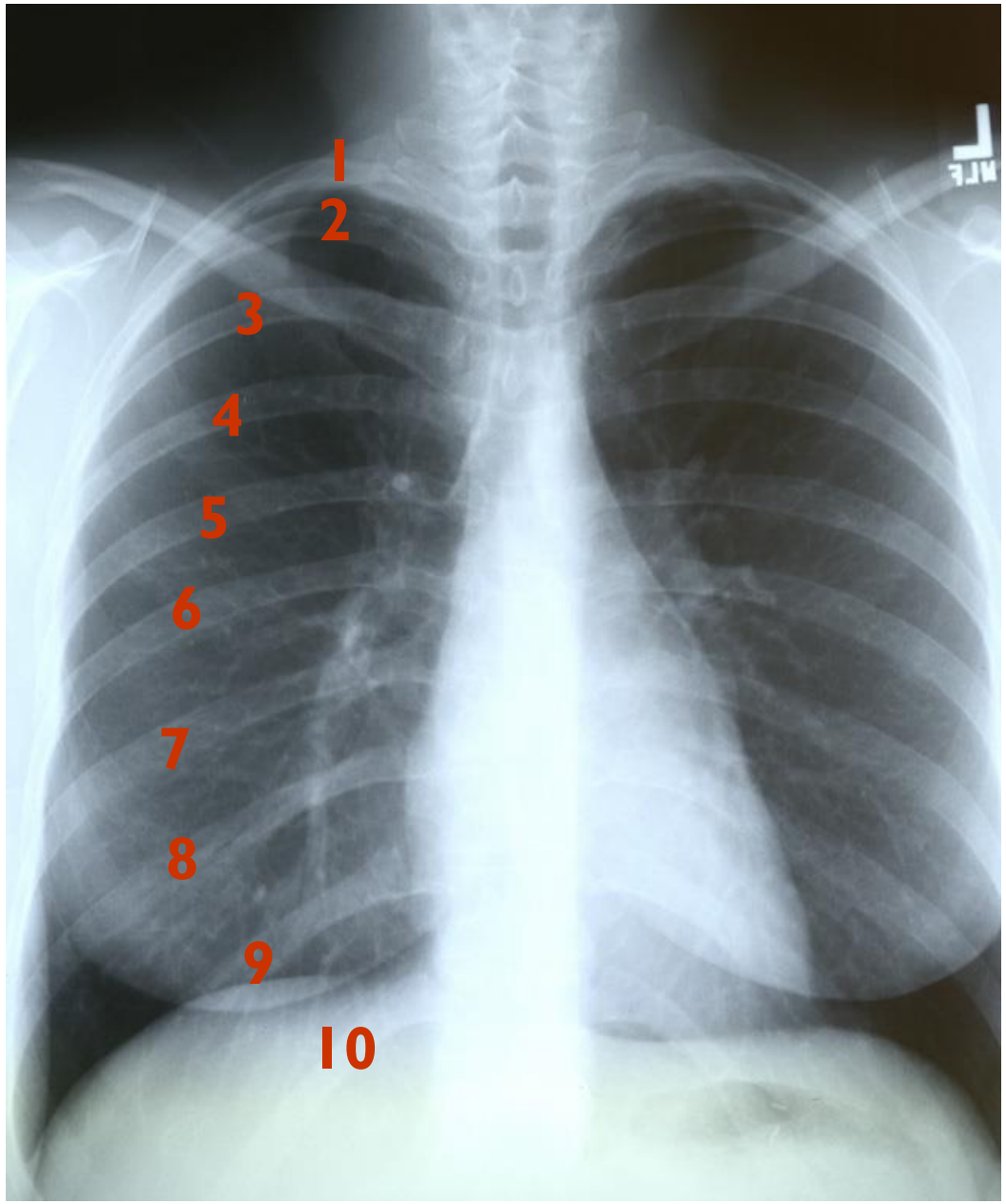
- Hemidiaphragms are obscured
- Pulmonary markings more prominent than they actually are



6. Inspiration

- The volume of air in the hemithorax will affect the configuration of the heart with question of cardiac enlargement with a shallow level of inspiration.
- The vascular pattern in the lung fields will be accentuated with a shallow inspiration since the same amount of blood flow is now distributed to a smaller volume of lung.
- The level of inspiration can be estimated by counting ribs.
- Visualization of nine posterior ribs, or seven anterior ribs on an upright PA radiograph projecting above the diaphragm would indicate a satisfactory inspiration.

- Should be able to count 9-10 posterior ribs
- Heart shadow should not be hidden by the diaphragm



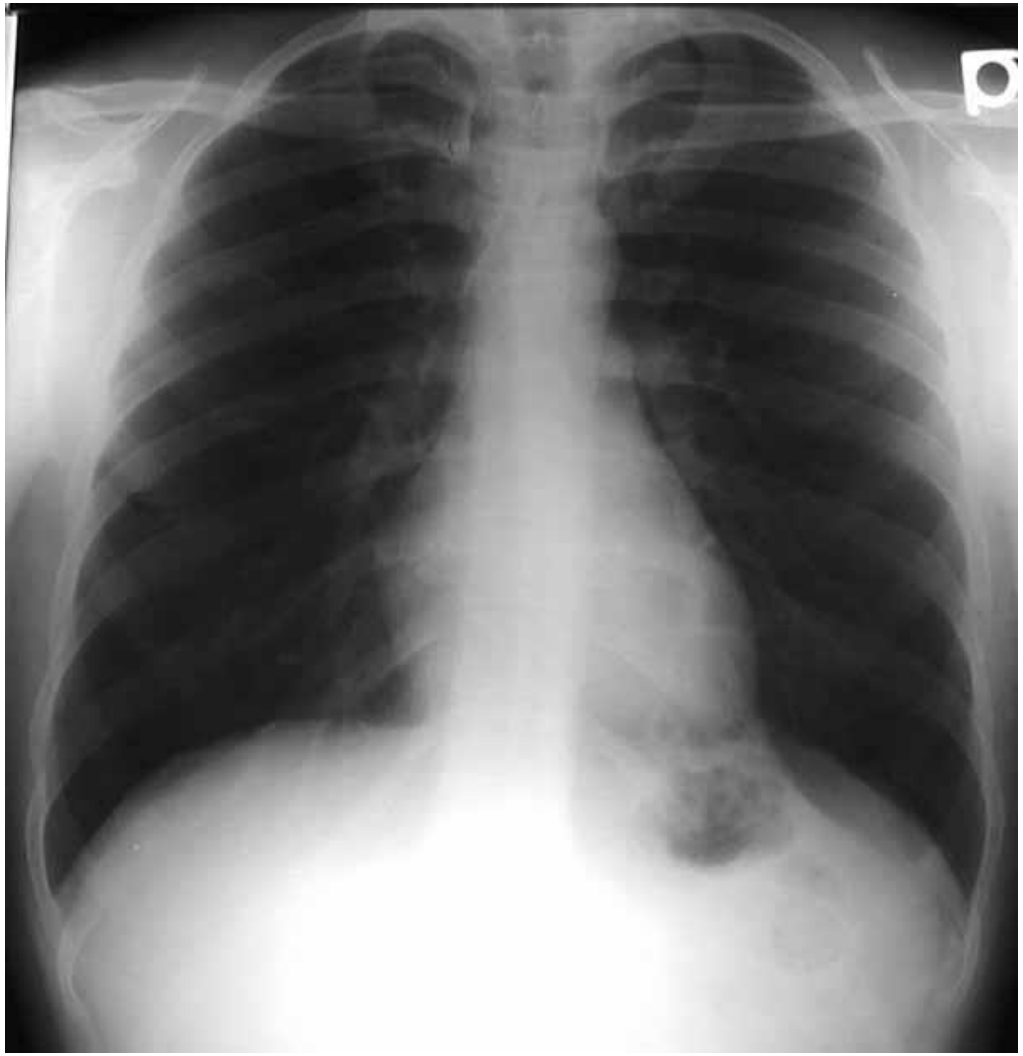
About 8 posterior ribs are showing

9-10 posterior ribs are showing

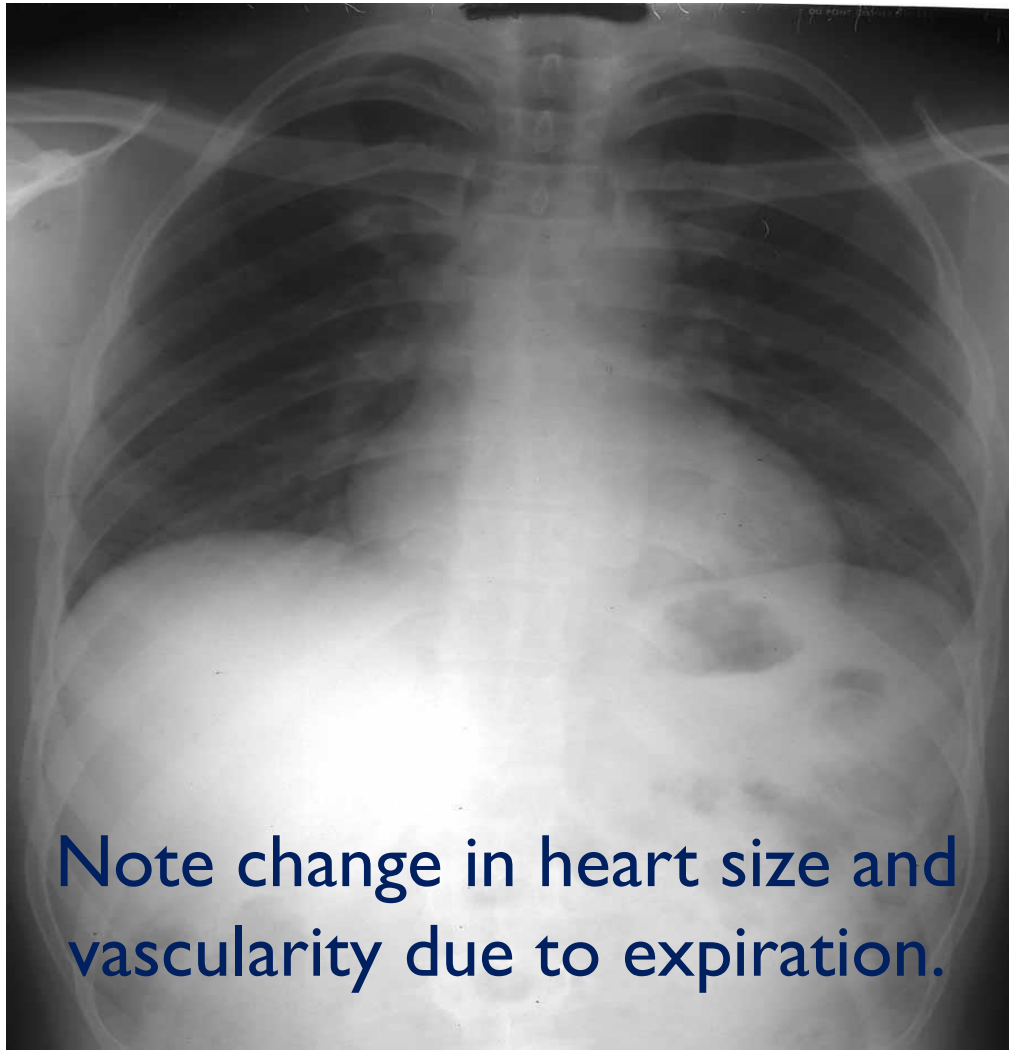
Poor inspiration can crowd lung markings producing pseudo-airspace disease

With better inspiration, the "disease process" at the lung bases has cleared

Inspiration



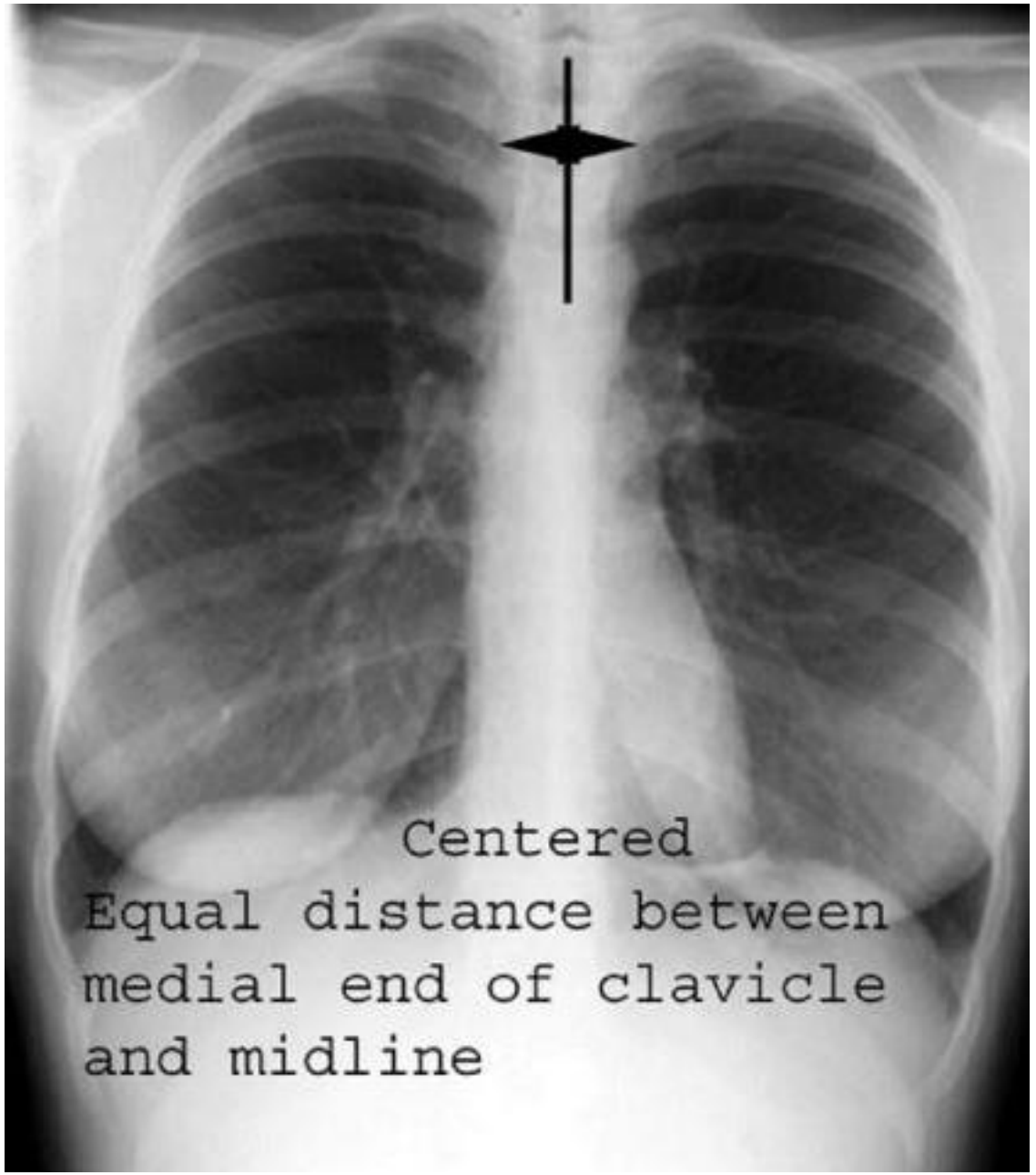
Expiration

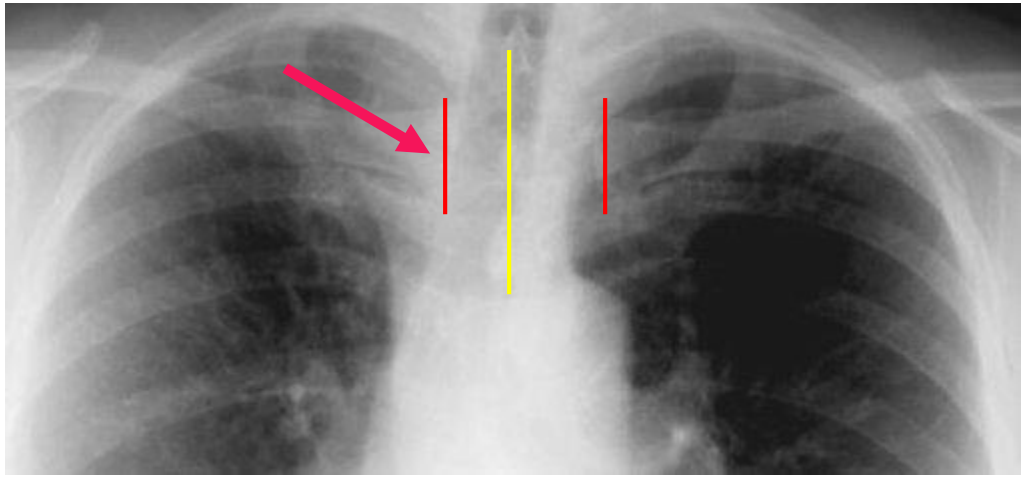


7. Rotation

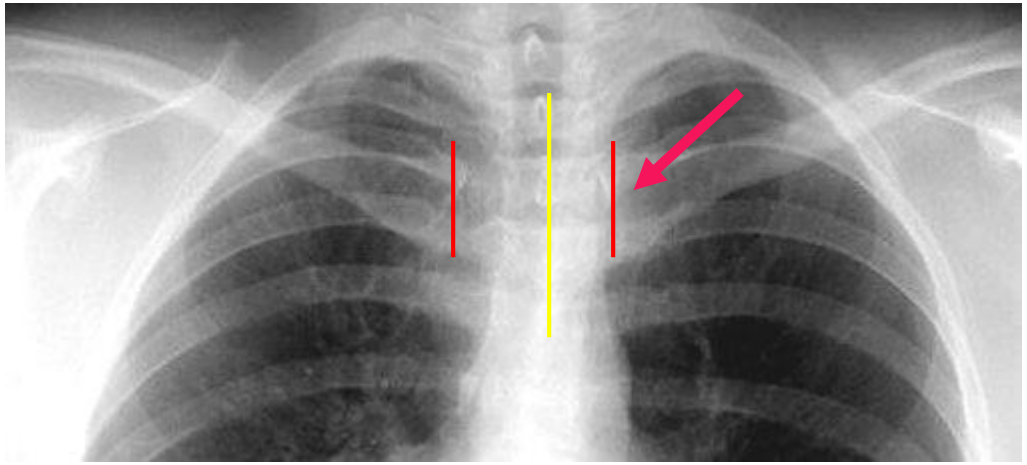
- Rotation of the patient distorts mediastinal anatomy and makes assessment of cardiac chambers and the hilar structures especially difficult.
- Chest wall tissue also contributes to increased density over the lower lobe fields simulating disease.
- Rotation of the radiograph is assessed by judging the position of the clavicle heads and the thoracic spinous process.
- Ideally the clavicle heads should be equidistant from the spinous process.

- Medial ends of bilateral clavicles are equidistant from the midline or vertebral bodies





If spinous process appears closer to the right clavicle (red arrow), the patient is rotated toward their own left side

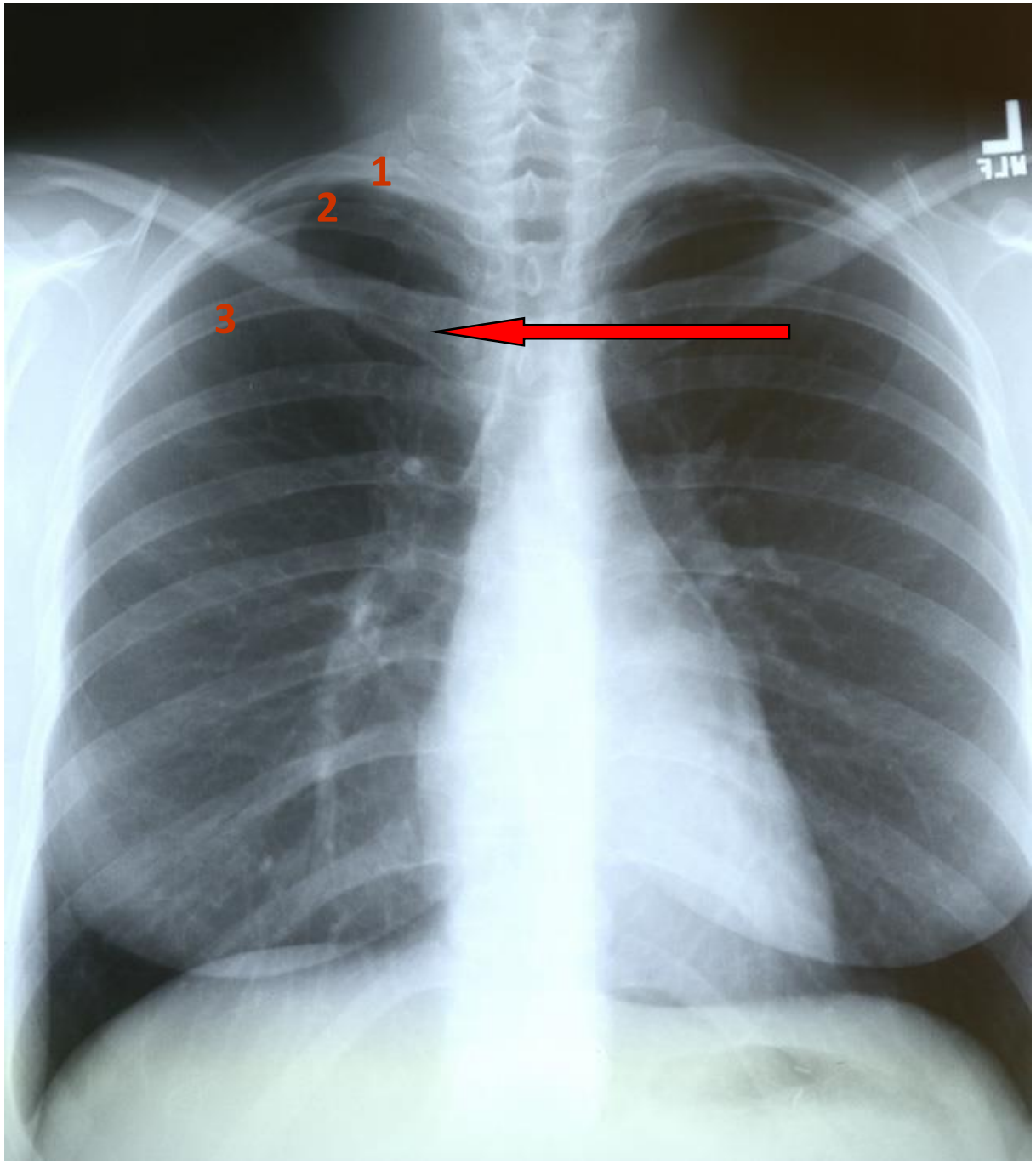


If spinous process appears closer to the left clavicle (red arrow), the patient is rotated toward their own right side

8. Angulation

- With the patient in a more lordotic projection the clavicles will project superiorly relative to the upper thorax again causing some distortion of the normal mediastinal anatomy.
- With the lordotic projection of the ribs assume a more horizontal orientation.
- Occasionally a lordotic x-ray can be obtained intentionally to better visualize structures in the thoracic apex obscured by overlying boney structures.

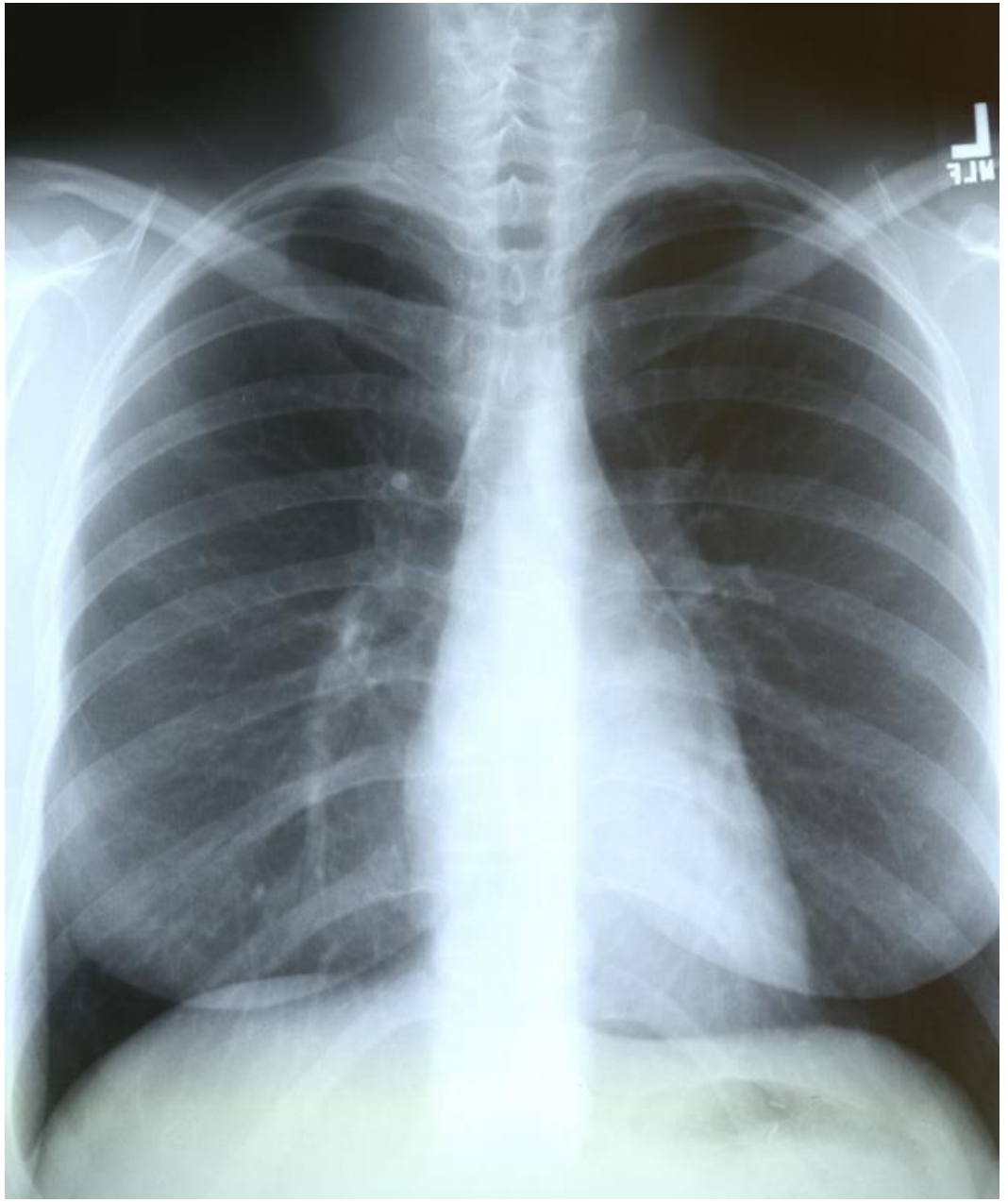
- Clavicle should lay over 3rd rib



Findings

9. Soft tissue and bony structures

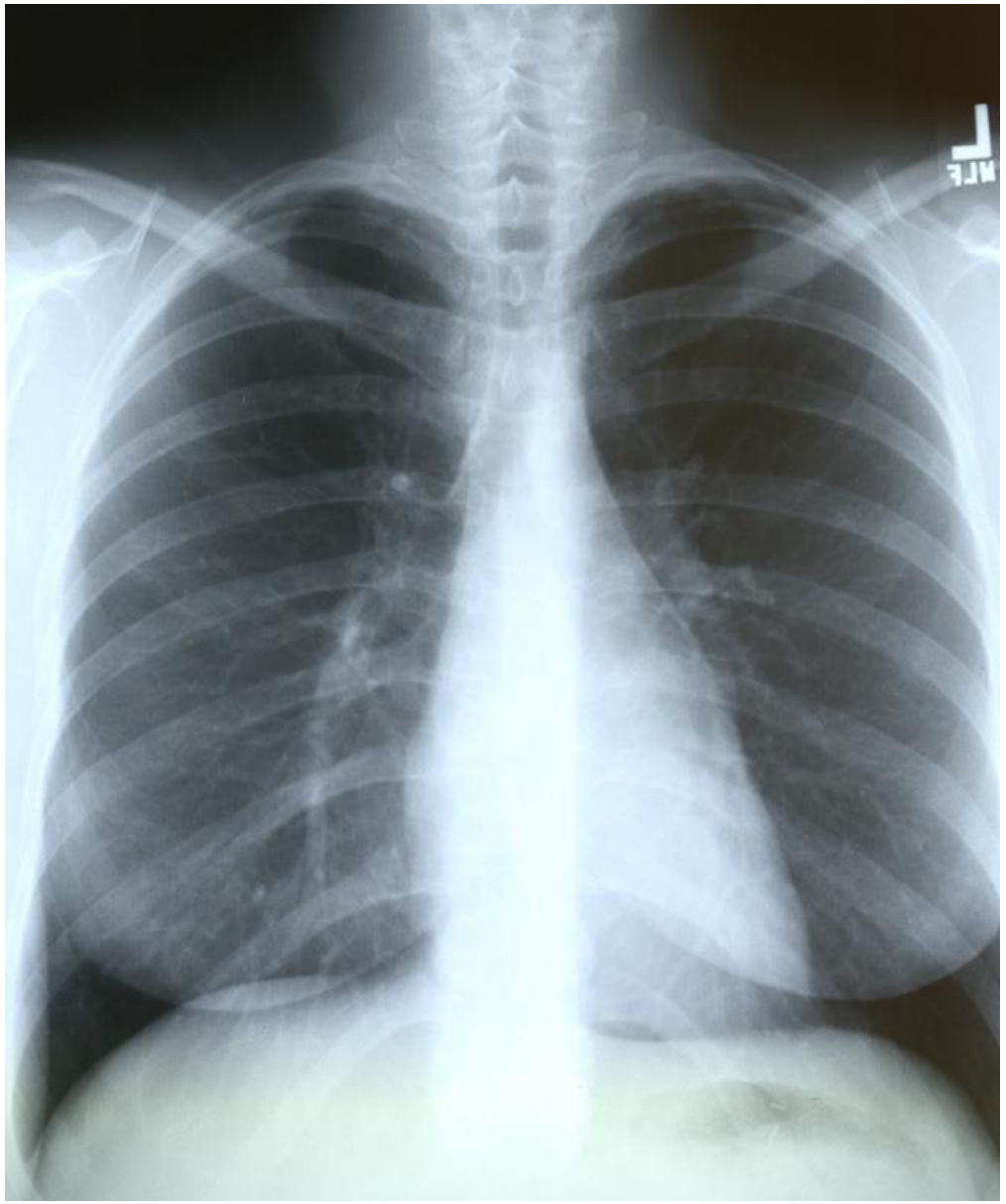
- Check for
  - Symmetry
  - Deformities
  - Fractures
  - Masses
  - Calcifications
  - Lytic lesions



Findings

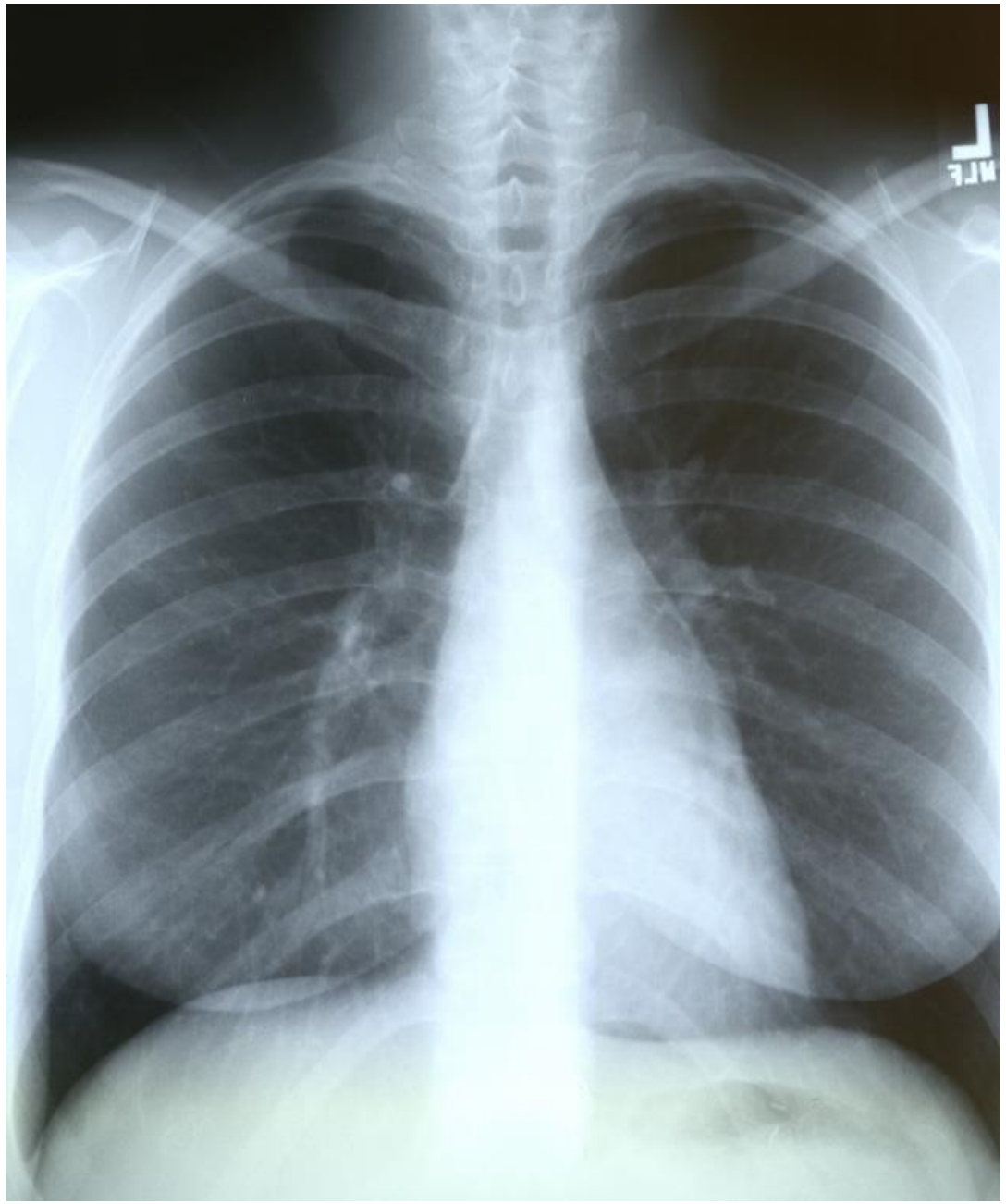
10. Mediastinum

- Check for
  - Cardiomegaly
  - Mediastinal and Hilar contours for increase densities or deformities



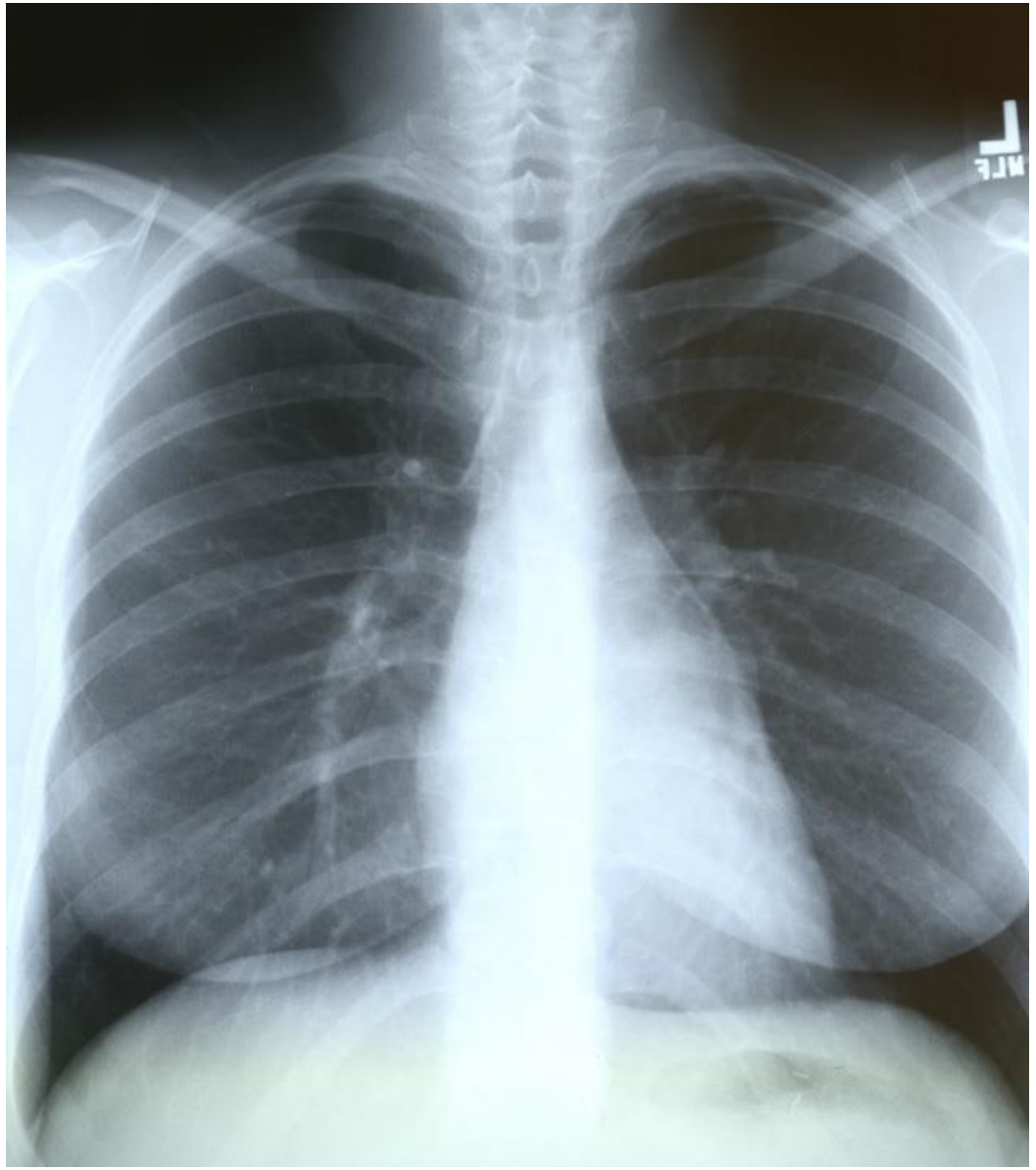
11. Diaphragms

- Check sharpness of borders
- Right is normally higher than left
- Check for free air, gastric bubble, pleural effusions



12. The Lung Fields!

- To help you determine abnormalities and their location…
- Use silhouettes of other thoracic structures
- Use fissures



Lung Fields: Fissures

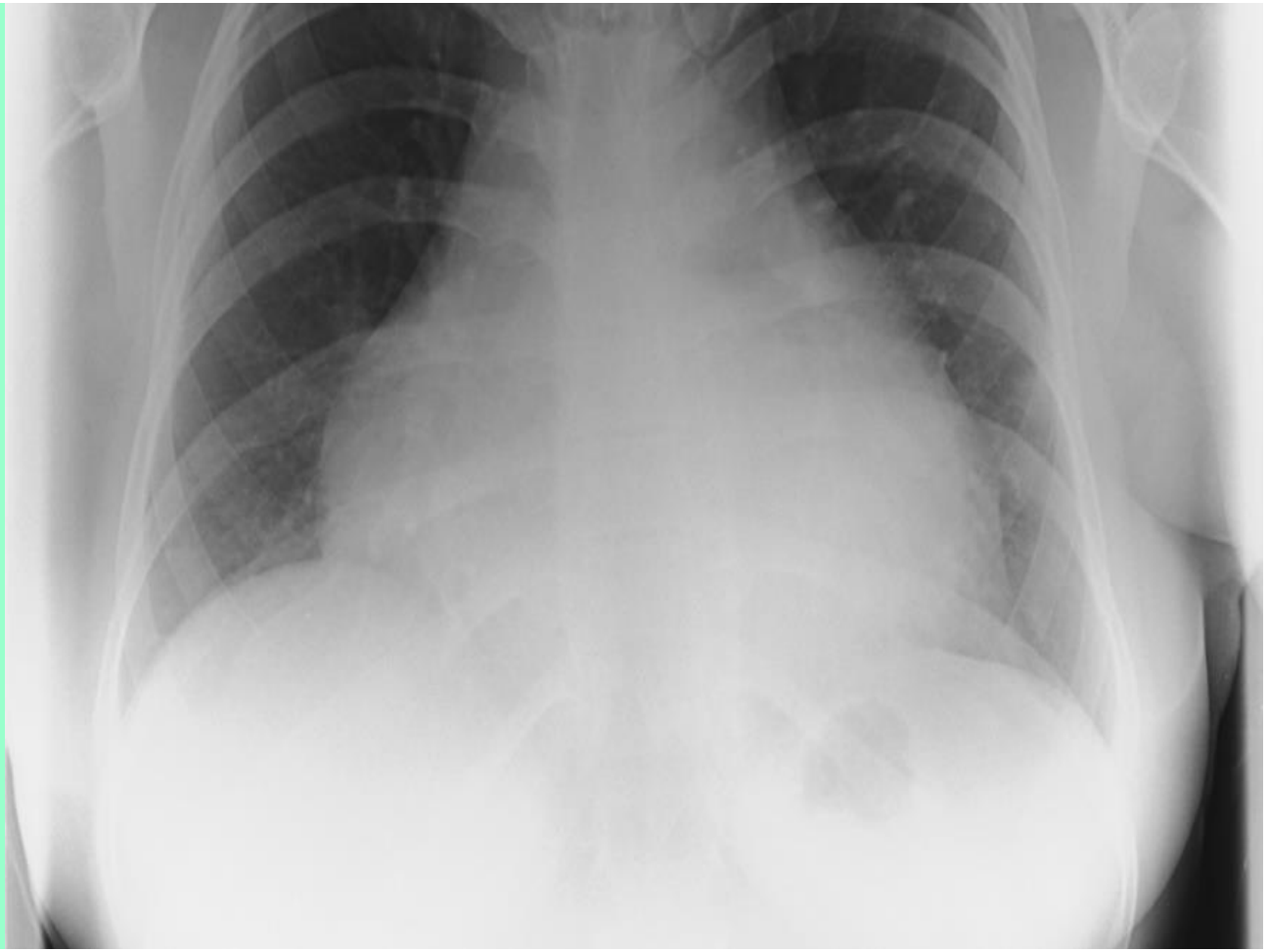
- The fissures can also help you to determine the boundaries of pathology

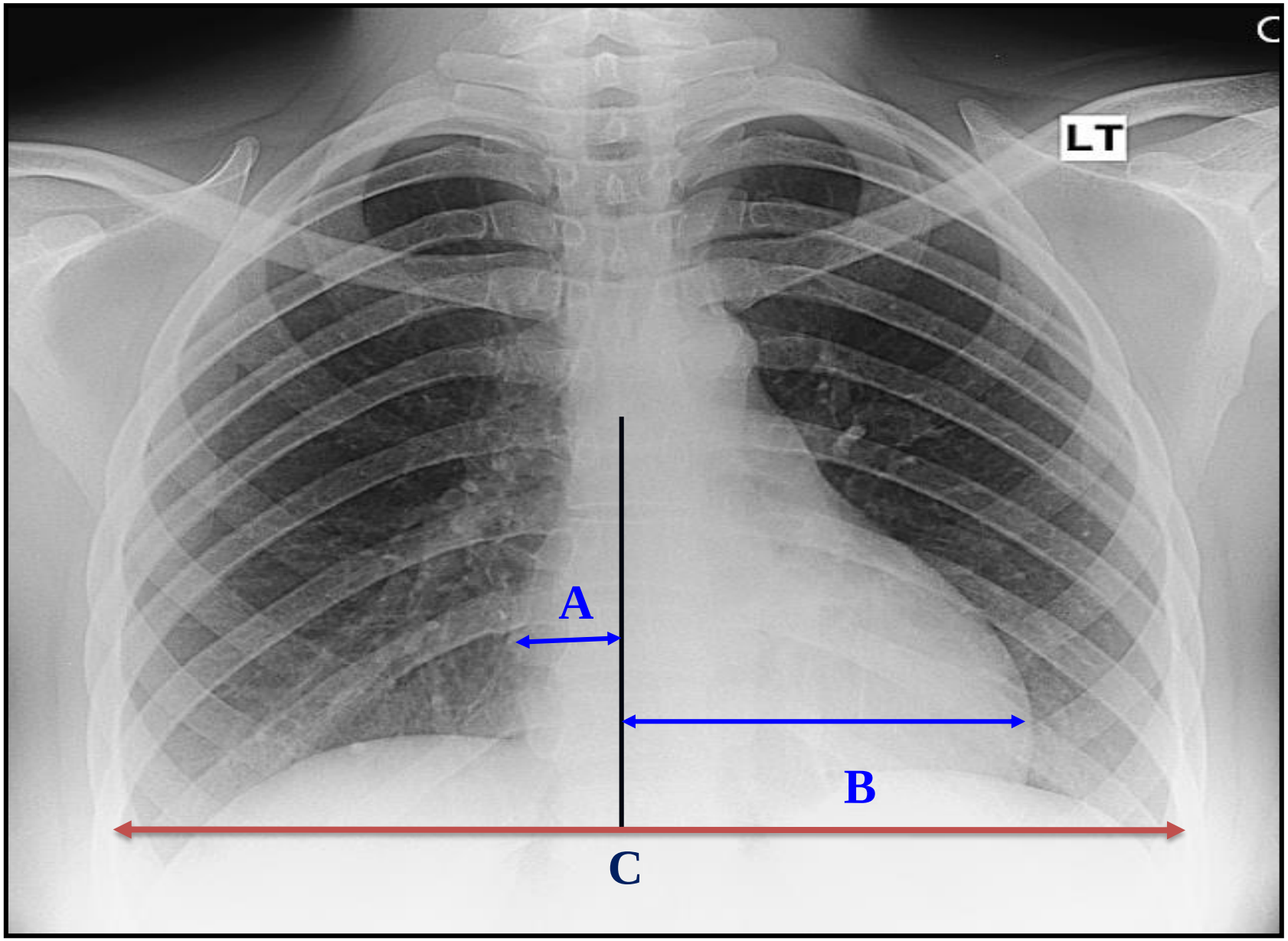
Major Oblique Fissure	Separates the LUL from the LLL
Right Major Fissure	Separates the RUL/RML from the RLL
Right Minor Fissure	Separates the RUL from the RML

Cardiomegaly

Causes:

1. Heart failure.
2. Cardiomyopathy.
3. Volvulus disease.
4. Congenital heart diseases.
5. Pericardial diseases.





Heart size - Cardiothoracic Ratio (CTR)

A+B/C

Air under diaphragm

Causes:

1. Post operative.
2. Chaladety sign (colon under RT diaphragm)
3. Perforated vescus.
4. Peritonitis with gas productive organism.
5. Subhepatic abscess